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Parton Physics on Euclidean Lattice

Xiangdong Ji^{1,2}

¹*INPAC, Department of Physics and Astronomy,
Shanghai Jiao Tong University, Shanghai, 200240, P. R. China*

²*Maryland Center for Fundamental Physics,
Department of Physics, University of Maryland,
College Park, Maryland 20742, USA*

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Abstract

I show that the parton physics related to correlations of quarks and gluons on the light-cone can be studied through the matrix elements of frame-dependent, equal-time correlators in the large momentum limit. This observation allows practical calculations of parton properties on an Euclidean lattice. As an example, I demonstrate how to recover the leading-twist quark distribution by boosting an equal-time correlator to a large momentum.

Since deep-inelastic scattering experiments at the Stanford Linear Accelerator Center in late 1960's, the proton structure has been probed in various hard scattering processes [1]. The results are the quark and gluon (parton) momentum distributions and correlations, as well as distribution amplitudes, etc. The parton physics involves intrinsically light-cone correlations in the sense that all quark and gluons fields are separated along the light-cone, defined with the real Minkowski time t . The information is extremely useful for understanding the non-perturbative structure of the proton and, for example, for calculating the Higgs production cross section at Large Hadron Collider [2].

Computing the parton physics from the fundamental theory, quantum chromodynamics (QCD), has been difficult. In principle, a light-cone correlation is best calculated using the proton's wave function in light-front coordinates [3]. However, despite many years of effort, a realistic light-front proton wave function has not yet been established. On the other hand, in the formulation of non-perturbative QCD on a Euclidean lattice, one cannot directly calculate time-dependent correlations. Instead, one can only compute moments of parton distributions and distribution amplitudes, which are matrix elements of local operators. However, the difficult grows considerably for higher moments for technical reasons [4].

In this paper, I present a direct approach to compute parton physics on a Euclidean lattice through Lorentz boost. To demonstrate this, let us consider the quark distribution in a proton,

$$q(x, \mu^2) = \int \frac{d\xi^-}{4\pi} e^{-ix\xi^- P^+} \langle P | \bar{\psi}(\xi^-) \gamma^+ \times \exp \left(-ig \int_0^{\xi^-} d\eta^- A^+(\eta^-) \right) \psi(0) | P \rangle , \quad (1)$$

where the nucleon momentum P^μ is along the z -direction, $P^\mu = (P^0, 0, 0, P^z)$, and $\xi^\pm = (t \pm z)/\sqrt{2}$ is the light-cone coordinates with t as the physical time, μ^2 is the renormalization scale, A^+ is a gluon potential matrix in the fundamental representation, ψ is a quark field of flavor q . The above expression is boost-invariant along z , i.e., independent P^z ; in particular it is valid in the rest frame where $P^z = 0$.

To find a way to calculate $q(x, \mu^2)$ on a Euclidean lattice, it is useful to exhibit the origin of the light-cone correlation: The parton distribution has also been formulated in terms of the local twist-2 operators, which are defined as

$$O^{\mu_1 \dots \mu_n} = \bar{\psi} \gamma^{(\mu_1} i D^{\mu_2} \dots i D^{\mu_n)} \psi - \text{trace} , \quad (2)$$

where $(\mu_1 \dots \mu_n)$ indicates that all the indices are symmetrized, the trace terms include operators with at least one factor of the metric tensor $g^{\mu_i \mu_j}$ multiplied by operators of dimension $(n+2)$ with $n-2$ Lorentz indices, etc. Its matrix elements in the proton state are,

$$\langle P | O^{\mu_1 \dots \mu_n}(\mu^2) | P \rangle = 2a_n(\mu^2) (P^{\mu_1} \dots P^{\mu_n} - \text{trace}) , \quad (3)$$

and the parton distribution is related to the local matrix elements through $\int dx x^{n-1} q(x, \mu^2) = a_n(\mu^2)$ with even n . The time-dependent correlation for the parton distribution in Eq. (1) is recovered by taking all the components as $+$ in Eq. (3),

$$\langle P | O^{+ \dots +}(\mu^2) | P \rangle = 2a_n(\mu^2) P^+ \dots P^+ . \quad (4)$$

Thus, the light-cone correlation is *kinematically* connected with $+$ components of the nucleon four-momentum.

To eliminate the time dependence, we consider the matrix elements of the twist-two operator with $\mu_1 = \mu_2 = \dots = \mu_n = z$, and in the nucleon state with large P^z . Then

$$O^{z \dots z} = \bar{\psi} \gamma^z i D^z \dots i D^z \psi - \text{trace} , \quad (5)$$

where again the trace terms contain operators with at most $n-2$ z 's. According to Lorentz symmetry, the matrix elements of the trace terms are at most $(P^z)^{n-2}$ times Λ_{QCD}^2 . Similarly the right hand side of Eq. (3) is $2a_n(\mu^2) [(P^z)^n - \lambda M^2 (P^z)^{n-2} - \dots]$, where λ is a numerical number of order 1, M is the nucleon mass, and $\dots$ represents terms with still lesser powers of P^z . Thus, we conclude that

$$\langle P | \bar{\psi} \gamma^z i D^z \dots i D^z \psi | P \rangle = 2a_n(\mu^2) (P^z)^n \times [1 + \mathcal{O}(\Lambda_{\text{QCD}}^2 / (P^z)^2, M^2 / (P^z)^2)] \quad (6)$$

where terms other than the first are power-suppressed in the large P^z limit.

Inverting the above result in term of a time-independent, non-local expression, we find in the large P^z limit,

$$\begin{aligned} q(x, \mu^2, P^z) &= \int \frac{dz}{4\pi} e^{izk^+} \langle P | \bar{\psi}(z) \gamma^z \\ &\times \exp \left(-ig \int_0^z dz' A^z(z') \right) \psi(0) | P \rangle \\ &+ \mathcal{O}(\Lambda^2 / (P^z)^2, M^2 / (P^z)^2) , \end{aligned} \quad (7)$$

where $x = k^z / P^z$. An intuitive way to understand the above result is to consider the Lorentz transformation of a line segment connecting $(0, 0, 0, z)$ with the origin of the coordinates. As the boost velocity approaches the speed of light, the space-like line segment is tilted to the light-cone direction. Of course, it cannot literally be on the light-cone because the invariant length cannot change for any amount of boost. However, this slight off-light-cone-ness only introduces power corrections which vanish asymptotically.

Intuitively, the parton distribution in Eq. (1) can be regarded for the nucleon in the rest frame but the probe is traveling at the speed of light, and hence the light-cone correlation; On the other hand, the distribution Eq. (7) is obtained from a static probe on a nucleon traveling at the speed of light. This late quantity is related to deep-inelastic scattering in a frame where the virtual photon momentum contains only the space component $q^\mu = (0, Q)$, and the nucleon momentum is $P^z \sim Q/2x$. One can also obtain a similar expression for the quark helicity distribution with the replacement of γ^z by $\gamma^z \gamma_5$. As for the gluon distribution, we have,

$$\begin{aligned} g(x, \mu^2, P_z) &= \int \frac{dz}{4\pi k^+} e^{izk^+} \langle P | F^{3\mu}(z) \\ &\times \exp \left(-ig \int_0^z dz' A^z(z') \right) F_\mu^3(0) | P \rangle \end{aligned} \quad (8)$$

in the large P^z limit, where A^z is a matrix in adjoint representation and i sums over the transverse directions. And similarly, we have the polarized gluon distribution,

$$\begin{aligned} \Delta g(x, \mu^2, P_z) &= i \int \frac{dz}{4\pi k^+} e^{izk^+} \langle P | F^{3\mu}(z) \\ &\times \exp \left(-ig \int_0^z dz' A^z(z') \right) \tilde{F}_\mu^3(0) | P \rangle , \end{aligned} \quad (9)$$

from which one can integrate to get ΔG ($\tilde{F}^{\mu\nu} = 1/2\epsilon^{\mu\alpha\beta}F_{\alpha\beta}$). However, we also find another way to calculate ΔG on lattice without explicitly doing this integral [6].



FIG. 1: One-loop correction to the parton distribution in Eq. (7) in $A^2 = 0$ gauge.

Taking the infinite-momentum limit is a subtle process in field theories, and the ultra-violet (UV) divergences must be taken care of properly in the loop integrals with a correct limiting procedure. In our case, we need to take the limit after the renormalization procedure, whereas the standard light-cone result in Eq. (1) has $P^z \rightarrow \infty$ taken before the renormalization. Thus in modern language, the light-cone distribution is an effective field theory of the quantity in Eq. (7), in which there are large logarithms $\ln P^z$ in perturbation theory, which can be transformed into the standard renormalization scale dependence in the light-cone distribution through matching conditions [?]. One can see the interplay of the different limits by computing the one-loop correction in the physical $A^2 = 0$ gauge (see Fig. 1),

$$q(x, \mu^2, P^z) = \frac{\alpha_s}{2\pi} T(x) \frac{\Lambda}{P^z} + \frac{\alpha_s}{2\pi} P(x) \ln \frac{\Lambda P^z}{\lambda^2} \quad (10)$$

where $\Lambda = \sqrt{\mu^2 + (1-x)^2(P^z)^2} - (1-x)P^z$, λ is an infrared mass cut-off and μ^2 is an UV cut-off (renormalization scale); $T(x) = C_F x/(1-x)^2$, and $P(x) = C_F(1+x^2)/(1-x)$ with $C_F = 4/3$ is the standard Altarelli-Parisi kernel [5]. In the limit of $P^z \rightarrow \infty$ first, $\Lambda \sim \mu^2/P^z$, the first term is power-suppressed and the second term is the standard AP result. However, in the limit of $\mu^2 \rightarrow \infty$ first, which is of our interest here, the first term is linearly divergent, whereas the second term becomes $\ln(\mu P^z/\lambda^2)$. The connection of the two limits is reflected through the following factorization theorem up to power corrections,

$$q(x, \mu^2, P^z) = \int_x^1 \frac{dy}{y} Z\left(\frac{x}{y}, \frac{\mu}{P^z}\right) q(y, \mu^2) \quad (11)$$

where Z has a perturbative expansion in α_s ,

$$Z(x, \mu/P^z) = \delta(x-1) + \frac{\alpha_s}{2\pi} Z^{(1)}(x, \mu/P^z) + \dots \quad (12)$$

with $Z^{(1)} = T(x)(\mu/P^z) + P(x)\ln(P^z/\mu)$. Taking $P^z \rightarrow \infty$ first yields $Z(x, \mu/P^z) = \delta(x-1)$. Thus the large logarithmic dependence on P^z in $q(x, \mu^2, P^z)$ can be transformed into the renormalization scale dependence through the above matching condition. On lattice, the matching must be recalculated up to a constant accuracy using the standard approach [8].

Therefore, Eq. (11) can be used to calculate the parton distribution $q(x, \mu^2)$ on lattice by measuring the time-independent, non-local quark correlator $\bar{\psi}(z)\gamma^z L(z, 0)\psi(0)$ with

$L(z, z_0) = \exp\left(-ig \int_{z_0}^z dz' A^z(z')\right)$ in a state with increasingly large P^z (maximum $\sim 1/a$ with a denoting lattice spacing). Because the smallest and the largest momenta on lattice is $1/L_z a$ and $1/a$, respectively, the smallest x that one can resolve is $1/L_z$, with L_z the number of lattice sites in the z -direction. Thus to reach $x = 10^{-2}$, one has to have 100 lattice sites. On the other hand, the present approach can get a reasonable description of large x partons, emphasized by higher moments of the distribution. Thus, the method is in a sense complementary to the traditional moment approach. On the other hand, for a fixed x , large P^z implies large k^z , and hence small z . Thus, a fast nucleon experiences a Lorentz contraction in the z -direction which requires increasing resolution (anisotropic lattices such as ones used for spectroscopy calculations [7]) to study its longitudinal structure.

The above approach can be generalized to any light-cone correlations in hadron physics. A partial list of interesting physical quantities includes:

- Generalized parton distributions (GPD's) [9], for which one has the same twist-two operators, but the external states are off-diagonal. Thus, for example, the GPD's $E(x, \xi, t)$ and $H(x, \xi, t)$ can be calculated from the following matrix elements,

$$F(x, \xi, t, P^z) = \int \frac{dz}{2\pi} e^{-izk^z} \langle P | \bar{\psi}(-z/2) \gamma^z \times L(-z/2, z/2) \psi(z/2) | P \rangle \quad (13)$$

with $\vec{P}^+ + \vec{P}^- = 2(0, 0, \vec{P}^z)$, in the $P^z \rightarrow \infty$, $\xi = (P^{z'} - P^z)/\vec{P}^z$, $x = k^z/\vec{P}^z$ -fixed limit. Similarly one can obtain other twist-two GPD's with different insertions of the Dirac matrix and nucleon polarization states. Thus one can now explore the three-dimensional dependence on x , ξ and t in full detail without model assumptions [10].

- Transverse-momentum dependent (TMD) parton distribution [11]. For the Dirac matrix γ^z , for example, we can obtain the following TMD,

$$q(x, k_\perp, P_z, \mu^2) = \int \frac{dz}{4\pi} d^2 \vec{r}_\perp e^{i(xk^z + \vec{k}_\perp \cdot \vec{r}_\perp)} \langle P | \bar{\psi}(\vec{r}_\perp, z) \times L^1(\pm\infty; (\vec{r}_\perp, z)) \gamma^z L(\pm\infty; 0) \psi(0) | P \rangle \quad (14)$$

where P^z must be large and $L(\pm\infty; (\vec{r}_\perp, z)) = e^{-ig \int_0^{\pm\infty} dz' A^z(\vec{r}_\perp, z')}$. The dependence on P^z now contains large logarithms $\ln P^z$ and does not disappear in the large P^z limit, and therefore is shown in the function explicitly. One can derive an evolution equation for P^z [11]. The gauge links here extend from the location of the quark field to $\pm\infty$ along the z -direction only, and therefore they involve only the A^z component of the gauge potential. Here $\pm\infty$ corresponds to two different kinematic conditions in which the distributions are probed. In gauge choices where gauge potential vanishes at infinity, the above expression is gauge invariant. However, for a finite lattice with all gauges averaged over, the two gauge links must be connected by another gauge link $L((\vec{r}_\perp, \pm\infty); (0, \pm\infty))$ in the transverse-space direction. Similar expressions work for TMDs defined with other Dirac structure. An exploratory study of x -moments of TMD distributions has been made on the lattice where gauge links are taken as staple in spatial directions [12]. The connection to light-cone kinematics is made through invariant decompositions, not through boosts, although the distributions themselves are defined at large P^z .

- Wigner distributions [13], which are generating functions for both GPD's and TMD's. The definition follows from a TMD operator sandwiched in a non-forward nucleon states. Thus one can calculate them in the large P^z limit as, for example,

$$\begin{aligned}
W(x, k_\perp, b_\perp, P^z, \mu^2) &= \int \frac{dz}{4\pi} i^2 \Delta_\perp d^2 \bar{r}_\perp e^{-i\Delta_\perp \cdot b_\perp} e^{i(xk^+ + \bar{k}_\perp \cdot \bar{r}_\perp)} \\
&\times \langle P' | \bar{\psi}(\bar{r}_\perp, z) L^1(\pm\infty; (\bar{r}_\perp, z)) \gamma^z L(\pm\infty; 0) \psi(0) | P \rangle
\end{aligned} \tag{15}$$

where $\Delta_\perp = (P' - P)_\perp$ and again a transverse gauge link is needed to ensure gauge invariance on a finite lattice. We also note that in the large P^z limit, the P^z dependence does not disappear.

- Light-cone amplitudes, which are the matrix elements between the hadron states and the QCD vacuum, of the light-cone correlations of the quark and gluon interpolating fields. We can now calculate them as the matrix element of spatial correlation in the large momentum limit. For example, the leading light-cone amplitude of the π^+ meson recovers as the large P^z limit of

$$\phi_0(x, P^z) 2P^z = \int \frac{dz}{2\pi} e^{izk^+} \langle 0 | \bar{d}(0) \gamma^z \gamma_5 L(0, z) u(z) | \pi^+(P) \rangle . \tag{16}$$

These light-cone amplitudes can be probed from the hard exclusive processes such as the form factors [14, 15]. There has been much discussion about the x -dependence of $\phi_0(x)$ amplitude in the literature.

- Light-cone wave functions. In the light-front coordinates, the nucleon state can be written as an infinite sum of Fock states with wave-function amplitudes [3]. In a series of publications [16], we have systematically classified the wave-function amplitudes for pion and proton up to four partons, and connected them with the matrix elements of quark and gluon correlations on the light-cone and transverse directions. All these are gauge-invariant when light-cone gauge links are inserted from the locations of the fields to the infinity along the light-cone direction. These amplitudes again can be calculated on lattice as the infinite-momentum limit of the similar correlation functions with fields connected by gauge links going to infinity along the z -direction. One important application is to the exclusive decays of B mesons where many factorization formulas have been derived with light-cone wave functions as a part of the decay amplitudes [17].
- Higher-twist parton distributions. The complete list of twist-two, -three and -four parton distributions in unpolarized, longitudinally-polarized and transversely-polarized nucleon has been worked out in the literature [18]. They all contains quark and gluons fields separated along the light-cone. In the spirit of this paper, they can be obtained from related spatial correlation functions with gauge links along the z -direction in the limit of $P^z \rightarrow \infty$. In particular, the jet quenching parameter $\hat{q}$ might be calculated this way [19].

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It will be interesting to see these quantities can now be explored in lattice QCD calculations.

In conclusion, I have shown that the light-cone correlations of quarks and gluons can be calculated by boosting the matrix elements of spatial correlations to a large momentum. This is particularly useful for lattice QCD calculations of these experimental observables which have been very difficult to tackle from the first principle in the past. To be sure, studying a large momentum hadron on lattice is computationally still challenging, but at least this could be achieved when computational power continues improving.

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P12: 大动量有效理论与部分子物理

主题归属	LaMET、quasi/pseudo 分布与因子化
底稿	arXiv:1404.6680
arXiv / DOI	1404.6680
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Parton Physics from Large-Momentum Effective Field Theory

Xiangdong Ji^{1,2}

¹*INPAC, Department of Physics and Astronomy,
Shanghai Jiao Tong University, Shanghai, 200240, P. R. China*

²*Maryland Center for Fundamental Physics,
Department of Physics, University of Maryland,
College Park, Maryland 20742, USA*

(Dated: April 29, 2014)

Abstract

Parton physics, when formulated as light-front correlations, are difficult to study non-perturbatively, despite the promise of light-front quantization. Recently an alternative approach to partons have been proposed by re-visiting original Feynman picture of a hadron moving at asymptotically large momentum. Here I formulate the approach in the language of an effective field theory for a large hadron momentum P in lattice QCD, LaMET for short. I show that using this new effective theory, parton properties, including light-front parton wave functions, can be extracted from lattice observables in a systematic expansion of $1/P$, much like that the parton distributions can be extracted from the hard scattering data at momentum scales of a few GeV.

One of the greatest simplifications in describing physics of hadron scattering at high energy, such as production of Higgs bosons at the Large Hadron Collider (LHC) [2], is the parton model introduced by R. Feynman [1]. According to this, a fast moving hadron, such as proton, can be viewed as a beam of noninteracting quarks and gluons (partons) characterized by their momentum density, $q(x)$ and $g(x)$, where x is the fraction of the longitudinal momentum carried by the parton, $x = k^z/P^z$ with the hadron momentum $P^z \rightarrow \infty$. Then the hard-scattering cross sections involving the hadrons can be calculated as the convolution of the basic parton scattering cross sections $\hat{\sigma}$ and parton densities. In the fundamental theory of strong interactions, quantum chromodynamics (QCD), this simple picture can be justified by the so-called factorization theorems [3–5]. The only corrections introduced by quantum field theory is that the parton densities are scheme and scale dependent, and the latter can be studied through renormalization group equations [6]. The scheme and scale dependence of the parton densities are of course cancelled by the similar dependences of parton scattering cross sections, leaving the physical quantities invariant under the perturbative definitions of partons.

While parton scattering cross sections can be computed in QCD perturbation theory thanks to asymptotic freedom, the parton densities are intrinsically non-perturbative. As mentioned above, Feynman’s definition of parton density was made in the infinite momentum frame (IMF), in which the parent hadrons have an infinite momentum [1]. This seems to be a mathematical limit difficult for intuitive understandings. The exact notion of the infinite-momentum limit can actually be understood by boosting Feynman diagrams in perturbation theory, as was done by Weinberg [7]. Over the years, however, one found that it is convenient to formulate the parton density in the formalism of light-cone correlation function [8]: Introduce the light-cone coordinates,

$$\xi^\pm = \frac{1}{2}(\xi^0 \pm \xi^3), \quad (1)$$

and similarly for other four-vectors, where ξ^μ ($\mu = 0, 1, 2, 3$) is the space-time coordinates (reserving x for the momentum fraction) and the hadron is moving in the 3 or z -direction. The parton density is now calculated as the matrix elements of the non-local correlator [9],

$$q(x, \mu^2) = \int \frac{d\xi^-}{4\pi} e^{-ix\xi^- P^+} \langle P | \bar{\psi}(\xi^-) \gamma^+ \times \exp \left(-ig \int_0^{\xi^-} d\eta^- A^+(\eta^-) \right) \psi(0) | P \rangle, \quad (2)$$

where ψ is the quark Dirac field, A^μ is the gluon potential, g is the strong coupling, μ^2 is the renormalization scale. Here the infinite boost factors all disappeared, and IMF physics is reflected entirely through the boosted quark and gluon operator. The above matrix element is now independent of the hadron external momentum P^μ ! The price one pays, however, is that the parton density no longer represents a equal-time correlation, as was originally defined in the IMF by Feynman.

Although a significant progress is made by getting rid of the infinite boost, the light-cone correlation functions by no means are easy to calculate. For instance, to use Wilson’s lattice QCD method [10], which is intrinsically Euclidean, one has to get rid of the real time dependence. One has to Taylor-expand the separation between the quark fields and obtain the local operators with multiples of derivatives, whose matrix elements are related

to the moments of parton distributions [11]. However, measuring operators matrix elements with many derivatives is intrinsically noisy, and as such, one can only evaluate the first few moments. Over the years, people have invented approaches to overcome this difficulty [12–14], but none has been promising enough for realistic numerical simulations.

An alternative approach that has been advocated strongly by some is light-front quantization [15–17], related to the form of the time-independent perturbation theory in the IMF [7]. Actually the light-front formulation of a dynamical theory goes back more than half a century to Dirac’s original paper [18]. The formalism goes roughly as follows: Consider ξ^+ as the new time and introduce the equal “time” commutators to quantize the field theory. The fields at $\xi^+ = 0$ have canonical plane-wave expansions in terms of the Fock particles. The new hamiltonian is $\hat{H}_{LC} = P^-$, which can be used to develop time-independent perturbation theory as in non-relativistic quantum mechanics. One can then go on to determine the spectrum of P^- and eigenstates by solving the non-perturbative Schrödinger-like equation,

$$\hat{P}^-|P\rangle = (M^2/2P^+)|P\rangle, \quad (3)$$

where eigenvalue M^2 is the hadron mass, and P^+ is the light-front momentum. In QCD, it is most convenient to work with $A^+ = 0$ gauge, the Fock particles are partons. The hadron states can be expressed in terms of the Fock expansion [17],

$$|P\rangle = \sum_{n\alpha\lambda_i} \Pi_i \frac{dx_i d^2k_{\perp i}}{\sqrt{2x_i(2\pi)^3}} \psi_{n\alpha}(x_i, k_{\perp i}, \lambda_i) |n\alpha : x_i P^+, k_{\perp i}, \lambda_i\rangle, \quad (4)$$

where n is the number of partons, λ_i are helicity labels. All partons have the longitudinal fraction x_i , and transverse momentum $k_{\perp i}$. Index α sums over possible amplitudes $\psi_{n\alpha}$ for a given parton number and parton helicities. The examples of the proton and meson Fock states can be found in Ref. [19, 20]. Once we have the light-front wave functions, one can calculate any parton physics observables of interest.

Solving a field theory on the light-cone front is notoriously difficult, despite the many nice features of the theory such as the vacuum becomes “trivial”, and one more generator of the Lorentz group becoming kinematical, etc. [17]. Apart from two-dimensional theories and perturbative expansion in terms of the number of Fock particles, no systematic approximation has been found for non-perturbative calculations in 3+1 dimension. Unlike the ordinary formulation of asymptotic field theories whose static properties can be simulated on Euclidean lattices, no such formalism exists for light-front theories. The difficulty might be related to the fact that they are intrinsically Minkowskian. A hybrid formulation of the light-front theories in terms of the transverse lattice has been proposed and explored [21], but so far it has not lead to successful simulations.

In recent publications, a new approach to calculating parton distributions using Euclidean lattice QCD has been proposed [22–26]. This approach essentially goes back to the original definition of the parton densities by Feynman, i.e., starting with the ordinary momentum distribution $n(\vec{k}, P^z)$ related to the space correlation function of the quark fields in a hadron. This quantity is calculable using ordinary lattice QCD. However, it depends on the hadron momentum P^z . The Feynman distribution is obtained in the limit of $P^z \rightarrow \infty$ limit, i. e.,

$$q(x) \sim \lim_{P^z \rightarrow \infty} \int d^2k_{\perp} dk^+ n(\vec{k}, P^z) \delta(x - k^+/P^z). \quad (5)$$

The lattice simulations, however, cannot provide an infinite P^z result directly. How can one then recover the $P^z = \infty$ limit from a moderately-large- P^z result that might be possible on

the lattice? An added subtlety is that field theory usually has ultraviolet (UV) divergences. In a typical Feynman diagram, the order of regularization UV divergences vs. $P^z \rightarrow \infty$ does not commute with each other. While defining the parton distribution requires P^z to be much larger than the cut-off scale, the lattice simulations only makes sense when the UV cut-off is much larger than the hadron momentum.

The solution to the above problems have been discussed in Refs. [22]: $n(\vec{k}, P^z)$ calculable in lattice QCD can be matched to the parton density $g(x)$ through perturbatively calculable multiplicative factor Z plus higher-order corrections in powers of $(1/P^z)^2$. The approach can be generalized to any other light-front quantities and can be formulated in terms of an effective field theory of lattice QCD in the presence of one (or more) large momentum scale P^z . The situation is very similar to that of the heavy quark effective field theory (HQET), which by now has been well-understood and widely used [27]. Here I discuss in some detail this new *large momentum effective field theory* (LaMET), which allows parton physics to be calculated in lattice QCD at finite P^z in a systematic approximation.

Let us first review how HQET works. Consider a physical system with one heavy quark, such as B^- meson with the $b\bar{u}$ quark content. A general physical observable O of the system, such as the weak decay constant, will depends on the b -quark mass parameter m_b . Because of the asymptotic freedom, $O(m_b)$ for a large $m_b \gg \Lambda_{\text{QCD}}$, where Λ_{QCD} is the QCD scale, can be expanded in terms of powers of $1/m_b$,

$$O(m_b/\Lambda) = Z(m_b/\Lambda, \Lambda/\mu) o(\mu) + \mathcal{O}(1/m_b) + \dots \quad (6)$$

where we have assumed O depends on some UV cutoff Λ , which can also be replaced by a renormalization scale after proper renormalization in the full theory. The leading term in the expansion contains the logarithmic dependence of the heavy quark mass in the matching coefficient Z , which has a perturbative expansion in α_s . $o(\mu)$ is a quantity defined in the effective theory, namely HQET, in which the b -quark is effectively infinitely heavy. μ is the renormalization scale in HQET. The sub-leading terms are suppressed by at least one power of $1/m_b$.

From perturbation theory, it is easy to understand what HQET does. Consider a Feynman diagram with quark-mass dependence in the external wave functions and internal propagators. In the full theory, these diagrams are well defined after proper UV regularization. In the effective theory, one simply takes $m_b \rightarrow \infty$ inside the integrals. This limit shall not change the infrared property of the quantity under calculation. Thus both O and o in the full and effective theories, respectively, contains the same non-perturbative physics. However, they do have different UV physics. And the difference is reflected in the multiplicative matching factor Z , which is infrared free and hence calculable in perturbation theory. The non-trivial aspect here is Z is multiplicative. This can be proved order by order in perturbation theory, just like proving perturbative renormalizability [28]. Of course, we have ignored the complication that there might be mixings of several operators of the same quantum numbers, and there is also a possibility of a convolution between Z and $o(\mu)$ in a certain parameter space.

The important physical observation in applications of HQET is that once m_b is in the perturbation region, the difference between $O(m_b, \Lambda)$ and its effective theory counter-part $o(\mu)$ is under control, namely we know how to calculate the matching between them. In certain cases, we need to take into account of the power corrections which can be categorized and studied in details. Thus to a certain extent, the physics of a heavy quark is similar to that of an infinitely-heavy one. When can we consider a quark to be heavy? For some

physical observables, the charm quark of mass 1.5 GeV is already a good candidate. For the bottom quark of mass 4.2 GeV, HQET is expected to work fairly well.

Now we can consider the formulation of LaMET. Consider a lattice (Euclidean) observable F which depends on a large hadron momentum P^z . Using asymptotic freedom, we can systematically expand the P_z dependence,

$$F(P^z/\Lambda) = Z(P^z/\Lambda, \Lambda/\mu) f(\mu) + \mathcal{O}(1/(P^z)^2) + \dots \quad (7)$$

The quantity $f(\mu)$ is defined in a theory with $P^z \rightarrow \infty$, exactly as in Feynman's parton model. In other words, $f(\mu)$ is a light-front correlation or parton observable, containing all the infrared collinear singularities. Therefore, Feynman's parton model is in fact an effective theory for the nucleon moving at large momentum! The important message of the expansion is that it may converge at moderately large P^z , allowing access to quantities defined at infinite P^z . The extraction of the parton physics can be made more precise by accurately calculating the matching factor Z and higher-order corrections.

The examples of the above expansion have been considered in Ref.[22, 23]. Since the large momentum scale comes from the external states, we do not attempt to define the effective theory through the lagrangian formalism, just like in the case of perturbative QCD discussion of factorization theorems.

Momentum dependence of the lattice quasi-observables can be studied through the renormalization group. Define the anomalous dimension through

$$\gamma(\alpha_s) = \frac{1}{Z} \frac{\partial Z}{\partial \ln P^z} \quad (8)$$

Then one has,

$$\frac{\partial F(P^z)}{\partial \ln P^z} = \gamma(\alpha_s) F(P^z) + \mathcal{O}(1/(P^z)^2) \quad (9)$$

up to power corrections. One can sum large logarithms involving P^z using the above equation.

The reason for the existence of the above expansion or effective description is similar to the existence of HQET. When taking $P^z \rightarrow \infty$ first in $F(P^z)$ before a UV regularization is imposed, one recovers from $\hat{F}$ the light-cone operator $\hat{f}$, by construction. On the other hand, the lattice matrix element is calculated at large P^z , with UV regularization (lattice cut-off) imposed first. Thus the difference between the matrix elements f and F is the matter of the orders of the limits. This is the standard set-up for an effective field theory. The different limits do not change the infrared physics. In fact the factorization in terms of Feynman diagrams can be proved order by order as in the renormalization program.

The above effective theory expansion gives a recipe to study parton physics in lattice QCD: First, start from a particular parton observable $\hat{f}$ which is an operator made of light-cone fields. Then construct an Euclidean version $\hat{F}$ which, under an infinite Lorentz boost, goes to $\hat{f}$. Calculate the lattice matrix element of $\hat{F}$ in a hadron with large momentum P^z and use Eq. 7 to extract the parton physics $\hat{f}$. Of course, matrix element F depends on P^z as well as all the lattice UV artifacts. All these physics will be captured in the matching factor Z .

While the previous works have shown examples how LaMET work for parton densities, here I would like to consider the light-front wave functions itself. One can calculate any parton physics once the complete light-front wave functions are known, and this is the main object the light-front quantization is after. For simplicity, let us consider the π^+ meson,

although a similar discussion applies for a proton. The leading light-front wave function is related to the matrix element,

$$\langle 0 | \bar{d}(0) \gamma^+ \gamma_5 u(\xi^-, \xi_\perp) | \pi^+(P) \rangle, \quad (10)$$

where u and d are up and down quark fields in coordinate space, defined in which is defined in the $A^+ = 0$ gauge. In fact all the Fock wave functions in the light-front theories can be expressed in terms of light-front correlations between the hadron state and QCD vacuum. The gauge-invariant version of the above operator needs some gauge links along the ξ^- direction to either $\xi^- = +\infty$ or $-\infty$. A choice of the ξ^- directions shall be made once and for all, and two choices are physically inequivalent, are related to each other by the complex conjugation [19, 20]. To ensure complete gauge-invariance, the gauge links at infinity must be connected by transverse gauge links as the transverse gauge potential does not vanish at $\xi^- = \pm\infty$ in this singular gauge. One needs to specify how the connections are made to make the above matrix element uniquely defined. This can be done with a straight-line gauge link along $\xi_\perp$ [29]. In the case of many field correlations, these end-point links can be chosen in an infinite number of possibilities. In a certain limiting process of the hadron momentum going to infinity, these end-point links shall not contribute as they are outside of the physical correlation length, as will be discussed below.

In LaMET, we need to find a quasi-operator, which under the infinite Lorentz boost will recover the operator in Eq. 10. We choose,

$$\hat{F}_\pm(z, \xi_\perp) = \bar{d}(0) W^\dagger(\pm\infty, 0; 0) \gamma^+ \gamma_5 W(\pm\infty, z; \xi_\perp) u(z, \xi_\perp) \quad (11)$$

where the all fields are at $\xi^0 \equiv t = 0$ and distributed along the $\xi^3 \equiv z$ direction, γ^+ is replaced by γ^z . To have gauge invariance, we insert a gauge link for every field, which goes off to $z = \pm\infty$ along the z -direction,

$$W(\pm\infty, z; \xi_\perp) = P \exp \left(\mp i g \int_z^{\pm\infty} dz' A^z(z', \xi_\perp) \right), \quad (12)$$

where P indicates path ordering. In gauges like the covariant one, $\hat{F}$ is already gauge-invariant. On the lattice, however, one has to insert a gauge link at large $\pm z$ to connect $W^\dagger(\infty, 0; 0)$ and $W(\infty, z; \xi_\perp)$. We likewise define the end-gauge links as needed in the case of the light-front quantization. Again, the result shall be independent of these end-point links as the physical correlation length is finite.

At this point, it is instructive to compare the extraction of parton physics from high-energy scattering with that of LaMET. In the high-energy scattering, one starts from scattering cross sections depending on some large momenta Q 's. One performs scale separation and proves factorization theorems, relating the these cross sections to parton properties; the latter can then be extracted finally from the experimental data [30]. On the other hand, on a Euclidean lattice, one calculates quasi observables which depend on the large hadron momentum P^z . One can use LaMET to perform scale separation, and relating these quasi observables to the parton physics. There is a complete analogy here. Thus, simply put, LaMET allows using lattice "data" to extract parton physics.

Just like the same parton distribution can be extracted from different hard scattering processes, the same light-front physics can be extracted from different lattice operators. All operators that yield the same light-front physics form a universality class. The existence of the universality class allows one exploring different operators $\hat{F}$ so that a result at finite

P^z can be made as close to that at large P^z as possible. An example of exploring different operators to calculate the total gluon polarization in the nucleon has been presented in a recent publication [26].

Finally, let us comment on the relationship between the present approach and light-front quantization. In Feynman's original language, the parton physics is described by the matrix elements of type

$$\langle P_\infty | \hat{F} | P_\infty \rangle, \quad (13)$$

where $\hat{F}$ is a non-local, time-independent operator and P_∞ indicates a momentum approaching infinity, with possibly a residual momentum as used in the paper by Weinberg [7]. On the other hand, one can write $|P_\infty\rangle = U_\infty |p_{\text{res}}\rangle$, where p_{res} is the residual momentum and U_∞ is a unitary transformation in the Hilbert space for the infinite boost. Thus, one can write the above quantity as

$$\langle p_{\text{res}} | U^\dagger \hat{F} U | p_{\text{res}} \rangle, \quad (14)$$

with the boost operator acting on F . This infinite boost will give rise to a light-front operator $\hat{f} \equiv U^\dagger \hat{F} U$,

$$\langle P_\infty | \hat{F} | P_\infty \rangle = \langle p_{\text{res}} | \hat{f} | p_{\text{res}} \rangle, \quad (15)$$

where the latter matrix element is independent of the residual momentum. While the present paper directly deals with the matrix element $\langle P_\infty | \hat{F} | P_\infty \rangle$ by taking the large P^z limit, the light-front quantization attempts to calculate the right-hand side of the equation with light-front wave functions.

The above discussion sheds important light on the space-time picture of the hadron states. Although one has often draw a round circle of radius about 1fm to represent a coordinate space picture of a hadron at rest, this runs into problem in quantum mechanics because in a plane wave state, the hadron can be anywhere in space. One may construct a wave packet to localize a hadron, then there is no clean separation between the center-of-mass motion and the internal motion in relativistic theory, unless the hadron is very heavy. More problems appear when considering the spatial picture of hadron moving with a certain velocity, when one often draws a Lorentz contracted circle. Physical meaning of the length contraction in a hadron state is often unclear.

One may consider instead the field correlation functions in a plan wave hadron state. Generally speaking, when the hadron is at rest (zero center-of-mass momentum), the field correlation functions shall have correlation length about 1fm $\sim \Lambda_{\text{QCD}}$ along all the spatial directions. Beyond that, the correlations shall gradually drop to zero. Now consider a hadron moving with velocity v in the z -direction. Introduce the Lorentz factor $\gamma = 1/\sqrt{1-v^2}$. The valence partons will have a fixed fraction of the hadron longitudinal momentum, $x = k^z/P^z$. Therefore, the typical longitudinal momentum of the valence parton will increase as xP^z . Correspondingly, the longitudinal correlation length will go as $1/xP^z$, inversely proportional to the γ factor. This may be considered as the Lorentz contraction effect. Thus to calculate the valence parton distributions using LaMET, one has to have the longitudinal spatial resolution increases with the nucleon momentum. If one only cares about the valence partons, the longitudinal lattice box size can also shrink by a factor of γ . Thus the total number of lattice sites can be similar to that in the transverse directions.

On the other hand, if one cares about the small x partons, one has to have the hadron momentum large enough. The smallest x one can reach is $x \sim \Lambda_{\text{QCD}}/P^z$. The correlation length in this case is about $1/\Lambda_{\text{QCD}}$ which is the normal size of a hadron. For example, at $x \sim 10^{-4}$, the hadron momentum must be around 3 TeV, and the normal valence partons

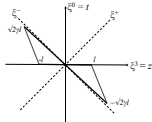


FIG. 1: The line segment in the space direction z in a frame with a high-momentum hadron is transformed into a line segment along the light-cone direction in light-front coordinates of the hadron at rest, the length is increased by the boost factor γ , which goes to infinity in the IMF.

will have correlation length of 10^{-3}fm . Clearly calculating parton density at such a small x will need thousands of lattice sites in the z -direction.

While one sees a contracted hadron in LaMET, the correlation lengths in the light-front calculations grow dramatically without bound. This can be seen in Fig. 1, where we have shown a correlation length ℓ along the z -axis. In the light-front calculations, the same correlation is now boosted to along ξ^- direction, increased by a factor of γ . Thus for valence partons, the correlation length in the light-front coordinates will be on the order of $\sim 1/\Lambda_{\text{QCD}}$. For small x -partons, the correlation length be as long as $P^+/(M\Lambda_{\text{QCD}})$, where M is the mass of the hadron. Therefore the light-front quantization calculations cannot be limited to a compact space-time region, which makes the Monte Carlo simulations difficult.

In summary, we have discussed in this paper the concept of a new effective field theory for hadrons with large momentum in lattice QCD simulations. The effective theory allows one to extract the light-front parton physics from practical calculations with hadron momentum of order a few GeV. The related expansion involves perturbatively-calculable matching coefficients and higher-power corrections. This makes the lattice data as useful as the real experimental data in studying the bound state properties in QCD.

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P13: 准部分子分布单圈匹配

主题归属	LaMET、quasi/pseudo 分布与因子化
底稿	arXiv:1310.7471
arXiv / DOI	1310.7471
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阅读方法

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One-Loop Matching for Parton Distributions: Non-Singlet Case

Xiaonu Xiong,¹ Xiangdong Ji,^{1,2,3} Jian-Hui Zhang,² and Yong Zhao³

¹*Center for High-Energy Physics, Peking University, Beijing, 100080, P. R. China*

²*INPAC, Department of Physics and Astronomy,
Shanghai Jiao Tong University, Shanghai, 200240, P. R. China*

³*Maryland Center for Fundamental Physics,
Department of Physics, University of Maryland,
College Park, Maryland 20742, USA*

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Abstract

We derive one-loop matching condition for non-singlet quark distributions in transverse-momentum cut-off scheme, including unpolarized, helicity and transversity distributions. The matching is between the quasi-distribution defined by static correlation at finite nucleon momentum and the light-cone distribution measurable in experiments. The result is useful for extracting the latter from the former in a lattice QCD calculation.

I. INTRODUCTION

Parton distributions characterize the structure of nucleons in terms of partons in high-energy scattering processes. They are an essential ingredient in making physical predictions for hadron-hadron or lepton-hadron collision experiments. Although much effort has been devoted to extracting parton distributions from various experimental data [1–6], the computation of parton distributions from the underlying theory of strong interactions, quantum chromodynamics (QCD), is a difficult task, due to their non-perturbative nature. One may wonder whether it is possible to evaluate parton distributions from lattice QCD, which is so far the only reliable framework for non-perturbative phenomena in QCD. In the field-theoretic language, the parton distribution is given in terms of non-local light-cone correlators, which can be viewed as the density of quarks and gluons in the infinite momentum limit (before ultraviolet (UV) cut-offs are imposed) or light-front correlations of partons at finite nucleon momentum [7]. Such correlations are time-dependent, and thus cannot be readily computed on the lattice. Past attempts have been mainly focused on the evaluation of local moments of the distributions. However, the evaluation of high moments of parton distributions on the lattice becomes considerably difficult for technical reasons [8].

Recently, a direct approach to compute parton physics on a Euclidean lattice has been proposed by one of the present authors. In this approach one computes, instead of the light-cone distribution, a related quantity, which may be called quasi-distribution [9]. In the case of unpolarized quark density, the quasi-distribution is [10]

$$\begin{aligned} \bar{q}(x, \mu^2, P^z) = & \int \frac{dz}{4\pi} e^{izk^+} \langle P | \bar{\psi}(z) \gamma^z \\ & \times \exp \left(-ig \int_0^z dz' A^z(z') \right) \psi(0) | P \rangle, \end{aligned} \quad (1)$$

where $x = k^+/P^z$. The above quantity is time-independent and non-local, and can be simulated on a lattice for any $P^z \ll 1/a$, where a is the lattice spacing. However, the result is not the light-cone distribution extracted from the experimental data, $q(x, \mu^2)$. To recover the latter from the former, one needs to find a matching condition of the type (for the non-singlet case)

$$\bar{q}_{\text{NS}}(x, \mu^2, P^z) = \int dy Z(x/y, \mu/P^z) q_{\text{NS}}(y, \mu^2) + \mathcal{O}((M/P^z)^2), \quad (2)$$

for large P^z , where the matching factor Z is entirely perturbative. The above relation can also be viewed as a factorization theorem: All the soft divergences are cancelled on both sides, and all the collinear divergences in $\bar{q}(x, \mu^2, P^z)$ are the same as those in the light-cone distributions. One has yet to prove that the above relation holds to all orders in perturbation theory. However, as a first step, we will test this in one-loop calculations in this paper. Of course the choice of quasi-distributions is not unique, one can define more than one possible quasi-distributions which have similar properties as $\bar{q}(x, \mu^2, P^z)$. Here we will focus on the simplest type suitable for lattice QCD calculations.

In this paper, we consider the non-singlet quark distribution, and prove the above factorization at one-loop level. Throughout the process we obtain the matching factor Z between the time-independent quasi-distribution and light-cone distribution. The result will be useful for recovering the light-cone distribution from lattice QCD simulations of the

quasi-distribution to the leading logarithmic accuracy. Power suppressed corrections of the type $(1/P^2)^n$ with $n \geq 1$ are ignored and will be considered in separate publications.

The rest of this paper is organized as follows. In Sec. 2, we present the result of one-loop calculation for unpolarized quasi-quark distribution. Based on this result, we propose in Sec. 3 a factorization theorem valid to one-loop level and extract the matching factor between quasi distribution and light-cone distribution. In Sec. 4, the results for quark helicity and transversity distributions are given. Finally we conclude in Sec. 5.

II. ONE-LOOP RESULT FOR UNPOLARIZED QUASI-QUARK DISTRIBUTION

In this section, we consider the one-loop correction in the case of unpolarized quasi-quark distribution $\bar{q}(x, \mu^2, P^z)$. The one-loop computation for non-singlet quark distribution is similar to QED because the non-Abelian property does not enter in the non-singlet case.

Since the one-loop result is gauge-invariant, we can perform the calculation in any gauge. The simplest choice is the axial gauge $A^2 = 0$ [13], as in this gauge the gauge link in Eq. (1) becomes unity. In the axial gauge, the relevant Feynman diagrams are shown in Fig. 1, where the non-local operator is depicted as a dashed line. The diagrams contain UV, soft and collinear divergences. We use the quark mass m to regulate the collinear divergence. The soft divergence is expected to cancel between the diagrams. The UV divergence is regulated by a transverse-momentum cut-off. Of course this cut-off violates rotational symmetry. However, it is expected to yield the correct leading-logarithmic behavior.



FIG. 1: One loop corrections to quasi quark distribution.

The one-loop diagrams generate the following result

$$\bar{q}(x, \mu^2, P^z) = (1 + \bar{Z}_P^{(1)} + \dots) \delta(x-1) + \bar{q}^{(1)}(x) + \dots \quad (3)$$

with

$$\bar{q}^{(1)}(x) = \frac{\alpha_S C_F}{2\pi} \begin{cases} \frac{1+x^2}{1-x} \ln \frac{x(\Lambda(x)-xP^z)}{(x-1)(\Lambda(1-x)+P^z(1-x))} + 1 - \frac{xP^z}{\Lambda(x)} + \frac{x\Lambda(1-x)+(1-x)\Lambda(x)}{(1-x)^2 P^z}, & x > 1, \\ \frac{1+x^2}{1-x} \ln \frac{(P^z)^2}{m^2} + \frac{1+x^2}{1-x} \ln \frac{4x(\Lambda(x)-xP^z)}{(1-x)(\Lambda(1-x)+(1-x)P^z)} - \frac{4x}{1-x} + 1 - \frac{xP^z}{\Lambda(x)} \\ + \frac{x\Lambda(1-x)+(1-x)\Lambda(x)}{(1-x)^2 P^z}, & 0 < x < 1, \\ \frac{1+x^2}{1-x} \ln \frac{(x-1)(\Lambda(x)-xP^z)}{x(\Lambda(1-x)+P^z(1-x))} - 1 - \frac{xP^z}{\Lambda(x)} + \frac{x\Lambda(1-x)+(1-x)\Lambda(x)}{(1-x)^2 P^z}, & x < 0 \end{cases} \quad (4)$$

for finite P^z , where $\Lambda(x) = \sqrt{\mu^2 + x^2(P^z)^2}$ and the logarithm with collinear divergences is related to the standard Altarelli-Parisi kernel [14]. The wave function renormalization

correction depends on the momentum P^z as well

$$\tilde{Z}_F^{(1)} = \frac{\alpha_S C_F}{2\pi} \int dy \begin{cases} -\frac{1+y^2}{1-y} \ln \frac{y(\Lambda(y)-yP^z)}{(y-1)(\Lambda(1-y)+P^z(1-y))} - 1 - \frac{y\Lambda(1-y)+(1-y)\Lambda(y)}{(1-y)^2 P^z} \\ + \frac{y^2 P^z}{\Lambda(y)} + \frac{y(1-y)P^z}{\Lambda(1-y)} + \frac{\Lambda(y)-\Lambda(1-y)}{P^z}, & y > 1, \\ -\frac{1+y^2}{1-y} \ln \frac{(P^z)^2}{y^2 P^z} - \frac{1+y^2}{1-y} \ln \frac{4y(\Lambda(y)-yP^z)}{(1-y)(\Lambda(1-y)+(1-y)P^z)} + \frac{4y^2}{1-y} + 1 \\ - \frac{y\Lambda(1-y)+(1-y)\Lambda(y)}{(1-y)^2 P^z} + \frac{y^2 P^z}{\Lambda(y)} + \frac{y(1-y)P^z}{\Lambda(1-y)} + \frac{\Lambda(y)-\Lambda(1-y)}{P^z}, & 0 < y < 1, \\ -\frac{1+y^2}{1-y} \ln \frac{y(\Lambda(1-y)+(1-y)P^z)}{y(\Lambda(1-y)+(1-y)P^z)} + 1 - \frac{y\Lambda(1-y)+(1-y)\Lambda(y)}{(1-y)^2 P^z} \\ + \frac{y(1-y)P^z}{\Lambda(1-y)} + \frac{y^2 P^z}{\Lambda(y)} + \frac{\Lambda(y)-\Lambda(1-y)}{P^z}, & y < 0. \end{cases} \quad (5)$$

It is easy to check that the result obeys the vector current conservation condition

$$\int_{-\infty}^{+\infty} dx \tilde{q}(x, \mu^2, P^z) = 1 \quad (6)$$

to one-loop order. Since the constituent of the quark in a quasi-distribution does not have a parton interpretation, the parton momentum fraction extends from $-\infty$ to $+\infty$. However, it is interesting to see that the collinear divergence exists only for $0 < x < 1$.

In field theory calculations, one often takes the ultraviolet cut-off to be larger than any other scale in the problem. In other words, one shall take the limit $\mu \rightarrow \infty$ and keep only the leading contribution and ignore the power-suppressed ones. This in principle shall also be the case in lattice QCD calculations. Thus the actual field theoretical result for the quasi-distribution shall be

$$\tilde{q}^{(1)}(x, \mu, P^z) = \frac{\alpha_S C_F}{2\pi} \begin{cases} \frac{1+x^2}{1-x} \ln \frac{x}{1-x} + 1 + \frac{\mu}{(1-x)^2 P^z}, & x > 1, \\ \frac{1+x^2}{1-x} \ln \frac{1}{m^2} + \frac{1+x^2}{1-x} \ln \frac{4x}{1-x} - \frac{4x}{1-x} + 1 + \frac{\mu}{(1-x)^2 P^z}, & 0 < x < 1, \\ \frac{1+x^2}{1-x} \ln \frac{x-1}{x} - 1 + \frac{\mu}{(1-x)^2 P^z}, & x < 0, \end{cases} \quad (7)$$

and

$$\tilde{Z}_F^{(1)} = \frac{\alpha_S C_F}{2\pi} \int dy \begin{cases} -\frac{1+y^2}{1-y} \ln \frac{y}{y-1} - 1 - \frac{\mu}{(1-y)^2 P^z}, & y > 1, \\ -\frac{1+y^2}{1-y} \ln \frac{1}{m^2} - \frac{1+y^2}{1-y} \ln \frac{4y}{1-y} + \frac{4y^2}{1-y} + 1 - \frac{\mu}{(1-y)^2 P^z}, & 0 < y < 1, \\ -\frac{1+y^2}{1-y} \ln \frac{y-1}{y} + 1 - \frac{\mu}{(1-y)^2 P^z}, & y < 0. \end{cases} \quad (8)$$

One shall note several interesting features of the above result: 1) There are contributions in the regions $x > 1$ and $x < 0$. The physics behind this is transparent: when the parent particle has a finite momentum P^z , the constituent parton can have momentum larger than P^z , and even negative. This is very different from the infinite momentum frame result, where the momentum fraction is restricted to $0 < x < 1$. 2) There is a linear divergence arising from the self-energy of the gauge link, which can be easily seen if one goes to a non-axial gauge, the Feynman gauge for example. This linear divergence is usually ignored in dimensional regularization. However, it is present in lattice regularization. 3) There is no logarithmic UV divergence. Instead there is a logarithmic dependence on P^z in the region $0 < x < 1$. We will see later on that this logarithmic dependence can be transformed into

the renormalization scale dependence of the light-cone parton distribution by a matching condition. 4) All soft divergences are cancelled. However, there are remaining collinear divergences reflected by the quark-mass dependence.

On the other hand, with the same regularization, one can calculate the light-cone parton distribution by taking the limit $P^z \rightarrow \infty$. This is done following the spirit of Ref. [15] and the result is

$$q(x, \mu^2) = (1 + Z_F^{(1)} + \dots) \delta(x-1) + q^{(1)}(x) + \dots \quad (9)$$

with

$$q^{(1)}(x) = \frac{\alpha_S C_F}{2\pi} \begin{cases} 0, & x > 1 \text{ or } x < 0, \\ \frac{1+x^2}{1-x} \ln \frac{x^2}{m^2} - \frac{1+x^2}{1-x} \ln(1-x)^2 - \frac{2x}{1-x}, & 0 < x < 1, \end{cases} \quad (10)$$

and

$$Z_F^{(1)} = \frac{\alpha_S C_F}{2\pi} \int dy \begin{cases} 0, & y > 1 \text{ or } y < 0, \\ -\frac{1+y^2}{1-y} \ln \frac{y^2}{m^2} + \frac{1+y^2}{1-y} \ln(1-y)^2 + \frac{2y}{1-y}, & 0 < y < 1, \end{cases} \quad (11)$$

where the integrand of $\delta Z^{(1)}$ is exactly opposite to that of $q^{(1)}(x)$. This result agrees exactly with that derived from the light-cone definition of parton distribution in transverse momentum cut-off scheme. Also the collinear or mass singularity is clearly the same as in the quasi-parton distribution. This shows that at one-loop level, the quasi-parton distribution captures all the collinear physics in the infinite momentum frame. Moreover, the contribution comes only from the diagram in which the intermediate gluon has a cut, which has a parton interpretation.

III. ONE-LOOP FACTORIZATION

Now we are ready to construct a factorization formula at one-loop order. In the infinite momentum frame or on the light-cone, the momentum fraction in parton distributions and splitting functions is limited to $[0, 1]$. However, in the present case, the splitting in the quasi-distribution is not constrained to this region, it can be in $[-\infty, \infty]$. Thus the connection of the two distributions is reflected through the following factorization theorem up to power corrections in the large P^z limit

$$\bar{q}(x, \mu^2, P^z) = \int_0^1 \frac{dy}{y} Z\left(\frac{x}{y}, \frac{\mu}{P^z}\right) q(y, \mu^2) + \mathcal{O}\left(\Lambda^2/(P^z)^2, M^2/(P^z)^2\right), \quad (12)$$

where the integration range is determined by the support of the quark distribution $q(y)$ on the light cone.

The Z factor has a perturbative expansion in α_s

$$Z(\xi, P^z/\mu) = \delta(\xi-1) + \frac{\alpha_s}{2\pi} Z^{(1)}(\xi, P^z/\mu) + \dots \quad (13)$$

Before we proceed, it is important to note the existence of a linear divergence coupled to the axial-gauge singularity, which is a double pole at $\xi = 1$, in the one-loop corrections to the quasi quark distribution, as one can see from Eqs. (7) and (8) above. As mentioned in the previous section, if one chooses a covariant gauge like the Feynman gauge, the diagrams in

Fig. 1 do not give axial singularities, but now one has extra diagrams with gauge link and the singularities come from these diagrams. Obviously the double pole structure is absent in dimensional regularization, but it is present in the cutoff regularization used here. Unlike in the case of light-cone parton distribution, where one encounters at most a single pole at $\xi = 1$ and can appropriately regularize it by a plus-prescription, the plus-prescription is unable to fully regularize the divergence of the double pole. In general, it only reduces the double pole to a single pole, which still yields singular contribution upon integration over ξ . However, it turns out that in our case this singularity can be removed with a principal value prescription, which corresponds to a regularization of the Wilson line self-energy. Although the linear divergence can also be singled out (this can be done on the lattice by varying the lattice spacing with fixed P^z and x) and subtracted [16], we propose to include this linearly divergent term within our factorization formula, in order to simplify the extraction of light-cone distribution from lattice data.

Now we are ready to write down the matching factor connecting the quasi quark distribution to the light-cone quark distribution. For $\xi > 1$, one has

$$Z^{(1)}(\xi)/C_F = \left(\frac{1+\xi^2}{1-\xi} \right) \ln \frac{\xi}{\xi-1} + 1 + \frac{1}{(1-\xi)^2} \frac{\mu}{P^z}, \quad (14)$$

whereas for $0 < \xi < 1$

$$Z^{(1)}(\xi)/C_F = \left(\frac{1+\xi^2}{1-\xi} \right) \ln \frac{(P^z)^2}{\mu^2} + \left(\frac{1+\xi^2}{1-\xi} \right) \ln [4\xi(1-\xi)] - \frac{2\xi}{1-\xi} + 1 + \frac{\mu}{(1-\xi)^2 P^z}, \quad (15)$$

and for $\xi < 0$

$$Z^{(1)}(\xi)/C_F = \left(\frac{1+\xi^2}{1-\xi} \right) \ln \frac{\xi-1}{\xi} - 1 + \frac{\mu}{(1-\xi)^2 P^z}. \quad (16)$$

Near $\xi = 1$, one has an additional term coming from the self energy correction

$$Z^{(1)}(\xi) = \delta Z^{(1)}(2\pi/\alpha_s)\delta(\xi-1), \quad (17)$$

which can be extracted from Eqs. (8) and (11) above, and provides a plus-regularization for the singularity at $\xi = 1$. Thus the large logarithmic dependence on P^z in $\bar{q}(x, \mu^2, P^z)$ can be transformed into the renormalization scale dependence through the above matching condition. On the lattice, the matching must be recalculated up to a constant accuracy using the standard approach [18].

So far, we have considered only the quark contribution. One can start with an antiquark to do the one-loop calculation. In this case, one also has a contribution to $\bar{q}(x, \mu^2, P^z)$ from $\bar{q}(x)$. However, the antiquark distribution has the property

$$\bar{q}(x) = -q(-x), \quad (18)$$

which is related to quark distribution at negative x . By including both quark and antiquark contribution, one obtains the following factorization with the integration extended from -1 to $+1$,

$$\bar{q}(x, \mu^2, P^z) = \int_{-1}^1 \frac{dy}{|y|} Z\left(\frac{x}{y}, \frac{\mu}{P^z}\right) q(y, \mu^2) + \mathcal{O}\left(\Lambda^2/(P^z)^2, M^2/(P^z)^2\right), \quad (19)$$

where the negative y region includes the antiquark contribution. The above is the complete one-loop factorization theorem, which replaces Eq. (11) and the Z -factor in Ref. [10].

We have constructed at one-loop level a factorization formula connecting the quasi parton distribution to the light-cone parton distribution. Of course it remains to be shown that there exists such a formula to all-loop orders. The factorization formula then allows one to extract the parton distribution $q(x, \mu^2)$ from calculating the quasi parton distribution on the lattice by measuring the time-independent, non-local quark correlator $\overline{\psi}(z)\gamma^z L(z, 0)\psi(0)$ with $L(z, z_0) = \exp\left(-ig \int_{z_0}^z dz' A^z(z')\right)$ in a state with increasingly large P^z (maximum $\sim 1/a$ with a denoting lattice spacing).

IV. HELICITY AND TRANSVERSITY DISTRIBUTIONS

In previous sections, we have considered the matching condition for unpolarized quark distribution. Here we present the results for the quark helicity and transversity distribution. For the quark helicity distribution in a longitudinally polarized quark, the quasi distribution $\Delta\tilde{q}^{(1)}(x)$ can be obtained by replacing γ^z with $\gamma^z\gamma^5$ in Eq. (1). The one-loop result then reads

$$\Delta\tilde{q}^{(1)}(x) = \frac{\alpha_S C_F}{2\pi} \begin{cases} \frac{1+x^2}{1-x} \ln \frac{x(\Lambda(x)-xP^+)}{(x-1)\Lambda(1-x)+P^+(1-x)} + 1 - \frac{xP^+}{\Lambda(x)} + \frac{x\Lambda(1-x)+(1-x)\Lambda(x)}{(1-x)^2 P^+}, & x > 1, \\ \frac{1+x^2}{1-x} \ln \frac{(P^+)^2}{m^2} + \frac{1+x^2}{1-x} \ln \frac{4x(\Lambda(x)-xP^+)}{(1-x)(\Lambda(1-x)+P^+(1-x))} - \frac{4}{1-x} + 2x + 3 \\ - \frac{xP^+}{\Lambda(x)} + \frac{x\Lambda(1-x)+(1-x)\Lambda(x)}{(1-x)^2 P^+}, & 0 < x < 1, \\ \frac{1+x^2}{1-x} \ln \frac{(x-1)\Lambda(x)-xP^+}{x(\Lambda(1-x)+P^+(1-x))} - 1 - \frac{xP^+}{\Lambda(x)} + \frac{x\Lambda(1-x)+(1-x)\Lambda(x)}{(1-x)^2 P^+}, & x < 0. \end{cases} \quad (20)$$

Taking the limit $\mu \rightarrow \infty$ yields

$$\Delta\tilde{q}^{(1)}(x) = \frac{\alpha_S C_F}{2\pi} \begin{cases} \frac{1+x^2}{1-x} \ln \frac{x}{x-1} + 1 + \frac{\mu}{(1-x)^2 P^+}, & x > 1, \\ \frac{1+x^2}{1-x} \ln \frac{(P^+)^2}{m^2} + \frac{1+x^2}{1-x} \ln \frac{x}{x-1} - \frac{4}{1-x} + 2x + 3 + \frac{\mu}{(1-x)^2 P^+}, & 0 < x < 1, \\ \frac{1+x^2}{1-x} \ln \frac{x-1}{x} - 1 + \frac{\mu}{(1-x)^2 P^+}, & x < 0. \end{cases} \quad (21)$$

The result for the light-cone distribution is again given by taking $P^z \rightarrow \infty$

$$\Delta\tilde{q}^{(1)}(x) = \frac{\alpha_S C_F}{2\pi} \begin{cases} 0, & x > 1 \text{ or } x < 0, \\ \frac{1+x^2}{1-x} \ln \frac{x^2}{m^2} - \frac{1+x^2}{1-x} \ln(1-x)^2 - \frac{2}{1-x} + 2x, & 0 < x < 1. \end{cases} \quad (22)$$

Note that as in the unpolarized case, the collinear singularity in the quasi quark helicity distribution is exactly the same as in the light-cone distribution.

Similarly, for the transversity distribution in a transversely polarized quark, the quasi distribution $\delta\tilde{q}^{(1)}(x)$ is obtained by replacing γ^z with $\gamma^z\gamma^z\gamma^5$ in Eq. (1). The one-loop

result is

$$\delta\bar{q}^{(1)}(x) = \frac{\alpha_S C_F}{2\pi} \begin{cases} \frac{2x}{1-x} \ln \frac{x(\Lambda(x)-xP^2)}{(x-1)(\Lambda(1-x)+P^2(1-x))} + \frac{x\Lambda(1-x)+(1-x)\Lambda(x)}{(1-x)^2 P^2}, & x > 1, \\ \frac{2x}{1-x} \ln \frac{(P^+)^2}{m^2} + \frac{2x}{1-x} \ln \frac{4x(\Lambda(x)-xP^2)}{(1-x)(\Lambda(1-x)+(1-x)P^2)} - \frac{4x}{1-x} + \frac{x\Lambda(1-x)+(1-x)\Lambda(x)}{(1-x)^2 P^2}, & 0 < x < 1, \\ \frac{2x}{1-x} \ln \frac{(x-1)(\Lambda(x)-xP^2)}{x(\Lambda(1-x)+(1-x)P^2)} + \frac{x\Lambda(1-x)+(1-x)\Lambda(x)}{(1-x)^2 P^2}, & x < 0. \end{cases} \quad (23)$$

The limit $\mu \rightarrow \infty$ gives

$$\delta\bar{q}^{(1)}(x) = \frac{\alpha_S C_F}{2\pi} \begin{cases} \frac{2x}{1-x} \ln \frac{x}{x-1} + \frac{\mu}{(1-x)^2 P^2}, & x > 1, \\ \frac{4x}{1-x} \ln \frac{(P^+)^2}{m^2} + \frac{2x}{1-x} \ln \frac{4x}{1-x} - \frac{4x}{1-x} + \frac{\mu}{(1-x)^2 P^2}, & 0 < x < 1, \\ \frac{2x}{1-x} \ln \frac{x-1}{x} + \frac{\mu}{(1-x)^2 P^2}, & x < 0, \end{cases} \quad (24)$$

and the result in the infinite momentum frame is

$$\delta q^{(1)}(x) = \frac{\alpha_S C_F}{2\pi} \begin{cases} 0, & x > 1 \text{ or } x < 0, \\ \frac{2x}{1-x} \ln \frac{\mu^2}{m^2} - \frac{2x}{1-x} \ln(1-x)^2 - \frac{2x}{1-x}, & 0 < x < 1. \end{cases} \quad (25)$$

One can construct similar matching conditions as in Eq. (12) for the helicity and transversity distributions. We just list the results for the matching factors here, noting that the quark self-energy is the same. For the quark helicity distribution, one has for $\xi > 1$

$$\Delta Z^{(1)}(\xi)/C_F = \left(\frac{1+\xi^2}{1-\xi} \right) \ln \frac{\xi}{\xi-1} + 1 + \frac{1}{(1-\xi)^2} \frac{\mu}{P^2}, \quad (26)$$

whereas for $0 < \xi < 1$

$$\Delta Z^{(1)}(\xi)/C_F = \left(\frac{1+\xi^2}{1-\xi} \right) \ln \frac{(P^+)^2}{\mu^2} + \left(\frac{1+\xi^2}{1-\xi} \right) \ln [4\xi(1-\xi)] - \frac{2}{1-\xi} + 3 + \frac{\mu}{(1-\xi)^2 P^2}, \quad (27)$$

and for $\xi < 0$

$$\Delta Z^{(1)}(\xi)/C_F = \left(\frac{1+\xi^2}{1-\xi} \right) \ln \frac{\xi-1}{\xi} - 1 + \frac{\mu}{(1-\xi)^2 P^2}. \quad (28)$$

The linearly divergent term is the same as in the unpolarized case.

Finally, in the factorization theorem for transversity distribution, one has the matching factor for $\xi > 1$,

$$\delta Z^{(1)}(\xi)/C_F = \left(\frac{2\xi}{1-\xi} \right) \ln \frac{\xi}{\xi-1} + \frac{1}{(1-\xi)^2} \frac{\mu}{P^2}, \quad (29)$$

whereas for $0 < \xi < 1$

$$\delta Z^{(1)}(\xi)/C_F = \left(\frac{2\xi}{1-\xi} \right) \ln \frac{(P^+)^2}{\mu^2} + \left(\frac{2\xi}{1-\xi} \right) \ln [4\xi(1-\xi)] - \frac{2\xi}{1-\xi} + \frac{\mu}{(1-\xi)^2 P^2}, \quad (30)$$

and for $\xi < 0$

$$\delta Z^{(1)}(\xi)/C_F = \left(\frac{2\xi}{1-\xi} \right) \ln \frac{\xi-1}{\xi} + \frac{\mu}{(1-\xi)^2 P^2}. \quad (31)$$

One again has a linearly divergent contribution. Near $\xi = 1$, one needs to include an extra contribution from self energy as before.

V. CONCLUSION

We have derived one-loop matching conditions for non-singlet quark distributions, including unpolarized, helicity and transversity distributions. The matching condition, which can also be viewed as a factorization, connects the quasi quark distribution to the light-cone distribution measurable in experiments, and thereby allows an extraction of the latter from the former, which is defined as a time-independent correlation at finite nucleon momentum and therefore can be evaluated on the lattice. To carry out a lattice simulation of the quasi quark distribution, one employs the transverse momentum cutoff regulator for UV divergences. This choice of UV regulator generates a linear divergence multiplied by a singular factor in the axial gauge $A^+ = 0$. The origin of this is the self energy of the Wilson line. In our one-loop factorization formula this divergence can be removed with a principal value prescription. However, to eventually achieve a factorization to all-loop orders, it is important to investigate how this divergence structure affects the matching between quasi distribution and light-cone distribution beyond one-loop level.

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阅读方法

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More On Large-Momentum Effective Theory Approach to Parton Physics

Xiangdong Ji,^{1,2} Jian-Hui Zhang,³ and Yong Zhao⁴

¹*Tsung-Dao Lee Institute, and College of Physics and Astronomy,
Shanghai Jiao Tong University, Shanghai, 200240, P. R. China*

²*Department of Physics, University of Maryland,
College Park, Maryland 20742, USA*

³*Institut für Theoretische Physik, Universität Regensburg, D-93040 Regensburg, Germany*

⁴*Center for Theoretical Physics,
Massachusetts Institute of Technology, Cambridge, MA 02139, USA*

Abstract

Large-Momentum Effective Theory (or LaMET) advocated by the present authors provides a direct approach to simulate parton physics in Euclidean lattice QCD theory. Recently, there has been much interest in this theory in the literature, with some questioning its validity and effectiveness. Here we provide some discussions aiming at a further exposition of this approach. In particular, we explain why it does not have the usual power divergence problem in lattice QCD calculations for the moments of parton distributions. The only power divergence in the LaMET approach comes from the self-energy of the Wilson lines which can be properly factorized. We show that although the Ioffe-time distribution provides an alternative way to extract the parton distribution from the same lattice observables, it also requires the same large momentum (or short distance) limit as in LaMET to obtain a precision calculation. With a proper quantification of errors, both extraction methods shall be compared with the same lattice data.

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I. INTRODUCTION

The parton model invented by Feynman has been one of the most useful languages in describing the physics of strong interactions at high energy [1]. Partons are idealized objects that exist when hadrons travel at the speed of light. Thus in modern language, they are effective degrees of freedoms in an effective theory, for instance, the soft-collinear effective theory [2]. The parton properties of a hadron, such as parton distributions, parton wave functions or amplitudes, parton correlations, etc., are low-energy properties which require solutions of quantum chromodynamics (QCD) in the strong coupling region.

When formulated in the rest frame of the hadron, parton physics represents the space-time correlations of QCD quarks or gluons along the light-cone direction ($\xi^- = (t - z)/\sqrt{2}$), which involves an explicit time dependence [3]. (We use $\xi^\mu = (t, x, y, z)$ ($\mu = 0, 1, 2, 3$) to denote space-time coordinates, although x , y , and z are commonly used to denote momentum fractions as well.) In physics, the time-dependent correlations are in a class by themselves because the time evolution calls for the presence of the interaction-dependent Hamiltonian H , and the observables are called “dynamical”.

Dynamical correlations are hard to calculate for at least two different reasons: 1) if one inserts a set of intermediate Hamiltonian eigenstates to simplify the evolution, one needs to know the wave functions of all the intermediate excited states. Thus knowing just the initial (often ground) state wave function is not sufficient to calculate the time-dependent correlations. Methods designed to approximate the ground state might not be applicable to the excited states, and thus dynamical correlations are usually calculated less reliably than time-independent or static observables for which the initial state wave function is sufficient. 2) Since the time-dependence involves the unitary evolution operator, e^{-iHt} , it is a problem for numerical Monte Carlo simulations as it contains a sign-changing phase. In fact, the majority of the Monte Carlo approaches designed to calculate static properties cannot be readily applied to dynamical correlations. There have been attempts to calculate instead an imaginary or Euclidean evolution with $e^{-H\tau}$, and to analytically continue the result to the real time [4, 5]. However, these approaches are hardly reliable because the analytical functions are unknown. An alternative trick is to Taylor-expand the time-evolution operator, obtaining the time dependence through the sum of an infinite series. Such an approach again is very limited because of the poor precision of calculations for higher moments [6]. For the above reasons, the time-dependent correlations are generally referred to as “Minkowskian” in the sense that time and space correlations are intrinsically different. On the other hand, the time-independent correlations may be called “Euclidean” because they can be readily calculated with Monte Carlo simulations, except that there may be additional sign problems.

For Minkowskian observables, the traditional approach is to use analytical approximations. For instance, the so-called Schwinger-Dyson approach solves the time dependence in terms of dynamical equations [7]. For parton physics, in the so-called light-front quantization approach, the coordinates are redefined such that the new time is a combination of space and the old time, and the light-cone correlations become time-independent [8]. However, the resulting theory cannot be solved readily using Monte Carlo methods because the metric is not positive definite.

Fortunately, there are certain “dynamical” problems that are not intrinsically “Minkowskian” in the sense that with some clever formulation, they can be solved using Monte Carlo methods. For instance, scattering phase-shifts usually involve complex S-matrix elements which cannot be calculated directly using a Monte Carlo approach. How-

ever, one can put the theory in a box, calculating the bound state levels as a function of the boundary condition and extracting the phase shifts from bound state energy levels [9–12]. A second example is certain Minkowskian correlators can be directly analytically continued to Euclidean region without encountering any “poles”, and therefore, can be calculated directly in Euclidean space [13].

Recently it was suggested by the authors that parton physics in non-perturbative QCD belongs to a class of problems that are not intrinsically Minkowskian [14–22]. This observation is based on two important points: 1) light-cone correlators can be approximated by near-light-cone correlators through QCD perturbation theory, thanks to asymptotic freedom. 2) the Lorentz boost is dynamical, and it can be used to absorb the dynamical evolution between the fields through boosting a hadron state. Therefore, all remainder dynamics is included in the boosted hadron wave function, not in the probe operators. In this way, all parton physics can be accessed from equal-time correlations in a large-momentum hadron state which are calculable in Euclidean Monte Carlo simulations.

The approach is an effective field theory in the following sense. For any time-dependent light-cone operator $O(\xi_1^-, \xi_2^-, \dots)$, one can construct an infinite number of Euclidean quasi-operators $\mathcal{O}_i(z_1, z_2, \dots)$, where i is a label for all such operators, and z_i are the spatial coordinates of the fields along the direction of hadron momentum, which is taken as the third coordinate direction. The matrix element of such an operator in a hadron state with a large momentum p_z , calculated for example in lattice QCD, has a large momentum expansion [18]

$$\langle p | \mathcal{O}_i(z_1, z_2, \dots) | p \rangle = C_0(\alpha_s, p_z) \otimes \langle p | O(\xi_1^-, \xi_2^-, \dots) | p \rangle + \mathcal{O}(1/p_z^2), \quad (1)$$

where we have omitted all the renormalization scales, and C_0 is a hard coefficient. The higher-order corrections can be calculated systematically. Since in this effective theory the large momentum is used to approach the light-cone, it is called the large momentum effective field theory or LaMET for short.

It is worthwhile to point out that the choice of the Euclidean quasi-operator above is not unique. Any quasi-operator that yields the same light-front physics in the infinite momentum limit can be used to probe the desired light-cone operator. All such quasi-operators form a universality class. In practice, the existence of the universality class allows one to make an optimized choice of the quasi-operators so that the $\mathcal{O}(1/p_z^2)$ correction can be made as small as possible. One such example is the calculation of the total gluon polarization in the nucleon [16].

Recently, there has been much interest about this theory in the literature, with some questioning its validity or effectiveness. Here we provide some discussions aiming at a further exposition of this approach. In Ref. [23], the factorization within the LaMET approach has been questioned. Following Ref. [24], we explicitly show that the problem arises from the procedure of analytical continuation. In Ref. [25], the issue of power divergences has been raised based on an analysis of moments. We show that this problem does not exist in the non-local approach where the only power divergence arises from the self-energy of the Wilson lines. Finally, in Refs. [26, 27], a new approach has been proposed to extract parton distributions. We show that the Ioffe-time distribution is an alternative way to extract parton distribution from the same lattice observable, and hence is entirely equivalent to the LaMET approach. Besides, the apparent scaling observed in that approach might or might not help with the convergence problem.

II. PERTURBATIVE MATCHING IN MINKOWSKI SPACE AND NON-PERTURBATIVE RENORMALIZATION IN EUCLIDEAN SPACE

In a recent paper [23], the validity about the factorization of the quasi-distribution into parton distribution as in Eq. (1) has been questioned. In the original one-loop factorization proof in Ref. [17], the quasi and physical parton distributions were calculated using the Lehmann-Symanzik-Zimmermann reduction formula in Minkowskian field theory [3] where all the internal loop integrals are Minkowskian. In this form, all the infrared (IR) properties are very clear. On the other hand, because the quasi-distribution has no time-dependence, it can be calculated from a Euclidean space state that evolves as

$$|p\rangle = e^{-\beta H}|p\rangle \quad (2)$$

in Monte Carlo simulations. Here $|p\rangle$ is a state with momentum p and correct quantum number but not an eigenstate, whereas H is the QCD Hamiltonian, β is an evolution parameter (“Euclidean time”). In lattice QCD, nucleon matrix elements are obtained from the source-operator-sink correlation where the source, operator, and sink are separated by imaginary time. For large separation, this three-point correlation function is dominated by the contribution from the lowest physical state [28]. Thus, the matrix elements are effectively calculated in an Euclidean-like field theory. However, the equivalence of the Euclidean calculations with the original Minkowskian field theory is guaranteed through the derivation of the formalism. This equivalence has been the basis for all the Euclidean lattice calculations in the past.

It was shown, however, that the Feynman integrals in lattice perturbation theory with external Minkowskian momenta do not have collinear divergences [23]. Therefore, it appears that the quasi-distributions from lattice QCD do not have proper IR properties. In a subsequent paper [24], it was explicitly shown that the lattice calculation procedure leads to the Minkowskian theory matrix element with correct collinear divergences. The question then becomes what the correct procedure is to recover the physical result (or collinear divergences) with Minkowskian external momenta from the Euclidean perturbation theory. The answer is that one cannot directly insert the Minkowskian external momentum into the Euclidean integrals, as this procedure will change the pole structures. The correct procedure is to finish the Euclidean integral with Euclidean external momenta and then analytically continue the result to Minkowski spacetime. To see that the nucleon matrix elements of the quasi-distribution calculated from lattice QCD does capture the correct IR physics in perturbation theory, one has to perform an analytical continuation of the matrix elements after the Euclidean loop integrals are done, not before.

In the following, we provide an example to illustrate the above point: We calculate a one-loop integral appearing in the Euclidean quasi-distribution and show that the collinear divergence becomes manifest after an analytic continuation from p_z^2 to p_{Mz}^2 with p_{Mz}^2 and p_E^2 being the Minkowskian and Euclidean momenta. Actually the result is identical to its Minkowskian counterpart.

We first start with the Minkowskian integral, where $p_M^0 = \sqrt{p_z^2 + m^2}$,

$$I_M = \int \frac{d^4 k_M}{(2\pi)^4} \frac{(2\pi) p^\tau \delta(k_M^\tau - x p^\tau)}{(k_M^2 - m^2 + i\epsilon) [(p_M - k_M)^2 + i\epsilon]}, \quad (3)$$

We focus on the physical region $0 < x < 1$ where the collinear divergence could exist. For I_M , we first integrate over k_M^0 with the residue theorem, and then over the transverse

components, to get the result

$$I_M = \frac{i}{8\pi} \frac{1}{\sqrt{1+\rho_M}} \left[\ln \frac{1+\sqrt{1+\rho_M}}{\rho_M} + \ln \frac{\sqrt{(1+\rho_M)(x^2+\rho_M)} + x + \rho_M}{1-x} \right], \quad (4)$$

where

$$\rho_M = \frac{m^2}{(p^x)^2} < 1. \quad (5)$$

For the Euclidean space integral, we first keep p_E^x real,

$$I_E = \int \frac{d^4 k_E}{(2\pi)^4} \frac{(2\pi) p^x \delta(k_E^x - x p^x)}{(k_E^2 + m^2)(p_E - k_E)^2}. \quad (6)$$

Integrating over the poles, we find

$$I_E = \frac{1}{8\pi \sqrt{\rho_E - 1 - i\epsilon}} \left[\arctan \frac{\rho_E + \rho_M - 2(1-x)}{2\sqrt{\rho_E - 1 - i\epsilon}(1-x)} + \arctan \frac{\rho_E - \rho_M - 2x}{2\sqrt{(\rho_E - 1 - i\epsilon)(x^2 + \rho_M)}} \right], \quad (7)$$

where

$$\rho_E = \frac{(p^x)^2 + (p^t)^2}{(p^x)^2} > 1. \quad (8)$$

We keep $-i\epsilon$ in the imaginary part of ρ_E so that we can analytically continue it to ρ_M through the lower half complex plane by making the replacement $p^x \rightarrow -ip^\theta$,

$$\sqrt{\rho_E - 1 - i\epsilon} \rightarrow -i\sqrt{1 + \rho_M}. \quad (9)$$

We find that

$$\begin{aligned} I_E' &= -\frac{1}{8\pi} \frac{1}{\sqrt{1+\rho_M}} \left[\frac{1}{2} \operatorname{arctanh} \left(-\frac{1}{\sqrt{1+\rho_M}} \right) \right. \\ &\quad \left. + \frac{1}{2} \operatorname{arctanh} \left(-\frac{x+\rho}{\sqrt{(1+\rho_M)(x^2+\rho_M)}} \right) \right] \\ &= -iI_M, \end{aligned} \quad (10)$$

which is exactly the Minkowskian matrix element with a factor of i that is to be canceled in $dk_M^0 = idk_E^4$. If we take the massless limit $m \rightarrow 0$, then both I_M and I_E' give

$$I_M = \frac{i}{8\pi} \left(\ln \frac{(p^x)^2}{m^2} + \ln \frac{4x}{1-x} \right), \quad (11)$$

with collinear divergence correctly characterized by $\ln m^2$.

Despite the analytical continuation to recover the correct collinear divergence, we find that the Euclidean matrix elements with Euclidean external momenta are still useful for renormalization. The quasi-distribution calculated in lattice QCD requires a non-perturbative renormalization, which can be done, e.g., in the RI/MOM scheme used in Refs. [29–31]. To compute the renormalization factors in such a scheme, one needs to compute the vertex functions with external quarks and gluons that have deep-Euclidean momenta. After renormalization, the quasi-distribution is expected to have a well-defined continuum limit. It can then be matched perturbatively to the normal distributions in the $\overline{\text{MS}}$ scheme, where the matching factor can be computed in Minkowski space.

III. POWER DIVERGENCES AND RENORMALIZATION OF QUASI-DISTRIBUTIONS IN LATTICE QCD

In a paper by Rossi and Resta [25], a question has been raised about the practical implementation of the factorization theorem, Eq. (1), in lattice QCD. In particular, they consider the quasi-distribution in Eq. (19) of their paper, and try to compute its second moment in the rest frame of the hadron. They point out that because of the mixing of the trace term with the lower dimensional operator, the $1/p_z^2$ suppressed terms have a power-divergent contribution as shown in Eq. (35) of their paper. Therefore, they conclude that “unless one performs a non-perturbative subtraction of the power divergent terms, the $p_z \rightarrow \infty$ limit does not exist.” However, a careful analysis shows that this problem does not exist in the present formalism.

First of all, the quasi-distribution used to extract the physical parton distribution, is a function of $y = k_z/p_z$. Formally, one can take the second moment and show that it results in a local operator with double derivatives. However, in practice, the moments of the quasi-distribution do not exist. As can be seen from the one-loop calculation [17], the result behaves like $1/|y|$ at large y , therefore all higher moments other than the zeroth one do not converge. It is exactly this large y divergence that calls for an extra renormalization in the local operators. However, without taking the moments, the divergence does not appear.

In fact, it is exactly this non-local quasi-distribution formulation of parton physics that avoids the power divergence problem in moment calculations using the old-fashioned approach. The non-local formulation has much simpler ultraviolet (UV) physics and hence makes it much easier to control the divergences.

For example, the renormalization of the quasi-distributions is straightforward in “heavy-quark” formalism. It can be shown that for quark non-singlets, except for the Wilson-line self-energy or mass correction, there are no other power divergences [32–36]. All other divergences are logarithmic in the lattice spacing. Therefore, the quasi-distributions can be renormalized by factoring out the power-divergent self-energy factor and logarithmic renormalization factor, and have a smooth limit when the lattice spacing vanishes. Note that without factoring out the power-divergent self-energy factor, a one-loop mass correction of form $\alpha_s C_1 |z|/a$ with lattice spacing a and Wilson-line length $|z|$ can generate a finite collinearly divergent term in the quasi-distribution at two-loop, which appears to break the IR factorization of the quasi-distribution [37]. However, such terms do not contribute when the linear divergence is factored out of the quasi-distributions before taking the lattice spacing to zero.

After renormalizing the quasi-distributions in an appropriate scheme, one can then use the factorization formula in Eq. (1) to match it to the $\overline{\text{MS}}$ parton distribution. Then the

higher-twist $1/p_z^2$ terms vanish smoothly as $p_z \rightarrow \infty$. This is exactly what RI/MOM scheme does in recent papers [29–31].

IV. QUASI-DISTRIBUTIONS VS. PSEUDO DISTRIBUTIONS, SIMILARITIES AND DIFFERENCES

In a recent paper [26], an alternative interpretation has been suggested for the Euclidean matrix elements proposed in LaMET, which provides another method to extract the physical parton distribution from the same lattice calculations. The starting point is the so-called Ioffe-time distribution $\mathcal{M}(\nu, \xi^2)$, which is defined as the matrix elements of bilinear quark operators, separated with arbitrary spacetime distances ξ^μ , and in a hadron state with four-momentum p^μ . As such, it is an invariant quantity of $\nu = -p \cdot \xi$ (Ioffe-time), and space-time invariant $-\xi^2$. In practice, such a Lorentz-invariant function can be obtained on the Euclidean lattice from the two-quark spatial correlation function,

$$h(zp_z, z^2, \Lambda) = \langle p | \bar{\psi}(z) \Gamma L(z, 0) \psi(0) | p \rangle, \quad (12)$$

where Γ is Dirac matrix, $L(z, 0)$ the straight gauge link, and Λ a renormalization scale or lattice cutoff scale. Thus, the matrix element $h(zp_z, z^2, \Lambda)$ in the quasi distribution calculations in LaMET can be viewed as a covariant Ioffe-time distribution $h(\nu, \xi^2, \Lambda)$ with $\nu = zp_z$, in space-like separation, $-\xi^2 = z^2$.

Radyushkin suggests to interpret the Ioffe-time distribution in a new frame, in which $\xi^\mu = (\xi^+ = 0, \xi^-/\gamma, \xi_\perp)$, where we have used the light-cone variable $\xi^\mu = (\xi^+, \xi^-, \xi_\perp)$, and $p^\mu = (p^+ \gamma, 0, p^- = M^2/(p^+ \gamma))$, M is the hadron mass, and γ is a boost parameter. With $\gamma \rightarrow \infty$, one recovers the infinite momentum frame. In this new frame, the Ioffe time is $\nu = -p^+ \xi^-$, and $-\xi^2 = \xi_\perp^2$ is again space-like, and they are γ independent. And h becomes a near light-cone correlation function,

$$h(p^+ \xi^-, \xi_\perp, \Lambda) = \langle p | \bar{\psi}(0, \xi^-, \xi_\perp) \Gamma L(0, \xi^-, \xi_\perp; 0) \psi(0) | p \rangle, \quad (13)$$

where now the gauge link has both components along the light-cone and transverse directions. Lorentz symmetry guarantees the quality of Eq. (11) and (12), with $\xi_\perp = z$, and $\xi^- = -zp_z/p^+$. The Fourier transformation with respect to the Ioffe-time ν has the direct interpretation as the light-cone momentum fraction x , therefore is bounded in $[-1, 1]$ as always the case in the infinite momentum frame.

One can then introduce a “pseudo”-distribution from the space-like correlation,

$$\mathcal{P}(y, \xi_\perp, \Lambda) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dy e^{-iy\nu} h(zp_z, z^2, \Lambda), \quad (14)$$

where $\xi_\perp^2 = z^2$. In this approach, the hadron momentum on lattice is re-interpreted as the Ioffe time at a fixed $\xi_\perp$. The physics of the pseudo-distribution is that it is a Fourier transformation of a type of transverse-momentum dependent parton distributions. Since there is a gauge link between the quark fields going along the transverse direction in Eq. (13), in $A^+ = 0$ gauge, there is an infinite number of transversely polarized gluons involved in the distributions. Therefore the parton picture of the pseudo-distribution is not simple. In particular, the momentum fraction y contains not only the contribution from a single quark, but also that from transversely-polarized physical gluons. y approaches the quark light-cone fraction only in the limit of $\xi_\perp = 0$.

In this alternative interpretation of the Euclidean matrix element, the physical quark distribution is recovered by putting the quarks entirely on the light-cone, i.e., when the transverse coordinate of the quark $\xi_\perp = z$ goes to zero. In this limit, to get finite Ioffe-time $\nu = zp_z$, one again has to let $p_z \rightarrow \infty$, the same as used in LaMET. Thus, the requirement to get a precision parton distribution is exactly the same in the two approaches.

In practice, one has to work with small but nonvanishing $\xi_\perp$ to avoid light-cone divergences. One can then establish a similar factorization theorem as in LaMET (“a small-distance effective theory”),

$$\mathcal{P}(y, \xi_\perp, \bar{\mu}) = \int \frac{dx}{x} C(y/x, \xi_\perp \bar{\mu}, \bar{\mu}/\mu) q(x, \mu) + \mathcal{O}(\xi_\perp^2), \quad (15)$$

where q is a physical parton distribution, C is a matching coefficient that depends on the specific lattice regularization, $\bar{\mu}$ and μ are the renormalization scales of the Ioffe-time distribution and physical PDF. To have continuum limit, $\xi_\perp$ shall be much larger than lattice spacing a , but much smaller than $1/\Lambda_{\text{QCD}}$. The corrections are in terms of small $\xi_\perp$ expansion. It would be interesting to extract the parton distributions through the above factorization formula at a fixed $\xi_\perp$.

To demonstrate the above factorization theorem, let us take the isovector unpolarized quark distribution as an example. In the Feynman gauge, we calculate the quark matrix elements of the quasi and physical PDFs at one-loop order in dimensional regularization with $D = 4 - 2\epsilon$. The external quark state is chosen to be onshell and massless, so the UV and collinear divergences are regularized by $1/\epsilon_{UV}$ ($\epsilon_{UV} > 0$) and $1/\epsilon_{IR}$ ($\epsilon_{IR} < 0$) respectively. We compute the same Feynman diagrams in Ref. [17] in the coordinate space to obtain $h^{(1)}(zp^+, z^2, \mu, \epsilon)$ (for $\Gamma = \gamma^0$) and $q^{(1)}(p^+ z^-, \mu, \epsilon)$, where μ can be regarded as the renormalization scale. The Feynman rules are not conventional as there is a Fourier transform of one loop momentum component in each of the irreducible diagrams, for instance,

$$\mu^{2\epsilon} \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2 (p-k)^2} e^{-ik^+ z} = i\mu^{2\epsilon} \int_0^1 du \int \frac{d^d k}{(2\pi)^d} \frac{e^{-ik^+ z}}{k_E^4} e^{-iup^+ z}. \quad (16)$$

Using Schwinger parametrization, we can evaluate the above integral analytically,

$$\begin{aligned} & i\mu^{2\epsilon} \int_0^1 du \int \frac{d^d k}{(2\pi)^d} \frac{e^{-ik^+ z}}{k_E^4} e^{-iup^+ z} \\ &= i\mu^{2\epsilon} \int_0^1 du \int_0^\infty d\alpha \alpha \int \frac{d^d k}{(2\pi)^d} e^{-\alpha k^2 - ik^+ z} e^{-iup^+ z} \\ &= \frac{i}{16\pi^2} (\pi z^2 \mu^2)^{\epsilon+1} \Gamma(-\epsilon_{IR}) \int_0^1 du e^{-iup^+ z}. \end{aligned} \quad (17)$$

The complete results for $h^{(1)}(zp^+, z^2, \mu, \epsilon)$ and $q^{(1)}(p^+z^-, \mu, \epsilon)$ are

$$\begin{aligned}
& h^{(1)}(zp^+, z^2, \mu, \epsilon) \\
&= \frac{\alpha_s C_F}{2\pi} \int_0^1 du \left[\left(1 - u + \left(\frac{2u}{1-u} \right)_+ \right) \left(-\ln(z^2 \mu^2) - \frac{1}{\epsilon_{IR}} \right) \right. \\
&\quad \left. + (1-u) \right] e^{-i\varphi^+ z} \\
&+ \frac{\alpha_s C_F}{2\pi} (i\varphi^+) \int_0^1 du \int_0^1 dt (2-u)(-\ln t^2) e^{-i(1-tu)\varphi^+ z} \\
&+ \left[(1 + \delta Z_\psi) + \frac{\alpha_s C_F}{2\pi} \left(2\ln(z^2 \mu^2) + \frac{2}{\epsilon'_{UV}} + 2 \right) \right] e^{-i\varphi^+ z}, \tag{18}
\end{aligned}$$

$$\begin{aligned}
& q^{(1)}(p^+z^-, \mu, \epsilon) \\
&= \frac{\alpha_s C_F}{2\pi} \int_0^1 du \left(1 - u + \left(\frac{2u}{1-u} \right)_+ \right) \left(\frac{1}{\epsilon'_{UV}} - \frac{1}{\epsilon_{IR}} \right) e^{i\varphi^+ z^-} \\
&+ (1 + \delta Z_\psi) e^{i\varphi^+ z^-}, \tag{19}
\end{aligned}$$

where $Z_\psi = 1 + \delta Z_\psi$ is the quark wavefunction renormalization constant, and we have made the substitutions $1/\epsilon'_{UV,IR} = 1/\epsilon_{UV,IR} + \gamma_E + \ln \pi$. To guarantee vector current conservation, Z_ψ is given in the on-shell scheme,

$$Z_\psi = 1 - \frac{\alpha_s C_F}{2\pi} \frac{1}{2} \left(\frac{1}{\epsilon'_{UV}} - \frac{1}{\epsilon_{IR}} \right) + O(\alpha_s^2). \tag{20}$$

Then we Fourier transform the Ioffe time p^+z or p^+z^- into x , and obtain

$$\begin{aligned}
& \mathcal{P}^{(1)}(x, \xi_\perp, \mu, \epsilon) \\
&= \frac{\alpha_s C_F}{2\pi} \left[\left(\frac{1+x^2}{1-x} \right)_+ \left(-\ln(\xi_\perp^2 \mu^2) - \frac{1}{\epsilon_{IR}} - 1 \right) - \left(\frac{4\ln(1-x)}{1-x} \right)_+ \right. \\
&\quad \left. + 2(1-x) \right] \theta(x) \theta(1-x) \\
&+ \left[1 + \frac{\alpha_s C_F}{2\pi} \left(\frac{3}{2} \ln(\xi_\perp^2 \mu^2) + \frac{3}{2} \frac{1}{\epsilon'_{UV}} + \frac{3}{2} \right) \right] \delta(1-x), \tag{21}
\end{aligned}$$

whereas the corresponding light-cone PDF is

$$q^{(1)}(x, \mu, \epsilon) = \delta(1-x) + \frac{\alpha_s}{2\pi} \left(\frac{1+x^2}{1-x} \right)_+ \left(\frac{1}{\epsilon'_{UV}} - \frac{1}{\epsilon_{IR}} \right). \tag{22}$$

The support in the pseudo distribution $\mathcal{P}^{(1)}(x, \xi_\perp, \mu, \epsilon)$ is restricted to $0 < x < 1$, as expected. If we renormalize both Eq. (21) and Eq. (22) in the $\overline{\text{MS}}$ scheme, then the matching

coefficient in Eq. (15) is read off as

$$\begin{aligned}
& C(y, \xi_\perp \mu) \\
&= \left[1 + \frac{\alpha_s C_F}{2\pi} \left(\frac{3}{2} \ln(\xi_\perp^2 \mu^2) + \frac{3}{2} \right) \right] \delta(1-y) \\
&+ \frac{\alpha_s C_F}{2\pi} \left[- \left(\frac{1+y^2}{1-y} \right)_+ (\ln(\xi_\perp^2 \mu^2) + 1) - \left(\frac{4 \ln(1-y)}{1-y} \right)_+ \right. \\
&\quad \left. + 2(1-y) \right] \theta(y) \theta(1-y) .
\end{aligned} \tag{23}$$

Similar factorization formula can be extended to the case of lattice regularization and non-perturbative renormalization.

Instead of using the factorization formula in Eq. (15) to extract parton distribution from lattice matrix element calculated at a fixed $\xi_\perp$, Ref. [26] focused on the possible approximate factorized property of the Ioffe-time distribution, i.e. $\mathcal{M}(\nu, z) \sim g(\nu)h(z)$. If so, $g(\nu) \sim \mathcal{M}(\nu, z)/h(z)$ will have approximate scaling at different z . One can then get the small z Ioffe-time distribution from a large z calculation, which does not need a very large momentum to produce a large Ioffe time. Indeed, an exploratory calculation on lattice shows that this special factorization or scaling works quite well [27].

While such a scaling or factorization behavior is interesting, just like parton-hadron duality in deep-inelastic scattering [38] allowing the extraction of parton distributions at small Q^2 , it is of limited use in actual precision calculations of parton distributions. The scaling in the exploratory calculations was observed in a limited kinematic domain with limited precision. The parton distributions involve Ioffe time at all sizes, particularly small x distribution involves very large Ioffe time, where the scaling becomes much harder to test and the present result is certainly incorrect. To get the correct large Ioffe-time behavior, one needs calculation at small $\xi_\perp$ and larger nucleon momenta, the same as required in LaMET. Moreover, the new $\xi_\perp$ dependence factorization must be corrected for precision calculations. Since there is no systematic approach to correct for the violation of this factorization, such a phenomenological approach will not be a replacement for rigorous approaches as in Eq. (15), where corrections can be quantified.

It of course will be interesting to compare the parton distributions from two seemingly different factorization approaches (large momentum vs. small distance), when extracted from the same lattice matrix elements. It is important to quantify the errors of extraction in both approaches consistently. We emphasize that LaMET is a much more general framework to calculate parton physics with simple physical pictures as provided by Feynman. It has a well-defined recipe on how to systematically calculate all parton observables with quantifiable errors, including multi-parton amplitudes and correlations.

V. CONCLUSION

We have presented some discussions aiming at a further exposition of the LaMET approach. We clarified the validity of the factorization in this approach, and stressed the importance of analytical continuation in recovering the correct IR behavior of Minkowski space integrals from their Euclidean counterparts. We also pointed out that power divergences plaguing the traditional moments approach do not pose a problem in the quasi-distribution approach, which employs a non-local formulation. Finally, we showed that the Ioffe-time

distribution method to extract the parton distribution from the same lattice observables used in LaMET requires exactly the same physical conditions. It is interesting to compare the results of both approaches after proper quantifications of errors.

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P15: 准分布动量分布与伪分布

主题归属	LaMET、quasi/pseudo 分布与因子化
底稿	arXiv:1705.01488
arXiv / DOI	1705.01488
完整原始 PDF	6 页; 来源 course-cache; SHA-256 a76dcf322909...
转排索引	EN 10 页; ZH 9 页 (只作书目对照)
备注	索引未记录额外备注。

阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

页码范围：P15-p001 – P15-p006

Quasi-PDFs, momentum distributions and pseudo-PDFs

A. V. Radyushkin

Old Dominion University, Norfolk, VA 23529, USA and
Thomas Jefferson National Accelerator Facility, Newport News, VA 23606, USA

We show that quasi-PDFs may be treated as hybrids of PDFs and primordial rest-frame momentum distributions of partons. This results in a complicated convolution nature of quasi-PDFs that necessitates using large $p_3 \gtrsim 3$ GeV momenta to get reasonably close to the PDF limit. As an alternative approach, we propose to use pseudo-PDFs $\tilde{\mathcal{P}}(x, z_3^2)$ that generalize the light-front PDFs onto spacelike intervals and are related to Ioffe-time distributions $\mathcal{M}(\nu, z_3^2)$, the functions of the Ioffe time $\nu = p_{3,23}$ and the distance parameter z_3^2 with respect to which it displays perturbative evolution for small z_3 . In this form, one may divide out the z_3^2 dependence coming from the primordial rest-frame distribution and from the problematic factor due to lattice renormalization of the gauge link. The ν -dependence remains intact and determines the shape of PDFs.

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I. INTRODUCTION

The parton distribution functions (PDFs) $f(x)$ [1] are related to matrix elements of bilocal operators on the light cone $z^2 = 0$, which prevents a straightforward calculation of these functions in the lattice gauge theory formulated in Euclidean space. The usual way is to calculate their moments. However, recently, X. Ji [2] suggested a method allowing to calculate PDFs as functions of x . To this end, he proposes to use purely space-like separations $z = (0, 0, 0, z_3)$. Then one deals with quasi-PDFs $Q(y, p_3)$ describing sharing of the p_3 hadron momentum component, and tending to PDFs $f(y)$ in the $p_3 \rightarrow \infty$ limit. The same method can be applied to distribution amplitudes (DAs). The results of lattice calculations of quasi-PDFs were reported in Refs. [3–5] and of the pion quasi-DA in Ref. [6].

In our recent papers [7, 8], we have studied nonperturbative p_3 -evolution of quasi-PDFs and quasi-DAs using the formalism of virtuality distribution functions [9, 10]. We found that quasi-PDFs can be obtained from the transverse momentum dependent distributions (TMDs) $\mathcal{F}(x, k_\perp^2)$. We built models for the nonperturbative evolution of quasi-PDFs using simple models for TMDs. Our results are in qualitative agreement with the p_3 -evolution patterns obtained in lattice calculations.

In the present paper, our first goal is to develop a picture for quasi-PDFs as hybrids of PDFs and primordial momentum distributions of partons in a hadron at rest. As an intermediate step, we demonstrate that the connection between TMDs and quasi-PDFs [7] is a mere consequence of Lorentz invariance. Then we show that, when a hadron is moving, the parton k_3 momentum comes from two sources. The motion of the hadron as a whole gives the $x p_3$ part, governed by the dependence of the TMD $\mathcal{F}(x, \kappa^2)$ on its x argument. The remaining part $k_3 - x p_3$ is governed by the dependence of the TMD on its second argument, κ^2 , dictating the primordial rest-frame momentum distribution. The convolution nature of quasi-PDFs results in a rather complicated pattern of their p_3 -evolution, necessitating rather large values $p_3 \sim 3$ GeV

for getting close to the PDF limit.

Thus, our second goal is to propose an alternative approach for lattice PDF extraction. To this end, we introduce *pseudo-PDFs* $\tilde{\mathcal{P}}(x, z_3^2)$ that generalize the light-cone PDFs $f(x)$ onto spacelike intervals like $z = (0, 0, 0, z_3)$. The pseudo-PDFs are Fourier transforms of the *Ioffe-time* [11] distributions [12] $\mathcal{M}(\nu, z_3^2)$ that are basically given by generic matrix elements like $\langle p | \phi(0) \phi(z) | p \rangle$ written as functions of $\nu = p_{3,23}$ and z_3^2 . Unlike quasi-PDFs, the pseudo-PDFs have the “canonical” $-1 \leq x \leq 1$ support for all z_3^2 . They tend to PDFs when $z_3 \rightarrow 0$, showing in this limit a usual perturbative evolution with $1/z_3$ serving as an evolution parameter. Finally, we discuss how these properties of pseudo-PDFs may be used for extraction of PDFs on the lattice.

II. PARTON DISTRIBUTIONS

A. Generic matrix element and Lorentz invariance

Historically [1], PDFs were introduced to describe spin-1/2 quarks. Since complications related to spin do not affect the very concept of parton distributions, we start with a simple example of a scalar theory. In that case, information about the target is accumulated in the generic matrix element $\langle p | \phi(0) \phi(z) | p \rangle$. By Lorentz invariance, it is a function of two scalars, $(p z) = -\nu$ and z^2 (or $-z^2$ if we want a positive value for spacelike z):

$$\langle p | \phi(0) \phi(z) | p \rangle = \mathcal{M}(-(p z), -z^2). \quad (1)$$

It can be shown [7, 13] that, for all contributing Feynman diagrams, its Fourier transform $\mathcal{P}(x, -z^2)$ with respect to $(p z)$ has the $-1 \leq x \leq 1$ support, i.e.,

$$\mathcal{M}(-(p z), -z^2) = \int_{-1}^1 dx e^{-ix(p z)} \mathcal{P}(x, -z^2). \quad (2)$$

Note that Eq. (2) gives a covariant definition of x . There is no need to assume that $p^2 = 0$ or $z^2 = 0$ to define x .

B. Collinear PDFs

Choosing some special cases of p and z , one can get expressions for various parton distributions, all in terms of the same function $\mathcal{M}(-(\underline{p}_\perp, -z^2)$. In particular, taking a light-like z , e.g., that having the light-front minus component z_- only, we parameterize the matrix element by the twist-2 parton distribution $f(x)$

$$\mathcal{M}(-p_+ z_-, 0) = \int_{-1}^1 dx f(x) e^{-ixp_+ z_-}, \quad (3)$$

with $f(x)$ having the usual interpretation of probability that the parton carries the fraction x of the target momentum component p_+ . The inverse relation is given by

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\nu e^{-i\nu x} \mathcal{M}(\nu, 0) = \mathcal{P}(x, 0). \quad (4)$$

Since $f(x) = \mathcal{P}(x, 0)$, the function $\mathcal{P}(x, -z^2)$ generalizes PDFs onto non-lightlike intervals z^2 , and we will call it *pseudo-PDF*. The variable $(p_\perp)$ is called the *Ioffe time* [11], and $\mathcal{M}(\nu, -z^2)$ is the *Ioffe-time distribution* [12].

Note that the definition of $\mathcal{P}(x, -z^2)$ is simpler than that of $f(x)$ because it does not require taking a subtle $z^2 \rightarrow 0$ limit. In renormalizable theories, the function $\mathcal{M}(\nu, z^2)$ has $\sim \ln z^2$ singularities generating perturbative evolution of parton densities. Within the operator product expansion (OPE) approach, the $\ln z^2$ singularities are subtracted using some prescription, say, dimensional renormalization, and the resulting PDFs depend on the renormalization scale μ , i.e., $f(x) \rightarrow f(x, \mu^2)$.

C. Transverse momentum dependent distributions

When z^2 is spacelike, one can treat $-z^2$ as the magnitude squared of a two-dimensional vector $\{z_1, z_2\}$, and introduce a two-dimensional Fourier transform with respect to its components, i.e., to write

$$\begin{aligned} \mathcal{P}(x, z_1^2 + z_2^2) &= \int_{-\infty}^{\infty} dk_1 e^{ik_1 z_1} \\ &\times \int_{-\infty}^{\infty} dk_2 e^{ik_2 z_2} \mathcal{F}(x, k_1^2 + k_2^2). \end{aligned} \quad (5)$$

Due to rotational invariance of $\mathcal{P}(x, z_1^2 + z_2^2)$ in $\{z_1, z_2\}$ plane, the function $\mathcal{F}(x, k_1^2 + k_2^2)$ depends on k_1, k_2 through $k_1^2 + k_2^2$, the fact already reflected in the notation. Combining this representation with Eq. (2), one has

$$\begin{aligned} \mathcal{M}(\nu, z_1^2 + z_2^2) &= \int_{-1}^1 dx e^{ix\nu} \int_{-\infty}^{\infty} dk_1 e^{ik_1 z_1} \\ &\times \int_{-\infty}^{\infty} dk_2 e^{ik_2 z_2} \mathcal{F}(x, k_1^2 + k_2^2). \end{aligned} \quad (6)$$

A physical interpretation of $\mathcal{F}(x, k_1^2 + k_2^2)$ may be given in the frame where the target momentum p is longitudinal, $p = (E, \mathbf{0}_\perp, P)$, while the vector $\{z_1, z_2\}$ is in the

transverse plane. Taking z that has z_- and $z_\perp$ components only, one can identify $\mathcal{F}(x, k_\perp^2)$ with the TMD and write

$$\mathcal{P}(x, z_\perp^2) = \int d^2 \mathbf{k}_\perp e^{i(\mathbf{k}_\perp, \mathbf{z}_\perp)} \mathcal{F}(x, k_\perp^2). \quad (7)$$

In this case, the pseudo-PDFs $\mathcal{P}(x, z_\perp^2)$ coincide with the *impact parameter distributions*, a well-known concept actively used in TMD studies.

The $\sim \ln z_\perp^2$ terms in $\mathcal{M}(\nu, z_\perp^2)$ are produced by the $\sim 1/k_\perp^2$ hard tail of $\mathcal{F}(x, k_\perp^2)$. Thus, it makes sense to visualize $\mathcal{M}(\nu, z_\perp^2)$ as a sum of a soft part $\mathcal{M}^{\text{soft}}(\nu, z_\perp^2)$, that has a finite $z_\perp^2 \rightarrow 0$ limit and a hard part reflecting the evolution. For TMDs, soft part decreases faster than $1/k_\perp^2$, say, like a Gaussian $e^{-k_\perp^2/\Lambda^2}$. In the $z_\perp$ space, the distributions are then concentrated in $z_\perp \sim 1/\Lambda$ region.

III. QUASI-DISTRIBUTIONS

A. Definition and relation to TMDs

Since one cannot have light-like separations on the lattice, it was proposed [2] to consider spacelike separations $z = (0, 0, 0, z_3)$ [or, for brevity, $z = z_3$]. Then, in the $p = (E, \mathbf{0}_\perp, P)$ frame, one introduces the quasi-PDF $Q(y, P)$ through a parametrization

$$\langle p | \phi(0) \phi(z_3) | p \rangle = \int_{-\infty}^{\infty} dy Q(y, P) e^{iyPz_3}. \quad (8)$$

According to this definition, the function $Q(y, p)$ characterizes the probability that the parton carries fraction y of hadron's third momentum component P . Viewing the matrix element as a function of the ν and $-z^2$ variables (they are given by Pz_3 and z_3^2 in this case), we have

$$\mathcal{M}(\nu, z_3^2) = \int_{-\infty}^{\infty} dy Q(y, P) e^{iy\nu}. \quad (9)$$

Noticing that $z_3^2 = \nu^2/P^2$, we get the inverse Fourier transformation in the form

$$Q(y, P) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\nu e^{-iy\nu} \mathcal{M}(\nu, \nu^2/P^2). \quad (10)$$

It indicates that $Q(y, P)$ tends to $f(y)$ in the $P \rightarrow \infty$ limit, as far as $\mathcal{M}(\nu, \nu^2/P^2) \rightarrow \mathcal{M}(\nu, 0)$.

Thus, the deviation of the quasi-PDF $Q(y, P)$ from the PDF $f(y)$ is determined by the dependence of $\mathcal{M}(\nu, z_\perp^2)$ with respect to its second argument. By Eq. (6), this dependence is related to the dependence of the TMD $\mathcal{F}(x, \kappa^2)$ on its second argument κ^2 . Hence, the difference between $Q(y, P)$ and $f(y)$ may be described in terms of TMDs.

To this end, we incorporate the fact that Eq. (6) is a *mathematical* relation between the function $\mathcal{M}(\nu, z_1^2 + z_2^2)$ and the function $\mathcal{F}(x, k_1^2 + k_2^2)$, no matter what is a *physical* meaning of the variables z_1, z_2 and

k_1, k_2 . Thus, we substitute Eq. (6) with $z_1 = 0$ and $z_2 = \nu/P$ into Eq. (10) to convert it into the expression for quasi-PDFs in terms of TMDs

$$Q(y, P)/P = \int_{-\infty}^{\infty} dk_1 \int_{-1}^1 dx \mathcal{F}(x, k_1^2 + (y-x)^2 P^2). \quad (11)$$

Originally, this relation was derived in Ref. [7] using a Nakanishi-type representation of Refs. [9, 10]. Now, we see that it is a mere consequence of Lorentz invariance.

B. Quantum chromodynamics (QCD) case

The formulas derived above are directly applicable for non-singlet parton densities in QCD. In that case, one deals with matrix elements of

$$\mathcal{M}^a(z, p) = \langle p | \bar{\psi}(0) \gamma^a \hat{E}(0, z; A) \psi(z) | p \rangle \quad (12)$$

type, where $\hat{E}(0, z; A)$ is the standard $0 \rightarrow z$ straight-line gauge link in the quark (fundamental) representation. These matrix elements may be decomposed into p^+ and z^α parts:

$$\begin{aligned} \mathcal{M}^a(z, p) = & 2p^+ \mathcal{M}_p(-zp, -z^2) \\ & + z^\alpha \mathcal{M}_z(-zp, -z^2). \end{aligned} \quad (13)$$

The $\mathcal{M}_p(-zp, -z^2)$ part gives the twist-2 distribution when $z^2 \rightarrow 0$, while $\mathcal{M}_z(-zp, -z^2)$ is a purely higher-twist contamination, and it is better to get rid of it.

If one takes $z = (z_-, z_\perp)$ in the $\alpha = +$ component of $\mathcal{M}^a$, the z^α -part drops out, and one can introduce a TMD $\mathcal{F}(x, k_\perp^2)$ that is related to $\mathcal{M}_p(\nu, z_\perp^2)$ by the scalar formula (6). For quasi-distributions, the easiest way to remove the z^α contamination is to take the time component of $\mathcal{M}^a(z = z_3, p)$ and define

$$\mathcal{M}^0(z_3, p) = 2p^0 \int_{-1}^1 dy Q(y, P) e^{iyPz_3}. \quad (14)$$

Then the connection between $Q(y, P)$ and $\mathcal{F}(x, k_\perp^2)$ is given by the scalar formula (11).

One may notice that the operator defining $\mathcal{M}^a(z, p)$ involves a straight-line link from 0 to z rather than a stapled link usually used in the definitions of TMDs appearing in the description of Drell-Yan and semi-inclusive DIS processes. As is well-known, the stapled links reflect initial or final state interactions inherent in these processes. The “straight-link” TMDs, in this sense, describe the structure of a hadron when it is in its non-disturbed or “primordial” state. While it is unlikely that such a TMD can be measured in a scattering experiment, it is a well-defined QFT object, and one may hope that it can be measured on the lattice.

C. Momentum distributions

The quasi-PDFs describe the distribution in the fraction $y = k_3/P$ of the third component k_3 of the parton momentum to that of the hadron. One can introduce distributions in k_3 itself: $R(k_3, P) = Q(k_3/P, P)/P$. Then we can rewrite Eq. (11) as

$$R(k_3, P) = \int_{-\infty}^{\infty} dk_1 \int_{-1}^1 dx \mathcal{F}(x, k_1^2 + (k_3 - xP)^2) \quad (15)$$

or, switching to the linear argument $k_3 - xP$,

$$R(k_3, P) = \int_{-1}^1 dx \mathcal{R}(x, k_3 - xP), \quad (16)$$

where

$$\mathcal{R}(x, k_3) = \int_{-\infty}^{\infty} dk_1 \mathcal{F}(x, k_1^2 + k_3^2) \quad (17)$$

is the TMD $\mathcal{F}(x, \kappa^2)$ integrated over the k_1 component of the two-dimensional vector $\kappa = \{k_1, k_3\}$. According to (17), $\mathcal{R}(x, k_3)$ depends on k_3 through k_3^2 .

For a hadron at rest, we have

$$R(k_3, P = 0) = r(k_3) = \int_{-1}^1 dx \mathcal{R}(x, k_3). \quad (18)$$

This one-dimensional distribution may be directly obtained through a parameterization of the density

$$\rho(z_\perp^2) = \mathcal{M}(0, z_\perp^2) = \int_{-\infty}^{\infty} dk_3 r(k_3) e^{ik_3 z_\perp} \quad (19)$$

given by $\langle p | \bar{\psi}(0) \psi(z_\perp) | p \rangle|_{p=0}$. Thus, $r(k_3)$ describes a primordial distribution of k_3 (or any other component of $\mathbf{k}$) in a rest-frame hadron.

The formula (16) has a straightforward interpretation. According to it, when the hadron is moving, the parton’s k_3 momentum has two sources.

The first part, xP comes from the motion of the hadron as a whole, and the probability to get xP is governed by the dependence of the TMD $\mathcal{F}(x, \kappa^2)$ on its first argument, x .

On the other hand, the probability to get the remaining part $k_3 - xP$ is governed by the dependence of the TMD on its second argument, κ^2 , describing the primordial rest-frame momentum distribution.

The parameter x appears in both arguments of $\mathcal{R}(x, k_3 - xP)$ in Eq. (16), i.e., $R(k, P)$ is given by a convolution. In this sense, the momentum distributions $R(k, P)$ and, hence, the quasi-PDFs have a hybrid structure influenced by the shape both of PDFs and rest-frame distributions.

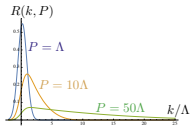


FIG. 1. Momentum distributions $R(k, P)$ in the factorized Gaussian model for $P/\Lambda = 1, 10, 50$.

D. Factorized models.

Since the two sources of k_3 look like independent, it is natural to demonstrate the hybrid nature of momentum distributions and quasi-PDFs using a factorized model $\mathcal{R}(x, k_3 - xP) = f(x)r(k_3 - xP)$ (the x integral of $f(x)$ is normalized to 1). For original $\mathcal{M}(\nu, -z^2)$ function, this Ansatz corresponds to the factorization assumption $\mathcal{M}(\nu, -z^2) = \mathcal{M}(\nu, 0)\mathcal{M}(0, -z^2)$.

For illustration, we take a Gaussian form $\rho_G(z_3^2) = e^{-z_3^2\Lambda^2/4}$ for the rest-frame density. It corresponds to

$$r_G(k_3) = \frac{1}{\sqrt{\pi}\Lambda} e^{-k_3^2/\Lambda^2}. \quad (20)$$

For $f(x)$, we take a simple PDF resembling nucleon valence densities $f(x) = 4(1-x)^3\theta(0 \leq x \leq 1)$. As one can see from Fig. 1, the curve for $R(k, P)$ changes from a Gaussian shape for small P to a shape resembling stretched PDF for large P .

This result is in perfect compliance with a known fact that wave functions of moving hadrons are not given by a mere kinematical “boost” of the rest-frame wave functions. Indeed, with increasing P , the impact of the rest-frame distribution $r(k)$ is less and less visible, and eventually the shape of $R(k, P)$ is determined by a completely different function $f(k/P)$.

Rescaling to the $y = k/P$ variable gives the quasi-PDF $Q(y, P)$ shown in Fig. 2. For large P , it clearly tends to the $f(y)$ PDF form. In particular, using a momentum $P \sim 10\Lambda$ one gets a quasi-PDF that is rather close to the $P \rightarrow \infty$ limiting shape. Still, since $\Lambda \sim \langle k_\perp \rangle$, assuming the folklore value $\langle k_\perp \rangle \sim 300$ MeV one translates the $P \sim 10\Lambda$ estimate into $P \sim 3$ GeV, which is uncomfortably large. Thus, a natural question is how to improve the convergence.

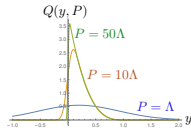


FIG. 2. Evolution of quasi-PDF $Q(y, P)$ in the factorized Gaussian model for $P/\Lambda = 1, 10, 50$.

E. Pseudo-PDFs

A formal reason for the complicated structure of a quasi-PDF $Q(y, P)$ is the fact that it is obtained by the ν -integral of $\mathcal{M}(\nu, z_3^2)e^{i\nu y}$ along a non-horizontal line $z_3 = \nu/P$ in the (ν, z_3) plane (see Eq. (10)). With increasing P , its slope decreases, the line becomes more horizontal, and quasi-PDFs convert into PDFs.

In contrast, pseudo-PDFs $\mathcal{P}(x, z_3^2)$, by definition, are given by integration of $\mathcal{M}(\nu, z_3^2)e^{i\nu x}$ over horizontal lines $z_3 = \text{const}$. A very attractive feature of the pseudo-PDFs is that they have the $-1 \leq x \leq 1$ support for all z_3 values. For small z_3 , they convert into PDFs.

More precisely, when z_3 is small, $1/z_3$ is analogous to the renormalization parameter μ of scale-dependent PDFs $f(x, \mu^2)$ of the standard OPE approach.

There is a subtlety, however, that while the μ^2 -dependence of PDFs $f(x, \mu^2)$ comes solely from the evolution logarithms $\ln(\mu^2/m^2)$, the z_3^2 -dependence of quasi-PDFs in QCD comes both from the evolution logarithms $\ln(z_3^2/m^2)$ and from the ultraviolet logarithms $\ln(z_3^2\mu_R^2)$, where μ_R is a cut-off parameter for divergences related to the gauge link renormalization (see Ref. [14]). At the leading logarithm level, these divergences do not depend on ν . As a result, the “reduced” Ioffe-time distribution

$$\mathfrak{R}(\nu, z_3^2) = \frac{\mathcal{M}(\nu, z_3^2)}{\mathcal{M}(0, z_3^2)} \quad (21)$$

satisfies, for small z_3 , the leading-order evolution equation

$$\frac{d}{d \ln z_3^2} \mathfrak{R}(\nu, z_3^2) = -\frac{\alpha_s}{2\pi} C_F \int_0^1 du B(u) \mathfrak{R}(u\nu, z_3^2) \quad (22)$$

with respect to $1/z_3$ that is similar to the evolution equation for $f(x, \mu^2)$ with respect to μ . The leading-order evolution kernel $B(u)$ for the non-singlet quark case is given [12] by

$$B(u) = \left[\frac{1+u^2}{1-u} \right]_+, \quad (23)$$

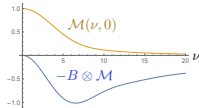


FIG. 3. Model Ioffe-time distribution $\mathcal{M}(\nu, 0)$ and the function $B \otimes \mathcal{M}$ governing its evolution.

with $[\dots]_+$ denoting the standard “plus” prescription.

For the model used above (and $x \rightarrow -x$ symmetrized, as required for non-singlet PDFs), we have $\mathcal{M}(\nu, 0) = 12 [\mu^2 - 4 \sin^2(\nu/2)] / \nu^2$. The shape of this function and of the convolution integral $B \otimes \mathcal{M}(\nu)$ are shown in Fig. 3. As one can see, $B \otimes \mathcal{M}(\nu)$ vanishes for $\nu = 0$, which reflects conservation of the vector current. Thus, the rest-frame density $\mathfrak{M}(0, z_\perp^2)$ is not affected by perturbative evolution.

F. Lattice implementation

A possible way to find the Ioffe-time distributions on the lattice (suggested by K. Orginos) is to calculate $\mathcal{M}(Pz, z_\perp^2)$ for several values of P , and then to fit the results by a function of ν and $z_\perp^2$.

Recalling our discussion of two apparently independent sources of obtaining k_3 for a moving hadron, one may hope that $\mathcal{M}(\nu, z_\perp^2)$ factorizes, i.e., $\mathcal{M}(\nu, z_\perp^2) = \mathcal{M}(\nu, 0)\mathcal{M}(0, z_\perp^2)$. Then the reduced function $\mathfrak{M}(\nu, z_\perp^2)$ defined by Eq. (21) is equal to $\mathcal{M}(\nu, 0)$, and the goal of obtaining $\mathcal{M}(\nu, 0)$ is reached. Formally, what remains is just to take its Fourier transform to get the PDF $f(x)$.

In fact, such a factorization has been already observed several years ago in the pioneering study [15] of the transverse momentum distributions in lattice QCD.

A serious disadvantage of quasi-PDFs is that they have the x -convolution structure (11) even in a favorable situation when the TMD [and $\mathcal{M}(\nu, z_\perp^2)$] factorizes. On the other hand, using pseudo-PDFs in the form of the ratio $\mathfrak{M}(\nu, z_\perp^2)$, one divides out the $z_\perp^2$ -dependence of the primordial distribution without affecting the ν -dependence that dictates the shape of PDF.

A further advantage of using the ratio (pointed out by K. Orginos) is the cancellation of the z_3 -dependence generated by the lattice renormalization of the gauge link $\tilde{E}(0, z_3; A)$. Such a renormalization is required by linear $|z_3|\delta m$ (where $\delta m \sim 1/a$, and a is the ultraviolet cut-off)

and logarithmic $\ln(z_\perp^2/a^2)$ divergences [16, 17]. Due to their local nature, they are expected to combine into a ν -independent factor $Z(z_\perp^2/a^2)$ that is the same in the numerator and denominator of the ratio $\mathfrak{M}(\nu, z_\perp^2)$.

The multiplicative renormalizability of the linear divergences of $\mathcal{M}(\nu, z_\perp^2)$ to all orders was recently argued in Refs. [18, 19]. A general proof for both linear and logarithmic divergences was claimed in Ref. [20] on the basis of a direct analysis of relevant Feynman graphs.

Another approach [21, 22] is to treat $\tilde{E}(0, z; A)$ as $h(0)h(z)$, where the auxiliary field $h(z)$ is analogous to the infinitely heavy quark field of the heavy quark effective theory (HQET). Since HQET is known to be multiplicatively renormalizable [23] this means that $\tilde{\psi}(0)\tilde{E}(0, z; A)\psi(z)$ is also multiplicatively renormalizable to all orders in perturbation theory.

In reality, $\mathfrak{M}(\nu, z_\perp^2)$ will have a residual $z_\perp^2$ -dependence. It comes both from a possible violation of factorization for the soft part (according to results of Ref. [15], it is expected to be rather mild) and from mandatory perturbative evolution. For a nonzero ν , the latter should be visible as a $\ln(1/z_\perp^2\Lambda^2)$ spike for small $z_\perp^2$.

Hence, a proposed strategy is to extrapolate $\mathfrak{M}(\nu, z_\perp^2)$ to $z_\perp^2 = 0$ from not too small values of $z_\perp^2$, say, from those above 0.5 fm^2 . The resulting function $\mathcal{M}^{\text{extr}}(\nu, 0)$ may be treated as the Ioffe-time distribution producing the PDF $f_0(x)$ “at low normalization point”. The remaining $\ln(1/z_\perp^2\Lambda^2)$ spikes at small $z_\perp^2$ will generate its evolution.

To convert $\mathfrak{M}(\nu, z_\perp^2)$ into a function of x , one should, in principle, know $\mathfrak{M}(\nu, z_\perp^2)$ for all ν , which is impossible. The maximal values of ν reached in existing lattice calculations range from 3π [3] to 5π [5] and 6π [24]. Taking a Fourier transform in these limited ranges produces unphysical oscillations in x . Thus, the idea is to avoid the Fourier transform in ν , and just compare the reduced Ioffe-time distributions obtained from the lattice with those derived from experimentally known parton distributions.

Of course, an actual technical implementation of this program should be discussed when the lattice data on $\mathfrak{M}(\nu, z_\perp^2)$ will become available.

IV. SUMMARY

In this paper, we showed that quasi-PDFs may be seen as hybrids of PDFs and the primordial rest-frame momentum distributions of partons. In this context, the parton’s k_3 momentum comes from the motion of the hadron as a whole and from the primordial rest-frame momentum distribution. The complicated convolution nature of quasi-PDFs necessitates using $p_3 \gtrsim 3 \text{ GeV}$ to wipe out the primordial momentum distribution effects and get reasonably close to the PDF limit.

As an alternative approach, we propose to use pseudo-PDFs $\mathcal{P}(x, z_\perp^2)$ that generalize the light-front PDFs onto spacelike intervals. By a Fourier transform, they are related to the Ioffe-time distributions $\mathcal{M}(\nu, z_\perp^2)$ given by

generic matrix elements written as functions of $\nu = p_{z3}$ and z_3^2 . The advantageous features of pseudo-PDFs are that they, first, have the same $-1 \leq x \leq 1$ support as PDFs, and second, their z_3^2 -dependence for small z_3^2 is governed by a usual evolution equation.

Forming the ratio $\mathcal{M}(\nu, z_3^2)/\mathcal{M}(0, z_3^2)$ of Ioffe-time distributions one divides out the bulk of z_3^2 dependence generated by the primordial rest-frame distribution. Furthermore, taking this ratio one can exclude the z_3^2 -dependent factor coming from the lattice renormalization of the $\bar{E}(0, z_3; A)$ link creating difficulties (see, e.g., [18]) for lattice calculations of quasi-PDFs.

Testing the efficiency of using pseudo-PDFs for lattice extractions of PDFs is a challenge for future studies.

In fact, while this paper was in the review process, an actual lattice calculation [24] based on the ideas of the present paper was performed. It has clearly demonstrated the presence of a linear component in the z_3 -dependence of the rest-frame function $\mathcal{M}(0, z_3^2)$, that may be attributed to the $Z(z_3^2/a^2) \sim e^{-c|z_3|/a}$ behavior generated by the gauge link. It was also observed that the ratio $\mathcal{M}(P_{z3}, z_3^2)/\mathcal{M}(0, z_3^2)$ has a Gaussian-type behavior with respect to z_3 , which indicates that the $Z(z_3^2/a^2)$ factors entering into the numerator and denominator of

the $\mathfrak{M}(P_{z3}, z_3^2)$ ratio have been canceled, as we expected.

Furthermore, it was found that when plotted as a function of ν and z_3 , the data for the reduced distribution $\mathfrak{M}(\nu, z_3^2)$ have a very mild dependence on z_3^2 . This observation indicates that the soft part of the z_3^2 -dependence of $\mathcal{M}(\nu, z_3^2)$ has been canceled by the rest-frame density $\mathcal{M}(0, z_3^2)$. This phenomenon corresponds to factorization of the x - and $k_\perp$ -dependence for the soft part of the TMD $\mathcal{F}(x, k_\perp^2)$.

It was also demonstrated that the residual z_3 -dependence for small $z_3 \lesssim 4a$, may be explained by perturbative evolution, with the α_s value corresponding to $\alpha_s/\pi = 0.1$.

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P16: 部分子伪分布的格点探索

主题归属	LaMET、quasi/pseudo 分布与因子化
底稿	arXiv:1706.05373
arXiv / DOI	1706.05373
完整原始 PDF	12 页; 来源 course-cache; SHA-256 12401554a993...
转排索引	EN 21 页; ZH 18 页 (只作书目对照)
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先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

页码范围：P16-p001 – P16-p012

Lattice QCD exploration of parton pseudo-distribution functions

Kostas Orginos,^{1,2} Anatoly Radyushkin,^{3,2} Joseph Karpie,^{1,2} and Savvas Zafeiropoulos^{1,2}

¹*Department of Physics, The College of William & Mary, Williamsburg, VA 23187, USA*

²*Thomas Jefferson National Accelerator Facility, Newport News, VA 23606, USA*

³*Physica Department, Old Dominion University, Norfolk, VA 23529, USA*

We demonstrate a new method of extracting parton distributions from lattice calculations. The starting idea is to treat the generic equal-time matrix element $M(Pz_3, z_3^2)$ as a function of the Ioffe time $\nu = Pz_3$ and the distance z_3 . The next step is to divide $M(Pz_3, z_3^2)$ by the rest-frame density $M(0, z_3^2)$. Our lattice calculation shows a linear exponential z_3 -dependence in the rest-frame function, expected from the $Z(z_3^2)$ factor generated by the gauge link. Still, we observe that the ratio $M(Pz_3, z_3^2)/M(0, z_3^2)$ has a Gaussian-type behavior with respect to z_3 for ν values of P used in the calculation. This means that $Z(z_3^2)$ factor was canceled in the ratio. When plotted as a function of ν and z_3 , the data are very close to z_3 -independent functions. This phenomenon corresponds to factorization of the x - and $k_\perp$ -dependence for the TMD $\mathcal{F}(x, k_\perp^2)$. For small $z_3 \leq 4a$, the residual z_3 -dependence is explained by perturbative evolution, with $\alpha_s/\pi = 0.1$.

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I. INTRODUCTION

Extraction of parton distribution functions (PDFs) $f(x)$ [1] on the lattice is a challenging problem attracting a lot of attention. The usual method to approach PDFs on the lattice is to calculate their moments. However, recently, X. Ji [2] suggested a method allowing a calculation of PDFs as functions of x .

Since the PDFs are related to matrix elements of bilocal operators on the light cone $z^2 = 0$, this was a stumbling block preventing a direct calculation of these functions in the lattice gauge theory formulated in Euclidean space.

To overcome this difficulty, X. Ji proposes to use purely space-like separations $z = (0, 0, 0, z_3)$. The functions in this case are quasi-PDFs $Q(y, p_3)$ describing the distribution of the p_3 hadron momentum component. The key point is that quasi-PDFs $Q(y, p_3)$ tend to usual PDFs $f(y)$ in the $p_3 \rightarrow \infty$ limit. The same method can be applied to distribution amplitudes (DAs). The results of quasi-PDF calculations on the lattice were reported in Refs. [3–5] and of the pion quasi-DA in Ref. [6].

Recent papers [7, 8] by one of the authors (A.R.) contain an investigation of the nonperturbative p_3 -evolution of quasi-PDFs and quasi-DAs. This study is based on the formalism of virtuality distribution functions [9, 10]. The approach developed in Refs. [7, 8] has established a connection between the quasi-PDFs and the “straight-link” transverse momentum dependent distributions (TMDs) $\mathcal{F}(x, k_\perp^2)$. Starting from simple models for TMDs, models were built for the nonperturbative evolution of quasi-PDFs. The derived curves agree qualitatively with the patterns of p_3 -evolution produced by lattice simulations.

The structure of quasi-PDFs was further studied in Ref. [11]. It was shown that, when a hadron is moving, the parton k_3 momentum may be treated as coming from two sources. The hadron’s motion as a whole yields the $x p_3$ part, which is governed by the dependence of the

TMD $\mathcal{F}(x, \kappa^2)$ on its first argument namely x . The residual part $k_3 - x p_3$ is controlled by the way that the TMD depends on its second argument, κ^2 , which dictates the shape of the primordial rest-frame momentum distribution. Quasi-PDFs due to their convolution nature possess a rather involved pattern of their p_3 -evolution, making mandatory relatively big values $p_3 \gtrsim 3$ GeV in order to safely approach the PDF limit.

To accelerate the convergence, a different approach for the PDF extraction from lattice calculations was proposed [11]. It is based on the concept of *pseudo-PDFs* $\mathcal{P}(x, z_3^2)$. They generalize the light-cone PDFs $f(x)$ onto spacelike intervals like $z = (0, 0, 0, z_3)$. The pseudo-PDFs are Fourier transforms of the *Ioffe-time* [12] distributions [13] $\mathcal{M}(\nu, z_3^2)$ which are generically given by matrix elements $\langle p | \phi(0) \phi(z) | p \rangle$ written as functions of $\nu = p_3 z_3$ and z_3^2 . In contrast to quasi-PDFs, the pseudo-PDFs have the “canonical” $-1 \leq x \leq 1$ support for all values of z_3^2 . In the limit $z_3 \rightarrow 0$ they tend to PDFs, showing, in this limit, a typical perturbative evolution with the scale $1/z_3$ being the parameter of evolution.

As discussed in [7, 8], the fast nonperturbative decrease with z_3^2 of the pseudo-PDFs $\mathcal{P}(x, z_3^2)$ or the Ioffe-time distribution $\mathcal{M}(\nu, z_3^2)$, is responsible for delaying the approach of quasi-PDFs $Q(y, p_3)$ to the PDF $f(y)$. An important observation is that one can strongly reduce the z_3^2 -dependence by simply dividing the Ioffe-time distribution $\mathcal{M}(\nu, z_3^2)$ by an appropriate factor $D(z_3^2)$ satisfying $D(0) = 1$ and having the z_3^2 -dependence close (on average) to that of $\mathcal{M}(\nu, z_3^2)$. The absence of the ν -dependence in this factor and its $D(0) = 1$ normalization guarantees that the ratio $\mathcal{M}(\nu, z_3^2)/D(z_3^2)$ taken in the $z_3^2 \rightarrow 0$ limit will produce the same PDF as the original function $\mathcal{M}(\nu, z_3^2)$ taken in the same limit.

The choice for $D(z_3^2)$ advocated in Ref. [11], is to take it to be equal to the rest-frame function $\mathcal{M}(0, z_3^2)$. An additional advantage of this choice is that both $\mathcal{M}(\nu, z_3^2)$ and $\mathcal{M}(0, z_3^2)$ contain the same multiplicative

factor $Z(z_3^2)$ generated by the renormalization of the gauge link. In the ratio, it should cancel out.

Our goal in the present work is an exploratory lattice calculation of the u - d proton PDF using the strategy outlined in Ref. [11]. To make this article self-contained, we reproduce in Sections II and III the main ideas of Ref. [11]. The description of the method used for the lattice extraction of the reduced Ioffe-time distribution is given in Section IV. The data analysis and interpretation is discussed in Section V. The summary of the paper is given in Section VI.

II. PARTON DISTRIBUTIONS

A. Generic matrix element and parton distributions

The basic object for defining parton distributions is a matrix element of a bilocal operator that (skipping inessential details of its spin structure) may be written generically like $\langle p|\phi(0)\phi(z)|p\rangle$. Due to invariance under Lorentz transformations, it is given by a function of two scalars, (pz) (which will be denoted by $-\nu$) and z^2 (or $-z^2$, in order to have a positive value for spacelike z)

$$\langle p|\phi(0)\phi(z)|p\rangle = \mathcal{M}(-(pz), -z^2) = \mathcal{M}(\nu, -z^2). \quad (1)$$

One can demonstrate [7, 14] that, for all relevant Feynman diagrams, its Fourier transform $\mathcal{P}(x, -z^2)$ with respect to (pz) has $-1 \leq x \leq 1$ as support, i.e.,

$$\mathcal{M}(-(pz), -z^2) = \int_{-1}^1 dx e^{-ix(pz)} \mathcal{P}(x, -z^2). \quad (2)$$

Eq. (2) serves as a covariant definition of x . In this definition of x , one does not need to assume that $p^2 = 0$ or $z^2 = 0$.

Choosing a light-like z , e.g., having solely the light-front component z_+ , we parametrize the matrix element by $f(x)$, the twist-2 parton distribution

$$\mathcal{M}(-p_+ z_+, 0) = \int_{-1}^1 dx f(x) e^{-ixp_+ z_+}. \quad (3)$$

One can rewrite this definition as

$$\mathcal{M}(\nu, 0) = \int_{-1}^1 dx f(x) e^{ix\nu}. \quad (4)$$

The inverse relation is given by

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\nu e^{-i\nu x} \mathcal{M}(\nu, 0) = \mathcal{P}(x, 0). \quad (5)$$

Due to the fact that $f(x) = \mathcal{P}(x, 0)$, the function $\mathcal{P}(x, -z^2)$ provides a generalization of the concept of PDFs onto non-lightlike intervals z^2 (in principle, z^2 may be even timelike). Following [11], we will be referring to

it as the *pseudo-PDF*. The variable $(pz) = -\nu$ is called often the *Ioffe time* [12], and consequently $\mathcal{M}(\nu, -z^2)$ is the *Ioffe-time distribution* [13].

In renormalizable theories (including QCD), the function $\mathcal{M}(\nu, -z^2)$ has logarithmic $\sim \ln(-z^2)$ singularities which generate the perturbative evolution of parton densities. In the approach based on the operator product expansion (OPE), the standard procedure is to remove these singularities with the help of some prescription. The most popular of them is the $\overline{\text{MS}}$ scheme based on dimensional regularization. Consequently the resulting PDFs have a dependence on the renormalization scale μ , and therefore one should write the PDFs as $f(x, \mu^2)$.

At small spacelike z^2 and at the leading logarithm level, the pseudo-PDFs are related to the $\overline{\text{MS}}$ distributions by a simple rescaling of their second arguments. In particular, when $z^2 = -z_3^2$, one has

$$\mathcal{P}(x, z_3^2) = f\left(x, (2e^{-\gamma_E}/z_3)^2\right), \quad (6)$$

where γ_E is the Euler's constant. The rescaling factor between μ and $1/z_3$ is very close to 1, since $2e^{-\gamma_E} = 1.12$.

B. Transverse momentum dependent- and quasidistributions

Treating the target momentum p as longitudinal, $p = (E, \mathbf{0}_\perp, P)$, one can introduce transverse degrees of freedom. In particular, taking z that has z_- and $z_\perp = \{z_1, z_2\}$ components only, one defines the *TMD* $\mathcal{F}(x, k_\perp^2)$

$$\mathcal{P}(x, z_\perp^2) = \int d^2\mathbf{k}_\perp e^{i(\mathbf{k}_\perp \cdot \mathbf{x}_\perp)} \mathcal{F}(x, k_\perp^2). \quad (7)$$

In this context, the pseudo-PDFs $\mathcal{P}(x, z_\perp^2)$ actually coincide with the *impact parameter distributions*, a familiar object used in many TMD studies.

Since one cannot arrange light-like separations on the lattice, it was proposed [2] to consider equal-time spacelike separations $z = (0, 0, 0, z_3)$ (or, for brevity, $z = z_3$). Then, in the $p = (E, \mathbf{0}_\perp, P)$ frame, one can introduce the quasi-PDF $Q(y, P)$ through a parametrization

$$\langle p|\phi(0)\phi(z_3)|p\rangle = \int_{-\infty}^{\infty} dy Q(y, P) e^{iyPz_3}. \quad (8)$$

According to this definition, the quasi-PDF $Q(y, P)$ describes the probability that the parton carries the fraction y of the parent hadron's third momentum component P . The variables ν and $-z^2$ in this case are given by Pz_3 and z_3^2 , so we have

$$\mathcal{M}(\nu, z_3^2) = \int_{-\infty}^{\infty} dy Q(y, P) e^{i\nu y}. \quad (9)$$

Since $z_3^2 = \nu^2/P^2$, the inverse Fourier transformation may be written as

$$Q(y, P) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\nu e^{-i\nu y} \mathcal{M}(\nu, \nu^2/P^2). \quad (10)$$

It shows that $Q(y, P)$ tends to $f(y)$ in the $P \rightarrow \infty$ limit, since formally $\mathcal{M}(\nu, \nu^2/P^2) \rightarrow \mathcal{M}(\nu, 0)$ when $P \rightarrow \infty$.

As established in Ref. [7], quasi-PDFs may be written in terms of TMDs

$$Q(y, P)/P = \int_{-\infty}^{\infty} dk_2 \int_{-1}^1 dx \mathcal{F}(x, k_2^2 + (y-x)^2 P^2). \quad (11)$$

C. Quantum chromodynamics (QCD) case

In case of the non-singlet parton densities of QCD, one is considering matrix elements

$$\mathcal{M}^a(z, p) = \langle p | \bar{\psi}(0) \gamma^a \hat{E}(0, z; A) \psi(z) | p \rangle, \quad (12)$$

where $\hat{E}(0, z; A)$ is the standard $0 \rightarrow z$ straight-line gauge link in the quark (fundamental) representation. These matrix elements can be decomposed into p^a and z^a parts

$$\mathcal{M}^a(z, p) = 2p^a \mathcal{M}_p(-\langle zp \rangle, -z^2) + z^a \mathcal{M}_z(-\langle zp \rangle, -z^2) \quad (13)$$

Only the $\mathcal{M}_p(-\langle zp \rangle, -z^2)$ part gives the twist-2 distribution when $z^2 \rightarrow 0$.

Introducing TMDs, one takes $z = (z_-, z_\perp)$ and the $\alpha = +$ component of $\mathcal{M}^a$. Hence, the z^a -part drops out. After that, $\mathcal{M}_p(\nu, z_\perp^2)$ is the only surviving part of $\mathcal{M}^a(z, p)$, and in the remaining discussion we use the short hand notation of $\mathcal{M} \equiv \mathcal{M}_p$.

In the case of quasidistributions $Q(y, P)$, we can avoid the z^a contamination by considering the time component of $\mathcal{M}^a(z = z_3, p)$ and defining

$$\mathcal{M}^0(z_3, p) = 2p^0 \int_{-1}^1 dy Q(y, P) e^{iyPz_3}. \quad (14)$$

D. Factorized models

The structure of the quasi-PDFs may be illustrated on the example of the simplest models in which the nonperturbative (or soft) part of the TMDs $\mathcal{F}(x, k_\perp^2)$ is represented by a product

$$\mathcal{F}^{\text{soft}}(x, k_\perp^2) = f(x) K(k_\perp^2) \quad (15)$$

of the collinear parton distribution $f(x)$ and a $k_\perp^2$ -dependent factor $K(k_\perp^2)$, usually modeled by a Gaussian. As we shall see, the quasi-PDFs have a rather complicated structure, even when they are built from these simple factorized models.

For the Ioffe-time distribution $\mathcal{M}(\nu, -z^2)$, this Ansatz corresponds to the factorization assumption

$$\mathcal{M}^{\text{soft}}(\nu, z_3^2) = \mathcal{M}^{\text{soft}}(\nu, 0) \mathcal{M}(0, z_3^2) \quad (16)$$

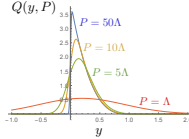


FIG. 1. Evolution of quasi-PDF $Q(y, P)$ in the factorized Gaussian model for $P/\Lambda = 1, 5, 10, 50$.

for its soft part. Still, even if the soft TMD factorizes, the soft part of the quasi-PDF has the convolution structure of Eq. (11). Taking, for example, a Gaussian form

$$K_G(k_\perp^2) = \frac{1}{\pi \Lambda^2} e^{-k_\perp^2/\Lambda^2}, \quad (17)$$

one gets the following model for the quasi-PDF

$$Q_G(y, P) = \frac{P}{\Lambda \sqrt{\pi}} \int_{-1}^1 dx f(x) e^{-(x-y)^2 P^2/\Lambda^2}. \quad (18)$$

Choosing for $f(x)$ a simple toy PDF resembling the nucleon valence densities $f(x) = 4(1-x)^3 \theta(0 \leq x \leq 1)$, one gets the curves shown in Fig. 1. For large P , the quasi-PDF clearly tends to the $f(y)$ PDF form. However, only for $P \sim 10\Lambda$ one gets a quasi-PDF that is rather close to the $P \rightarrow \infty$ limiting shape. Still, since $\Lambda \sim (k_\perp)$, one translates the $P \sim 10\Lambda$ estimate into $P \sim 3$ GeV, which is rather large.

III. PSEUDO-PDFS

The involved structure of a quasi-PDF $Q(y, P)$ can be attributed to the formal fact that it is given by the Fourier ν -transform of the function $\mathcal{M}(\nu, \nu^2/P^2)$, in which ν appears both in the first and second argument of the Ioffe-time distribution. One should take P -values that are sufficiently large to neglect the ν -dependence coming from the second argument.

Another way [11] is to try to *eliminate* the z_3^2 -dependence induced by $\mathcal{M}(\nu, z_3^2)$. The main idea is based on the observation that if one takes the ν -Fourier transform of the modified function $\mathcal{M}(\nu, z_3^2)/D(z_3^2)$, the $z_3 \rightarrow 0$ limit will give the same PDF as the original Ioffe-time distribution, provided that $D(z_3^2)$ is a function of z_3^2 only (but not of ν) and is equal to 1 for $z_3^2 = 0$.

Thus, the strategy is to find a function $D(z_3^2)$ whose z_3^2 -dependence would compensate, as much as possible,

the z_3^2 -dependence of $\mathcal{M}(\nu, z_3^2)$. The next step is to fit the residual polynomial z_3^2 -dependence by polynomials of z_3^2 (they may be different for different values of ν), and in this way extrapolate the data to $z_3^2 = 0$ limit. The Fourier transform of the resulting function would correspond to the same PDF as the z_3^2 limit of the original loffe-time distribution $\mathcal{M}(\nu, z_3^2)$.

In the most lucky situation, the ratio $\mathcal{M}(\nu, z_3^2)/D(z_3^2)$ would have no polynomial z_3^2 -dependence (or just a very mild one). In particular, when $\mathcal{M}(\nu, z_3^2)$ factorizes, i.e., $\mathcal{M}(\nu, z_3^2) = \mathcal{M}(\nu, 0)\mathcal{M}(0, z_3^2)$, one should take $D(z_3^2) = \mathcal{M}(0, z_3^2)$. In this case, the reduced function

$$\mathfrak{M}(\nu, z_3^2) = \frac{\mathcal{M}(\nu, z_3^2)}{\mathcal{M}(0, z_3^2)} \quad (19)$$

is equal to $\mathcal{M}(\nu, 0)$, and the task of obtaining the $z_3 \rightarrow 0$ limit is accomplished.

While there is no “first principle” reason for such a factorization, one may expect that the functions $\mathcal{M}(\nu, z_3^2)$ for different ν have more or less similar dependence on z_3 , basically reflecting the finite size of the nucleon.

As we mentioned already, the soft part of $\mathcal{M}(\nu, z_3^2)$ factorizes if the soft part of TMD $\mathcal{F}(x, k_\perp^2)$ factorizes. That this happens, is a standard assumption of the TMD practitioners (see, e.g., Ref. [15]). So, there are good chances that this part of the z_3^2 -dependence of $\mathcal{M}(\nu, z_3^2)$ will be canceled or strongly reduced by the rest-frame function $\mathcal{M}(0, z_3^2)$.

On the lattice, there is another (and troublesome, see, e.g., Ref. [16]) source of z_3 -dependence: the $Z(z_3^2)$ factor generated by the renormalization of the gauge link $\bar{E}(0, z_3; A)$. Fortunately, this problematic factor $Z(z_3^2)$ does not depend on ν and is the same for the numerator and denominator of the ratio $\mathfrak{M}(\nu, z_3^2)$. This provides another motivation for using $\mathcal{M}(0, z_3^2)$ as a factor $D(z_3^2)$.

Thus, the proposal is to perform a lattice study of the reduced loffe-time function $\mathfrak{M}(\nu, z_3^2)$. Even if it would have a residual polynomial z_3^2 -dependence, it should be much easier to extrapolate this dependence to $z_3 = 0$, than the z_3 -dependence of the original loffe-time distribution $\mathcal{M}(\nu, z_3^2)$.

Furthermore, if one observes that the ratio $\mathfrak{M}(\nu, z_3^2)$ does not have z_3 -dependence, one should conclude that $\mathcal{M}(\nu, z_3^2)$ factorizes. In fact, such a factorization has been already observed several years ago in the pioneering study [17] of the transverse momentum distributions in lattice QCD.

Still, there is an unavoidable source of factorization breaking. When z_3 is small, $\mathcal{M}(\nu, z_3^2)$ has logarithmic $\ln z_3^2$ singularities generating the perturbative evolution of PDFs. As we discussed, $1/z_3$ is analogous then to the renormalization parameter μ of the scale-dependent PDFs $f(x, \mu^2)$ within the standard OPE approach.

More specifically, for small values of z_3 , the pseudo-PDF $\mathcal{P}(x, z_3^2)$ satisfies a leading-order evolution equation with respect to $1/z_3$ that is identical to the evolution equation for $f(x, \mu^2)$ with respect to μ . The evolution

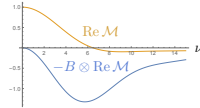


FIG. 2. Real part of model distribution $\mathcal{M}(\nu)$ and the function $-B \otimes \text{Re } \mathcal{M}$ that governs its evolution (the minus sign here is for convenience of placing two curves on one figure).

equation for the reduced loffe-time distribution $\mathfrak{M}(\nu, z_3^2)$ can also be written [11]

$$\frac{d}{d \ln z_3^2} \mathfrak{M}(\nu, z_3^2) = -\frac{\alpha_s}{2\pi} C_F \int_0^1 du B(u) \mathfrak{M}(u\nu, z_3^2), \quad (20)$$

where $C_F = 4/3$, and the leading-order evolution kernel $B(u)$ for the non-singlet quark case is given [13] by

$$B(u) = \left[\frac{1+u^2}{1-u} \right]_+, \quad (21)$$

where $[\dots]_+$ denotes the “plus” prescription, i.e.

$$\begin{aligned} & \int_0^1 du \left[\frac{1+u^2}{1-u} \right]_+ \mathfrak{M}(u\nu) \\ &= \int_0^1 du \frac{1+u^2}{1-u} [\mathfrak{M}(\nu) - \mathfrak{M}(u\nu)]. \end{aligned} \quad (22)$$

Note that being a Fourier transform,

$$\mathcal{M}(\nu) = \int_{-1}^1 dx f(x) e^{ix\nu}, \quad (23)$$

the loffe-time distribution has real and imaginary parts even if the function $f(x)$ is real (which is the case with parton distributions). In particular,

$$\text{Re } \mathcal{M}(\nu) = \int_{-1}^1 dx f(x) \cos(x\nu), \quad (24)$$

and

$$\text{Im } \mathcal{M}(\nu) = \int_{-1}^1 dx f(x) \sin(x\nu). \quad (25)$$

In Fig. 2, we show the function $\text{Re } \mathcal{M}(\nu)$ for a model PDF

$$q(x) = \frac{315}{32} \sqrt{x} (1-x)^3 \theta(0 \leq x \leq 1). \quad (26)$$

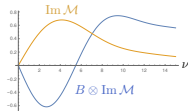


FIG. 3. Imaginary part of model Ioffe-time distribution $\mathcal{M}(\nu)$ and the function $B \otimes \text{Im} \mathcal{M}$ that governs its evolution.

Its integral is normalized to 1, and it is nonzero for positive x only, which corresponds to the absence of antiquarks. As we shall see, this particular form appears in the description of actual lattice data. In Fig. 3, we show the function $\text{Im} \mathcal{M}(\nu)$ for the same model PDF.

We also show in these figures the convolution integrals governing the evolution, namely $-B \otimes \text{Re} \mathcal{M}(\nu)$ and $B \otimes \text{Im} \mathcal{M}(\nu)$. The reader can notice that $B \otimes \mathcal{M}(\nu)$ is zero for $\nu = 0$, resulting from the vector current conservation. As a consequence, the perturbative evolution leaves the rest-frame density $\mathcal{M}(0, z_1^2)$ (which is always real) unaffected. In other words, the $\ln z_1^2$ terms are present only in the numerator $\mathcal{M}(\nu, z_1^2)$ of the $\mathfrak{M}(\nu, z_1^2)$ ratio, but not in its $\mathcal{M}(0, z_1^2)$ denominator.

Note also that the evolution of the real part always leads to a decrease of $\text{Re} \mathcal{M}(\nu, z_1^2)$ when z_1^2 increases. For the imaginary part, the evolution pattern is more complicated. Namely, below $\nu \sim 5.5$, the function $\text{Im} \mathcal{M}(\nu, z_1^2)$ increases when z_1^2 increases. Only above $\nu \sim 5.5$, the evolution leads to a decrease of $\text{Im} \mathcal{M}(\nu, z_1^2)$ with z_1^2 , and the evolution pattern becomes similar to that of the real part.

IV. NUMERICAL INVESTIGATION

In order to check numerically the ideas discussed above we performed lattice QCD calculations in the quenched approximation at $\beta = 6.0$ on $32^3 \times 64$ lattices (lattice spacing $a = 0.093$ fm). We used the non-perturbatively tuned clover fermion action with the clover coefficients computed by the Alpha collaboration [18].

We used a total of 500 configurations separated by 1000 updates each one consisting of four over-relaxation and one heatbath sweeps. On each configuration we computed correlation functions from 6 randomly selected point sources. The pion and nucleon masses in this setup were determined to be 601(1) MeV and 1411(4) MeV respectively. Conversion to physical energy units was per-

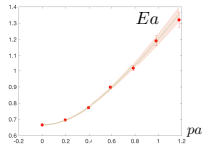


FIG. 4. Nucleon dispersion relation. Energies and momenta are in lattice units. The solid line is the continuum dispersion relation (not a fit) while the errorband is an indication of the statistical error of the lattice nucleon energies.

formed using the Alpha collaboration scale setting for quenched QCD [19].

Our nucleon states were boosted up to a total momentum of 2.5 GeV (corresponding to the 6th lattice momentum). Inside this momentum range, the continuum dispersion relation for the nucleon was satisfied within the errors of the calculation, indicating small lattice artifacts of $\mathcal{O}(aP)$. In Fig. 4 we plot the nucleon energy as a function of momentum along with the continuum dispersion relation corresponding to our lattice nucleon zero momentum energy.

The computation of the matrix elements was performed using the methodology described in [20] with an operator insertion given by Eq. (12). Taking the time component of the current we can isolate $\mathcal{M}_0(-z \cdot p, -z^2)$ which as discussed above is directly related to PDFs.

Following [20] we need to compute two types of correlation functions. The first is a regular nucleon two point function given by

$$C_P(t) = \langle \mathcal{N}_P(t) \overline{\mathcal{N}}_P(0) \rangle, \quad (27)$$

where $\mathcal{N}_P(t)$ is a helicity averaged, non-relativistic nucleon interpolating field with momentum p . The quark fields in $\mathcal{N}_P(t)$ are smeared with a gauge invariant Gaussian smearing. This choice of an interpolation field is known to couple well to the nucleon ground state (see discussion in [20]). The quark smearing width was optimized to give good overlap with the nucleon ground state within the range of momenta in our calculation. The second correlator is given by

$$C_P^{C^0(z)}(t) = \sum_{\tau} \langle \mathcal{N}_P(t) C^0(z, \tau) \overline{\mathcal{N}}_P(0) \rangle, \quad (28)$$

where

$$C^0(z, t) = \overline{\psi}(0, t) \gamma^0 \gamma_5 E(0, z; A) \psi(z, t), \quad (29)$$

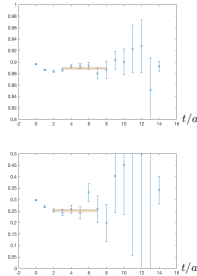


FIG. 5. Typical fits used to extract the reduced matrix element. The upper panel corresponds to $p = 2\pi/L \cdot 2$ and $z = 4$ and the lower panel to $p = 2\pi/L \cdot 3$ and $z = 8$, where momentum and position are in lattice units.

with γ_3 being the flavor Pauli matrix. The proton momentum and the displacement of the quark fields were both taken along the $\hat{z}$ axis ($\vec{\tau} = z_3 \hat{z}$ and $\vec{p} = P\hat{z}$). We define the effective matrix element as

$$\mathcal{M}_{eff}(z_3 P, z_3^2; t) = \frac{C_P^{(z)}(t+1)}{C_P(t+1)} - \frac{C_P^{(z)}(z)(t)}{C_P(t)}. \quad (30)$$

As it was shown in [20], our matrix element $\mathcal{J}$ can then be extracted at the large Euclidean time separation as

$$\frac{\mathcal{J}(z_3 P, z_3^2)}{2E} = \lim_{t \rightarrow \infty} \mathcal{M}_{eff}(z_3 P, z_3^2; t), \quad (31)$$

where E is the energy of the nucleon. This method of extracting the matrix element, contrary to the traditional sequential source approach, allows for the computation of the matrix element using all source-sink separations for the nucleon creation and annihilation operators.

The resulting effective matrix element has contaminations from excited states that scale as $e^{-t\Delta E}$, where t is the Euclidean time separation of the nucleon creation and annihilation operators, and ΔE is the mass gap to the first excited state of the nucleon. Furthermore, it allows for the computation of all nucleon matrix elements

that correspond to different nucleon momentum spin polarization and flavor structure without additional computational cost.

As a result, the total computational cost of this approach is less than the equivalent cost of performing the calculations with the sequential source method, especially because in our approach we put emphasis on having as many nucleon momentum states as possible. This approach has recently been successfully used for both single and multi-nucleon matrix element calculations [21–23].

In order to normalize our lattice matrix elements we note that, for $z_3 = 0$, the matrix element $\mathcal{M}(z_3 P, z_3^2)$ corresponds to a local vector (iso-vector) current, and therefore should be equal to 1. However, on the lattice this is not the case due to lattice artifacts. Therefore we introduce a renormalization constant

$$Z_P = \frac{1}{\mathcal{J}(z_3 P, z_3^2)|_{z_3=0}}. \quad (32)$$

The factor Z_P has to be independent from P . However, again due to lattice artifacts or potential fitting systematics, this is not the case. For this reason, we renormalize the matrix element for each momentum with its own Z_P factor taking this way advantage of maximal statistical correlations to reduce statistical errors, as well as the cancellation of lattice artifacts in the ratio. Therefore, our matrix element is extracted using the ratio

$$\mathcal{M}(z_3 P, z_3^2) = \lim_{t \rightarrow \infty} \frac{\mathcal{M}_{eff}(z_3 P, z_3^2; t)}{\mathcal{M}_{eff}(z_3 P, z_3^2; t)|_{z_3=0}}. \quad (33)$$

In order to determine the reduced matrix element $\mathfrak{M}(\nu, z_3^2)$ we introduce the double ratio

$$\mathfrak{M}(\nu, z_3^2) = \lim_{t \rightarrow \infty} \frac{\mathcal{M}_{eff}(z_3 P, z_3^2; t)}{\mathcal{M}_{eff}(z_3 P, z_3^2; t)|_{z_3=0}} \times \frac{\mathcal{M}_{eff}(z_3 P, z_3^2; t)|_{P=0, z_3=0}}{\mathcal{M}_{eff}(z_3 P, z_3^2; t)|_{P=0}}, \quad (34)$$

which takes care of the renormalization of the vector current according to Eq. (32). In practice, the infinite t limit is obtained with a fit to a constant for a suitable choice of a fitting range. In all cases we studied, the average χ^2 per degree of freedom was $\mathcal{O}(1)$. Typical fits used to extract the reduced matrix element are presented in Fig. 5. All fits are performed with the full covariance matrix and the error bars are determined with the jackknife method.

We note here that that the reduced matrix element defined in Eq. (34) has a well defined continuum limit and no additional renormalization is required. This continuum limit is obtained at fixed ν and z^2 as well as at fixed quark mass.

In this calculation we used momenta up to $6 \cdot 2\pi/L$ along the z -axis. This corresponds to a physical momentum of about 2.5 GeV.

V. DISCUSSION OF RESULTS

A. Rest-frame density and Z factor

An important object is the rest-frame density $\mathcal{M}(0, z_3^2)$. It is produced by data at $P = 0$. The results for its imaginary part are compatible with zero, as required. The real part, shown in Fig. 6, is a symmetric function of z_3 , and has a clearly visible linear component in its fall-off with $|z_3|$ for small and middle values of $|z_3|$. In fact, a linear exponential factor $Z(z_3^2) \sim e^{-c|z_3|/a}$ is expected as a manifestation of the nonperturbative effects generated by the straight-line gauge link.

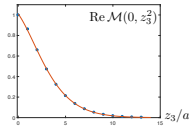


FIG. 6. Real part of the rest-frame density $\mathcal{M}(0, z_3^2)$

B. Reduced Ioffe-time distributions

In Fig. 7, we plot the results for the real part of the ratio $\mathcal{M}(Pz_3, z_3^2)/\mathcal{M}(0, z_3^2)$ as a function of z_3 taken at six fixed values of the momentum P . One can see that all the curves have a Gaussian-like shape. Thus, the $Z(z_3^2)$ link renormalization factor has been canceled in the ratio, as expected.

Furthermore, the curves look similar to each other, differing only by a decreasing width with P . In Fig. 8, we plot the same data, but change the axis to $\nu = Pz_3$. As one can see, now the data practically fall on the same curve. For the imaginary part, the situation is similar.

This phenomenon corresponds to factorization of the x - and $k_\perp$ -dependence for the soft TMD $\mathcal{F}(x, k_\perp^2)$, as discussed in previous sections.

C. Quark-antiquark decomposition

The real part of the Ioffe-time distribution is obtained from the cosine Fourier transform

$$\mathfrak{M}_R(\nu) = \int_0^1 dx \cos(\nu x) q_v(x) \quad (35)$$

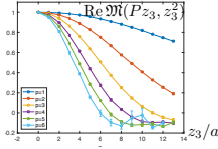


FIG. 7. Real part of the reduced distribution $\mathfrak{M}(Pz_3, z_3^2)$ plotted as a function of z_3 . Here, $P = 2\pi p/L$.

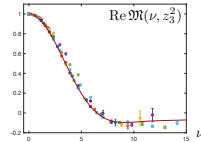


FIG. 8. Real part of $\mathfrak{M}(\nu, z_3^2)$ plotted as a function of $\nu = Pz_3$ and compared to the curve given by Eqs. (35), (36).

of the function $q_v(x)$ given by the difference $q_v(x) = q(x) - \bar{q}(x)$ of quark and antiquark distributions. In our case, q is $u - d$ and $\bar{q} = \bar{u} - \bar{d}$. The x -integral of $u - \bar{u}$ equals to the number of u -quarks in the proton, which is 2, while the x -integral of $d - \bar{d}$ equals 1. Thus, the x -integral of $q_v(x)$ should be equal to 1.

We found that our data for the real part are well described if one chooses the function

$$q_v(x) = \frac{315}{32} \sqrt{x} (1-x)^3, \quad (36)$$

whose x -integral is normalized to 1. To get it, we formed cosine Fourier transforms $\mathcal{M}(\nu; a, b)$ of the normalized $x^a(1-x)^b$ -type functions and found the parameters a, b by fitting our data. The comparison of the data with the curve based on Eqs. (35), (36) is shown in Fig. 8.

While all the data points were used in the fit, the latter is clearly dominated by the points with the smaller

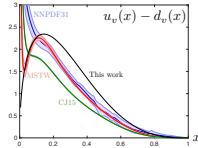


FIG. 9. Valence distribution $q_v(x)$ as given by Eq. (36) compared with the $Q^2 = 1 \text{ GeV}^2$ NNLO global fits NNPDF31_mlo_pch_as_0118_mc_164 [24] and MSTW2008mlo68cl_n4 [25]; and the NLO global fit CJ15nlo [26], all extracted using the LHAPDF library [27]. The bands around the global fits indicate their experimental and systematic uncertainties.

values of $\text{Re } \mathfrak{M}(\nu, z_3^2)$. For $\nu < 10$, the data points lying above the curve, correspond to values of $z_3 = 3a$ to $5a$. As we will see later, they reflect the perturbative evolution: $\text{Re } \mathfrak{M}(\nu, z_3)$ increases when z_3 decreases. In this context, the overall curve (36) corresponds to PDF “at low normalization point”, i.e., in the region, where the perturbative evolution stops.

In general, it is more appropriate to fit $\text{Re } \mathfrak{M}(\nu, z_3^2)$ as a function of two variables, ν and z_3 , even though the dependence on z_3 is rather weak and noticeable just for a few points. Since we made a fit of $\text{Re } \mathfrak{M}(\nu, z_3)$ as a function of just one variable ν , there are points that visibly deviate from the curve, but we do not think that it makes sense to translate the evolution z_3 -dependence of small- z_3 points into an error band to our curve in Fig. 8. In the Section V.D, we evolve the data points to a common reference scale $z_0 = 2a$ and show the error band for the results obtained in this way.

We realize that our lattice setup is rather crude (quenched approximation, very large pion mass), and for this reason we do not attempt to perform a thorough comparison of our results with experimental data. Still, we think that some kind of comparison is rather useful as an illustration.

Thus, we compare our $q_v(x)$ with three global fits for the difference $u_v(x) - d_v(x)$ of the valence distributions, see Fig. 9. These global fits curves correspond to $\mu = 1 \text{ GeV}$ scale, while our “low normalization point” curve corresponds to $\mu \lesssim 0.3 \text{ GeV}$. Still, one can see that our curve is not very far from the NNPDF31 [24] NNLO fit down to $x = 0.1$ and from the MSTW [25] NLO fit down to $x = 0.05$. We also show the NLO fit CJ15 [26].

Since the areas under each curve are equal to 1, our curve compensates the strong deficiency in the $x < 0.1$ region by exceeding the NNLO curves at $x > 0.1$ values.

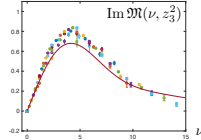


FIG. 10. Imaginary part of $\mathfrak{M}(\nu, z_3^2)$ compared to the curve $\mathfrak{M}_I(\nu)$ based on $\bar{q}(x) = 0$.

In other words, if our curve would better describe data in the $x < 0.1$ region, it would necessarily be smaller in the $x > 0.1$ region.

The sine Fourier transform

$$\mathfrak{M}_I(\nu) = \int_0^1 dx \sin(\nu x) q_+(x) \quad (37)$$

is built from the function $q_+(x) = q(x) + \bar{q}(x)$, which may be also represented as $q_+(x) = q_v(x) + 2\bar{q}(x)$. If we neglect the antiquark contribution and use $q_+(x) = q_v(x)$, we get the curve shown in Fig. 10 (call it $\mathfrak{M}_I^v(\nu)$). The agreement with the data is strongly improved if we use a non-vanishing antiquark contribution, namely

$$\bar{q}(x) = u(x) - d(x) = 0.07 [20x(1-x)^2], \quad (38)$$

see Fig. 11. This function was obtained by fitting the data for the difference $\text{Im } \mathfrak{M}(\nu, z_3^2) - \mathfrak{M}_I^v(\nu)$ by sine

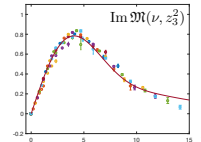
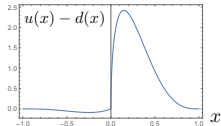


FIG. 11. Imaginary part of $\mathfrak{M}(\nu, z_3^2)$ compared to the curve based on $\bar{q}(x)$ given by Eq. (38).

FIG. 12. Overall distribution $q(x)$ as defined by Eq. (40).

Fourier transforms of $Ax^a(1-x)^b$ functions. This result corresponds to

$$\int_0^1 dx [u(x) - \bar{d}(x)] = 0.07. \quad (39)$$

The combined distribution

$$q(x) = u(x) - d(x) = [q_u(x) + q(x)]\theta(x > 0) - q(-x)\theta(x < 0) \quad (40)$$

defined on the $-1 \leq x \leq 1$ interval is shown in Fig. 12.

D. Evolution

While an overall agreement of the data with a z_3 -independent curve looks satisfactory, one can easily notice a residual z_3 -dependence in the data. It is especially visible when, for a particular ν , there are several data points corresponding to different values of z_3 . It is interesting to check if this dependence corresponds to perturbative evolution.

To begin with, the evolution of the real part should lead to its *decrease* when z_3^2 increases. On the other hand, as pointed out at the end of section III, the function $\ln \mathcal{M}(\nu, z_3^2)$ *increases* when z_3^2 *increases* as long as $\nu \lesssim 5.5$. Our data follow these patterns.

As we discussed, the evolution corresponds to $\ln z_3^2$ singularities of the Ioffe-time distributions for small z_3^2 . Thus, a natural idea is to check if the data corresponding to small z_3^2 and z_3 may be related by

$$\Re(\nu, z_3'^2) = \Re(\nu, z_3^2) - \frac{2}{3} \frac{\alpha_s}{\pi} \ln(z_3'^2/z_3^2) B \otimes \Re(\nu, z_3^2) \quad (41)$$

for some value of α_s . Here B is the evolution kernel (21).

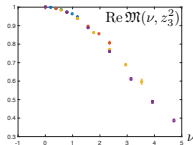
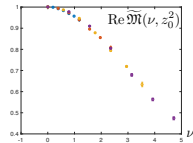
FIG. 13. Real part of $\Re(\nu, z_3^2)$ for $z_3/a = 1, 2, 3$, and 4.

FIG. 14. Evolved data points for the real part.

In our case,

$$B \otimes \Re(\nu) = \int_0^1 du \frac{1+u^2}{1-u} [\Re(\nu) - \Re(u\nu)]. \quad (42)$$

More specifically, we fix the point z_3' at the value $z_0 = 2a$ corresponding, at the leading logarithm level, to the $\overline{\text{MS}}$ -scheme scale $\mu_0 = 1$ GeV and build the function

$$\widetilde{\Re}(\nu, z_3^2) = \Re(\nu, z_3^2) - \frac{2}{3} \frac{\alpha_s}{\pi} \ln(z_0^2/z_3^2) B \otimes \Re(\nu, z_3^2) \quad (43)$$

from the data points for $\Re(\nu, z_3^2)$ using various values for α_s .

Since the perturbative evolution is expected for small z_3 , we include in this analysis the data with z_3 up to 4 lattice spacings, which corresponds to energy scales $\mu = 2, 1, 0.7$ and 0.5 GeV.

For the real part, these data points are shown in Fig. 13. As one can see, there is a visible scatter of the

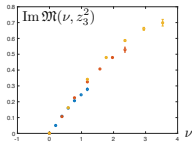


FIG. 15. Imaginary part of $\mathfrak{M}(\nu, z_3^2)$ for $z_3/a = 1, 2, 3$, and 4.

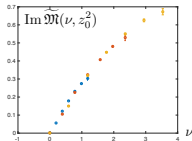


FIG. 16. Evolved data points for the imaginary part.

data points. Using $\alpha_s/\pi = 0.1$, we calculate the “evolved” data points corresponding to the function $\mathfrak{M}(\nu, z_0^2)$. The results are shown in Fig. 14. The evolved data points are now very close to a universal curve.

In Fig. 15, we show the initial data points for the imaginary part. The evolved data points constructed using the same $\alpha_s/\pi = 0.1$ value are shown in Fig. 16. Again, they are close to a universal curve. This analysis indicates that the residual z_3^2 -dependence of $\mathfrak{M}(\nu, z_3^2)$ at fixed ν is compatible with the expected logarithmic evolution at small z_3^2 . Clearly this is an important feature of our calculation which needs to be further studied as it will play an essential role in reliable extraction of renormalized PDFs from this type of lattice calculations.

With a smaller lattice spacing, the use of perturbative evolution may be justified in a wider region of ν . While our data extend to rather large separations ~ 1 fm, we find it instructive to use them as an example to illustrate the trends generated by the perturbative evolution.

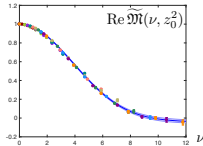


FIG. 17. Data points for $\text{Re } \mathfrak{M}(\nu, z_0^2)$ with $z_3 \leq 10a$ evolved to $z_0 = 2a$ as described in the text.

To this end, we applied the leading logarithm formula (43) with $z_0 = 2a$ and $\alpha_s/\pi = 0.1$ to our data points with $z_3 \leq 6a$. Assuming that evolution stops for $z_3 \gtrsim 6a$ (as indicated by our data), the data points with $7a \leq z_3 \leq 10a$ were evolved to z_0 using Eq. (43) with $z_3 = 6a$. The data points evolved in this way are shown in Fig. 17.

Fitting the evolved points by cosine Fourier transforms $\mathcal{M}(\nu; a, b)$ of the normalized $N(a, b)x^a(1-x)^b$ -type functions, we found that they may be described if one takes $a = 0.96(6)$ and $b = 3.95(22)$. Treating $z_0 = 2a$ as the $\overline{\text{MS}}$ scale $\mu = 1$ GeV, one can further evolve the curve to

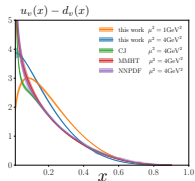


FIG. 18. Curve for $u_v(x) - d_v(x)$ built from the evolved data shown in Fig. 17, and treated as corresponding to the $\mu^2 = 1$ GeV² scale; then evolved to the reference point $\mu^2 = 4$ GeV² of the global fits.

the standard reference scale $\mu^2 = 4 \text{ GeV}^2$ of the global fits, see Fig. 18. Comparing with Fig. 9, we see that the perturbative evolution shifts our curves, moving them closer to the global fits.

VI. SUMMARY

In this paper, we demonstrated a new method of extracting parton distributions from lattice calculations. It is based on the ideas, formulated in Ref. [11].

First, we treat the generic equal-time matrix element as a function $\mathcal{M}(\nu, z_3^2)$ of the Ioffe time $\nu = Pz_3$ and the distance z_3 . The next idea is to form the ratio $\mathfrak{M}(\nu, z_3^2) \equiv \mathcal{M}(\nu, z_3^2)/\mathcal{M}(0, z_3^2)$ of the Ioffe-time distribution $\mathcal{M}(\nu, z_3^2)$ and the rest-frame density given by $\mathcal{M}(0, z_3^2)$.

Our lattice calculation clearly shows the presence of a linear component in the z_3 -dependence of the rest-frame function, that may be attributed to the expected $Z(z_3^2) \sim e^{-c|z_3|/a}$ behavior generated by the gauge link. On the next step, we observe that the ratio $\mathcal{M}(Pz_3, z_3^2)/\mathcal{M}(0, z_3^2)$ has a Gaussian-type behavior with respect to z_3 for all 6 values of P that were used in the calculation. This means that $Z(z_3^2)$ factors entering into the numerator and denominator of the $\mathfrak{M}(Pz_3, z_3^2)$ ratio have been canceled, as expected.

Still, there is no *a priori* principle predicting that the remaining non-logarithmic z_3^2 -dependence cancels between the numerator and the denominator of the ratio $\mathcal{M}(\nu, z_3^2)/\mathcal{M}(0, z_3^2)$. Such a z_3^2 -dependence can be removed if needed with a systematic fitting procedure from which the Ioffe time PDF will be extracted in the $z_3^2 \rightarrow 0$ limit.

However, we found that when plotted as a function of ν and z_3 , the data both for the real and imaginary parts of $\mathfrak{M}(\nu, z_3^2)$ are very close to the respective universal functions. This observation indicates that the soft part of the z_3^2 -dependence of $\mathcal{M}(\nu, z_3^2)$ has been canceled by the rest-frame density $\mathcal{M}(0, z_3^2)$. This phenomenon corresponds to factorization of the x - and $k_\perp$ -dependence for the TMD $\mathcal{F}(x, k_\perp^2)$.

While this evidence in favor of the factorization property is an important result on its own, we want to stress that our approach is *not* based on the factorization. It is based on the use of the ratio $\mathcal{M}(\nu, z_3^2)/\mathcal{M}(0, z_3^2)$. Its residual soft z_3^2 -dependence may be systematically analyzed and fitted, so that the $z_3^2 \rightarrow 0$ -limit may be taken in a controllable way.

Luckily, the data do not show a visible polynomial dependence on z_3^2 within our current statistical and systematic errors. In future work we intend to carefully study the residual polynomial z_3^2 effects and incorporate them in the extraction of PDFs using the lattice methodology introduced here.

In addition, we have checked that, for small $z_3 \leq 4a$, the residual z_3 -dependence may be explained by perturbative evolution, with the α_s value corresponding to

$\alpha_s/\pi = 0.1$. We have evolved these small- z_3 data points to the $z_3 = 2a$ scale, which corresponds to $\mu^2 = 1 \text{ GeV}^2$. The evolved data better approximate universal curves both for real and imaginary parts of $\mathcal{M}$, supporting the argument that perturbative evolution is observed.

Thus, these $\nu \lesssim 4$ parts of the universal curves may be treated as corresponding to the $\mu = 1 \text{ GeV}$ scale. Other data points correspond to $z_3 > 4a$ values, and formally should be treated as corresponding to scales $\mu \lesssim 0.3 \text{ GeV}$. All these data points basically lie on the same universal curve. This indicates that evolution stops at such scales. We compared this “low normalization point” curve with three global fits evolved to the $\mu = 1 \text{ GeV}$ scale, and observed that our curve (36) for the valence $u_v(x) - d_v(x)$ distribution shows the $(1-x)^3$ behavior for $x \rightarrow 1$ in accord with usual expectations. Also, it rather closely follows the NNPDF31 and, especially, MSTW NNLO global fits down to rather small x values.

Still, our curve strongly deviates from the global fits for $x < 0.1$ in the NNPDF31 case and for $x < 0.05$ in the MSTW case. However, the shape of PDFs is affected by the perturbative evolution. To illustrate the scope of these effects, we evolved all our points with $z_3 \lesssim 10a$ to a universal scale $z_0 = 2a$ corresponding to $\mu = 1 \text{ GeV}$, and then further evolved the resulting PDF to $\mu = 2 \text{ GeV}$, that is the standard reference scale for global fits. Our final curve is rather close to these fits, which demonstrates that the perturbative evolution plays an important role in comparison of lattice results with the data. Again, one needs smaller lattice spacings to justify the use of the perturbative evolution equation in a sufficiently wide interval of Ioffe time parameters ν .

The data also indicate a nonzero *positive* antiquark distribution $\bar{q}(x) = \bar{u}(x) - \bar{d}(x)$. It changes the x -integral of $q(x)$ by 7% and has $\sim x(1-x)^3$ behavior. Since we are using the quenched approximation, these antiquarks come from “connected diagrams”. Hence, one should expect that the ratio $\bar{u}/\bar{d}$ must follow the flavor content of the proton, i.e. $\bar{u}/\bar{d} \sim 2$ and $\bar{u} > \bar{d}$. Our data agree with this expectation.

The present study has an exploratory nature, and its main goal was to develop techniques for lattice extraction of PDFs based on the ideas of Ref. [11]. Our results indicate that the basic method we put forward has a strong potential for obtaining reliable PDFs from lattice QCD. In future work we will refine our methods for incorporating evolution and controlling residual polynomial z_3^2 effects in the extraction of the Ioffe time distributions.

To achieve this, it is evident that smaller lattice spacings are required as well as a larger range of nucleon momenta. Furthermore, we need to study finite volume effects as well as to incorporate dynamical fermions with pion masses closer to the physical point. We plan to address all these issues in our future work.

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P17: 欧氏与光锥分布因子化定理

主题归属	LaMET、quasi/pseudo 分布与因子化
底稿	arXiv:1801.03917
arXiv / DOI	1801.03917
完整原始 PDF	21 页; 来源 PyQCD; SHA-256 c5398a1bd179...
转排索引	EN 31 页; ZH 28 页 (只作书目对照)
备注	索引未记录额外备注。

阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

页码范围：P17-p001 – P17-p021

Factorization Theorem Relating Euclidean and Light-Cone Parton Distributions

Taku Izubuchi,^{1,2} Xiangdong Ji,^{3,4} Luchang Jin,^{1,2,5} Iain W. Stewart,⁶ and Yong Zhao⁶¹Physics Department, Brookhaven National Laboratory, Upton, NY, 11973, USA²RIKEN-BNL Research Center, Brookhaven National Laboratory, Upton, NY 11973, USA³Tsung-Dao Lee Institute, and College of Physics and Astronomy,

Shanghai Jiao Tong University, Shanghai, 200240, P. R. China

⁴Maryland Center for Fundamental Physics, Department of Physics,

University of Maryland, College Park, Maryland 20742, USA

⁵Physics Department, University of Connecticut, Storrs, Connecticut 06269-3046, USA⁶Center for Theoretical Physics, Massachusetts Institute of Technology, Cambridge, MA 02139, USA

In a large-momentum nucleon state, the matrix element of a gauge-invariant Euclidean Wilson line operator accessible from lattice QCD can be related to the standard light-cone parton distribution function through the large-momentum effective theory (LaMET) expansion. This relation is given by a factorization formula with a non-trivial matching coefficient. Using the operator product expansion we derive the large-momentum factorization of the quasi-parton distribution function in LaMET, and show that the more recently discussed pseudo-parton distribution approach also obeys an equivalent factorization formula. Explicit results for the coefficients are obtained and compared at one-loop. We also prove that the matching coefficients in the $\overline{\text{MS}}$ scheme depend on the large partonic momentum rather than the nucleon momentum.

I. INTRODUCTION

Parton distribution functions (PDFs) are key quantities for gaining an understanding of hadron structure and for making predictions for the cross sections in high-energy scattering experiments. In QCD factorization theorems for hard scattering processes [1], the relevant PDFs are defined in terms of the nucleon matrix elements of light-cone correlation operators. For example, in dimensional regularization with $d = 4 - 2\epsilon$, the bare unpolarized quark PDF is

$$q(x, \epsilon) = \int \frac{d\xi^-}{4\pi} e^{-ixP^+\xi^-} \langle P | \bar{\psi}(\xi^-) \gamma^+ U(\xi^-, 0) \psi(0) | P \rangle, \quad (1)$$

where x is the momentum fraction, the nucleon momentum $P^\mu = (P^0, 0, 0, P^3)$, $\xi^\pm = (t \pm z)/\sqrt{2}$ are the light-cone coordinates, and the Wilson line is

$$U(\xi^-, 0) = P \exp \left(-ig \int_0^{\xi^-} d\eta^- A^+(\eta^-) \right). \quad (2)$$

Most often the bare PDF is renormalized in the $\overline{\text{MS}}$ scheme to obtain $q(x, \mu)$, and this renormalized PDF is used to make predictions for experiment. The relation is

$$q(x, \epsilon) = \int_x^1 \frac{dy}{y} Z^{\overline{\text{MS}}} \left(\frac{x}{y}, \epsilon, \mu \right) q(y, \mu), \quad (3)$$

where μ is the renormalization scale, and we have suppressed the flavor indices in the renormalization constant $Z^{\overline{\text{MS}}}$ and PDFs. In light-cone quantization with $A^+ = 0$, the $\overline{\text{MS}}$ definition has an interpretation as a parton number density.

So far our main knowledge of the PDF is obtained from global fits to deep inelastic scattering and jet data, see

for example [2–6]. On the other hand, calculating the PDF from first principles with QCD has been an attractive subject, which can for example provide access to spin and momentum distributions that are hard to determine experimentally. Several different approaches to this have been considered using the lattice theory which is a non-perturbative method to solve QCD. Since the lattice theory is defined in a discretized Euclidean space with imaginary time, it is very difficult to calculate Minkowskian quantities with real-time dependence such as the PDF. The first and most well explored option is calculating the moments of the PDF [7–11] that are matrix elements of local gauge-invariant operators. However, since the lattice regularization breaks $O(4)$ rotational symmetry, the consequent mixing between operators of different dimensions makes it difficult to compute higher moments, which in practice has limited the amount of information that can be extracted from this approach. A method to improve this situation by restoring the rotational symmetry has been proposed in Ref. [12]. Other proposals include extracting the PDF from the hadronic tensor [13–16] and the forward Compton amplitude [17], possibly with flavor changing currents [18], and the more general “lattice cross sections” [19, 20]. Systematic lattice analyses of these approaches are under investigation, but challenges remain.

In Ref [21] Ji proposed that the x -dependence of the PDF can be extracted from a Euclidean distribution on the lattice, which can be understood in the language of the large momentum effective theory (LaMET) [22]. This Euclidean distribution is referred to as the quasi-PDF, whose bare matrix element is defined using a spatial cor-

relation of quarks along the z direction,

$$\tilde{q}(x, P^z, \epsilon) = \int_{-\infty}^{\infty} \frac{dz}{4\pi} e^{izP^z} \langle P | \bar{\psi}(z) \Gamma U(z, 0) \psi(0) | P \rangle, \quad (4)$$

where $\Gamma = \gamma^z$, $z^\mu = z e^\mu$, $e^\mu = (0, 0, 0, 1)$, and the Wilson line is

$$U(z, 0) = P \exp \left(-ig \int_0^z dz' A^z(z') \right). \quad (5)$$

For finite momentum P^z , $\tilde{q}(x, P^z, \epsilon)$ has support in $-\infty < x < \infty$. According to Ref. [23], there is a universality class of operators that can be considered. For example, for the quasi-PDF, one could also replace $\Gamma = \gamma^z$ by $\Gamma = \gamma^0$ in Eq. (4) as both definitions reduce to the PDF under an infinite Lorentz boost along the z direction. Unlike the PDF in Eq. (1) that is invariant under the Lorentz boost, the quasi-PDF depends dynamically on it through the nucleon momentum P^z . When the nucleon momentum P^z is much larger than the nucleon mass M and Λ_{QCD} , which is an attainable window on the lattice, the quasi-PDF can be factorized into a matching coefficient and the PDF [21, 22]. The factorization formula is

$$\begin{aligned} \tilde{q}(x, P^z, \mu_R) = & \int_{-1}^1 \frac{dy}{|y|} C \left(\frac{x}{y}, \frac{\mu_R}{\mu}, \frac{\mu}{P^z} \right) q(y, \mu) \\ & + O \left(\frac{M^2}{P_z^2}, \frac{\Lambda_{\text{QCD}}^2}{P_z^2} \right), \end{aligned} \quad (6)$$

where the renormalized quasi-PDF $\tilde{q}(x, P^z, \mu_R)$ is defined in a particular scheme at renormalization scale μ_R , and $O(M^2/P_z^2, \Lambda_{\text{QCD}}^2/P_z^2)$ are power corrections suppressed by the nucleon momentum. In general the result for C will depend on the choice of $\Gamma = \gamma^z$ or γ^0 and renormalization schemes. For $\Gamma = \gamma^z$ the matching coefficient C has been computed for the iso-vector quark quasi-PDF at one-loop level, first with a transverse momentum cutoff in [24], confirmed in [19, 25], and also recently determined in the regularization-invariant momentum subtraction (RI/MOM) scheme [26]. Matching for the gluon quasi-PDF is calculated in [27, 28]. The matching coefficient C is independent of the choice of states used for $\tilde{q}$ and q .¹ Since matching calculations are carried out with quark states of momentum p^z , it can be tricky to know what the right choice to make is for C , and in some of the literature the choice of $p^z = P^z$ has been suggested when utilizing C for the hadronic nucleon state. This is for example the case in the original quasi-PDF papers [21, 22, 24] and in the pioneering lattice calculations

of the PDF from the quasi-PDF in [25, 29–36], which was summarized in Ref. [37]. In the quasi-generalized parton distribution analysis in [38] it was observed that one should take $p^z = |y|P^z$. Through our rigorous analysis of Eq. (6) we show that the correct result for this equation is indeed $p^z = |y|P^z$.

Recently, a different procedure [39] to extract PDFs from the same lattice QCD matrix element as in [21] has been proposed based on the Lorentz invariant variables of the spatial correlator (or pseudo-PDF), in place of the quasi-PDF. In this approach, one starts from the spatial correlator $\tilde{Q}_{\gamma^0}$ defined for $\mu = 0$ or $\mu = z$ by

$$\frac{1}{2} \langle P | \tilde{O}_{\gamma^0}(z, \epsilon) | P \rangle = P^0 \tilde{Q}_{\gamma^0}(\zeta = P^z z, z^2, \epsilon), \quad (7)$$

which depends on the two Lorentz invariants z^2 and $\zeta = -z \cdot P = P^z z$; the latter is also called the Ioffe time. For an arbitrary Dirac matrix Γ the operator $\tilde{O}_\Gamma$ is defined as

$$\tilde{O}_\Gamma(z, \epsilon) = \bar{\psi}(z) \Gamma U(z, 0) \psi(0). \quad (8)$$

This is the same spatial correlator (calculable on lattice) used to define the quasi-PDF in Eq. (4), where P^z is fixed and one Fourier transforms with respect to z . If instead z^2 is fixed, and we Fourier transform from the Ioffe time ζ —which is in principle integrating over P^z —to the momentum fraction x , then one obtains the pseudo-PDF [39],

$$\mathcal{P}(x, z^2, \epsilon) = \int_{-\infty}^{\infty} \frac{d\zeta}{2\pi} e^{ix\zeta} \tilde{Q}_{\gamma^0}(\zeta, z^2, \epsilon). \quad (9)$$

For arbitrary finite z , the pseudo-PDF only has support in $-1 \leq x \leq 1$ [40, 41], but has no parton model interpretation. (Again the pseudo-PDF can equally well be considered for $\Gamma = \gamma^z$.) The spatial correlator or pseudo-PDF approach has been explored on the lattice [42, 43], where the short distance behavior was explored. The PDF corresponds to the situation when z^μ is light-like, in which case the space-time correlator is referred to as the Ioffe-time distribution [44].

$$Q(\zeta, \epsilon) = \tilde{Q}_{\gamma^+}(\zeta = -P^+ \xi^-, z^2 = 0, \epsilon). \quad (10)$$

When Fourier transformed this correlation gives the PDF

$$q(y, \epsilon) = \int_{-\infty}^{\infty} \frac{d\zeta}{2\pi} e^{i\zeta y} Q(\zeta, \epsilon). \quad (11)$$

In short the quasi-PDF and pseudo-PDF are different representations of the Euclidean spatial correlator, as summarized in Table. I.

It was pointed out in Ref. [45] that to obtain enough information to extract the PDF for the spatial correlator with small z^2 , one has to do lattice calculations with large momenta P^z , which is the same requirement as for the quasi-PDF. Ref. [45] also proposed that the renormalized

¹ The scheme definition itself may separately involve a choice of state, such as in the RI/MOM scheme, but the result is still an operator renormalization that can be used for different choices of hadronic states for $\tilde{q}$ and q . The transverse cutoff and $\overline{\text{MS}}$ schemes do not require even this choice.

Distribution	Fourier transform from spatial correlator	Arguments
Spatial correlation		$\zeta = zP^z, z^2$
Quasi-PDF	$z \rightarrow zP^z$	x, P^z
Pseudo-PDF	$\zeta \rightarrow z$	x, z^2

TABLE I. Summary of the relationship between different Euclidean distributions.

pseudo-PDF satisfies the following small z^2 factorization,

$$\mathcal{P}(x, z^2 \mu_R^2) = \int \frac{dy}{|y|} C\left(\frac{x}{y}, \frac{\mu_R^2}{\mu^2}, z^2 \mu_R^2\right) q(y, \mu) + \mathcal{O}(z^2 \Lambda_{\text{QCD}}^2, z^2 M^2), \quad (12)$$

which they verified at order $\mathcal{O}(\alpha_s)$ for the unpolarized iso-vector case with $\Gamma = \gamma^0$. (Again the coefficient C will depend on the choice of $\Gamma = \gamma^0$ or γ^z .)

In Ref.[19] a diagrammatic derivation of the factorization formula in Eq. (6) for the quasi-PDF was given. Here we derive this factorization formula for the quasi-PDF in an alternate manner, and also show that spatial correlator and pseudo-PDF are different representations of the same fundamental factorization. Our approach is based on the operator product expansion (OPE) for spacelike separated local operators [46]. For such operators the OPE has been proven for scalar field theory to all orders in perturbation theory [47–49], and is widely assumed to hold for any renormalizable quantum field theory including QCD. By introducing auxiliary fields in place of the Wilson line [50], the correlator in Eq. (8) is known to be equivalent to a product of local renormalizable operators of this type. Through our derivation we find the explicit form of the large P^z and small z^2 factorization formulas in Eq.(6) and Eq. (12) respectively, as well as the relationship between the matching coefficients C and $\bar{C}$. Since the requirement for large P^z and small z^2 is the same for both the quasi-PDF and pseudo-PDF approaches, there is in principle no fundamental difference in applying either one to lattice calculations of the proton matrix element of $\bar{O}_T(z)$. It is interesting to compare both approaches utilizing the same lattice data, although they shall not yield different result in principle.

The rest of this paper is organized as follows: In Sec. II, we use an OPE of $\bar{O}_T(z)$ to derive the large P^z factorization of LaMET in Eq. (6) and small z^2 factorization of the pseudo-PDF in Eq. (12). We prove that one must take $p^z = |y|P^z$ in Eq. (6), so the corresponding argument in C is $\mu/|y|P^z$. (This OPE approach was used recently in Ref. [20] to prove the factorization theorem for the “lattice cross sections”, and the OPE proof carried out here was done independently and first presented in Ref. [51].) In Sec. III, we derive the spatial correlator, pseudo-PDF, and quasi-PDF distributions and matching coefficients at one-loop in $\overline{\text{MS}}$ and analyze the Fourier-transform relation between the quasi-PDF and pseudo-PDF. Unlike earlier results for the quasi-PDF in

$\overline{\text{MS}}$, we also use dimensional regularization with minimal subtraction to renormalize divergences at $x = \pm\infty$. In Sec. IV, we discuss how renormalization schemes other than $\overline{\text{MS}}$ are easily incorporated into the factorization formulas. In Sec. V we carry out a numerical analysis of the matching coefficients, by computing the convolution in Eq. (6) numerically using the PDF determined by global fits [4]. We show that the difference between using $p^z = P^z$ and $p^z = |y|P^z$ in Eq. (6) is an important effect, and that our $\overline{\text{MS}}$ matching coefficients are insensitive to cutoffs in the convolution integral. In Sec. VI, we discuss the implications of our OPE analysis for the lattice calculation of the PDF in both the quasi-PDF and pseudo-PDF approaches. Finally, we conclude in Sec. VII.

II. FACTORIZATION FROM THE OPE

In this section we make use of the operator product expansion to derive the matching relation for the quasi-PDF, as well as the equivalent matching relations for the spatial correlator and pseudo-PDF. For simplicity these three equivalent cases are presented in separate subsections.

A. OPE and Factorization for the Spatial Correlator

The OPE is a technique to expand nonlocal operators with separation z^μ in terms of local ones in the Euclidean limit of $z^2 \rightarrow 0$. It can be applied to both bare regulated operators as well as renormalized operators, and our focus will be on the latter. For the gauge-invariant Wilson operator $\bar{O}_T(z)$, it was proven that it can be multiplicatively renormalized in coordinate space as [52, 53]

$$\bar{O}_T(z, \mu) = Z_{\psi,z} e^{\delta m|z|} \bar{O}_T(z, \epsilon), \quad (13)$$

where δm captures the power divergence from the Wilson line self-energy, $Z_{\psi,z}$ only depends on the end points $z, 0$ and renormalizes the logarithmic divergences. This multiplicative renormalization was also discussed earlier in Refs. [54–56]. For simplicity, in this section we take $\Gamma = \gamma^z$ for Eq. (13).

In the $\overline{\text{MS}}$ scheme, the power divergence vanishes, and using the OPE the renormalized $\bar{O}_T(z, \mu)$ can be expanded in terms of local gauge-invariant operators as $z^2 \rightarrow 0$ giving

$$\bar{O}_{\gamma^z}(z, \mu) = \sum_{n=0}^{\infty} \left[C_n(\mu^2 z^2) \frac{(-iz)^n}{n!} e_{\mu_1} \cdots e_{\mu_n} O_1^{\mu_0 \mu_1 \cdots \mu_n}(\mu) + C'_n(\mu^2 z^2) \frac{(-iz)^n}{n!} e_{\mu_1} \cdots e_{\mu_n} O_2^{\mu_0 \mu_1 \cdots \mu_n}(\mu) + \text{higher-twist operators} \right], \quad (14)$$

where $\mu_0 = z$, $C_n = 1 + \mathcal{O}(\alpha_s)$ and $C'_n = \mathcal{O}(\alpha_s)$ are Wilson coefficients, and $O_1^{\mu_0 \mu_1 \cdots \mu_n}(\mu)$ and $O_2^{\mu_0 \mu_1 \cdots \mu_n}(\mu)$

are the only allowed renormalized traceless symmetric twist-2 quark and gluon operators at leading power in the OPE,

$$\begin{aligned} O_1^{\mu_1 \mu_2 \dots \mu_n}(\mu) &= Z_{n+1}^{\eta_1} \left(\bar{\psi} \gamma^{\mu_1} D^{\mu_2} \dots i D^{\mu_n} \psi - \text{trace} \right), \\ O_2^{\mu_1 \mu_2 \dots \mu_n}(\mu) &= Z_{n+1}^{\eta_2} \left(F^{\mu_1 \mu_2} D^{\mu_3} \dots i D^{\mu_{n-1}} F_{\rho}^{\mu_n} - \text{trace} \right). \end{aligned} \quad (15)$$

Here $Z_{n+1}^{ij} = Z_{n+1}^{ij}(\mu, \epsilon)$ are multiplicative $\overline{\text{MS}}$ renormalization factors and $(\mu_1, \dots, \mu_n)$ stands for the symmetrization of these Lorentz indices.

The above OPE is valid for the operator itself, where we implicitly constrain ourselves to the subspace of matrix elements for which the twist expansion is appropriate. In the iso-vector case, the mixing with the gluon operators is absent, which we will stick to for the rest of the paper. When $O_1^{\mu_1 \mu_2 \dots \mu_n}$ is evaluated in the nucleon state,

$$\langle P | O_1^{\mu_1 \mu_2 \dots \mu_n} | P \rangle = 2a_{n+1}(\mu) (P^{\mu_1} P^{\mu_2} \dots P^{\mu_n} - \text{trace}), \quad (16)$$

where $a_{n+1}(\mu)$ is the $(n+1)$ -th moment of the PDF,

$$a_{n+1}(\mu) = \int_{-1}^1 dx x^n q(x, \mu), \quad (17)$$

and the explicit expression of the trace term have been derived in Ref. [30, 57, 58]. The inverse relation to Eq. (17) is that $q(x, \mu)$ has an expansion with terms proportional to the n 'th derivative of the δ -function, as in $\delta^{(n)}(x) a_n(\mu)$, without any nontrivial short distance Wilson coefficient.

As pointed out in Ref. [45], to obtain enough information for the spatial correlator at $|\zeta| = |P^2 z| \sim 1$ at small z^2 , we have to choose P^2 large compared to the scale Λ_{QCD} . When $P^2 \gg \{\Lambda_{\text{QCD}}^2, M^2\}$, the trace terms in Eq.(16) are suppressed by powers of M^2/P^2 , while the contributions from higher-twist operators in Eq. (14) are suppressed by powers of $\Lambda_{\text{QCD}}^2/P^2$ or $z^2 \Lambda_{\text{QCD}}^2$. Therefore, the twist-2 contribution is the leading approximation of the nucleon matrix element $\langle P | \bar{O}_{\gamma^*}(z) | P \rangle$ at large momentum. From now on we will drop all the power corrections from our discussion.

The Wilson coefficients $C_n(\mu^2 z^2)$ in the OPE of $\bar{O}_{\gamma^*}(z)$ can be computed in perturbation theory for $\mu \sim 1/|z| \gg \Lambda_{\text{QCD}}$. In the $\overline{\text{MS}}$ scheme, the C_n are log-singular near $z^2 = 0$, and so is $\langle P | \bar{O}_{\gamma^*}(z, \mu) | P \rangle$. For this reason the x -moments of the quasi-PDF $\bar{q}(x, P^2, \mu)$ are proportional to $C_n|_{z=0}$ which is divergent, $\bar{q}(x, P^2, \mu)$ will not simply become the PDF in the infinite P^2 limit. Instead, we need a factorization formula which matches the quasi-PDF to the PDF. In contrast, for the pseudo-PDF the moments do exist since we hold $\mu^2 z^2$ fixed when taking the x -moment. However, we still need a factorization formula to match the pseudo-PDF to the PDF. We will comment further about this below.

Based on Eqs. (14-17), we can write down the leading-twist approximation to the spatial correlator as

$$\begin{aligned} \bar{Q}_{\gamma^*}(\zeta, \mu^2 z^2) &= \sum_n C_n(\mu^2 z^2) \frac{(-i\zeta)^n}{n!} a_{n+1}(\mu) \\ &= \sum_n C_n(\mu^2 z^2) \frac{(-i\zeta)^n}{n!} \int_{-1}^1 dy y^n q(y, \mu). \end{aligned} \quad (18)$$

It should be noted that the only approximation we have made so far is ignoring the higher-twist effects that are suppressed by small z^2 and the large momentum P^2 of the nucleon. In the limit of $P^2 \gg M^2, \Lambda_{\text{QCD}}^2$, we have $P^0 \sim P^z$, so even if $\mu_0 = 0$ in Eq. (14), the leading approximation of $\bar{O}_{\gamma^*}(z)$ is still given by the twist-2 contributions in Eq. (18), just with modified coefficients C_n .

Based on the OPE results in Eq. (18), we can derive a factorization formula for the Euclidean spatial correlator. First of all, let us define a function $\mathcal{C}(\alpha, \mu^2 z^2)$:

$$\mathcal{C}(\alpha, \mu^2 z^2) = \int \frac{d\zeta}{2\pi} e^{i\alpha\zeta} \sum_n C_n(\mu^2 z^2) \frac{(-i\zeta)^n}{n!}. \quad (19)$$

From Eq. (18) and the renormalized analog of the Fourier-transform relation in Eq. (9) $\mathcal{C}(\alpha, \mu^2 z^2)$ corresponds to a pseudo-PDF in the special case where $a_{n+1}(\mu) = 1$. The analysis of Refs. [40, 41] implies that the support of $\mathcal{C}(\alpha, \mu^2 z^2)$ is $-1 \leq \alpha \leq 1$. Noting that

$$\int d\alpha e^{-i\alpha(y\zeta)} \mathcal{C}(\alpha, \mu^2 z^2) = \sum_n C_n(\mu^2 z^2) \frac{(-i\zeta y)^n}{n!}, \quad (20)$$

we find from Eq. (18) that

$$\bar{Q}(\zeta, \mu^2 z^2) = \int_{-1}^1 dy \int_{-1}^1 d\alpha e^{-i\alpha(y\zeta)} \mathcal{C}(\alpha, \mu^2 z^2) q(y, \mu). \quad (21)$$

Finally, using the inverse transform of the renormalized analog of Eq. (11),

$$Q(\zeta, \mu) = \int_{-1}^1 dy e^{-iy\zeta} q(y, \mu). \quad (22)$$

we obtain

$$\bar{Q}(\zeta, \mu^2 z^2) = \int_{-1}^1 d\alpha \mathcal{C}(\alpha, \mu^2 z^2) Q(\alpha\zeta, \mu) + \mathcal{O}(z^2 \Lambda_{\text{QCD}}^2). \quad (23)$$

The result in Eq. (23) is the factorization formula for the lattice calculable spatial correlator $\bar{Q}(\zeta, \mu^2 z^2)$ which expresses it in terms of the light-cone correlation $Q(\zeta, \mu)$ that defines the PDF. It has the same structure as the factorization formula for the spatial correlator used for the calculation of the pion distribution amplitude in Ref. [39, 60].

B. Factorization for the quasi-PDF

The renormalized quasi-PDF is defined as a Fourier transform of the renormalized spatial correlator,

$$\bar{q}\left(x, \frac{\mu}{P_z}\right) = \int_{-\infty}^{\infty} \frac{d\zeta}{2\pi} e^{i\zeta x} \bar{Q}\left(\zeta, \frac{\mu^2 \zeta^2}{P_z^2}\right). \quad (24)$$

Note that we could use either $\bar{Q}_{\zeta'}$ or $\bar{Q}_{\zeta''}$ here. Using the result for the spatial correlator in Eq. (18) this gives

$$\begin{aligned} \bar{q}\left(x, \frac{\mu}{P_z}\right) &= \int_{-1}^1 dy \left[\int_{-\infty}^{\infty} \frac{d\zeta}{2\pi} e^{i\zeta x} \sum_{n=0}^{\infty} C_n \left(\frac{\mu^2 \zeta^2}{P_z^2} \right) \frac{(-i\zeta)^n}{n!} y^n \right] q(y, \mu) \\ &= \int_{-1}^1 \frac{dy}{|y|} \left[\int_{-\infty}^{\infty} \frac{d\zeta}{2\pi} e^{i\zeta x} \sum_{n=0}^{\infty} C_n \left(\frac{\mu^2 \zeta^2}{(yP_z)^2} \right) \frac{(-i\zeta)^n}{n!} \right] q(y, \mu). \end{aligned} \quad (25)$$

Already, one can see that the matching kernel is a function of x/y and $\mu/(|y|P_z)$. We define the kernel as

$$C\left(\frac{x}{y}, \frac{\mu}{|y|P_z}\right) = \int_{-\infty}^{\infty} \frac{d\zeta}{2\pi} e^{i\zeta x} \sum_{n=0}^{\infty} C_n \left(\frac{\mu^2 \zeta^2}{(yP_z)^2} \right) \frac{(-i\zeta)^n}{n!}, \quad (26)$$

and then Eq. (25) can be rewritten as

$$\bar{q}\left(x, \frac{\mu}{P_z}\right) = \int_{-1}^1 \frac{dy}{|y|} C\left(\frac{x}{y}, \frac{\mu}{|y|P_z}\right) q(y, \mu), \quad (27)$$

which is the $\overline{\text{MS}}$ factorization formula for the quasi-PDF. This result shows that the factorization formula in Eq. (6) must have $p^\perp = |y|P_z$ for the quasi-PDF in the $\overline{\text{MS}}$ scheme. We will show that this remains true for any quasi-PDF renormalization scheme in Sec. IV. This differs from the choice $p^\perp = P_z$ which had been conjectured and used in the early papers on the quasi-PDF [21, 22, 24]. Physically the correct result in Eq. (27) can be understood as the fact that the matching coefficient is only sensitive to the perturbative partonic dynamics, and hence it is the magnitude of the partonic momentum $|y|P_z$ which appears, rather than the hadronic momentum P_z .

Taking the moment of the quasi-PDF using Eq. (24) gives

$$\begin{aligned} \int_0^1 dx x^n \bar{q}\left(x, \frac{\mu}{P_z}\right) &= \left(i \frac{d}{d\zeta} \right)^n \bar{Q}\left(\zeta, \frac{\mu^2 \zeta^2}{P_z^2}\right) \Big|_{\zeta \rightarrow 0} \\ &= \sum_{n'} \left(i \frac{d}{d\zeta} \right)^n \left[C_{n'} \left(\frac{\mu^2 \zeta^2}{P_z^2} \right) \frac{(-i\zeta)^{n'}}{n'!} \right] \Big|_{\zeta \rightarrow 0} a_{n'+1}(\mu). \end{aligned} \quad (28)$$

Since the $C_{n'}$ coefficients have $\ln(\zeta^2)$ dependence, the derivative for $n' = n$ will always have a logarithmic singularity as $\zeta \rightarrow 0$, and there will be even more singular terms for $n' < n$. This explains why the short distance Wilson coefficient causes the moments not to exist for the quasi-PDF.

C. Factorization for the pseudo-PDF

The renormalized pseudo-PDF is the Fourier transform of the renormalized spatial correlator

$$\mathcal{P}(x, \mu^2 z^2) = \int_{-\infty}^{\infty} \frac{d\zeta}{2\pi} e^{i\zeta x} \bar{Q}(\zeta, \mu^2 z^2). \quad (29)$$

Since both the pseudo-PDF and spatial correlator are multiplicatively renormalized in a ζ -independent manner, this follows immediately from Eq. (9). If we take Eq. (23) and Fourier transform the spatial correlator $\bar{Q}(\zeta, \mu^2 z^2)$ into the pseudo-PDF, and light-cone correlation $Q(\alpha\zeta, \mu)$ into the PDF, then we immediately obtain the factorization formula for the pseudo-PDF,

$$\begin{aligned} \mathcal{P}(x, z^2 \mu^2) &= \int_{|x|}^1 \frac{dy}{|y|} C\left(\frac{x}{y}, \mu^2 z^2\right) q(y, \mu) \\ &\quad + \int_{-1}^{-|x|} \frac{dy}{|y|} C\left(\frac{x}{y}, \mu^2 z^2\right) q(y, \mu) \\ &\quad + \mathcal{O}(z^2 \Lambda_{\text{QCD}}^2), \end{aligned} \quad (30)$$

which is the small z^2 factorization formula in Eq. (12). The upper and lower limits of the integrals in Eq. (30) follow immediately from the support $-1 \leq \alpha \leq 1$ of the matching coefficient $\mathcal{C}(\alpha, z^2 \mu^2)$, and we recall that we also have $-1 \leq x \leq 1$ for the pseudo-PDF on the LHS.

Since the range of x is bounded for the pseudo-PDF the terms in the series expansion of the exponential in Eq. (29) exist,

$$\bar{Q}(\zeta, \mu^2 z^2) = \sum_{n=0}^{\infty} \int_{-1}^1 dx \frac{(-i\zeta)^n x^n}{n!} \mathcal{P}(x, \mu^2 z^2). \quad (31)$$

Comparing with Eq. (18) this implies that the moments of the pseudo-PDF are given by

$$\int_{-1}^1 dx x^n \mathcal{P}(x, \mu^2 z^2) = C_n(\mu^2 z^2) a_{n+1}(\mu). \quad (32)$$

So far we have proven the large P_z factorization of the quasi-PDF and small z^2 factorization of the spatial correlation and pseudo-PDFs. After deriving one factorization, it immediately leads to the others, since they are just different representations of the same spatial correlator. Indeed, we see that the quasi-PDF and pseudo-PDF are related at leading power by their definitions:

$$\bar{q}\left(x, \frac{\mu}{P_z}\right) = \int \frac{d\zeta}{2\pi} e^{i\zeta x} \int_{-1}^1 dy e^{-iy\zeta} \mathcal{P}\left(y, \frac{\mu^2 \zeta^2}{P_z^2}\right), \quad (33)$$

where we have used $z = \zeta/P_z$. Based on Eq. (19) and Eq. (26), the Wilson coefficients in their factorization theorems also maintain the same relationship,

$$C\left(\zeta, \frac{\mu}{|y|P_z}\right) = \int \frac{d\zeta'}{2\pi} e^{i\zeta'\zeta} \int_{-1}^1 d\alpha e^{-i\alpha\zeta'} C\left(\alpha, \frac{\mu^2 \zeta'^2}{(yP_z)^2}\right). \quad (34)$$

For the relations in Eqs. (33) and (34) the same choice of $\Gamma = \gamma^0$ or γ^z should be used in the quasi- and pseudo-PDFs, or their corresponding coefficients.

In summary, there is a unique factorization formula that matches the quasi-PDF, spatial correlator and pseudo-PDF to the PDF. Since their factorizations into the PDF all require small distances and have large nucleon momentum, the setup for their lattice calculations must also be the same. Therefore, the LaMET and pseudo distribution approaches are in principle equivalent to each other, and they differ perhaps only by effects related to their implementation on the lattice.

In Ref. [39] it was speculated that one can study a ratio function

$$\hat{Q}(\zeta, z^2, a^{-1})/\hat{Q}(0, z^2, a^{-1}) \quad (35)$$

on a lattice with spacing a , and the $O(z^2)$ corrections may cancel approximately. This idea was tested in Ref. [42] in lattice QCD, and the results show that the ratio evolves slowly in z^2 at small values. It is then interesting to consider what type of non-perturbative information can be extracted from this ratio.

This question can be answered using the small z^2 factorization for the spatial correlator. According to Eq. (18),

$$\hat{Q}(0, \mu^2 z^2) = C_0(\mu^2 z^2) + O(z^2 \Lambda_{\text{QCD}}^2), \quad (36)$$

where in the $\overline{\text{MS}}$ scheme to one-loop

$$C_0(\mu^2 z^2) = 1 + \frac{\alpha_s C_F}{2\pi} \left[\frac{3}{2} \ln(\mu^2 z^2 e^{2\gamma_E}/4) + \frac{5}{2} \right], \quad (37)$$

which was also derived recently in [61]. Then the ratio becomes

$$\frac{\hat{Q}(\zeta, \mu^2 z^2)}{\hat{Q}(0, \mu^2 z^2)} = \sum_n \frac{C_n(\mu^2 z^2)}{\bar{C}_0(\mu^2 z^2)} \frac{(-\zeta)^n}{n!} a_{n+1}(\mu) + O(z^2 \Lambda_{\text{QCD}}^2). \quad (38)$$

Using Eq. (32) and our $\overline{\text{MS}}$ one-loop perturbative pseudo-

PDF result in Eq. (54) below we find for $\Gamma = \gamma^0$ that

$$C_n(\mu^2 z^2) = 1 + \frac{\alpha_s C_F}{2\pi} \left[\left(\frac{3+2n}{2+3n+n^2} + 2H_n \right) \ln \frac{\mu^2 z^2 e^{2\gamma_E}}{4} + \frac{5+2n}{2+3n+n^2} + 2(1-H_n)H_n - 2H_n^{(2)} \right], \quad (39)$$

where the Harmonic numbers are $H_n = \sum_{i=1}^n 1/i$ and $H_n^{(2)} = \sum_{i=1}^n 1/i^2$. For the case $\Gamma = \gamma^z$ we have $C_n^z(\mu^2 z^2) = C_n(\mu^2 z^2) + \Delta C_n^z(\mu^2 z^2)$ with

$$\Delta C_n^z(\mu^2 z^2) = \frac{\alpha_s C_F}{2\pi} \frac{2}{2+3n+n^2}, \quad (40)$$

which also modifies Eq. (37) for $n = 0$. At small z^2 where the perturbative expansion with $\mu \simeq 1/|z|$ is valid, the ratio in Eq. (38) has a weak logarithmic dependence on $|z|$, which is consistent with the lattice findings in Refs. [42, 43]. The weak dependence on $|z|$ can be quantitatively described by an evolution equation in $\ln z^2$ [39, 62]. According to our OPE analysis, here $\hat{Q}(0, \mu^2 z^2)$ only serves as an overall normalization factor which is contaminated by higher-twist corrections, and the $\ln z^2$ evolution can be put in accurate terms with the factorization formula in Eq. (23) that enables us extract the PDF from the ratio function. The same point was demonstrated by work done very recently in [61], which appeared simultaneously with our paper.

III. EQUIVALENCE AT ONE-LOOP ORDER

As has been proven in Sec. II, the quasi-PDF and pseudo-PDF as well as their matching coefficients are related by a simple Fourier transform in Eq. (33) and Eq. (34). This relation is valid to all orders in perturbation theory. In this section we check the relations in Eq. (33) and Eq. (34) at one-loop order. We choose $\Gamma = \gamma^0$ for our main presentation, but also quote final results for the case $\Gamma = \gamma^z$.

In the Feynman gauge, we calculate the quark matrix elements of the unpolarized iso-vector quasi-PDF, pseudo-PDF, and light-cone PDF at one-loop order in dimensional regularization with $d = 4 - 2\epsilon$. The external quark state is chosen to be on-shell and massless, and we regularize the UV and collinear divergences by $1/\epsilon_{\text{UV}}$ ($\epsilon_{\text{UV}} > 0$) and $1/\epsilon_{\text{IR}}$ ($\epsilon_{\text{IR}} < 0$) respectively. The one-loop order Feynman diagrams are shown in Fig. 1.

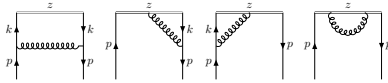


FIG. 1. One-loop Feynman diagrams for the quasi-PDF, spatial correlator and pseudo-PDFs. The first one is named “vertex”, the second and third ones are named “sail”, and the last one “tadpole”. The standard quark self energy wavefunction is also included.

In an on-shell quark state with momentum $p^\mu = (p^0 = p^2, 0, 0, p^2)$, for $\Gamma = \gamma^0$, each diagram gives

$$\begin{aligned}\tilde{Q}_{\text{vertex}}^{(1)}(\zeta, z^2, \epsilon) &= \frac{\mu^{2\epsilon}}{2p^0} \bar{u}(p) \int \frac{d^d k}{(2\pi)^d} (-igT^a \gamma^\mu) \frac{i}{k} \gamma^0 \frac{i}{k} (-igT^a \gamma^\mu) \frac{-ig_{\mu\nu}}{(p-k)^2} u(p) e^{-ik^* z}, \\ \tilde{Q}_{\text{sail}}^{(1)}(\zeta, z^2, \epsilon) &= \frac{\mu^{2\epsilon}}{2p^0} \bar{u}(p) \int \frac{d^d k}{(2\pi)^d} (igT^a \gamma^0) \frac{1}{i(p^2 - k^2)} \left(e^{-ip^* z} - e^{-ik^* z} \right) \delta^{\mu z} \frac{i}{k} (-igT^a \gamma^\mu) \frac{-ig_{\mu\nu}}{(p-k)^2} u(p) \\ &\quad + \frac{\mu^{2\epsilon}}{2p^0} \bar{u}(p) \int \frac{d^d k}{(2\pi)^d} (-igT^a \gamma^\mu) \frac{i}{k} (igT^a \gamma^0) \frac{1}{i(p^2 - k^2)} \left(e^{-ip^* z} - e^{-ik^* z} \right) \delta^{\mu z} \frac{-ig_{\mu\nu}}{(p-k)^2} u(p), \\ \tilde{Q}_{\text{tadpole}}^{(1)}(\zeta, z^2, \epsilon) &= \frac{\mu^{2\epsilon}}{2p^0} \bar{u}(p) \int \frac{d^d k}{(2\pi)^d} (-g^2) C_F \gamma^0 \delta^{\mu z} \delta^{\nu z} \left(\frac{e^{-ip^* z} - e^{-ik^* z}}{(p^2 - k^2)^2} - \frac{ze^{-ip^* z}}{i(p^2 - k^2)} \right) \frac{-ig_{\mu\nu}}{(p-k)^2} u(p),\end{aligned}\quad (41)$$

where $z = e^{\gamma_E/(4\pi)}$ is included to implement μ in the $\overline{\text{MS}}$ scheme, $C_F = 4/3$, and T^a is the SU(3) color matrix in the fundamental representation. The second term in the brackets in the last line, which is proportional z , does not contribute to the loop integral as it is odd under the exchange of $p^2 - k^2 \rightarrow -(p^2 - k^2)$. The quark self-energy correction is $\tilde{Q}_{\text{self}}^{(1)}(\zeta, z^2, \epsilon) = \delta Z_\psi \tilde{Q}^{(0)}(\zeta, z^2)$ with the tree level matrix element $\tilde{Q}^{(0)}(\zeta, z^2) = e^{-i\zeta}$ and on-shell renormalization constant δZ_ψ ,

$$\delta Z_\psi = \frac{\alpha_s C_F}{2\pi} \left(-\frac{1}{2} \right) \left(\frac{1}{\epsilon_{\text{uv}}} - \frac{1}{\epsilon_{\text{in}}} \right). \quad (42)$$

After carrying out the loop integrals in Eq. (41) according to the method in Ref. [45], we obtain

$$\begin{aligned}\tilde{Q}_{\text{vertex}}^{(1)}(\zeta, z^2, \epsilon) &= \frac{\alpha_s C_F}{2\pi} e^{\gamma_E} \int_0^1 du (1-u) e^{-i\kappa\zeta} \Gamma(-\epsilon) 4^{-\epsilon} (\mu|z|)^{2\epsilon} \\ &= \frac{\alpha_s C_F}{2\pi} e^{\gamma_E} \left(\frac{\mu|z|}{2} \right)^{2\epsilon} \frac{(-1)\Gamma(2-\epsilon)}{\epsilon_{\text{in}}} \frac{1 - i\zeta - e^{-i\zeta}}{\zeta^2}, \\ \tilde{Q}_{\text{sail}}^{(1)}(\zeta, z^2, \epsilon) &= \frac{\alpha_s C_F}{2\pi} e^{\gamma_E} \left[(i\zeta) \int_0^1 du \int_0^1 dt (2-u) e^{-i(1-ut)\zeta} \Gamma(-\epsilon) 4^{-\epsilon} (t^2 z^2 \mu^2)^\epsilon \right. \\ &\quad \left. - \int_0^1 du e^{-i\kappa\zeta} \Gamma(-\epsilon) 4^{-\epsilon} (z^2 \mu^2)^\epsilon + \left(\frac{1}{\epsilon_{\text{uv}}} - \frac{1}{\epsilon_{\text{in}}} \right) e^{-i\zeta} \right] \\ &= \frac{\alpha_s C_F}{2\pi} e^{\gamma_E} \left(\frac{\mu|z|}{2} \right)^{2\epsilon} \left\{ \frac{-\Gamma(1-\epsilon)}{\epsilon_{\text{in}}} \frac{2(1-\epsilon)}{1-2\epsilon} \left[\frac{1 - e^{-i\zeta}}{-i\zeta} + e^{-i\zeta} (-i\zeta)^{-2\epsilon} (\Gamma(2\epsilon) - \Gamma(2\epsilon - i\zeta)) \right] \right. \\ &\quad \left. + \frac{\Gamma(1-\epsilon)}{\epsilon_{\text{in}}} \frac{e^{-i\zeta}}{\epsilon} + \left(\frac{1}{\epsilon_{\text{uv}}} - \frac{1}{\epsilon_{\text{in}}} \right) e^{-i\zeta} \right\}, \\ \tilde{Q}_{\text{tadpole}}^{(1)}(\zeta, z^2, \epsilon) &= \frac{\alpha_s C_F}{2\pi} e^{\gamma_E} \left(\frac{\mu|z|}{2} \right)^{2\epsilon} \frac{\Gamma(1-\epsilon)}{\epsilon_{\text{uv}}(1-2\epsilon)} e^{-i\zeta}.\end{aligned}\quad (43)$$

For simplicity we have left out the tree level multiplicative spinor factor $\bar{u}\gamma^0 u$ when quoting one-loop results in Eq. (43), and will continue to do so for the spatial correlator, quasi-PDF, and pseudo-PDF results quoted below. Since $\alpha_s^{\text{bare}} = \alpha_s(\mu)\mu^{2\epsilon} Z_g^2 = \alpha_s(\mu)\mu^{2\epsilon} + \mathcal{O}(\alpha_s^2)$ is μ -independent we do not include μ as an argument in the bare

functions. In the final result for each term we have also specified whether $1/\epsilon$ factors (that remain after expanding about $\epsilon \rightarrow 0$) are IR or UV divergences. Combined with the wavefunction correction, the bare spatial correlator $\tilde{Q}^{(1)}(\zeta, z^2, \epsilon)$ is

$$\tilde{Q}^{(1)}(\zeta, z^2, \epsilon) = \frac{\alpha_s C_F}{2\pi} e^{\gamma_E} \left(\frac{|\mu|z|}{2} \right)^{2\epsilon} \left\{ \frac{3}{2} \left(\frac{1}{\epsilon_{uv}} - \frac{1}{\epsilon_m} \right) e^{-i\zeta} + \frac{(-1)\Gamma(2-\epsilon)}{\epsilon_m} \left[\frac{(1-i\zeta - e^{-i\zeta})}{\zeta^2} \right. \right. \\ \left. \left. + \frac{2}{1-2\epsilon} \left(\frac{1-e^{-i\zeta}}{-i\zeta} + e^{-i\zeta}(-i\zeta)^{-2\epsilon} (\Gamma(2\epsilon) - \Gamma(2\epsilon, -i\zeta)) - \frac{e^{-i\zeta}}{2\epsilon} \right) \right] \right\}. \quad (44)$$

Note that as $\epsilon \rightarrow 0$ the terms in the innermost curly brackets have no $1/\epsilon$ term. Also we can verify that in the local limit of the operator that the bare one-loop correction vanishes as expected by conservation of the vector current:

$$\lim_{z \rightarrow 0} \tilde{Q}^{(1)}(zP^2, z^2, \epsilon) = \lim_{z \rightarrow 0} \frac{\alpha_s C_F}{2\pi} e^{\gamma_E} \left(\frac{|\mu|z|}{2} \right)^{2\epsilon} \left\{ \frac{3}{2} \left(\frac{1}{\epsilon_{uv}} - \frac{1}{\epsilon_m} \right) + \frac{3}{2\epsilon_m} \right\} = 0, \quad (45)$$

where we note that it is important that the $1/\epsilon_m$ terms cancel since the assumption $\epsilon > 0$ is only valid for the $1/\epsilon_{uv}$ term. The corresponding bare pseudo-PDF is

$$\mathcal{P}^{(1)}(x, z^2, \epsilon) = \frac{\alpha_s C_F}{2\pi} \left[-1 + x + \frac{2}{(1-2\epsilon)} - \left(\frac{2}{(1-2\epsilon)} \frac{1}{(1-x)^{1-2\epsilon}} \right)_{+(1)}^{[0,1]} \right] e^{\gamma_E} \left(\frac{|\mu|z|}{2} \right)^{2\epsilon} \frac{\Gamma(2-\epsilon)}{\epsilon_m} \theta(x) \theta(1-x) \\ + \frac{\alpha_s C_F}{2\pi} e^{\gamma_E} \left(\frac{|\mu|z|}{2} \right)^{2\epsilon} \frac{3}{2} \left(\frac{1}{\epsilon_{uv}} - \frac{1}{\epsilon_m} \right) \delta(1-x). \quad (46)$$

Since we will encounter plus functions over different domains below, we define a plus function at $x = x_0$ within a given domain D so that

$$\int_D dx [g(x)]_{+(x_0)}^D h(x) = \int_D dx g(x) [h(x) - h(x_0)]. \quad (47)$$

(See App. C for more details.) It is straightforward to confirm that the bare pseudo-PDF satisfies the local vector current conservation, $\lim_{z \rightarrow 0} \int dx \mathcal{P}^{(1)}(x, z^2 \mu^2, \epsilon) = 0$, with the same cancellation as in Eq. (45).

Now, according to the relations between the quasi-PDF and the spatial correlator or pseudo-PDFs in Eqs. (24,33), we can do a Fourier or double Fourier transform of the results in Eqs. (44,46) to get the quasi-PDF. Despite its straightforwardness, the Fourier transform is subtle and the details are provided in App. A. Here we simply quote the result for the bare quasi-PDF,

$$\bar{q}^{(1)}(x, p^2, \epsilon) = \frac{\alpha_s C_F}{2\pi} \left\{ \frac{3}{2} \left(\frac{1}{\epsilon_{uv}} - \frac{1}{\epsilon_m} \right) \delta(1-x) + \frac{\Gamma(\epsilon + \frac{1}{2}) e^{\gamma_E} \mu^{2\epsilon}}{\sqrt{\pi}} \frac{1-\epsilon}{p_{\perp}^2 \epsilon_m (1-2\epsilon)} \right. \\ \left. \times \left[|x|^{-1-2\epsilon} \left(1+x + \frac{x}{2} (x-1+2\epsilon) \right) - |1-x|^{-1-2\epsilon} \left(x + \frac{1}{2} (1-x)^2 \right) + I_3(x) \right] \right\}, \quad (48)$$

where

$$I_3(x) = \theta(x-1) \left(\frac{x^{-1-2\epsilon}}{x-1} \right)_{+(1)}^{[1,\infty]} - \theta(x) \theta(1-x) \left(\frac{x^{-1-2\epsilon}}{1-x} \right)_{+(1)}^{[0,1]} - \delta(1-x) \pi \csc(2\pi\epsilon) + \theta(-x) \frac{|x|^{-1-2\epsilon}}{x-1}. \quad (49)$$

After some algebra one can confirm that the bare quasi-PDF satisfies local vector current conservation, with $\int dx \bar{q}^{(1)}(x, p^2, \epsilon) = 0$. To verify this result one must carefully separate out $1/\epsilon_{uv}$ factors arising from requiring $\epsilon > 0$ to obtain convergence at $x = \pm\infty$, and $1/\epsilon_m$ factors that arise from requiring $\epsilon < 0$ to obtain convergence at $x = 1$.

An alternate method of obtaining the quasi-PDF is to directly calculate it from the Feynman diagrams by first Fourier transforming z into xp^2 . As a result, the factors $(e^{-ip^2 z} - e^{-ik^2 z})$ are transformed into $[\delta(p^2 - xp^2) - \delta(k^2 - xp^2)]$, and all the loop integrals reduce to $(d-1)$ -dimensional ones. This is the procedure for the matching calculations of the quasi-PDF used in Refs. [24, 26, 27], and is distinct from doing the Fourier transformation after fully carrying out the integrals as in Eqs. (43-48). As a cross-check we have confirmed in App. B that we obtain the exact same bare quasi-PDF in Eq. (48) from both procedures.

Now we consider the ϵ expansion to obtain $\overline{\text{MS}}$ renormalized results for the spatial correlator, pseudo-PDF, and quasi-PDF. Expanding the spatial correlator in ϵ we obtain

$$\tilde{Q}^{(1)}(\zeta, z^2, \epsilon) = \delta\tilde{Q}^{(1)}(\zeta, z^2, \mu, \epsilon_{uv}) + \tilde{Q}^{(1)}(\zeta, z^2, \mu, \epsilon_m) + \mathcal{O}(\epsilon) \quad (50)$$

with the $\overline{\text{MS}}$ counterterm and renormalized spatial correlator given by

$$\begin{aligned} \delta\tilde{Q}^{(1)}(\zeta, z^2, \mu, \epsilon_{uv}) &= \frac{\alpha_s C_F}{2\pi} e^{-i\zeta} \frac{3}{2} \frac{1}{\epsilon_{uv}}, \\ \tilde{Q}^{(1)}(\zeta, z^2, \mu, \epsilon_m) &= \frac{\alpha_s C_F}{2\pi} \left\{ \frac{3}{2} \left(\ln \frac{\mu^2 z^2 e^{2\gamma_E}}{4} + 1 \right) e^{-i\zeta} + \left(-\frac{1}{\epsilon_m} - \ln \frac{z^2 \mu^2 e^{2\gamma_E}}{4} - 1 \right) h(\zeta) + \frac{2(1-i\zeta-e^{-i\zeta})}{\zeta^2} \right. \\ &\quad \left. + 4i\zeta e^{-i\zeta} {}_3F_3(1, 1, 1, 2, 2, 2, i\zeta) \right\}. \end{aligned} \quad (51)$$

Here ${}_3F_3$ is a hypergeometric function and the Fourier transform of $[(1+x^2)/(1-x)]_{+(1)}^{[0,1]}$ gives the function

$$h(\zeta) = \frac{3}{2} e^{-i\zeta} + \frac{1+i\zeta-e^{-i\zeta}-2i\zeta e^{-i\zeta}}{\zeta^2} - 2e^{-i\zeta} [\Gamma(0, -i\zeta) + \gamma_E + \ln(-i\zeta)]. \quad (52)$$

For the position space PDF we have

$$Q^{(1)}(\zeta, \mu, \epsilon_m) = -\frac{\alpha_s C_F}{2\pi} \frac{1}{\epsilon_m} h(\zeta). \quad (53)$$

Next we expand the bare pseudo-PDF from Eq. (46) in ϵ to obtain the $\overline{\text{MS}}$ counterterm and renormalized pseudo-PDF as $\mathcal{P}^{(1)}(x, z^2, \epsilon) = \delta\mathcal{P}^{(1)}(x, z^2\mu^2, \epsilon_{uv}) + \mathcal{P}^{(1)}(x, z^2\mu^2, \epsilon_m) + \mathcal{O}(\epsilon)$ with

$$\begin{aligned} \delta\mathcal{P}^{(1)}(x, z^2\mu^2, \epsilon_{uv}) &= \frac{\alpha_s C_F}{2\pi} \frac{3}{2\epsilon_{uv}} \delta(1-x), \\ \mathcal{P}^{(1)}(x, z^2\mu^2, \epsilon_m) &= \frac{\alpha_s C_F}{2\pi} \left\{ \left(\frac{1+x^2}{1-x} \right)_{+(1)}^{[0,1]} \left[-\left(\frac{1}{\epsilon_m} + \ln \frac{e^{2\gamma_E}}{4} \right) - \ln(z^2\mu^2) - 1 \right] - \left(\frac{4\ln(1-x)}{1-x} \right)_{+(1)}^{[0,1]} + 2(1-x) \right\} \theta(x)\theta(1-x) \\ &\quad + \frac{\alpha_s C_F}{2\pi} \left[\frac{3}{2} \ln(z^2\mu^2) + \frac{3}{2} \ln \frac{e^{2\gamma_E}}{4} + \frac{3}{2} \right] \delta(1-x). \end{aligned} \quad (54)$$

Note that the renormalized $\overline{\text{MS}}$ pseudo-PDF depends explicitly on μ^2 , and satisfies the relation to the renormalized $\overline{\text{MS}}$ spatial correlator given in Eq. (29). It is also interesting to note that having expanded in ϵ , local vector current conservation is no longer satisfied by the limit of the renormalized $\overline{\text{MS}}$ pseudo-PDF, since

$$\lim_{z \rightarrow 0} \int dx \mathcal{P}^{(1)}(x, z^2\mu^2, \epsilon_m) \simeq (3\alpha_s C_F/4\pi) \lim_{z \rightarrow 0} \ln(z^2\mu^2) \quad (55)$$

gives a divergent result. The same divergence is present in the one-loop $\overline{\text{MS}}$ renormalized spatial correlator in Eq. (51). Although this is the case in $\overline{\text{MS}}$, it does not need to be the case in other renormalization schemes.

For the quasi-PDF there are two methods that we can consider for the renormalized calculation, either expanding the bare result in Eq. (48) and renormalizing in (x, p^2) space, or following our preferred definition in Eq. (24) and Fourier transforming the renormalized spatial correlator in Eq. (51). Although these two approaches will lead to the same final result for C for practical applications, there is a subtle difference that we will explain.

First consider the renormalization of the quasi-PDF done in (x, p^2) space. Expanding Eq. (48) in ϵ , and writing $\tilde{q}^{(1)}(x, p^2, \epsilon) = \delta\tilde{q}^{(1)}(x, \mu/|p^2|, \epsilon_{uv}) + \tilde{q}^{(1)}(x, \mu/|p^2|, \epsilon_m) + \mathcal{O}(\epsilon)$ allows us to identify the $\overline{\text{MS}}$ counterterm and renomal-

ized quasi-PDF as

$$\begin{aligned} \delta\tilde{q}^{(1)}(x, \mu/p^2, \epsilon_{\text{uv}}) &= \frac{\alpha_s C_F}{2\pi} \frac{3}{2\epsilon_{\text{uv}}} \left[\delta(1-x) - \frac{1}{2} \frac{1}{x^2} \delta^+\left(\frac{1}{x}\right) - \frac{1}{2} \frac{1}{(1-x)^2} \delta^+\left(\frac{1}{1-x}\right) \right], \\ \tilde{q}^{(1)}(x, \mu/p^2, \epsilon_{\text{in}}) &= \frac{\alpha_s C_F}{2\pi} \left\{ \begin{aligned} &\left(\frac{1+x^2}{1-x} \ln \frac{x}{x-1} + 1 + \frac{3}{2x} \right)_{+(1)}^{[1,\infty]} - \left(\frac{3}{2x} \right)_{+(\infty)}^{[1,\infty]} & x > 1 \\ &\frac{1+x^2}{1-x} \left[-\frac{1}{\epsilon_{\text{in}}} - \ln \frac{\mu^2}{4p_z^2} + \ln(x(1-x)) \right] - \frac{x(1+x)}{1-x} \Big|_{+(1)}^{[0,1]} & 0 < x < 1 \\ &\left(-\frac{1+x^2}{1-x} \ln \frac{-x}{1-x} - 1 + \frac{3}{2(1-x)} \right)_{+(1)}^{[-\infty,0]} - \left(\frac{3}{2(1-x)} \right)_{+(-\infty)}^{[-\infty,0]} & x < 0 \end{aligned} \right. \\ &+ \frac{\alpha_s C_F}{2\pi} \left[\delta(1-x) - \frac{1}{2} \frac{1}{x^2} \delta^+\left(\frac{1}{x}\right) - \frac{1}{2} \frac{1}{(1-x)^2} \delta^+\left(\frac{1}{1-x}\right) \right] \left(\frac{3}{2} \ln \frac{\mu^2}{4p_z^2} + \frac{5}{2} \right). \end{aligned} \quad (56)$$

The details of working out the ϵ expansion of Eq. (48) are provided in App. C, including definitions of the plus functions and δ -functions at $x_0 = \pm\infty$ that appear in the result quoted here. The $\overline{\text{MS}}$ quasi-PDF obtained in Eq. (56) still satisfies vector current conservation

$$\int dx \tilde{q}^{(1)}(x, \mu/p^2, \epsilon_{\text{in}}) = 0. \quad (57)$$

This is obviously the case for the plus function terms which individually integrate to zero, and is also true for the combination of δ -functions which appears in Eq. (56).

The renormalized $\overline{\text{MS}}$ quasi-PDF in Eq. (56) differs slightly from that obtained using our definition in Eq. (24). Using Eq. (24) and the renormalized spatial correlator in Eq. (51) we instead obtain

$$\begin{aligned} \delta\tilde{q}^{(1)}(x, \mu/p^2, \epsilon_{\text{uv}}) &= \frac{\alpha_s C_F}{2\pi} \frac{3}{2\epsilon_{\text{uv}}} \delta(1-x), \\ \tilde{q}^{(1)}(x, \mu/p^2, \epsilon_{\text{in}}) &= \frac{\alpha_s C_F}{2\pi} \left\{ \begin{aligned} &\left(\frac{1+x^2}{1-x} \ln \frac{x}{x-1} + 1 + \frac{3}{2x} \right)_{+(1)}^{[1,\infty]} - \left(\frac{3}{2x} \right)_{+(\infty)}^{[1,\infty]} & x > 1 \\ &\frac{1+x^2}{1-x} \left[-\frac{1}{\epsilon_{\text{in}}} - \ln \frac{\mu^2}{4p_z^2} + \ln(x(1-x)) \right] - \frac{x(1+x)}{1-x} \Big|_{+(1)}^{[0,1]} & 0 < x < 1 \\ &\left(-\frac{1+x^2}{1-x} \ln \frac{-x}{1-x} - 1 + \frac{3}{2(1-x)} \right)_{+(1)}^{[-\infty,0]} - \left(\frac{3}{2(1-x)} \right)_{+(-\infty)}^{[-\infty,0]} & x < 0 \end{aligned} \right. \\ &+ \frac{\alpha_s C_F}{2\pi} \left[\delta(1-x) \left(\frac{3}{2} \ln \frac{\mu^2}{4p_z^2} + \frac{5}{2} \right) + \frac{3}{2} \gamma_E \left(\frac{1}{(x-1)^2} \delta^+\left(\frac{1}{x-1}\right) + \frac{1}{(1-x)^2} \delta^+\left(\frac{1}{1-x}\right) \right) \right]. \end{aligned} \quad (58)$$

To carry out this calculation we defined the Fourier transformation of the singular function $\ln(\zeta^2)$ as

$$\begin{aligned} \int \frac{d\zeta}{2\pi} e^{i\omega\zeta} \ln \zeta^2 &= \left[\frac{d}{d\eta} \int \frac{d\zeta}{2\pi} e^{i\omega\zeta} (\zeta^2)^\eta \right] \Big|_{\eta=0} = \left[\frac{d}{d\eta} \frac{4^{\eta}}{\Gamma(-\eta)} \frac{\Gamma(\eta+1/2)}{\sqrt{\pi}} \frac{\theta(x) + \theta(-x)}{|x|^{1+2\eta}} \right] \Big|_{\eta=0} \\ &= \gamma_E \left[\left(-\delta(x) + \frac{1}{x^2} \delta^+\left(\frac{1}{x}\right) \right) + \left(-\delta(x) + \frac{1}{x^2} \delta^+\left(\frac{1}{-x}\right) \right) \right] \\ &\quad - \left[\left(\frac{1}{x} \right)_{+(0)}^{[0,1]} + \left(\frac{1}{x} \right)_{+(\infty)}^{[1,\infty]} \right] \theta(x) - \left[\left(\frac{1}{-x} \right)_{+(0)}^{[-1,0]} + \left(\frac{1}{-x} \right)_{+(\infty)}^{[-\infty,-1]} \right] \theta(-x), \end{aligned} \quad (59)$$

where we have used the results in Eqs. (A1,C7) to derive the second and last equalities, and took the limit $\eta \rightarrow 0^+$ or $\eta \rightarrow 0^-$ when needed. This $\tilde{q}^{(1)}(x, \mu/p^2, \epsilon_{\text{in}})$ does not satisfy vector-current conservation, and is different from $\tilde{q}^{(1)}(x, \mu/p^2, \epsilon_{\text{in}})$ in Eq. (56) only by the δ -functions at $x_0 = \pm\infty$. Within the function domain $-\infty < x < \infty$, they are exactly the same. We will see below that both Eq. (56) and Eq. (58) eventually lead to the same result for the one-loop matching coefficient.

The final ingredient we need for the matching calculations is the PDF, whose one-loop bare matrix element can be written as a sum of an $\overline{\text{MS}}$ counterterm and renormalized matrix element, $q^{(1)}(x, \epsilon) = \delta q^{(1)}(x, \epsilon_{\text{uv}}) + q^{(1)}(x, \epsilon_{\text{in}})$,

where

$$\begin{aligned}\delta q^{(1)}(x, \epsilon_{uv}) &= \frac{\alpha_s C_F}{2\pi} \frac{1}{\epsilon_{uv}} \left(\frac{1+x^2}{1-x} \right)_{+(1)}^{[0,1]} \theta(x) \theta(1-x), \\ q^{(1)}(x, \epsilon_m) &= \frac{\alpha_s C_F}{2\pi} \frac{(-1)}{\epsilon_m} \left(\frac{1+x^2}{1-x} \right)_{+(1)}^{[0,1]} \theta(x) \theta(1-x).\end{aligned}\quad (60)$$

With the above results in hand we can now determine the matching coefficients up to one-loop order. Using Eq. (30) we find

$$C^{(1)}(\alpha, z^2 \mu^2) = \mathcal{P}^{(1)}(\alpha, z^2 \mu^2, \epsilon_m) - q^{(1)}(\alpha, \epsilon_m). \quad (61)$$

Therefore the matching coefficient relating the pseudo-PDF and PDF in the $\overline{\text{MS}}$ scheme with $\Gamma = \gamma^0$ is²

$$\begin{aligned}C(\alpha, z^2 \mu^2) &= \left[1 + \frac{\alpha_s C_F}{2\pi} \left(\frac{3}{2} \ln(z^2 \mu^2) + \frac{3}{2} \ln \frac{e^{2\gamma_E}}{4} + \frac{3}{2} \right) \right] \delta(1-\alpha) \\ &+ \frac{\alpha_s C_F}{2\pi} \left\{ \left(\frac{1+\alpha^2}{1-\alpha} \right)_{+(1)}^{[0,1]} \left[-\ln(z^2 \mu^2) - \ln \frac{e^{2\gamma_E}}{4} - 1 \right] - \left(\frac{4 \ln(1-\alpha)}{1-\alpha} \right)_{+(1)}^{[0,1]} + 2(1-\alpha) \right\} \theta(\alpha) \theta(1-\alpha).\end{aligned}\quad (62)$$

This result is independent of the infrared regulator as it must be. We have also computed the matching coefficient for the $\Gamma = \gamma^z$ case, and it is $C_{\gamma^z}(\alpha, z^2 \mu^2) = C(\alpha, z^2 \mu^2) + \Delta C_{\gamma^z}(\alpha, z^2 \mu^2)$ with

$$\Delta C_{\gamma^z}(\alpha, z^2 \mu^2) = \frac{\alpha_s C_F}{2\pi} 2(1-\alpha) \theta(\alpha) \theta(1-\alpha). \quad (63)$$

Due to the $\ln(z^2 \mu^2) \delta(1-\alpha)$ term in Eq. (62), the matching coefficient for the $\overline{\text{MS}}$ pseudo-PDF again displays the fact that there is not a smooth local limit as $z \rightarrow 0$. It is possible to define a scheme other than $\overline{\text{MS}}$ to ensure that this limit is smooth, reproducing a renormalization for $z \rightarrow 0$ that agrees with the fact that the local operator corresponds with a conserved current. One such scheme would be to simply multiply all $\overline{\text{MS}}$ renormalization constants by $C_0(\mu^2, z^2)$, which would lead to a spatial correlator renormalized in a different scheme, and a corresponding different matching coefficient in Eq. (62) with a smooth $z \rightarrow 0$ limit. This is equivalent to studying the ratio of Eq. (38) from the start as advocated in Ref. [39, 62]. We will give explicit results for this scheme choice below. This modified scheme should not be confused with the strict definition of the $\overline{\text{MS}}$ scheme.

From Eq. (27) the corresponding relation for the matching coefficient for the quasi-PDF defined in Eq. (24) is

$$C^{(1)}(\xi, \mu/|y|P^z) = \tilde{q}^{(1)}(\xi, \mu/|y|P^z, \epsilon_m) - q^{(1)}(\xi, \epsilon_m). \quad (64)$$

Therefore using Eq. (58) the matching coefficient relating the quasi-PDF and PDF is

$$\begin{aligned}C\left(\xi, \frac{\mu}{|y|P^z}\right) &= \delta(1-\xi) + \frac{\alpha_s C_F}{2\pi} \left\{ \begin{aligned} &\left(\frac{1+\xi^2}{1-\xi} \ln \frac{\xi}{\xi-1} + 1 + \frac{3}{2\xi} \right)_{+(1)}^{[1,\infty]} - \left(\frac{3}{2\xi} \right)_{+(\infty)}^{[1,\infty]} && \xi > 1 \\ &\left(\frac{1+\xi^2}{1-\xi} \left[-\ln \frac{\mu^2}{y^2 P_z^2} + \ln(4\xi(1-\xi)) \right] - \frac{\xi(1+\xi)}{1-\xi} \right)_{+(1)}^{[0,1]} && 0 < \xi < 1 \\ &\left(-\frac{1+\xi^2}{1-\xi} \ln \frac{-\xi}{1-\xi} - 1 + \frac{3}{2(1-\xi)} \right)_{+(1)}^{[-\infty,0]} - \left(\frac{3}{2(1-\xi)} \right)_{+(-\infty)}^{[-\infty,0]} && \xi < 0 \end{aligned} \right. \\ &+ \frac{\alpha_s C_F}{2\pi} \left[\delta(1-\xi) \left(\frac{3}{2} \ln \frac{\mu^2}{4y^2 P_z^2} + \frac{5}{2} \right) + \frac{3}{2} \gamma_E \left(\frac{1}{(\xi-1)^2} \delta^+\left(\frac{1}{\xi-1}\right) + \frac{1}{(1-\xi)^2} \delta^+\left(\frac{1}{1-\xi}\right) \right) \right]. \quad (65)\end{aligned}$$

² A one-loop analysis of the spatial correlator in the coordinate space also recently appeared in Refs. [62, 63]. Our factorization result for the spatial correlator in Eq. (23) has a similar form to the hard part of the reduced spatial correlator found in Eq. (3.35) of Ref. [62] and Eq. (17) of Ref. [63]. It is therefore interesting to compare our $C(\alpha, z^2 \mu^2)/C_0(\mu^2, z^2)$ and this hard part. Our $\overline{\text{MS}}$

result Eq. (62) differs from Refs. [62] due to the presence of the $2(1-\alpha)$ term. The result in the final version of Ref. [63] agrees with ours. Eq. (62) also agrees with the original result derived in Ref. [45], up to the addition of our $e^{2\gamma_E}$ terms. The result for $C(\alpha, z^2 \mu^2)$ in Eq. (62) should be used to extract an $\overline{\text{MS}}$ PDF from an $\overline{\text{MS}}$ result for the pseudo-PDF.

Again this result is independent of the IR regulator as it must be. Here the plus function terms $[g_1(\xi)]_{+(1)}^{[1,\infty]}$ and $[g_2(\xi)]_{+(1)}^{[-\infty,0]}$ have integrands that converge for $\xi \rightarrow \pm\infty$, behaving as $g_i(\xi) \sim 1/\xi^2$. Note that if we had instead used the renormalized $\overline{\text{MS}}$ quasi-PDF calculated in Eq. (56), we would obtain a different matching coefficient C with different δ -functions at $\xi = \pm\infty$. However, the δ -functions do not contribute to the convolution integral in Eq. (27) for any integrable PDFs. For example, to carry out the convolution with $1/\xi^2 \delta^+(1/\xi)$ we can use $\delta^+(\frac{1}{\xi}) = \lim_{\beta \rightarrow 0^+} \delta(\frac{1}{\xi} - \beta)$, which when plugged into the factorization formula gives

$$\lim_{\beta \rightarrow 0^+} \int \frac{dy}{|y|} \frac{y^2}{x^2} \delta\left(\frac{y}{x} - \beta\right) f_{u-d}(y) = \lim_{\beta \rightarrow 0^+} \beta f_{u-d}(\beta x). \quad (66)$$

For the plus-function at ∞ using Eqs. (C9) and (C3) we have

$$\begin{aligned} \int_{-1}^{+1} \frac{dy}{|y|} \left[\frac{1}{(x/y)_{+(1)}} \right]_{+(1)}^{[1,\infty]} f_{u-d}(y) &= \lim_{\beta \rightarrow 0^+} \int_{-1}^{+1} \frac{dy}{|y|} \left[\frac{\theta(x/y - \beta)}{x/y} + \frac{y^2}{x^2} \delta\left(\frac{y}{x} - \beta\right) \ln \beta \right] f_{u-d}(y) \\ &= \int_{-1}^{+1} \frac{dy}{x} \frac{y}{|y|} f_{u-d}(y) + \lim_{\beta \rightarrow 0^+} \beta f_{u-d}(\beta x) \ln \beta. \end{aligned}$$

In the last line we dropped the $\theta(x/y - \beta)$ since at small y our PDF behaves as $f_{u-d}(y) \sim y^{-1+a}$ with $0 < a < 1$. This also implies

$$\begin{aligned} \lim_{\beta \rightarrow 0} \beta f_{u-d}(\beta x) &\propto x^{-1+a} \lim_{\beta \rightarrow 0} \beta^a = 0, \\ \lim_{\beta \rightarrow 0} \beta f_{u-d}(\beta x) \ln \beta &\propto x^{-1+a} \lim_{\beta \rightarrow 0} \beta^a \ln \beta = 0, \end{aligned} \quad (67)$$

which means that the distribution contributions evaluated at $\xi = \pm\infty$ in the matching coefficient C give zero contribution.

Therefore, the matching coefficients calculated from the quasi-PDFs in Eq. (56) and Eq. (58) are the same in effect, and we can simply drop all the δ -functions at $\xi = \pm\infty$ when plugging them into the factorization formula:

$$\begin{aligned} C^{\overline{\text{MS}}} \left(\xi, \frac{\mu}{|y|P^+} \right) &= \delta(1 - \xi) + \frac{\alpha_s C_F}{2\pi} \left\{ \begin{aligned} &\left(\frac{1 + \xi^2}{1 - \xi} \ln \frac{\xi}{\xi - 1} + 1 + \frac{3}{2\xi} \right)_{+(1)}^{[1,\infty]} - \frac{3}{2\xi} & \xi > 1 \\ &\left(\frac{1 + \xi^2}{1 - \xi} \left[-\ln \frac{\mu^2}{y^2 P^2} + \ln(4\xi(1 - \xi)) \right] - \frac{\xi(1 + \xi)}{1 - \xi} \right)_{+(1)}^{[0,1]} & 0 < \xi < 1 \\ &\left(\frac{1 + \xi^2}{1 - \xi} \ln \frac{-\xi}{1 - \xi} - 1 + \frac{3}{2(1 - \xi)} \right)_{+(1)}^{[-\infty,0]} - \frac{3}{2(1 - \xi)} & \xi < 0 \end{aligned} \right. \\ &+ \frac{\alpha_s C_F}{2\pi} \delta(1 - \xi) \left(\frac{3}{2} \ln \frac{\mu^2}{4y^2 P^2} + \frac{5}{2} \right). \end{aligned} \quad (68)$$

The use of Eq. (68) in the factorization formula is valid for any PDF that behaves as $\lim_{y \rightarrow 0} f(y, \mu) \sim y^{-1+a}$ with $a > 0$. We have also computed the matching coefficient for the $\Gamma = \gamma^+$ case, and it is given by $C_{\gamma^+}(\xi, \mu/(|y|P^+)) = C(\xi, \mu/(|y|P^+)) + \Delta C_{\gamma^+}(\xi, \mu/(|y|P^+))$ with

$$\Delta C_{\gamma^+}(\xi, \mu/(|y|P^+)) = \frac{\alpha_s C_F}{2\pi} 2(1 - \xi) \theta(\xi) \theta(1 - \xi). \quad (69)$$

Note that our result for the quark matching coefficient in $\overline{\text{MS}}$ differs from that of Ref. [27] which is a pure plus function, but gives a convolution that does not converge, just as in the case of the quasi-PDF with a transverse momentum cutoff, see Ref. [26].

Since the renormalized pseudo-PDF and quasi-PDF satisfy the relation in Eq. (33) by definition, $C(\xi, \mu/(|y|P^+))$ and $C(\alpha_s, x^2 \mu^2)$ that are given by Eqs. (62, 65) automatically satisfy the relation in Eq. (34).

Besides, if one uses a scheme other than $\overline{\text{MS}}$ for the quasi-PDF, such as the scheme obtained by absorbing C_0 into the $\overline{\text{MS}}$ renormalization constant, then this will lead to a result for the matching coefficient that is a pure plus function and hence satisfies current conservation. Starting with Eq. (65) and using Eq. (37) together with Eq. (39)

we obtain

$$C^{\pi\pi\pi\pi}\left(\xi, \frac{\mu}{|y|P^2}\right) = \delta(1-\xi) + \frac{\alpha_s C_F}{2\pi} \begin{cases} \left(\frac{1+\xi^2}{1-\xi} \ln \frac{\xi}{\xi-1} + 1 - \frac{3}{2(1-\xi)}\right)_{+(1)}^{[1,\infty]} & \xi > 1 \\ \left(\frac{1+\xi^2}{1-\xi} \left[-\ln \frac{\mu^2}{y^2 P^2} + \ln(4\xi(1-\xi)) - 1\right] + 1 + \frac{3}{2(1-\xi)}\right)_{+(1)}^{[0,1]} & 0 < \xi < 1 \\ \left(-\frac{1+\xi^2}{1-\xi} \ln \frac{-\xi}{1-\xi} - 1 + \frac{3}{2(1-\xi)}\right)_{+(1)}^{[-\infty,0]} & \xi < 0 \end{cases}, \quad (70)$$

and for the $\Gamma = \gamma^2$ case,

$$\Delta C_{\gamma^2}^{\pi\pi\pi\pi}(\xi, \mu/(|y|P^2)) = \frac{\alpha_s C_F}{2\pi} [2(1-\xi)]_{+(1)}^{[0,1]}. \quad (71)$$

While retaining current conservation in the renormalized quasi-PDF, Eq. (70) can be used for example as input to the two-step matching procedure in the lattice calculation of PDF in Refs. [33]. For the matching step, an equivalent procedure is to study the ratio given in Eq. (38) in the $\overline{\text{MS}}$ scheme from the start, as advocated in Ref. [39, 62], performing its matching onto the PDF, which will yield Eq. (70). This concludes our discussion of matching results and the equivalence between the quasi-PDF and pseudo-PDF at one-loop order.

IV. OTHER RENORMALIZATION SCHEMES

Although we derive the above matching formula assuming that the quasi-PDF is renormalized in the $\overline{\text{MS}}$ scheme, this is not a limitation to our result. Since the gauge-invariant Wilson line operator $\hat{Q}_T(z)$ has been proven to be multiplicatively renormalizable in the coordinate space [52, 53], one can convert $\hat{Q}_T(z)$ from any other scheme to the $\overline{\text{MS}}$ scheme before using the above factorization formula. The renormalization of the quasi-PDF has been studied in many recent papers [26, 28, 33, 34, 54–56, 64–66]. We will discuss some of these results and show how they can be incorporated into the factorization formula in Eq. (27).

The $\overline{\text{MS}}$ scheme is convenient for our discussion of the OPE as it guarantees Lorentz and gauge invariances, but it is not practical for lattice renormalization. Since the lattice theory has a natural UV cut-off $1/a$ with a being the lattice spacing, the unrenormalized spatial correlator $\hat{Q}$ inherits the power divergence from the Wilson line self-energy according to Eq. (13). For an arbitrary scheme X , the renormalized spatial correlator

$$\hat{Q}^X(\zeta, z^2 \mu_R^2) = \lim_{a \rightarrow 0} Z_X^{-1}(z^2 \mu_R^2, a^2 \mu_R^2) \hat{Q}(\zeta, z^2/a^2) \quad (72)$$

should be free of all the UV divergences and have a well-defined continuum limit as $a \rightarrow 0$. This continuum limit, in particular, is independent of the UV regulator, so

$$\begin{aligned} \lim_{a \rightarrow 0} Z_X^{-1}(z^2 \mu_R^2, a^2 \mu_R^2) \hat{Q}(\zeta, z^2/a^2) \\ = Z_X^{-1}(z^2 \mu_R^2, \epsilon) \hat{Q}(\zeta, z^2, \epsilon). \end{aligned} \quad (73)$$

As a result, we can relate $\hat{Q}^X(\zeta, z^2 \mu_R^2)$ to the $\overline{\text{MS}}$ scheme

by the conversion

$$\begin{aligned} \hat{Q}^X(\zeta, z^2 \mu_R^2) &= \frac{Z_{\overline{\text{MS}}}(\epsilon, \mu)}{Z_X(z^2 \mu_R^2, \epsilon)} \hat{Q}^{\overline{\text{MS}}}(\zeta, z^2 \mu^2) \\ &= Z_X'(z^2 \mu_R^2, \mu_R^2/\mu^2) \hat{Q}^{\overline{\text{MS}}}(\zeta, z^2 \mu^2), \end{aligned} \quad (74)$$

where the regulator ϵ dependence is completely canceled out between $Z_{\overline{\text{MS}}}$ and Z_X . The ratio Z_X' can be calculated perturbatively in QCD, which was done in [56] for several lattice schemes and the RI/MOM scheme. Thus the factorization formula we have proven in Sec. II still applies to $\hat{Q}^X$ with a slight modification to the coefficient function,

$$\hat{Q}^X(\zeta, z^2 \mu_R^2) = \int_{-1}^1 d\alpha C^X(\alpha, \mu_R^2/\mu^2, \mu^2 z^2) Q(\alpha\zeta, \mu), \quad (75)$$

where the matching coefficient for the scheme X is related to that of $\overline{\text{MS}}$ by

$$C^X(\alpha, \mu_R^2/\mu^2, \mu^2 z^2) = Z_X'(z^2 \mu_R^2, \mu_R^2/\mu^2) C(\alpha, \mu^2 z^2). \quad (76)$$

For the pseudo-PDF the modified result also involves this same coefficient

$$\begin{aligned} \mathcal{P}^X(x, z^2 \mu_R^2) &= \int_{|x|}^1 \frac{dy}{|y|} C^X\left(\frac{x}{y}, \frac{\mu_R^2}{\mu^2}, \mu^2 z^2\right) q(y, \mu) \\ &\quad + \int_{-1}^{-|x|} \frac{dy}{|y|} C^X\left(\frac{x}{y}, \frac{\mu_R^2}{\mu^2}, \mu^2 z^2\right) q(y, \mu). \end{aligned} \quad (77)$$

Meanwhile, for the quasi-PDF we have,

$$\begin{aligned}\bar{q}_X\left(x, \frac{\mu_R^2}{P^2}\right) &= \int \frac{d\zeta}{2\pi} e^{i x \zeta} \bar{Q}^X\left(\zeta, \frac{\mu^2 \zeta^2}{P^2}\right) \\ &= \int_{-1}^1 \frac{dy}{|y|} C^X\left(\frac{x}{y}, \frac{\mu_R}{\mu}, \frac{\mu}{|y| P^2}\right) q(y, \mu).\end{aligned}\quad (78)$$

Here the modified coefficient for the X scheme is related to coefficient in the $\overline{\text{MS}}$ scheme by

$$\begin{aligned}C^X\left(\frac{x}{y}, \frac{\mu_R}{\mu}, \frac{\mu}{|y| P^2}\right) \\ = \int d\eta Z'_X\left(\eta^2, \frac{\mu_R^2}{\mu^2}\right) C\left(\frac{x}{y} - \frac{\eta}{|y|} \frac{\mu_R}{P^2}, \frac{\mu}{|y| P^2}\right),\end{aligned}\quad (79)$$

where here Z'_X is defined by the Fourier transform

$$Z'_X\left(\eta^2, \frac{\mu_R^2}{\mu^2}\right) = \int \frac{d\tau}{2\pi} e^{i\eta\tau} Z_X\left(\tau^2, \frac{\mu_R^2}{\mu^2}\right). \quad (80)$$

Depending on the scheme X we note that slightly modified definitions of Z_X may be more appropriate.

One undesirable feature of the $\overline{\text{MS}}$ scheme for the renormalized spatial correlator is that it does not have a smooth $z \rightarrow 0$ limit, and hence no simple connection with the fact that the local operator for $z = 0$ is a conserved current. To avoid this one can simply make use of a different scheme that has a simple relation to $\overline{\text{MS}}$, such as by adding $C_0(\mu^2 z^2)$ to the $\overline{\text{MS}}$ renormalization constant. This removes the offending $\ln(\mu^2 z^2)$ terms and yields a scheme with a smooth connection to the conserved current.

Besides the $\overline{\text{MS}}$ scheme, the quasi-PDF has also been defined with a transverse momentum cut-off [19, 24, 25, 29] and in the RI/MOM scheme [26, 33, 34, 56, 66]. The RI/MOM scheme has attracted strong interest recently as it can be implemented nonperturbatively on the lattice, so we consider it as an explicit example of the above relations. In this scheme, the renormalization constant Z_{OM} is determined by imposing a condition on the spatial correlator in an off-shell quark state,

$$\begin{aligned}Z_{\text{OM}}^{-1} \bar{Q}(\zeta) &= z p^2, z^2/a^2, -p^2 a^2 \Big|_{p^2 = -\mu_R^2, p^+ = p_R^+} \\ &= \bar{Q}^{(0)}(z p_R^2, z^2/a^2, z^2 \mu_R^2) = e^{-i z p_R^+},\end{aligned}\quad (81)$$

where q denotes the quark state, p^μ is the external momentum, and $\bar{Q}^{(0)}$ in the superscript stands for the tree-level matrix element. As a result,

$$\begin{aligned}Z_{\text{OM}} &= Z_{\text{OM}}(z p_R^2, z^2/a^2, a^2 \mu_R^2), \\ Z'_{\text{OM}} &= Z'_{\text{OM}}(z p_R^2, z^2 \mu_R^2, \mu_R^2/\mu^2),\end{aligned}\quad (82)$$

and here we define

$$\begin{aligned}Z'_{\text{OM}}\left(\eta, \frac{\mu_R^2}{(p_R^2)^2}, \frac{\mu_R^2}{\mu^2}\right) \\ = p_R^2 \int \frac{d\tau}{2\pi} e^{i\eta p_R^+} Z_{\text{OM}}(z p_R^2, z^2 \mu_R^2, \mu_R^2/\mu^2).\end{aligned}\quad (83)$$

Then the matching coefficient in Eq. (78) becomes

$$\begin{aligned}C^{\text{OM}}\left(\frac{x}{y}, \frac{\mu_R}{\mu}, \frac{\mu}{|y| P^2}\right) \\ = \int d\eta Z'_{\text{OM}}\left(\eta, \frac{\mu_R^2}{(p_R^2)^2}, \frac{\mu_R^2}{\mu^2}\right) C\left(\frac{x}{y} - \frac{\eta}{|y|} \frac{p_R^2}{P^2}, \frac{\mu}{|y| P^2}\right).\end{aligned}\quad (84)$$

The choices of μ_R and p_R^2 are independent of μ and P^2 , and $p_R^2 = P^2$ was used in Refs. [26, 34].

It should be noted that on the lattice, due to the breaking of chiral symmetry, the vector-like quark Wilson line operator $\bar{O}_V(z)$ can mix with the scalar operator $\bar{O}_1(z)$, as has been discussed in Refs. [33, 34, 56, 66, 67]. After considering the mixing effects, the same factorization formula can still be applied to the RI/MOM quasi-PDF from lattice QCD.

V. NUMERICAL RESULTS

In this section we numerically analyze the quasi-PDF, spatial correlator and pseudo-PDF by studying how the matching coefficients in Eqs. (27,30) change the PDF. The quasi-PDF has already been studied in this manner for the $\overline{\text{MS}}$, transverse momentum cut-off, and RI/MOM schemes in Ref. [26]. Our new $\overline{\text{MS}}$ result for the matching is given in Eq. (68), and leads to stable convolution integrals. We also compare the differences between using hadron momentum $p^2 = P^2$ and the parton momentum $p^2 = |y| P^2$ for the matching coefficient in the $\overline{\text{MS}}$ scheme. We take $\Gamma = \gamma^5$ for the results here.

As an example we use for our analysis the unpolarized iso-vector parton distribution,

$$f_{u-d}(x, \mu) = f_u(x, \mu) - f_d(x, \mu) - f_u(-x, \mu) + f_d(-x, \mu), \quad (85)$$

where we include $f_u(-x, \mu) = -f_u(x, \mu)$ and $f_d(-x, \mu) = -f_d(x, \mu)$, the anti-parton distributions. For ease of comparison, we use the next-to-leading-order iso-vector PDF f_{u-d} from MSTW 2008 [4] with the corresponding running coupling $\alpha_s(\mu)$.

To implement the plus functions in the numerical calculation, we impose a soft cutoff $|y-x| < 10^{-m}$ and test the sensitivity of results to m . Since the limit of $y \rightarrow 0$ corresponds to the asymptotic region $|x/y| \rightarrow \infty$, we also impose a UV cutoff $|y| > 10^{-n}$ to test the convergence of the convolution integral. We find that all the results presented below are insensitive to m and n . The fact that our result in Eq. (68) has terms outside the plus function at 1 in each of the $\xi \in [1, \infty]$ and $\xi \in [-\infty, 0]$ intervals is important for ensuring that our $\overline{\text{MS}}$ result for C is insensitive to the $|y| > 10^{-n}$ cutoff. This was not the case for the quasi-PDF that was defined with a transverse momentum cutoff [24]. The RI/MOM scheme result [26] also does not suffer from this issue.

In Fig. 2 we compare the PDF with the quasi-PDF in the $\overline{\text{MS}}$ scheme obtained from the convolution in Eq. (27)

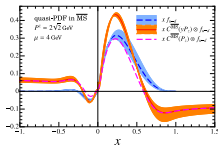


FIG. 2. The $\overline{\text{MS}}$ scheme PDF $x f_{u-d}$ and the $\overline{\text{MS}}$ quasi-PDF obtained from $x C^{\overline{\text{MS}}}(p^2) \otimes f_{u-d}$, comparing results obtained with $p^2 = y P^2$ and $p^2 = P^2$.

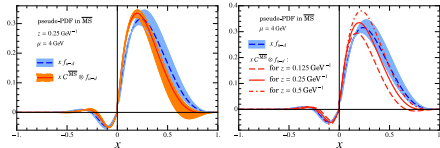


FIG. 3. (Left) Comparison between the PDF $x f_{u-d}$ and the pseudo-PDF $x(C^{\overline{\text{MS}}} \otimes f_{u-d})$ in the $\overline{\text{MS}}$ scheme. The orange and blue bands indicate the results from varying the factorization scale $\mu = 4 \text{ GeV}$ by a factor of two. (Right) Same but now showing only central pseudo-PDF curves for different values of z .

using our one-loop result in Eq. (68). We observe that changing from $p^2 = P^2$ to the correct $p^2 = |y|P^2$ shifts the result in the physical region by a considerable amount.

The same type of comparison can be made for the pseudo-PDF in the $\overline{\text{MS}}$ scheme by applying the factorization formula in Eq. (30) and matching coefficient in Eq. (62). In Fig. 3a we compare the PDF and pseudo-PDF and their dependence on the factorization scale μ , while in Fig. 3b we include the dependence of the pseudo-PDF on the distance $|z|$. Since the matching coefficient in Eq. (62) is similar to the parton splitting function except for the nontrivial finite constants, matching the PDF to the pseudo-PDF is analogous to evolving the PDF from μ to the scale of $1/|z|$. This evolution has been calculated in Refs. [39]. The variation of $|z|$ has a similar effect to the PDF evolution, as is observed in the right panel of Fig. 3. When $|z|\mu = 1$, the logarithm is zero, and the matching effect from C is determined by the nontrivial

constants in Eq. (62), which shifts the PDF downward in the large- x region and upward in the small- x region.

Finally, we can make a similar comparison for the spatial correlator in the $\overline{\text{MS}}$ scheme obtained with Eq. (23) and Eq. (62). Its real and imaginary parts are even and odd functions of ζ respectively, and are shown in Fig. 4. Again we show the residual dependence on μ and $|z|$ which are similar to that for the pseudo-PDF. The matching broadens the curves in the coordinate space. The spatial correlator renormalized in the $\overline{\text{MS}}$ scheme does not exhibit vector current (or particle number) conservation, which can be clearly seen from the fact that the real part of the distribution is not equal to 1 at $\zeta = 0$ (except for the special case where $|z|\mu$ is tuned to cancel the constant terms in the one-loop C).

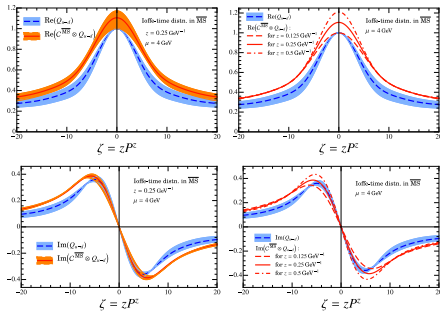


FIG. 4. (left) Comparison between the light-cone time distribution Q_{a-d} and spatial correlator from $(C^{TSS} \otimes Q_{a-d})$ in the $\overline{\text{MS}}$ scheme. The orange and blue bands indicate the results from varying the factorization scale $\mu = 4 \text{ GeV}$ by a factor of two. (right) Same but now showing only central spatial correlator curves for different values of z . The top panels show the real part, while bottom panels show the imaginary part.

VI. IMPLICATIONS FOR LATTICE CALCULATIONS

Our proof in Sec. II makes clear the relationship between the renormalized quasi-PDF, spatial correlator, and pseudo-PDF distributions. As a practical matter there are a few different ways in which these equations can be used to convert a lattice calculation of the spatial correlator $\bar{Q}$ into a PDF. Three examples are 1) first Fourier transform to the quasi-PDF with Eq. (24), and then use Eq. (27), 2) first Fourier transform to the pseudo-PDF with Eq. (29), and then use Eq. (30), and 3) first match to the Fourier transform the position space PDF $Q(\zeta, \mu)$ using Eq. (23), and then transform it to the PDF with the inverse of Eq. (22). Since the numerical implementation of these steps may have slightly different systematics it is interesting to compare them, or to use more than one approach in order to reduce uncertainties.

According to the analysis in Sec. II, for the factorization formula of the Euclidean distributions to work, one

must calculate the same spatial correlator with small distance z^2 and large momentum P^z so that the dynamical and kinematic higher-twist effects are suppressed. For practical lattice calculations, this means that there is only a finite number of useful data points in (z, P^z) that we can use to extract the PDF.

To illustrate this, consider a $48^3 \times 64$ lattice with spacing $a = 0.09 \text{ fm}$. The distance of the spatial correlation z is in units of $a \sim 1/2.2 \text{ GeV}^{-1}$ and the nucleon momentum P^z is in units of $2\pi/(48a) \sim 0.29 \text{ GeV}$. Let us take $\Lambda_{\text{QCD}} \sim 0.3 \text{ GeV}$. In principle the target mass corrections can be subtracted. If we consider various values $z = m a$ and $P^z = n \pi / (48a)$ for integer m and n , then to satisfy $z\Lambda_{\text{QCD}} \ll 1$ and $P^z \gg \Lambda_{\text{QCD}}$, we must have

$$m \ll 11, \quad n \gg 1. \quad (86)$$

To control the higher-twist correction at 20%, i.e. $z^2\Lambda_{\text{QCD}}^2 \sim 0.2$, $\Lambda_{\text{QCD}}^2/P^2 \sim 0.2$, we can only choose

$$m = \{0, 1, 2, 3, 4\}, \quad n = \{3, 4, 5, 6, \dots\}, \quad (87)$$

where the largest value for n is limited by what is practical in current lattice simulations. Six is the largest number of units attained in Ref. [42]. For quasi-PDF calculations, there are $4 \times 2 + 1 = 9$ useful data points for each fixed momentum $|P^z|$; for pseudo-PDF calculation, there are only $4 \times 2 = 8$ useful data points for each fixed $|z|$. In either case, it is anticipated that a direct Fourier transform with respect to z or $\zeta = zP^z$ will lead to oscillation in x -space and incorrect prediction for the small- z region due to the truncation in coordinate space. This has been observed in a recent lattice calculation of the quasi-PDF in Ref. [34]. Methods have been developed in recent works to eliminate the oscillation from the truncation effect [35, 68] in the quasi-PDF, while the higher-twist contributions at large z still need to be systematically corrected. It should be noted that the above is a rough estimate of the higher-twist corrections since the prefactor of $z^2 \Lambda_{\text{QCD}}^2$ could be smaller than 1. Their actual significance can only be quantitatively determined from lattice simulations.

To fully take advantage of all the useful data points, we can evolve them to either the same z^2 or P^z according to the perturbative analysis, which has been studied in Refs. [42, 43, 62] for the spatial correlator. However, since the evolution equation of the spatial correlator in z^2 or P^z follows a nonlocal convolution in $\zeta = zP^z$ or z , one has to know the full information in coordinate space to do the evolution. With limited number of data points, either large uncertainties or adopting a model-dependent assumption about the shape is inevitable.

To improve the precision of either approach, the only way forward is to have finer lattice spacing a so that we could have more data points which satisfy $|z| \ll \Lambda_{\text{QCD}}^{-1}$ and larger nucleon momentum P^z . With increasing P^z , the valence distribution of the nucleon is contracted in the z direction, so the spatial correlation of valence quarks is shrunk into smaller distance in z . If P^z is large enough, the spatial correlation will fall off quickly within $|z| \ll \Lambda_{\text{QCD}}^{-1}$, then the truncation error from Fourier transform will be significantly reduced. On the other hand, if we interpret the spatial correlation as the spatial correlator, its shape will not change under a Lorentz boost because it is a scalar function of $\zeta = zP^z$ and z^2 . Nevertheless, finer lattice spacing a allows for calculation with a wider range of P^z , thus covering larger values of $\zeta = zP^z$ to reduce the truncation error. Since the number of useful data points increases quadratically with $1/a$, a more precise lattice calculation with controlled systematic errors will be available in the future.

VII. CONCLUSIONS

Starting with a Euclidean operator product expansion for products of gauge invariant operators in QCD, we have derived the factorization formulas for the quasi-PDF, spatial correlator and pseudo-PDF. The three Euclidean distribution functions are related observables,

and all follow from the same fundamental factorization. For the spatial correlator this derivation implies that the ratio in Eq. (35) does scale in z^2 , but needs the small z^2 factorization formula in Eq. (23) to extract the PDF. Our derivation for the factorization formula applies when the renormalized spatial correlator is defined in any scheme. The OPE used here could also be used to systematically derive factorization formulas for power corrections to Eq. (6), which will involve matching onto higher-twist parton distributions. (The numerical relevance of these corrections is considered in Ref. [30].) Note that LaMET is not equivalent to the expansion from the OPE, as the former is more general and can be applied to the lattice calculations of other quantities, for example the TMD-PDF where a simple OPE does not exist.

Our derivation of the factorization formula for the quasi-PDF also verifies that the parton momentum p^z in the matching coefficient in Eq. (6) has to be $p^z = yP^z$, which makes a considerable difference for the MS matching result when compared with $p^z = P^z$ (see Fig. 2). The proper p^z should therefore be used in lattice calculations of the PDF in the LaMET approach.

As a non-trivial test of relations between the various distributions and factorization formulas we have considered results at one-loop in the MS scheme. We have derived a corrected results for the coefficient C for the MS scheme, given in Eq. (68), which leads to convergent results in the convolution integral. We have also computed the one-loop MS result for the Wilson coefficient C appearing in the spatial correlator and pseudo-PDF factorizations. A numerical analysis of these one-loop corrections in MS has also been provided. The one-loop matching coefficient C has a smaller effect for the pseudo-PDF than C does for quasi-PDF, as can be seen by comparing Figs. 2 and 3. Given systematic uncertainties in manipulating the lattice data, it is potentially interesting to consider using the same lattice data on the spatial correlator to extract the parton distribution function using both the quasi-PDF and pseudo-PDF approaches.

There are several different ways of implementing the factorization formula to calculate the PDF from lattice data for the spatial correlator $\bar{Q}$, which we have discussed in Sec. VI. One always has to work with short distance correlation and large nucleon momentum to reduce higher-twist corrections. This limits the number of useful data points from lattice calculations as described in Sec. VI. To achieve precision calculations without making model assumptions it will be highly desirable to move towards finer lattice spacing to increase the number of effective data points.

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Appendix A: Fourier Transform

To Fourier transform the bare pseudo-PDF in Eq. (46) into the bare quasi-PDF, we use the identity

$$\begin{aligned} & \int \frac{d\zeta}{2\pi} e^{i\alpha\zeta} (z^2\mu^2)^{\epsilon} \Gamma(-\epsilon) 4^{-\epsilon} \\ &= \left(\frac{\mu^2}{p_z^2}\right)^{\epsilon} \int \frac{d\zeta}{2\pi} e^{i\alpha\zeta} \int_0^{\infty} d\alpha \alpha^{-1-\epsilon} e^{-\alpha\zeta^2} 4^{-\epsilon} \\ &= \left(\frac{\mu^2}{p_z^2}\right)^{\epsilon} \int_0^{\infty} \frac{d\alpha}{2\sqrt{\pi}} \alpha^{-3/2-\epsilon} e^{-\alpha^2/(4\alpha)} 4^{-\epsilon} \\ &= \left(\frac{\mu^2}{p_z^2}\right)^{\epsilon} \frac{\Gamma(\epsilon+1/2)}{\sqrt{\pi}} \frac{1}{|x|^{1+2\epsilon}}, \end{aligned} \quad (\text{A1})$$

which is true for $\epsilon < 0$.

Now let us turn to a plus function $g(y)_{+(1)}^{[0,1]}$. The double Fourier transform

$$\begin{aligned} & \int \frac{d\zeta}{2\pi} e^{i\alpha\zeta} \int_0^1 dy e^{-i\alpha\zeta} g(y)_{+(1)}^{[0,1]} (z^2\mu^2)^{\epsilon} \Gamma(-\epsilon) 4^{-\epsilon} \\ &= (i\zeta) \int \frac{d\zeta}{2\pi} e^{i(\alpha-1)\zeta} \int_0^1 dy \int_0^1 dt y g(1-y) \\ & \quad \times e^{i\alpha\zeta} (z^2\mu^2)^{\epsilon} \Gamma(-\epsilon) 4^{-\epsilon} \\ &= \left(\frac{\mu^2}{p_z^2}\right)^{\epsilon} \frac{\Gamma(\epsilon+\frac{1}{2})}{\sqrt{\pi}} \frac{\partial}{\partial x} \int_0^1 dy \int_0^1 dt \frac{y g(1-y)}{|x-1+ty|^{1+2\epsilon}}. \end{aligned} \quad (\text{A2})$$

Since the integration over y and t leads to a piecewise function of x , the derivative will end up with a plus function as the discontinuity at $x = 1$ gives $\delta(1-x)$.

Appendix B: Quasi-PDF Calculation

Here we quote the pure dimensional regularization results obtained when we carry out the quasi-PDF calculation following Refs. [24, 26] by Fourier transforming from z to $x p^z$ for the integrand to obtain $[\delta(p^z - x p^z) - \delta(k^z - x p^z)]$. For the vertex and wavefunction renormalization graphs we obtain

$$\begin{aligned} & \bar{q}_{\text{vertex}}^{(1)}(x, p^z, \epsilon) + \bar{q}_{\text{w.f.}}^{(1)}(x, p^z, \epsilon) \\ &= \frac{\alpha_s C_F}{2\pi} \left(\frac{\mu^2}{p_z^2}\right)^{\epsilon} \left[(1-\epsilon) \int_0^1 dy (1-y) \frac{1}{|x-y|^{1+2\epsilon}} \frac{\Gamma(\epsilon+\frac{1}{2})}{\sqrt{\pi}} - \delta(1-x) \frac{1}{2} \left(\frac{1}{\epsilon_{\text{uv}}} - \frac{1}{\epsilon_{\text{in}}} \right) \right], \end{aligned} \quad (\text{B1})$$

while for the sail diagram

$$\begin{aligned}
& \hat{q}_{\text{sail}}^{(1)}(x, p^z, \epsilon) \\
&= \frac{\alpha_s C_F}{2\pi} \left(\frac{4\pi\mu^2}{p_z^2} \right)^\epsilon \left[\int_0^1 dy \frac{x+y}{1-x} \frac{1}{|x-y|^{1+2\epsilon}} \frac{\Gamma(\epsilon+\frac{1}{2})}{\sqrt{\pi}} - \delta(1-x) \int dx' \int_0^1 dy \frac{x'+y}{1-x'} \frac{1}{|x'-y|^{1+2\epsilon}} \frac{\Gamma(\epsilon+\frac{1}{2})}{\sqrt{\pi}} \right] \\
&= \frac{\alpha_s C_F}{2\pi} \left(\frac{\mu^2}{p_z^2} \right)^\epsilon \left[\int_0^1 dy \frac{x+y}{1-x} \frac{1}{|x-y|^{1+2\epsilon}} \frac{\Gamma(\epsilon+\frac{1}{2})}{\sqrt{\pi}} + \delta(1-x) \left(\frac{1}{\epsilon_{\text{uv}}} - \frac{1}{\epsilon_{\text{IR}}} \right) \right. \\
&\quad \left. - \delta(1-x) \int dx' \int_0^1 dy \frac{1+y}{1-x'} \frac{1}{|x'-y|^{1+2\epsilon}} \frac{\Gamma(\epsilon+\frac{1}{2})}{\sqrt{\pi}} \right], \tag{B2}
\end{aligned}$$

and for the tadpole diagram

$$\begin{aligned}
& \hat{q}_{\text{tadpole}}^{(1)}(x, p^z, \epsilon) \\
&= \frac{\alpha_s C_F}{2\pi} \left(\frac{4\pi\mu^2}{p_z^2} \right)^\epsilon \left[-\frac{1}{1-2\epsilon} \frac{1}{|1-x|^{1+2\epsilon}} \frac{\Gamma(\epsilon+\frac{1}{2})}{\sqrt{\pi}} - \delta(1-x) \frac{1}{1-2\epsilon} \frac{1}{|1-x|^{1+2\epsilon}} \int_{-\infty}^{\infty} dx' \frac{1}{|1-x'|^{1+2\epsilon}} \frac{\Gamma(\epsilon+\frac{1}{2})}{\sqrt{\pi}} \right] \\
&= \frac{\alpha_s C_F}{2\pi} \left(\frac{4\pi\mu^2}{p_z^2} \right)^\epsilon \left[-\frac{1}{1-2\epsilon} \frac{1}{|1-x|^{1+2\epsilon}} \frac{\Gamma(\epsilon+\frac{1}{2})}{\sqrt{\pi}} + \delta(1-x) \left(\frac{1}{\epsilon_{\text{uv}}} - \frac{1}{\epsilon_{\text{IR}}} \right) \right]. \tag{B3}
\end{aligned}$$

After adding these expressions and carrying out the remaining integrations over y , we obtain the same result as in Eq. (48).

Appendix C: ϵ Expansion and Plus Functions

Since the support of the quasi-PDF ranges from $-\infty$ to ∞ , its asymptotic behavior as $\sim 1/|x|$ at $|x| \rightarrow \infty$ implies a UV divergence which can be regularized by dimensional regularization. Therefore, the ϵ expansion of the quasi-PDF should account for this feature.

In general, we need to expand

$$\frac{\theta(x)}{x^{1+\epsilon}} = \frac{\theta(x)\theta(1-x)}{x^{1+\epsilon}} + \frac{\theta(x-1)}{x^{1+\epsilon}}, \tag{C1}$$

and it is well known that

$$\frac{\theta(x)\theta(1-x)}{x^{1+\epsilon}} = -\frac{1}{\epsilon} \delta(x) + \tilde{\mathcal{L}}_0(x) - \epsilon \tilde{\mathcal{L}}_1(x) + O(\epsilon^2), \tag{C2}$$

where $\epsilon < 0$. Here we follow [69] and the plus functions $\mathcal{L}_n(x)$ are defined as

$$\begin{aligned}
\mathcal{L}_n(x) &= \left[\frac{\theta(x) \ln^n x}{x} \right]_+ \\
&= \lim_{\beta \rightarrow 0} \left[\frac{\theta(x-\beta) \ln^n x}{x} + \delta(x-\beta) \frac{\ln^{n+1} \beta}{n+1} \right], \tag{C3}
\end{aligned}$$

and we let

$$\tilde{\mathcal{L}}_n(x) = \theta(1-x) \mathcal{L}_n(x). \tag{C4}$$

Note that $\int_0^1 dx \mathcal{L}_n(x) = 0$.

To define the expansion in the range $x \in [1, \infty]$ we simply map this interval to $[0, 1]$ via $t = 1/x$. Taking an arbitrary smooth test function $g(x)$ we have

$$\begin{aligned}
& \int_1^\infty dx \frac{1}{x^{1+\epsilon}} g(x) \\
&= \int_0^1 dt \frac{1}{t^{1-\epsilon}} g(1/t) \\
&= \int_0^1 dt \left[\frac{1}{\epsilon} \delta(t) + \mathcal{L}_0(t) + \epsilon \mathcal{L}_1(t) + O(\epsilon^2) \right] g(1/t) \\
&= \int_1^\infty dx \left[\frac{1}{\epsilon} \delta^+ \left(\frac{1}{x} \right) + \mathcal{L}_0 \left(\frac{1}{x} \right) + \epsilon \mathcal{L}_1 \left(\frac{1}{x} \right) + O(\epsilon^2) \right] g(x). \tag{C5}
\end{aligned}$$

Here $\epsilon > 0$ and the superscript $+$ on the δ^+ function indicates that its argument should be positive. Therefore $\delta^+(1/x)$ has its support at $x = +\infty$, not $x = -\infty$. Since g is arbitrary we can identify

$$\frac{\theta(x-1)}{x^{1+\epsilon}} = \frac{1}{\epsilon} \frac{1}{x^2} \delta^+ \left(\frac{1}{x} \right) + \frac{1}{x^2} \tilde{\mathcal{L}}_0 \left(\frac{1}{x} \right) + \frac{1}{x^2} \tilde{\mathcal{L}}_1 \left(\frac{1}{x} \right) + O(\epsilon^2). \tag{C6}$$

Combining the above results and denoting which $1/\epsilon$ poles are UV or IR divergences we have

$$\begin{aligned}
\frac{\theta(x)}{x^{1+\epsilon}} &= \left[-\frac{1}{\epsilon_{\text{IR}}} \delta(x) + \frac{1}{\epsilon_{\text{UV}}} \frac{1}{x^2} \delta^+ \left(\frac{1}{x} \right) \right] \\
&\quad + \left(\frac{1}{x} \right)_{+(0)}^{[0,1]} + \left(\frac{1}{x} \right)_{+(-\infty)}^{[1,\infty]} \\
&\quad - \epsilon \left[\left(\frac{\ln x}{x} \right)_{+(0)}^{[0,1]} + \left(\frac{\ln x}{x} \right)_{+(-\infty)}^{[1,\infty]} \right] + O(\epsilon^2), \tag{C7}
\end{aligned}$$

where we have defined the distributions

$$(1/x)_{+(0)}^{[0,1]} = \mathcal{L}_0(x), \quad (\ln x/x)_{+(-\infty)}^{[0,1]} = \tilde{\mathcal{L}}_1(x), \tag{C8}$$

and

$$\begin{aligned} (1/x)_{+^{(\infty)}}^{[1,\infty]} &= (1/x^2)\tilde{\mathcal{L}}_0(1/x), \\ (\ln x/x)_{+^{(\infty)}}^{[1,\infty]} &= -(1/x^2)\tilde{\mathcal{L}}_1(1/x). \end{aligned} \quad (\text{C9})$$

Note that Eq. (C7) is consistent with the expected result

that

$$\int_0^\infty \frac{dx}{x^{1+\epsilon}} = \frac{1}{\epsilon_{\text{uv}}} - \frac{1}{\epsilon_{\text{in}}}. \quad (\text{C10})$$

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P18: 大动量有效理论综述

主题归属	LaMET、quasi/pseudo 分布与因子化
底稿	arXiv:2004.03543
arXiv / DOI	2004.03543
完整原始 PDF	75 页；来源 course-cache；SHA-256 bfc7dc7b4100d...
转排索引	EN 121 页；ZH 105 页（只作书目对照）
备注	索引未记录额外备注。

阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

页码范围：P18-p001 – P18-p075

Large-Momentum Effective Theory

Xiangdong Ji*

Maryland Center for Fundamental Physics,
Department of Physics,
University of Maryland,
College Park, Maryland 20742,
USA

Tsung-Dao Lee Institute,
Shanghai Jiao Tong University,
Shanghai, 200240,
China

Yizhuang Liu†

Tsung-Dao Lee Institute,
Shanghai Jiao Tong University,
Shanghai, 200240,
China
Institut für Theoretische Physik,
Universität Regensburg,
D-93040 Regensburg,
Germany

Institute of Theoretical Physics,
Jagiellonian University, 30-348 Kraków,
Poland

Yu-Sheng Liu‡

Tsung-Dao Lee Institute,
Shanghai Jiao Tong University,
Shanghai, 200240,
China

Jian-Hui Zhang§

Center of Advanced Quantum Studies,
Department of Physics,
Beijing Normal University, Beijing 100875,
China

Yong Zhao¶

Physics Department,
Brookhaven National Laboratory Bldg. 510A,
Upton, NY 11973,
USA
Physics Division,
Argonne National Laboratory,
Lemont, IL 60439,
USA

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Since the parton model was introduced by Feynman more than fifty years ago, we have learned much about the partonic structure of the proton through a large body of high-energy experimental data and dedicated global fits. However, calculating the partonic observables such as parton distribution function (PDFs) from the fundamental theory of strong interactions, QCD, has made limited progress. Recently, the authors have advocated a formalism, large-momentum effective theory (LaMET), through which one can extract parton physics from the properties of the proton travelling at a moderate boost-factor, e.g., $\gamma \sim (2-5)$. The key observation behind this approach is that Lorentz symmetry allows the standard formalism of partons in terms of light-front operators to be replaced by an equivalent one with large-momentum states and time-independent operators of a universality class. With LaMET, the PDFs, generalized PDFs or GPDs, transverse-momentum-dependent PDFs, and light-front wave functions can all be extracted in principle from lattice simulations of QCD (or other non-perturbative methods)

through standard effective field theory matching and running. Future lattice QCD calculations with exa-scale computational facilities can help to understand the experimental data related to the hadronic structure, including those from the upcoming Electron-Ion Colliders dedicated to exploring the partonic landscape of the proton. Here we review the progress made in the past few years in development of the LaMET formalism and its applications, particularly on the demonstration of its effectiveness from initial lattice QCD simulations.

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I. INTRODUCTION

The proton and neutron, collectively called the nucleon, are the basic building blocks of visible matter in the universe today. Ever since they were discovered in laboratories nearly a century ago (Chadwick, 1932; Rutherford, 1919), their fundamental properties have been vigorously explored: from the determination of the spin through the specific heat of liquid hydrogen (Dennison, 1927), to the measurement of the magnetic moments (Estermann *et al.*, 1933), and the extraction of their electromagnetic sizes through elastic electron scattering (Hofstadter, 1956). The most revealing discovery, however, came from the electron deep-inelastic scattering (DIS) on the proton and nuclei at Stanford Linear Accelerator Center (SLAC) in the late 1960s, in which the constituents of the proton and neutron, quarks (and later gluons), were discovered (Bloom *et al.*, 1969). Soon after, quantum chromodynamics (QCD), a quantum field theory (QFT) based on “color” SU(3) gauge symmetry, was established as the fundamental theory of strong interactions (Fritzsch *et al.*, 1973; Gross and Wilczek, 1973; Politzer, 1973), and of the internal structure of the nucleon as well (Thomas and Weise, 2001).

During the last fifty years, significant progress has been made in understanding the nucleon’s internal structure in both experiment and theory. Multiple experimental facilities have been built to study high-energy collisions involving protons and nuclei, from which a large amount

* xji@umd.edu

† yizhuang.lin@jnu.edu.cn

‡ mostel@uremail.com

§ zhangjianshui@bnu.edu.cn

¶ yong.zhao@ucl.ac.uk

of experimental data has been accumulated. Based on the QCD factorization theorems (Collins, 2011a), derived from perturbative QCD analyses beyond Feynman's parton model (Feynman, 1972), the parton distribution functions (PDFs), which characterize the longitudinal momentum distributions of quarks and gluons in hadrons moving at infinite momentum, have been obtained from global fits to these data (Ball *et al.*, 2017; Gao *et al.*, 2018; Harland-Lang *et al.*, 2015; Hou *et al.*, 2019). A recent result of the phenomenological proton PDFs is shown in Fig. 1 where x is the momentum fraction of the proton carried by partons. The PDFs provide a comprehensive description of the quark and gluon content of the nucleon. On the theoretical frontier, the Euclidean path-integral formalism of QCD, combined with the lattice regularization and Monte Carlo simulations (Wilson, 1974), has offered a systematic way of performing *ab initio* calculations of non-perturbative strong interactions. The rapid rise in computational power and development of intelligent numerical algorithms have made such a lattice QCD approach extremely successful in computing hadron spectroscopy, the strong coupling, hadronic form factors, etc., and even scattering phase shifts (Aoki *et al.*, 2020; Briceño *et al.*, 2018; Tanabashi *et al.*, 2018).

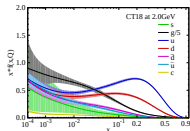


FIG. 1: Phenomenological parton distributions obtained by the CTEQ-TEA collaboration (CT18) from fits to global high-energy scattering data (Hou *et al.*, 2019), where $0 \leq x \leq 1$ is the fraction of the proton's infinite momentum carried in a parton.

Despite these impressive achievements, we have not been able to systematically explain the partonic structure of the proton from first principles, or more explicitly, we have not made fundamental progress in computing the quark and gluon distributions starting from the QCD Lagrangian (see Sec. I C for a brief summary). There is actually a good reason behind it: The standard formulation of parton physics in the textbooks (Collins, 2011a; Sterman, 1993) is accomplished through the *dynamical correlators* of quark and gluon fields on the light-front (LF) defined by $t - z = \text{const.}$, which has the important feature of being independent of the proton's momentum. On the other hand, lattice QCD is formulated

in the Euclidean space with imaginary time, and cannot be used to directly calculate the dynamical correlations that depend on real time. The standard lattice approach to parton physics has been to calculate the lower moments of parton distributions, which are matrix elements of local operators (Lin *et al.*, 2018c). However, the limitations to the first few moments prohibit practitioners from reproducing reliably the x -dependent structure as shown in Fig. 1, other than fitting model functional forms. Over the years, Hamiltonian diagonalization in LF quantization (LFQ) (Brodsky *et al.*, 1998) and Schwinger-Dyson equations (Maris and Roberts, 2003) have been proposed to solve the nucleon structure as Minkowskian approaches. Although significant advances have been made phenomenologically, a systematic approximation to calculate the nucleon PDFs is still missing.

A few years ago, some of the present authors proposed a general approach to calculate x -dependent parton distributions based on Feynman's original idea about partons: They are the infinite-momentum limit of *static properties* of the proton at large momentum, and therefore are intrinsically Euclidean quantities accessible through lattice QCD (Ji, 2013, 2014; Ji *et al.*, 2013b). According to this, parton physics in an intermediate range of $x_{\text{min}} \sim 0.1 < x < x_{\text{max}} \sim 0.9$ can be calculated from the physical properties of the proton at a moderately-large momentum, e.g., with a Lorentz boost factor $\gamma = 2 - 5$. The theory has been named as *large-momentum effective theory* (LaMET) because a rigorous connection between the infinite-momentum frame (IMF) partons and quarks and gluons at a finite momentum requires proper account of the ultraviolet (UV) modes with large momentum in effective field theory (EFT) and systematic power counting.

The basic principle for LaMET comes from an implicit observation in the naive parton model: The structure of the proton is approximately independent of its momentum so long as it is much larger than a typical strong-interaction scale Λ_{QCD} , or its mass. For example, the quark momentum distribution at moderate x in the proton at $P = |\vec{P}| = 5 \text{ GeV}$ is not very different from that at $P = 50 \text{ GeV}$ or $P = 5 \text{ TeV}$. One might call this phenomenon *large-momentum symmetry*, the nature of which is similar to that of the electronic structure of the hydrogen atom is not sensitive to the proton mass, so long as it is much larger than that of the electron. The asymptotic behavior of the proton structure might be controlled by an expansion in Λ_{QCD}/P , but a justification would require a better understanding of the underlying dynamics. Assuming this, Feynman replaced the protons probed at large but finite momenta in high-energy scattering with the one at infinite momentum $P = \infty$, corresponding to the leading term in the Λ_{QCD}/P expansion, and therefore the idealized concepts of the proton in the IMF and its constituents—*partons*—were born.

In QFTs, however, the existence of the $P \rightarrow \infty$ limit depends on their UV behavior. In general, the infinite-momentum limit does not commute with the UV cut-off limit $\Lambda_{UV} \rightarrow \infty$. While the physical limit is $(\Lambda_{UV} \gg P) \rightarrow \infty$, the parton model and subsequent QCD factorization theorems use $(P \gg \Lambda_{UV}) \rightarrow \infty$, keeping all PDFs with the finite support $|x| \leq 1$ where negative x is for antiquarks. Thus partons are an idealized concept which does not exist in the real world. Fortunately, because of asymptotic freedom, the above differences can be calculated in perturbative QCD. Therefore, LaMET is an effective theory of partons, which uses the ordinary field theoretical calculations $(\Lambda_{UV} \gg P) \rightarrow \infty$ and systematically takes into account non-commuting $P \rightarrow \infty$ limits through EFT matching and running and finite P effects by power corrections. Thus, the PDFs defined in the IMF or on the LF can be accessed at moderate x from the structure calculations at $P \sim$ a few GeVs.

The first application of LaMET was to the total gluon helicity ΔG in the polarized proton, a quantity of significant experimental interest at the polarized RHIC (Bunce *et al.*, 2000), but not within theoretical reach for many years. In (Ji *et al.*, 2013b), we have shown that from a large-momentum matrix element of the gluon spin operator in a physical gauge, ΔG can be obtained through an EFT matching. Following this success, LaMET was applied to the collinear quark PDFs (Ji, 2013). This latter application has generated considerable theoretical as well as numerical activities, particularly for the flavor non-singlet $u - d$ distributions in the proton and other hadrons. A general LaMET framework was subsequently introduced in (Ji, 2014). More recently, the approach has been extended to the gluons as well (Li *et al.*, 2019; Zhang *et al.*, 2019b). Therefore, the PDFs can now be computed directly in lattice QCD at specific Feynman variable x , without using LFQ. Besides, the partonic landscape of the proton is extremely rich, and LaMET holds the promise of computing parton physics beyond the collinear PDFs.

In recent years, tremendous progress has been made in formulating new parton observables for the proton. In particular, two parallel concepts have been developed in characterizing the transverse structure of the proton. The first is the generalized parton distributions (GPDs) (Ji, 1997b; Müller *et al.*, 1994; Radyushkin, 1999). The GPDs combine the features of the proton's elastic form factors, which provide the transverse-space density of partons (Miller, 2007), and Feynman PDFs, and interpolate them. Given the joint longitudinal-momentum and transverse-space distributions, one can construct the orbital angular momentum (OAM) of partons, among others (Ji, 1997b). In general, the GPDs can be used to generate momentum-dissected transverse space images of the proton (Burkardt, 2000). A new class of experimental processes, deeply-virtual exclusive processes (DVES), including deeply-virtual Compton scat-

tering (DVCS) in which the final state is a diffractive real photon plus a recoiling proton, has been found to measure them (Ji, 1997a,b). The second concept is the transverse-momentum-dependent (TMD) PDFs (or TMDPDFs), in which the parton's transverse momentum is explicit (Collins, 2011a; Collins and Soper, 1981). Much theoretical progress has been made in recent years regarding their proper definitions, factorizations, and spin correlations (Collins and Rogers, 2017, 2013; Echevarría *et al.*, 2013). TMDPDFs can be measured in experimental processes by observing the transverse momentum of the final-state particles.

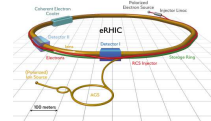


FIG. 2: A realization of Electron-Ion Collider at BNL (figure credit to BNL), which can be used to probe the partonic landscape of the proton.

Over the years, it has gradually become clear that a dedicated experimental facility to fully explore the partonic landscape of the proton is required. To meet this requirement, the US nuclear science community has proposed, to build a high-energy, high-luminosity Electron-Ion Collider (EIC) (Aprahamian *et al.*, 2015), which has been recently approved by the US Department of Energy. The new collider accelerates electrons to 10-30 GeV and ions—including the proton and heavy nuclei all the way up to Pb or U—up to 100 GeV per nucleon, realizing the center-of-mass collision energy E_{cm} from 40 to 170 GeV. The corresponding electron energy in fixed-target experiments would be 100 GeV to 10 TeV. The beams are polarized, with high-luminosity up to 10^{33-34} collisions/(cm²·s), which are critical for studying exclusive processes such as DVCS. The kinematic range of the collisions covers the Bjorken x_B (which coincides with the parton momentum fraction x in the naive parton model to be discussed in the next section) down to sub- 10^{-4} , and Q^2 as high as 10^4 GeV². Much of the EIC science has been discussed in a dedicated study (Accardi *et al.*, 2016b).

Of course, the EIC and lattice QCD efforts will not stop at the precision parton physics of the proton. More importantly, we need to develop ways or languages to describe the nucleon as a strongly-coupled relativistic quantum system, in much the same way as we understand, for

example, the quantum Hall effects in condensed matter physics. Without a deep understanding of the mechanisms of strongly-coupled QCD physics, we cannot claim a fundamental understanding of the structure of the proton and neutron, in particular, the origin of their mass and spin. This is one of the most challenging goals facing the standard model of particle and nuclear physics today.

This review is to systematically expose the idea, formalism, and results of the LaMET approach to parton physics. We do not claim to be entirely complete because the field is rapidly developing. References in the related fields are not meant to be complete either, and we apologize for any important omissions. Closely-related reviews on lattice parton physics can be found in (Cichy and Constantinou, 2019; Zhao, 2019). There have been studies on the effectiveness of LaMET in various models (Bhattacharya *et al.*, 2019a,b; Broniowski and Ruiz Arriola, 2017, 2018; Del Debbio *et al.*, 2020b; Gamberg *et al.*, 2015; Hobbs, 2018; Ji *et al.*, 2019c; Jia and Xiong, 2016; Kock *et al.*, 2020; Ma *et al.*, 2019; Nam, 2017; Radyushkin, 2017d; Son *et al.*, 2019; Xu *et al.*, 2018), some of which we will mention in the following for illustrative purposes. There have been also papers questioning the validity of LaMET method (Carlson and Freid, 2017; Rossi and Testa, 2017, 2018) and some got clarified later in the literature (Briceño *et al.*, 2017; Ji *et al.*, 2017b; Radyushkin, 2019c). We will not discuss them here and interested readers may refer to the above references. We have used *proton* in most places in the text to emphasize its importance in nuclear and particle physics. However, the discussions apply equally to the neutron and other hadrons as well.

The plan for the presentation of this review is as follows. In the remainder of the Introduction, we explain the nature of parton physics as an effective description of the internal structure of the proton at large momentum, as well as other existing methods in the literature for solving the parton structure. In Sec. II, we introduce the LaMET method starting from momentum renormalization group equation (RGE) of physical observables in a moving hadron, followed by the matching between momentum distributions and PDFs. We then formulate an EFT expansion to compute parton physics from theoretical methods suitable for the structure of a large-momentum proton. In Sec. III, we discuss some important details for collinear PDFs: renormalization of the nonlocal operators, particularly power divergences in lattice regularization, and matching to all orders in perturbation theory. Sec. IV is devoted to applications to general collinear parton observables including GPDs, parton distribution amplitudes and higher-order parton correlations. We also discuss applications for the OAM of the partons in a polarized proton. In Sec. V, we consider the application to TMDPDFs, a new class of parton observables. We study matching of the quasi-TMDPDFs to

the physical ones, and explore the lattice calculation of the soft function. Finally, Sec. VI summarizes the recent lattice calculations relevant to the LaMET applications, and the conclusion is given in Sec. VII. The review is completed with an Appendix with a list of acronyms and glossaries, as well as notations and conventions.

A. Partons through Infinite-Momentum States

Although partons have become a ubiquitous language for high-energy scattering, their role as effective degrees of freedom of QCD for describing the internal structure of the nucleon is less emphasized in the literature. In applications within QCD factorization theorems, they are—following Feynman—objects arising from the limit of infinite momentum, with the potential UV divergences regulated and renormalized after the limit. Thus, the partons are an idealized concept, referring to the quark and gluon Fock components of the nucleon or other hadrons only in the context of IMF and LF gauge $A^+ = (A^0 + A^3)/\sqrt{2} = 0$. They are in the same category of concepts as the *infinitely-heavy quark* in heavy-quark effective field theory (HQET) (Manohar and Wise, 2000). To motivate LaMET, it is important to understand this origin and nature of partons.

Built from the knowledge of electron scattering in non-relativistic systems (atoms and molecules) (West, 1975), Feynman introduced the *naive parton model* to describe deep-inelastic scattering (DIS) on the proton, and to explain the observed phenomenon of Bjorken scaling (Bjorken and Paschos, 1969; Feynman, 1972, 1969).

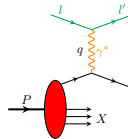


FIG. 3: Deep-inelastic scattering in which partons are probed in the proton.

Shown in Fig. 3 is the DIS process in which a virtual photon with large momentum q^μ is absorbed by a proton of momentum P^μ and mass M . The invariant variables are $Q^2 = -q^\mu q_\mu$ and $P \cdot q = M\nu$, and Bjorken $x_B = Q^2/(2P \cdot q)$ fixed in the scaling (or Bjorken) limit $Q^2 \rightarrow \infty$, $P \cdot q \rightarrow \infty$. The inclusive DIS cross section can

be factored into a product of leptonic and hadronic tensors, where the former is associated with the electromagnetic current of the lepton, while the latter contains all information about the electromagnetic interaction with the target proton.

To learn about the proton structure, it is best to consider the scattering in the Breit frame where

$$q^\mu = (0, 0, 0, -Q),$$

$$P^\mu = \left(\sqrt{\frac{Q^2}{4x_B^2} + M^2}, 0, 0, \frac{Q}{2x_B} \right), \quad (1)$$

and the virtual photon has zero energy. The probe is sensitive only to the spatial structure as in non-relativistic electron scattering. However, relativity now constrains the proton to move at a large momentum $P^z = Q/(2x_B)$ with boost factor $\gamma = Q/(2x_B M)$, which approaches $P^z = \infty$ in the Bjorken limit.

Feynman made intuitive assumptions about the proton structure and scattering mechanism, without QFT subtleties (Feynman, 1972): The proton structure at different large P^z should be similar, and can be approximated by that at $P^z = \infty$, or in the IMF. The interactions between constituents (partons) are infinitely time-dilated, and the wave function configurations are frozen. The proton in high-energy scattering can be seen as being made of non-interacting partons, each with a longitudinal momentum xP^z with $0 < x < 1$.

The internal structure of non-relativistic systems is independent of their overall momentum. However, relativistic systems is different as they least experience the Lorentz contraction. The structures of such systems are inextricably mixed with the overall motion, and their dependence on the external momentum is a dynamical problem. On the other hand, if the internal structure depends on a particular hadron scale Λ_{QCD} , the protons at all large-momentum with $P^z \gg \Lambda_{\text{QCD}}$ have a similar structure, corresponding to the $P^z \rightarrow \infty$ limit. This means that if $f(k^2, P^z)$ is the constituent momentum- k^2 distribution in a proton of momentum P^z , it might be analytical at $P^z = \infty$ and admits Taylor series expansions in $1/P^z$,

$$f(k^2, P^z) = f(x) + f_2(x)(\Lambda_{\text{QCD}}/P^z)^2 + \dots, \quad (2)$$

where $x = k^2/P^z$. If so, one may find a large-momentum symmetry of the proton properties up to power corrections $\mathcal{O}(1/P^z)$ (we omit the upper index z sometimes for simplicity), and $f(x)$ is the parton distribution.

The above picture can be shown to hold in certain simple QFT models, where the dynamical frame dependence of wave functions for composite systems can be studied straightforwardly. There are many interesting examples of two-dimensional systems, for which solutions can be found. One of the much studied cases is the large N_c QCD, also called 't Hooft model ('t Hooft, 1974),

in which the bound states have a well-defined large-momentum limit. The wave functions can be expanded in $1/P^z$ with the corrections starting from $(1/P^z)^2$. The momenta of the constituents, k and $P-k$, scale in this limit. When plotted as a function of $x = k/P^z$, the change in the wave function with the magnitude of the momentum can be found in Figs. 8–11 in (Jia *et al.*, 2017). This is the type of example in which Feynman's intuition applies.

However, such an intuition fails in many 3+1 dimensional QFTs, such as QCD. When a bound state travels at increasingly large momentum, more and more high-momentum modes of a field theory are needed to build up its internal structure. Lorentz contraction indicates that the range of constituent momentum important for the structure also increases. If these high-momentum modes do not decouple effectively from the low-momentum ones, large logarithms of the form $\ln P^z$ will develop in the structural quantities. Hence a singularity (cut) at $P^z = \infty$ can exist in these theories, making $P^z \rightarrow \infty$ limit ill-defined and the large momentum expansion impossible. This situation is intimately related to UV properties of the theories, for which the limits of taking the UV cutoff $\Lambda_{\text{UV}} \rightarrow \infty$ and $P^z \rightarrow \infty$ do not commute. While the physically-relevant one is $(\Lambda_{\text{UV}} \gg P^z) \rightarrow \infty$, partons in QCD factorizations are obtained in the other limit $(\Lambda_{\text{UV}} \ll P^z) \rightarrow \infty$ when the UV divergences are ignored. Thus one can formally write the parton distribution as

$$f(x) = \int \frac{d\lambda}{2\pi} e^{i\lambda x} \langle P^z = \infty | \psi^\dagger(z) \psi(0) | P^z = \infty \rangle, \quad (3)$$

where $\lambda = \lim_{P^z \rightarrow \infty, z \rightarrow 0} (zP^z)$, and ψ is a quantum field.

Historically, the IMF limit of field theories has been studied first at the level of diagrammatic rules for perturbation theory (Weinberg, 1966). It was found that taking $P^z \rightarrow \infty$ by ignoring the UV divergences considerably simplifies the perturbation theory rules: Many time-ordered diagrams vanish and only few have finite contributions. Moreover, scattering in this limit resembles that in non-relativistic quantum mechanics, and the wave function description becomes useful. The Fock states define the partons which have the proper kinematic support ($0 < x < 1$). After the limit is taken, all physical quantities are now independent of P^z , and large-momentum symmetry is exact before UV divergences are regulated. Therefore, it is the “naive” limit, $\Lambda_{\text{UV}} \ll P^z \rightarrow \infty$, that corresponds to Feynman's naive parton model.

In the standard QCD study of high-energy scattering, the above concept of partons as effective degrees of freedom has been used implicitly. The PDFs are defined in terms of the naive $P^z = \infty$ limit, and are used to match the experimental cross sections, resulting in QCD factorization theorems (Collins, 2011a).

B. Partons through Light-Front Correlators

In the literature and textbooks, parton distributions are not traditionally represented in terms of the Euclidean matrix elements as in Eq. (3). Rather, they are represented by the so-called LF correlators of quantum fields (“operator formalism”) (Brodsky *et al.*, 1998; Collins, 2011a). A more explicit formulation in terms of collinear quantum fields and effective Lagrangian is made in the soft-collinear effective theory (SCET) (Bauer *et al.*, 2001, 2002; Bauer and Stewart, 2001).

There is a physical way to see that the parton description of high-energy scattering results in the light-front correlations. Consider DIS in the rest frame of the proton, where the virtual photon has momentum

$$q^\mu = (\nu, 0, 0, -\sqrt{\nu^2 + 2x_B M\nu}). \quad (4)$$

In the Bjorken limit $\nu \rightarrow \infty$, although the invariant mass Q of the photon goes to infinity, the photon momentum becomes actually light-like in the sense that it approaches the light front. Therefore, in inclusive DIS cross section, the separation of the two electromagnetic currents in the hadronic tensor, which is Fourier conjugate to the photon momentum, also approaches the light-cone direction.

Thus, it appears natural that all the structural physics of the proton in the IMF can also be expressed in terms of time-dependent LF correlators or correlations of quantum fields on the LF. Formally, this is simple to see if one writes

$$|P \rightarrow \infty) = U(\Lambda_\infty)|P = 0). \quad (5)$$

The boost operator $U(\Lambda_\infty)$ can be applied to the static nonlocal operators in the ordinary momentum distributions. In doing so, all static correlations become light-cone ones. The boost process is then similar to shifting the Hamiltonian evolution in quantum mechanics from Schrödinger to Heisenberg picture where time-dependence is now in the operators.

To express light-cone correlations, it is convenient to introduce two conjugate light-like (or light-cone) vectors, $p^\mu = (\Lambda, 0, 0, \Lambda)$ and $n^\mu = (1/2\Lambda, 0, 0, -1/2\Lambda)$, with the following properties, $n^2 = p^2 = 0$, and $n \cdot p = 1$, where Λ is a parameter. Then any four-vector can be expanded as,

$$k^\mu = k \cdot n p^\mu + k \cdot p n^\mu + k_\perp^\mu. \quad (6)$$

In particular, the momentum P^μ of a proton moving in the z -direction can be expressed as

$$P^\mu = p^\mu + (M^2/2)n^\mu, \quad (7)$$

where M is the proton mass.

Using the above notation, one can express the unpolarized quark distribution in the proton as (Collins, 2011a),

$$q(x) = \frac{1}{2} \int \frac{d\lambda}{2\pi} e^{i\lambda x} \langle P | \bar{\psi}(0) \not{n} W(0, \lambda n) \psi(\lambda n) | P \rangle, \quad (8)$$

where ψ is the quark field and W is a gauge-link defined as

$$W(x_2, x_1) = \mathcal{P} \exp[i - ig \int_0^1 dt (x_2 - x_1)_\mu A^\mu(x_1 + (x_2 - x_1)t)] \quad (9)$$

to ensure gauge invariance with $\mathcal{P}$ denoting the path ordering. c indicates the connected contributions only, and will be suppressed in the rest of this work. It is a property of gauge theories in which the charge fields are not gauge-invariant, and the physical distributions must include a beam of collinear gauge particles. Note that the above expression is true for any momentum P (a residual momentum symmetry), in particular, in the rest frame of the nucleon. The x -support of the above distribution is $[-1, 1]$. For negative x , one defines the antiquark distribution with $-q(-x) = \bar{q}(x)$. The above expression has been more familiar in the literature than Feynman’s original formulation of PDFs. In the single quark target, one finds $q(x) = \delta(x - 1)$.

To expose the partons in the above equation, one can follow the QCD light-front quantization (Chang and Ma, 1969; Drell and Yan, 1971; Kogut and Soper, 1970), suggested by Dirac in 1949 (Dirac, 1949). In LFQ (Brodsky *et al.*, 1998), one defines the LF coordinates,

$$\xi^\pm = (\xi^0 \pm \xi^3)/\sqrt{2}, \quad (10)$$

where ξ^+ is the LF “time”, and ξ^- is the LF “spatial coordinate”. And any four-vector A^μ will be now written as $(A^+, A^-, \vec{A}_\perp)$. Dynamical degrees of freedom are defined on the $\xi^+ = 0$ plane with arbitrary ξ^- and $\vec{\xi}_\perp$, with conjugate momentum k^+ and $\vec{k}_\perp$. Dynamics is generated by the light-cone Hamiltonian $H_{LC} = P^-$. For a free particle with three-momentum $(k^+, k_\perp)$ and mass m , the on-shell LF energy is $k^- = (\vec{k}_\perp^2 + m^2)/(2k^+)$.

For QCD, one can define the Dirac matrices $\gamma^\pm = (\gamma^0 \pm \gamma^3)/\sqrt{2}$, and the projection operators for the quark fields as $P_\pm = (1/2)\gamma^\pm \gamma^\pm$, so that any ψ can be decomposed into $\psi = \psi_+ + \psi_-$ with $\psi_\pm = P_\pm \psi$, where ψ_+ is considered as a dynamical degree of freedom. For the gauge field, A^+ is fixed by the LF gauge $A^+ = 0$. $A_\perp$ are dynamical degrees of freedom. ψ_- and A^- are dependent variables, which can be expressed in terms of ψ_+ and $A_\perp$ using equations of motion (Kogut and Soper, 1970).

The physics of the LF correlations becomes manifest if

one introduces the canonical expansion,

$$\begin{aligned} \psi_+(\xi^+ = 0, \xi^-, \tilde{\xi}_\perp) = & \int \frac{d^2 \tilde{k}_\perp}{(2\pi)^2} \frac{d\tilde{k}^+}{2\tilde{k}^+} \sum_\sigma \left[b_\sigma(k) u(k, \sigma) \right. \\ & \left. \times e^{-i(k^+ \xi^- - \tilde{E}_\perp \cdot \tilde{\xi}_\perp)} + d_\sigma^\dagger(k) v(k, \sigma) e^{i(k^+ \xi^- - \tilde{E}_\perp \cdot \tilde{\xi}_\perp)} \right], \quad (11) \end{aligned}$$

where $b^\dagger(k)$ and $d^\dagger(k)$ ($b(k)$ and $d(k)$) are quark and anti-quark creation (annihilation) operators, respectively. σ is the light-cone helicity of the quarks which can take $+1/2$ or $-1/2$. Covariant normalization is adopted for the particle states and the creation and annihilation operators, i.e.,

$$\begin{aligned} \{b_\sigma(k), b_{\sigma'}^\dagger(k')\} &= \{d_\sigma(k), d_{\sigma'}^\dagger(k')\} \\ &= (2\pi)^3 \delta_{\sigma\sigma'} 2k^+ \delta(k^+ - k'^+) \delta^{(2)}(\tilde{k}_\perp - \tilde{k}'_\perp). \quad (12) \end{aligned}$$

Substituting the above expansion into Eq. (8), one finds the quark distribution as

$$q(x) = \frac{1}{2x} \sum_\sigma \int \frac{d^2 \tilde{k}_\perp}{(2\pi)^2} \langle P | b_\sigma^\dagger(x, \tilde{k}_\perp) b_\sigma(x, \tilde{k}_\perp) | P \rangle / \langle P | P \rangle \quad (13)$$

for $x > 0$, and similarly for $x < 0$ for which one gets the antiquark distribution. The factor $1/x$ comes from the normalization of the creation and annihilation operators. The matrix element above should be interpreted as the matrix element in a wave packet state, in the limit of a state of definite momentum (Collins, 2011a). This way, one recovers the physical meaning of PDFs in the LF correlator (operator) formalism.

C. Other Approaches to Parton Structure

Calculating the partonic structure of the hadrons from QCD has always been an important goal in hadronic physics. There have been two main approaches apart from various phenomenology and models: light-front quantization and lattice QCD. Here the authors give a very brief review on LFQ and lattice approaches that are different from the main subject of this review.

Although LFQ explicitly uses the parton degrees of freedom, it has not been very successful in practical calculations. First of all, LF perturbation theory, like the standard Hamiltonian perturbation theory, breaks Lorentz symmetry manifestly and requires a sophisticated renormalization scheme to restore it. A potential renormalization scheme must deal with the long-range correlations in the ξ^- direction which require functional dependence on the renormalization counterterms (Wilson *et al.*, 1994). Thus LF perturbation theory has not been used for any calculations beyond one loop, except for the two-loop anomalous magnetic moment in QED (Langnau and Burkardt, 1993). In fact, the common wisdom of using dimensional regularization

(DR) for the transverse integrals and cut-off regularization for the longitudinal one has not been proven useful for multi-loop calculations, although it has been successfully used to derive the BFKL evolution by Mueller from the quarkonium wave functions (Mueller, 1994a).

The enthusiasm for using LFQ in QCD is not about perturbation theory, but to solve the hadron states. Discretized LFQ was proposed in (Pauli and Brodsky, 1985) to make practical calculations for the bound state problems. This non-perturbative method turns out to be successful for models in 1+1 dimension, such as the Schwinger model (Harada *et al.*, 1996; McCartor, 1994), the 1+1 QCD (Burkardt, 1989; Srivastava and Brodsky, 2001), the 1+1 ϕ^4 theory (Harindranath and Vary, 1987) and the sine-Gordon model (Burkardt, 1993). For 3+1 dimensional theories, simple approximations have been considered, like the Tamm-Dancoff approximation (Perry *et al.*, 1990). For QCD itself, one again has to use severe truncations in the number of Fock states. Some recent works of this type include (Jia and Vary, 2019; Lan *et al.*, 2019; Vary *et al.*, 2010). However, to derive a fully-renormalized hamiltonian is difficult and moreover, there has been no demonstration so far that the Fock-space truncation actually converges (Wilson *et al.*, 1994). Therefore a systematic approximation for QCD bound states in LFQ has yet to be found.

Given the rapid development in lattice QCD, it is natural to use it to compute parton physics. However, simulating real-time evolution directly is numerically challenging, which runs into the so-called sign problem or more generally NP-hard problem. Over the years, a number of methods have been proposed previously to indirectly calculate the PDFs, which includes well-studied moment methods, hadronic tensor and Compton amplitude method, coordinate space factorization, etc. These approaches calculate lattice observables that can be related to the PDFs/structure functions through OPE or the dispersion relation, and thus can be used to probe certain information on the partonic structure of hadrons. However, their aims are mainly to get the lower moments of PDFs and/or segments of certain coordinate correlations, not directly in parton degrees of freedom.

The most-adopted approach on the lattice has been to calculate the moments of PDFs as the matrix elements of local operators (Kronfeld and Photiadis, 1985; Martinelli and Sachrajda, 1987). In the moments approach, one starts with the so-called twist-two operators (Christ *et al.*, 1972),

$$O^{\mu_1 \dots \mu_n} = \bar{\psi} \gamma^{\{\mu_1} D^{\mu_2} \dots D^{\mu_n\}} \psi - \text{trace} \quad (14)$$

in the quark case, where $(\mu_1 \dots \mu_n)$ indicates that all the indices are symmetrized, the trace terms are those with at least one factor of the metric tensor $g^{\mu_i \mu_j}$ multiplied by operators of dimension $(n+2)$ with $n-2$ Lorentz

indices, etc. Their matrix elements in the proton state are

$$\langle P | O^{\mu_1 \dots \mu_n}(\mu) | P \rangle = 2a_n(\mu) (P^{\mu_1} \dots P^{\mu_n} - \text{trace}), \quad (15)$$

and the PDFs are related to the local matrix elements through

$$\begin{aligned} a_n(\mu) &= \int_{-1}^1 dx x^{n-1} q(x, \mu^2) \\ &= \int_0^1 x^{n-1} [q(x, \mu^2) + (-1)^n \bar{q}(x, \mu^2)] \end{aligned} \quad (16)$$

with $n = 1, 2, \dots$. The time-dependent correlation for the PDF in Eq. (8) is recovered by taking all the components as + in Eq. (15),

$$\langle P | O^{+ \dots +}(\mu) | P \rangle = 2a_n(\mu) P^+ \dots P^+, \quad (17)$$

and packaging all the moments into a distribution. Likewise, for the gluon PDF, its moments are again related to the matrix elements of local operators,

$$O^{\mu_1 \dots \mu_n} = -g^{\mu_1 \mu_2} D^{\mu_2 \dots \mu_n} + i D^{\mu_2 \dots \mu_{n-1}} P^{\mu_n}, \quad (18)$$

with $n = 2, 4, 6, \dots$

A large number of lattice QCD calculations of PDF moments have been done so far with various degrees of control in systematics (Lin *et al.*, 2018c), which include discretization errors, physical pion mass, finite volume effects, excited state contaminations, and proper renormalization. Most of the lattice calculations have been focused on the first and second moments, $\langle x \rangle$ (Alexandrou *et al.*, 2017a; Bali *et al.*, 2014; Green *et al.*, 2014), and $\langle x^2 \rangle$ (Deka *et al.*, 2009; Dolgov *et al.*, 2002) for the unpolarized distributions, and the zero-th and first moments, $\langle 1 \rangle$ (Alexandrou *et al.*, 2019a, 2017a; Chang *et al.*, 2018; Gong *et al.*, 2017), and $\langle x \rangle$ (Abdel-Rehim *et al.*, 2015; Aoki *et al.*, 2010) for the polarized distributions. However, it has been difficult to calculate higher moments, due to power divergences and rapid decay in signals. Nonetheless, moment calculations can provide a useful calibration for any comprehensive lattice approach to PDFs.

To get more information about the PDFs, it was proposed to calculate the hadronic tensor of DIS in Euclidean space, and analytically continue the result to Minkowski space (Liu, 2000, 2016, 2017, 2020; Liu and Dong, 1994; Liu *et al.*, 1999). Since numerical methods for analytical continuation are known to be difficult for precision control (similar to NP-hard or sign problem mentioned earlier), the approach is useful mainly for the nucleon low-lying excitations. It is very challenging to obtain parton physics this way.

A similar approach called “operator product expansion (OPE) without OPE” was suggested in (Aglietti *et al.*, 1998; Martinelli, 1999), see also (Capitani *et al.*, 1999a,b;

Dawson *et al.*, 1998). The point is that the Compton amplitude in the non-dispersive region can be calculated in the Euclidean space (Ji and Jung, 2001). Through dispersion relation and Taylor-expansion at $\nu = P \cdot q = 0$, one can extract the higher moments of structure functions from the lattice Compton amplitude. The recent works and references for parton structure from this approach can be found in (Chambers *et al.*, 2017; Hannafor-Gunn *et al.*, 2020; Horsley *et al.*, 2020). A similar method has been adopted for Compton amplitude with heavy-light currents (Detmold and Lin, 2006). This approach has been used to calculate the second moment of pion distribution amplitude (Detmold *et al.*, 2020, 2018).

The current-current correlators can also be studied through OPE in the coordinate space without momentum insertion into the currents (Braun and Müller, 2008). The spatial correlation at small distances can be used to calculate higher-moments of distribution amplitudes of the mesons. A number of lattice studies have been performed in (Bali *et al.*, 2019, 2018a; Braun *et al.*, 2015). Similar strategy has been suggested more recently by Qiu *et al.* (Ma and Qiu, 2018a) for parton distributions, and has been used in lattice simulations (Sufian *et al.*, 2020, 2019). The pseudo-PDF has been proposed based on the equal-time correlation—or the quasi-PDF in Fourier space—used in LaMET (Radyushkin, 2017a, 2019b), and uses a coordinate-space factorization or OPE at small distance as in (Braun and Müller, 2008). Because of its close connection with the quasi-PDF, we will discuss comparisons of the pseudo-PDF data analysis method with that for the quasi-PDF in Sec. III C.

There have been pioneering studies on moments of the “quasi” quark TMDPDFs on lattice (Engelhardt *et al.*, 2016; Hagler *et al.*, 2009; Musch *et al.*, 2012, 2011; Yoon *et al.*, 2017). The staples-shaped gauge link operators have been used to connect the quark fields separated in the spatial direction to simulate the moments of TMDPDF. The ratios of these moments are presumed independent of the unknown soft function and may be compared with experimental data. However, a rigorous relation of these constructions to the physical moments of TMDPDFs had not been investigated before LaMET, particularly the relationship between large momentum limit and the rapidity cutoff which is an essential ingredient of TMD physics. Comparison of this approach and LaMET will be made in Sec. V B.

II. LARGE-MOMENTUM EFFECTIVE THEORY

As has been explained in Sec. I A, Feynman’s partons were motivated from describing the structure of a bound state travelling at large momentum P . On the other hand, in QCD factorizations, they appear as effective degrees of freedom arising in infinite momentum

limit disregarding UV divergences. Reconciling these two pictures results in large-momentum effective theory (LaMET) for the parton structure of hadrons.

In this section, we start by considering the structure of the proton at finite momentum. We define the ordinary momentum distributions of the constituents, and trying to illustrate their dependence on the proton momentum. We demonstrate that the large- P momentum-dependence follow a RGE, similar to the well-known RGE for partons. In Sec. II.C, we show that momentum distributions at large P , are related to PDFs through a matching between different orders of $P \rightarrow \infty$ and UV cut-off limits. This matching process has a standard EFT explanation: Parton physics or observables can be obtained from an effective theory in which $P \ll \Lambda_{UV}$ are calculated non-perturbatively in the so-called $\mathcal{P}$ space (Messiah, 1979), after “integrating out” degrees of freedom between P and ∞ (or $\mathcal{Q} = 1 - \mathcal{P}$ space) through perturbation theory. Therefore, the LaMET approach to partons is in some sense similar to lattice QCD as an EFT approach for continuum field theories, in which all active degrees of freedom ($\mathcal{P}$ space) are bounded by $|k| \leq \pi/a$, where a is lattice spacing, whereas those at $|k| \geq \pi/a$ ($\mathcal{Q}$ space) are taken into account through perturbative coefficients and higher dimensional operators.

In Sec. II.D, we outline the formalism of LaMET for a general parton observable. The method can in principle be used also to calculate any LF correlations in terms of large momentum external states (see in particular the application to soft function in Sec. V). The strategy is also applicable for the components of the LF wave functions. Thus, LaMET offers a practical and systematic way to carry out the program of LFQ. Instead of working with the LF coordinates directly, one uses the instant form of dynamics and large momentum or boost factor γ as a regulator for the LF divergences. In a certain sense, the quantization using tilted light-cone coordinates (Lenz *et al.*, 1991) is similar to the spirit of the LaMET approach.

At present, the only systematic approach to solve non-perturbative QCD is lattice field theory (Wilson, 1974). Therefore, a practical implementation of LaMET can be done through lattice calculations. It can also be done with other bound-state methods using Euclidean approaches, such as the instanton liquid model (Schäfer and Shuryak, 1998). While LFQ may provide an attractive physical picture for the proton, the Euclidean equal-time formulation is more practical for carrying out the calculations, and LaMET serves to bridge them.

A. Structure of the Proton at Finite Momentum

In relativistic theories, the internal structure of a composite system is frame-dependent (we always refer to the

total momentum eigenstates), and we are interested in the properties of the proton at a momentum much larger than its rest mass.

We start from the quark momentum density in a fast-moving proton, assuming that it moves in the z -direction. A straightforward definition is

$$N_P(\vec{k}) = \sum_{\sigma} \langle P | b_{\sigma}^{\dagger}(\vec{k}) b_{\sigma}(\vec{k}) | P \rangle / (P^0 P^z), \quad (19)$$

where the quark helicity, color, and other implicit indices are summed over. This equation should be compared with the parton density in Eq. (13). To make it gauge invariant, it is convenient to consider the definition from a coordinate-space correlator,

$$N_{P,W}(\vec{k}) = \int \frac{d^3\xi}{(2\pi)^3} e^{-i\vec{k}\cdot\vec{\xi}} \langle P | \bar{\psi}(0) \gamma^0 W(0, \vec{\xi}) \psi(\vec{\xi}) | P \rangle, \quad (20)$$

where the Dirac matrix γ^0 ensures that it is a number density. Clearly, it is a static quantity without time-dependence and can be calculated in Euclidean field theories, in contrast to Eq. (8) for partons. The gauge invariance is ensured by the Wilson line $W(0, \vec{\xi})$ between the quark fields separated by $\vec{\xi}$, which is defined in the fundamental representation of the color SU(3) group. There are infinitely many choices for the Wilson line, generating infinitely many momentum densities. For example, one can choose a straight-line link between 0 and $\vec{\xi}$. One can also let the Wilson line run from the fields along the z -direction for a long distance (if not infinity) before joining them together along the transverse direction (a staple).

For its obvious connection to the PDFs, we consider a transverse-momentum integrated, longitudinal-momentum distribution,

$$\begin{aligned} N_P(k^z) &= \int d^2\vec{k}_{\perp} N_{P,W}(\vec{k}) \\ &= \int \frac{dz}{2\pi} e^{-ik^z z} \langle P | \bar{\psi}(0) \gamma^0 W(0, z) \psi(z) | P \rangle, \end{aligned} \quad (21)$$

where we ignore the question of convergence at large $|\vec{k}_{\perp}|$. Now the gauge-link $W(0, z)$ is naturally taken as a straight-line,

$$\begin{aligned} W(0, z) &= \exp(-i \int_0^z A^z(z') dz') \\ &= \exp(i \int_{\infty}^0 A^z(z') dz') \exp(i \int_z^{\infty} A^z(z') dz') \\ &= W^{\dagger}(\infty, 0) W(\infty, z), \end{aligned} \quad (22)$$

where in the second line we have split the gauge link into two, going from z to the infinity and coming back from the infinity to zero. We can define a “gauge-invariant” quark field

$$\Psi(\vec{\xi}) = W(\infty, \vec{\xi}) \psi(\vec{\xi}), \quad (23)$$

and the above density becomes,

$$N_P(k^2) = \int \frac{dz}{2\pi} e^{-ik^+z} \langle P | \bar{\Psi}(0) \gamma^0 \Psi(z) | P \rangle, \quad (24)$$

where again we have not considered UV divergences. The momentum distribution defined above has been called *quasi-PDF*, but it is really a physical momentum distribution in a proton of momentum P .

In the rest frame of the proton, $N_{P=0}(k^2)$ is symmetric in positive and negative k^2 , probably peaks around $k^2 = 0$ and decays away as $k^2 \rightarrow \pm\infty$. Due to the perturbative QCD effects, it decays algebraically at large k^2 , instead of exponentially. Because of this property, the high moments of the distribution, $\int dk^2 (k^2)^n N_0(k^2)$ with $n > 0$, have the standard QFT UV divergences.

As P becomes non-zero and large, the peak $N_P(k^2)$ will be around αP^2 , where α is a constant of order one. The density at negative k^2 becomes smaller, but not zero. This is due to the so-called backward-moving particles from the large momentum kick in perturbation theory. For the same reason, the density at $k^2 > P^2$ is not zero either.

$N_P(k^2)$ has a renormalization scale dependence because the quark fields must be renormalized. One can choose DR and modified minimal subtraction ($\overline{\text{MS}}$) scheme. Any other regularization scheme can be converted into this one perturbatively. For $z \neq 0$, the only renormalization necessary is the quark wave function (with anomalous dimension γ_P) in the $A^2 = 0$ gauge, because the linear divergence associated with the gauge link vanishes in the $\overline{\text{MS}}$ scheme. More extensive discussions on the renormalization issue, particularly about non-perturbative renormalization, will be made in the following section.

As an example showing how the parton momentum density depends on P , we depict in Fig. 4 the quark wave function amplitude of a meson in the 't Hooft model (1+1 dimensional QCD with $N_c \rightarrow \infty$) ('t Hooft, 1974), the square of which yields the quark momentum density. In this model, a meson of momentum P^0 can be built as

$$|P^0\rangle = \int \frac{dk}{2\pi|P|} [M(k-P, k) \phi_a^+(k, P) + M^1(k, k-P) \phi_a^-(k, P)] |0\rangle, \quad (25)$$

where $M(p, k) = \sum_i d_{-p}^\dagger b_k^\dagger / \sqrt{N_c}$, and $M^1(p, k) = \sum_i b_k^\dagger d_{-p}^\dagger / \sqrt{N_c}$ are annihilation and creation operators for quark-antiquark pairs. The corresponding wave function amplitudes, $\phi_a^+(k, P)$ and $\phi_a^-(k, P)$, satisfy a pair of equations first derived in (Bars and Green, 1978).

The meson bound state defined above has a well-defined large-momentum limit. The wave functions can be expanded in $1/P$, with the corrections starting from $(1/P)^2$. The momenta of the constituents, k and $P-k$, scale in this limit. When plotted as a function of $x = k/P$, the change in the wave function with the magnitude of the momentum is shown in Fig. 4.

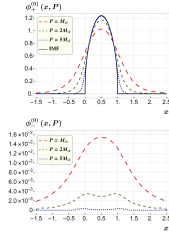


FIG. 4: Wave function amplitudes of a meson in the 't Hooft model at different external momenta (Jia et al., 2017).

B. Momentum Renormalization Group

In this subsection, we consider how to calculate the external momentum P dependence of physical observables discussed in the previous subsection. Clearly, the dependence is related to the boost properties of the operators under consideration, namely their commutation relations with the boost generators, $\hat{K}^1$. We argue that in the large momentum limit, one has a *momentum RGE* which is a differential equation relating properties of the system at different momenta. Momentum RGE will be, in the end, related to the renormalization properties of the observables on the L.F.

Consider a generic operator $\hat{O}$, and its matrix element in a state with momentum P ,

$$O(P) = \langle P | \hat{O} | P \rangle. \quad (26)$$

We calculate the momentum dependence by writing $|P\rangle = \exp(-i\omega(P)\hat{K}^1)|P=0\rangle$, where $\hat{K}^1$ is the boost operator along the momentum direction and ω is a boost parameter depending on P . Taking a derivative with respect to the boost parameter gives

$$\frac{dO(P)}{dP} = i \frac{d\omega(P)}{dP} \langle P | [\hat{O}, \hat{K}^1] | P \rangle. \quad (27)$$

The r.h.s. of the equation depends on the commutator $[\hat{O}, \hat{K}^1]$, i.e., the boost properties of the operator. For a scalar operator, the commutation relation vanishes, and $O(P)$ is frame independent. For a vector operator,

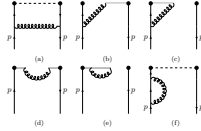


FIG. 5: One-loop diagrams for the quasi-PDF in a free quark state in the Feynman gauge. The conjugate diagrams of (b), (c), (e), (f) do contribute but are not shown here.

the commutation relation resembles that of an energy-momentum four-vector, and the result is the standard Lorentz transformation of a four-vector. For nonlocal operators, the commutation relation requires the elementary formula,

$$[J^{\mu\nu}, \phi_i(x)] = i [t^{\mu\nu} \delta_{ij} + S^{\mu\nu}_{ij}] \phi_j(x), \quad (28)$$

where $t^{\mu\nu} = -i(x^\mu \partial^\nu - x^\nu \partial^\mu)$ is the OAM operator and $S^{\mu\nu}$ is the intrinsic spin matrix. Thus one of the fields is now $\phi_i(t = \sinh\omega z, 0, \cosh\omega z)$ which generates a time-dependent correlation function.

In the large-momentum limit, because of asymptotic freedom, the P -dependence is calculable in perturbation theory, and Eq. (27) simplifies. One obtains the momentum or boost RGE (Ji, 2014),

$$\frac{dO(P)}{dP^2} = \lim_{\Delta P^2 \rightarrow 0} [O(P + \Delta P) - O(P)]/\Delta P^2 \quad (29)$$

$$\xrightarrow{P^2 \gg M^2} C(\alpha_s(P)) \otimes O(P) + O(M^2/P^2). \quad (30)$$

where $C(\alpha_s(P))$ is a perturbative expansion in the strong coupling α_s . The symbol “ $\otimes$ ” can be a simple multiplication or certain form of convolution, depending on the observable $O(P)$ studied. The proof of the above equation is non-trivial, and it can be analyzed on a case-by-case basis. There can be mixings among a set of independent operators with the same quantum numbers. The momentum RGEs are very similar to those for scale transformation or that for the coarse graining of a Hamiltonian. That the two are connected in some cases may be traced to Lorentz symmetry.

As an example of the momentum RGE, we calculate the quark momentum distribution in a perturbative quark state using Eq. (24). Since it is gauge invariant, we can calculate it in any gauge, for example, the Feynman gauge. The one-loop diagrams in QCD are shown in Fig. 5. There are two sources of UV divergences, one is the logarithmic divergences from the vertex and self-energy diagrams, and the other is the linear divergence

in the self-energy of the Wilson line. For the moment, we will use transverse momentum cut-off, Λ_{UV} , as the UV regulator. Using $y = k^2/P^2$, the one-loop result reads for a large momentum quark (Xiong *et al.*, 2014),

$$\begin{aligned} \hat{q}^{(1)}(y, P^2, \Lambda_{UV}) &= \frac{\alpha_s C_F}{2\pi} \\ &\times \begin{cases} \frac{1+y^2}{1-y} \ln \frac{1}{1-y} + 1 + \frac{\Lambda_{UV}^2}{(1-y)^2 P^2}, & y > 1 \\ \frac{1+y^2}{1-y} \ln \frac{P^2}{m^2} + \frac{1+y^2}{1-y} \ln \frac{4m}{1-y} \\ - \frac{2}{1-y} + 1 + \frac{\Lambda_{UV}^2}{(1-y)^2 P^2}, & 0 < y < 1 \\ \frac{1+y^2}{1-y} \ln \frac{1}{1-y} - 1 + \frac{\Lambda_{UV}^2}{(1-y)^2 P^2}, & y < 0 \end{cases} \quad (31) \end{aligned}$$

where we have ignored all power-suppressed contributions and keep the leading P^2 dependence only. There is an additional contribution of the form $\delta Z_1(\Lambda_{UV}/P^2)\delta(y-1)$.

The above result has several interesting features:

- The distribution does not vanish outside $[0, 1]$. The radiative gluon can carry a large negative momentum fraction, resulting in a recoiling quark carrying larger momentum than the parent quark, and thus $y > 1$. The same gluon can also carry a momentum larger than P^2 , making the active quark have $y < 0$.
- While the above effect is easy to understand perturbatively, it is surprising that a scaling contribution remains outside $[0, 1]$ in the IMF. As the proton travels faster, one might think any constituent has a momentum k^+ positive from Lorentz transformation. However, the order of limits matters because no matter how large the parent-quark momentum is, there are always quarks with much larger momentum, i.e., $k^+ \gg P^2 \gg \Lambda_{QCD}$. In this sense, Feynman’s parton model does not describe the exact properties of the momentum distribution in a large-momentum nucleus.
- The contribution outside $[0, 1]$ at one-loop is entirely perturbative because of the absence of any infrared (IR) divergence. This is no longer true at two-loop level, but the contribution depends only on the same one-loop IR physics in $[0, 1]$.
- The distribution for y in $[0, 1]$ has a term depending on $\ln P^2$. This dependence reflects that the quark substructure is resolved as a function of P^2 , an interesting feature of boost. This dependence is perturbative in the sense that the derivative is IR safe,

$$\begin{aligned} P^2 \frac{d\hat{q}(y, P^2, \Lambda_{UV})}{dP^2} &= \frac{\alpha_s C_F}{\pi} \left[\left(\frac{1+y^2}{1-y} \right)_+ - \frac{3}{2} \delta(1-y) \right]. \quad (32) \end{aligned}$$

Apart from the δ -function term, the r.h.s. is similar to the one-loop quark splitting function in DGLAP evolution (Altarelli and Parisi, 1977; Dokshitzer, 1977; Gribov and Lipatov, 1972). Therefore one might suspect that the momentum dependence is closely related to the familiar renormalization scale evolution in the PDFs. In fact, the physics is just the other way around: *It is the hadron-momentum dependence of the physical momentum distribution that generates the DGLAP evolution in the infinite-momentum limit.* One can derive an all-order momentum RGE for the momentum distribution function. Momentum RGE also provides a method to sum over the large logarithms of the momentum.

- There is a singularity at $y = 1$. This singularity is generated from soft-gluon radiation. Fortunately, this singularity combined with the virtual contribution yields a finite result.
- There is a linear divergence in the cut-off regulator, leading to Λ_{UV}/P^2 term, which is absent in DR. Thus, to keep $1/(P^2)^2$ power counting, it is important to work in a renormalization scheme where this term does not exist.

We can also move on to study the hadron momentum RGEs of other structural properties considered in the previous subsection. In particular, the RGE for TMD distributions will lead to the familiar rapidity RGE in the literature. We reserve these discussions to Sec. V.

C. Effective Field Theory Matching to PDFs

As we have seen in the previous subsection, the momentum distributions of the constituents (now called quasi-PDFs in the literature) in a proton at large P are different from the PDFs or LF distributions in many ways. In particular, the momentum fraction y in a physical momentum distribution is not limited to $[0,1]$ due to backward moving particles, which is the case even in the $P \rightarrow \infty$ limit. In fact, the infinite-momentum limit is not analytical due to the presence of $\ln P$.

However, partons are effective objects arising from a different limit $\Lambda_{UV} \ll P \rightarrow \infty$. There is also an important computational advantage in taking the naive limit $P \gg \Lambda_{UV}$ in perturbative calculations: Feynman integrals have one fewer four-momentum. Therefore, this limit of QFTs serves as a reference system where the structure of the bound states is manifestly independent of the hadron momentum, and is similar to scale-invariant critical points at which second-order phase transitions occur in condensed matter systems. However, the theory in the naive IMF limit introduces additional UV divergences.

Therefore, the partons in QCD are very similar to the infinitely heavy quarks in HQET (Manohar and Wise,

2000). In certain QCD systems, heavy quarks such as the bottom quark are present, and their masses are much larger than the typical QCD scale Λ_{QCD} . In this case, one might study the dependence on the heavy quark mass by expanding around $m_Q = \infty$. This expansion will generally produce a power series in $1/m_Q$. However, the limits of taking $\Lambda_{UV} \rightarrow \infty$ and infinite heavy-quark mass limits are not interchangeable, due to the presence of the large logarithms $\ln m_Q$. In an EFT approach, one takes the $m_Q \rightarrow \infty$ limit first, this will result in a new theory with different UV behavior, but without the heavy-quark mass, and symmetries among very different heavy-quark systems become manifest. The renormalization of the extra UV divergences yields a RGE which can be used to resum large quark-mass logarithms.

Therefore, the momentum distribution at large- P differs from the parton distributions only in the order of limits, their IR non-perturbative physics is the same. In asymptotically free theories such as QCD, differences (or discontinuities) in taking the limits of $P \gg \Lambda_{UV}$ and $\Lambda_{UV} \gg P \rightarrow \infty$ are perturbatively calculable, as only the high-momentum modes matter. The differences are called *matching coefficients*. Therefore, one is able to write down a power expansion for the momentum-dependent distribution (quasi-PDF) in terms of the PDF (Izubuchi *et al.*, 2018; Ma and Qiu, 2018a,b; Xiong *et al.*, 2014),

$$\bar{q}(y, P^2, \mu) = \int_{-1}^1 \frac{dx}{|x|} C\left(\frac{y}{x}, \frac{\mu}{xP^2}\right) q(x, \mu) + \mathcal{O}\left(\frac{\Lambda_{QCD}^2}{(yP^2)^2}, \frac{\Lambda_{QCD}^2}{((1-y)P^2)^2}\right), \quad (33)$$

where the power correction is suppressed by the parton momentum yP^2 and the spectator momentum $(1-y)P^2$ (Ji *et al.*, 2020b). This expansion may be also called a factorization formula, as the quasi-PDF contains all the IR physics in the PDF, and C involves only UV physics. As we will discuss extensively in the next section, this factorization formula is true to all orders in perturbation theory. The above relation allows us to calculate the LF parton physics from the momentum distribution at large P . Since the expansion parameter is $\Lambda_{QCD}^2/(yP^2)^2$ and $\Lambda_{QCD}^2/((1-y)P^2)^2$, for intermediate y one might not need very large P^2 to neglect the power corrections.

The above relation between the two quantities has a simple explanation in terms of the Lorentz boost: Consider the spatial correlation along z shown in Fig. 6 in a large momentum state. It can be seen as approaching the LF one in the rest frame of the proton. In other words, we are using a near-LF correlation to approximate a LF correlation. Accordingly, we can invert the above equation recursively to express the PDF in terms of quasi-PDF with their differences being taken care of through

the perturbative matching $\tilde{C}$ and power corrections,

$$q(x, \mu) = \int_{-\infty}^{\infty} \frac{dy}{|y|} \tilde{C}\left(\frac{x}{y}, \frac{\mu}{yP^z}, \mu\right) \tilde{q}(y, P^z, \mu) + \mathcal{O}\left(\frac{\Lambda_{\text{QCD}}^2}{(xP^z)^2}, \frac{\Lambda_{\text{QCD}}^2}{((1-x)P^z)^2}\right). \quad (34)$$

The above equation has an EFT interpretation: The parton physics is calculated in an effective field theory with physical momentum scale from 0 to P_z whereas the physics from degrees of freedom from P to ∞ can be integrated out to generate the perturbative coefficients $\tilde{C}$ and the high-order terms in $1/(P^z)^2$. In contrast to HQET, the full QCD degrees of freedom are used in LaMET calculations. In other words, the effective Lagrangian of LaMET is the standard QCD one, while the large momentum P for expansions appears only in the external states.

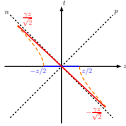


FIG. 6: The line segment in the z -direction in the frame of a large-momentum hadron. Through Lorentz boost, it is equivalent to a line segment of length $\sim \gamma z$ close to the light-cone in the hadron state of zero momentum. Here $\gamma z/\sqrt{2}$ is the length of projection of the boosted line segment to the light-cone direction n . Thus, we call the dimensionless variable $\lambda \equiv zP^z \sim \gamma zM$ as the quasi light-cone distance.

D. Recipe for Parton Physics in LaMET

The principle of LaMET is to simulate the time-dependence of parton observables through external states at large momentum. Thus, we can generalize the discussions in the previous subsection to any type of physical observables for the large momentum proton, which will be generally called quasi-parton observables. Examples will be given in the later sections including transverse-momentum dependent distributions and LF wave functions.

Consider any Euclidean quasi-observable O which depends on a large hadron momentum P^z and UV cut-off $\Lambda_{\text{UV}} \gg P^z$. Using asymptotic freedom, we can system-

atically expand the P^z dependence,

$$O(P^z, \Lambda_{\text{UV}}) = Z\left(\frac{P^z}{\Lambda_{\text{UV}}}, \frac{P^z}{\mu}\right) \otimes o(\mu) + \mathcal{O}\left(\frac{\Lambda_{\text{QCD}}^2}{(P^z)^2}\right) + \dots, \quad (35)$$

where $\otimes$ refers to a convolution if appropriate, and Z factorizes all the perturbative dependence on P^z and does not contain any IR divergence. The quantity $o(\mu)$ is defined in a theory with $P^z \rightarrow \infty$, exactly as in Feynman's parton model. In fact, $o(\mu)$ is a LF correlation containing all the IR collinear (and soft) singularities. The important point of the expansion is that it may converge at moderately large P^z , say a few GeV, allowing access to quantities needed for very large P^z (a few TeV). One can also use the large boost-factor $\gamma = P^z/M$ as the expansion parameter $1/\gamma$.

Momentum dependence of the quasi-observables can be studied through momentum RGEs. Defining the anomalous dimension through

$$\gamma_P(\alpha_s) = \frac{1}{Z} \frac{\partial Z}{\partial \ln P^z}, \quad (36)$$

it follows that

$$\frac{\partial O(P^z)}{\partial \ln P^z} = \gamma_P(\alpha_s) \otimes O(P^z), \quad (37)$$

up to power corrections. One can resum large logarithms involving P^z using the above equation.

When taking $P^z \rightarrow \infty$ first in $O(P^z)$ before a UV regularization is imposed, one recovers from $\hat{O}$ the light-cone operator $\hat{o}$ by construction. On the other hand, the physical matrix element is calculated at a large P^z with UV regularization such as the lattice cut-off imposed first. Thus the difference between the matrix elements of $\hat{o}$ and $\hat{O}$ is a matter of the order of limits. This is the standard set-up for an EFT. The different limits do not change the IR physics. In fact, the factorization in terms of Feynman diagrams can be proved order by order as in the renormalization program, as discussed in the following section.

The parton physics can be calculated more directly by reversing Eq. (35) to produce an EFT expansion,

$$o(\mu) = \tilde{Z}\left(\frac{P^z}{\Lambda_{\text{UV}}}, \frac{P^z}{\mu}\right) \otimes O(P^z, \Lambda_{\text{UV}}) + \mathcal{O}\left(\frac{\Lambda_{\text{QCD}}^2}{(P^z)^2}\right) + \dots \quad (38)$$

Thus, to compute any parton observable defined by an operator made of LF dynamical fields, $\hat{o}$, one constructs a time-independent version $\hat{O}$ which, under an infinite Lorentz boost, approaches $\hat{o}$. Then, one calculates the matrix element of $\hat{O}$ in a hadron with large momentum P^z using whatever approach (lattice QCD is an obvious choice for the time-independent operator $\hat{O}$) and then uses Eq. (38) to systematically approximate the parton

observable. Usually the matrix element of $\hat{O}$ depends on P^2 as well as all the lattice UV artifacts. In principle, the latter does not affect the EFT expansion and will be cancelled by the matching coefficient $\hat{Z}$ and higher order terms in the expansion. However, in practical applications such as the quasi-PDF calculations, a nonperturbative renormalization is still necessary to remove all the power divergences to ensure a continuum limit.

E. Universality

LaMET provides a framework to systematically compute partonic observables on the LF from the properties of a large-momentum proton. However, the relationship is not one-to-one. There can be infinitely many possible Euclidean operators in the large-momentum proton which generate the same LF observable. This is because the large-momentum physical states have built-in collinear (as well as soft) parton modes, and upon acting on a Euclidean operator, they help to *project out* the leading LF physics. All operators projecting out the same LF physics form a *universality class*. Accordingly, in the operator formulation for parton physics such as SCET, one uses LF operators to project out parton physics off the external states of any momentum, including $P = 0$.

The concepts such as universality class have been used in critical phenomena in condensed matter physics, where systems with different microscopic Hamiltonians can have the same scaling properties near their critical points. Critical phenomena correspond to the IR fixed points of the scale transformation, and are dominated by physics at long-distance scales. In the present case, parton physics arises from the infinite-momentum limit, $P = \infty$, which is a UV fixed point of the momentum RGEs. It is the longitudinal short distance (or large momentum) physics that is relevant at the fixed point. However, the short distance here does not mean everything is perturbative. The part that is non-perturbative characterizes the partonic structure of the proton. The critical region near $P = \infty$ acts as a filter to select only the physics that is relevant, so universality classes emerge.

In the case of unpolarized PDFs, the initial proposal in LaMET starts from the matrix element of the following operator (Ji, 2013),

$$O_1(z) = \bar{\psi}(0)\gamma^+W(0,z)\psi(z) . \quad (39)$$

However, one can equally start from (Constantinou and Panagopoulos, 2017; Radyushkin, 2017b),

$$O_2(z) = \bar{\psi}(0)\gamma^0W(0,z)\psi(z) , \quad (40)$$

and the leading contributions in the large-momentum expansion are the same. One can also consider any linear combination of the two. In (Jia *et al.*, 2018), the calculations have been done with these two operators in the

't Hooft model, and the results have been compared at different hadron momenta. For lattice simulations, an important issue is about the operator mixing, which depends on specific choices of the operators in the universality class.

Another example of Euclidean operators for PDFs is the current-current correlators in a pure space separation,

$$O_3(z) = J^\mu(0)J^\mu(z) , \quad (41)$$

where J^μ is, for example, an electromagnetic current. This type of correlator was first considered in (Bali *et al.*, 2018b; Braun and Müller, 2008) for calculating pion DA, and recently has been suggested to calculate PDFs with generalized bilocal ‘currents’ (Ma and Qiu, 2018a). When the matrix elements are calculated in the large-momentum states, $O(z)$ falls into the same universality class as the operators in Eqs. (39) and (40). Instead of using light quarks as the intermediate propagator in $O(z)$, one can have a number of other choices for LaMET applications, including scalars (Abada *et al.*, 2001; Aglietti *et al.*, 1998) and heavy-quarks (Detmold and Lin, 2006). One can also similarly work with quark bilinear operators in any physical gauge which become the light-front one in the large momentum limit (Gupta *et al.*, 1993).

Another important example of universality class is the gluon helicity contribution to the spin of the proton, as we will discuss in detail in Sec. IV.D. The gluon spin operator $\vec{E} \times \vec{A}$ is gauge-dependent. However, in physical gauges where the transverse degrees of freedom are dynamical, its matrix element is the same in the large-momentum limit. Therefore, one can potentially choose different gauges to perform calculations at finite momentum on lattice, such as Coulomb gauge $\vec{\nabla} \cdot \vec{A} = 0$, axial gauge $A^+ = 0$ or temporal gauge $A^0 = 0$. Different gauge choices will have different UV properties (in P) and hence different matching conditions. However, the IR part of the matrix element is the same (Hatta *et al.*, 2014).

At a practical level, it is very useful to find which operator has the fastest convergence in the LaMET expansion. The current-current correlators use the light-quark propagator to simulate the light-like Wilson line (sometimes called light-ray). The quasi-PDF approach not only starts from a quantity with clear physical meaning (a momentum distribution), but also generates the needed Wilson line simply by rotating a space-like one, shown in Fig. 6. Thus, it is plausible that the quasi-PDF will provide mathematically the fastest large- P convergence than the other choices.

III. RENORMALIZATION AND MATCHING FOR PDFS

In this section, we consider the LaMET application to calculating the simplest collinear PDFs, which have been most extensively studied in the literature so far.

Although universality allows one to extract the collinear PDFs from the matrix elements of a wide class of operators evaluated at large momentum, we will focus on physical observables closely resembling the collinear PDFs, i.e., the quark and gluon momentum distributions or the *quasi-PDFs*. We also discuss the coordinate-space factorization approach in which the pseudo-PDF and current-current correlators have been studied.

We mainly review the technical progress made in renormalization and matching using the quasi-PDFs. The matching can be done in principle at the bare matrix elements level, since the factorization formula like Eq. (33) is valid for both bare and renormalized momentum distributions. All the UV divergences in the bare quasi-PDF can be factorized into the matching coefficient C , and the latter automatically renormalizes the bare lattice matrix elements, so the continuum limit can be taken afterwards. However, such a matching coefficient then has to be calculated in lattice perturbation theory, which is computationally challenging and converges slowly (Lepage and Mackenzie, 1993). More importantly, the quasi-PDF contains linear power divergence under UV cutoff regularization due to the Wilson line self-energy (Ji, 2013; Xiong *et al.*, 2014), which makes it impossible to take the continuum limit with fixed-order calculations in lattice perturbation theory. Though the latter problem can be improved by resumming the linear and possibly logarithmic divergences, it is usually preferred to nonperturbatively renormalize the quasi-PDFs on the lattice, after which a continuum limit can be taken and a perturbative matching can be done in the continuum theory. To this end, a thorough understanding of the renormalizability of Wilson-line operators defining the quasi-PDFs is required. In addition to renormalization, the applications of LaMET rely on the validity of the large-momentum factorization formula Eq. (33), which can be proven in perturbation theory to all orders by showing that the collinear divergences are the same in the momentum distributions and light-cone PDFs.

We begin in Sec. III.A with the proof of multiplicative renormalizability of the Wilson-line operators that define the quasi-PDFs. We first work in the continuum theory with $\overline{\text{MS}}$ scheme, and then generalize the conclusion to lattice theory. Next, in Sec. III.B we outline the factorization theorem for momentum distributions to all orders in perturbation theory, and state the form of convolution between the matching coefficient and the PDF. In Sec. III.C we show that the factorization theorem has an equivalent form in coordinate space, which can be used as an alternate route to extract PDFs from lattice matrix elements. Finally, we discuss the nonperturbative renormalization of quasi-PDFs on the lattice and their matching to the $\overline{\text{MS}}$ PDF in Sec. III.D.

A. Renormalization of Nonlocal Wilson-Line Operators

The momentum distributions of the proton are defined from equal-time nonlocal Wilson line operators of the form in Eq. (21). In this subsection, we review the renormalization of these spacelike nonlocal operators (the renormalization of lightlike nonlocal operators defining the PDFs can be found in (Collins, 2011a; Collins and Soper, 1982b)). We first discuss their renormalization in DR using an auxiliary field approach, followed by the discussion on similar gluon operators. We then consider power divergences in the momentum cutoff type of UV regularization. The conclusion is that they are all multiplicatively renormalizable with a finite number of mixings with other operators.

1. Renormalization of nonlocal quark operators

We are interested in operators of the following kind,

$$O_T(z) = \bar{\psi}\left(\frac{z}{2}\right) \Gamma W\left(\frac{z}{2}, -\frac{z}{2}\right) \psi\left(-\frac{z}{2}\right). \quad (42)$$

Since the Wilson line $W(z_1, z_2)$ is a path-ordered integral of gauge fields, it is not obvious that such operators are multiplicatively renormalizable. The renormalization of non-lightlike Wilson loops and Wilson lines has been studied in early literature (Craigie and Dorn, 1981; Dotsenko and Vergèes, 1980), and the all-order proof of their multiplicative renormalizability was first made using diagrammatic methods in (Craigie and Dorn, 1981; Dotsenko and Vergèes, 1980) and then the functional formalism of gauge theories in (Dorn, 1986). The same conclusion was conjectured to hold also for the quark bilinear operator $O_T(z)$, whose renormalization takes the following form (Chen *et al.*, 2017; Ishikawa *et al.*, 2016; Musch *et al.*, 2011),

$$O_T^B(z, \Lambda) = Z_{\psi, z}(\Lambda, \mu) e^{\delta m(\Lambda) |z|} O_T^B(z, \mu), \quad (43)$$

where “ B ” and “ R ” stand for bare and renormalized operators respectively, and all the fields and couplings in $O_T^B(z, \Lambda)$ are bare ones which depend on the UV cutoff Λ . $\delta m(\Lambda)$ is the “mass correction” of the Wilson line, which includes all the linear power divergences of its self-energy. $Z_{\psi, z}(\Lambda, \mu)$ includes all the logarithmic divergences from wavefunction and vertex renormalizations.

An early two-loop study of the quasi-PDF in the $\overline{\text{MS}}$ scheme indeed indicated the multiplicative renormalizability of $O_T(z)$ (Ji and Zhang, 2015). The first rigorous proof of Eq. (43) was given in the auxiliary “heavy quark” field formalism (Green *et al.*, 2018; Ji *et al.*, 2018) which was used to prove the renormalizability of Wilson lines (Arefeva, 1986; Dorn, 1986; Gervais and Neveu, 1980; Samuel, 1979). This formalism is defined by extending the QCD Lagrangian to include the auxiliary

“heavy quark” fields $Q, \bar{Q}$ and their gauge interaction,

$$\mathcal{L} = \mathcal{L}_{\text{QCD}} + \bar{Q}_0 \text{in}_z \cdot D_0 Q_0, \quad (44)$$

where the subscript “0” denotes bare quantities. $n_z^\mu = (0, 0, 0, 1)$ is the direction vector of the spacelike Wilson line $W(z, 0)$, $D_0^\mu = \partial^\mu + ig_0 A_0^\mu$, and Q_0 is a color-triplet scalar Grassmann field in the fundamental representation of $\text{SU}(3)$. Note that if we replace n_z^μ with the timelike vector $n_t^\mu = (1, 0, 0, 0)$, then Eq. (44) yields the leading order HQET Lagrangian.

In the theory defined by Eq. (44), the Wilson line can be expressed as the connected two-point function of the “heavy-quark” fields,

$$\langle Q_0(\xi) \bar{Q}_0(\eta) \rangle_Q = S_Q^0(\xi, \eta), \quad (45)$$

where ξ and η are space-time coordinates, and $\langle \dots \rangle_Q$ stands for integrating out the auxiliary fields. The above equation is valid up to the determinant of $\text{in}_z \cdot D_0$, which is a constant and can be absorbed into the normalization of the generating function (Mannel *et al.*, 1992). The Green’s function $S_Q^0(\xi, \eta)$ satisfies

$$\text{in}_z \cdot D_0(\xi) S_Q^0(\xi, \eta) = \delta^{(4)}(\xi - \eta), \quad (46)$$

which has the solution

$$S_Q^0(\xi, \eta) = W(\xi^3, \eta^3) \theta(\xi^3 - \eta^3) \delta(\xi^\perp - \eta^\perp) \quad (47)$$

with a proper choice of boundary condition. In this way, the Wilson-line operator $O_Q^B(z)$ can be replaced by the product of two local composite operators averaged over all the “heavy-quark” field configurations (Dorn, 1986),

$$O_Q^B(z) = \int d^4\xi \, \delta(\xi^3 - z) \times \langle \bar{\psi}_0(\frac{\xi}{2}) Q_0(\frac{\xi}{2}) \Gamma Q_0(-\frac{\xi}{2}) \bar{\psi}_0(-\frac{\xi}{2}) \rangle_Q, \quad (48)$$

where the UV regulator is suppressed.

Consequently, the renormalization of $O_Q^B(z)$ is reduced to that of the two local “heavy-to-light” currents

$$J^B = Q_0 \bar{\psi}_0. \quad (49)$$

The renormalizability of this auxiliary field theory has been proven using the standard functional techniques for gauge theories (Dorn, 1986). After fixing the covariant gauge and introducing the ghost fields, the theory including the auxiliary “heavy-quark” has a residual BRST symmetry, from which one can derive the Ward-Takahashi identities to show that all the UV divergences of the Green’s functions can be subtracted with a finite number of local counterterms. In analogy, the same method has also been used to prove the all-order renormalization of HQET in perturbation theory (Bagan and Goschinsky, 1994).

According to (Dorn, 1986), the “heavy-quark” Lagrangian can be renormalized in a covariant gauge as

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{QCD}}[g_0, \bar{\psi}_0, A_0, c_0] + \bar{Q}_0 \text{in}_z \cdot D_0 Q_0 \\ &= \mathcal{L}_{\text{QCD}}[g, \bar{\psi}, A, c] + \mathcal{L}_{c, \perp}[g, \bar{\psi}, A, c] \\ &\quad + Z_Q Q(\text{in}_z \cdot \partial - i\delta m) Q - g Z_1^{Q\bar{Q}g} \bar{Q} m_z \cdot A_\perp \epsilon^a Q, \end{aligned} \quad (50)$$

where $\mathcal{L}_{c, \perp}[g, \bar{\psi}, A, c]$ are the QCD counterterms, and the bare fields and coupling are related to the renormalized ones through

$$\bar{\psi}_0 = Z_\psi^\dagger \bar{\psi}, \quad A_0 = Z_A^\dagger A, \quad Q_0 = Z_Q^\dagger Q, \quad g_0 = Z_g g. \quad (51)$$

The heavy-quark-ghon vertex renormalization constant $Z_1^{Q\bar{Q}g}$ is related to Z_g through the Slavnov-Taylor identities of the auxiliary field theory (Dorn, 1986),

$$Z_g = Z_1^{Q\bar{Q}g} Z_A^{-\frac{1}{2}} Z_Q^{-1}. \quad (52)$$

The $i\delta m$ can be regarded as the mass correction of the “heavy quark” except that it is imaginary. For Dirac fermions, the mass correction is logarithmically divergent and proportional to the bare mass, as a result of chiral symmetry; for HQET, the mass correction of the heavy quark is proportional to the UV cutoff, i.e. linearly divergent, which is also expected for the auxiliary field here. Since the proof of renormalizability for this auxiliary field theory is carried out in the $\overline{\text{MS}}$ scheme with DR ($d = 4 - 2\epsilon$), all power divergences vanish, so does $i\delta m$. Nevertheless, $i\delta m$ may include $\mathcal{O}(\Lambda_{\text{QCD}})$ contributions due to the renormalon ambiguities which are known to exist in HQET (Beneke and Braun, 1994; Bigi *et al.*, 1994).

Since the auxiliary field theory is renormalizable, the renormalization of the operator product in Eq. (48) amounts to the renormalizations of the two “heavy-to-light” currents. Using the standard techniques in quantum field theory (Collins, 1986), one can show recursively that the overall UV divergence of the insertion of J^B into Green’s functions is absorbed into a renormalization factor Z_J to all orders in perturbation theory,

$$J^B = Z_J J^B = Z_\psi^{1/2} Z_A^{1/2} Z_V J^B, \quad (53)$$

where Z_V is the vertex renormalization constant of the “heavy-to-light” current. The renormalization of heavy-to-light currents in HQET has been calculated up to three-loop order in perturbative QCD (Broadhurst and Grozin, 1991; Chetyrkin and Grozin, 2003; Ji and Musolf, 1991; Politzer and Wise, 1988; Shifman and Voloshin, 1987). More recently, it has been argued that the anomalous dimension of the “heavy-to-light” current is identical to that in HQET to all orders (Braun *et al.*, 2020), which is also the case for the “heavy-to-ghon” current that will be discussed below, so the renormalization factors for

the spacelike and timelike Wilson line operators should be exactly the same.

Using the above results, we can show that

$$\begin{aligned} O_F^B(z) &= Z_F^2 \int d^4\xi \delta(\xi^3 - z) \langle J^B(\frac{\xi}{2}) \Gamma J^B(-\frac{\xi}{2}) \rangle_Q \\ &= Z_F^2 e^{\delta m|z|} O_F^B(z), \end{aligned} \quad (54)$$

where δm arises from the determinant of $(in_z \cdot \partial - \delta m)$ in Eq. (50). In this way, we identify that $Z_{\psi,z} = Z_F^2$ in Eq. (43) which is independent of Γ . At one-loop order (Stefanis, 1984),

$$Z_{\psi,z} = 1 + \frac{\alpha_s C_F}{4\pi} \frac{3}{\epsilon_{UV}}, \quad (55)$$

where the UV regulator ϵ_{UV} is to be distinguished from the IR regulator ϵ_{IR} in DR.

The multiplicative renormalizability of $O_F^B(z)$ has also been proven with a recursive analysis of all-order Feynman diagrams (Ishikawa *et al.*, 2017). In addition to Eq. (43), it was found that $O_F^B(z)$ does not mix with gluons or quarks of other flavors. This can also be easily understood within the auxiliary field formalism, as the flavor-changing “heavy-to-light” current does not mix with other operators (Green *et al.*, 2020).

Finally, under lattice regularization we can still use the above techniques to prove Eq. (54), where the mass correction δm is now nonvanishing and equal to the lattice UV cutoff $1/a$ multiplied by a perturbative series in the coupling constant α_s .

2. Renormalization of nonlocal gluon operators

Using the same “heavy-quark” auxiliary field formalism, it has also been proven that the Wilson-line operators for the gluon quasi-PDF are multiplicatively renormalizable (Zhang *et al.*, 2019b), which is echoed by the diagrammatical proof in (Li *et al.*, 2019).

According to LaMET, the gluon quasi-PDF can be defined as (Ji, 2013)

$$\tilde{g}(x, P^+) = N \int \frac{d\lambda}{4\pi x(P^+)^2} e^{i\lambda z} \langle P | O_g(z) | P \rangle, \quad (56)$$

where N is a normalization factor, and

$$O_g^B(z) = g_{\perp,\mu\nu} F_{0,a}^{n_1,\mu}(\frac{z}{2}) W^{ab}(\frac{z}{2}, -\frac{z}{2}) F_{0,b}^{n_2,\nu}(-\frac{z}{2}) \quad (57)$$

with $F_{0,a}^{\alpha\mu} = n_\mu F_{0,a}^{\alpha\mu}$ and n_1^μ, n_2^μ being either n_c^μ or n_s^μ . a, b are color indices in the adjoint representation. The transverse metric tensor

$$g_\perp^{\mu\nu} = g^{\mu\nu} - n_c^\mu n_c^\nu / n_c^2 - n_s^\mu n_s^\nu / n_s^2, \quad (58)$$

and $N = (n_z \cdot P / n_\perp \cdot P)^{n_1(n_1+n_2)+n_4}$. For lattice implementation, $O_F^B(z)$ can also be defined as (Dorn, 1986;

Zhang *et al.*, 2019b)

$$O_g^B(z) = 2g_\perp^{\mu\nu} \text{tr} \left[F_{0,\mu}^{n_1}(\frac{z}{2}) W(\frac{z}{2}, -\frac{z}{2}) F_{0,\nu}^{n_2}(-\frac{z}{2}) W(-\frac{z}{2}, \frac{z}{2}) \right], \quad (59)$$

where $F^{\mu\nu} = F_a^{\mu\nu} t^a$ and W are in the fundamental representation. Similar to Eq. (48), we can express $O_g^B(z)$ as a product of two local composite operators,

$$\begin{aligned} \tilde{O}_g^B(z) &= \int d^4\xi \delta(\xi^3 - z) \\ &\times g_{\perp,\mu\nu} \langle F_{0,a}^{n_1,\mu}(\frac{\xi}{2}) Q_0^a(\frac{\xi}{2}) Q_0^b(-\frac{\xi}{2}) F_{0,b}^{n_2,\nu}(-\frac{\xi}{2}) \rangle_Q \\ &= \int d^4\xi \delta(\xi^3 - z) g_\perp^{\mu\nu} \langle J_{n_1,\mu}^B(\frac{\xi}{2}) J_{n_2,\nu}^B(-\frac{\xi}{2}) \rangle_Q, \end{aligned} \quad (60)$$

where the auxiliary “heavy” quark fields are in the adjoint representation, and

$$J_B^{\mu\nu} = F_{0,a}^{\mu\nu} Q_0^a, \quad \bar{J}_B^{\mu\nu} = Q_0^a F_{0,a}^{\mu\nu}. \quad (61)$$

The renormalization of $J_B^{\mu\nu}$ and $\bar{J}_B^{\mu\nu}$ is more involved than the quark case, as they can mix with other composite operators of the same or less dimensions. In DR, BRST symmetry allows $J_B^{\mu\nu}$ to mix with (Dorn, 1986; Zhang *et al.*, 2019b)

$$J_{2B}^{\mu\nu} = (n_z^\mu F_{0,a}^{n_1,\nu} - n_z^\nu F_{0,a}^{n_1,\mu}) Q_0^a / n_z^2, \quad (62)$$

$$J_{3B}^{\mu\nu} = (-in_z^\mu A_{0,a}^{\nu\mu} + in_z^\nu A_{0,a}^{\mu\mu}) [(in_z \cdot D_0 - i\delta m) Q_0]^\mu / n_z^2. \quad (63)$$

Their renormalization matrix is given by (Dorn, 1986)

$$\begin{pmatrix} J_{2B}^{\mu\nu} \\ J_{3B}^{\mu\nu} \end{pmatrix} = \begin{pmatrix} Z_{11} & Z_{12} & Z_{13} \\ 0 & Z_{22} & Z_{23} \\ 0 & 0 & Z_{33} \end{pmatrix} \begin{pmatrix} J_{2B}^{\mu\nu} \\ J_{3B}^{\mu\nu} \end{pmatrix}, \quad (64)$$

where $J_{2B}^{\mu\nu}$ is gauge invariant while $J_{3B}^{\mu\nu}$ is gauge dependent and proportional to the equation of motion (EOM) for the auxiliary field. The Green’s functions of the EOM operator will result in a δ -function,

$$(in_z \cdot D_0(\xi) - i\delta m) \langle Q_0(\xi) \bar{Q}_0(0) \rangle_Q = \delta^{(4)}(\xi), \quad (65)$$

which only contributes a contact term $\delta(z)$ after integrating over the auxiliary fields. As long as $z \neq 0$, such mixing vanishes in all Green’s functions of $O_g^B(z)$, so we can ignore the mixing between $J_B^{\mu\nu}$ and $J_{3B}^{\mu\nu}$ in the renormalization of $O_F^B(z)$. At $z = 0$, $O_F^B(z)$ becomes a local operator and it is known to mix with BRST-exact and EOM operators (Collins and Scalise, 1994), whose renormalization can be performed in the standard way.

Note that when contracted with n_z ,

$$J_{2B}^{n_s,\mu} = J_B^{n_s,\mu} = F_{0,a}^{n_s,\mu} Q_0^a, \quad (66)$$

$$J_{3B}^{\mu\nu} = i(-A_0^{\mu,\mu} + \frac{n_z^\mu}{n_z^2} n_z \cdot A_0^\mu) [(in_z \cdot D_0 - i\delta m) Q_0]^\mu_a,$$

the $J_B^{n\mu}$ only mixes with the EOM operator $J_{B,3}^{n\mu}$. As has been argued above, we can ignore such mixing for $z \neq 0$. Moreover, this degeneracy also leads to relations among elements in the renormalization matrix (Dorn, 1986),

$$Z_{11} + Z_{12} = Z_{22}, \quad Z_{13} = Z_{23}. \quad (67)$$

When contracted with n_i ,

$$\begin{aligned} J_B^{n\mu} &= F_{a,0}^{n\mu} Q_0^a, \\ J_{2B}^{n\mu} &= n_z^2 F_{a,0}^{n\mu} Q_0^a/n_z^2, \\ J_{3B}^{n\mu} &= -\frac{n_z^2}{n_z^2} n_i \cdot A_0^a [(in_z \cdot D_0 - im) Q_0]_a. \end{aligned} \quad (68)$$

As one can see, $J_{2B}^{n\mu}$ and $J_{3B}^{n\mu}$ vanish after contraction with $g^{\mu\nu}$, so $J_B^{n\mu}$ with transverse Lorentz index μ is multiplicatively renormalizable.

To summarize, for $z \neq 0$ and transverse Lorentz index μ , both $J_B^{n\mu}$ and $J_{B,3}^{n\mu}$ are multiplicatively renormalizable in coordinate space, thus proving the renormalizability of the gluon Wilson-line operator $O_g^B(z)$,

$$\begin{aligned} O_g^B(z) &= Z_J Z_J \int d^4\xi \delta(\xi^3 - z) g_\perp^\mu \langle J_{n\mu}^R(\frac{\xi}{2}) J_{n\nu}^R(-\frac{\xi}{2}) \rangle_Q \\ &= e^{\delta m|z|} Z_J Z_J O_g^B(z), \end{aligned} \quad (69)$$

where

$$J_B^{n\mu} = Z_J J_R^{n\mu} = (Z_Q^0)^{\frac{1}{2}} Z_A^{\frac{1}{2}} Z_V^{\frac{1}{2}} J_R^{n\mu}, \quad (70)$$

$$J_{B,3}^{n\mu} = Z_J J_R^{n\mu} = (Z_Q^0)^{\frac{1}{2}} Z_A^{\frac{1}{2}} Z_V^{\frac{1}{2}} J_R^{n\mu}, \quad (71)$$

with Z_V^0 and Z_Q^0 being the renormalization constants for the vertex involving one gluon and one “heavy quark” fields. The wavefunction renormalization constant for the auxiliary “heavy quark”, Z_Q^0 , is different from the quark case as it is in the adjoint representation.

In addition, since $J_B^{n\mu}$ and $J_{B,3}^{n\mu}$ do not mix with “heavy-to-light” quark currents due to the mismatch of quantum numbers, it implies that the nonlocal gluon Wilson-line operator does not mix with the singlet quark one under renormalization.

For the polarized gluon quasi-PDF, its definition is the same as Eq. (56), except that the gluon Wilson-line operator becomes

$$\Delta O_g^B(z) = e_{\perp,\mu\nu} F_{0,3}^{n\mu}(z) W^{ab}(z, 0) F_{0,3}^{n\nu}(0), \quad (72)$$

where $e_{\perp}^{\mu\nu} = \epsilon^{\alpha\beta\mu\nu}$. Since $e_{\perp}^{\mu\nu}$ only contracts with the transverse Lorentz indices, one can use the same proof for $O_g^B(z)$ to show that $\Delta O_g^B(z)$ is also multiplicatively renormalizable and does not mix with singlet quark case (Zhang *et al.*, 2019b).

Finally, one can also prove that Eq. (69) is valid under lattice regularization with δm being linearly divergent (Zhang *et al.*, 2019b). This completes the proof of renormalizability of the gluon Wilson-line operators.

One-loop renormalization. Now we demonstrate the above result by an explicit one-loop example. For the nonlocal Wilson-line operators to be multiplicatively renormalizable, it is important that all linear divergences associated with diagrams other than the Wilson line self-energy cancel out among themselves. To see this, a gauge symmetry preserving regularization is crucial. We use DR and keep poles around $d = 3$ to identify the linear divergences (Wang *et al.*, 2019c; Zhang *et al.*, 2019b).

The one-loop vertex correction to the “heavy-to-gluon” current is shown in Fig. 7. Each diagram contributes

$$\begin{aligned} I_a^{\mu\nu} &= \frac{\alpha_s C_A}{\pi} \left[\frac{1}{4-d} \frac{3}{4} F_a^{\mu\nu} Q_a + \text{finite terms} \right], \\ I_b^{\mu\nu} &= \frac{\alpha_s C_A}{\pi} \left[\frac{1}{d-4} (A_a^\mu n_z^\mu - A_a^\nu n_z^\nu) \cdot \partial Q_a / n_z^2 \right. \\ &\quad \left. + \frac{\pi\mu}{d-3} (n_z^\mu A_a^\nu - n_z^\nu A_a^\mu) Q_a + \text{finite terms} \right], \\ I_c^{\mu\nu} &= \frac{\alpha_s C_A}{\pi} \left[\frac{1}{d-4} \left(\frac{1}{2} (F_a^{\mu\nu} n_z^\nu - F_a^{\nu\mu} n_z^\mu) Q_a / n_z^2 \right. \right. \\ &\quad \left. \left. + \frac{1}{4} F_a^{\mu\nu} Q_a + \frac{1}{2} (A_a^\mu n_z^\nu - A_a^\nu n_z^\mu) n_z \cdot \partial Q_a / n_z^2 \right) \right. \\ &\quad \left. - \frac{\pi\mu}{d-3} (n_z^\mu A_a^\nu - n_z^\nu A_a^\mu) Q_a + \text{finite terms} \right]. \end{aligned} \quad (73)$$

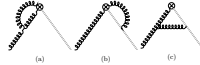


FIG. 7: One-loop vertex corrections to the “heavy-to-gluon” current.

Both Fig. 7b and Fig. 7c include a linear divergence that is evident as the $\mu/(d-3)$ term, but they cancel among themselves. This guarantees that the overall UV divergence in the vertex correction is logarithmic, thus the renormalization of the “heavy-to-gluon” current is multiplicative. Combining the one-loop results in Eq. (73) and wavefunction renormalizations, we have

$$\begin{aligned} Z_{11} &= 1 + \frac{\alpha_s C_A}{4\pi} \frac{1}{\epsilon_{UV}}, & Z_{12} &= 1 - \frac{\alpha_s C_A}{4\pi} \frac{1}{\epsilon_{UV}}, \\ Z_{13} &= Z_{23} = 1 - \frac{\alpha_s C_A}{4\pi} \frac{1}{\epsilon_{UV}}, & Z_{22} &= 0, \end{aligned} \quad (74)$$

where $C_A = N_c = 3$ for QCD. If we ignore the mixing to the EOM operator,

$$\begin{aligned} Z_V^{J^{n\mu\nu}} &= Z_V^{J^{n\mu\nu}} = 0, \\ Z_V^{J^{n\mu\nu}} &= Z_V^{J^{n\mu\nu}} = Z_V^{J^{n\mu\nu}} = 1 + \frac{\alpha_s C_A}{4\pi} \frac{1}{\epsilon_{UV}}, \end{aligned} \quad (75)$$

where $i, j = 1, 2$. As a result, the one-loop current renormalization constant is

$$\begin{aligned} Z_{J^{\nu a^*}} &= Z_{J^{\nu a}} = 1 + \frac{\alpha_s}{4\pi} \left(\frac{1}{6} C_A - \frac{4}{3} n_f T_F \right) \frac{1}{\epsilon_{UV}}, \\ Z_{J^{\nu a^*}} &= Z_{J^{\nu a}} = Z_{J^{\nu i}} = Z_{J^{\nu j}} \\ &= 1 + \frac{\alpha_s}{4\pi} \left(\frac{7}{6} C_A - \frac{4}{3} n_f T_F \right) \frac{1}{\epsilon_{UV}}, \end{aligned} \quad (76)$$

where $T_F = 1/2$, and n_f is the number of active quark flavors. The two-loop results can be found in (Braun *et al.*, 2020).

As one can see, the anomalous dimension of the “heavy-to-gluon” current is the same for $\mu, \nu = 0, 1, 2$, which is due to $SO(2, 1)$ (or $SO(3)$ in Euclidean space) symmetry around the z -axis.

B. Factorization of Quasi-PDFs

The key to LaMET applications for collinear parton physics is the factorization formula that relates the quasi-PDFs to light-cone PDFs (Ji, 2013). Here we use the perturbative properties of the matching coefficients to write the factorization form in the $\overline{\text{MS}}$ scheme in a way consistent with a direct EFT calculations of PDFs at any given x (Izubuchi *et al.*, 2018; Wang *et al.*, 2019b)

$$q_i(x, \mu) = \int_{-\infty}^{\infty} \frac{dy}{|y|} \left[\sum_j \tilde{C}_{q_i q_j} \left(\frac{x}{y}, \frac{\mu}{yP^+} \right) \bar{q}_j(y, P^+, \mu) \right. \quad (77)$$

$$\left. + \tilde{C}_{q_i g} \left(\frac{x}{y}, \frac{\mu}{yP^+} \right) \bar{g}(x, P^+, \mu) \right] + \dots, \\ g(x, \mu) = \int_{-\infty}^{\infty} \frac{dy}{|y|} \left[\sum_j \tilde{C}_{g q_j} \left(\frac{x}{y}, \frac{\mu}{yP^+} \right) \bar{q}_j(y, P^+, \mu) \right. \quad (78)$$

$$\left. + \tilde{C}_{g g} \left(\frac{x}{y}, \frac{\mu}{yP^+} \right) \bar{g}(y, P^+, \mu) \right] + \dots,$$

where i, j runs over quark and anti-quark flavors. The “...” term includes mass corrections whose analytical forms have been derived to all orders of $M^2/(P^+)^2$ (Chen *et al.*, 2016), and higher-twist contributions of order $\mathcal{O}(\Lambda_{\text{QCD}}^2/(xP^+)^2, \Lambda_{\text{QCD}}^2/((1-x)P^+)^2)$ (see Eq. (33)). All P^+ -dependence on the right hand side cancels out, just like a renormalization scale.

As we have explained in the Sec. II, the above factorization is guaranteed on the physics ground because the difference between quasi-PDFs and light-cone PDFs is the order of limits in $P^+ \rightarrow \infty$ and $\Lambda_{\text{UV}} \rightarrow \infty$, and the IR physics in both quantities must be the same. An all-order factorization proof for the quark quasi-PDF in perturbation theory was first given with a diagrammatic approach (Ma and Qiu, 2018b). The formula has also been derived using the operator product expansion (OPE) of nonlocal Wilson-line operators (Izubuchi *et al.*,

2018; Ma and Qiu, 2018a; Wang *et al.*, 2019b). Here we outline the diagrammatic proof similar to (Ma and Qiu, 2018b), showing that the collinear divergences of the quasi-PDFs do factorize and are equal to those of the light-cone PDFs. Since the collinear divergence is a concept in perturbation theory, we will show the factorization using a massless external quark state with lightlike momentum $P^\mu = (P^+, 0, 0, P^+)$. While the proof is only for perturbative free quark states, the factorization formulas are widely believed to be true nonperturbatively as well. We use DR to regulate both UV and collinear divergences and only consider bare quantities, since UV renormalization does not change the leading collinear divergences.

Before the analysis, we should mention that all the soft divergences cancel between the real and virtual contributions to the quasi-PDFs, as discussed in Sec. II.B, thus we only need to focus on the collinear divergences. To obtain an intuitive understanding of the structure for collinear divergences, we start from the one-loop diagram in Fig. 5a in the Feynman gauge. The integral reads

$$\int \frac{d^{4-2\epsilon} k}{(2\pi)^{4-2\epsilon}} \frac{\text{tr}[\not{P} \not{k} \gamma^+ \not{k}] \delta(k^+ - yP^+)}{(k^2 + i0)^2 ((P-k)^2 + i0)}. \quad (79)$$

The internal quark momentum is $k^\mu = (k^+, k^-, \vec{k}_\perp)$ and the gluon momentum is $P - k$. When k^+ and $k_\perp = |\vec{k}_\perp|$ are very small, the internal quark and gluon become collinear to the external quark, i.e. $k^\mu \sim (k^+, 0, 0_\perp)$ and $(P-k)^\mu \sim (P^+ - k^+, 0, 0_\perp)$. In this case, the denominator of the quark and gluon propagators, $(k^2)^2$ and $(P-k)^2$, both vanish, which leads to collinear divergence. Conversely, for $k^2 = (P-k)^2 = 0$, k must be collinear to P since the condition requires $k^2 = k \cdot P = P^2 = 0$. For small k^+ and $k_\perp$, the δ function is dominated by the k^+ term of $k^2 = (k^+ - k^-)/\sqrt{2}$ and reduces to $\sqrt{2}\delta(k^+ - yP^+)$. This is just the vertex which restricts $k^+ = yP^+$ for the light-cone PDF, up to the factor $\sqrt{2}$. Furthermore, for collinear k and $(P-k)$, the spinor trace in the numerator is dominated by the γ^+ part of $\gamma^2 = (\gamma^+ - \gamma^-)/\sqrt{2}$, $\text{tr}[\not{P} \not{k} \gamma^+ \not{k}] \sim \text{tr}[\not{P} \not{k} \gamma^+ \not{k}]/\sqrt{2}$. Thus in the collinear region $k^\mu \sim (k^+, 0, 0, 0)$ the above integral reduces to that for the light-cone PDF:

$$\int_c \frac{d^{4-2\epsilon} k}{(2\pi)^{4-2\epsilon}} \frac{\text{tr}[\not{P} \not{k} \gamma^+ \not{k}] \delta(k^+ - yP^+)}{(k^2 + i0)^2 ((P-k)^2 + i0)}, \quad (80)$$

where the subscript “c” denotes the collinear region.

The above picture naturally arises in a highly boosted hadron state where the quark is approximately onshell. Therefore, as explained in Sec. II.E, *although the operator contains no light-cone information, the large-momentum external hadron state can still generate collinear divergences equivalent to those in the light-cone PDFs*. By subtracting the full integral for light-cone PDF from that for the quasi-PDF, the logarithmic collinear divergence

cancels, and the remaining difference is perturbative and can be absorbed into the matching kernel.

Similarly, for the vertex diagram in Fig. 5b, the loop integral is proportional to

$$\int \frac{d^{4-2\epsilon}k}{(2\pi)^{4-2\epsilon}} \frac{1}{P^2 - k^2} \text{tr}[\not{P}\gamma^2 \not{k}\gamma^+] \delta(k^+ - yP^+) \quad (81)$$

The whole integral in the collinear region reduces to

$$\int_c \frac{d^{4-2\epsilon}k}{(2\pi)^{4-2\epsilon}} \frac{1}{P^+ - k^+} \text{tr}[\not{P}\gamma^+ \not{k}\gamma^+] \delta(k^+ - yP^+) \quad (82)$$

which is the corresponding integral for the light-cone PDF. One key feature of the diagram is that while the gauge link probes the z -component of the gluon field $A^+ = (A^+ - A^-)/\sqrt{2}$, only the A^+ component (longitudinal polarization) contributes to the leading collinear divergence. While attaching a new collinear gluon to the gauge-link induces a power suppression from the link propagator of $\mathcal{O}(1/P^+)$, the A^+ component of the collinear gluon radiated from fast-moving color charges receives enhancement from Lorentz boost factor γ that compensates for the suppression.

The above result can be generalized to all orders. Similar to the one-loop diagrams, in the leading region of collinear divergence there are an arbitrary number of longitudinally polarized A^+ gluons, which are emitted dynamically from the fast-moving state instead of being put in by hand using the lightlike gauge link, in contrast to the standard collinear PDF. The existence of the A^+ gluons clearly increases the level of complication in showing the equivalence of collinear divergences between the quasi- and light-cone PDFs. For simplification, from now on we choose to work in the light-cone gauge $A^+ = 0$ to eliminate all the A^+ gluons. Therefore, the vertex diagrams no longer contribute to the leading collinear divergence, thus making its structure much simpler.

In a general diagram, we decompose the potential leading region of the quasi-PDF into the ladder structure shown in Fig. 8. The upper two-particle-irreducible (2PI) kernel that contains the nonlocal operator defining the quasi-PDF is H . The 2PI kernel in the ladder is K . K contains the upper two external quark lines but not the lower ones. The momentum flowing out of the ladders are labeled as k_1 to k_n from bottom to top when there are n 2PI kernels. We write H and K as matrices in spinor and momentum space. $H = H_{\alpha\beta}(yP^+; k)$ where k denotes the momentum flowing into H and $K = K_{\alpha\beta, \alpha'\beta'}(k, k')$ where k, k' are the momenta of the upper and lower external legs, respectively. Here $\alpha\beta$ and $\alpha'\beta'$ are the spinor indices for the upper and lower two external legs, respectively. Following the method in (Collins, 2011a; Curci *et al.*, 1980), we find that:

1. There are no collinear divergences in the upper part H in the light-cone gauge.

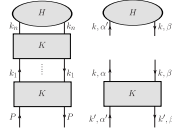


FIG. 8: The ladder decomposition of the quasi-PDF (left). The upper 2PI kernel H contains the operator defining the quasi-PDF, and external two legs at the bottom of the diagram is the external large P^+ state. The kernels H and K are shown on the right.

2. If none of $k_1, \dots, k_n$ is collinear, there will be no leading collinear divergence. More generally, for the i 'th 2PI kernel, if either of k_{i-1} and k_i is not collinear, then the sub-integrals inside the kernel are finite and it does not contribute to leading collinear divergence.
3. If k_i is not collinear, then there are no collinear divergences for the upper part of the diagram above the i 'th ladder.

Therefore, the collinear divergences are generated in the momentum regions R_i in which k_1 to k_i are collinear while k_{i+1} to k_n are not. We can construct counter terms that subtract out the collinear divergences in each of the regions R_i . For this we keep only the $+$ component of k_i in the convergent upper part HK^{n-i} as in the one-loop example, namely $k_i \rightarrow (k_i^+, 0, 0_\perp)$ in the upper part. This will clearly leave the collinear divergence unchanged. Also notice that $[HK^{n-i}]_{\alpha\beta} = H_{\alpha\beta'} K_{\alpha'\beta'', \alpha\beta}^{n-i-1}$ should be understood as a 4×4 Dirac matrix with indices $\alpha\beta$, while the lower part is $[K'P]_{\alpha\beta} = K'_{\alpha\beta, \alpha'\beta'} P_{\alpha'\beta'}$. In the leading region of collinear divergence, HK^{n-i} and $K'P$ are proportional to γ^+ and γ^- respectively. Therefore, to obtain the leading collinear divergence, we can disentangle the spinor traces for the upper and lower parts by contracting them with $\gamma^-/2$ and $\gamma^+/2$ separately. The only communication between them is the k^+ integration. The collinear divergence is contained in the lower part

$$q^d(x, \epsilon_n) = \int \frac{dk^- d^{d-2}k_\perp}{2(2\pi)^d} \text{tr}[\gamma^+ K^+(xP^+, k^-, k_\perp; P) \not{P}], \quad (83)$$

where $d = 4 - 2\epsilon$, $k^+ = xP^+$, and the subtraction for the region R_i can be written effectively as a convolution

$$\int \frac{dx}{x} \tilde{C}^{n-i}(y, x, P^+) q^d(x, \epsilon_n), \quad (84)$$

where

$$\hat{C}^{n-i}(y, x, P^2) = \frac{1}{2} \text{tr} [H K^{n-i}(y P^2; x P^2, 0, 0_\perp) (x P^+) \gamma^-] \quad (85)$$

is the naive matching kernel. Here the y dependence comes from the operator in H . However, the naive matching kernel still suffers from collinear sub-divergences that need to be subtracted. This can be achieved using the subtracted matching kernels $C^{n-i}(y, x)$ defined recursively in a way similar to the BPHZ relation for UV renormalization (Collins, 2011a). Summing over n and i , the recursive relation leads to

$$\begin{aligned} \tilde{q}(y, P^2, \epsilon_m) &= \sum_{n=0}^{\infty} \sum_{i=0}^n \int \frac{dx}{x} C^{n-i}(y, x, P^2) q^i(x, \epsilon_m) \\ &= \int \frac{dx}{x} C(y, x, P^2) q(x, \epsilon_m), \end{aligned} \quad (86)$$

where $\tilde{q}(y, P^2, \epsilon_m)$ is the quasi-PDF, $C(y, x, P^2) = \sum_{n=0}^{\infty} C^n(y, x, P^2)$ is the all-order matching kernel and $q(x, \epsilon_m) = \sum_{i=0}^{\infty} q^i(x, \epsilon_m)$. Based on the definition of $q^i(x, \epsilon_m)$, it is clear that q^i equals the light-cone PDF with i 2PI kernels and q is the full light-cone PDF with natural support $0 < x < 1$. The light-cone PDF $q(x)$ is independent of the operator defining the quasi-PDF, as it is only sensitive to the explicit form of the collinear divergence. The r.h.s. of Eq. (86) contains all the collinear divergences from the quasi-PDF $\tilde{q}$. Thus the matching relation for bare quantities is established. A similar matching can be written down for the renormalized quantities, where the renormalization only affects the matching kernel $C(y, x, P^2)$. Note that the explicit solution for $C^{n-i}(y, x, P^2)$, which leads to Eq. (33), can be given based on a subtraction operator defined similar to that in (Collins, 2011a). Besides, Eq. (86) can be inverted order by order in α_s , thus proving Eq. (33), which can also be generalized to Eqs. (77) and (78).

Now we present the matching coefficient in the $\overline{\text{MS}}$ scheme at one-loop order. The one-loop expansion of the $\overline{\text{MS}}$ quasi- and light-cone PDFs in a free massless quark state with momentum $p^\mu = (p^2, 0, 0, p^2)$ are

$$\tilde{q}(y, \mu/p^2, \epsilon_m) = \tilde{q}^{(0)}(y) + \frac{\alpha_s C_F}{2\pi} \tilde{q}^{(1)}(y, \mu/p^2, \epsilon_m), \quad (87)$$

$$q(x, \epsilon_m) = q^{(0)}(x) + \frac{\alpha_s C_F}{2\pi} q^{(1)}(x, \epsilon_m). \quad (88)$$

At tree level, $\tilde{q}^{(0)}(y) = q^{(0)}(y) = \delta(1-y)$. At one loop, the $\overline{\text{MS}}$ quasi-PDF and its counterterm

are (Izubuchi *et al.*, 2018)

$$\begin{aligned} \tilde{q}^{(1)}(y, \mu/p^2, \epsilon_m) &= \left(\left(\frac{1+y^2}{1-y} \ln \frac{y}{1-y} + 1 + \frac{3}{2} \right)_{+(1)}^{[1,\infty]} - \frac{3}{2} \right) y > 1 \\ &= \left(\frac{1+y^2}{1-y} \left[-\frac{1}{\epsilon_m} - \ln \frac{\mu^2}{4(p^2)^2} + \ln(y(1-y)) \right]_{+(1)}^{[0,1]} \right. \\ &\quad \left. - \frac{3(1+y)}{1-y} + 2\sigma(1-y) \right)_{+(1)}^{[0,1]} 0 < y < 1 \\ &= \left(-\frac{1+y^2}{2(1-y)} \ln \frac{y}{1-y} - 1 + \frac{3}{2} \right)_{+(1)}^{[-\infty,0]} y < 0 \\ &\quad + \delta(1-y) \left[\frac{3}{2} \ln \frac{\mu^2}{4(p^2)^2} + \frac{5+2\sigma}{2} \right], \end{aligned} \quad (89)$$

$$\delta \tilde{q}^{(1)}(y, \mu/p^2, \epsilon_{\text{UV}}) = \frac{3}{2\epsilon_{\text{UV}}} \delta(1-y), \quad (90)$$

where ϵ_m regulates the collinear divergence, $\sigma = 0$ for $\Gamma = \gamma^t$ and 1 for $\Gamma = \gamma^z$. The plus function at $y = y_0$ with support in a given domain D is defined as

$$\int_D dy [g(y)]_{+(y_0)}^D h(y) = \int_D dy g(y) [h(y) - h(y_0)] \quad (91)$$

with arbitrary $g(y)$ and $h(y)$. Note that the $\overline{\text{MS}}$ renormalization of the quasi-PDF actually requires a subtle treatment of vector current conservation (Izubuchi *et al.*, 2018). We only present results in the form that is sufficient for our discussion, which differs slightly from that in (Izubuchi *et al.*, 2018) by the δ -functions at $y = \pm\infty$ and from the treatment in (Alexandrou *et al.*, 2019b).

On the other hand,

$$q^{(1)}(x, \epsilon_m) = \frac{\alpha_s C_F}{2\pi} \frac{(-1)}{\epsilon_m} \left(\frac{1+x^2}{1-x} \right)_{+(1)}^{[0,1]}, \quad (92)$$

which is limited to the physical region as expected.

By comparing the quasi- and light-cone PDFs in Eqs. (89) and (92), we find that both of them have the same collinear divergence, or in other words, they share the same IR physics, thus validating the factorization formula at one-loop order. Setting $p^2 = x P^2$ and plugging the one-loop results into Eq. (33), we extract the matching coefficient for the hadron matrix element which only depends on the perturbative scales μ and P^2 :

$$\begin{aligned} C^{\overline{\text{MS}}} \left(y, \frac{\mu}{x P^2} \right) &= \delta(1-y) + \frac{\alpha_s C_F}{2\pi} \left[\tilde{q}^{(1)} \left(y, \frac{\mu}{x P^2}, \epsilon_m \right) \right. \\ &\quad \left. - q^{(1)}(y, \epsilon_m) \right]. \end{aligned} \quad (93)$$

The complete one-loop matching coefficients in Eq. (33) in the transverse-momentum cutoff and $\overline{\text{MS}}$ schemes can be found in (Wang *et al.*, 2019b; Wang and Zhao, 2018; Wang *et al.*, 2018). The two-loop results were obtained recently in (Clen *et al.*, 2020a,b,c; Li *et al.*, 2020).

C. Coordinate-Space Factorization of Bilinear Operators

Although the LaMET application to PDFs concerns the expansion of momentum densities in the $P^z \rightarrow \infty$ limit, lattice QCD calculations actually start from computing coordinate-space correlations, for example,

$$\tilde{h}(z, P^z) = \frac{1}{N_T} \langle P^z | O_T(z) | P^z \rangle, \quad (94)$$

at all z and do Fourier transform with respect to $\lambda = zP^z$ at a fixed P^z . Here the normalization factor $N_T = 2P^z$ for $\Gamma = \gamma^z$ and $N_T = 2P^z$ for $\Gamma = \gamma^t$. The $\tilde{h}(z, P^z)$ is a function of two independent variables z and P^z , and in LaMET analysis the relevant combinations are quasi-LF distance λ (see Fig. 6) and P^z , hence $\tilde{h}(\lambda, P^z)$ will be called quasi-LF correlation, which is distinguished from the LF correlation $h(\lambda, \mu)$ below.

The coordinate-space factorization approach in (Braun *et al.*, 1995) has been suggested as an alternative way to extract the PDFs from $\tilde{h}(z, P^z)$ (Orginos *et al.*, 2017; Radyushkin, 2017a, 2019a), which is closely related to the OPE. Instead of working with variables λ and P^z , one may consider $\tilde{h}$ as a function of λ and z^2 , i.e., $\tilde{h}(\lambda, z^2)$. The Fourier transform of $\tilde{h}(\lambda, z^2)$ with respect to λ is no longer the momentum distribution of the proton at a fixed momentum. Instead, it is called a pseudo-distribution (Radyushkin, 2017a). At small $|z| \ll \Lambda_{\text{QCD}}^{-1}$, $\tilde{h}(\lambda, z^2)$ can be factorized into the light-cone correlation (Izubuchi *et al.*, 2018; Radyushkin, 2018a),

$$\tilde{h}(\lambda, z^2 \mu^2) = \int_{-1}^1 d\alpha C(\alpha, z^2 \mu^2) h(\alpha \lambda, \mu) + \dots, \quad (95)$$

where ... are the power corrections in $z^2 \Lambda_{\text{QCD}}^2$, and the matching coefficient C is related to C in Eq. (33) by

$$C\left(\eta, \frac{\mu}{xP^z}\right) = \int \frac{d\lambda}{2\pi} e^{i\eta\lambda} \int_{-1}^1 d\alpha e^{-i\alpha\lambda} C\left(\alpha, \frac{\mu^2\lambda^2}{(xP^z)^2}\right). \quad (96)$$

To illustrate the connection between the above factorization in Eq. (95) and OPE, let us take the non-singlet quark case as an example (Izubuchi *et al.*, 2018; Wang *et al.*, 2019b). In the $\overline{\text{MS}}$ scheme, the renormalized $O_{\gamma^0}(z, \mu)$ can be expanded in terms of local gauge-invariant twist-2 operators as $z^2 \rightarrow 0$,

$$O_{\gamma^0}(z, \mu) = \sum_{n=0}^{\infty} \left[C_n(\mu^2 z^2) \frac{(iz)^n}{n!} (n_z)_{\mu_1} \cdots (n_z)_{\mu_n} \right. \\ \left. \times O^{\mu_0 \mu_1 \cdots \mu_n}(\mu) + \text{higher-twist} \right], \quad (97)$$

where $\mu_0 = 0$ or 3, $C_n = 1 + \mathcal{O}(\alpha_s)$ is the Wilson coefficient, and $O^{\mu_0 \mu_1 \cdots \mu_n}(\mu)$ is the twist-two operator in Eq. (14).

Using the hadron matrix elements in Eq. (15) and their relation to the light-cone PDF in Eq. (16), we write down

the small- $|z|$ expansion of the hadron matrix element of $O_{\gamma^0}(z, \mu)$ (Izubuchi *et al.*, 2018),

$$\begin{aligned} \tilde{h}(\lambda, z^2 \mu^2) &= \langle P | O_{\gamma^0}(z, \mu) | P \rangle / (2P^0 \mu) \\ &= \sum_{n=0}^{\infty} C_n(z^2 \mu^2) \frac{(-i\lambda)^n}{n!} \left[1 + \mathcal{O}\left(\frac{M^2}{(P^z)^2}\right) \right] \\ &\quad \times \int_{-1}^1 dx x^n q(x, \mu) + \mathcal{O}\left(z^2 \Lambda_{\text{QCD}}^2\right), \end{aligned} \quad (98)$$

where the $\mathcal{O}(M^2/(P^z)^2)$ term comes from the kinematic trace contribution and the $\mathcal{O}(z^2 \Lambda_{\text{QCD}}^2)$ term from higher-twist. The Wilson coefficients $C_n(z^2 \mu^2)$ have been calculated at one-loop (Izubuchi *et al.*, 2018) and two-loop (Li *et al.*, 2020) orders. Comparing the above equation with Eq. (95), we identify

$$C(\alpha, \mu^2 z^2) = \int \frac{d\lambda}{2\pi} e^{i\lambda\alpha} \sum_n C_n(\mu^2 z^2) \frac{(-i\lambda)^n}{n!}. \quad (99)$$

Since z^2 is fixed in $C(\alpha, \mu^2 z^2)$, the integration in Eq. (99) is actually over P^z from $-\infty$ to $+\infty$. $C(\alpha, z^2 \mu^2)$ has support $-1 \leq \alpha \leq 1$, and its one-loop result is

$$\begin{aligned} C(\alpha, z^2 \mu^2) &= \int \frac{d\lambda}{2\pi} e^{i\lambda\alpha} \sum_n C_n(\mu^2 z^2) \frac{(-i\lambda)^n}{n!} \\ &= \left[1 + \frac{\alpha_s C_F}{2\pi} \left(\frac{3}{2} \ln \frac{z^2 \mu^2 e^{2\gamma_E}}{4} + \frac{3}{2} \right) \right] \delta(1 - \alpha) \\ &\quad + \frac{\alpha_s C_F}{2\pi} \left\{ \left(\frac{1 + \alpha^2}{1 - \alpha} \right)_{+(1)}^{[0,1]} \left[-\ln \frac{z^2 \mu^2 e^{2\gamma_E}}{4} - 1 \right] \right. \\ &\quad \left. - \left(\frac{4 \ln(1 - \alpha)}{1 - \alpha} \right)_{+(1)}^{[0,1]} + 2(1 + \sigma)(1 - \alpha) \right\} \theta(\alpha) \theta(1 - \alpha), \end{aligned} \quad (100)$$

which was also calculated and further studied in (Ji *et al.*, 2017b; Li *et al.*, 2020; Radyushkin, 2018a,b; Zhang *et al.*, 2018). One can check that the above result is indeed related to one-loop momentum-space matching by Eq. (96). Since we are interested in the relation between the quasi-LF correlation with the matrix element of the light-ray operator $O_{\gamma^+}(\lambda n)$, Eq. (95) can also be obtained by using the light-ray operator expansion in (Balitsky and Braun, 1989; Braun *et al.*, 1995; Braun and Müller, 2008).

Using OPE or short-distance expansion, the exact factorization formula for the gluon and singlet quark quasi-PDFs, which includes their mixings, has also been derived in coordinate space and studied at one-loop order (Balitsky *et al.*, 2019; Wang *et al.*, 2019b).

It is easy to see that the limits $P^z \rightarrow \infty$ in LaMET expansion and $z \rightarrow 0$ in coordinate-space factorization, keeping finite $\lambda = zP^z$, are equivalent. However, in practical lattice QCD calculations, one is limited by the largest momentum P_{max}^z in a specific setup, and the two approaches start to differ.

In LaMET systematic approximation, one should calculate $\tilde{h}(z, P_{\text{max}}^z)$ with all possible z or λ , but in prac-

tice the largest $\lambda_{\text{max}} = z_{\text{max}} P_{\text{max}}^z$ is limited by the lattice volume as well as data quality at large z . Due to QCD confinement, $\tilde{h}(z, P_{\text{max}}^z)$ has a correlation length $\sim 1/\Lambda_{\text{QCD}}$, leading to an exponential decay at large z (Ji *et al.*, 2020b). Therefore, if z_{max} is sufficiently large (e.g., the proton size ~ 1 fm) for $\tilde{h}(z, P_{\text{max}}^z)$ to fall to almost zero, then the truncated Fourier transform of $\tilde{h}(z, P_{\text{max}}^z)$ should converge quickly, and the truncation effects mainly affect results at small $x \lesssim 1/\lambda_{\text{max}}$. If $\tilde{h}(z, P_{\text{max}}^z)$ exhibits exponential decay but still has a nonzero value at z_{max} , then one can perform a physically motivated extrapolation beyond z_{max} (Ji *et al.*, 2020b) to do the Fourier transform, which removes the unphysical oscillation from truncation and only affects the small- x region. In the momentum space, one can use LaMET expansion to calculate the PDF point by point in x with systematic error controlled by $\Lambda_{\text{QCD}}^2/(xP_{\text{max}}^z)^2$ and $\Lambda_{\text{QCD}}^2/((1-x)P_{\text{max}}^z)^2$, which gives the prediction for a certain region of x , $[x_{\text{min}}, x_{\text{max}}]$, with a target error.

In coordinate-space factorization, one expands $\tilde{h}(\lambda, z^2)$ in $z^2 \Lambda_{\text{QCD}}^2$. For the factorization formula to be valid, z must remain in the perturbative region. For example, an estimate in (Ji *et al.*, 2020b) gives $z_{\text{max}} \sim 0.3$ – 0.4 fm. Although there have been observations that forming ratios of $\tilde{h}(\lambda, z^2)$ may cancel the higher-twist contributions at $z > 0.4$ fm (Orginos *et al.*, 2017), this cancellation needs to be quantified for precision calculations. With a finite range of quasi-LF correlations, the PDFs can be extracted through modelling the x -dependence or more advanced techniques such as Bayesian analysis (Bringewatt *et al.*, 2020) or neural network (Cichy *et al.*, 2019; Del Debbio *et al.*, 2020a; Karpie *et al.*, 2019), which is similar to extracting the PDFs from experimental data (Ma and Qiu, 2018a), although quantifying the systematic error from fitting can be challenging. The coordinate-space factorization can also provide the extraction of the lowest moments of PDFs (Gao *et al.*, 2020; Joó *et al.*, 2020; Karpie *et al.*, 2018; Shagert *et al.*, 2020), where the main systematic error is controlled by $z^2 \Lambda_{\text{QCD}}^2$.

So far, there have been very limited studies about the comparison between quasi- and pseudo-PDF analysis (Alexandrou *et al.*, 2020; Bhat *et al.*, 2020). It remains to be seen how systematic errors in the two strategies are compared and contrasted.

D. Nonperturbative Renormalization and Matching

The multiplicative renormalizability of the non-local Wilson-line operators for quasi-PDFs allows a nonperturbative renormalization on the lattice, after which the continuum limit can be taken. This is an important step in the application of LaMET. One way of doing so is to perform a mass subtraction of the Wilson line first (Chen *et al.*, 2017; Green *et al.*,

2018, 2020; Ishikawa *et al.*, 2016; Musch *et al.*, 2011; Zhang *et al.*, 2017), and then renormalize the remnant UV divergences with lattice perturbation theory or other nonperturbative schemes. Another scheme which has gained more popularity in recent years is the regularization-independent momentum subtraction (RI/MOM) scheme (Alexandrou *et al.*, 2017b; Chen *et al.*, 2018; Constantinou and Panagopoulos, 2017; Liu *et al.*, 2020; Stewart and Zhao, 2018). In the coordinate space approach where $|z| \ll \Lambda_{\text{QCD}}^{-1}$, the ratios of quasi-LF correlations in different states (Braun *et al.*, 2019; Li *et al.*, 2020; Orginos *et al.*, 2017; Radyushkin, 2017a) have also been proposed as a renormalization scheme. At large z , the RI/MOM and ratio schemes introduce extra nonperturbative effects at different levels, which may distort the IR property of the original quasi-LF correlations. Due to the suppression of long-range contributions by large P^z in the Fourier transform, this nonperturbative contamination mainly affects the end-point region in x -space, while the existing LaMET calculations with RI/MOM scheme at moderate x , for example in (Alexandrou *et al.*, 2018b; Lin *et al.*, 2018a), suffers less from such systematics. Nevertheless, the above complication can be avoided by switching to the hybrid scheme (Ji *et al.*, 2020b) where one utilizes the advantages of different schemes at short and large distances. In the following, we discuss the above schemes in order, with a particular focus on the hybrid renormalization scheme.

Before we proceed, it should be noted that the current-current correlators in (Braun and Müller, 2008; Detmold and Lin, 2006; Ma and Qiu, 2018a) do not need or have simple renormalization on the lattice, though it might be more costly to simulate them. Besides, there is another distinct method based on a redefinition of the quasi-PDF with smeared fermion and gauge fields via the gradient flow (Monahan and Orginos, 2017). The smeared quasi-PDF is free from UV divergences and remains finite in the continuum limit, which can be perturbatively matched onto the PDF (Monahan, 2018b). Nevertheless, this method awaits to be implemented on the lattice.

1. Wilson-line mass-subtraction scheme

Since the mass correction δm includes all the linear UV divergences, it is highly favored to nonperturbatively subtract it from the quasi-PDFs. It is well known that the Wilson line renormalization is related to the additive renormalization of the static quark-antiquark potential, i.e., δm , especially in the context of finite temperature field theory. For a rectangle-shaped Wilson loop of dimension $L \times T$ in the spatial and temporal directions, its

vacuum expectation value for large T scales as

$$\lim_{T \rightarrow \infty} W(L, T) = c(L)e^{-V(L)T}. \quad (101)$$

The renormalized static potential is

$$V^R(L) = V(L) + 2\delta m, \quad (102)$$

and δm can be fixed by imposing the condition $V^R(L_0) = 0$ for a particular value of L_0 . Alternatively, one can also fit δm from the famous string potential model,

$$V(L) = \sigma L - \frac{\pi}{12L} - 2\delta m. \quad (103)$$

Apart from using the static potential to determine δm , it was also proposed to calculate this quantity in the auxiliary “heavy quark” field theory with the following condition (Green *et al.*, 2018),

$$\delta m = \frac{d}{dz} \ln \text{Tr} \langle Q(x + z n_z) Q(x) \rangle_{\text{QCD}+Q} \Big|_{z=z_0}. \quad (104)$$

Other suggestions have also been made for a nonperturbative calculation of δm (Ji *et al.*, 2020b). For example, one can investigate the asymptotic large- z behavior of the hadron matrix element or the single quark Green’s function, of the vacuum expectation value of $O_T(z, a)$ in a fixed gauge. The δm calculated from all these matrix elements will have the following dependence on the lattice spacing a ,

$$\delta m = m_{-1}(a)/a + m_0, \quad (105)$$

where $m_{-1}(a)$ is the coefficient of the power divergence which is independent of the specific matrix element, while $m_0 \sim O(\Lambda_{\text{QCD}})$ is finite and depends on the external state. The determination of m_0 can be rather nontrivial, and in practical calculations one could adopt a fine-tuning method, such as that for the Wilson-fermion mass, to find the critical value of m_0 at which the final result converges fastest in the large P^2 limit.

After the Wilson-line mass subtraction, there are still logarithmic UV divergences in $O_T(z, a)$. One can use lattice perturbation theory to match δm -subtracted $O_T(z, a)$ to the $\overline{\text{MS}}$ scheme (Constantinou and Panagopoulos, 2017; Ishikawa *et al.*, 2016; Xiong *et al.*, 2017), but the convergence still needs to be examined at higher orders. In (Green *et al.*, 2018, 2020), the logarithmic divergences were nonperturbatively renormalized with RI/MOM-like schemes.

The Wilson-line mass-subtraction has been implemented on the lattice in (Alexandrou *et al.*, 2020c; Green *et al.*, 2018; Musch *et al.*, 2011; Zhang *et al.*, 2017, 2019c).

2. RI/MOM scheme

The RI/MOM scheme has been widely used in lattice QCD for the renormalization of local composite quark operators that are free from power-divergent mixings (Martinelli *et al.*, 1995). It is essentially a momentum subtraction scheme in QFT and can be nonperturbatively implemented on the lattice. For an arbitrary composite quark bilinear operator O^B that is multiplicatively renormalized as $O^B = Z_O O^B$, the RI/MOM scheme is defined by imposing the following condition on its off-shell quark matrix element at a subtraction scale μ_R ,

$$Z_O^{-1} \langle p | O^B | p \rangle \Big|_{p^2 = -\mu_R^2} = \langle p | O | p \rangle_{\text{tree}}. \quad (106)$$

where the subscript “tree” means the tree-level matrix element in perturbation theory. If $\mu_R \gg \Lambda_{\text{QCD}}$, Z_O defined in Eq. (106) is in the perturbative region, and we can convert it to the $\overline{\text{MS}}$ scheme order by order in perturbation theory. In this sense, Z_O is not literally nonperturbative, but an all-order calculable quantity.

Since the nonlocal quark bilinear operator $O_T(z)$ has been proven to be multiplicatively renormalizable in the coordinate space, one can also renormalize it in the RI/MOM scheme and then match the result to PDF in the $\overline{\text{MS}}$ scheme (Constantinou and Panagopoulos, 2017; Stewart and Zhao, 2018). On the lattice, the off-shell matrix element of an operator is defined from its amputated Green’s function, or vertex function, with off-shell quarks. For the nonlocal Wilson-line operator, the latter is

$$\Lambda_0^\Gamma(z, a, p) = [S_0^{-1}(p, a)]^\dagger \sum_{x, y} e^{ip \cdot (x-y)} \times \langle 0 | T [\psi_0(x, a) O_T^\Gamma(z, a) \bar{\psi}_0(y, a)] | 0 \rangle S_0^{-1}(p, a), \quad (107)$$

where $S_0(p, a)$ is the bare quark propagator, and the external momentum p is Euclidean on the lattice. Since Green’s functions are not gauge invariant, one needs to fix a gauge (usually Landau gauge $\partial \cdot A = 0$ is chosen), and the gauge dependence is expected to be canceled by the matching or scheme conversion order by order in perturbation theory.

After including the quark wavefunction renormalization Z_q , which can be determined independently on the lattice (Martinelli *et al.*, 1995), Eq. (106) is revised as

$$Z_q Z_{O_T} \Lambda_0^\Gamma(z, a, p) \Big|_{p^2 = \mu_R^2} = \Lambda_0^\Gamma(z, a, p) = \Gamma e^{ip \cdot z}. \quad (108)$$

Since $O_T(z, a)$ is not $O(4)$ covariant, one needs to define the RI/MOM scheme with two scales, one is $\mu_R = |p_R|$, and the other $\bar{\mu}_R^-$. For convenience we simply denote them as $p = p_R$. To work in the perturbative region and control the lattice discretization effects that are of order $O(a^2 \mu_R^2, a^4 (p_R^2)^2)$, one must work in the window $\Lambda_{\text{QCD}} \ll \mu_R \ll a^{-1}$, $\bar{\mu}_R^- \ll a^{-1}$, which is attainable if the lattice spacing is small enough.

Since the quarks are off-shell, also finite mixings with the EOM operators can appear. As a result, Eq. (108) in general cannot be satisfied as a matrix equation. Instead, one usually needs a projection operator $\mathcal{P}$ to define the off-shell matrix elements, i.e.

$$\langle p | O_V^B | p \rangle = \text{tr} [\Lambda_0^B(z, a, p) \mathcal{P}], \quad (109)$$

so as to calculate the renormalization factor Z_{O_V} .

Then, the bare hadron matrix element $\tilde{h}_B(z, P^2, a)$ can be renormalized in coordinate space as

$$\tilde{h}_B(z, P^2, p_R^-, \mu_R, a) = Z_O^{-1}(z, p_R^-, \mu_R, a) \tilde{h}_B(z, P^2, a), \quad (110)$$

In the continuum limit, the renormalized matrix element is independent of the UV regulator, so we should obtain the same result in DR under RI/MOM scheme, i.e.,

$$\begin{aligned} \tilde{h}_B(z, P^2, p_R^-, \mu_R) &= \lim_{a \rightarrow 0} \tilde{h}_B(z, P^2, p_R^-, \mu_R, a) \\ &= \lim_{\epsilon \rightarrow 0} Z_O^{-1}(z, p_R^-, \mu_R, \epsilon) \tilde{h}_B(z, P^2, \epsilon), \end{aligned} \quad (111)$$

which allows us to compute the matching coefficients in continuum perturbation theory. Note that δm vanishes in Z_O due to the use of DR.

By Fourier transforming the above renormalized matrix element to momentum space, one can then work out the RI/MOM matching coefficient for the quasi-PDFs (Stewart and Zhao, 2018). The one-loop matching coefficient for different spin structures has been obtained in (Liu *et al.*, 2018, 2020; Stewart and Zhao, 2018), and the two-loop result for the unpolarized case can be found in (Chen *et al.*, 2020b). Alternatively, one can also first convert the RI/MOM matrix element to the $\overline{\text{MS}}$ or modified $\overline{\text{MS}}$ schemes (Alexandrou *et al.*, 2019b; Constantinou and Panagopoulos, 2017), and then do the Fourier transform and momentum-space matching.

3. Ratio scheme

In the coordinate-space factorization, $|z| \ll \Lambda_{\text{QCD}}^{-1}$ must be small, whereas P^2 can be of any value. In this case, the ratio scheme in (Orginos *et al.*, 2017; Radyushkin, 2017a) can be an effective choice for lattice renormalization. Consider the ratio

$$\tilde{h}(\lambda, z^2, a) / \tilde{h}(0, z^2, a), \quad (112)$$

where the denominator is a nonperturbative matrix element at $P^2 = 0$. Since $\tilde{h}(\lambda, z^2, a)$ and $\tilde{h}(0, z^2, a)$ calculated from the same lattice ensemble are correlated with each other, the error in the ratio can be reduced. Besides, the ratio does not need further renormalization on the lattice, so one can directly take the continuum limit

$$\lim_{a \rightarrow 0} \frac{\tilde{h}(\lambda, z^2, a)}{\tilde{h}(0, z^2, a)} = \frac{\tilde{h}(\lambda, z^2)}{\tilde{h}(0, z^2)}, \quad (113)$$

which has referred to as the “reduced Ioffe-time pseudo” distribution in (Orginos *et al.*, 2017; Radyushkin, 2017a). In the $\overline{\text{MS}}$ scheme, $\tilde{h}(0, z^2, \mu^2)$ has a small- z expansion,

$$\tilde{h}(0, z^2, \mu^2) = C_0(z^2 \mu^2) + \mathcal{O}(z^2 M^2, z^2 \Lambda_{\text{QCD}}^2), \quad (114)$$

where the lowest moment of the iso-vector quark PDF a_0 is trivially one. If we ignore all the power corrections, then $\tilde{h}(0, z^2, \mu^2)$ is perturbative and can be regarded as a renormalization factor. Therefore, the ratio in Eq. (113) still satisfies a similar OPE or factorization formula to Eqs. (98) and (95), except that the matching coefficient must be modified correspondingly (Izubuchi *et al.*, 2018; Radyushkin, 2018a),

$$\mathcal{C}^{\text{ratio}}(\alpha, z^2 \mu^2) = \mathcal{C}(\alpha, z^2 \mu^2) - \delta(1 - \alpha) C_0(z^2 \mu^2). \quad (115)$$

In other variants of the ratio scheme, it has also been suggested that one replaces $\tilde{h}(0, z^2, a)$ by the vacuum matrix element of the nonlocal Wilson line operator (Braun *et al.*, 2019; Li *et al.*, 2020), as the UV divergence does not depend on the external state.

4. Hybrid scheme

Since the factorization formula for the quasi-PDF is only proven in the $\overline{\text{MS}}$ scheme, it is not legitimate to use momentum-space factorization for any other scheme that differ from $\overline{\text{MS}}$ nonperturbatively. The RI/MOM and ratio schemes fall into this category as the conversion factors that match them to $\overline{\text{MS}}$ includes logarithms of z^2 (Constantinou and Panagopoulos, 2017; Izubuchi *et al.*, 2018), which requires running α_s to the IR region when $z \sim \Lambda_{\text{QCD}}^{-1}$. In contrast, the Wilson-line mass-subtraction scheme with wavefunction renormalizations is essentially the same as $\overline{\text{MS}}$, so it will not introduce extra IR effects.

However, the Wilson-line mass-subtraction scheme also has disadvantages. On the lattice, due to discretization effects at $z \sim a$, the lattice scheme cannot reproduce the short-distance $\ln z^2$ behavior of the $\overline{\text{MS}}$ matrix elements of the nonlocal operator. Such discretization effects, however, are cancelled in the RI/MOM or ratio scheme. To take advantages of both types of schemes, the hybrid scheme was proposed in (Ji *et al.*, 2020b) which provides a viable approach to renormalize the quasi-LF correlations at all z .

To begin with, for $|z| \leq z_8$ where z_8 is smaller than the distance at which the leading-twist approximation in the OPE becomes unreliable, one renormalizes the quasi-LF correlation as

$$\frac{\tilde{h}(z, a, P^2)}{Z_X(z, a)}, \quad (116)$$

where “ X ” can be the RI/MOM or ratio scheme.

For $|z| > z_8$, one applies the Wilson-line mass subtraction

$$\tilde{h}(z, a, P^z) e^{-\delta m |z|} Z_{\text{hybrid}}, \quad (117)$$

where Z_{hybrid} denotes the wavefunction and vertex renormalizations, which can be nonperturbatively determined by imposing a continuity condition at $z = z_8$,

$$Z_{\text{hybrid}} e^{-\delta m |z|} \tilde{h}(z, a, P^z) = \frac{\tilde{h}(z, a, P^z)}{Z_X(z_8, a)}, \quad (118)$$

leading to

$$Z_{\text{hybrid}}(z_8, a) = e^{\delta m |z|} / Z_X(z_8, a). \quad (119)$$

In this way, one only has to calculate δm . Note that the final result should be independent of z_8 , so one should try multiple values and find the optimal one around which the result changes the most slightly.

The perturbative matching for the hybrid renormalized quasi-PDF can be derived accordingly. Taking Z_X being the zero-momentum matrix element in the ratio scheme as an example, the $O(\alpha_s)$ matching has been derived as (Ji *et al.*, 2020b)

$$C_{\text{hybrid}}(\xi, \mu^2 / (p^z)^2, z_8^2 \mu^2) = C_{\text{ratio}}(\xi, \mu^2 / (p^z)^2) + \frac{\alpha_s C_F}{2\pi} \frac{3}{2} \left[-\frac{1}{|1-\xi|_+} + \frac{2\text{Si}((1-\xi)\lambda_8)}{\pi(1-\xi)} \right], \quad (120)$$

where C_{ratio} can be found in (Izubuchi *et al.*, 2018), $\xi = y/x$, and $\lambda_8 = z_8 p^z$ with $p^z = x P^z$ being the parton momentum. The plus function is defined as

$$\frac{1}{|1-\xi|_+} = \lim_{\beta \rightarrow 0^+} \left[\frac{\theta(1-\xi-\beta)}{|1-\xi|} + 2\delta(1-\xi) \ln \beta \right]. \quad (121)$$

Due to finite lattice volume and deteriorating signal-to-noise ratios at large z , the available lattice data have to be truncated at z_L . As we have discussed in Sec. II C, the quasi-LF correlation has a correlation length $\xi_z \sim \Lambda_{\text{QCD}}$ and exhibits an exponential decay at large z (~ 1 fm). If z_L is not sufficiently large and the quasi-LF correlation still has a considerable nonzero value, then a direct Fourier transform truncated at z_L will lead to unphysical oscillations and other systematics in the quasi-PDF.

To improve this situation, it is suggested in the hybrid scheme to perform an extrapolation to $z \rightarrow \infty$ (Ji *et al.*, 2020b). When P^z is not very large and the lattice matrix elements exhibit the exponential behavior near z_L , one can use the form $\sim e^{-z/L_z}$ to do the extrapolation, although some algebraic behavior can be added on top to better reflect the z -dependence. If P^z is very large, then the signal-to-noise ratio gets worse, so z_L is smaller. In this case, the quasi-LF correlation is yet to show exponential decay and dominated by the leading-twist contributions, so one can use the algebraic decay form to

do the extrapolation. Since $\lambda_L = z_L P^z$ can reach reasonably large values with contemporary computing resources, the extrapolation will only affect very small- x region, for which the LaMET expansion is not well under control after all.

To summarize, the hybrid scheme provides a proper renormalization of the quasi-LF correlations at all z , which allows for a controlled calculation of the PDF for $x \in [x_{\text{min}}, x_{\text{max}}]$ through LaMET expansion in momentum space. Therefore, we expect it to play a dominant role in the LaMET calculation of PDFs in the future.

E. Total Gluon Helicity ΔG and Transversity PDF

Apart from the collinear PDFs, the first application of LaMET is the gluon helicity contribution ΔG to the proton spin (Ji *et al.*, 2013b). In the naive sum rule for the proton spin (Jaffe and Manohar, 1990), ΔG is related to the matrix element of a nonlocal light-cone correlation operator (Manohar, 1991),

$$\Delta G = \langle PS | i \int \frac{dx d\lambda}{2\pi x P^+} e^{i\lambda x F^+ \alpha} (0) W(0, \lambda n) \tilde{F}_\alpha^+ (\lambda n) | PS \rangle, \quad (122)$$

which in the light-cone gauge $A^+ = 0$ reduces to

$$\Delta G = \langle PS | (\vec{E} \times \vec{A})^z | PS \rangle / (2P^+). \quad (123)$$

Within the LaMET framework, one can start from a static “gluon spin” operator, which is defined as $\vec{E} \times \vec{A}$ fixed in a time-independent gauge which maintains the transverse polarizations of the gluon field in the IMF limit. For example, the Coulomb gauge $\vec{\nabla} \cdot \vec{A} = 0$, axial gauges $A^z = 0$ and $A^0 = 0$ are viable options (Hatta *et al.*, 2014).

In the Coulomb gauge and $\overline{\text{MS}}$ scheme, the static “gluon spin” $\Delta \vec{G}$ in a massive on-shell quark state at one-loop order is (Chen *et al.*, 2011; Ji *et al.*, 2013b)

$$\Delta \vec{G}(P^z, \mu)(2S^z) = \langle PS | e_L^i F^{\alpha 0} A^j | PS \rangle_q \Big|_{\vec{q} \cdot \vec{A} = 0} \quad (124) \\ = \frac{\alpha_s C_F}{4\pi} \left[\frac{5}{3} \ln \frac{\mu^2}{m^2} - \frac{1}{9} + \frac{4}{3} \ln \frac{(2P^z)^2}{m^2} \right] (2S^z),$$

where the subscript q denotes a quark, and S^0 is the spin vector. The collinear divergence is regulated by the finite quark mass m .

If we follow the procedure in (Weinberg, 1966) and take $P^z \rightarrow \infty$ limit before UV regularization (Ji *et al.*, 2013b), then

$$\Delta \vec{G}(\infty, \mu)(2S^z) = \langle PS | e_L^i F^{\alpha 0} A^j | PS \rangle_q \Big|_{\vec{q} \cdot \vec{A} = 0} \quad (125) \\ = \frac{\alpha_s C_F}{4\pi} \left(3 \ln \frac{\mu^2}{m^2} + 7 \right) (2S^z),$$

which is exactly the same as the light-cone gluon helicity $\Delta G(\mu)$ (Hoodbhoy *et al.*, 1999a). Therefore, despite the difference in the UV divergence, the collinear divergences of $\Delta\bar{G}(P^z, \mu)$ and $\Delta G(\mu)$ are exactly the same, which allows for a perturbative matching between them.

The complete factorization formula that relates $\Delta\bar{G}(P^z, \mu)$ to ΔG and $\Delta\Sigma$ is

$$\Delta\bar{G}(P^z, \mu) = Z_{gg}(P^z/\mu)\Delta G(\mu) + Z_{gq}(P^z/\mu)\Delta\Sigma(\mu) + \dots, \quad (126)$$

where ... are power corrections suppressed by $1/P^z$, and the matching coefficients Z_{gg} and Z_{gq} have been calculated for the Coulomb gauge at one-loop (Ji *et al.*, 2015c).

Besides, one can also calculate the gluon helicity PDF $\Delta g(x)$ according to the factorization formula in Sec. III, and then integrate it over x to obtain ΔG .

At leading-twist, apart from the unpolarized and helicity PDFs that we have discussed before, there is also the transversity PDF defined as (Jaffe and Ji, 1991, 1992)

$$h_1(x) = \frac{1}{2P^+} \int \frac{d\lambda}{2\pi} e^{-i\lambda x} \langle PS_\perp | \bar{\psi}(0) \gamma^+ \gamma_\perp \gamma_5 \psi(\lambda n) | PS_\perp \rangle. \quad (127)$$

The $h_1(x)$ simply counts the number of transversely polarized quarks carrying the momentum fraction x in a transversely polarized proton. The first moment of this distribution corresponds to the so-called tensor charge δq , which is the matrix element of a chiral-odd operator. $h_1(x)$ can be accessed through the transverse-transverse spin asymmetry in Drell-Yan processes (Jaffe and Ji, 1991, 1992; Ralston and Soper, 1979) or the Collins single-spin asymmetry in SIDIS where the transversity TMDPDF couples to a chiral-odd TMD fragmentation function (Collins, 1993). At present, experimental results on the transversity PDF are very limited (Barone *et al.*, 2002; Cammarota *et al.*, 2020; Kang *et al.*, 2016; Lin *et al.*, 2018b; Radici and Bacchetta, 2018), especially for the sea quark contributions (Chang and Peng, 2014), so this is one scenario where lattice QCD calculation can make an important difference.

The LaMET calculation of $h_1(x)$ is straightforward as the nonlocal operator has the same renormalization as the unpolarized case, and its one-loop matching has been calculated in the $\overline{\text{MS}}$ and RI/MOM schemes at one-loop order (Alexandrou *et al.*, 2018b; Liu *et al.*, 2018). First lattice calculations of $h_1(x)$ have been done in (Alexandrou *et al.*, 2018b; Chen *et al.*, 2016; Liu *et al.*, 2018), which will be discussed with more details in Sec. VI.

IV. GENERALIZED COLLINEAR PARTON OBSERVABLES

In the previous section, we have extensively discussed the leading-twist collinear PDFs that characterize the 1D structure of the proton in longitudinal momentum space. There exist various other parton observables that provide complementary information. In this section, we focus on observables defined by collinear parton correlators, in the sense that only the collinear quark and gluon mode contribute, corresponding to the so-called *collinear expansion* in QCD factorizations (Collins, 2011a; Sterman, 1993). We call them “generalized collinear parton observables” (GCPOs), and discuss their calculations through LaMET framework. For observables defined by parton correlators involving transverse separations, in particular, the TMDPDFs, Wigner functions, and LFWFs, we will consider them in the following sections.

One of the important GCPOs is the GPDs introduced in (Müller *et al.*, 1994), and rediscovered (Ji, 1997b) from their connection to the spin structure of the proton. They describe the correlation between the transverse position and longitudinal momentum of partons inside the proton, and thus provide important information for 3D imaging of the proton. A proton spin sum rule was derived in terms of the moments of the GPDs, which has stimulated considerable general interest in the GPDs. It was also found that in the so-called zero skewness limit or when the longitudinal momentum transfer vanishes, the GPD has a probability interpretation in the impact parameter space (Burkardt, 2000). In general case, it is related to the quantum phase-space distributions or Wigner functions (Belitsky *et al.*, 2004; Ji, 2003). Experimentally, the GPDs can be measured through hard exclusive processes such as deeply virtual Compton scattering (DVCS) or meson production (DVMP) that were first proposed in (Ji, 1997a,b). Much effort has been devoted to measuring such processes at completed and ongoing experiments, including HERA, COMPASS and JLab. For a more comprehensive discussion on the GPDs, we refer the readers to the review articles (Belitsky and Radyushkin, 2005; Diehl, 2003; Ji, 2004, 1998). Despite that the GPDs have more complicated kinematic dependence and relation to experimental observables, various fitting methods have been proposed in the literature to fit available DVCS and DVMP data (Fwart *et al.*, 2016; Kumericki *et al.*, 2016). In parallel, one can also extract certain information on the GPDs from lattice calculations of their moments (Alexandrou *et al.*, 2020b; Gockeler *et al.*, 2004; Hagler *et al.*, 2008), which, however, is again very limited due to the same difficulties existing in lattice calculations of the PDF moments. For JLab 12 GeV program and future EIC, it is critically important to have first-principle calculations of GPDs with much better understanding of the physical landscape in different kinematic variables.

A simpler but closely related GCPO is the parton distribution amplitudes (DAs), which are collinear ma-

trix elements of light-cone operators between a hadron state and the QCD vacuum, representing the probability amplitude of finding a given Fock state in the hadron. They can be probed in certain exclusive processes (Brodsky, 2002), and are crucial inputs for processes relevant to measuring fundamental parameters of the Standard Model and probing new physics. There exists a vast amount of literature on this subject, particularly about the pion DA. For a review see e.g. (Braun, 2006; Brodsky and Lepage, 1989; Grozin, 2005).

Another type of GCPO is the higher-twist parton distributions. They are defined by multi-parton correlation functions, and quantify the proton structure in terms of longitudinal momentum correlations (Ellis *et al.*, 1983; Jaffe and Ji, 1992; Jaffe and Siskind, 1982). Although physically interesting, they are hard to separate theoretically due to mixing with the leading-twist ones (Ji, 1995; Mueller, 1985), and difficult to extract experimentally because they are power-suppressed (Ji, 1993). Higher-twist effects can become important in kinematic regions where the suppression is relaxed. Moreover, some twist-three distributions, g_T and h_L , are different; they have no leading-twist to mix with and are dominant in spin-related observables (Jaffe and Ji, 1992). Twist-three GPDs are also relevant for studying parton OAM in the proton (Courtney *et al.*, 2014; Hatta and Yoshida, 2012; Ji *et al.*, 2013a) and can be accessed through DVCS process (Kiptily and Polyakov, 2004; Penttinen *et al.*, 2000).

In principle, all the GCPOs discussed above can be computed within LaMET. In addition, an accurate LaMET expansion for the leading-twist PDFs requires calculations of quasi higher-twist matrix elements. In the following, we begin with the flavor non-singlet quark GPDs and hadronic DAs for which the computational procedure has been well established, and then give some generic discussions on higher-twist distributions, followed by the discussion on power-suppressed contributions required to extract the leading-twist quark PDFs, which have been investigated using different approaches though not yet implemented in numerical computations.

A. Generalized Parton Distributions

The operators defining the GPDs are the same as those defining the PDFs. Thus, the LaMET calculation of PDFs can be rather straightforwardly generalized to the GPDs by taking into account the non-forward kinematics (Liu *et al.*, 2019b). To illustrate how it works, let us take the nonsinglet unpolarized quark GPDs in the nucleon as an example.

The unpolarized quark GPDs are defined through the

following matrix element (Ji, 2004)

$$F = \frac{1}{2P^+} \int \frac{d\lambda}{2\pi} e^{-ix\lambda} \langle P' S' | O_{\gamma^+}(\lambda n) | P S \rangle \\ = \frac{1}{2P^+} \bar{u}(P' S') \left[H \gamma^+ + E \frac{i\sigma^{\mu\nu} \Delta_\mu}{2M} \right] u(P S), \quad (128)$$

where we have suppressed the arguments (x, ξ, t, μ) of F , H and E for simplicity. The operator

$$O_{\gamma^+}(\lambda n) = \bar{\psi}\left(\frac{\lambda n}{2}\right) \gamma^+ W\left(\frac{\lambda n}{2}, -\frac{\lambda n}{2}\right) \psi\left(-\frac{\lambda n}{2}\right) \quad (129)$$

with $n^\mu = 1/\sqrt{2}(1/P^+, 0, 0, -1/P^+)$ is the same operator used to define the unpolarized quark PDF, M is the nucleon mass. The momentum fraction $x \in [-1, 1]$, and

$$\Delta = P' - P, \quad t = \Delta^2, \quad \xi = -\frac{P'^+ - P^+}{P'^+ + P^+} = -\frac{\Delta^+}{2P^+}, \quad (130)$$

where without loss of generality we have chosen a Lorentz frame in which the average momentum takes the following form

$$\bar{P}^\mu = \frac{P'^\mu + P^\mu}{2} = (P^0, 0, 0, P^z). \quad (131)$$

The skewness parameter $\xi \in [-1, 1]$ since $P^+, P'^+ \geq 0$. Besides, there exists another kinematic constraint on ξ , which follows from $\bar{\Delta}_\perp^2 \geq 0$,

$$\xi \leq \xi_{\max}(t) = \sqrt{\frac{-t}{-t + 4M^2}}. \quad (132)$$

In the following, we will also assume $\xi > 0$ without loss of generality. With these kinematic constraints, the GPDs can be divided into several kinematic regions that have different physical interpretations. As shown in Fig. 9, in the region $\xi < x < 1$ ($-1 < x < -\xi$) the distribution describes the emission and reabsorption of a quark (antiquark), while in the region $-\xi < x < \xi$ it represents the creation of a quark and antiquark pair. The first region is similar to that present in usual PDFs and referred to as the DGLAP region, whereas the second is similar to that in a meson DA, which will be discussed later in this section, and referred to as the Efremov-Radyushkin-Brodsky-Lepage (ERBL) region. The easiest way to see this is in light-cone quantization and light-cone gauge where the matrix element defining the GPDs can be rewritten in terms of parton creation and annihilation operators, for details see e.g. (Ji, 2004).

The quark GPDs defined above have a number of remarkable properties, see, e.g., (Belitsky and Radyushkin, 2005; Diehl, 2003; Ji, 2004, 1998), which either hold or have similar counterparts for the quark quasi-GPDs to be defined below. Apart from their physical significance, these properties also serve as useful checks on calculations related to GPDs.



FIG. 9: Parton interpretation of the GPDs in different kinematic regions.

According to LaMET, the unpolarized quark GPDs defined above can be determined by calculating the following quasi-GPDs

$$\begin{aligned}\bar{F} &= \frac{1}{2P^0} \int \frac{d\lambda}{2\pi} e^{i\lambda z} \langle P' S' | O_{\gamma^0}(z) | P S \rangle \\ &= \frac{1}{2P^0} u(P' S') \left\{ \hat{H} \gamma^0 + E \frac{i\sigma^{0\mu} \Delta_\mu}{2M} \right\} u(P S),\end{aligned}\quad (133)$$

where we have again suppressed the arguments (y, ξ, t, P^+, μ) of F , H , and E . The operator $O_{\gamma^0}(z) = \bar{\psi}(\frac{z}{2}) \gamma^0 W(\frac{z}{2}, -\frac{z}{2}) \psi(-\frac{z}{2})$ is the same operator defining the unpolarized quark quasi-PDF, and $\lambda = zP^+$. As in the quasi-PDF case, the momentum fraction y extends from $-\infty$ to ∞ . The skewness parameter for the quasi-GPD

$$\tilde{\xi} = -\frac{P'^+ - P^+}{P'^+ + P^+} = -\frac{\Delta^+}{2P^+} = \xi + \mathcal{O}\left(\frac{M^2}{(P^+)^2}, \frac{t}{(P^+)^2}\right)\quad (134)$$

differs from the light-cone skewness ξ by power suppressed corrections. Moreover, the constraint from $\Delta_\perp^2 \geq 0$ becomes (Ji *et al.*, 2015a)

$$\tilde{\xi} \leq \frac{1}{2P^+} \sqrt{\frac{-t[(P^+)^2 + M^2 - t/4]}{M^2 - t/4}},\quad (135)$$

which differs from the constraint in Eq. (132) by corrections of $\mathcal{O}(M^2/(P^+)^2, t/(P^+)^2)$. We can replace ξ with $\tilde{\xi}$ and attribute the difference to generic power suppressed contributions.

The quasi-GPDs defined above can be renormalized by observing that their UV divergence depends only on the operators defining them, but not on the external states. Since $O_{\gamma^0}(z)$ is multiplicatively renormalized, we can choose the same renormalization factor as that for the quasi-PDF (Liu *et al.*, 2020; Stewart and Zhao, 2018) to renormalize the quasi-GPD. After renormalization, the quasi-GPD can then be matched to the usual GPD through a factorization formula.

The factorization of quasi-GPDs was first proposed and verified at one-loop order in (Ji *et al.*, 2015a; Xiong and Zhang, 2015), where a transverse momentum cutoff and a quark mass were used as the UV and IR regulator, respectively. Later on, a detailed derivation based on OPE was given in (Liu *et al.*, 2019a). In contrast with the OPE for the quasi-PDF, a crucial difference here is that the total derivative of operators can come into play,

as it simply gives momentum transfer factors when sandwiched between non-forward external states, and therefore is non-vanishing. In other words, the local twist-two operators as those in Eq. (97) will mix under renormalization with operators with total derivatives. The RGE that governs the mixing reads (Braun *et al.*, 2003),

$$\begin{aligned}\mu^2 \frac{d}{d\mu^2} \langle \mathcal{O}^{\mu_1 \mu_2 \dots \mu_n}(\mu) \rangle &= \sum_{m=0}^{\lfloor n/2 \rfloor} \Gamma_{nm} \\ &\times \left[i \bar{g}^{\mu_1 \mu_2} \dots i \bar{g}^{\mu_{2m} \mu_{2m+1}} i \overleftrightarrow{D}^{\mu_{2m+1}} \dots i \overleftrightarrow{D}^{\mu_n} \psi - \text{trace} \right],\end{aligned}\quad (136)$$

where Γ_{nm} is the anomalous dimension of the associated operators, $\overleftrightarrow{D} = (\overrightarrow{D} - \overleftarrow{D})/2$ with $\overrightarrow{D}(\overleftarrow{D})$ denoting the covariant derivative acting to the right (left). The above equation can be diagonalized by choosing an appropriate operator basis. Such an operator basis has been studied in the literature and known as the “renormalization group improved” conformal operators (Braun *et al.*, 2003; Mueller, 1994b). In terms of the matrix elements of these operators, we have

$$\begin{aligned}\langle P' | O_{\gamma^0}(z) | P \rangle &= 2P^0 \sum_{n=0}^{\infty} C_n(\mu^2 z^2) \mathcal{F}_n(-\lambda) \sum_{m=0}^n B_{nm}(\mu) \\ &\times \xi^n \int_{-1}^1 dx C_m^{\alpha/2}\left(\frac{x}{\xi}\right) F(x, \xi, t, \mu) + \dots,\end{aligned}\quad (137)$$

where $\mathcal{F}_n(-\lambda)$ are partial wave polynomials whose explicit forms are known in the conformal OPE of current-current correlators for the hadronic light-cone DAs (Braun and Müller, 2008). B_{nm} can be found in (Braun *et al.*, 2003; Mueller, 1994b), and $\dots$ denotes the higher-twist contributions $\mathcal{O}(M^2/(P^+)^2, t/(P^+)^2, z^2 \Lambda_{\text{QCD}}^2)$.

Fourier transforming the l.h.s of the above equation to momentum space and invert it order by order in α_s , we then obtain the following EFT expansion of the unpolarized quark GPD,

$$\begin{aligned}F(x, \xi, t, \mu) &= \int_{-\infty}^{\infty} \frac{dy}{|y|} C\left(\frac{x}{\xi}, \frac{y}{\xi}, \frac{\mu}{\xi P^+}\right) \bar{F}(y, \xi, t, P^+, \mu) + \dots \\ &= \int_{-\infty}^{\infty} \frac{dy}{|y|} C\left(\frac{x}{y}, \frac{\xi}{y}, \frac{\mu}{y P^+}\right) \bar{F}(y, \xi, t, P^+, \mu) + \dots,\end{aligned}\quad (138)$$

which has been organized following the same spirit as the factorization of PDFs in previous sections. Both forms have been used in the literature (Ji *et al.*, 2015a; Liu *et al.*, 2019a; Xiong and Zhang, 2015) with the matching coefficients being related by

$$C\left(\frac{x}{y}, \frac{\xi}{y}, \frac{\mu}{y P^+}\right) = \frac{|y|}{|\xi|} C\left(\frac{x}{\xi}, \frac{y}{\xi}, \frac{\mu}{\xi P^+}\right),\quad (139)$$

and $\dots$ denotes the higher-twist contributions which have the same power-counting as in Eq. (137) except that z^2

is replaced by $1/(xP^2)^2$. For the helicity and transversity quark quasi-GPDs, the factorization formula has the same form as Eq. (138) (Liu *et al.*, 2019a).

The matching coefficient can be obtained by replacing the hadron states in Eqs. (128) and (133) with the quark states carrying momentum $p + \Delta/2$ and $p - \Delta/2$ with $p^\mu = (p^0, 0, 0, p^3)$, and calculating the quark matrix element in perturbation theory. The explicit expression for the $\mathcal{O}(\alpha_s)$ matching coefficients can be found in (Liu *et al.*, 2019a). An important feature of the result is: The quasi-GPDs do not vanish in all y range, but the collinear singularities only show up in DGLAP and ERL regions at one-loop. They are exactly the same as those in light-cone GPDs, and thus cancel in the matching coefficient. Moreover, one can derive momentum RGEs for the quasi-GPDs, which are turned into RGE for the scale dependence of the GPDs by the matching procedure.

To conclude this subsection, let us make some remarks on the EFT formula for the quark GPD above. First, at zero skewness $\xi = 0$, we have

$$F(x, 0, t, \mu) = \int_{-\infty}^{\infty} \frac{dy}{|y|} C\left(\frac{x}{y}, 0, \frac{\mu}{yP^2}\right) \tilde{F}(y, 0, t, P^2, \mu) + \dots, \quad (140)$$

where the matching kernel $C(x/y, 0, \mu/(yP^2))$ is exactly the same as the matching coefficient for the quasi-PDF (Izubuchi *et al.*, 2018), even when $t \neq 0$. This can be understood as follows: At zero skewness, both the longitudinal momentum transfer and the energy transfer vanish, the momentum transfer is purely transverse and thus is not affected by Lorentz boost along the longitudinal z direction. As a result, no extra matching related to t is required in the large P^2 limit, and the matching remains the same as in the quasi-PDF case. If we take the forward limit $\Delta \rightarrow 0$, then Eq. (140) reduces exactly to the EFT expansion formula for the PDF (Izubuchi *et al.*, 2018; Ji *et al.*, 2015a).

Second, in the limit $\xi \rightarrow 1$ and $t \rightarrow 0$, the quasi-GPD reduces to the quasi-DA that will be discussed in the next subsection, and the corresponding matching kernel also reduces to that for the quasi-DA.

B. Hadronic Distribution Amplitudes

Within LaMET, the DAs of protons as well as other hadrons can also be extracted from lattice simulations of appropriately chosen quasi-DAs. In this subsection, we show how this can be done in practice. For illustration, we take the leading-twist pion DA as an example. The application to other hadrons (Wang *et al.*, 2019a; Zhang *et al.*, 2019c) is analogous.

The leading-twist DA of the pion is the simplest and most extensively studied hadronic DA. It represents the probability amplitude of finding the valence $q\bar{q}$ Fock state

in the pion with the quark carrying a fraction x of the total pion momentum, and is defined as

$$\phi_\pi(x) = \frac{1}{i f_\pi} \int \frac{d\lambda}{2\pi P^+} e^{-i(x-\frac{1}{2})\lambda} \langle 0 | O_{\gamma^+ \gamma_5}(\lambda n) | \pi(P) \rangle, \quad (141)$$

with normalization $\int_0^1 dx \phi_\pi(x) = 1$. Here f_π denotes the decay constant, and $O_{\gamma^+ \gamma_5}(\lambda n)$ has the same structure as that used in Eq. (128) with γ^+ replaced by $\gamma^+ \gamma_5$. The pion DA can be constrained from experimental measurements of, e.g., $\gamma^* \gamma^* \rightarrow \pi^0$ from BaBar and Belle (Aubert *et al.*, 2009; Uehara *et al.*, 2012), and then used as an input to test QCD in other measurements such as the pion form factor (Efremov and Radyushkin, 1980; Farrar and Jackson, 1979). In the asymptotic limit, it is well known that the pion DA takes the form $6x(1-x)$ (Efremov and Radyushkin, 1980; Lepage and Brodsky, 1979). However, how it behaves at lower scales remains under debate (see e.g. (Chernyak and Zhitnitsky, 1982)). Calculating the pion DA with controllable systematics in LaMET will be able to shed new lights on its shape and thus on our understanding of pion structure.

Following the same strategy as before, we can access the x -dependence of the pion DA by studying the following quasi-DA (Ji *et al.*, 2015a; Zhang *et al.*, 2017)

$$\tilde{\phi}_\pi(y, P^2) = \frac{1}{i f_\pi} \int \frac{d\lambda}{2\pi P^+} e^{i(y-\frac{1}{2})\lambda} \langle 0 | O_{\gamma^+ \gamma_5}(z) | \pi(P) \rangle, \quad (142)$$

The longitudinally and transversely polarized vector meson quasi-DAs can be defined analogously by replacing $\gamma^+ \gamma_5$ in the above equation with γ^0 , $\gamma^+ \gamma_\perp$, respectively (Liu *et al.*, 2019b).

The quark bilinear operators defining quasi-DAs follow the same renormalization pattern as those defining the quasi-PDFs or quasi-GPDs. In the literature, the Wilson-line mass-subtraction scheme was used in the first LaMET calculations of the meson DAs (Zhang *et al.*, 2017, 2019c), and the RI/MOM scheme has been adopted in more recent works (Zhang *et al.*, 2020b).

The LaMET expression for DAs takes the following form in the $\overline{\text{MS}}$ scheme (Ji *et al.*, 2015a; Liu *et al.*, 2019b)

$$\phi_\pi(x, \mu) = \int_{-\infty}^{\infty} dy C_\pi(x, y, P^2/\mu) \tilde{\phi}_\pi(y, P^2, \mu) + \dots \quad (143)$$

The matching coefficient for the quasi-DAs can be obtained by replacing the meson state $|\pi(P)\rangle$ in Eqs. (141) and (142) with the lowest Fock state $|q(y)P\bar{q}((1-y)P)\rangle$ and calculating the quark matrix elements, where yP and $(1-y)P$ are the momenta of the quark q and anti-quark $\bar{q}$, respectively. Its one-loop results have been calculated in both $\overline{\text{MS}}$ and RI/MOM

schemes (Liu *et al.*, 2019b), which agrees with matching coefficient for the quasi-GPDs (Ji *et al.*, 2015a; Liu *et al.*, 2019a; Xiong and Zhang, 2015) in Eq. (138) with the replacement of $\xi \rightarrow 1/(2y-1)$, $x/\xi \rightarrow 2x-1$, and the external momentum p^2 to $p^2/2$.

Apart from the LaMET approach in momentum space, the shape of the pion DA has also been studied using equal-time current-current correlation in coordinate space approach (Bali *et al.*, 2018a,b),

$$\begin{aligned} \langle 0 | T \left\{ J_\mu \left(\frac{z}{2} \right) J_\nu \left(-\frac{z}{2} \right) \right\} | \pi^0(P) \rangle \\ = \frac{2i f_\pi}{3\pi^2 z^4} \epsilon_{\mu\nu\alpha\beta} P^\alpha z^\beta \Phi_\pi(\lambda, z^2), \end{aligned} \quad (144)$$

where $\Phi_\pi(\lambda, z^2)$ can be factorized as

$$\Phi_\pi(\lambda, z^2) = C_2(\lambda, z^2 \mu^2, x) \otimes \phi_\pi(x, \mu) + \dots \quad (145)$$

Here the matching coefficient C_2 depends on the choice of the currents. Its explicit expression can be found in (Bali *et al.*, 2018a). The above factorization is controlled by $\mathcal{O}(z^2 \Lambda_{\text{QCD}}^2)$, with power corrections denoted by "...". In (Bali *et al.*, 2018a), a combined analysis of several current-current correlations has been performed where twist-four contributions were also included using the model estimate in (Ball *et al.*, 2006; Braun and Filyanov, 1990). The leading-twist pion DA was then extracted from a global fit to the data, and the second moment of the pion DA has been fitted with controlled precision, both of which favor a considerably broader shape than the asymptotic DA at a scale of 2 GeV. A large pion momentum is required to access information at large λ so that we can extract wider range of x or higher moments of the pion DA (Bali *et al.*, 2018b).

C. Higher-Twist Distributions

Higher-twist distributions are quantities of great interest because they describe the coherent quark-gluon correlations in the proton. In contrast with the leading-twist distributions, our understanding of the higher-twist ones is rather poor. On one hand, they often depend on more than one parton momentum fractions; on the other hand, there is no physical intuition about what they may look like, in particular, about how they behave asymptotically at small and large x (Braun *et al.*, 2011). There have been studies on the higher-twist distributions in the context of their connection to the DIS structure function, the transverse single-spin asymmetries in various hadron productions, GPDs related to quark and gluon OAM, parton DAs, etc. LaMET will be able to shed new lights by providing a possibility to access them from the lattice.

Higher-twist contributions also appear in LaMET expansion, where the suppression is provided by powers of the hadron momentum squared. In all factorizations presented in previous sections, only the leading-twist terms

that capture the logarithmic dependence on hadron momentum are taken into account. The higher-twist contributions have been assumed to be small. If the hadron momentum is not sufficiently large compared and/or one is close to the endpoint region ($x \rightarrow 0$ and $x \rightarrow 1$), the higher-twist contributions can become non-negligible, whose structure and impact require understanding.

1. Higher-twist collinear-parton observables

Beyond leading-twist, there exist three simplest twist-three quark distributions $e(x)$, $g_T(x)$ and $h_L(x)$ related to the unpolarized, transversely and longitudinally polarized proton (Jaffe and Ji, 1992),

$$e(x) = \frac{1}{2M} \int \frac{d\lambda}{2\pi} e^{i\lambda x} \times \langle PS | \psi_\perp^\dagger(0) \gamma_0 \psi_\perp(\lambda n) | PS \rangle + \text{h.c.}, \quad (146)$$

$$g_T(x) = \frac{1}{2M} \int \frac{d\lambda}{2\pi} e^{i\lambda x} \times \langle PS_\perp | \psi_\perp^\dagger(0) \gamma_0 \gamma_\perp \gamma_5 \psi_\perp(\lambda n) | PS_\perp \rangle + \text{h.c.}, \quad (147)$$

$$h_L(x) = \frac{1}{2M} \int \frac{d\lambda}{2\pi} e^{i\lambda x} \times \langle PS_z | \psi_\perp^\dagger(0) \gamma_0 \gamma_5 \psi_\perp(\lambda n) | PS_z \rangle + \text{h.c.}, \quad (148)$$

where we have again employed the decomposition of quark fields $\psi = \psi_+ + \psi_-$ in Sec. I A and the light-cone gauge $A^+ = 0$, and "h.c." stands for Hermitian conjugate.

The twist-three distributions can contribute as leading effects in certain experimental observables. For example, $g_T(x)$ and $h_L(x)$ can be measured as the leading effects in the longitudinal-transverse spin asymmetry in polarized Drell-Yan process.

Since ψ_- is a non-dynamical component depending on ψ_+ , all the above distributions can be shown to be related to more complicated quark-gluon correlation functions (Balitsky *et al.*, 1996; Ji and Chou, 1990). A complete set of such correlation functions has been given in (Ji, 1992; Ji and Osborne, 2001; Kang and Qiu, 2009; Qiu and Sterman, 1991), where the quark-gluon correlations in a transversely-polarized proton take the following form

$$T_q(x_1, x_2) = \frac{1}{(P^+)^2} \int \frac{d\lambda d\zeta}{(2\pi)^2} e^{i\lambda x_1 + i\zeta(x_2 - x_1)} \times \langle PS_\perp | \bar{\psi}(0) \gamma_1^+ e^{+S \cdot \hat{z}} g F^{+i}(\zeta n) \psi(\lambda n) | PS_\perp \rangle, \quad (149)$$

$$T_{Aq}(x_1, x_2) = \frac{1}{(P^+)^2} \int \frac{d\lambda d\zeta}{(2\pi)^2} e^{i\lambda x_1 + i\zeta(x_2 - x_1)} \times \langle PS_\perp | \bar{\psi}(0) \gamma_1^+ \gamma_5 S_z^i g F^{+i}(\zeta n) \psi(\lambda n) | PS_\perp \rangle. \quad (150)$$

There are also ones in an unpolarized and longitudinally-polarized proton. Generalizing to off-forward kinematics, the resulting twist-three GPDs are also related

to quark and gluon OAM contribution to the proton spin (Hatta and Yoshida, 2012; Ji *et al.*, 2013a).

One can also define twist-four distributions in a similar way as in Eq. (149) by using ψ_- for both quark fields. More general twist-four distributions will involve three light-cone variables, which will contribute to, e.g., $1/Q^2$ term in DIS (Ellis *et al.*, 1983; Jaffe and Ji, 1992; Jaffe and Soldate, 1982; Ji, 1993).

In principle, all the above higher-twist distributions, as well as others that have not been listed here, can be computed using the LaMET approach by choosing appropriate quasi-LF correlations. For example, the first exploratory lattice calculation of $g_T(x)$ has been done in (Bhattacharya *et al.*, 2020a), which will be discussed in Sec. V LC. However, extra complications are expected due to their complex structure. For instance, the light-cone zero modes that do not enter in dealing with leading-twist distributions come into play here. Recently, one of the authors has shown how to study the properties of these zero modes from lattice simulations in LaMET (Ji, 2020). In addition, the higher-twist distributions will have a more complex mixing pattern (Balitsky *et al.*, 1996; Ji and Chou, 1990). Thus, their matching from the corresponding quasi distributions must take into account such mixings, making them more challenging than calculating the twist-two PDFs. One-loop studies of the matching for twist-three distributions have been carried out in (Bhattacharya *et al.*, 2020b,c).

2. Higher-twist contributions to quasi-PDFs

Let us turn to the power suppressed higher-twist contributions appearing in the extraction of leading-twist quark PDFs using LaMET. Such contributions have two distinct origins. To understand them, let us recall the OPE for the quasi-LF correlation in Eq. (98). For simplicity, we ignore the renormalization here. Recovering the leading-twist quark PDF requires removing the contributions of both trace terms in that equation. The trace terms on the r.h.s. of Eq. (98), which lead to contributions suppressed by powers of $M^2/(P^+)^2$, are known as kinematic power contributions or target mass corrections. In DIS, they can be accounted for by changing the scaling variable x to the Nachtmann variable (Nachtmann, 1973). In the case of LaMET, it behaves slightly differently, as shown in the following. The second type of power corrections come from the trace terms in the operators on the r.h.s. of Eq. (97), and in general leads to contributions of $O(\Lambda_{\text{CD}}^2/(P^+)^2)$. These are genuine higher-twist contributions that involve multi-parton correlations, sometimes also known as dynamical higher-twist contributions. The target mass corrections have been computed to all orders in $M^2/(P^+)^2$ for the quark quasi-PDFs in (Chen *et al.*, 2016; Radyushkin, 2017c). The genuine higher-twist contributions have been investi-

gated using two different approaches (Braun *et al.*, 2019; Chen *et al.*, 2016).

According to (Chen *et al.*, 2016), the $M^2/(P^+)^2$ corrections can be obtained from the ratio

$$K_m = \frac{n_{i_{\text{max}}} \cdots n_{i_{\text{min}}} P^{n_{i_1}} \cdots P^{n_{i_m}}}{n_{\mu_1} \cdots n_{\mu_m} P^{\mu_1} \cdots P^{\mu_m}} = \sum_{c=0}^{i_{\text{max}}} C_m^c c^c, \quad (151)$$

where $i_{\text{max}} = (m - \text{Mod}[m, 2])/2$, C is the binomial function and $c = -n^2 M^2/4(n \cdot P)^2 = M^2/4(P^+)^2$ with $n^\mu = (0, 0, 0, -1)$ and $n \cdot P = P^+$.

Plugged into the tree-level OPE formula in Eq. (98), the above factors can then be converted to the following relation between unpolarized PDF and quasi-PDF (Chen *et al.*, 2016)

$$q(x) = \sqrt{1+4c} \sum_{n=0}^{\infty} \frac{(4c)^n}{f_+^{n+1}} \left[(1 + (-1)^n) \bar{q} \left(\frac{f_+^{2n+1} x}{2(4c)^n} \right) + (1 - (-1)^n) \bar{q} \left(\frac{-f_+^{2n+1} x}{2(4c)^n} \right) \right], \quad (152)$$

where $f_+ = \sqrt{1+4c} + 1$. It is worth noting that quark number conservation is preserved in the above result. The target mass corrections for the longitudinally and transversely polarized quasi-PDFs can be derived analogously.

The trace part on the r.h.s. of Eq. (97) is a genuine higher-twist effect. One may try to construct a non-local form of the higher-twist operators from OPE. The leading trace term, which is a twist-four effect, has been studied in (Chen *et al.*, 2016) (see also (Balitsky and Braun, 1989)) and shown to give rise to a twist-four PDF

$$q_4(x, P^2) = \int_{-\infty}^{\infty} \frac{d\lambda}{8\pi P^+} \Gamma_0(-ix\lambda) \langle P | O_{4T}(z) | P \rangle, \quad (153)$$

with

$$O_{4T}(z) = \int_0^z dz_1 \bar{\psi}(0) \left[\Gamma^{\mu} W(0, z_1) D_{\mu} W(z_1, z) \right. \\ \left. + \int_0^{z_1} dz_2 n \cdot \Gamma W(0, z_2) D^{\mu} W(z_2, z_1) D_{\mu} W(z_1, z) \right] \psi(zn), \quad (154)$$

where one has $\Gamma^{\mu} = \gamma^{\mu}, \gamma^{\mu}\gamma^5, \gamma^{\perp}\gamma^{\mu}\gamma^5$ for the unpolarized, helicity and transversity PDFs, respectively. Γ_0 is the incomplete Gamma function

$$\Gamma_0(-ix) = \int_0^1 \frac{dt}{t} e^{ix/t}. \quad (155)$$

The above twist-four contribution needs to be removed from the quasi-PDF to recover the leading-twist PDF. It also provides a possibility for practical computations on the lattice. However, as a multi-parton correlation involving more gauge links and covariant derivatives, its lattice computation is rather challenging and has not been carried out in any existing work yet.

Another approach that has been used to estimate power corrections related to quark quasi-PDFs is the renormalon model (see (Beneke, 1999) for a comprehensive review). It is based on the observation that the perturbative expansion of the matching coefficient for the quasi-PDF diverges factorially with the loop order, implying that it is only well defined up to a power accuracy. This is known as the renormalon ambiguity, which must be cancelled by terms in the higher-twist contributions.

In (Braun *et al.*, 2019), it was shown that the cancellation of renormalon ambiguity requires that the leading higher-twist or twist-four contribution takes the following form

$$q_4(y, P^2, \mu) = \mu^2 \int_{-1/|\mu|}^1 \frac{dx}{|x|} D\left(\frac{y}{x}\right) q(x, \mu) + q'_4(y, P^2, \mu), \quad (156)$$

where the first term on the r.h.s. cancels the renormalon ambiguity from the leading-twist matching coefficient, and q'_4 depends on μ at most logarithmically. Since the first term is to merely cancel similar contributions in the matching coefficient, it does not contribute to any physical observable. The renormalon model of power corrections (Beneke and Braun, 1995; Beneke *et al.*, 1997; Braun *et al.*, 2004; Dasgupta and Webber, 1996, 1997; Dokshitzer *et al.*, 1996) is based on the assumption that, by replacing μ with a suitable nonperturbative scale, this contribution reflects the order and the functional form of actual power-suppressed contributions. This was known as “ultraviolet dominance” in (Beneke, 1999; Beneke and Braun, 2000; Braun, 1995). Under this assumption, we obtain the following estimate,

$$q_4(y, P^2, \mu) = \kappa \Lambda_{\text{QCD}}^2 \int_{-1/|\mu|}^1 \frac{dx}{|x|} D\left(\frac{y}{x}\right) q(x, \mu), \quad (157)$$

where κ is a dimensionless coefficient of $\mathcal{O}(1)$ that cannot be fixed within theory and remains a free parameter.

A detailed analysis (Braun *et al.*, 2019) showed that for the quasi-PDF we have

$$q_4(y, P^2) = \frac{\kappa \Lambda_{\text{QCD}}^2}{y^2(1-y)(P^2)^2} \times (1-y) \left[\int_{|y|}^1 \frac{dx}{x} \left[\frac{x^2}{(1-x)_+} - 2x^2 \right] q\left(\frac{y}{x}\right) + 2q(y) - |y|q'(y) \right], \quad (158)$$

where the first term in the integral was reproduced in a recent analysis of the renormalon effects in the quasi-PDF (Liu and Chen, 2020). As one can see, the second row vanishes as $q(y)$ when $y \rightarrow 1$ if $\lim_{y \rightarrow 1} q(y) \sim (1-y)^a$ with $a > 0$. This gives an estimate of the twist-four contribution on the r.h.s. of Eq. (33), which implies that the higher-twist contributions are enhanced as $1/y^2$ and $1/(1-y)$ for $y \sim 0$ and $y \sim 1$, respectively. Similar analysis can also be done for the pseudo-PDF. The above result can be used as a way to model the twist-four contribution with κ as the only parameter.

D. Orbital Angular Momentum of Partons in the Proton

Over the past three decades, much experimental and theoretical work has been done on the origin and structure of proton spin, which has been covered in depth in the review articles (Aidala *et al.*, 2013; Bass, 2005; Deur *et al.*, 2019; Filippone and Ji, 2001; Ji, 2017; Ji *et al.*, 2020c; Leader and Lorcé, 2014).

In addition to the spin-dependent PDFs and TMDs, the GCPs—in particular the GPDs—also play an important role in understanding the spin structure of the proton. Since GPDs describe the correlation between the transverse position and longitudinal momentum of quarks and gluons inside the proton, they offer a unique channel to study the orbital angular momentum (OAM) from experiments.

There are two widely known definitions of OAM in literature. One is the kinetic OAM in the gauge-invariant and frame-independent sum rule for the proton spin (Ji, 1997a,b), which is related to the first moment of twist-two GPDs and can be calculated from the form factors of the symmetric QCD energy-momentum tensor. A review of the lattice calculations of kinetic OAM can be found in (Ji *et al.*, 2020c). The other definition, which has a clear partonic interpretation in comparison to the kinetic OAM, is the canonical OAM in the naive partonic sum rule (Jaffe and Manohar, 1990) based on the free-field form of the QCD angular momentum,

$$\vec{J} = \int d^3\xi \, \psi^\dagger \frac{\vec{\Sigma}}{2} \psi + \int d^3\xi \, \psi^\dagger \left[\vec{\xi} \times (-i\vec{\nabla}) \right] \psi + \int d^3\xi \, \vec{E} \times \vec{A} + \int d^3\xi \, E^i \left(\vec{\xi} \times \vec{\nabla} \right)^i A^i, \quad (159)$$

where i is the spatial Lorentz index. Except for the first one, the other three operators are gauge dependent, and their matrix elements are generally gauge dependent. In high-energy scattering, there is one frame and gauge that are special: the IMF and light-front gauge, $A^+ = 0$. Therefore, the naive partonic sum rule for proton spin can be expressed as (Jaffe and Manohar, 1990)

$$\frac{1}{2} = \frac{1}{2} \Delta\Sigma(\mu) + I_q^z(\mu) + \Delta G(\mu) + I_g^z(\mu), \quad (160)$$

where $I_q^z(\mu)$ and $I_g^z(\mu)$ are the canonical OAM of the quark and gluon partons, respectively. Both I_q^z and I_g^z can be related to twist-three GPDs (Hatta, 2012; Hatta and Yoshida, 2012; Ji *et al.*, 2013a), which can be accessed through spin-asymmetries in hard exclusive processes (Bhattacharya *et al.*, 2018, 2017; Hatta *et al.*, 2017; Ji *et al.*, 2017a) (see the recent review (Ji *et al.*, 2020c)).

To fully understand the partonic spin structure of the proton, one also needs to determine the quark and gluon canonical OAM, I_q^z and I_g^z . LaMET allows extraction of I_q^z and I_g^z from lattice calculation in the same way as the gluon helicity that was reviewed in Sec. III E.

The quasi-partonic OAM operators can be chosen as the free-field operators fixed in gauges that belong to the universality class (Hatta *et al.*, 2014). Their matrix elements $\tilde{I}_q^z$ and $\tilde{I}_g^z$ can be calculated from the off-forward matrix elements of the relevant energy-momentum tensors (Zhao *et al.*, 2016), for example,

$$\tilde{I}_q^z(2S^z) = \lim_{\Delta \rightarrow 0} \epsilon^{\mu\nu} \frac{\partial}{\partial \Delta^\nu} \langle P^z S | \psi^\dagger(0) i \partial^\mu \psi(0) | P S \rangle \Big|_{\varphi, \vec{x} \rightarrow 0}. \quad (161)$$

where the kinematics is the same as Eq. (130).

Along with ΔG , $\tilde{I}_q^z$ and $\tilde{I}_g^z$ can be matched to the partonic quantities defined in the Jaffe-Manohar sum rule through the factorization formulas,

$$\tilde{I}_q^z(P^z, \mu) = P_{q\bar{q}I_q^z}(\mu) + P_{q\bar{q}I_g^z}(\mu) + p_{q\bar{q}}\Delta\Sigma(\mu) + p_{q\bar{q}}\Delta G(\mu) + \dots, \quad (162)$$

$$\tilde{I}_g^z(P^z, \mu) = P_{q\bar{q}I_q^z}(\mu) + P_{q\bar{q}I_g^z}(\mu) + p_{q\bar{q}}\Delta\Sigma(\mu) + p_{q\bar{q}}\Delta G(\mu) + \dots, \quad (163)$$

where $\dots$ are power corrections suppressed by the momentum P^z , and the one-loop matching coefficients in front of each term on the r.h.s. have been calculated in the Coulomb gauge (Ji *et al.*, 2015c). Since the quasi-partonic operators are gauge-variant and need to be fixed in a particular gauge, they can mix with new operators that are not allowed by Lorentz or gauge symmetries. For example, the gauge-dependent potential angular momentum $\psi^\dagger(\vec{r} \times \vec{A})\psi$ comes into play (Ji *et al.*, 2016; Wakamatsu, 2014). Such mixings must be taken into account in lattice renormalization to have a controlled calculation of the canonical OAM.

Apart from the above approach, it has also been proposed to calculate the ratio of $\tilde{I}_q^z$ and the valence quark number from the derivatives of off-forward matrix elements of staple-shaped quark Wilson line operators (Engelhardt, 2017), whose definition can be found in Eq. (195) below. The first lattice calculations with this approach have been carried out in (Engelhardt, 2017; Engelhardt *et al.*, 2018), which shows different size of effects between the kinetic and canonical OAM. For systematic improvement in this calculation, one should include the matching of such matrix elements to the physical I_q^z in the limit when the transverse separation of the quark fields approaches zero.

For the transverse polarization, it is natural to define a twist-two partonic transverse angular momentum density of quarks (Hoodbhoy *et al.*, 1999b; Ji *et al.*, 2012; Ji and Yuan, 2020; Ji *et al.*, 2020c),

$$J_{q\perp}^i(x) = x[q(x) + E_{q,g}(x)]/2, \quad (164)$$

and similarly for the gluons, where $q(x)$ is the unpolarized quark/antiquark distributions, and $E_{q,g}(x)$ is the GPDs defined earlier in this section. Thus to get a simple partonic picture of the proton transverse spin from

the first principles, it is important to calculate the GPD $E(x)$ using LaMET.

V. TRANSVERSE-MOMENTUM DEPENDENT PDFs

The transverse-momentum-dependent (TMD) parton distribution functions (TMDPDFs) are a natural generalization of the collinear PDFs to include both longitudinal and transverse momentum of partons. They are in principle probability distributions $f_i(x, \vec{k}_\perp, \sigma)$ of finding a parton of given species i , longitudinal and transverse momentum $(xP^+, k_\perp)$, and polarization σ inside the hadron state. TMDPDFs are playing an increasingly important role in understanding the partonic structure of hadrons and high-energy scattering.

The TMD parton densities were firstly introduced by Collins and Soper in 1980s (Bodwin, 1985; Collins and Soper, 1981, 1982a; Collins *et al.*, 1983, 1985a,b) to understand the Drell-Yan (DY) and e^+e^- annihilation process, and generalized in (Ji *et al.*, 2004, 2005) to semi-inclusive deep-inelastic scattering (SIDIS) process. The TMD factorization has been reanalyzed in the framework of SCET in which modes are made manifest by effective fields (Bauer *et al.*, 2001, 2002; Becher and Neubert, 2011; Chiu *et al.*, 2012; Echevarria *et al.*, 2013; Echevarria *et al.*, 2012; Manohar and Stewart, 2007). Various TMD factorization formalisms finally converged to the standard one where a scheme-independent TMDPDF can be defined (Collins and Rogers, 2017, 2013; Echevarria *et al.*, 2013).

The TMD parton densities are important in understanding the experimental processes where the transverse momenta of final state particles are measured. For example, in DY pair and W, Z production it is known that the differential cross section $de/dQ_\perp^2$ normally peaks at relatively small transverse momentum. For $Q \sim 10$ GeV, the peak is typically located at $Q_\perp \sim 1$ GeV where non-perturbative effects are important (Collins *et al.*, 1985b). A good knowledge of TMD parton densities is therefore crucial for the determination of the cross sections and precision test of perturbative QCD predictions.

Besides their importance in understanding the high-energy experimental data, the TMD parton densities are also important by themselves for their crucial role in describing hadron structures. With them, one can simultaneously study the fast-moving collinear physics through the longitudinal x -dependencies, and the non-perturbative effect from the transverse $\vec{k}_\perp$ -dependencies. Moreover, the TMDPDFs are sensitive to effects such as soft radiations. Therefore, the physics in the presence of transverse degrees of freedom is rather rich. This is particularly true in studies of spin-dependent phenomena where one can define various TMDPDFs through Lorentz decompositions (see Sec. V.B). One

example is the Sivvers function for a transversely polarized proton, $e_{ij} k_{\perp}^i S_{\perp}^j f_{1T}^{\perp}(x, k_{\perp})$, which is naive-time-reversal odd and is predicted to change sign between the DY and SIDIS processes (Collins, 2011a). Similar properties also exist in the Boer-Mulders function (Boer and Mulders, 1998) concerning a transversely-polarized parton distribution in an unpolarized hadron. These two functions are related to the single transverse spin asymmetry. If we generalize the TMDPDFs to include the impact parameter dependence, we can further define the Wigner function, the parton orbital angular momentum distributions, etc (Belitsky *et al.*, 2004; Lorce *et al.*, 2012). Therefore, the TMDPDFs allow for a more complete and refined 3D description (or tomography) of the hadron structure (Boer *et al.*, 2011; Burkardt, 2000). The 3D tomography of the proton is a major physical goal of the EIC program. The TMDPDFs are also important in understanding small- x physics (Balitsky, 1996; Balitsky and Lipatov, 1978; Kovchegov, 1999; Kovchegov and Levin, 2012; Kuraev *et al.*, 1977).

Our current knowledge on TMDPDFs mainly comes from fitting to the experimental data (Bacchetta *et al.*, 2019, 2017; Bertone *et al.*, 2019; Echevarria *et al.*, 2014; Kang *et al.*, 2016; Konychev and Nadelsohn, 2006; Landry *et al.*, 2001; Scimemi and Vladimirov, 2018a, 2019; Sun *et al.*, 2018). This is, however, rather primitive due to the paucity of data. Although the future EIC will make up the gap and produce more data for TMD measurements, it is still important to develop first-principle methods for the determination of nonperturbative TMDPDFs, which can serve as a test or provide useful inputs to constrain the global fits. LaMET provides a systematic way to extract TMDPDFs from the lattice calculations. Early studies (Ebert *et al.*, 2019a,b; Ji *et al.*, 2019a, 2015b) have tried to construct a quasi-TMDPDF on the lattice, but its relation to the physical TMDPDF is expected to be nonperturbative due to complications in the soft function (Ebert *et al.*, 2019b). The recent works in (Ji *et al.*, 2019b, 2020a) provide a formulation to calculate the soft function so that a perturbative matching formula can be established between the quasi- and physical TMDPDFs, allowing for a complete determination of the latter from lattice QCD. In this section we review the application of LaMET to the nonperturbative TMDPDFs. The investigation is still in its early stage and a lot remains to be explored, particularly in lattice calculations and matching.

In the first subsection we introduce the TMDPDFs and discuss the associated rapidity divergences. In the following subsections, we define the quasi-TMDPDFs or TMD momentum distributions in a proton of finite momentum, and study their momentum RGEs and UV renormalization properties. In the process, we introduce the off-light-cone soft functions. We then present the factorization of the quasi-TMDPDFs into the light-cone TMDPDFs and

the off-light-cone soft function, where various one-loop results and the relevant RGEs are also given. The properties of the off-light-cone soft function are discussed in the last subsection, where it is shown to be related to the form factor of a pair of charged color sources, which paves the way for its calculation on a Euclidean lattice.

A. Introduction to TMDPDFs and Rapidity Divergence

As explained in Sec. II, we can define various TMDPDFs by choosing different gauge-links between the quark or gluon bilinears. The one relevant to high-energy phenomena is defined with light-like Wilson lines. The links represent the propagation of high-energy color-charged particles, and are crucial in forming gauge-invariant nonlocal operators (Belitsky *et al.*, 2003). As argued in previous sections, such operators are the result of an EFT description (more explicitly so in SCET) arising from taking the infinite-momentum limit of the proton. Thus, it is natural to expect that they require additional regularization and renormalization.

Let us take the non-singlet quark unpolarized TMDPDF as an example. Without the field theoretic subtleties, the distribution is

$$f(x, \vec{k}_{\perp}) = \frac{1}{2P^+} \int \frac{d\lambda}{2\pi} \frac{d^2 \vec{b}_{\perp}}{(2\pi)^2} e^{-i\lambda x + i\vec{k}_{\perp} \cdot \vec{b}_{\perp}} \quad (165)$$

$$\times \langle P | \bar{\psi}(\lambda n/2 + \vec{b}_{\perp}) \gamma^+ W_n(\lambda n/2 + \vec{b}_{\perp}) \psi(-\lambda n/2) | P \rangle,$$

where $W_n(\lambda n + \vec{b}_{\perp})$ is the staple-shaped gauge-link of the form

$$W_n(\xi) = W_n^+(\xi) W_{\perp} W_n(-\xi \cdot pn), \quad (166)$$

$$W_n(\xi) = \mathcal{P} \exp \left[-ig \int_0^{\infty} d\lambda n \cdot A(\xi + \lambda n) \right], \quad (167)$$

along the light-cone direction n^μ , as shown in Fig. 10. The $W_{\perp}$ is a transverse gauge-link at light-cone infinity to maintain gauge-invariance. If one uses LFQ and ignores the transverse gauge-link, the above distribution is just $\langle P | b^i(x, \vec{k}_{\perp}) b(x, \vec{k}_{\perp}) | P \rangle$ for $x > 0$, as expected.

However, there are a number of qualifications in the above definition. First, the light-like gauge-links W_n are chosen to be past-pointing in accordance with the DY kinematics, but for SIDIS they should be chosen as future-pointing, as shown in Fig. 10. For unpolarized TMDPDFs there is no distinction between the two choices, but for spin-dependent TMDPDFs there are physical consequences associated with the direction of gauge-links.

Second, there exists a new type of divergence associated with the infinitely-long light-like gauge-links. These divergences are due to radiation of gluons collinear to the light-like gauge-link and cannot be regularized by the

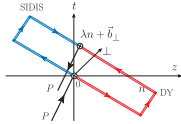


FIG. 16: The space-time picture of TMDPDF for DY and SIDIS process. The circled crosses denote the staple-link vertices. Notice that the vertices are placed at $\lambda n + \vec{b}_\perp$ and 0 which gives the same result as the symmetric choice in Eq. (165).

standard UV regulators. An example is the following integral in dimensional regularization (DR) (Ebert *et al.*, 2019b),

$$I = \int dk^+ dk^- \frac{f(k^+ k^-)}{(k^+ k^-)^{1+\epsilon}} = \frac{1}{2} \int \frac{dy}{y} \int dm^2 \frac{f(m^2)}{m^{2+2\epsilon}}, \quad (168)$$

where $m^2 = k^+ k^-$ and $y = k^+ / k^-$ is the rapidity-related variable. The divergences in y arise from large and small y where the integral is unregulated. The contribution from $k^+ = 0$ is called the light-zero mode in LFQ, where it is also called light-cone divergence which causes considerable problems.

To regulate the light-cone or rapidity divergences, a number of methods have been introduced in the literature (for a review see (Ebert *et al.*, 2019b)). They can be put into two classes: on-light-cone regulators and off-light-cone regulators. In the former case, the gauge-links are kept along the light-cone direction n^μ after regularization. For example, the so-called δ regulator (Echevarria *et al.*, 2016a,b) regularizes the gauge-link as:

$$W_n(\xi) \rightarrow W_n(\xi)_{|\delta^-} = \mathcal{P} \exp \left[-ig \int_0^\infty d\lambda A^+((\xi + \lambda n) e^{-\frac{\delta^-}{2\epsilon} |\lambda|}) \right], \quad (169)$$

and similarly for the conjugate direction. The δ regulator breaks gauge-invariance, but preserves the boost invariance $\delta^\pm \rightarrow e^{\pm Y} \delta^\pm$ where Y is the rapidity of the Lorentz boost. Other on-light-cone regulators include the exponential regulator (Li *et al.*, 2016), η regulator (Chiu *et al.*, 2012), analytical regulator (Becher and Neubert, 2011), etc. In the remainder of this section, we will use the δ regulator as a representative whenever we need an on-light-cone regulator.

The off-light-cone regulator was introduced in (Collins, 2011a; Collins and Soper, 1981; Ji *et al.*, 2004,

2005), and also used in (Ji *et al.*, 2005). This type of regulator chooses off-light-cone directions to avoid the rapidity divergence. One can choose, for instance, to deform the gauge-links into the space-like region:

$$n \rightarrow n_Y = n - e^{-2Y} \frac{P}{(P^+)^2}. \quad (170)$$

Here Y plays the role of a rapidity regulator, as when $Y \rightarrow \infty$, $n_Y \rightarrow n$. In certain cases one can also deform n_Y into time-like region (Collins and Metz, 2004).

The on-light-cone regulators are consistent with the spirit of parton physics, and therefore are useful to define COM-momentum-independent parton densities. The off-light-cone regulators, on the other hand, follow a similar spirit as LaMET, and therefore can be exploited for practical lattice QCD calculations, as we shall see in the next subsection.

To avoid light-cone divergences, from now on we include the rapidity regulator in the definition of the light-cone TMDPDFs. Using the same label f for the TMDPDFs in both momentum and coordinate spaces, we have

$$f(\lambda, b_\perp, \mu, \delta^- / P^+) \quad (171)$$

$$= \langle P | \bar{\psi}(\lambda n / 2 + \vec{b}_\perp) \not{n} W_n(\lambda n / 2 + \vec{b}_\perp)_{|\delta^-} \psi(-\lambda n / 2) | P \rangle,$$

where μ is the $\overline{\text{MS}}$ scale for UV renormalization. Due to rotational invariance, the bare TMDPDF defined above is a function of $b_\perp = |\vec{b}_\perp|$, so we have omitted the vector arrow for $b_\perp$ in f and will do so throughout the discussion for the soft functions, quasi-TMDPDFs, etc. The subscript δ^- denotes that the staple-shaped gauge-link W is regulated by the δ regulator in the light-cone minus direction. f diverges logarithmically as $\delta^- \rightarrow 0$, and the finite part also depends on the rapidity regulator. To define the physical TMDPDF, we need to remove all divergences and rapidity regularization scheme dependencies in f , in a way similar to removing UV divergences in physical quantities.

The rapidity divergence for TMDPDFs can be removed by the soft function, which also plays an important role in TMD factorization. Intuitively, the soft function represents a cross section for fast-moving charged particles emitting soft gluons into final states. It has rapidity divergence associated with the light-cone direction, which is ultimately related to the mass singularity. The TMD soft function corresponding to Drell-Yan process is defined (Collins, 2011b; Echevarria *et al.*, 2016b) as

$$S(b_\perp, \mu, \delta^+, \delta^-) = \frac{\text{Tr}(\langle 0 | \bar{T} W_P(\vec{b}_\perp)_{|\delta^+} W_n^{\dagger}(\vec{b}_\perp)_{|\delta^-} T W_n(0)_{|\delta^-} W_P^{\dagger}(0)_{|\delta^+} | 0 \rangle)}{N_c} = \frac{\text{tr}(0 | W_n(\vec{b}_\perp)_{|\delta^+} W_P^{\dagger}(\vec{b}_\perp)_{|\delta^-} | 0 \rangle)}{N_c}, \quad (172)$$

where $T/\bar{T}$ stands for time/anti-time ordering. The first equality defines the soft function in terms of cut-diagrams

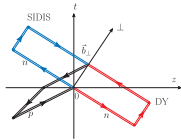


FIG. 11: The soft function $S(b_\perp, \mu, \delta^+, \delta^-)$ as space-time Wilson-loop arising in the factorization of DY and SIDIS process.

as an amplitude square. Since the soft function for DY process is independent of time ordering, one can also define it with a single time ordering or no time ordering, leading to the second equality. The staple-shaped gauge-link W_s is defined in Eq. (166), while the staple-shaped gauge-link W_p is defined similarly as:

$$W_p(\xi) = W_s^\dagger(\xi) W_\perp W_p(0), \quad (173)$$

$$W_p(\xi) = \mathcal{P} \exp \left[-ig \int_0^{-\infty} d\lambda p \cdot A(\xi + p\lambda) \right]. \quad (174)$$

The soft function is shown in Fig. 11 as a Wilson loop in Minkowski space.

If the rapidity divergences are multiplicative, one can use S as the rapidity renormalization factor for the TMDPDF defined in Eq. (165). In on-light-cone schemes such as the δ regularization, it has been argued in (Vladimirov, 2018) based on conformal transformation that the rapidity divergences are indeed multiplicative in position space. For each of the staple-shaped light-like gauge-links, the rapidity divergence is proportional to $\exp[-(1/2)K(b_\perp, \mu) \ln(\mu^2/2(\delta^\pm)^2)]$ where $K(b_\perp, \mu)$ is the nonperturbative Collins-Soper evolution kernel (Collins and Soper, 1981). Thus at small $\delta^\pm$, we can write

$$S(b_\perp, \mu, \delta^+, \delta^-) = e^{\ln \frac{\mu^2}{2\delta^+\delta^-} K(b_\perp, \mu) + \mathcal{D}_2(b_\perp, \mu)}, \quad (175)$$

where $\mathcal{D}_2(b_\perp, \mu)$ is a $b_\perp$ -dependent but rapidity-independent function. Notice that our definitions of $\delta^\pm$ differ from those in Ref. (Echevarria *et al.*, 2016b) by a factor of $\sqrt{2}$ due to our normalization of light-cone vectors.

The soft-function in δ regularization satisfies the renormalization group equation

$$\begin{aligned} \mu^2 \frac{d}{d\mu^2} \ln S(b_\perp, \mu, \delta^+, \delta^-) \\ = -\Gamma_{\text{cusp}}(\alpha_s) \ln \frac{\mu^2}{2\delta^+\delta^-} + \gamma_s(\alpha_s), \end{aligned} \quad (176)$$

where $\Gamma_{\text{cusp}}(\alpha_s)$ is the light-like cusp anomalous dimension (Korchinsky and Radyushkin, 1987; Polyakov, 1980) and the $\gamma_s(\alpha_s)$ is the soft anomalous dimension (Korchinskaya and Korchinsky, 1992). The Collins-Soper kernel and the rapidity-independent part $\mathcal{D}_2$ satisfy the RGEs:

$$\mu^2 \frac{d}{d\mu^2} K(b_\perp, \mu) = -\Gamma_{\text{cusp}}(\alpha_s), \quad (177)$$

$$\mu^2 \frac{d}{d\mu^2} \mathcal{D}_2(b_\perp, \mu) = \gamma_s(\alpha_s) - K(b_\perp, \mu). \quad (178)$$

At one-loop, the soft function $S(b_\perp, \mu, \delta^+, \delta^-)$ is given by (Echevarria *et al.*, 2013):

$$\begin{aligned} S(b_\perp, \mu, \delta^+, \delta^-) \\ = 1 + \frac{\alpha_s C_F}{2\pi} \left(L_b^2 - 2L_b \ln \frac{\mu^2}{2\delta^+\delta^-} + \frac{\pi^2}{6} \right), \end{aligned} \quad (179)$$

where $L_b = \ln(\mu^2 b_\perp^2 e^{2\gamma_E}/4)$. Therefore, we have at the leading order,

$$K(b_\perp, \mu) = -\frac{\alpha_s C_F}{\pi} L_b, \quad (180)$$

$$\mathcal{D}_2(b_\perp, \mu) = \frac{\alpha_s C_F}{2\pi} \left(L_b^2 + \frac{\pi^2}{6} \right), \quad (181)$$

and $\Gamma_{\text{cusp}} = \alpha_s C_F/\pi + \mathcal{O}(\alpha_s^2)$, $\gamma_s = \mathcal{O}(\alpha_s^2)$. It is worth pointing out that K (Li and Zhu, 2017; Vladimirov, 2017) and $\mathcal{D}_2$ (Li and Zhu, 2017) are known to 3-loop order in the exponential regularization scheme.

With the above soft function, we can take its square root to perform rapidity renormalization for the bare TMD correlator. The square root can be explained as follows: S contains two staples, while f contains one, thus the rapidity divergences as well as scheme dependencies in S are twice as those in f . This leads to the following definition of the renormalized physical TMDPDF (Collins and Rogers, 2013; Echevarria *et al.*, 2013):

$$f^{\text{TMD}}(x, b_\perp, \mu, \zeta) = \lim_{\delta^- \rightarrow 0} \frac{f(x, b_\perp, \mu, \delta^-/P^+)}{\sqrt{S(b_\perp, \mu, \delta^- e^{-\gamma_E}, \delta^-)}}, \quad (182)$$

where the rapidity scale reads

$$\zeta = 2(xP^+)^2 e^{2\gamma_E}. \quad (183)$$

The rapidity dependence in the numerator of the right-hand side of Eq. (182) has the form $\exp[-\frac{1}{2}K(b_\perp, \mu) \ln \frac{(\delta^-)^2}{(xP^+)^2}]$, while in the denominator it behaves as $\exp[\frac{1}{2}K(b_\perp, \mu) \ln \frac{\mu^2}{2(\delta^- e^{-\gamma_E})^2}]$. The δ^- dependence thus cancels out in the ratio, leaving a dependence on the rapidity scale as $\exp[-\frac{1}{2}K(b_\perp, \mu) \ln \frac{\mu^2}{\zeta^2}]$, which is controlled by the so-called Collins-Soper evolution equation:

$$2\zeta \frac{d}{d\zeta} \ln f^{\text{TMD}}(x, b_\perp, \mu, \zeta) = K(b_\perp, \mu). \quad (184)$$

The ζ -dependence comes from the initial-state quark radiation and is intrinsically nonperturbative for large $b_\perp$. $f^{\text{TMD}}(x, b_\perp, \mu, \zeta)$ is the standard object to be matched to in LaMET.

We should emphasize that although f^{TMD} is free from rapidity divergences, it does contain soft radiation from the charged particles in the initial state. This can be seen clearly by considering Feynman diagrams for the unsubtracted f and applying soft approximation to gluons. “One-half” of the soft contribution in f is subtracted to define the physical f^{TMD} due to the requirement of factorization of physical processes. The remaining soft radiation also has a natural rapidity cut-off associated with $\ln(xP^+)$, reflected in the ζ -dependence. What is remarkable, however, is that f^{TMD} is rapidity-regulator independent. Although a general proof to all orders in perturbation theory is beyond the scope of this review, it is due to factorization and exponentiation of the soft physics in f and thus the scheme cancellation can be done systematically in the exponent. It worth mentioning that in old-fashioned or SCET-like approaches, one can define the “subtracted” TMDPDF or “beam functions” that contains only collinear physics. However, they are generically scheme dependent and must be combined with an extra soft functions in factorization theorems. At one-loop level, the scheme-independent one-loop TMDPDF for an external quark state reads,

$$f^{\text{TMD}}(x, b_\perp, \mu, \zeta) = \delta(1-x) + \frac{\alpha_s C_F}{2\pi} F(x, \epsilon_{\text{in}}, b_\perp, \mu) \theta(x) \theta(1-x) + \frac{\alpha_s C_F}{2\pi} \delta(1-x) \times \left[-\frac{1}{2} L_b^2 + \left(\frac{3}{2} - \ln \frac{\zeta}{\mu^2} \right) L_b + \frac{1}{2} - \frac{\pi^2}{12} \right], \quad (185)$$

where

$$F(x, \epsilon_{\text{in}}, b_\perp, \mu) = \left[- \left(\frac{1}{\epsilon_{\text{in}}} + L_b \right) \frac{1+x^2}{1-x} + 1-x \right]_+ . \quad (186)$$

Two-loop order results for the TMDPDFs can be found in (Catani and Grazzini, 2012; Catani *et al.*, 2012; Echevarria *et al.*, 2016c; Gehrmann *et al.*, 2014; Lübbert *et al.*, 2016; Luo *et al.*, 2019) and three-loop order results can be found in (Luo *et al.*, 2020).

The physical TMDPDF also satisfies the RG equation,

$$\gamma_\mu(\mu, \zeta) = \mu^2 \frac{d}{d\mu^2} \ln f^{\text{TMD}}(x, b_\perp, \mu, \zeta) = \frac{1}{2} \Gamma_{\text{cusp}}(\alpha_s) \ln \frac{\mu^2}{\zeta} - \gamma_H(\alpha_s), \quad (187)$$

where γ_H is called the hard anomalous dimension. At one-loop, the cusp and hard anomalous dimensions read

$$\Gamma_{\text{cusp}}(\alpha_s) = \frac{\alpha_s C_F}{\pi}; \quad \gamma_H(\alpha_s) = -\frac{3\alpha_s C_F}{4\pi}. \quad (188)$$

Recently the cusp anomalous dimension have been calculated to 4-loops (Henn *et al.*, 2019; von Manteuffel *et al.*, 2020).

Combining the RGE and the rapidity evolution equation for the TMDPDF, one obtains the consistency condition :

$$\mu^2 \frac{d}{d\mu^2} K(b_\perp, \mu) = -2\zeta \frac{d}{d\zeta} \gamma_\mu(\mu, \zeta) = -\Gamma_{\text{cusp}}(\alpha_s(\mu)), \quad (189)$$

from which one finds a resummed form for the Collins-Soper kernel:

$$K(b_\perp, \mu) = -2 \int_{1/b_\perp}^{\mu} \frac{d\mu'}{\mu'} \Gamma_{\text{cusp}}(\alpha_s(\mu')) + K(\alpha_s(1/b_\perp)). \quad (190)$$

Here $K(\alpha_s(1/b_\perp))$ contains both perturbative and non-perturbative contributions. The TMDPDFs at different scales are then related by

$$f^{\text{TMD}}(x, b_\perp, \mu, \zeta) = f^{\text{TMD}}(x, b_\perp, \mu_0, \zeta_0) \times \exp \left[\int_{\mu_0}^{\mu} \frac{d\mu'}{\mu'} \gamma_\mu(\mu', \zeta_0) \right] \exp \left[\frac{1}{2} K(b_\perp, \mu) \ln \frac{\zeta}{\zeta_0} \right]. \quad (191)$$

The double-scale evolution in the $\mu - \zeta$ plane for phenomenology has been recently studied in (Scimemi and Vladimirov, 2018b). With the scheme-independent physical TMDPDF defined above, the DY cross section at $s = (P_A + P_B)^2$ and small $Q_\perp$ can be factorized as

$$\frac{d\sigma}{dQ_\perp^2} = \int dx_A dx_B d^2 b_\perp e^{iQ_\perp \cdot Q_\perp} \hat{\sigma}(x_A x_B s, \mu) \times f_A^{\text{TMD}}(x_A, b_\perp, \mu, \zeta_A) f_B^{\text{TMD}}(x_B, b_\perp, \mu, \zeta_B) + \dots \quad (192)$$

The rapidity scales satisfy $\zeta_A \zeta_B = Q^2 = (x_A x_B s)^2$. The remaining term at large but finite Q^2 are called power corrections or “higher-twist” contributions. A detailed study of the power corrections to TMD factorization is beyond the scope of this review. Without mention we will omit all the power-corrections in equations. The QCD part of the hard cross section $\hat{\sigma}$ at one-loop level reads

$$\hat{\sigma}(x_A, x_B) = \left| 1 + \frac{\alpha_s C_F}{4\pi} \left(-L_Q^2 + 3L_Q - 8 + \frac{\pi^2}{6} \right) \right|^2, \quad (193)$$

where $L_Q = \ln \frac{Q^2 - i0}{s}$, and the result is now known up to three loops (see (Baikov *et al.*, 2009; Gehrmann *et al.*, 2010; Lee *et al.*, 2010; Moch *et al.*, 2005) and the references therein). Similarly for the SIDIS process we have

$$\frac{d\sigma}{dQ_\perp^2} = \int dx dz d^2 b_\perp e^{iQ_\perp \cdot Q_\perp} H(x, z, \mu, Q) \times f^{\text{TMD}}(x, b_\perp, \mu, \zeta_A) d^{\text{TMD}}(z, b_\perp, \mu, \zeta_B), \quad (194)$$

where $d^{\text{TMD}}(z, b_\perp, \mu, \zeta_B)$ is the TMD fragmentation function and H is the hard kernel.

B. Lattice Quasi-TMDPDFs and Matching

Before LaMET, there had been efforts to access TMD physics from lattice QCD by calculating the ratios of the x -moments of TMDPDFs (Engelhardt *et al.*, 2016; Hagler *et al.*, 2009; Musch *et al.*, 2012, 2011; Yoon *et al.*, 2017), which are free from complications associated with the soft function and can be compared to certain experimental observables. In LaMET, we are more interested in obtaining the full x and $k_\perp$ dependence of the TMDPDFs (Ebert *et al.*, 2019a,b; Ji *et al.*, 2019a,b, 2020a, 2015b). Therefore, a proper treatment of the soft function subtraction and matching is essential. The earliest suggestion of a bent soft function in (Ji *et al.*, 2015b) and the follow-up work (Ebert *et al.*, 2019b) has the correct IR logarithms at one-loop order, but this is expected to break down at higher-loop orders (Ji *et al.*, 2020a), thus not allowing for a perturbative matching. Another suggestion which uses a naive rectangle-shaped Wilson loop (Ebert *et al.*, 2019b; Ji *et al.*, 2019a) does not possess the correct IR physics, either. Nevertheless, in (Ebert *et al.*, 2019a) important progress was made for calculating the nonperturbative Collins-Soper kernel $K(b_\perp, \mu)$ from the ratio of quasi-TMDPDFs at two different large momenta. Recently, some of the authors showed (Ji *et al.*, 2019b, 2020a) that the quasi-TMDPDF combined with a reduced soft function capture the correct IR physics to all-orders and thus allow for a perturbative matching to the physical TMDPDF.

To construct such quasi-TMDPDFs, the collinear part can be treated in a way similar to the collinear PDFs, while the soft piece is more challenging. Our starting point is that the physical f^{TMD} is independent of the rapidity regulator, so one can use a scheme in which the gauge-links in both f and S are off the light-cone, such as that used in (Collins, 2011a). In this case, one can use Lorentz symmetry to boost the staple-shaped gauge-link W_n in f to a purely space-like staple with no time dependence. However, one can only use this trick for one of the staples in S , say W_n , whereas the other one W_p is still time-dependent. In other words, there is no way to get rid of the time dependence in S entirely with Lorentz boost alone. This is natural because S in fact represents the square of an S -matrix, which appears to be intrinsically Minkowskian. However, using the LaMET principle that time dependence of an operator can be simulated through external physical states at large momentum, we find that S can indeed be calculated on the lattice in the off-light-cone scheme as a form factor. A detailed discussion will be given in the next subsection. Here we assume that this is true, and discuss the matching between quasi- and physical TMDPDFs.

First, we define the quasi-TMDPDF with a staple-shaped gauge-link along the z direction (Ebert *et al.*,

2019b; Ji *et al.*, 2019a,b, 2015b) as

$$\begin{aligned} \bar{f}(\lambda, b_\perp, \mu, \zeta_z) \\ = \lim_{L \rightarrow \infty} \frac{\langle P | \bar{\psi} \left(\frac{\not{b}_\perp}{2} + \not{b}_\perp \right) \gamma^\gamma W_z \left(\frac{\not{b}_\perp}{2} + \not{b}_\perp; L \right) \psi \left(-\frac{\not{b}_\perp}{2} \right) | P \rangle}{\sqrt{Z_E(2L, b_\perp, \mu)}}, \end{aligned} \quad (195)$$

where the $\overline{\text{MS}}$ renormalization is implied, and

$$W_z(\xi; L) = W_z^+(\xi; L) W_z^-(-\xi^z n_z; L), \quad (196)$$

$$W_z(\xi; L) = \mathcal{P} \exp \left[-ig \int_{\xi^z}^L d\lambda n_z \cdot A(\xi_\perp + n_z \lambda) \right]. \quad (197)$$

Here $\xi^z = -\xi \cdot n_z$ and $\zeta_z = (2xP^z)^2$ is the Collins-Soper scale of the quasi-TMDPDF. $W_\perp$ is inserted at $z = L$ to maintain explicit gauge invariance. $\sqrt{Z_E(2L, b_\perp, \mu, 0)}$ is the square root of the vacuum expectation value of a flat rectangular Euclidean Wilson-loop along the n_z direction with length $2L$ and width $b_\perp$:

$$Z_E(2L, b_\perp, \mu) = \frac{1}{N_c} \text{Tr} \langle 0 | W_\perp(\vec{b}_\perp; 2L) | 0 \rangle. \quad (198)$$

Again, γ^z can be replaced by γ^t as in the collinear quasi-PDF. For a depiction of $\bar{f}$ and Z_E see Fig. 12.

The purpose of the factor Z_E is as follows. At large L , the naive quasi-TMD correlator in the numerator of Eq. (195) contains divergences that go as $e^{-L E(b_\perp, \mu)}$ where $E(b_\perp)$ is the ground state energy of a pair of static heavy-quarks. $E(b_\perp, \mu) = 2\delta m + V(b_\perp, \mu)$ contains both the linear divergent mass corrections $2\delta m$ and the heavy-quark potential $V(b_\perp, \mu)$ due to mutual interactions. In literature the $LV(b_\perp, \mu)$ part was sometimes called the “pinch pole singularity.” Therefore, we introduce the square root of a rectangular Wilson-loop $Z_E(2L, b_\perp, \mu)$ with twice the length to cancel all these divergences and guarantee the existence of the $L \rightarrow \infty$ limit after the subtraction. The introduction of $\sqrt{Z_E}$ also removes additional contributions from the transverse gauge link. An alternative approach to avoid the pinch-pole singularity was proposed in (Li, 2016). We should mention that although the $\sqrt{Z_E}$ subtraction removes all the linear divergences, the logarithmic UV divergences are still present. Therefore, a non-perturbative renormalization of $\bar{f}$ on the lattice is still required, which has been studied in the RI/MOM scheme (Shanahan *et al.*, 2019), and its matching to the $\overline{\text{MS}}$ scheme has been calculated at one-loop order (Constantinou *et al.*, 2019; Ebert *et al.*, 2020b).

The quasi-TMDPDFs defined above satisfy the following RGE (Collins and Soper, 1981; Ji *et al.*, 2019b, 2015b)

$$\mu^2 \frac{d}{d\mu^2} \ln \bar{f}(x, b_\perp, \mu, \zeta_z) = \gamma_F(\alpha_s(\mu)), \quad (199)$$

where γ_F is the anomalous dimension for the heavy-to-light current in Sec. III A. This is due to the fact that the

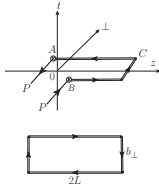


FIG. 12: The quasi-TMDPDF (upper) and the Euclidean Wilson-loop $Z_E(2L, b_\perp, 0)$ (lower). In the figure, $A = \lambda n_+/2 + \tilde{b}_\perp/2$, $B = -\lambda n_+/2 - \tilde{b}_\perp/2$ and $C = L n_+ + \tilde{b}_\perp$. The crosses denote the quark-link vertices.

quasi-TMDPDF, after the self-energy subtraction, contains only logarithmic UV divergences associated with quark-Wilson-line vertices. In the $\overline{\text{MS}}$ scheme, the one-loop quasi-TMDPDF in an external quark state with momentum $(p^2, 0, 0, p^2)$ reads (Ebert *et al.*, 2019b; Ji *et al.*, 2019a)

$$\begin{aligned} \tilde{f}(x, b_\perp, \mu, \zeta_z) = & 1 + \frac{\alpha_s C_F}{2\pi} F(x, \epsilon_n, b_\perp, \mu) \theta(x) \theta(1-x) + \frac{\alpha_s C_F}{2\pi} \delta(1-x) \\ & \times \left[-\frac{1}{2} L_b^2 + L_b \left(\frac{5}{2} - L_z \right) - \frac{3}{2} - \frac{1}{2} L_z^2 + L_z \right], \quad (200) \end{aligned}$$

where $L_z = \ln(\zeta_z/\mu^2)$. As expected, the L dependence has been cancelled in the large L limit.

As there is no light-like gauge-link in $\tilde{f}$, no additional rapidity regulator is needed. Instead, there is an explicit dependence on the hadron momentum (or energy), which is similar to the momentum RGE for collinear quasi-PDF. The momentum (rapidity) evolution equation for $\tilde{f}$ reads (Collins and Soper, 1981; Ji *et al.*, 2019b, 2015b),

$$P^2 \frac{d}{dP^2} \ln \tilde{f}(x, b_\perp, \mu, \zeta_z) = K(b_\perp, \mu) + \mathcal{G}\left(\frac{(P^2)^2}{\mu^2}\right), \quad (201)$$

where $\mathcal{G}(\zeta_z/\mu^2)$ is perturbative and $K(b_\perp, \mu)$ is the Collins-Soper kernel. A similar equation was proven for off-light-cone TMD-fragmentation functions in (Collins and Soper, 1981). From this equation, it is clear that a correct matching to $f^{\text{TMD}}(x, b_\perp, \mu, \zeta)$ with arbitrary ζ must include $K(b_\perp, \mu)$ to compensate the P^2 dependence.

There is actually one more requirement for the matching: there is a rapidity scheme dependence which must be removed, since the quasi-TMDPDF can be viewed as defined with an off-light-cone regulator along the z direction. To understand this dependence, let us consider f again in the off-light-cone regularization, where there are rapidity divergences. The divergence is cancelled by the square root of an off-light-cone soft function $S_{\text{DY}}(b_\perp, \mu, Y, Y')$, with Y, Y' being the rapidities of the off-light-cone space-like vectors $p \rightarrow p_Y = p - e^{-2Y}(p^+)n$ and $n \rightarrow n_{Y'} = n - e^{-2Y'}p/(p^+)^2$. Schematically, we have:

$$S_{\text{DY}}(b_\perp, \mu, Y, Y') = \frac{\text{tr}(\langle 0 | W_{n_{Y'}}(\tilde{b}_\perp) W_{p_Y}^\dagger(\tilde{b}_\perp) | 0 \rangle)}{N_c \sqrt{Z_E} \sqrt{Z_E}}, \quad (202)$$

where $W_{n_{Y'}}(\tilde{b}_\perp)$ and $W_{p_Y}^\dagger(\tilde{b}_\perp)$ are staple-shaped gauge-links in $n_{Y'}$, p_Y directions, respectively. $\sqrt{Z_E}$ is introduced to subtract the pinch pole singularities for the off-light-cone staple-shaped gauge-links. In terms of $\ln p^2 = 2(Y + Y')$ sometimes we also write this soft function as $S_{\text{DY}}(b_\perp, \mu, \rho)$. At large ρ , we have

$$S_{\text{DY}}(b_\perp, \mu, Y, Y') = e^{(Y+Y')K(b_\perp, \mu) + \mathcal{D}(b_\perp, \mu)} + \dots \quad (203)$$

We can perform a Lorentz boost of $W_{n_{Y'}}(\tilde{b}_\perp) W_{p_Y}^\dagger(\tilde{b}_\perp)$ in Eq. (202) such that one of the gauge-links, say $W_{n_{Y'}}$, is boosted to the equal-time version W_t in $\tilde{f}$, whereas the other gauge-link W_{p_Y} is boosted to $W_{n_{Y'+Y'}}$. The soft function becomes $S_{\text{DY}}(b_\perp, \mu, Y + Y', 0)$ which contains light-cone divergence for the $p_{Y+Y'}$ direction, but is still the same $S_{\text{DY}}(b_\perp, \mu, Y, Y')$ due to boost invariance. The square-root of the finite part $e^{\mathcal{D}(b_\perp, \mu)}$ is exactly what is needed to cancel the rapidity-scheme dependence. We define the rapidity-independent part as the reduced soft function:

$$S_r(b_\perp, \mu) = e^{-\mathcal{D}(b_\perp, \mu)}. \quad (204)$$

Based on the renormalization property of non-light-like Wilson-loops, the reduced soft function satisfies the RG equation

$$\mu^2 \frac{d}{d\mu^2} \ln S_r(b_\perp, \mu) = \Gamma_S(\alpha_s), \quad (205)$$

where Γ_S is the constant part of the cusp-anomalous dimension at large hyperbolic cusp angle $Y + Y'$ for the off-light-cone soft function:

$$\begin{aligned} \mu^2 \frac{d}{d\mu^2} \ln S_{\text{DY}}(b_\perp, \mu, Y, Y') \\ = -(Y + Y') \Gamma_{\text{cusp}}(\alpha_s) - \Gamma_S(\alpha_s). \quad (206) \end{aligned}$$

At one-loop level (Ebert *et al.*, 2019b),

$$S_{\text{DY}}^{(1)}(b_\perp, \mu, Y, Y') = \frac{\alpha_s C_F}{2\pi} [2 - 2(Y + Y')] L_b, \quad (207)$$

and $\Gamma_S^{(1)}(\alpha_s) = \alpha_s C_F / \pi$. Based on RGE, at two-loop level $\mathcal{D}(b_{\perp}, \mu)$ can be predicted to be

$$\mathcal{D}^{(2)}(b_{\perp}, \mu) = c_2 + \Gamma_S^{(2)} L_b - \frac{\alpha_s^2 \beta_0 C_F}{2\pi} L_b^2, \quad (208)$$

where

$$\Gamma_S^{(2)} = -\frac{\alpha_s^2}{\pi^2} \left[C_F C_A \left(-\frac{49}{36} + \frac{\pi^2}{12} - \frac{\zeta_3}{2} \right) + C_F N_F \frac{5}{18} \right]$$

is the two-loop anomalous dimension for S_F which can be extracted from (Grozin *et al.*, 2016), $\beta_0 = -(\frac{11}{3}C_A - \frac{2}{3}N_F)/(2\pi)$ is the coefficient of one-loop β -function, and c_2 is a constant to be determined by explicit calculation.

After taking into account the reduced soft function, we can now write down the matching formula between the quasi-TMDPDF and the scheme-independent TMDPDF (Ji *et al.*, 2019b):

$$\begin{aligned} f^{\text{TMD}}(x, b_{\perp}, \mu, \zeta) \\ = H \left(\frac{\zeta}{\mu^2} \right) e^{-\ln^2 \left(\frac{\zeta}{\mu^2} \right) K(b_{\perp}, \mu)} \bar{f}(x, b_{\perp}, \mu, \zeta) S_F^{\dagger}(b_{\perp}, \mu) + \dots, \end{aligned} \quad (209)$$

where the power-corrections of order $\mathcal{O}(\Lambda_{\text{QCD}}^2/\zeta, M^2/(P^+)^2, 1/(b_{\perp}^2 \zeta))$. The above relation except for the definition of $S_F(b_{\perp}, \mu)$ was argued to hold in (Ebert *et al.*, 2019b), where the unknown function g_S^0 in Eq. (5.3) should be identified as the reduced soft function here; it has also been confirmed recently in (Vladimirov and Schäfer, 2020).

We now explain the individual factors of the formula.

1. The factor $H(\zeta/\mu^2)$ is the perturbative matching kernel, which is a function of $\zeta/\mu^2 = (2xP^+)^2/\mu^2$. The kernel is responsible for the large logarithms of P^2 generated by the $\mathcal{G}(\zeta/\mu^2)$ term of the momentum RG equation. Unlike the case of quasi-PDFs, the momentum fractions of the quasi-TMDPDF and the TMDPDF are the same. This is due to the fact that at leading power in $1/\zeta$ expansion, the $k_{\perp}$ integral is naturally cut off by the transverse separation around $k_{\perp} \sim 1/b_{\perp} \ll P^2$. Therefore, the momentum fraction can only be modified by collinear modes for which there are no distinction between $x = k^+/P^2$ and $x = k^+/P^+$. In comparison, for the $k_{\perp}$ integrated quasi-PDF, the $k_{\perp} \geq P^2$ region leads to non-trivial x dependence outside the physical region. This is also consistent with the fact that the momentum evolution equation for quasi-TMDPDF is local in x instead of being a convolution.
2. The factor $\exp[\ln(\frac{\zeta}{\mu^2})K(b_{\perp}, \mu)]$ is the part involving the Collins-Soper evolution kernel. From the momentum evolution equation, it is clear that at large P^+ there are logarithms of the form

$K(b_{\perp}, \mu) \ln \frac{\zeta}{\mu^2}$ with ζ_s being the natural Collins-Soper scale. Therefore, to match to the TMDPDF at arbitrary ζ , a factor $\exp[\ln(\frac{\zeta}{\mu^2})K(b_{\perp}, \mu)]$ is required to compensate the difference. An important implication of this property is that one can obtain the Collins-Soper kernel $K(b_{\perp}, \mu)$ by constructing the ratio of quasi-TMDPDFs at two different momenta or ζ_s 's (Ebert *et al.*, 2019a),

$$\frac{\bar{f}(x, b_{\perp}, \mu, \zeta_{s,1})}{\bar{f}(x, b_{\perp}, \mu, \zeta_{s,2})} = \frac{H \left(\frac{\zeta_{s,1}}{\mu^2} \right)}{H \left(\frac{\zeta_{s,2}}{\mu^2} \right)} \left(\frac{\zeta_{s,1}}{\zeta_{s,2}} \right)^{K(b_{\perp}, \mu)}. \quad (210)$$

Thus given the $\bar{f}$'s at the two rapidity scales, the Collins-Soper kernel $K(b_{\perp})$ can be obtained.

Combining the RGEs of the quasi-TMDPDF $\bar{f}$, reduced soft function S_F and physical TMDPDF f^{TMD} , we obtain the RGE of the matching kernel $H \left(\frac{\zeta}{\mu^2} \right)$ (Ji *et al.*, 2019b),

$$\mu^2 \frac{d}{d\mu^2} \ln H^{-1} \left(\frac{\zeta}{\mu^2} \right) = \frac{1}{2} \Gamma_{\text{cusp}}(\alpha_s) \ln \frac{\zeta}{\mu^2} + \frac{\gamma_C(\alpha_s)}{2}, \quad (211)$$

where $\gamma_C(\alpha_s) = 2\gamma_F(\alpha_s) + \Gamma_S(\alpha_s) + 2\gamma_H(\alpha_s)$. The matching kernel is closely related to the perturbative part of the rapidity evolution kernel $\mathcal{G} \left(\frac{\zeta}{\mu^2} \right)$ through

$$2\zeta_s \frac{d}{d\zeta_s} \ln H^{-1} \left(\frac{\zeta}{\mu^2} \right) = \mathcal{G} \left(\frac{\zeta}{\mu^2} \right). \quad (212)$$

Again, we can see that the anomalous dimension of $\mathcal{G} \left(\frac{\zeta}{\mu^2} \right)$ is $\Gamma_{\text{cusp}}(\alpha_s)$.

It is convenient to write H in the exponential form, $H = e^{-\lambda}$. Collecting all the above results, one obtains at one-loop level (Ebert *et al.*, 2019b; Ji *et al.*, 2019a)

$$h^{(1)} \left(\frac{\zeta}{\mu^2} \right) = \frac{\alpha_s C_F}{2\pi} \left(-2 + \frac{\pi^2}{12} - \frac{L_z^2}{2} + L_z \right). \quad (213)$$

Similar as before, the two loop contribution $h^{(2)}$ is predicted to be

$$\begin{aligned} h^{(2)} \left(\frac{\zeta}{\mu^2} \right) = & c'_2 - \frac{1}{2} \left(\gamma_C^{(2)} - \alpha_s^2 \beta_0 c_1 \right) \ln \frac{\zeta}{\mu^2} \\ & - \frac{1}{4} \left(\Gamma_{\text{cusp}}^{(2)} - \frac{\alpha_s^2 \beta_0 C_F}{2\pi} \right) \ln^2 \frac{\zeta}{\mu^2} - \frac{\alpha_s^2 \beta_0 C_F}{24\pi} \ln^3 \frac{\zeta}{\mu^2}, \end{aligned} \quad (214)$$

where $c_1 = \frac{C_F}{2\pi} \left(-2 + \frac{\pi^2}{12} \right)$ and c'_2 is again a constant to be determined in perturbation theory at two-loop level.

Finally, we compare the current formulation with previous approaches to lattice TMDPDF. First, we comment on the developments in (Engelhardt *et al.*, 2016; Hagler *et al.*, 2009; Musch *et al.*, 2012, 2011; Yoon *et al.*, 2017) in which the x -moments of TMDPDF are extracted

from ratio of quasi-TMDPDF. From Eq. (209), it is clear that both the matching kernel H and the exponential factor of Collins-Soper kernel depends on x nontrivially. Therefore, simply taking the ratio of moments for quasi-TMDPDF will not be sufficient to reproduce the same ratio for TMDPDF, although the soft function does cancel. This observation is also made recently in Ref. (Ebert *et al.*, 2020a). Second, the quasi-TMDPDF defined with the naive rectangle-shaped soft function, i.e. Z_E , is f in Eq. (195), so it is obvious that it still needs the reduced soft function S_r to be matched to f^{TMD} . As for the other proposal in (Ebert *et al.*, 2019b; Ji *et al.*, 2015b), it replaces Z_E in f with S_{best} which is the vacuum matrix element of a spacelike bent-shaped Wilson loop with angle $\pi/2$ at each junction, and does not include the function S_r^{π} in Eq. (209). Although $\sqrt{S_{\text{best}}/Z_E}$ agrees with S_r^{π} at one-loop order (Ebert *et al.*, 2019b; Ji *et al.*, 2019a), it is expected to be different at higher orders. In fact, for the anomalous dimension $\Gamma_{\frac{\pi}{2}}$ defined through

$$\Gamma_{\frac{\pi}{2}}(\alpha_s) = \mu^2 \frac{d}{d\mu^2} \ln \left(\frac{S_{\text{best}}(L, b_{\perp}, \mu)}{Z_E(2L, b_{\perp}, \mu)} \right), \quad (215)$$

it starts to deviate from $\Gamma_S(\alpha_s)$ at two-loop order (Grozin *et al.*, 2016), as

$$-\Gamma_S(\alpha_s) = \frac{\alpha_s C_F}{\pi} \quad (216)$$

$$+ \frac{\alpha_s^2}{\pi^2} \left[C_F C_A \left(-\frac{49}{36} + \frac{\pi^2}{12} - \frac{\zeta_3}{2} \right) + C_F N_F \frac{5}{18} \right],$$

$$\Gamma_{\frac{\pi}{2}}(\alpha_s) = \frac{\alpha_s C_F}{\pi} \quad (217)$$

$$+ \frac{\alpha_s^2}{\pi^2} \left[C_F C_A \left(-\frac{49}{36} + \frac{\pi^2}{24} \right) + C_F N_F \frac{5}{18} \right].$$

In the equation, $\zeta_3 = \sum_{n=1}^{\infty} (1/n^3) \neq \pi^2/12$, therefore the two anomalous dimensions are different. The differences in the anomalous dimension will result in different logarithmic behaviors in $b_{\perp}$, as the soft functions are dimensionless and depend on $b_{\perp}$ and μ only. At large $b_{\perp}$, it will lead to different IR physics that cannot be controlled by perturbation theory.

Combining the reduced soft function and the quasi-TMDPDF, one can effectively factorize the DY cross section,

$$\sigma = \int dx_A dx_B d^2 b_{\perp} e^{i\vec{Q}_{\perp} \cdot \vec{b}_{\perp}} \hat{\sigma}(x_A, x_B, Q^2, \mu) \times \tilde{f}(x_A, b_{\perp}, \mu, \zeta_A) \tilde{f}(x_B, b_{\perp}, \mu, \zeta_B) S_r(b_{\perp}, \mu). \quad (218)$$

where all non-perturbative quantities do not involve the light cone, and can be calculated on lattice.

Spin-dependent TMDPDFs are also physically important. They can be computed using LaMET theory (Ebert *et al.*, 2020a). Again one can define quasi

distributions just like the spin-independent ones. For a general proton target $|PS\rangle$ and the general spin structure Γ of the parton, the parent TMDPDF can be defined as :

$$f_{[\Gamma]}^{\text{TMD}}(x, \vec{k}_{\perp}, \mu, \zeta) = \frac{1}{2^{P^+}} \int \frac{d\lambda}{2\pi} \int \frac{d^2 \vec{b}_{\perp}}{(2\pi)^2} e^{-i\lambda x + i\vec{k}_{\perp} \cdot \vec{b}_{\perp}} \times \lim_{\delta \rightarrow 0} \frac{\langle PS | \psi(\lambda n + \vec{b}_{\perp}) \Gamma W_n(\lambda n + \vec{b}_{\perp}) |_{\delta} \psi(0) | PS \rangle}{\sqrt{S(b_{\perp}, \mu, \delta^- e^{2i\mu n}, \delta^-)}}, \quad (219)$$

where the $\zeta = 2(P^+)^2 e^{2i\mu n}$ is the rapidity scale, see Sec. V for more detail of the soft function subtraction. The individual spin-dependent TMD distributions can then be obtained through Lorentz decompositions (Mulders and Tangerman, 1996; Ralston and Soper, 1979; Tangerman and Mulders, 1995):

$$f_{[\gamma^+]}^{\text{TMD}} = f_1 - \frac{e^{ij} k^i S^j}{M} f_{1T}^{\perp}, \quad (220)$$

$$f_{[\gamma^+ \gamma_3]}^{\text{TMD}} = S^+ g_1 + \frac{\vec{k}_{\perp} \cdot \vec{S}_{\perp}}{M} g_{1T}, \quad (221)$$

$$f_{[i\sigma^+ \gamma_3]}^{\text{TMD}} = S_1^{\perp} h_1 + \frac{(2k^i k^j - \vec{k}_{\perp}^2 \delta^{ij}) S_1^j}{2M^2} h_{1T}^{\perp} + \frac{S^+ k^i}{M} h_{1L}^{\perp} + \frac{e^{ij} k^j}{M} h_1^{\perp}, \quad (222)$$

where we suppress the arguments $(x, \vec{k}_{\perp}, \mu, \zeta)$ in all distributions; f_1 , g_1 , and h_1 are unpolarized, helicity and transversity TMDPDFs, respectively; the indices i and j are in transverse space of $k_{\perp}$; S^+ and $S_1^{\perp}$ are longitudinal and transverse spin components.

Note that the Sivers function $f_{1T}^{\perp}$ (Sivers, 1990) and the Boer-Mulders function $h_1^{\perp}$ (Boer and Mulders, 1998) are T -odd. The orientation of the gauge-link have important effects on these two functions (Collins, 2011a, 2002), such that they change sign between the DY and SIDIS processes. In the light-cone gauge, these contributions arise from the transversal gauge-link at infinities (Belitsky *et al.*, 2003). They are related to the phenomenologically interesting single transverse-spin asymmetry (Boer and Mulders, 1998; Collins *et al.*, 2005, 2006; Efremov *et al.*, 2005).

C. Off-light-cone Soft Function

In previous subsections, the soft function has been introduced to define rapidity-scheme-independent TMDPDFs. The major motivation of introducing the soft function is to capture nonperturbative effects due to soft-gluon radiations from fast moving color-charges. For many inclusive processes the soft radiations cancel in the total cross section, but for certain processes where a small transverse momentum is measured, such cancellation can be incomplete and result in measurable con-

sequences. In such cases, the TMD soft function is introduced to account for the soft-gluon effects and appears in factorization theorems for the Drell-Yan (DY) process (Collins *et al.*, 1985b, 1988) and semi-inclusive DIS (SIDIS) (Ji *et al.*, 2004, 2005).

To calculate the TMD physics nonperturbatively, formulating a Euclidean version of the soft function is critical. Since the soft function in fact is a cross section and hence real and positive, it satisfies the necessary condition for a Monte Carlo simulation. In this subsection, we present an approach to calculate it in heavy-quark effective theory (HQET) (Ji *et al.*, 2020a). There is also another method proposed to extract the reduced soft function S_r from a light-meson form factor (Ji *et al.*, 2020a), where many subtleties of HQET can be avoided. The first lattice calculation of the reduced soft function based on the light-meson formalism has been performed in the recent work (Zhang *et al.*, 2020a).

Due to the different space-time pictures of the DY and SIDIS processes, the soft functions for the two processes also differ from each other as shown in Fig. 11. To define the soft function, one also needs to specify a time-ordering prescription. Since it is a cross section, it involves a time order and an anti-time order (or cut diagrams). However, in the light-cone limit, the time order does not matter. What really matters is the rapidity regularization scheme. It has been proven for the δ regulator in (Vladimirov, 2018) that the time ordering is not quite relevant up to overall phase factors, and the soft functions for the two processes are equal. The method therein can be modified to apply to the off-light-cone scheme too. Therefore, our first step is to convert the cut-diagrams into Feynman diagrams by imposing just the single time order. In this way, the soft function can be viewed as a scattering amplitude.

In the off-light-cone scheme, there are further complications caused by the space-like or time-like choices for off-light-cone vectors. Fortunately, one can show that in the light-cone limit, the space-like and time-like choices are equivalent up to overall phase factors (Ji *et al.*, in preparation). Thus we will use the notation $S(b_\perp, \mu, Y, Y')$ to denote a generic off-light-cone soft function that satisfies our demands.

With these in mind, we show that the off-light-cone soft function $S(b_\perp, \mu, Y, Y')$ is equivalent to an equal-time form factor of fast-moving color sources and can be formulated on the Euclidean lattice. From the matching formula Eq. (209) in the last subsection, once the off-light-cone soft function is known, we can combine it with the lattice calculated quasi-TMDPDF to obtain the physical TMDPDF. Therefore, the cross section of DY processes in the low transverse-momentum region (Collins *et al.*, 1985b) becomes predictable from first principles (Ji *et al.*, 2020a).

To begin with, we define the scattering amplitude of a

Wilson loop as shown in Fig. 13:

$$W(t, t', b_\perp, Y, Y') = \frac{1}{N_c} \text{Tr} \left[W_v^\dagger(\vec{b}_\perp, t') W_v(\vec{b}_\perp, t) \right] |0\rangle \quad (223)$$

where $|0\rangle$ is the QCD vacuum state and N_c is number of colors and Tr is the color-trace. Timelike four-vectors $v^\mu = \gamma(1, \beta, \vec{0}_\perp)$ and $v'^\mu = \gamma'(1, -\beta', \vec{0}_\perp)$ approach the lightcone as β and $\beta' \rightarrow 1$. The rapidity Y and the speed β are related through $\beta = \tanh Y$, in terms of the light-cone vectors p and n , the velocities read $v = \frac{v^\mu}{\sqrt{2}} \left(\frac{p^\mu}{p^+} + e^{-2Y} p^+ n \right)$ and $v' = \frac{v'^\mu}{\sqrt{2}} \left(e^{-2Y'} \frac{p^\mu}{p^+} + p^+ n \right)$. The $W_v(\vec{b}_\perp, t)$ is a staple-shaped gauge-link along v direction similar to those defined in Eqs. (173) and (196). t and t' are the lengths of the t -components of the staples. The single time-order prescription for S allows physical interpretation as a chronological process. Similar to the

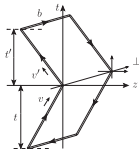


FIG. 13: The Wilson-loop W showing a pair of quark and antiquark scattering at $t = 0$.

quasi-TMDPDF, the Wilson-loop in Eq. (223) contains pinch-pole singularities associated to time evolution of initial and final states at large t and t' . Therefore, we need to subtract them out in Eq. (223) with rectangular Wilson-loops (Collins, 2008; Ji *et al.*, 2019a). This leads to an off-light-cone realization of the soft function:

$$S(b_\perp, \mu, Y, Y') = \lim_{\substack{t \rightarrow \infty \\ t' \rightarrow \infty}} \frac{W(t, t', b_\perp, \mu, Y, Y')}{\sqrt{Z(2t, b_\perp, \mu, Y) Z(2t', b_\perp, \mu, Y')}} \quad (224)$$

where $Z(2t, b_\perp, Y)$ is the vacuum expectation value of rectangular Wilson loop which is similar to W by setting $v' = v$ and $t' = t$, i.e. $Z(2t, b_\perp, Y) = W(t, t, b_\perp, Y, -Y)$. The factor Z has a clear physical interpretation: It can be viewed as the wave function renormalization for incoming or outgoing color sources. After the subtraction through Z , the only remaining UV divergences for $S(b_\perp, \mu, Y, Y')$ are the cusp divergences with hyperbolic angle $Y + Y'$.

We should mention that a more common definition of the soft function $S_{\text{DY}}(b_\perp, \mu, Y, Y')$ for the DY process was proposed in (Collins, 2011a,b). The space-like vectors $u^\mu = \gamma(\beta, 1, 0, 0)$ and $u'^\mu = \gamma'(-\beta', 1, 0, 0)$ were chosen instead of time-like v and v' to define the soft function for the DY process. This soft function has already been defined in the last subsection in Eq. (202). u and u' are equal to py , my up to overall normalization factors.

While S and S_{DY} are defined differently, we can show that

$$S(b_\perp, \mu, Y, Y') = S_{\text{DY}}(b_\perp, \mu, Y, Y') \quad (225)$$

using analyticity property (Ji *et al.*, in preparation). Here we focus on S in Eq. (224), which has a simple Euclidean realization.

After defining the soft function S , we now show that it is equal to a form factor. In HQET, the propagator of a color source is equivalent to a gauge-link along its moving direction. Thus $W(t, t', b_\perp, \mu, Y, Y')$ can be expressed by fields in HQET with the Lagrangian

$$\mathcal{L}_{\text{HQET}} = Q_s^\dagger(x)(i\bar{v} \cdot D)Q_s(x) + \bar{Q}_s^\dagger(x)(i\bar{v} \cdot D)Q_s(x), \quad (226)$$

where Q_s and $\bar{Q}_s$ are quark and anti-quark in the fundamental and anti-fundamental representations, respectively; $v^\mu = \gamma(1, \beta, \vec{0}_\perp)$ is the four velocity; D is the covariant derivative. Note that quarks in HQET can be viewed as color sources. If the gluon soft function is considered, the heavy quarks should be in the adjoint representation.

In HQET, a color-singlet heavy-quark pair separated by $\vec{b}$ generates a heavy quark potential $V(|\vec{b}|)$ in the ground state, and the spectrum includes a gapped continuum above it. The state can also have a residual momentum $\delta\vec{P}$, which is arbitrary due to reparameterization invariance (Luke and Manohar, 1992; Manohar and Wise, 2000), and for simplicity we always consider $\delta\vec{P} = 0$. When the sources move with a velocity v , the ground state can be labeled by $|\bar{Q}Q, \vec{b}, \delta\vec{P}\rangle_v$, where the residue momentum $\delta\vec{P} = \vec{P}_{\text{total}} - 2m_Q\gamma\vec{\beta}$ is the difference between the total momentum $\vec{P}_{\text{total}}$ and the kinetic momentum of the heavy-quarks. The residual energy of the state is $E = \gamma^{-1}V(|\vec{b}_\perp|) + \vec{\beta} \cdot \delta\vec{P}$.

Consider a process with incoming and outgoing states being heavy-quark pairs separated by $\vec{b}_\perp$ and at velocity v and v' , respectively. Such a state is created by the interpolating fields

$$\mathcal{O}_v(t, \vec{b}_\perp) = \int d^2\vec{r} Q_s^\dagger(t, \vec{r}) U(\vec{r}, \vec{r}', t) \bar{Q}_s^\dagger(t, \vec{r}'), \quad (227)$$

where $\vec{r}' = \vec{r} + \vec{b}_\perp$ and $U(\vec{r}, \vec{r}', t)$ is a gauge-link connecting $\vec{r}'$ to $\vec{r}$ at time t . The heavy-quark pair created by $\mathcal{O}_v$ is forced to be at relative separation $\vec{b}_\perp$ and to have vanishing residual momentum $\delta\vec{P} = 0$. Between

the incoming and outgoing states, a product of two local equal-time operators

$$J(v, v', \vec{b}_\perp) = \bar{Q}_{v'}^\dagger(\vec{b}_\perp) \bar{Q}_v(\vec{b}_\perp) Q_{v'}^\dagger(0) Q_v(0) \quad (228)$$

is inserted at $t = 0$. Then W can be expressed in terms of HQET propagators which are gauge-links in the v, v' directions. After integrating out the heavy-quark fields, we obtain up to an overall volume factor (Ji *et al.*, 2020a)

$$\begin{aligned} W(t, t', b_\perp, \mu, Y, Y') &= \frac{1}{N_c} \langle 0 | \mathcal{O}_{v'}^\dagger(t', \vec{b}_\perp) J(v, v', \vec{b}_\perp) \mathcal{O}_v(-t, \vec{b}_\perp) | 0 \rangle \\ &\xrightarrow[t' \rightarrow -\infty]{t \rightarrow \infty} \frac{1}{N_c} \Phi^\dagger(\vec{b}_\perp, \mu) S(b_\perp, \mu, Y, Y') \Phi(\vec{b}_\perp, \mu) e^{-iE't - iEt}, \end{aligned} \quad (229)$$

where

$$\begin{aligned} \Phi(b_\perp, \mu) &= \lim_{T \rightarrow \infty} {}_v \langle \bar{Q}Q, \vec{b}_\perp | \mathcal{O}_v(T, \vec{b}_\perp) | 0 \rangle, \quad (230) \\ S(b_\perp, \mu, Y, Y') &= {}_{v'} \langle \bar{Q}Q, \vec{b}_\perp | J(v, v', \vec{b}_\perp) | \bar{Q}Q, \vec{b}_\perp \rangle_v. \end{aligned}$$

In the last line of Eq. (229), we have inserted a complete set of heavy-quark pair states before and after J . At large t and t' , the contribution from the continuum spectrum is damped out due to the Riemann-Lebesgue lemma (Zuazo, 2001), while the contribution from $|\bar{Q}Q, \vec{b}_\perp, \delta\vec{P} = 0\rangle_v$ with residual energy $E = \gamma^{-1}V(|\vec{b}_\perp|)$ survives. As a result we obtain Eqs. (229)—(230), where we have omitted the state label $\delta\vec{P} = 0$ for simplicity. Alternatively, we can also give t and t' a small negative imaginary part, which is consistent with the time order, to damp out all states except $|\bar{Q}Q, \vec{b}_\perp\rangle_v$ at large t and t' . Note that $\Phi(\vec{b}_\perp, \mu)$ is independent of Y because it is boost invariant.

Similarly, Z can also be formulated in HQET as

$$\begin{aligned} Z(2t, b_\perp, Y) &= \frac{1}{N_c} \langle 0 | \mathcal{O}_{v'}^\dagger(t, \vec{b}_\perp) \mathcal{O}_v(-t, \vec{b}_\perp) | 0 \rangle \\ &\xrightarrow[t \rightarrow \infty]{} \frac{1}{N_c} \Phi^\dagger(\vec{b}_\perp, \mu) \Phi(\vec{b}_\perp, \mu) e^{-2iEt}, \quad (231) \end{aligned}$$

whose t -component has length $2t$. The Y dependence of Z is implicit in the energy E . Combining Eqs. (229) and (231), we obtain S defined in Eq. (224). We emphasize that Eq. (224) can be seen as a LSZ reduction formula, in which we amputate the external heavy-quark pair states $|\bar{Q}Q, \vec{b}_\perp\rangle_v$.

Being an equal-time observable, $S(b_\perp, \mu, Y, Y')$ can be straightforwardly realized in Euclidean time as:

$$\begin{aligned} S(b_\perp, \mu, Y, Y') &= \lim_{T \rightarrow \infty} \frac{W_E(T, T', b_\perp, \mu, Y, Y')}{\sqrt{Z_E(2T, b_\perp, \mu, Y) Z_E(2T', b_\perp, \mu, Y')}}, \quad (232) \end{aligned}$$

where the subscript E indicates the quantity is defined in Euclidean time, with corresponding variables

T and T' . Due to boost invariance, the factor $Z_E(T, b_\perp, \mu, Y)$ relates to the rectangular Wilson-loop defined in Eq. (198) along the n_z direction through the relation $Z_E(2T, b_\perp, \mu, Y) = Z_E(2\gamma^{-1}T, b_\perp, 0)$. The relevant matrix elements are now calculated by a lattice version of HQET with the Lagrangian (Aglietti, 1994; Hashimoto and Matsufuru, 1996; Horgan *et al.*, 2009)

$$\begin{aligned} \mathcal{L}_{\text{HQET}}^E &= Q_v^\dagger(x)(i\vec{v} \cdot D_E)Q_v(x) + Q_v^\dagger(x)(i\vec{v} \cdot D_E)Q_v(x), \end{aligned} \quad (233)$$

where the subscript E denotes the Euclidean space, $i\vec{v} \cdot D_E = \gamma(D^T - i\beta)D^\perp$ with $\vec{v}^\mu = \gamma(-i, -\beta, \vec{0}_\perp)$. We have explicitly verified Eq. (232) in Euclidean perturbation theory to the one-loop order.

The soft function cannot be calculated on the lattice by simply replacing the Minkowskian gauge-links in Eq. (223) with a finite number of Euclidean gauge-links. Through HQET, we find a time-independent formulation of the soft function, which opens up the possibility of direct lattice calculations.

D. Light-Front Wave-Function Amplitudes And Soft Function from Meson Form Factor

Light-front quantization (LFQ) or formalism is a natural language for parton physics in which partons are made manifest at all stages of calculations. It favors a Hamiltonian approach to QCD like for a non-relativistic quantum mechanical system, i.e., to diagonalize the Hamiltonian

$$\hat{P}^- |\Psi_n\rangle = \frac{M_n^2 + \vec{P}_\perp^2}{2P^+} |\Psi_n\rangle, \quad (234)$$

to obtain wave functions for the QCD bound states (Brodsky *et al.*, 1998). The light-front wave functions (LFWFs) thus obtained can, in principle, be used to calculate all the partonic densities and correlations functions. Moreover, like in condensed matter systems, knowing quantum many-body wave-functions allows one to understand interesting aspects of quantum coherence and entanglement, as well as the fundamental nature of quantum systems. Therefore, a practical realization of light-front quantization program clearly would be a big step forward in understanding the fundamental structure of the proton.

However, from a field theory point of view, wave functions are not the most natural objects to consider due to the non-trivial vacuum, UV divergences as well as the requirement of Lorentz symmetry, according to which the space and time should be treated on equal footing. The proton or other hadrons are excitations of the QCD vacuum which by itself is very complicated because of the well-known phenomena of chiral symmetry breaking and color confinement. To build a proton on top of this vacuum, one naturally has a question of what part of

the wave-function reflects the property of the proton and what reflects the vacuum: It is the difference that yields the properties of the proton that are experimentally measurable. There is no clean way to make this separation unless one builds the proton out of elementary excitations or quasi-particles that do not exist in the vacuum, as often done in condensed matter systems.

The partons in the IMF avoids the above problems to a certain extent. In fact, due to the kinematic effects, in the IMF all partons in the vacuum have longitudinal momentum $k^+ = 0$, and to some degree of accuracy, the proton is made of partons with $k^+ \neq 0$. This natural separation of degrees of freedom (DOF) is particularly welcome, making a wave-function description of the proton more natural and interesting in IMF than in any other frame.

To implement the above DOF separation, one possibility is to assume triviality of the light-front vacuum. The question that to what extent this holds has been continuously debated over the years. One knows *a priori* that in relativistic QFT, the vacuum state is boost invariant and frame-independent. In fact, it was proven in (Nakanishi and Yabuki, 1977; Nakanishi and Yamawaki, 1977) that not only the vacuum can not be trivial, even the Green's functions of the full theory cannot pose generic meaningful restrictions to the null-planes $\xi^+ = c$. In fact, the vacuum zero modes do contain non-trivial dynamics and contribute to the properties of the proton (Ji, 2020). Nevertheless, one can adopt an effective theory point of view to simply cut off the zero-modes and relegate their physics to renormalization constants. In some simple cases, these zero-modes can be treated explicitly (Heinzl *et al.*, 1991; Yamawaki, 1998).

By imposing an IR cut-off on the $k^+ \geq \epsilon$ in the effective Hilbert space, all physics below $k^+ = \epsilon$ are taken into account through renormalization constants. We then obtain an effective LF theory with trivial vacuum,

$$a_{k\lambda}|0\rangle = b_{p\sigma}|0\rangle = d_{p\sigma}|0\rangle = 0. \quad (235)$$

where $|0\rangle$ is the vacuum of LFQ. Therefore, the proton can be expanded in terms of the superposition of Fock states in the LF gauge $A^+ = 0$ (Brodsky *et al.*, 1998),

$$|P\rangle = \sum_{n=1}^{\infty} \int dT_n \psi_n^0(x_i, \vec{k}_{i\perp}) \prod a_i^\dagger(x_i, \vec{k}_{i\perp}) |0\rangle. \quad (236)$$

where $a^\dagger$ are generic quarks and gluon quanta on the light-front, the phase-space integral reads $dT_n = \prod \frac{dx_i d^2k_{i\perp}}{2\pi(2\pi)^2}$. The $\psi_n(x_i, \vec{k}_{i\perp})$ are LFWF amplitudes or simply WF amplitudes, where x_i to denote the set of momentum fractions from x_1 to x_n . They are a complete set of non-perturbative quantities which describe the partonic landscape of the proton. The above amplitudes can in principle be calculated through Hamiltonian

diagonalization. However, as explained in Sec. II A, a direct systematic solution in LFQ is impractical.

LaMET offers an alternate route to calculate these WF amplitudes. Thanks to the triviality of the vacuum after the truncation $k^+ \geq \epsilon$, they can then be written in terms of the invariant matrix elements by inverting the above expansion,

$$\psi_n^R(x_i, \vec{k}_{i\perp}) = \langle 0 | \prod a_i(x_i, \vec{k}_{i\perp}) | P \rangle. \quad (237)$$

After properly restoring gauge-invariance and imposing regularizations, they become the matrix elements of light-cone correlators, the same type as those in the TMD-PDFs. Therefore, the LaMET method applies to them, which allows one to effectively obtain the results of light-front quantization through instant quantization in a large momentum frame.

To realize the goal, the LFWF amplitudes also need a rapidity renormalization, as in the case of TMDPDFs. In this section, we explain how the reduced soft function S_r can be obtained by combining the LFWF amplitudes and a special light-meson form factor, instead of as the form factor in HQET discussed in the previous section. A lattice calculation based on the light-meson framework has been performed in (Zhang *et al.*, 2020b).

Let us consider the following form factor of a pseudoscalar light-meson state with constituents $\bar{\psi}\psi$,

$$F(b_\perp, P, P', \mu) = \langle P' | \bar{\eta}(\vec{b}_\perp) \Gamma' \eta(\vec{b}_\perp) \bar{\psi}(0) \Gamma \psi(0) | P \rangle \quad (238)$$

where ψ and η are light quark fields of different flavors; $P^\mu = (P^+, 0, 0, P^z)$ and $P'^\mu = (P', 0, 0, -P^z)$ are two large momenta which approach two opposite light-like directions in the limit $P^z \rightarrow \infty$; Γ and Γ' are Dirac gamma matrices, which can be chosen as $\Gamma = \Gamma' = 1, \gamma_5$ or $\gamma_\perp$ and $\gamma_\perp \gamma_5$, so that the quark fields have leading components on the respective light-cones.

At large momentum, the form factor factorizes through TMD factorization into LFWF amplitudes. To motivate the factorization, we need to consider the leading region of IR divergences in a similar way for SIDIS and Drell-Yan (Collins, 2011b; Ji *et al.*, 2005), and the result is shown in Fig. 14. There are two collinear sub-diagrams responsible for collinear modes in $+$ and $-$ directions, and a soft sub-diagram responsible for soft contributions. Besides, there are two IR-free hard cores localized around $(0, 0, 0, 0)$ and $(0, \vec{b}_\perp, 0)$. In the covariant gauge, there are arbitrary numbers of longitudinally-polarized collinear and soft gluons that can connect to the hard and collinear sub-diagrams. Based on the region decomposition, we now follow the standard procedure to make factorization into LF quantities (Collins, 2011b).

We first factorize the soft divergences. This can be done with the soft function $S(b_\perp, \mu, \delta^+, \delta^-)$. It re-sums the soft gluon radiations from fast-moving color-charges. Intuitively, soft gluons have no impact on the velocity of

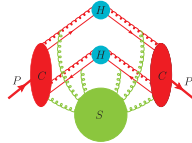


FIG. 14: The reduced diagram for the large-momentum form factor F of a meson. Two H denote the two hard cores separated in space by $b_\perp$, C are collinear sub-diagrams and S denotes the soft sub-diagram.

the fast-moving color charged partons, and the propagators of partons eikonalize to straight gauge links along their moving trajectory.

We then factorize the collinear divergences. For the incoming direction, the collinear divergences is captured by the LFWF amplitude for the incoming parton $\psi_{q_0}(x, b_\perp, \mu, \delta^-)$ defined with future-pointing gauge-links.

$$\psi_{q_0}(x, b_\perp, \mu, \delta^-) = \int \frac{d\lambda}{4\pi} e^{-i\lambda x} \quad (239)$$

$$\langle 0 | \bar{\psi}(\lambda n/2 + \vec{b}_\perp) \gamma^+ W_n(\lambda n/2 + \vec{b}_\perp)_{|x'} \psi(-\lambda n/2) | P \rangle,$$

where the staple-shaped gauge-link W_n is defined similar to that in Eq. (166), the only exception being the gauge-links W_n should point to $+\infty$ instead of $-\infty$.

However, the naive LFWF amplitude contains soft divergences as well, to avoid double-counting, we must subtract out the soft contribution from the bared collinear WF amplitude with the soft function $S(b_\perp, \mu, \delta^+, \delta^-)$. This leads to the collinear function for the incoming direction: $\psi_{q_0}(x, b_\perp, \mu, \delta^-)/S(b_\perp, \mu, \delta^+, \delta^-)$. Similarly, for the out-going direction one obtains the collinear function $\psi^1(x', b_\perp, \mu, \delta^+)/S(b_\perp, \mu, \delta^+, \delta^-)$.

Here we briefly comment on the choices for the gauge-link directions in the soft functions and the WF amplitudes. Naively, the gauge-links along the p direction have to be past-pointing. However, similar to the arguments in (Collins and Metz, 2004) for the SIDIS process, based on the space-time picture of collinear divergences, one can chose future pointing gauge-links along p direction as well. With all the gauge-links being future pointing, the soft function equals to S^- which is manifestly real, and the WF amplitudes for the incoming and outgoing hadrons are in complex conjugation to each other.

Besides the collinear and soft functions, we still need the hard core $H_P(Q^2, Q^2, \mu^2)$ where $Q^2 = x x' P \cdot P'$, $Q^2 =$

x, x', P, P' and an integral over the momentum fractions x, x' is assumed. Taking together, we have the TMD factorization of the form factor into hard, collinear and soft functions:

$$F(b_\perp, P, P', \mu) = \int dx dx' H_F(Q^2, Q^2, \mu^2) \quad (240)$$

$$\times \left[\frac{\psi_{b_1}(x', b_\perp, \mu, \delta^+)}{S(b_\perp, \mu, \delta^+, \delta^-)} \right] \left[\frac{\psi_{b_1}(x, b_\perp, \mu, \delta^-)}{S(b_\perp, \mu, \delta^+, \delta^-)} \right]$$

$$\times S(b_\perp, \mu, \delta^+, \delta^-).$$

All the rapidity regulators in all the WF amplitudes and the soft functions are cancelled.

Let us consider a one-loop example. The incoming hadron state consists of a free quark with momentum $x_0 P^+$ and a free anti-quark with momentum $x_0' P^+$. Similarly the outgoing state consists of a pair of free quark and anti-quark with momentum $x_0' P'^-$, $x_0 P'^-$, respectively. The spin projection operator for the incoming state is proportional to γ^+ and for the out-going state is proportional to γ^+ . The tree level form factor is normalized to 1. At one-loop level, the pseudo-scalar form factor with vector currents $\Gamma = \gamma^\mu$, $\Gamma^* = \gamma_\mu$, where a summation over μ is assumed reads:

$$F(b_\perp, P, P', \mu) = 1 + \frac{\alpha_s C_F}{2\pi} F^{(1)}(b_\perp, Q^2, Q^2, \mu^2), \quad (241)$$

where $Q^2 = 2x_0 x_0' P^+ P'^-$, $Q^2 = 2x_0 x_0' P^+ P'^-$ and

$$F^{(1)}(b_\perp, Q^2, Q^2, \mu^2) \quad (242)$$

$$= -7 + \left(-\frac{1}{2} \ln^2 b_\perp^2 Q^2 + \frac{3}{2} \ln b_\perp^2 Q^2 + (Q \rightarrow Q) \right).$$

This result can be obtained from the one-loop DY structure function (D'Alesio *et al.*, 2014) using the substitution $\ln^2(-Q^2 b_\perp^2) \rightarrow \frac{1}{2} \ln^2 Q^2 b_\perp^2 + \ln^2 Q^2 b_\perp^2$ and $\ln(-Q^2 b_\perp^2) \rightarrow \frac{1}{2} \ln Q^2 b_\perp^2 + \ln Q^2 b_\perp^2$. Similar to the TMD factorization for SIDIS and DY process, one should also notice that the hard kernel $H_F(Q^2, Q^2, \mu^2)$ can be obtained from that of the space-like Sudakov form factor:

$$H_F(Q^2, Q^2, \mu^2) = H^{rad}(-Q^2) H^{rad}(-Q^2), \quad (243)$$

where $H^{rad}(-Q^2)$ is given in (Collins and Rogers, 2017). At one-loop level, we then obtain:

$$H_F(Q^2, Q^2, \mu^2) = 1$$

$$+ \frac{\alpha_s}{4\pi} \left(-16 + \frac{\pi^2}{3} + 3L_Q + 3L_{\bar{Q}} - L_Q^2 - L_{\bar{Q}}^2 \right), \quad (244)$$

where $L_Q = \ln \frac{Q^2}{\mu^2}$ and $L_{\bar{Q}} = \frac{Q^2}{\mu^2}$.

Now we construct the Euclidean version of the factorization in terms of the quasi-WF amplitudes, the reduced

soft function, and hard contribution. The quasi-WF amplitudes are defined in a way similar to Eq. (195):

$$\tilde{\psi}_{b_1}(x, b_\perp, \mu, \zeta_\pm) = \int \frac{d\lambda}{4\pi} e^{i\lambda x} \quad (245)$$

$$\frac{\langle 0 | \bar{\psi} \left(\frac{\lambda}{2} + \bar{b}_\perp \right) \gamma^+ W_z \left(\frac{\lambda}{2} + \bar{b}_\perp; L \right) \psi \left(-\frac{\lambda}{2} \right) | P \rangle}{Z_F(2L, b_\perp, \mu)},$$

in which the staple-shaped gauge-link W_z is defined in Eq. (196). The gauge-links should point to $+z$ direction in accordance to the $+\infty$ choice on the light-cone side.

The factorization to the LFWF amplitude follows a similar reasoning to that of the quasi-TMDPDFs presented in previous sections. Alternatively, we can factorize it using quantities defined in on-light-cone rapidity scheme,

$$\tilde{\psi}_{b_1}(x, b_\perp, \mu, \zeta_\pm) = H_1^+ \left(\zeta_\pm / \mu^2, \zeta_\pm / \mu^2 \right) \quad (246)$$

$$\times \left[\frac{\psi_{b_1}(x, b_\perp, \mu, \delta^-)}{S(b_\perp, \mu, \delta^+, \delta^-)} \right] S(b_\perp, \mu, \delta^+).$$

This factorization is the result of applying a similar leading-region analysis to the quasi-WF amplitude. The $\tilde{\psi}_{b_1}(x, b_\perp, \mu, \delta^-)/S(b_\perp, \mu, \delta^+, \delta^-)$ re-sums all the collinear divergences, while the soft function $S(b_\perp, \mu, \delta^+)$ contains an off-light-cone direction along n_- . It resums the soft divergences of the quasi-WF amplitude. The soft functions $S(b_\perp, \mu, \delta^+, \delta^-)$ and $S(b_\perp, \mu, \delta^+)$ subtract away the regulator dependencies introduced in the bare LFWF amplitude. The overall combination in the right-hand side of Eq. (246) is rapidity-scheme independent. Similar to the case of the form factor, we can chose all the gauge-links along the incoming collinear direction to be future-pointing.

Combining together Eqs. (240) and (246) and using the relation $\zeta_\ell' = \zeta_\pm \zeta_\pm'$, one obtains the form factor factorization,

$$F(b_\perp, P, P', \mu) \quad (247)$$

$$= \int dx dx' H(x, x') \tilde{\psi}_{b_1}^\dagger(x', b_\perp) \tilde{\psi}_{b_1}(x, b_\perp) S_F(b_\perp, \mu),$$

where we have only kept the $x, b_\perp$ dependencies of the WF amplitudes with other variables being omitted, and the hard kernel H is given by:

$$H(x, x') = H(\zeta_\pm, \zeta_\pm', \zeta_\pm, \zeta_\pm', \mu^2)$$

$$= \frac{H_F(Q^2, Q^2, \mu^2)}{H_1^+ \left(\zeta_\pm / \mu^2, \zeta_\pm / \mu^2 \right) H_1^+ \left(\zeta_\pm' / \mu^2, \zeta_\pm' / \mu^2 \right)}, \quad (248)$$

where $Q^2 = \sqrt{\zeta_\pm \zeta_\pm'}$ and $\bar{Q}^2 = \sqrt{\zeta_\pm \zeta_\pm'}$. And the reduced soft function is

$$S_F(b_\perp, \mu) = \lim_{\delta^+, \delta^- \rightarrow 0} \frac{S(b_\perp, \mu, \delta^+, \delta^-)}{S(b_\perp, \mu, \delta^+) S(b_\perp, \mu, \delta^-)}. \quad (249)$$

It can be shown based on property of off-light-cone soft functions that S_F defined here agrees with the one defined in Eq. (204).

Therefore, with non-perturbative quantities F and ψ^+ , we obtain the reduced soft function,

$$S_v(b_\perp, \mu) = \frac{F(b_\perp, P^+, P^0, \mu)}{\int dx dx' H(x, x') \psi_{q\bar{q}}^+(x', b_\perp) \psi_{q\bar{q}}(x, b_\perp)}, \quad (250)$$

where H can be obtained perturbatively.

Based on the one-loop results for the form factor, the quasi-WF amplitudes and the reduced soft function, the one-loop matching kernel for the vector current can be extracted as:

$$H(\zeta_z, \zeta_z', \zeta_z, \zeta_z', \mu^2) = 1 + \frac{\alpha_s C_F}{2} \ln \frac{\sqrt{\zeta_z \zeta_z'}}{\sqrt{\zeta_z' \zeta_z}} \quad (251) \\ + \frac{\alpha_s C_F}{4\pi} \left(-8 + \ln^2 \frac{\sqrt{\zeta_z}}{\sqrt{\zeta_z'}} + \ln \frac{\sqrt{\zeta_z \zeta_z'}}{\mu^2} + (\zeta \rightarrow \zeta') \right),$$

and the renormalization group equation for H reads:

$$\mu^2 \frac{d}{d\mu^2} \ln H(\zeta_z, \zeta_z', \zeta_z, \zeta_z', \mu^2) = -2\gamma_F(\alpha_s) - \Gamma_S(\alpha_s), \quad (252)$$

where γ_F and Γ_S have been defined before.

Here we briefly comment on the end-point behavior. As $x \sim 0$, the hard kernel diverges logarithmically near the end point as $1 + \alpha_s \ln^2 x$, but the quasi-WF amplitudes approach zero at large or small x linearly, thus the end point regions behave as $x \ln^2 x$, which is free from those problems for the k_T factorization for electromagnetic form factor (Li and Sternman, 1992). Moreover, we can fix the z -component momentum transfer at each of the vertices to be P^z , which indicating that $x + x' = 1$. In this case the end-point behavior is improved to $x^2 \ln^2 x$.

VI. LATTICE PARTON PHYSICS WITH LAMET

Lattice gauge theory simulates continuum QCD in imaginary time on a discretized 4D Euclidean lattice. The method is characterized by the finite lattice spacing a and volume $L_4 \times L_2 \times L_3 \times T$, and input parameters such as the strong coupling and quark masses. To calculate physical quantities, one usually expects to take the continuum ($a \rightarrow 0$) and infinite volume $L_i, T \rightarrow \infty$ limits, as well as tuning the quark masses so that observables such as the pion mass m_π agrees with the physical value of ~ 140 MeV. There are different methods to implement the fermions on the lattice (Rothe, 1992), which leads to different properties of the lattice action such as chiral symmetry breaking for Wilson fermions. In the lattice calculation of hadron matrix elements, the initial and final states are generated by acting the source and sink interpolation operators on the vacuum, and the ground-state contributions are filtered out by propagating over a sufficiently large Euclidean time. A boosted hadron state

can be obtained by inserting momentum into the source and sink operators through Fourier transform in the 3D spatial coordinates.

The lattice QCD calculations of parton physics using LaMET started with the exploratory studies on the simplest PDFs and the gluon helicity (Alexandrou *et al.*, 2015; Lin *et al.*, 2015; Yang *et al.*, 2017), which yielded fairly encouraging results, demonstrating that LaMET is a viable approach. In subsequent studies, more attention has been paid to the systematics, including establishing a proper renormalization and matching procedure, simulating at the physical pion mass, removing the excited-state contamination, etc. Such studies have greatly improved the precision of the calculations, with the latest results exhibiting a reasonable agreement with phenomenological PDFs (Alexandrou *et al.*, 2018a,b; Gao *et al.*, 2020; Eubuchi *et al.*, 2019; Lin *et al.*, 2018a). In the meantime, explorations have also been made on similar large momentum data using the coordinate-space factorization methods including the pseudo-PDF (Joó *et al.*, 2019a,b, 2020; Orginos *et al.*, 2017) and current-current correlation (Bali *et al.*, 2019; Sufian *et al.*, 2020, 2019). Nevertheless, dedicated large-scale efforts with the state-of-art resources are yet to be seen. Lattice parton physics with LaMET is just at its dawn. With EIC in the US going forward, a new era of lattice calculations is to come.

In this section, we summarize the current status of lattice calculations using LaMET and discuss future prospects. We will begin with a general discussion on what kind of lattice setups are best suited for LaMET calculations, and then briefly summarize relevant lattice techniques that facilitate such calculations. After that, we review the lattice calculations that have been carried out so far and point out future improvements. A nice complementary discussion about lattice calculations has been made in (Cichy and Constantinou, 2019). Other reviews that summarize the recent developments in the lattice calculation of PDFs can be found in (Constantinou, 2020; Monahan, 2018a; Zhao, 2020).

A. Special Considerations for Lattice Calculations

In this subsection, we discuss the challenges for lattice calculations in LaMET, and estimate the required lattice requirements by taking the collinear PDFs as an example.

1. Challenges due to large momentum

In addition to common challenges with other lattice calculations, such as taking the continuum and infinite volume limits, simulating at or extrapolating to the physical pion mass, etc., LaMET applications require generating large-momentum hadron on lattice. For LaMET expansion, $1/(xP^z)$ is the expansion parameter, and

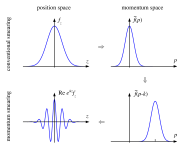


FIG. 15: Conventional smearing (left) versus momentum smearing (right) (Bali *et al.*, 2016). Conventional smearing has small overlap with high momentum state. Momentum smearing shifts momentum to peak at nonzero value in momentum space.

for the coordinate-space factorizations, large quasi-light-cone distance λ requires even bigger hadron momentum. However, realizing this faces a number of practical challenges. First, it has been difficult to generate large-momentum hadron states on the lattice, until the technique of momentum smearing (Bali *et al.*, 2016) was proposed. The conventional smearing method in coordinate space is designed to increase the overlap with ground state hadron at rest. Thus, it is not surprising that such a smearing is not efficient when the hadron has a large momentum. The momentum smearing technique introduces an extra phase factor $e^{ik \cdot x}$ to the quark field, such that it is peaked at nonzero momentum k in Fourier space, as shown in Fig. 15. In this way, the overlap with high momentum state is vastly increased after Euclidean time evolution. Recently, the momentum smearing technique has been incorporated into the framework of distillation (Egerer *et al.*, 2020) to improve the extraction of ground-state energy and matrix elements at momentum $\lesssim 3$ GeV. Although there are other proposed methods to generate large momentum (Wu *et al.*, 2018), the momentum smearing has become a standard technique in LaMET applications.

Second, the proton size is frame-dependent and changes with its momentum. In the proton's rest frame, simulating its structure requires that the lattice spacing be much smaller than the QCD confinement scale, i.e. $a \ll \Lambda_{\text{QCD}}^{-1}$. When the proton is moving fast, it undergoes Lorentz contraction by a boost factor γ in the momentum direction, thus a finer lattice spacing $a \ll (\gamma \Lambda_{\text{QCD}})^{-1}$ is needed. If $a \lesssim 0.2$ fm is the minimum requirement to investigate a static proton, one will need at least $a \lesssim 0.04$ fm to have the same resolution for a proton at 5 GeV. A smaller lattice spacing is difficult to achieve with current computing resources, for it suf-

fices from the well-known critical slowing down problem, i.e., the auto-correlation times of observables such as the topological charge increase when approaching the continuum limit (Schaefer *et al.*, 2011), which can be much longer than the Monte-Carlo simulation times. A lattice with open (Neumann) boundary condition on gauge fields in the Euclidean time direction (Lüscher and Schaefer, 2011), which allows topological charge to flow in and out at boundaries of time, may overcome this problem.

Third, the gaps between the ground state and the excited state energies become smaller because of the time dilation effect. In the proton's rest frame, the excited state contamination exponentially decays with the mass gap ΔM and evolution time τ in the form of $e^{-\Delta M \tau}$. In the boosted frame, the mass gap ΔM in the decay factor is replaced by the energy gap $\Delta E \sim \Delta M/\gamma$, and the decay changes like $e^{-\Delta M \tau} \rightarrow e^{-\Delta E \tau} = e^{-\Delta M \tau/\gamma}$ under Euclidean time evolution. Therefore, with a boosted state, a longer time evolution (source-sink separation) is needed. For example, if a source-sink separation of 1 fm is needed to separate the excited state of proton with 2 GeV momentum, a proton with 5 GeV momentum will require a source-sink separation of 2.5 fm. Even if the two-state fit technique is used, a longer time evolution is still required so that only the ground and first excited states dominate.

Last but not the least, lattice calculation requires $P^2 \ll 1/a$, so that discretization effects of $\mathcal{O}((aP^2)^n)$ are under control. Therefore, one has to go to smaller lattice spacing in order to reach larger momentum. The quantification of $\mathcal{O}((aP^2)^n)$ effects alone in LaMET calculations has not been done, as all discretization errors are treated on equal footing in continuum extrapolation.

To summarize, to achieve a precision calculation of boosted hadron structure on lattice, a fine lattice spacing (at least in the longitudinal direction) and a large box size in the time direction are essential, which of course will also require control over the signal-to-noise ratio at large Euclidean times.

2. Considerations for lattice setup

In practical calculations, a correlation function is first obtained on lattice in coordinate space, and then Fourier transformed to momentum space with the phase factor $e^{i\lambda x}$ where $\lambda = zP^2$. Therefore, the smallest x one can reach can be roughly estimated from the largest λ as $x \sim 1/\lambda$. However, a more stringent constraint comes from requiring that the higher-twist contribution $\mathcal{O}(\Lambda_{\text{QCD}}^2/(xP^2)^2)$ be small so that the factorization is still valid, which implies $x \gg \Lambda_{\text{QCD}}/P^2$. This also provides a rough estimate for the largest attainable x ($x \ll 1 - \Lambda_{\text{QCD}}/P^2$) since the momentum fraction carried by other partons is $\sim (1-x)$ which should also be bounded from below by the above estimate.

For the current state-of-the-art simulations, the lattice spacing can reach 0.04 fm (Fan *et al.*, 2020a; Gao *et al.*, 2020), which implies $P_{\text{max}}^z \sim 5$ GeV and the effective resolution in longitudinal direction is about $\gamma a \sim 0.2$ fm. Thus the valid z region that can be extracted from lattice is roughly 0.1 to 0.9. On the other hand, to avoid finite volume effects, it is believed that $m_\pi L \gtrsim 4$. For physical pion mass, the box size in spatial direction L should be at least 6 fm, which means the box size is 150 lattice spacing. So far, the largest box size in LaMET calculations is 5.8 fm (Lin *et al.*, 2018a). As discussed in Sec. VI A.1, the source-sink separation of 2.5 fm is needed for $P^z = 5$ GeV. So the box size in time direction T does not need to be particularly longer than L , and $T = L$ is sufficient in this lattice setup. In summary, with $a = 0.04$ fm at physical pion mass, one need a $L^3 \times T = 150^3 \times 150$ lattice to reliably extract $0.1 < x < 0.9$ region, which could be possible in an exa-scale computer.

There are potential tricks to reduce the computational cost. First, the required source-sink separation can be shorter if one uses a multi-state instead of two-state fit with enough statistics. However, since the number of fitting parameters in n -state fit grows as n^2 , such a fitting will become infeasible for too large n . Second, note that the resolution required for transverse proton structures is not affected by the Lorentz boost, one may use a coarse lattice in the transverse directions, $a_\perp = 0.1$ fm. The required box size is then $L_\perp \times L_\perp^2 \times T = 150 \times 60^2 \times 150$. This asymmetric lattice can greatly reduce the resources needed for large momentum since the transverse box size is fixed. However, generating configurations and renormalization on such a lattice might bring new problems and should be further studied.

In the near future, exascale supercomputers may help to reach higher momentum, as large as 5 GeV for the proton, and improve the precision of LaMET calculations. Further theoretical developments and new ideas on the technique and algorithms are also needed to overcome the simulation difficulties.

B. Non-Singlet PDFs

In this subsection, we review current status of lattice calculations of flavor non-singlet (isovector) PDFs in the proton and pion. The non-singlet case has the advantage that the mixing with gluons as well as the lattice calculation of disconnected diagrams can be avoided, thus greatly reducing the computational challenge. It is the most extensively studied parton observable with LaMET so far.

1. Proton

The pioneering lattice studies for the isovector quark PDF in the proton were carried out in (Alexandrou *et al.*, 2015; Lin *et al.*, 2015). These are proof-of-principle studies as the renormalization of quasi-PDFs was not well understood at that time. Nevertheless, their results encouraged the follow-up theoretical works on LaMET, including a proper renormalization and matching suitable for lattice implementations.

Certain lattice artifacts have also been studied. For example, although there is no power-divergent mixing for the quasi-PDF operators on the lattice, additional operator mixings that are not seen in the continuum can still occur if a non-chiral lattice fermion such as the Wilson-type fermion is used. In (Chen *et al.*, 2019a; Constantinou and Panagopoulos, 2017) it was shown that at $\mathcal{O}(a^0)$ the operator for the unpolarized quark quasi-PDF, $O_{\gamma^3}(z)$, can mix with the scalar operator $O_1(z)$, whereas $O_{\gamma^*}(z)$ does not. To reduce the systematic uncertainty from such mixing, $\Gamma = \gamma^3$ has been used since then for lattice calculations of the unpolarized quark PDF, e.g. in (Alexandrou *et al.*, 2017b; Chen *et al.*, 2018; Green *et al.*, 2018). Similarly, for helicity and transversity cases, one should choose $\Gamma = \gamma^3\gamma^5$ and $\Gamma = i\sigma^{3\pm} = \gamma^3\gamma^\pm$, respectively, in order to avoid the mixing. It should be noted that at $\mathcal{O}(a)$ all $\tilde{O}_i(z)$'s can mix with others (Chen *et al.*, 2019a). Nevertheless, a fine lattice spacing can reduce these effects.

In (Alexandrou *et al.*, 2017b; Chen *et al.*, 2018; Green *et al.*, 2018), the nonperturbative renormalization (NPR) of the quasi-PDFs was studied in the RI/MOM scheme (Martinelli *et al.*, 1995). This scheme has several advantages: The lattice regularization scheme can be converted to $\overline{\text{MS}}$ scheme through RI/MOM renormalization condition, the computation cost is affordable, the systematic errors can be reduced or quantified more easily, etc. The works before 2018 did not include NPR and the systematics were not accurately quantified. The later works have implemented the RI/MOM scheme and the corresponding perturbative matching (Constantinou and Panagopoulos, 2017; Lin *et al.*, 2020; Stewart and Zhao, 2018). The coordinate-space method is also developed in parallel in (Bhat *et al.*, 2020; Cichy *et al.*, 2019; Joó *et al.*, 2019a, 2020; Orginos *et al.*, 2017). In Figs. 16 and 17, we select some most recent lattice results. ETMC published the proton unpolarized, helicity and transversity PDFs with $P^z = 1.4$ GeV at physical pion mass (Alexandrou *et al.*, 2018a,b), and LP³ published the proton helicity PDF with unprecedented momentum $P^z = 3.0$ GeV at physical pion mass (Lin *et al.*, 2018a). Recently, calculations on fine lattices (Alexandrou *et al.*, 2020c; Fan *et al.*, 2020a) and an extrapolation to the continuum limit (Alexandrou *et al.*, 2020c) have become available. The finite volume effects, which was first studied in a

model (Briceño *et al.*, 2018), have also been investigated on the lattice lately in (Lin and Zhang, 2019), where no sizeable volume-dependence was observed at $P^z = 1.3$ and 2.6 GeV.

The PDFs extracted from LaMET can be useful for phenomenology by providing input in kinematic regions that are difficult to measure in experiments. It has attracted attention from global fit community (Bringewatt *et al.*, 2020; Constantinou *et al.*, 2020; Hobbs *et al.*, 2019; Lin *et al.*, 2018c). For example, it has been found that in the large- x region of unpolarized PDF the lattice result will lead to significant improvement on global fit result if it reaches an accuracy of about 10% (Lin *et al.*, 2018c). The sea quark asymmetry (Geesaman and Reimer, 2019) is also possible to be investigated now directly on lattice. For the transversity PDF, due to the difficulty of measurement in experiment, lattice results can already have impact on improving global fit and even making predictions. In addition to the isovector cases, calculations of the strange and charm unpolarized distributions (Zhang *et al.*, 2020c), as well as the flavor separation of light quarks in the helicity PDF (Alexandrou *et al.*, 2020a), have also been carried out recently. From early exploratory results showing qualitative behavior of PDFs to the latest results which are comparable with global fits, it has come a long way in developing new techniques (momentum smearing, renormalization, matching, etc.) and the computation resources have been steadily increased over time. The systematic uncertainties in the lattice calculation of PDFs have been thoroughly investigated by the ETMC (Alexandrou *et al.*, 2019b). Further studies on systematics such as the discretization effects and finite volume effects on various lattice ensembles are still necessary. In the future, lattice QCD is expected to make a significant impact on nucleon structure.

To conclude this subsection, we would like to mention that there are also lattice studies of the isovector PDF of other baryons, Δ^+ to be more concrete, using LaMET (Chai *et al.*, 2020).

2. Pion

The pion valence quark distribution has been extracted from various Drell-Yan data for pion-nucleon/pion-nucleus scattering, while theoretical predictions do not yield consistent results with the experimental extraction, especially in large- x region (Holt and Roberts, 2010). LaMET calculations will be able to shed valuable light on how to resolve this disagreement, provided that all systematics are well under control.

In principle, calculating the pion valence PDF is easier than the proton PDF. First, the pion state is easier to produce and the quark contractions are fewer. Second, the energy gap between the first excited and ground

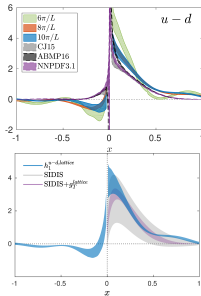


FIG. 16: Proton isovector quark PDF (Alexandrou *et al.*, 2018a,b): The unpolarized PDF with P^z from 0.82 to 1.4 GeV and the transversity PDF with $P^z = 1.4$ GeV are in upper and lower figures. CJ15 (Accardi *et al.*, 2016a), ABMP16 (Alekhin *et al.*, 2017), and NNPDF3.1 (Ball *et al.*, 2017) are global fits. SIDIS is global fit and SIDIS+ g_1^{lattice} is global fit with lattice constraint on tensor charge g_1^{lattice} (Lin *et al.*, 2018b).

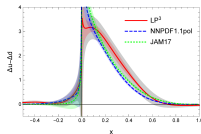


FIG. 17: The proton isovector quark helicity PDF ($P^z = 3.0$ GeV) (Lin *et al.*, 2018a) with red band for statistic error and grey band for statistic and systematic errors. NNPDF1.1pol (Nocera *et al.*, 2014) and JAM17 (Ehmer *et al.*, 2017) are global fits.

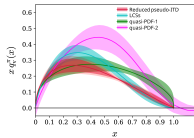


FIG. 18: Pion valence quark PDFs in various approach: Compare the results of pseudo-PDF [Reduced pseudo-ITD (Joo *et al.*, 2019b)], quasi-PDF [quasi-PDF-1 (Izubuchi *et al.*, 2019) and quasi-PDF-2 (Zhang *et al.*, 2019a)], and the current-current correlator approach [LCSs (Stefan *et al.*, 2019)].

state of the pion is much bigger than the energy gap of the proton. Therefore, the excited-state contamination is easier to control. The simulation was first performed in (Zhang *et al.*, 2019a) with the same lattice setup and procedure used in exploratory studies of the proton PDF. A more thorough study on the pion valence quark PDF was done by the lattice QCD group of BNL (Izubuchi *et al.*, 2019). It is worth pointing out that the excited state contamination was thoroughly studied using multi-state fits, with the ground and first excited states both agreeing with the expected dispersion relations, indicating that the excited contamination is well under control. The comparison of the lattice results from quasi-PDF, pseudo-PDF and current-current correlator approach are shown in Fig. 18. Note that the LP^3 (Zhang *et al.*, 2019a) result was obtained using Fourier transformation and inversion of factorization formula, while other three groups used parameterization models to fit the lattice data. More dedicated effort is needed to reduce the errors, and a meaningful comparison between different operators and analysis methods should be made.

For other mesons, we would like to mention that there is a study of kaon valence quark PDF using MILC configurations (Lin *et al.*, 2020).

C. Gluon Helicity and Other Collinear Parton Properties

In this subsection, we summarize the applications of LaMET to other collinear parton observables, including the gluon helicity, the gluon PDFs, meson DAs and GPDs.

1. Total gluon helicity

The total gluon helicity ΔG is a key component in understanding the proton spin structure. It has been intensively explored at RHIC and will be dedicatedly pursued at EIC in the future. However, a theoretical lattice calculation of ΔG had not been possible until the proposal of LaMET.

The first such effort was made by χ QCD collaboration in (Yang *et al.*, 2017). The calculation was carried out with valence overlap fermions on $2+1$ flavor domain-wall fermion gauge configurations, using ensembles with multiple lattice spacings and volumes including one with physical pion mass. The authors simulated proton matrix elements of the free-field operator $(\vec{E} \times \vec{A})^3$ in the Coulomb gauge at various momenta, and then converted them to the $\overline{\text{MS}}$ scheme with one-loop lattice perturbation theory. The $\overline{\text{MS}}$ matrix elements at each lattice momentum are shown in Fig. 19. Though a LaMET matching is necessary to match the results to the physical gluon helicity, the authors did not apply it due to the concern of perturbative convergence of the matching coefficient (Ji *et al.*, 2015c). Instead, as the $\overline{\text{MS}}$ matrix elements show rather mild momentum dependence up to the maximum momentum ~ 1.5 GeV, they extrapolated the results to infinite momentum, as well as physical pion mass and continuum limits, with a model motivated by chiral EFT. Their final result is $\Delta G(\mu^2 = 10 \text{ GeV}^2) = 0.251(47)(16)$, or $50(9)(3)\%$ of the total proton spin, which agrees with the truncated moment of $\Delta g(x)$ (de Florian *et al.*, 2014; Nocera *et al.*, 2014) within uncertainties.

Despite such progress, one should be cautious that this calculation still needs further improvements in the future. Among others, the most important ones are simulations at larger proton momentum, performing an NPR and investigating perturbative convergence of LaMET matching and its implementation.

2. Gluon PDF

The gluon PDF is of great interest not only for precision physics at LHC, but also for understanding the gluonic structure of the proton and nuclei—as well as the small- x dynamics—at the future EIC. With the recent progress on the renormalization and matching for gluon quasi-PDFs (Li *et al.*, 2019; Wang *et al.*, 2019b; Wang and Zhao, 2018; Wang *et al.*, 2018; Zhang *et al.*, 2019b) or the coordinate-space “pseudo distributions” (Balitsky *et al.*, 2019), a systematic lattice calculation of the gluon PDFs can be carried out in principle.

Before the above theoretical developments, an exploratory lattice study of the proton and pion unpolarized gluon PDFs were carried out in (Fan *et al.*, 2018).

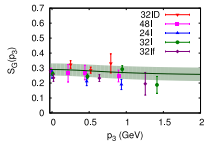


FIG. 19: Total gluon helicity (Yang *et al.*, 2017): The results are extrapolated to the physical pion mass and continuum as a function of the proton momentum p_3 on all the five ensembles indicated by different colors of the data points.

The authors calculated quasi gluon LF correlations and compared them to the LF correlations for the gluon PDFs. Later on, based on the multiplicative renormalizability of certain choice of the quasi gluon LF correlator (Zhang *et al.*, 2019b), the same authors used the ratio scheme (Balitsky *et al.*, 2019) in coordinate space to renormalize the lattice matrix elements, and fitted the proton unpolarized gluon PDF with a simple two-parameter model (Fan *et al.*, 2020b). Although the results show agreement with the global analyses in the large- x region, the systematics from the model-dependence of the fit remains to be quantified for a controlled calculation of the gluon PDF.

3. DA

According to Sec. IV B, LaMET can be readily applied to calculating DAs, and the lattice resource needed is expected to be cheaper than that for PDFs since there is one less external state, which reduces the number of contractions for the quark propagators. So far there are a few exploratory investigations on meson DAs, in particular, on pion (Zhang *et al.*, 2017) and kaon DAs (Zhang *et al.*, 2019c). The lattice calculations of pion (Zhang *et al.*, 2017) and kaon DAs (Zhang *et al.*, 2019c) were first explored without the NPR and the corresponding matching. Recently, the pion and kaon DAs from the RI/MOM scheme analysis are extrapolated to the continuum limit (Zhang *et al.*, 2020b), where the authors eventually adopted a two-parameter model to fit the final result. The above results are shown in Fig. 20. Apart from LaMET, the current-current correlation methods (Braun and Müller, 2008; Braun *et al.*, 2015; Detmold and Lin, 2006) have also made much progress on the pion DA (Bali *et al.*, 2019; Detmold *et al.*, 2020, 2018).

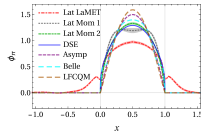


FIG. 20: Pion DA (Zhang *et al.*, 2019c): Comparison of ϕ_π (Lat LaMET) to previous determinations in the literature. Lat Mom 1 and 2 are parameterized fits to the lattice moments (Braun *et al.*, 2015); DSE is Dyson-Schwinger equation calculations (Chang *et al.*, 2013); Asymp is the asymptotic form $6x(1-x)$; Belle is a fit to the Belle data (Agar *et al.*, 2012); LFCQM is light-front constituent quark model (de Melo *et al.*, 2016).

4. GPD

As discussed in Sec. IV, the global fitting of GPDs still faces challenges from their complicated kinematic dependence and limited information from the experimental observables despite the progress made (Favart *et al.*, 2016; Kumericki *et al.*, 2016). On the other hand, previous lattice QCD method is only able to calculate the lowest few moments of the GPDs (Hagler, 2010), which is far from sufficient to reconstruct their full kinematic dependence. Applying LaMET to GPD calculations will provide important information on the GPDs, especially in kinematic regions that are not accessible in currently available experiments. In addition, on the lattice one can study the GPD dependence on one kinematic variable by fixing the others. All these will help differentiate commonly used models in GPD parameterization.

Calculating the quasi-GPDs requires more resources than quasi-PDF, but does not need further techniques in principle. Besides, the lattice renormalization factors for the quasi-PDFs can be used here, as has been argued in Sec. IV. The first lattice calculation of the pion unpolarized isovector quark GPD was carried out in (Chen *et al.*, 2019b), though the results are not yet able to differentiate models or compare to experiments. Recently, ETMC completed the first proof-of-principle calculation of the proton unpolarized and helicity GPDs (Alexandrou *et al.*, 2020d), as shown in Fig. 21, which demonstrates that it is feasible to extract the GPDs with controlled systematics on available computational resources.

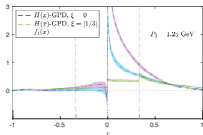


FIG. 21: Proton unpolarized isovector quark GPD (Alexandron *et al.*, 2020d) $H(x, \xi, t)$ for $t = -0.49 \text{ GeV}^2$ extracted from quasi-GPDs at $P_3 = 1.25 \text{ GeV}$, which is compared to the unpolarized PDF $f_1(x)$.

5. Higher-twist PDF

The higher-twist PDFs probe multi-parton correlations, and their contribution at $x = 0$ can shed light on the LF zero modes (Ji, 2020). As we discussed in Sec. IV, such distributions can also be calculated on the lattice with the LaMET approach.

The first attempt to calculate the isovector twist-three PDF $g_T(x)$ has been carried out by ETMC (Bhattacharya *et al.*, 2020a) using the one-loop matching coefficient they computed in (Bhattacharya *et al.*, 2020b,c). Their results show agreement with the Wandzura-Wilczek approximation (Wandzura and Wilczek, 1977), which ignores the contribution from dynamical twist-three contributions, and the Burkhardt-Cottingham sum rule (Burkhardt and Cottingham, 1970). Nevertheless, the mixing between $g_T(x)$ and other twist-three distributions was not considered, and further study is still required for an accurate matching to the light-cone PDF.

D. TMDs

With tremendous experimental focus on the TMD-PDFs for studying 3D proton structures and gluon saturation at EIC, their first-principle calculation from lattice QCD will significantly boost this direction by providing useful nonperturbative inputs for all the phenomenological analyses.

In this subsection, we discuss the status and prospects of calculating the quasi-TMDPDF and soft function with LaMET. Besides, we note that before LaMET there had already been efforts to extract information of TMDs by studying ratios of the lattice correlators (Engelhardt *et al.*, 2016; Hagler *et al.*, 2009;

Musch *et al.*, 2012, 2011; Yoon *et al.*, 2017), which has made a series of progress in the past decade. We begin with a brief review of them.

1. Pre-LaMET study — ratio of lattice correlators

By employing Lorentz covariance, the x -moments of TMDPDFs are related to the form factors of space-like staple-shaped gauge link operators, which can be directly simulated on the lattice. Although the lattice calculation of the soft function was not available during that time, ratios of the spin-dependent and the unpolarized matrix elements were formed to cancel it, thus providing useful information of different TMDPDFs. For example, the time-reversal odd TMD-PDFs can be studied with the staple-shaped gauge link operator in a transversely polarized proton state, thus helping understand properties related to single-spin asymmetry (SSA), which was measured experimentally at STAR (Adamczyk *et al.*, 2016) and COMPASS (Aghasyan *et al.*, 2017). In (Engelhardt *et al.*, 2016; Musch *et al.*, 2012), the Sivers and Boer-Mulders functions of proton and pion were studied; Other time-reversal even functions, such as the worm-gear function g_{1T} (Tangerman and Mulders, 1995), were also studied (Yoon *et al.*, 2017).

2. Quasi-TMDPDF and Collins-Soper kernel

The lattice calculation of the quasi-TMDPDF defined in Eq. (195) is straightforward. The matrix element of the staple-shaped quark Wilson line operator can be simulated the same way as the quasi-PDF case, except that the geometry of the gauge-link is different, while the calculation of Wilson loop Z_G is standard practice in lattice QCD. The more challenging part, however, is the renormalization of the quasi-TMDPDF and its matching to the $\overline{\text{MS}}$ scheme.

Using the auxiliary field theory formalism, one can argue that staple-shaped quark Wilson line operator is also multiplicatively renormalizable (Ebert *et al.*, 2020b; Green *et al.*, 2020). On a non-chiral lattice, it suffers from finite mixing with other quark bilinear operators, as was predicted by one-loop lattice perturbation theory (Constantinou *et al.*, 2019). The full mixing pattern for such operators with different Dirac matrices have been studied in the RI/MOM scheme on three quenched lattice ensembles with different spacings (Shanahan *et al.*, 2019), and a diagonalization of the mixing matrix is adopted to renormalize these operators. Meanwhile, the one-loop conversion factors that convert the RI/MOM matrix elements to the $\overline{\text{MS}}$ scheme have been calculated in continuum perturbation theory for both the $z = 0$ (Constantinou *et al.*, 2019) and $z \neq 0$ (Ebert *et al.*,

2020b) cases.

Although the soft function is still needed to fully determine the physical TMDPDF, the $\overline{\text{MS}}$ quasi-TMDPDF can already be used to extract the Collins-Soper kernel according to Eq. (210) (Ebert *et al.*, 2019a,b; Ji *et al.*, 2015b). Since the Collins-Soper kernel can be defined from both the bare TMDPDF and the soft function, it is independent of the external state and can be calculated in a pion which is the least expensive on the lattice. Up to mass corrections suppressed by the momentum in Eq. (209), this calculation also allows for using an unphysical valence pion mass, as long as the sea quark masses are physical.

With the method developed in (Ebert *et al.*, 2019a), the first exploratory lattice calculation of the Collins-Soper kernel was performed in (Shanahan *et al.*, 2020) on a quenched lattice with heavy valence pion mass $m_\pi \sim 1.2$ GeV, and the result is shown in Fig. 22. As one can see, the lattice prediction is robust for $0.1 \text{ fm} < b_\perp < 0.8 \text{ fm}$, which covers the nonperturbative region that is important for TMD evolution in global analyses. Besides, at small $b_\perp$, the perturbative calculation can serve as a calibration for estimating the systematic uncertainties, as there are power corrections of $\mathcal{O}(1/(P^2 b_\perp))$ which can only be reduced with larger P^2 . In (Zhang *et al.*, 2020a), the Collins-Soper kernel has also been extracted from a pion quasi-TMD DA, where the lattice renormalization was left out, and the result is in agreement with (Shanahan *et al.*, 2020) within errors for a wide range of $b_\perp$. With improved lattice ensembles and systematic corrections in the future, it is promising to have a precise determination of the Collins-Soper kernel for TMD phenomenology.

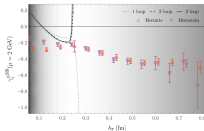


FIG. 22: The Collins-Soper kernel from the first exploratory calculation on a quenched lattice (Shanahan *et al.*, 2020). The results are obtained by using fits to the $\overline{\text{MS}}$ unsubtracted quasi-TMDPDFs with Borel and Borel+ polynomial bases. The solid and dashed lines are the perturbative predictions (Li and Zhu, 2017; Vladimirov, 2018), which is hit the Landau pole near $b_\perp \sim 0.25 \text{ fm}$. The background shading density is proportional to a naive estimate of the power corrections $1/(\bar{b}_\perp P^2) + b_\perp/L$.

3. Soft function

As the remaining piece towards physical TMDPDFs, the soft function must be calculated in lattice QCD. In particular, the reduced soft function in Eq. (204) eliminates the regulator-scheme-dependence of the off-the-light-cone quasi-TMDPDF, so its calculation alone has great physical significance. According to Secs. (V.C) and (V.D), two methods have been proposed to calculate the off-the-light-cone soft function or reduced soft function on the lattice (Ji *et al.*, 2020a), as we discuss in the following. One relies on simulating HQET on the lattice, while the other requires calculating a light-meson form factor of transversely-separated current products.

The latter method has been implemented in the first exploratory lattice calculation of the reduced soft function (Zhang *et al.*, 2020a), which includes simulations of the pion form factor in two external states with opposite large momenta, as well as the pion quasi-TMD DA. The results for the reduced soft function, which are obtained with tree-level matching and omission of lattice renormalization, are shown in Fig. 23. As one can see, they agree with the perturbative prediction for small $b_\perp$ within errors, as expected, and start to deviate when $b_\perp$ becomes large. Since the quasi-TMD DA depends on the momentum P^2 , the stability of results at different P^2 suggests the validity of Eq. (250). In the future, larger statistics and improved systematics in both lattice and perturbative matching will be necessary to achieve a precision calculation of this quantity.

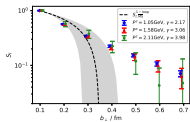


FIG. 23: The reduced soft factor as a function of $b_\perp$ extracted from the light-meson form factor in Sec. V (Zhang *et al.*, 2020a). The results are obtained with quasi-TMD DAs at different pion momenta P^2 , with perturbative matching and power corrections ignored. The dashed line is one-loop prediction in perturbation theory, which hits the Landau pole at $b_\perp \sim 0.3 \text{ fm}$, and the grey band is the error by varying μ by a factor of $1/\sqrt{2}$ and $\sqrt{2}$.

VII. CONCLUSION AND OUTLOOK

Since Feynman proposed the parton model more than fifty years ago, our understanding of the partonic structure of the proton has been greatly advanced. On one hand, a number of high-energy experiments carried out at facilities worldwide including SLAC, DESY, CERN, Fermi Lab, JLab, BNL, etc. allowed us to probe various aspects of hadronic structures at different energies and polarizations. On the other hand, many parton observables have been proposed in parallel that provide a multi-dimensional description of the proton structure, including the collinear PDFs, TMDPDFs, GPDs, parton DAs, LFWFs and so on.

Although QCD factorization theorems with RG improvement allow us to extract these parton observables through their connection to experimental observables, it is highly desirable to predict them from *ab initio* calculations such as lattice QCD. Developments along this line have been rather slow due to difficulties in simulating real-time dynamics. The situation, however, has changed since the proposal of LaMET a few years ago, which provides a systematically improvable method to calculate parton physics from first principles.

In this paper, we give an overview of LaMET formalism and its applications to observables which can be accessed in lattice QCD and other non-perturbative methods. By investigating the frame dependence of the structure of bound state hadrons, we explain how the IMF physics or parton physics naturally arises as an EFT description of the proton structure. Such an EFT description is most naturally formulated in SCET and LFQ, but practical non-perturbative calculations of the proton matrix elements have been difficult. LaMET in effect provides what is needed to realize LFQ. This is achieved by forming appropriate quasi parton observables in a large momentum state and match them to the true parton observables on the LF through factorization. In the case of PDFs, the former corresponds to finite-momentum distributions whose running is controlled by the momentum RGE, whereas the latter corresponds to IMF PDFs whose running is controlled by the usual RGE. It should be pointed out that LaMET is a very general framework which can be applied to large-momentum physical quantities calculated with any non-perturbative methods, either Euclidean (with imaginary time) or Minkowskian (with real time). Moreover, given a large momentum state, the same parton physics can be determined from different quasi observables that form a universality class.

We then present how to calculate the parton observables in practice, with a particular focus on the collinear PDFs, GPDs, DAs, TMDPDFs and LFWFs. We also discuss the proton spin structure and show how the partonic contributions to proton spin can be obtained following the same approach. We finally summarize the lattice studies carried out so far with LaMET which, on

one hand, demonstrate that LaMET is a promising approach to compute partonic structures of the proton, and on the other hand, clearly indicate that a lot of improvements are still required to reach such an accuracy that the lattice results can have considerable impact on phenomenology.

We complete this review with a few comments on improvements of lattice calculations for the future. We recommend (Alexandrou *et al.*, 2019b) for more systematic discussion on some of the issues, for example, the continuum, infinite volume, and physical pion mass limits.

- **Large hadron momentum.** Since the future of LaMET lies in larger momenta which naturally require smaller lattice spacings, it will be critical to address the challenges from using large momenta and small spacings for exa-scale computations, such as the excited state contamination or topological charge freezing problem.
- **Renormalization.** As discussed in Sec. III.D, the mass renormalization of Wilson line operators is favored for it is gauge invariant and does not introduce extra higher-twist effects or large statistical errors at long distance. However, its matching to the $\overline{\text{MS}}$ scheme, especially the renormalon ambiguities, still needs to be resolved for a full systematic application. Moreover, alternative schemes that include the above features are also highly desirable.
- **Higher-order perturbative matching.** In current LaMET calculations, one-loop perturbative matching has brought considerable corrections. Higher-order matching kernels will be necessary to control the systematics from this procedure.
- **Power corrections.** They are important if the hadron momentum is not very large or when x is close to 0 or 1. Little progress has been made toward a model-independent determination of the power corrections so far. One contingent strategy is to extrapolate to $P^2 \rightarrow \infty$ limit after implementing matching and target-mass corrections, but the ultimate solution relies on the lattice calculation of higher-twist distributions that has been discussed in Sec. IV.C.

The above discussion of systematics is generic and applies to all quasi-observables. The rich theoretical developments in the past years have paved the way for calculating a wide range of parton observables using LaMET. With the rapid increase in computing resources and progress in developing new techniques and algorithms, we expect to see the above systematics to be kept under control step by step in the future. That would be important in establishing LaMET as a systematic approach to computing parton physics, and making lattice calculations play a crucial role in the EIC era.

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APPENDIX A: ACRONYMS, ABBREVIATIONS AND TERMINOLOGIES

Here we list some acronyms, abbreviations and terminologies used throughout this review:

AM	angular momentum
BFKL	Balitsky-Fadin-Kuraev-Lipatov
BPHZ	Bogoliubov-Parasiuk-Hepp-Zimmermann
BRST	Becchi-Rouet-Stora-Tyutin
DA	distribution amplitude
DGLAP	Dokshitzer-Gribov-Lipatov-Altarelli-Parisi
DIS	deep-inelastic scattering
DR	dimensional regularization
DVCS	deeply-virtual Compton scattering
DVMP	deeply-virtual meson production
DY	Drell-Yan
EFT	effective field theory
EIC	Electron-Ion Collider
EOM	equation of motion
ERBL	Efremov-Radyushkin-Brodsky-Lepage
GCPO	generalized collinear parton observable
GPD	generalized parton distribution
GTMD	generalized transverse-momentum-dependent distribution
HQET	heavy-quark effective theory
IMF	infinite-momentum frame
IR	infrared
LaMET	large momentum effective theory
LC	light-cone
LF	light-front
LFWF	light-front wave function
$\overline{\text{MS}}$	modified minimal subtraction
NPR	non-perturbative renormalization
OAM	orbital angular momentum
OPE	operator product expansion
PDF	parton distribution function
QCD	quantum chromodynamics
QED	quantum electrodynamics
QFT	quantum field theory
RGE	renormalization group equation
RI/MOM	regularization-independent momentum subtraction
SCET	soft-collinear effective theory
SIDIS	semi-inclusive deep-inelastic scattering
TMD	transverse-momentum-dependent
UV	ultraviolet

Parton model: a model proposed by R. Feynman in which hadrons are viewed as a collection of point-like quasi-free partons.

Parton distribution function: a probability function describing how the longitudinal momentum is distributed among the partons (quarks and gluons) in a hadron.

Factorization theorem: a theorem that separates hadronic observables into process-dependent short-

distance partonic observables and universal long-distance functions characterizing the hadron structure.

Light-front quantization: a quantization program that is carried out at equal light-front time and yields a relativistic description of QCD bound states in terms of light-front wave functions.

Bjorken x_B : The variable proposed by J. D. Bjorken to characterize the kinematics in DIS. Its definition is given above Eq. (1).

Scaling: The behavior that an observable is independent of the scale at which it is probed.

Effective field theory: a theory framework that describes physical phenomena at a given length scale using only active degrees of freedom at that scale, while integrating out degrees of freedom at other length scales.

Renormalization group equation: an equation that describes how a physical system can be viewed and interpreted at different scales.

BQET: an effective field theory obtained from QCD by taking the infinite heavy quark mass limit.

Gauge link or Wilson line: a nonlocal quantity constructed as exponentials of integrals of gauge fields along a given path, used to connect fields at different spacetime points to maintain gauge invariance.

Compton amplitude: the quantum amplitude for scattering of a (virtual) photon by the proton.

Auxiliary field approach: an approach in which the nonlocal gauge link can be replaced by the two-point function of the auxiliary field.

Matching: a procedure used to relate full theory operators to effective field theory operators.

Nonsinglet: a combination accounting for the difference between quark distributions, e.g., the isovector combination $u - d$ discussed extensively in the context of this review.

Universality class: a collection of operators that flow into the same fixed point under momentum renormalization group running.

Quasi-light-front correlations: spatial correlations defining the finite momentum distributions.

Collinear divergence: divergence in a Feynman diagram when loop momentum of the internal line is collinear to that of the external massless particle.

Two-particle-irreducible diagram: a Feynman diagram that cannot be divided into disconnected parts by cutting two internal lines.

Wilson fermion: a way to discretize the QCD fermion action on the lattice, which breaks down the chiral symmetry.

Generalized parton distribution: generalization of PDFs to non-forward kinematics, i.e., the initial and final states have different momenta.

Skewness: defined to characterize the longitudinal momentum transfer in GPDs.

Distribution amplitude: transition matrix element between vacuum and hadron state, representing the probability amplitude of finding a given Fock state in the hadron.

Twist: defined as dimension - spin of the operator. Leading-twist (higher-twist) denotes the leading (nonleading) power behavior in the quantity under investigation.

Zero-mode: the degrees of freedom with zero longitudinal momentum in LFP.

Transverse momentum dependent(TMD) PDF: defined in Eq. (220), the distribution function of both longitudinal and transverse momentum for partons.

Staple-shaped gauge-link: the pair of gauge-links separated along transverse directions that appear in the definition of TMD-PDFs, they are defined in Eq. (166), Eq. (173) and Eq. (196).

Rapidity divergence: the divergence of TMDPDF and soft functions due to the presence of infinite rapidity scale introduced by the infinite-long gauge-links.

Light-cone regulator: regulators that regulates the rapidity divergence. First appeared in the paragraph of Eq. (168).

On-light-cone: rapidity regulator that maintain the presence of light-like separations in the gauge-link.

Off-light-cone: rapidity regulator that makes the separations of the gauge-link non-light-like.

Soft function: functions that capture the factorable soft radiations of TMDPDF. Defined in Eq. (172) for on-light-cone and Eq. (202) for off-light-cone.

Collins Soper kernel: the kernel for rapidity evolution of TMDPDF, see Eq. (191).

Quasi-TMDPDF: defined in Eq. (195), similar to TMDPDF but with light-like separations replaced by space-like ones.

Pinch-pole singularity: the divergence due to infinite long gauge-link pair in the quasi-TMDPDF. Can be subtracted out by the factor Z_E , see discussion below Eq. (198).

Off-light-cone soft function: soft function using off-light-cone regulator. Defined in Eq. (202) and Eq. (224). Required for matching quasi-TMDPDF to TMDPDF.

Reduced soft function: the rapidity independent part of off-light-cone soft function, see Eq. (204).

Light-Front wave function: the wave function for hadron state in light-front quantization, expanded in the Free-Fock state.

Reduced diagram: the diagram showing the power-leading region of IR divergences. All the IR safe propagators are shrunk to blobs.

Momentum smearing: a lattice technique to increase the overlap of the field and nonzero-momentum state.

Non-singlet: transforms under the fundamental representation of $SU(N_f)$ with N_f the quark flavor number.

APPENDIX B: CONVENTIONS

We use the following convention for the metric tensor

$$g^{\mu\nu} = \text{diag}(1, -1, -1, -1). \quad (253)$$

In ordinary coordinates, a generic four-vector is denoted as $v^\mu = (v^0, v^x, v^y, v^z)$ or $v^\mu = (v^0, \vec{v}_\perp, v^z)$. For example, the spacelike and timelike direction vector are written as $n_z = (0, 0, 0, 1)$ and $n_t = (1, 0, 0, 0)$, respectively. In light-cone coordinates $\xi^\pm = \frac{1}{\sqrt{2}}(\xi^0 \pm \xi^z)$, a vector is denoted as $v^\mu = (v^+, v^-, \vec{v}_\perp)$.

The hadron state $|P\rangle$ is normalized as

$$\langle P'|P\rangle = (2\pi)^3 2P^0 \delta^{(3)}(\vec{P} - \vec{P}'). \quad (254)$$

The covariant derivative and the Wilson line gauge link in the fundamental representation are defined as

$$D^\mu \psi = (\partial^\mu + igA^\mu)\psi = (\partial^\mu + igt^a A_a^\mu)\psi, \quad (255)$$

and

$$W(x_2, x_1) = \exp \left[-ig \int_0^1 dt (x_2 - x_1)_\mu A^\mu(x_1 + (x_2 - x_1)t) \right], \quad (256)$$

The ones in the adjoint representation are completely analogous.

We use $O_T(s)$ to generically denote an operator defining the corresponding (quasi) parton observable, where s can be a lightlike (for parton observables) or spacelike (for quasi parton observables) separation, and Γ is a Dirac structure. The momentum fraction in a quasi-observable is denoted as y , while that in the usual parton observable is denoted as x .

The lightcone operator that defines the quark parton observable is

$$O_T(\lambda n) = \bar{\psi}(0) \Gamma W(0, \lambda n) \psi(\lambda n) \quad (257)$$

with Γ denoting a Dirac matrix. If we take $\Gamma = \not{n} = \gamma^+$, the unpolarized quark PDF is then given by

$$q(x) = \frac{1}{2P^+} \int \frac{d\lambda}{2\pi} e^{ix\lambda} \langle P | O_{\gamma^+}(\lambda n) | P \rangle \quad (258)$$

with $n^\mu = 1/\sqrt{2}(1/P^+, 0, 0, -1/P^+)$.

Accordingly, the quark quasi-observable is defined by

$$O_T(z) = \bar{\psi}(zn_z/2) \Gamma W(zn_z/2, -zn_z/2) \psi(-zn_z/2). \quad (259)$$

If we choose $\Gamma = \gamma^t$, the unpolarized quark quasi-PDF is then defined as

$$\bar{q}(y) = \frac{1}{2P^0} \int \frac{d\lambda}{2\pi} e^{iy\lambda} \langle P | O_{\gamma^t}(z) | P \rangle \quad (260)$$

with the quasi light-cone distance $\lambda = zP^z$.

The staple-shaped gauge link required for the TMD-PDFs is defined as:

$$W_n(\lambda n/2 + \vec{b}_\perp) = W_n^\dagger(\lambda n/2 + \vec{b}_\perp) W_\perp W_n(-\lambda n/2), \quad (261)$$

where

$$W_n(\xi) = W(\xi + \infty n, \xi). \quad (262)$$

The un-subtracted unpolarized quark TMDPDF is then defined as:

$$f(x, \vec{k}_\perp, \mu, \delta^-/P^+) = \frac{1}{2P^+} \int \frac{d\lambda}{2\pi} \frac{d^2 \vec{b}_\perp}{(2\pi)^2} e^{-i\lambda x + i\vec{k}_\perp \cdot \vec{b}_\perp} \times \langle P | \bar{\psi}(\lambda n/2 + \vec{b}_\perp) \not{n} W_n(\lambda n/2 + \vec{b}_\perp) \psi(-\lambda n/2) | P \rangle, \quad (263)$$

and the TMD soft function for DY process is defined as:

$$S(b_\perp, \mu, \delta^+, \delta^-) = \frac{\text{Tr}(0 | \bar{\mathcal{T}} W_p(\vec{b}_\perp) |_\delta W_n^\dagger(\vec{b}_\perp) |_\delta \mathcal{T} W_n(0) |_\delta W_p^\dagger(0) |_\delta | 0 \rangle}{N_c} = \frac{\text{Tr}(0 | W_n(\vec{b}_\perp) |_\delta W_p^\dagger(\vec{b}_\perp) |_\delta | 0 \rangle}{N_c}, \quad (264)$$

where $|_\delta$ denotes the rapidity regulator for the gauge links involved. In terms of these, the physical scheme independent TMDPDF is defined as:

$$f^{\text{TMD}}(x, b_\perp, \mu, \zeta) = \lim_{\delta^- \rightarrow 0} \frac{f(x, b_\perp, \mu, \delta^-/P^+)}{\sqrt{S(b_\perp, \mu, \delta^- e^{2b_\perp}, \delta^-)}}, \quad (265)$$

where $\zeta \equiv 2(xP^+)^2 e^{2b_\perp}$ is the rapidity scale.

The staple-shaped gauge link for the quasi-TMDPDF is defined as:

$$W_z(\frac{\lambda n_z}{2} + \vec{b}_\perp; L) = W_z^\dagger(\xi; L) W_\perp W_z(-\xi^z n_z; L), \quad (266)$$

where

$$W_2(\xi) = W(\xi + (L - \xi^2)n_z, \xi) . \quad (267)$$

The quasi-TMDPDF is then defined using $W_2(\frac{\lambda n}{2} + \vec{b}_\perp; L)$ in exactly the same way as that for the unsubtracted TMDPDF:

$$\tilde{f}(\lambda, b_\perp, \mu, \zeta_c) = \lim_{L \rightarrow \infty} \frac{\langle P | \bar{\psi} \left(\frac{\lambda n}{2} + \vec{b}_\perp \right) \gamma^2 W_2 \left(\frac{\lambda n}{2} + \vec{b}_\perp; L \right) \psi \left(-\frac{\lambda n}{2} \right) | P \rangle}{\sqrt{Z_E(2L, \vec{b}_\perp, \mu)}} , \quad (268)$$

where $Z_E(2L, \vec{b}_\perp, \mu)$ is a flat rectangular Euclidean Wilson-loop along the n_z direction with length $2L$ and width $b_\perp$:

$$Z_E(2L, \vec{b}_\perp, \mu) = \frac{1}{N_c} \text{Tr} \langle 0 | W_\perp W_2(\vec{b}_\perp; 2L) | 0 \rangle . \quad (269)$$

The staple-shaped operators for LFWFs and quasi-LFWFs are the same as those for TMD-PDFs and quasi-TMDPDFs, and can be found in Sec. V.D.

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Moments of parton distribution functions of any order from lattice QCD

Andrea Shindler*

*Institute for Theoretical Particle Physics and Cosmology, TTK, RWTH Aachen University
Nuclear Science Division, Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA and
Department of Physics, University of California, Berkeley, California 94720, USA*

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We describe a procedure to determine moments of parton distribution functions of any order in lattice quantum chromodynamics (QCD). The procedure is based on the gradient flow for fermion and gauge fields. The flowed matrix elements of twist-2 operators renormalize multiplicatively, and the matching with the physical matrix elements can be obtained using continuum symmetries and the irreducible representations of Euclidean 4-dimensional rotations. We calculate the matching coefficients at one-loop in perturbation theory for moments of any order in the flavor nonsinglet case. We also give specific examples of operators that could be used in lattice QCD computations. It turns out that it is possible to choose operators with identical Lorentz indices and still have a multiplicative matching. One can thus use twist-2 operators exclusively with temporal indices, thus substantially improving the signal-to-noise ratio in the computation of the hadronic matrix elements.

Introduction. Quantum chromodynamics (QCD) describes quark-gluon interactions over a wide energy range, posing nonperturbative challenges at lower energies. Hadron structure studies depend on parton distribution functions (PDFs), with PDF determinations relying on precise QCD calculations, influencing investigations at the LHC and future projects (EIC, HL-LHC, LHeC). The quest for new physics hinges on accurate experimental-theoretical comparisons, emphasizing PDF precision.

QCD factorization in collider processes, beyond PDFs, involves generalized parton distributions (GPDs) and transverse-momentum-dependent distributions (TMDs). While not directly measurable, these distribution functions can be determined by factorizing the process's cross section into hard and soft components.

In deep inelastic scattering (DIS), a key role is played by the structure functions, $F_k(x, Q^2)$ ($k = 1, 2, 3$), which depend on $Q^2 = -q^2$, where q is the 4-momentum transferred to the nucleon, and $x = Q^2/2(pq)$ represents the fraction of nucleon momentum carried by the parton. Here, p denotes the nucleon momentum. For Q^2 much larger than the nucleon mass, the factorization theorem [1] relates the structure functions to the convolution of the scale-dependent parton distribution functions $f_a(x, \mu^2)$ with the Wilson coefficients $C_k^a(Q^2/\mu^2)$ (summed over quark flavors and gluons, a). The Wilson coefficients are calculable in perturbation theory, and the evolution of the PDFs in μ is governed by the Dokshitzer-Gribov-Lipatov-Altarelli-Parisi (DGLAP) equations [2–5]. The DGLAP equations can be solved once we specify an initial condition at a certain scale $\mu = \mu_0$. The evolution of the PDFs is calculable in perturbation theory, but the precision of experimental data always necessitates higher-order calculations. The standard choice for renormalization and factorization scheme is the $\overline{\text{MS}}$ scheme. PDFs can be determined experimentally by analyzing a set of hard-scattering processes involving nucleons. There

is a long-standing community effort to determine PDFs using well-defined parametrizations for their x -dependence and starting the evolution around $\mu_0^2 = 1 - 4 \text{ GeV}^2$.

Lattice QCD can offer a theoretical input for the determination of the PDFs and their evolution. The direct calculation of PDFs using lattice QCD poses particular challenges due to the Euclidean geometry and the light-cone dominance of the kinematics. The connection between PDFs and hadronic matrix elements, which are calculable in lattice QCD, is established through the moments of the PDFs, denoted as $\langle x^n \rangle$ [6, 7]. Lattice QCD calculations of the moments of the PDFs, pioneered in Refs. [8–10], provide, in principle, a means for the complete reconstruction of the PDFs. This possibility has remained impractical due to the theoretical and numerical challenges associated with computing high moments. Alternative approaches have been developed to determine the x -dependence of the PDFs, the heavy-quarks OPE [11, 12], the quasi-PDF [13], the pseudo-PDF [14, 15], the OPE-based method of [16], the current-current approach [17, 18], and the hadron tensor method of [19]. It is worth noticing that these approaches allow in principle an indirect determination of the moments of PDF of nucleons [20, 21] and pions [22, 23].

Different high-energy scatterings are associated with different PDFs: unpolarized, helicity, and transversity. Certain kinematical regions, as well as specific PDFs, pose challenges for extraction from experimental data. This emphasizes the significance of reconstructing the PDFs from lattice QCD simulation data.

In this work, we revisit the possibility of calculating moments of PDFs, which, if successful, would provide an important and complementary approach to determine distribution functions from QCD. We describe a method that addresses both the theoretical and numerical challenges faced in the past, which hindered the calculation of moments of any order from lattice QCD.

Twist-2 operators. QCD factorization relates the moments of the structure functions and the moments of the parton distribution functions. Taking the example of F_1 ,

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* shindler@physik.rwth-aachen.de

the relation is given by e.g.

$$\int_0^1 dx x^{n-1} F_1(x, Q^2) = \sum_a C_{1,n}^a(Q^2/\mu^2) A_n^{a/h}(\mu) \quad (1)$$

where $A_n^{a/h}(\mu)$ represents the reduced matrix element of certain local operators, denoted by $O_{a,n}$, between the hadron states, denoted as h , and $C_{1,n}^a(Q^2/\mu^2)$ are the corresponding Wilson coefficients, calculable in perturbation theory. In the context of DIS kinematics, the leading contribution, given in Eq. (1), arises from local operators with the lowest twist, $\tau = d_O - n = 2$, where d_O is the physical dimension of the local operator and n is its spin. In the parton model the reduced matrix element is related to the moments of the PDFs, $A_n^{a/h} = \langle x^{n-1} \rangle_{a/h}$ [6, 7].

The form of the local operators O_n can be constrained by requiring that they belong to irreducible representations of the Lorentz group. This guarantees that the renormalization procedure does not generate mixing with operators of the same or lower dimensions. Lattice QCD simulations are performed in a Euclidean setup though, and the symmetry group in a Euclidean lattice is the hyper-cubic group $H(4)$. The reduced symmetry does not protect the operators from a complicated mixing pattern, that gets more complicated for larger n . Once the continuum limit has been performed, one recovers $O(4)$ symmetry.

The construction of irreducible representations of $O(4)$ symmetry group is standard material (see, for example, Ref. [24]). For the purpose of this work, the only relevant result is that traceless and symmetrized rank- n tensors are an irreducible representation of $O(4)$.

We restrict ourselves to unpolarized structure functions, meaning that we consider hadronic matrix elements averaged over the hadron polarization. Other moments of different distributions can be studied using the same procedure described here. We further limit our consideration to flavor non-singlet operators to avoid mixing with gluonic operators, but the same method is applicable to singlet operators.

We consider the following set of operators

$$O_n^{*s}(x) = O_{\mu_1 \dots \mu_n}^{*s}(x) = \bar{\psi}^s(x) \gamma_{(\mu_1} \overleftrightarrow{D}_{\mu_2} \dots \overleftrightarrow{D}_{\mu_n)} \psi^s(x), \quad (2)$$

with quark fields, ψ^s , of different flavors, $r \neq s$, and the covariant derivative $D_\mu = \partial_\mu + G_\mu$, where G_μ represent the gluon field. The symmetrization of the covariant derivative $\overleftrightarrow{D}_\mu = \frac{1}{2} (\overrightarrow{D}_\mu - \overleftarrow{D}_\mu)$ guarantees a definite transformation under charge conjugation.

In the continuum, traceless non-singlet twist-2 operators, with symmetrized Lorentz indices,¹ renormalize multiplicatively. For example, in the MS scheme, the bare

operator $\tilde{O}_{n,n}^{*s}$ renormalizes as follows

$$\tilde{O}_n^{*s}(x) = Z_n^{\text{MS}} \tilde{O}_{n,n}^{*s}(x), \quad (3)$$

with renormalization constant Z_n^{MS} . The intrinsic non-perturbative nature of the hadronic matrix elements suggests the use of lattice QCD. However, the breaking of the rotational group symmetry into the hypercubic group $H(4)$ adds complications to the renormalization.

On a hypercubic lattice, the Lorentz indices of the twist-2 operators must be chosen while considering the reduced hypercubic symmetry group $H(4)$. Irreducible representations of $O(4)$ generally become reducible representations of $H(4)$ inducing unwanted mixings under renormalization [8, 9, 25, 26]. The irreducible representations of $H(4)$ allow mixings with lower-dimensional operators, and, additionally, complicate the mixing with operators of the same dimensions. A notable example of this complication occurs with the operator O_3 . In the continuum, after symmetrizing over the Lorentz indices and subtracting the traces, the renormalization is multiplicative (e.g. in the MS scheme). However, on the lattice, terms like $1/a^2 \delta_{\mu\nu\rho\sigma} \cos(ap_\nu)$ are allowed by $H(4)$ symmetry and cannot be removed even after subtracting the trace [8]. This pertains to the fact that O_3 belongs to different irreducible representations of $H(4)$ depending on the choice of the Lorentz indices. The operator O_3 with all the indices equal, i.e. $\mu_1 = \mu_2 = \mu_3$, mixes with the lower-dimensional vector current, causing a power divergence proportional to $1/a^2$. For the operators with $\mu_1 \neq \mu_2 = \mu_3$, such as the irreducible representation $O_{411} - O_{433}$, the renormalization is multiplicative, up to a small mixing with a different operator of the same dimension.² We note that while O_{411} is also affected by power divergences, the subtraction $O_{411} - O_{433}$ guarantees their removal. Another choice is $\mu_1 \neq \mu_2 \neq \mu_3$. In this case, the operator belongs to another irreducible representation of $H(4)$ and renormalizes multiplicatively without introducing power divergences. However, this last example, while optimal in terms of renormalization, is far from optimal when calculating hadronic matrix elements³

$$\langle h(p) | \tilde{O}_n | h(p) \rangle = p_{\mu_1} \dots p_{\mu_n} A_n^h(\mu), \quad (4)$$

because it requires nonvanishing external momentum in at least 2 spatial directions. This substantially degrades the quality of the signal-to-noise ratio that is realized in numerical lattice QCD simulations. For $O_{411} - O_{433}$, one needs a nonvanishing external momentum in at least 1 spatial directions, and despite the subtraction of power

² The operator $O_{411} - O_{433}$ belongs to an irreducible representation of $H(4)$ that appears more than once. Therefore, a mixing between operators belonging to these equivalent representations can occur [25]. In perturbation theory the mixing seems to be numerically small.

³ In Eq. (4) for simplicity we omit the flavor indices.

¹ We denote the traceless operators with $\tilde{O}_n$.

divergences, one might hope to achieve a better signal-to-noise ratio. The operator with all the same indices has a power divergence that requires a complicated nonperturbative procedure to be subtracted and is not a viable solution. This is unfortunate because the matrix element of O_{444} , for example, does not require any external spatial momentum and should be optimal in terms of the signal-to-noise ratio.

The mixing pattern becomes more cumbersome for higher dimensional operators, $n \geq 4$. For $n = 4$, choosing all 4 indices differently results in the only 2 irreducible representations: the totally symmetric and antisymmetric representations. One then needs a nonvanishing external momentum in all 3 spatial directions to calculate the hadronic matrix elements [25]. For $n > 4$, mixings are unavoidable. The complicated renormalization pattern just described, together with the need of an external momentum to calculate the hadronic matrix elements, has, *de facto*, prevented the lattice calculation of the moments of parton distribution functions for $n > 4$ and rendered the calculations for $n = 3, 4$ challenging.

A very interesting idea to circumvent the renormalization issue, based on the recovery of the continuum rotational symmetry, was proposed in Ref. [27]. Even though the method was not further pursued numerically, it represents a first important step toward resolving the theoretical challenges encountered when calculating moments of PDFs.

Gradient flow. The gradient flow (GF) for gauge and fermion fields [28–30] provides a different way to regulate short-distance singularities of the twist-2 operators. The connection with the physical renormalized matrix elements at vanishing flow time, $t = 0$, is obtained with a short-flow-time expansion (SFTX) after performing the continuum limit of the hadronic matrix elements at a fixed physical value of the flow time t [31].

The flowed twist-2 operators, written in terms of flowed fermions, $\chi^a(x, t)$ and $\bar{\chi}^a(x, t)$, and gauge fields, $B_\mu(x, t)$, are given by

$$O_n^a(x, t) = \bar{\chi}^a(x, t) \gamma_{[\mu_1} \vec{D}_{\mu_2} \cdots \vec{D}_{\mu_n]} \chi^a(x, t), \quad (5)$$

where $\{\cdots\}$ denotes symmetrization over the included indices. These operators are particularly advantageous because they renormalize multiplicatively also on the lattice with a renormalization factor that depends only on the fermion content of the operator,

$$O_n^a(t) = Z_n O_{n,B}^a(t), \quad Z_n = Z_\chi, \quad (6)$$

where $Z_\chi^{1/2}$ is the renormalization constant of the flowed fermion fields [30]. For correlation functions containing flowed twist-2 operators, the flow-time t provides a regulator for short-distance singularities. Beside the renormalization of the bare parameters of the theory, the flowed fermion fields, and any local field at $t = 0$, correlation functions do not require any additional renormalization. For lattice QCD applications, it is convenient to define Z_χ in a scheme that is regularization independent, as then

the matching and the renormalization can be performed in the same scheme. The choice of the specific scheme is not relevant for the general discussion, but it becomes important when calculating the matching coefficients. In the calculations of the matching coefficients described below, we renormalize the twist-2 operators using the so-called ringed fields [32], defined by the $SU(N_c)$ gauge invariant and regularization independent condition

$$\left\langle \bar{\chi}_r(x, t) \vec{D} \chi_r(x, t) \right\rangle = -\frac{N_c}{(4\pi)^2 t^2}. \quad (7)$$

Imposing this condition in dimensional regularization, with $D = 4 - 2\epsilon$, leads to a finite renormalization between the ringed fields and renormalized fields in the $\overline{\text{MS}}$ scheme

$$\begin{aligned} \chi_r(x, t) &= (8\pi t)^{\epsilon/2} \chi_\chi^{1/2}(x, t) \\ \bar{\chi}_r(x, t) &= (8\pi t)^{\epsilon/2} \bar{\chi}_\chi^{1/2}(x, t). \end{aligned} \quad (8)$$

The 1-loop result for the finite renormalization is given by [32]

$$\zeta_\chi = 1 - \frac{\bar{g}^2(\mu)}{(4\pi)^2} C_F (3 \log(8\pi\mu^2 t) - \log(432)), \quad (9)$$

where $\bar{g}$ is the strong coupling renormalized in the $\overline{\text{MS}}$ scheme at the renormalization scale μ , and $C_F = N_c^2 - 1/2N_c$. This result has been extended to $\mathcal{O}(\bar{g}^4)$ in Refs. [33, 34]. Results obtained in the $\overline{\text{MS}}$ scheme can be converted to the $\overline{\text{MS}}$ scheme by replacing the renormalization scale μ with $\bar{\mu}$ related by $\log \bar{\mu} = \log \mu^2 + \gamma_E - \log 4\pi$.

$\mathcal{O}(a)$ improvement. The lattice QCD calculation of hadronic matrix elements of twist-2 operators is affected by cutoff effects.

For nonperturbatively improved clover fermions, the hadronic matrix elements of the standard twist-2 operators (with $t = 0$) are still affected by $\mathcal{O}(a)$ cutoff effects that could be removed by adopting an $\mathcal{O}(a)$ improved definition of the twist-2 operators. Excluding the case $n = 2$, where the corresponding improvement coefficients have been determined at 1-loop in perturbation theory [35], the continuum limit is reached with $\mathcal{O}(a)$ discretization effects.

If we consider flowed twist-2 operators for a generic n , the hadronic matrix elements are affected only by $\mathcal{O}(am)$ cutoff effects, where m is the quark mass. Additional improvement coefficients specific to the twist-2 operators are not needed. In the discussion of the method's applications below, we also consider ratios of flowed matrix elements where both Z_χ and the $\mathcal{O}(am)$ cutoff effects are eliminated. In summary, regardless of the lattice setup adopted, the use of flowed twist-2 operators not only provides a significant simplification in the renormalization pattern, but also improves the parametric scaling in the lattice spacing towards the continuum limit.

Matching coefficients. After performing the continuum limit of the hadronic matrix element for a fixed $t > 0$ value of the flow time, the connection with physical renormalized matrix element at $t = 0$ is obtained with a SFTX [31].

The SFTX of the twist-2 operators $O_n(x, t)$, in general contains power divergences with terms like $1/t^m \delta_{\mu_1 \mu_2}$, where $m = (d_n - d_p)/2$ depends on the dimensions of O_n , d_n , and on the dimension of the lower-dimensional operator, d_p . These terms, though, are classified using the continuum $O(4)$ symmetry and, as a result, the SFTX of the symmetrized and traceless operators, $\tilde{O}_n$, receives contributions only from the corresponding traceless operators at $t = 0$.

We renormalize the flowed fermion fields adopting ringed fields, introduced in the previous section, and define the ringed twist-2 operators

$$\tilde{O}_n^{\tau s}(x, t) = \bar{\chi}^{\tau}(x, t) \gamma_{(\mu_1} \overleftrightarrow{D}_{\mu_2} \cdots \overleftrightarrow{D}_{\mu_n}) \chi^s(x, t) - \text{terms with } \delta_{\mu_i \mu_j}, \quad (10)$$

where the subtraction guarantees that the resulting operator has vanishing traces. The SFTX of $\tilde{O}_n^{\tau s}$ is

$$\tilde{O}_n^{\tau s}(t) = c_n(t, \mu) \tilde{O}_{n, \text{MS}}^{\tau s}(\mu) + \cdots, \quad (11)$$

where we make explicit only the dependence on the flow time and the renormalization scale. The neglected terms are higher-order contributions of positive powers of t .

The matching coefficients are determined considering off-shell amputated one-particle irreducible (1PI) Green's functions containing the flowed operators $\tilde{O}_n^{\tau s}(t)$. The matching equations with massless fermions

$$\left\langle \psi^{\tau} \tilde{O}_n^{\tau s}(t) \bar{\psi}^s \right\rangle = c_n(t, \mu) \left\langle \psi^{\tau} \tilde{O}_{n, \text{MS}}^{\tau s}(t = 0, \mu) \bar{\psi}^s \right\rangle \quad (12)$$

are solved in $D = 4 - 2\epsilon$ at 1-loop in perturbation theory with

$$c_n(t, \mu) = 1 + \frac{\mathcal{B}(\mu)}{(4\pi)^2} c_n^{(1)}(t, \mu) + O(\mathcal{B}^4). \quad (13)$$

The matching coefficients $c_n(t, \mu)$ depend on the renormalized coupling and are independent of the soft scales the calculation of the matching coefficients. In the Supplemental Material [36] we describe their calculation and, based on the observation that they satisfy a renormalization group equation, we also provide a resummed expression at next-to-leading log (NLL).

The final result for the matching coefficients is

$$c_n^{(1)}(t, \mu) = C_F \left[\gamma_n \log(8\pi\mu^2 t) + B_n \right], \quad (14)$$

where $\gamma_n = 1 + 4 \sum_{j=2}^n \frac{1}{j} - \frac{2}{n(n+1)}$, and

$$\begin{aligned} B_n = & \frac{4}{n(n+1)} + 4 \frac{n-1}{n} \log 2 + \frac{2-4n^2}{n(n+1)} \gamma_E - \\ & - \frac{2}{n(n+1)} \psi(n+2) + \frac{4}{n} \psi(n+1) - 4\psi(2) - \\ & - 4 \sum_{j=2}^n \frac{1}{j(j-1)} \frac{1}{2^j} \phi(1/2, 1, j) - \log(432). \end{aligned} \quad (15)$$

In this expression $\gamma_E = 0.57721 \dots$ is the Euler's constant, $\psi(z)$ is the digamma function, and $\phi(z, s, a)$ is the Lerch transcendent defined as $\phi(z, s, a) = \sum_{k=0}^{\infty} \frac{z^k}{(k+a)^s}$.

The expression for γ_n matches the 1-loop anomalous dimension of the twist-2 operators [47] and provides a welcome check of the calculation. For $n > 2$ the finite part, B_n , in Eq. (15) provides a new result. The result with $n = 2$ was already obtained in Ref. [32] in the analysis of the energy-momentum tensor, and we reproduce their result.

The fact that the matching of the flowed matrix elements is so simple opens the possibility of computing moments of the parton distribution functions of any order. A possible operational procedure is the following: After fixing n , one constructs the symmetric and traceless operator $\tilde{O}_n^{\tau s}$. One proceeds calculating the matrix elements between hadron states of $\tilde{O}_n^{\tau s}(t)$ using standard techniques based on the spectral decomposition. To apply the decomposition in the lattice calculation of the 3-point function, one needs to ensure that the physical distance between the interpolators of the hadron states and of the flowed operators is much larger than the flow-time radius $\sqrt{8t}$. After performing the continuum limit, the renormalized matrix element in the MS scheme is calculated simply by

$$A_n^{\text{MS}}(\mu) = c_n(t, \mu)^{-1} A_n(t). \quad (16)$$

This matching is correct only if the flowed fields are defined as ringed fields. If the flowed fields are renormalized in a different scheme, the finite part of the matching coefficient has to be modified.

If the matching is successful, the l.h.s of Eq. (16) should be independent of the flow time. Violations stem from higher dimensional operators as linear terms in t , and from higher order terms in the perturbative expansion of the matching coefficients. To mitigate these possible systematics it is important to extend to $O(\mathcal{B}^3)$ the calculation presented here and conduct a thorough numerical study of the residual flow-time dependence of the hadronic matrix element after the matching.

Applications. The simplicity of the matching is guaranteed once the operators are symmetrized on the Lorentz indices, and all the traces have been subtracted. Traceless operators for any n are not difficult to write down. For example, for $n = 4$, we have

$$\tilde{O}_{4444} = O_{4444} - \frac{3}{4} O_{(\alpha\alpha\beta\beta)} + \frac{1}{16} O_{(\alpha\alpha\beta\beta)}. \quad (17)$$

The hadronic matrix element of $\tilde{O}_{4444}(t)$ is calculable with no external spatial momentum, and the matching with the MS scheme at renormalization scale μ is obtained using Eq. (16).

The continuum limit of the reduced hadronic matrix element of $\tilde{O}_n$, denoted as $A_n^h(t) = \langle x^{n-1} \rangle^h(t)$, would require a determination of Z_ψ from lattice QCD simulations. However, this can be avoided using the second

moment $A_{\text{MS}}^h(\mu) = \langle x \rangle_{\text{MS}}^h(\mu)$ as an observable input already computed with any method of choice. Then all the other moments for $n > 2$ are calculable from

$$\langle x^{n-1} \rangle_{\text{MS}}^h(\mu) = \langle x^{n-2} \rangle_{\text{MS}}^h(\mu) \frac{c_{n-1}(t, \mu)}{c_n(t, \mu)} R_n^h(t), \quad (18)$$

where $c_n(t, \mu)$ is given by Eqs. (13-15). The ratios

$$R_n^h(t) = \frac{\langle x^{n-1} \rangle^h(t)}{\langle x^{n-2} \rangle^h(t)}, \quad n > 2, \quad (19)$$

have a finite continuum limit, and there is no need to renormalize the flowed fields. Moreover, these ratios exhibit $\mathcal{O}(a^2)$ scaling violations once the lattice theory has been nonperturbatively improved. If convenient numerically, it is also possible to use different hadrons for the calculation of the moment matrix elements in Eq. (18) because the matching coefficients are independent on the hadron for which we calculate the matrix element.

The method proposed in this paper can be also generalized to study distribution amplitudes, but care should be taken when studying the short flow time expansion due to the presence of operators with vanishing contributions for forward matrix elements [48, 49].

The gradient flow has previously been employed to propose a locally smeared operator product expansion in scalar field theory [50] and to define quasi-PDF in [51], with the 1-loop matching calculated in [52]. For a more realistic application of the gradient flow in quasi-PDF calculations see Ref. [53]. While, in principle, the determination of moments of PDFs should allow a precise reconstruction of the PDFs, it would be intriguing to explore whether the determination of the PDF moments outlined in this work can be combined with quasi-PDF calculations.

Summary. We have described a method that resolves both the theoretical and numerical challenges faced when calculating moments of parton distribution functions from lattice QCD. A lattice regulator breaks the Euclidean rotational symmetry into the hypercubic group. As a result, the renormalization of twist-2 operators on the lattice is a substantial hurdle to determine moments of parton distribution functions $\langle x^{n-1} \rangle$. While for $n > 4$, power divergences in the lattice spacing are unavoidable, for

$n = 3, 4$ the cancellation of power divergences is still possible by choosing suitable irreducible representations of the hypercubic group. The price to pay is that the calculation of the hadronic matrix elements necessitates the use of a spatial external momentum, worsening the signal-to-noise ratio of the correlation functions. At finite flow time t , the correlation functions of twist-2 operators are finite once the bare parameters of the theory and the flowed fermion fields are renormalized. After performing the continuum limit, the matching with the physical matrix element can be done at short flow time using the continuum symmetries of the theory. Using traceless and symmetrized operators guarantees that the matching is multiplicative, and we have calculated, at 1-loop in perturbation theory, the matching coefficients.

It would be also very interesting to investigate how the approach described in this work can complement recent efforts aimed at the direct determination of PDFs from lattice QCD correlators.

While, the method proposed in this paper is applicable, at least in principle, to calculate any moments of parton distribution functions, in practice lattice QCD calculations of hadronic matrix elements of the flowed twist-2 operators in Eq. (10) are subject to several systematic uncertainties.

In the Supplemental Material [36] there is a discussion of the systematics, but only a thorough numerical investigation provides the final response about the range of applicability, and whether this result opens the way to determine moments of any order of the parton distribution functions.

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I. SUPPLEMENTAL MATERIAL

The method proposed in this paper is applicable, at least in principle, to calculate any moments of parton distribution functions, resolving the long-standing problem of renormalizing twist-2 operators on the lattice. In practice lattice QCD calculations of hadronic matrix elements of the flowed twist-2 operators in Eq. (10) are subject to several systematic uncertainties.

Finite volume effects. In the continuum limit the space-time extension of the flowed twist-2 operators in Eq. (10) is governed by the flow-time radius $\sqrt{8t}$, and the finite volume effects of the hadronic matrix elements are calculable using effective field theory methods [37]. At finite lattice spacing a the finite volume provides a bound on the number of covariant derivatives, thus a bound on the order of the moments that can be simulated. The operator $\hat{O}_n(x, t)$ contains $n - 1$ derivatives and the maximal distance from x reached by the gauge links is $(n - 1)a$. We now consider the example of the pion matrix element of $\hat{O}_n(x, t)$ calculated on a lattice with time extent T . A reasonable estimate for the maximal extension of the flowed operator is $\sim T/8$. For state-of-the-art lattice QCD simulations this implies that up to $n \sim 10$ there should be no obstruction for the calculation of the hadronic matrix elements. For nucleon matrix elements, the signal-to-noise ratio can provide a stronger constraint, that we address below.

Discretization errors. For the calculation of $\hat{o}(x^n)(t)$ for values of $\sqrt{8t}$ smaller than $\simeq na$ one can expect substantial discretization errors. For larger values of $\sqrt{8t}$ there is no reason to expect different discretization effects than in other similar calculations. Additionally, it is common practice in lattice QCD calculations to determine specific ratios, for which some of the potential systematic effects are reduced.

A typical application for ratios like $R_n^h(t)$, defined in Eq. (19), is to substantially reduce cutoff effects with respect to the standard twist-2 hadronic matrix elements. In the case of non-perturbatively improved Wilson fermions, this can be also confirmed from a theoretical analysis based on the Symanzik effective theory approach. In Ref. [30] it has been shown that, beside the usual $O(a)$ cutoff effects removed with the non-perturbative $O(a)$ improvement of the theory, the only additional cutoff effects are of $O(am)$, where m is the quark mass⁴. These cutoff effects depend only on the fermion content of the observable and thus cancel out in the ratio $R_n^h(t)$. One is left with a ratio of lattice correlators affected only by $O(a^2)$ effect.

⁴ Potentially short-distance $O(a)$ cutoff effects could affect the lattice correlators involving flowed fields, but they are expected to be insignificant in the calculation of the hadronic matrix elements (see Ref. [36] for another example where these short-distance $O(a)$ cutoff effects are negligible).

Perturbative matching. The perturbative calculation of the matching coefficients follows standard strategies (see [39, 40] and Refs. therein), expanding the integrands of the loop integrals of the matching equations (12) in all soft scales keeping t fixed. The $t = 0$ 1-loop contribution on the r.h.s of Eq. (12) vanishes because the expansion leads to scaleless, i.e. vanishing, integrals in D-dimensions. After renormalizing the gauge coupling and the flowed fermion fields, the left-hand side is UV finite. The infrared singularities stemming from the expansion in the soft scales, and regulated in dimensional regularization, match exactly the UV poles of $Z_n^{1\text{L}}$. The finite matching coefficients are the result of this procedure.

The Feynman diagrams contributing are depicted⁵ in Fig. 1. The cross indicates the insertion of the twist-2 operators, the single straight and wavy lines respectively represent the fermion and gluon propagator, and the double straight lines represent the kernel of the fermionic GF operator. The open circle represents a GF vertex, and the rest are standard QCD vertices. The Feynman rules for the operators are the same as in the unflowed case, and the remaining Feynman rules related to the expansion of the GF equation can be found, for example, in Refs. [40, 42].

The final result of the calculation is given by Eqs. (14). The matching coefficients satisfy a renormalization group (RG) equation that can be determined using the fact that the "ringed" matrix elements, or the ratios of flowed matrix elements in Eq. (18) are renormalization group invariant. The solutions of the RG equation, $c_n^{\text{RG}}(t, \mu)$, are given by

$$c_n^{\text{RG}}(t, \mu) = c_n(\bar{g}(q)) \exp \left\{ - \int_{\bar{g}(\mu)}^{\bar{g}(q)} dx \frac{\gamma_n(x)}{\beta(x)} \right\}, \quad (20)$$

where β is the beta function and γ_n the anomalous dimension of the twist-2 operators in Eq. (2). The matching coefficients $c_n(\bar{g}(q))$ are the matching coefficients in Eq. (13) with $q = 1/\sqrt{t}$. We consider the perturbative expansion of $\beta(\bar{g})$ up to order $\bar{g}^4$ [43], $\gamma_n(\bar{g})$ up to order $\bar{g}^3$ [6, 44, 45] and evaluate the next-to-leading-log (NLL) expression in Eq. (20), $c_n^{\text{NLL}}(t, \mu)$. Fixing $\mu = 2$ GeV for the second moment $\langle x \rangle$ the perturbative corrections are between 6 – 9% in the region where the flow-time radius $\sqrt{8t} = 0.2 - 0.3$ fm. In the left plot of Fig. 2 we show the matching coefficient of the second moment comparing the result of Eq. (14) with the resummed result of Eq. (20) for $\mu = 2$ GeV as a function of $\sqrt{8t}$. The range of the flow-time radius is chosen to be in the range where for lattice spacings $a = 0.05 - 0.08$ fm large cutoff effects are not expected. We observe that the resummation guarantees that the coupling goes smoothly to zero in the $t \rightarrow 0$ limit. In a lattice QCD simulation, after the perturbative matching, the residual flow time dependence could be

⁵ To draw the Feynman diagrams we have used FeynGame [41].

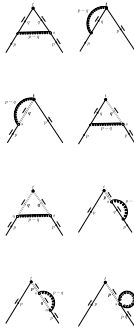


FIG. 1. Feynman diagram for the $O(5^2)$ calculations of the matching coefficients in Eq. (12). The circle with the cross represents the twist-2 operator and the empty circles the GF vertices. The double line represents the fermionic kernel of the GF differential operator. The arrows beside the fermion lines and the kernel lines indicate the direction of increasing flow time.

removed extrapolating to $t \rightarrow 0$ the numerical data. This is a standard procedure (see for example Refs. [46]) for calculations involving flowed observables.

For higher moments we consider the ratios in Eq. (19) that can be used to calculate the moments in the $\overline{\text{MS}}$ scheme using Eq. (18). Fixing again the renormalization scale at $\mu = 2$ GeV in the right plot of Fig. 2 we show the dependence of the ratios $\frac{R_n^{\text{lat}}(t, \mu, \beta[\mu])}{R_n^{\text{th}}(t, \mu, \beta[\mu])}$ on $\sqrt{8t}$, for $n = 3, \dots, 7$. We observe that the perturbative corrections decrease for increasing n . To give a first estimate of what uncertainty to expect with a NLL perturbative matching, we have calculated the perturbative corrections, δ_{PT} , for higher moments evaluated at $\sqrt{8t} = 2(n-1)a$, i.e. at a distance twice as large as the farthest possible contribution from the twist-2 local operators. The estimate has been obtained at a lattice spacing of $a = 0.05$ fm which is a lattice spacing reachable with state-of-the-art lattice QCD simulations. The results are listed in Table I.

n	$\langle x^{n-1} \rangle$	δ_{PT}
2	$\langle x^1 \rangle$	2%
3	$\langle x^2 \rangle$	11%
4	$\langle x^3 \rangle$	8%
5	$\langle x^4 \rangle$	7%
6	$\langle x^5 \rangle$	6%
7	$\langle x^6 \rangle$	6%

TABLE I. Perturbative corrections for moments of PDF at a matching scale of $\mu = 2$ GeV, $\sqrt{8t} = 2(n-1)a$, and $a = 0.05$ fm.

From this exercise one can then conclude that the flow time dependence of the ratio $R_n^{\text{lat}}(t)$ is very small, at least at NLL of perturbation theory, and this is true also in the region of $\sqrt{8t}$ where lattice discretization effects should be under control. After the perturbative matching is performed, it is conceivable that higher dimensional operators will contribute to the flow time dependence of the matrix element. This has been observed practically in all calculations of matrix elements of local flowed operators. It is common practice to extrapolate the matched results to $t \rightarrow 0$. The extrapolation is usually very smooth and it removes any residual flow time dependence. Only a thorough numerical investigation can confirm these expectations, but it is conceivable that the use of the ratios $R_n^{\text{lat}}(t)$ will be beneficial also to weaken the contributions from higher dimensional operators. In conclusion, choosing a renormalization scale of $\mu = 2$ GeV, it is possible to determine ratios of matrix elements where the perturbative corrections are very small, and where the higher order contributions in flow time could be smoothly extrapolated to zero.

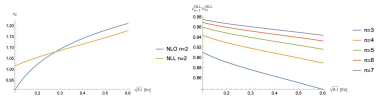


FIG. 2: Left Plot: flow-time dependence of the matching coefficient for the second moment calculated at $\mu = 2$ GeV comparing the NLO result with the resummed one. Right Plot: flow time dependence of the ratios $c_2^{NLO}(\mu, t)/c_2^{NLO}(\mu, t)$ at $\mu = 2$ GeV.

P48: 准分布的幂修正与 renormalon

主题归属	LaMET、quasi/pseudo 分布与因子化
底稿	arXiv:1810.00048
arXiv / DOI	1810.00048
完整原始 PDF	14 页; 来源 PyQCD; SHA-256 978bb7d735bf...
转排索引	EN 24 页; ZH 24 页 (只作书目对照)
备注	索引未记录额外备注。

阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

页码范围：P48-p001 – P48-p014

Power corrections and renormalons in parton quasi-distributions

Vladimir M. Braun,¹ Alexey Vladimirov,¹ and Jian-Hui Zhang¹

¹*Institut für Theoretische Physik, Universität Regensburg,
Universitätsstraße 31, 93040 Regensburg, Germany*

Perturbative expansions for short-distance quantities in QCD are factorially divergent and this deficiency can be turned into a useful tool to investigate nonperturbative corrections. In this work, we use this approach to study the structure of power corrections to parton quasi-distributions and pseudo-distributions which appear in lattice calculations of parton distribution functions. As the main result, we predict the functional dependence of the leading power corrections to quasi(pseudo)-distributions on the Bjorken x variable. We also show that these corrections can be strongly affected by the normalization procedure.

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I. INTRODUCTION

Lattice calculations in QCD have demonstrated the ability to complement, and in certain cases with the exceeding precision, significant amount of experimental measurements. Now, the lattice evaluation of parton distribution functions (PDFs) are coming on the agenda. New techniques are being explored aiming at the access of PDFs directly in the momentum fraction space, in addition to the standard approach that allows one to calculate first Mellin moments of PDFs. The existing actual proposals [1–6] differ in details but have a common general scheme: PDFs are extracted from the lattice calculations of suitable Euclidean correlation functions using QCD collinear factorization in the continuum theory.

A particularly popular suggestion [5] that has triggered a lot of recent activity [7–20], introduces a concept of a parton *quasi-distribution* (qPDF) defined as a Fourier transform of the nonlocal quark-antiquark operator connected by the Wilson line. QPDF is then matched to PDF either directly in the $\overline{\text{MS}}$ scheme or using the large-momentum factorization scheme at the intermediate step (“Large Momentum Effective Theory” (LaMET) [21, 22]). The latter technique is useful to emphasize that, after the Fourier transform, the hadron momentum p remains to be the only dimensional parameter. So, the relevant scale of the QCD coupling is related to p up to a dimensionless constant and also the higher-twist corrections are generically suppressed by powers of the momentum p . A closely related quantity, a *pseudo-distribution* (pPDF) was introduced in [23–25].

Lattice calculations of PDFs using these new methods are presently moving from exploratory stage towards precision calculations, therefore questions like whether the higher-twist (power suppressed) corrections are well under control have to be addressed. One possibility to investigate the impact of higher-twist corrections is to extract PDFs from the global analysis of many Euclidean correlation functions introducing such corrections as free parameters. This approach is showcased in [26] using the position-space strategy [4, 27] and the pion light-cone distribution amplitude (LCDA) as an example. In

this case, the analytic structure of higher-twist effects is well understood, so they could have been modelled by just one free parameter. The general situation may be more complicated. Having in mind the multitude of observables that can potentially be employed in lattice calculations, see e.g. [6], it is desirable to have a general method to estimate the corresponding higher-twist corrections and their expected Bjorken- x dependence. The purpose of this work is to point out that the problem of power-suppressed contributions for such observables can be addressed using the concept of renormalons [28, 29]. In what follows we apply this technique to qPDFs and pPDFs.

The renormalon approach to the investigation of power corrections is founded on the fact that operators of different twist mix with each other under renormalization, due to the violation of QCD scale invariance through the running of the coupling constant. In cutoff schemes, this mixing is explicit, whereas in dimensional regularization, it manifests itself in factorial divergence of the perturbative series. Independence of a physical observable on the factorization scale implies intricate cancellations between different twists — the cancellation of renormalon ambiguities. In turn, the existence of these ambiguities in the leading-twist expressions can be used to estimate the size of power-suppressed corrections. Conceptually, it is similar to the estimation of the accuracy of fixed-order perturbative results by the logarithmic scale dependence. The renormalon approach was used before for the study of Bjorken- x dependence of higher-twist corrections in deep-inelastic scattering [30–32], fragmentation functions [33, 34], pion LCDA [35] and transverse-momentum-dependent parton distributions [36].

In order to explain how the concept of renormalons can be used to get insight in the structure of power corrections, let us consider the usual expression for the quasi-distribution [22, 37],

$$\mathcal{Q}(x, p) = \int_{-1}^1 \frac{dy}{|y|} C_{\mathcal{Q}}\left(\frac{p}{y}, xp, \mu_F\right) q(y, \mu_F) + \frac{1}{p^2} \mathcal{Q}_2(x, p) + \dots \quad (1)$$

where $q(y, \mu_F)$ is the quark PDF, p is the hadron mo-

mentum, x refers to the momentum fraction. For brevity we do not show the dependence on the renormalization scale. The factorization scale μ_F has to be taken of the order of $|x|p$ to avoid large logarithms. The coefficient function $C(x, p, \mu_F) \sim \delta(1-x) + \mathcal{O}(\alpha_s)$ is given by the perturbative expansion. The correction of $\mathcal{O}(\alpha_s)$ was first computed for the flavor-nonsinglet case in [38], see also [39].

To understand the role of renormalons, it is necessary to examine carefully the separation made in (1) between the leading term and the higher twist addenda. Let us assume for a moment that the factorization is done using a hard cutoff $\Lambda_{\text{QCD}} \ll \mu_F \ll p$, i.e. the contributions with loop momenta $|k| > \mu_F$ are included in the coefficient function, whereas the contributions with $|k| < \mu_F$ are included in PDF. In this scheme, the coefficient function has the following expansion at $p \rightarrow \infty$

$$C_0(x, p, \mu_F) = \delta(1-x) + c_1 \alpha_s + c_2 \alpha_s^2 + \dots - \frac{\mu_F}{p^2} D_0(x) + \dots, \quad (2)$$

where $c_k = c_k(x, \ln p^2/\mu_F^2)$ are the perturbative coefficients depending logarithmically on the scales and the D_0 -term represents the leading power correction. Since the left-hand-side of Eq. (1) does not depend on μ_F , any such dependence should cancel on the right-hand-side. In particular, the logarithmic dependence on the scale in $c_k(x, \ln p^2/\mu_F^2)$ is canceled by the scale-dependence of PDF $q(x, \mu_F)$. The cancellation of the power dependence, on the other hand, must involve the twist-four contribution $Q_4(x, p)$. Thus, in this factorization scheme one expects that

$$Q_4(x, p, \mu_F) = \mu_F^2 \int_{-1}^1 \frac{dy}{|y|} D_0\left(\frac{x}{y}\right) q(y, \mu_F) + \tilde{Q}_4(x, p, \mu_F), \quad (3)$$

where $\tilde{Q}_4$ depends on μ_F at most logarithmically. Appearance of the term $\sim \mu_F^2$ can be traced to quadratic ultraviolet divergence (in addition to the logarithmic ultraviolet divergence) of the twist-four operators that are responsible for the power correction, in the cutoff scheme. One can prove that the cutoff dependence $\sim \mu_F^2$ of the higher-twist operators is indeed that of Eq. (2).

In practice, perturbative calculations are usually done using dimensional regularization. In this case, power-like terms as in (2) do not appear. The price to pay is that the coefficients c_k computed in a $\overline{\text{MS}}$ scheme grow factorially with the order k . The factorial growth implies that the sum of the perturbative series is only defined to a power accuracy and this ambiguity (renormalon ambiguity) must be compensated by adding a non-perturbative higher-twist correction. The detailed analysis shows [28, 29] that the divergent large- k behavior (the renormalon) of the coefficients is in the one-to-one correspondence with the sensitivity to extreme (small or large) loop momenta. In particular, infrared renormalons in the

leading-twist coefficient function are compensated by ultraviolet renormalons in the matrix elements of twist-four operators. In this way the same picture as in the cutoff scheme re-appears in dimension regularization.

Returning to (3), we observe that the quadratic term in μ_F is spurious since its sole purpose is to cancel the similar contribution to the coefficient function. Therefore, it does not contribute to any physical observable. The idea of the renormalon model of the power corrections [30–35] is that, with a replacement of μ_F by a suitable non-perturbative scale, this contribution reflects the order and the functional form of actual power-suppressed contribution. Assuming this “ultraviolet dominance” [28, 29, 40] one obtains the following model:

$$Q_4(x, p, \mu_F) = \kappa \Lambda_{\text{QCD}}^2 \int_{-1}^1 \frac{dy}{|y|} D_0\left(\frac{x}{y}\right) q(y, \mu_F), \quad (4)$$

with the dimensionless coefficient $\kappa = \mathcal{O}(1)$ which cannot be fixed within theory and remains a free parameter.

In this work we calculate the function $D_0(x)$ for different versions of the qPDFs (pPDFs) in the so-called bubble-chain approximation [28], which is our main result. This calculation reveals that the power corrections to quasi- and pseudo-distributions have the following generic structure

$$Q(x, p) = q(x) \left\{ 1 + \mathcal{O}\left(\frac{\Lambda^2}{p^2 x^2 (1-x)}\right) \right\}, \\ P(x, z) = q(x) \left\{ 1 + \mathcal{O}\left(z^2 \Lambda^2 (1-x)\right) \right\}, \quad (5)$$

respectively, and can be affected significantly by normalization. In particular, the normalization of the involved matrix elements to their value at zero momentum considerably reduces the power correction for the qPDFs at smaller- x at the cost of a strong enhancement at larger values. We emphasize that the leading power correction to qPDF at $x \rightarrow 0$ is enhanced by two powers of the Bjorken x variable. For pPDFs the power corrections are suppressed at $x \rightarrow 1$, which, unfortunately, does not hold after the normalization procedure of Ref. [24] (but can be upheld with a different choice). Additionally, as a byproduct of bubble-chain calculation, we have obtained the large- n_F part, $n_F^2 \alpha_s^{n_F+1}$, of the leading twist coefficient function to all orders in perturbation theory.

The presentation is organized as follows. In Sect. 2 we formulate our program in more precise terms. We define qPDFs and pPDFs as particular Fourier transforms of the position space (Ioffe-time) distributions, discuss briefly the light-ray operator product expansion (OPE) and the target mass corrections, and introduce the relevant techniques (Borel transform) and the systematic approximation (large- n_F expansion) that will be used throughout the rest of the work. In Sect. 3 we present our result for the Borel transform of the leading twist coefficient function and discuss the structure of its singularities. The leading power corrections to various versions of the quasi-distributions are obtained in Sect. 4. We

collect there the relevant analytic expressions and also present the results of a numerical study using realistic parametrizations for the valence quark PDFs. The final Sect. 5 is reserved for the summary and conclusions.

II. GENERAL FORMALISM

A. Parton quasi-distributions

Let us start with the following nucleon matrix element

$$\begin{aligned} \langle N(p) | \bar{q}(zv) \gamma_\alpha [zv, 0] q(0) | N(p) \rangle = \\ = \frac{2v_\alpha}{v^2} \langle \mu v | \mathcal{I}^1(v^2, z^2, \mu v z, \mu) \\ + 2 \left(p_\alpha - \frac{v_\alpha}{v^2} \langle \mu v \rangle \right) \mathcal{I}^1(v^2, z^2, \mu v z, \mu), \end{aligned} \quad (6)$$

where

$$[zv, 0] = \text{Pexp} \left[ig \int_0^z du v^\mu A_\mu(uv) \right], \quad (7)$$

p^μ is the nucleon momentum, $\mu^2 = m^2$, v^μ is a given four-vector and z a real number. All notations correspond to Minkowski space. In the following, we keep the normalization of the four-vector v_μ arbitrary, keeping in mind that $v^2 < 0$ in the lattice calculation. We suppress flavor indices and tacitly assume considering the flavor-non-singlet combination of the matrix elements in what follows. We also neglect quark masses.

The operator product in Eq. (6) suffers from ultraviolet (UV) divergences and has to be renormalized. The argument μ of matrix elements $\mathcal{I}$ refers to the renormalization scale. The renormalization-scale dependence can be studied by going over to an effective theory [41–43] such that the Wilson line is substituted by the propagator of an auxiliary field. For time-like separations the resulting theory is the heavy-quark effective theory (HQET) and the renormalization factor of interest is the renormalization factor (squared) for the heavy-light quark current, see e.g. [44]. We are not aware of a calculation for such a renormalization factor at space-like separations beyond one-loop order, however, to this accuracy there is no difference from the time-like case.

The “longitudinal” and “transverse” invariant functions in Eq. (6) (with respect to v_μ) correspond to particular projections of the matrix element that are employed in lattice calculations:

$$\begin{aligned} \langle N(p) | \bar{q}(zv) \not{v} [zv, 0] q(0) | N(p) \rangle &= 2 \langle \mu v | \mathcal{I}^1(v^2, z^2, \mu v z, \mu), \\ \langle N(p) | \bar{q}(zv) \not{v} [zv, 0] q(0) | N(p) \rangle &= 2 \langle \mu v | \mathcal{I}^1(v^2, z^2, \mu v z, \mu), \end{aligned} \quad (8)$$

where $(\epsilon \cdot v) = 0$. Eq. (8) is the starting point for the construction of quasi-distributions.

The invariant functions $\mathcal{I}^1, \mathcal{I}^2$ of Eq. (6) coincide at the tree level. Assuming the power counting

$$z = \mathcal{O}(\eta), \quad p = \mathcal{O}(\eta^{-1}), \quad \eta \rightarrow 0, \quad (9)$$

they can be written in terms of the position-space quark PDF [45–47]:

$$\begin{aligned} \mathcal{I}^{(1)}(z^2 v^2, \mu v z, \mu \sim 1/|vz|) &= I(\mu v z, \mu_F \sim 1/|vz|) \\ &+ \mathcal{O}(\alpha_s, \eta^2), \end{aligned} \quad (10)$$

where μ_F is the factorization scale, and

$$I(\mu v z, \mu_F) = \int_1^1 dx e^{ixz \mu v} q(x, \mu_F). \quad (11)$$

Here $q(x, \mu_F)$ for $x > 0$ is the quark PDF, and for $x < 0$ is the antiquark PDF. The position-space PDFs I are known as the Ioffe-time distribution (ITD). To distinguish ITD I from the functions $\mathcal{I}$, we refer to the functions $\mathcal{I}^1, \mathcal{I}^2$ as the Ioffe-time quasi-distributions (qITDs), following the terminology introduced in [5].

The qPDFs $\mathcal{Q}^1, \mathcal{Q}^2$ [3] and the pPDF $\mathcal{P}$ [23] are defined in terms of the qITDs by the Fourier transform

$$\begin{aligned} \mathcal{Q}^{(1)}(x, p, \mu) &= |\langle \mu v | \int_{-\infty}^{\infty} \frac{dz}{2\pi} e^{-ixz \mu v} \mathcal{I}^{(1)}(z^2 v^2, \mu v z, \mu), \\ \mathcal{P}(x, z, \mu) &= |z| \int_{-\infty}^{\infty} \frac{dz' \langle \mu v \rangle}{2\pi} e^{-ixz \mu v} \mathcal{I}^1(z'^2 v^2, \mu v z', \mu). \end{aligned} \quad (12)$$

In what follows we will tacitly assume that $\langle \mu v \rangle > 0$ in the discussion of qPDFs and $z > 0$ for pPDFs and drop the absolute value sign, for brevity.

In renormalization schemes with an explicit regularization scale, the Wilson line in Eq. (6) suffers from an additional linear UV divergence that has to be removed. In dimensional regularization, this UV linear divergence reveals itself as a factorial growth of high orders of perturbative series [48]. The linear UV divergence can be removed by a mass renormalization associated with the Wilson line [41, 42, 49] or by a regularization-independent renormalization [14, 39, 50]. Given the multiplicative renormalizability of the quasi-PDF operator [43, 51, 52], it can also be removed by considering a suitable ratio of matrix elements involving the same operator, e.g. by normalizing to the same matrix element at zero proton momentum [24],

$$\mathbf{I}(v^2 z^2, \mu v z) = \mathcal{I}(v^2 z^2, \mu v z, \mu) / \mathcal{I}(v^2 z^2, 0, \mu), \quad (13)$$

or, alternatively, to the vacuum expectation value

$$\bar{\mathbf{Y}}(v^2 z^2, \mu v z) = \mathcal{I}(v^2 z^2, \mu v z, \mu) / \mathcal{N}(v^2 z^2, \mu), \quad (14)$$

where

$$\mathcal{N}(v^2 z^2, \mu) = \left(\frac{2iN_c}{\pi^2 z^2 v^2} \right)^{-1} \langle 0 | \bar{q}(zv) \not{v} [zv, 0] q(0) | 0 \rangle. \quad (15)$$

Eqs. (13) and (14) will be our main focus in the present paper. By forming the ratio, the scale dependence cancels out (including the usual logarithmic renormalization) and one can define the scale-independent qPDF/pPDF

$$\mathbf{Q}(x, p) = |\langle \mu v | \int_{-\infty}^{\infty} \frac{dz}{2\pi} e^{-ixz \mu v} \mathbf{I}(z, p, v) |$$

$$\mathbf{P}(x, z) = |z| \int_{-\infty}^{\infty} \frac{d(\mu\nu)}{2\pi} e^{-izx(\mu\nu)} \mathbf{I}(z, \mu\nu), \quad (16)$$

and similarly for $\tilde{\mathbf{Q}}(x, p)$ and $\tilde{\mathbf{P}}(x, z)$. The difference between these two options in the present context is that the normalization to the vacuum correlator does not affect the leading $\mathcal{O}(v^2 z^2)$ power corrections that are subject of this work (since there is no gauge-invariant operator), whereas the normalization to the value at zero momentum, as we will see, has a substantial effect.

B. Light-ray OPE and target mass corrections

The general approach to collinear factorization of QCD amplitudes in the position space is provided by the light-ray OPE [53–58]. For illustration consider the “longitudinal” projection. Specializing to the present case (forward matrix elements) we write

$$\begin{aligned} \tilde{q}(zv) \# [zv, 0] q(0) &= \int_0^1 d\alpha \tilde{H}^1(z, \alpha, \mu x) \\ &\times \Pi_{L,1}^{\mu\nu} [\tilde{q}(\alpha zv) \# q(0)] + \dots, \end{aligned} \quad (17)$$

where we include the renormalization factor in the coefficient function $\tilde{H}^1 \sim \delta(1-\alpha) \circ \mathcal{O}(\alpha_*)$ and set the renormalization scale μ to be equal to the factorization scale μx . Finally, $\Pi_{L,1}^{\mu\nu}[\dots]$ is the leading-twist projection operator defined below and ellipses stand for the higher-twist contributions.

The leading-twist projection of the nonlocal quark-antiquark operator is defined as the generating function of renormalized local leading-twist operators (traceless and symmetrized over all indices)

$$\Pi_{L,1}^{\mu\nu} [\tilde{q}(zv) \# q(0)] = \sum_{n=1}^{\infty} \frac{z^{n-1}}{(n-1)!} v^{\mu\nu_1} \dots v^{\mu\nu_n} O_{\mu_1 \dots \mu_n}^n(0), \quad (18)$$

where

$$O_{\mu_1 \dots \mu_n}^n(z) = \tilde{q}(0) \gamma_{(\mu_1} \tilde{D}_{\mu_2} \dots \tilde{D}_{\mu_n)} q(0). \quad (19)$$

Here we indicate trace subtraction and symmetrization by enclosing the involved Lorentz indices in parentheses, for example $O_{(\mu\nu)} = \frac{1}{2}(O_{\mu\nu} + O_{\nu\mu}) - \frac{1}{2}g_{\mu\nu} O_\lambda^\lambda$.

The light-ray OPE differs from the usual short-distance Wilson expansion in local operators by imposing a different power counting. In the short-distance expansion one assumes that the distance between the quarks is small, $|z| \sim \eta \Lambda_{\text{QCD}}^{-1}$ with $\eta \rightarrow 0$, and the operator matrix elements are of order unity in this limit, $\langle O_{\mu_1 \dots \mu_n}^n \rangle \sim \Lambda_{\text{QCD}}^n$. In this case only a finite number of terms contribute to the r.h.s. of Eq. (18), whereas the rest as well as the higher-twist operators must be added to higher orders of OPE, starting from $\mathcal{O}(\eta^2)$. In other words, the relevant expansion parameter is the mass-dimension of an operator. The light-ray OPE assumes instead that the

leading-twist operators scale as $\langle O_{\mu_1 \dots \mu_n}^n \rangle \sim \eta^{-n} \Lambda_{\text{QCD}}^n$ so that $z^n v^{\mu_1} \dots v^{\mu_n} \langle O_{\mu_1 \dots \mu_n}^n \rangle \sim \mathcal{O}(1)$. In this case, the series in (18) must be resummed to all orders, revealing its non-local “light-ray” structure. For a generic hadronic matrix element of leading twist operators one has

$$\langle p | O_{\mu_1 \dots \mu_n}^n | p \rangle \sim p_{(\mu_1} \dots p_{\mu_n)} \langle\langle O^n \rangle\rangle, \quad (20)$$

where the reduced matrix element $\langle\langle O^n \rangle\rangle \sim \mathcal{O}(1)$. Therefore, the light-ray OPE is an adequate approximation if the hadron has large momentum, $|p| \sim \mathcal{O}(\eta^{-1})$ and hence $z p^\mu \sim \mathcal{O}(1)$. Higher-twist operators of the same dimension have smaller spin (by definition) and as a consequence, their matrix elements are power-suppressed [59].

Note that the above power counting is applicable both in Minkowski and Euclidean space. In Minkowski space, one can go over to a different reference frame where all components of the momentum are of order Λ_{QCD} and simultaneously the separation between the quarks is almost light-like, $|v_\mu| \sim \mathcal{O}(1)$ but $v^2 \sim \mathcal{O}(\eta^2) \rightarrow 0$.

On the calculation level, the light-ray OPE provides one with a convenient framework to operate with the leading-twist projected operators (18) avoiding the local expansion. Light-ray operators can be viewed as analytic operator functions of the separation between the quarks (all short-distance and light-cone singularities are subtracted). They satisfy the Laplace equation [55]

$$\frac{\partial}{\partial v^\mu} \frac{\partial}{\partial v_\mu} \Pi_{L,1}^{\mu\nu} [\tilde{q}(zv) \# q(0)] = 0 \quad (21)$$

with the boundary condition $\Pi_{L,1}^{\mu\nu} \rightarrow 1$ at $v^2 \rightarrow 0$. Explicit expressions for the projection operator $\Pi_{L,1}^{\mu\nu}$ can be found in [55, 58, 60, 61]. The light-ray OPE combined with the background field method is the standard technique, e.g., in light-cone sum rules [62], where it is used for the calculation of higher-twist contributions, and for the derivation of the evolution equations for off-forward parton distributions [63, 64]. The renormalization group kernel for the evolution of flavor-singlet light-ray operators in the general off-forward kinematics is known to three-loop accuracy [65].

The nucleon matrix element of the leading-twist projected operator (18) defines the leading-twist quark PDF,

$$\begin{aligned} &\langle N(p) | \Pi_{L,1}^{\mu\nu} [\tilde{q}(zv) \# q(0)] | N(p) \rangle \\ &= 2 \int_{-1}^1 dx \Pi_{L,1}[(p \cdot v) e^{ixz(\mu\nu)}] q(x, \mu x), \end{aligned} \quad (22)$$

where [57]

$$\begin{aligned} \Pi_{L,1}[(\mu\nu) e^{ixz(\mu\nu)}] &\sim (\mu\nu) \left[1 - \frac{1}{4} \frac{m^2 x^2}{(\mu\nu)^2} z \frac{d}{dz} \right] e^{ixz(\mu\nu)} \\ &+ \mathcal{O}(m^4). \end{aligned} \quad (23)$$

The second term in the square brackets is the leading nucleon mass correction, which is the position-space counterpart of the Nachtmann target mass correction [66] in

the deep-inelastic scattering (DIS). The all-order expression in powers of the nucleon mass for the leading-twist projected exponential function can be found in [57].

Taking the nucleon matrix element of the operator relation in Eq. (17) we obtain the factorization theorem for the “longitudinal” qITD in the form

$$T^A = \int_0^1 d\alpha H^A(z, \alpha, \mu_F) \left[1 - \frac{1}{4} \frac{m^2 v^2}{(\mu v)^2} z \frac{d}{dz} + \dots \right] \times I(\alpha z p v, \mu_F) + \dots, \quad (24)$$

where $I(\alpha z p v, \mu_F)$ is ITD (11) and ellipses stand for the higher-power target mass and “genuine” higher-twist corrections.

Making the Fourier transformation (12) one obtains, to the leading twist accuracy, the factorization theorem for the “longitudinal” qPDF,

$$Q^A(x, p) = \int_{-1}^1 \frac{dy}{|y|} C_S^A\left(\frac{y}{2}, xp, \mu_F\right) \left[1 + \frac{1}{4} \frac{m^2 v^2}{(\mu v)^2} \frac{d}{dy} y + \dots \right] \times q(y, \mu_F), \quad (25)$$

where the coefficient function is given by

$$C_S^A\left(\frac{y}{2}, xp, \mu_F\right) = (\mu v) |y| \int_{-\infty}^{\infty} \frac{dz}{2\pi} \int_0^1 d\alpha e^{i(\mu v)z(x-\alpha y)} \times H^A(z, \alpha, \mu_F). \quad (26)$$

In particular, at the leading order

$$Q^A(x, p) = q(x) + \frac{1}{4} \frac{m^2 v^2}{(\mu v)^2} [x q'(x) + q(x)] + \mathcal{O}(m^4/\mu^4), \quad (27)$$

where $q'(x) = (d/dx)q(x)$. Note that the derivative applied to the quark PDF lowers the power $q(x) \sim x^{-1} (1-x)^p$ by one unit so that the mass correction is effectively enhanced by the factor $1/(1-x)$ as compared to the quark distribution itself at large Bjorken x . The similar enhancement of the target mass correction at $x \rightarrow 1$ is familiar from DIS [66].

The target mass correction for the “transverse” qPDF can be calculated in a similar way, starting from the non-local operator with an open Lorentz index (6) and using the operator identity [55]

$$\Pi_{\Gamma\Gamma'}^A[q(zv)\gamma_\mu q(0)] = \int_0^\infty dt \frac{\partial}{\partial u^\mu} \Pi_{\Gamma\Gamma'}^A[q(tzv)\not{u} q(0)], \quad (28)$$

where the Wilson line between the quarks is implied. One obtains, at the tree level,

$$Q^A(x, p) = q(x) + \frac{1}{4} \frac{m^2 v^2}{(\mu v)^2} [x q'(x) + 3q(x)] - \frac{1}{2} \frac{m^2 v^2}{(\mu v)^2} \theta(|x| < 1) \int_{|x|}^1 \frac{dy}{y} q(x/y) + \mathcal{O}(m^4/\mu^4), \quad (29)$$

and the target mass correction to the pPDF

$$\mathcal{P}(x, z) = q(x) + \frac{1}{4} z^2 v^2 m^2 x^2 \theta(|x| < 1) \int_{|x|}^1 \frac{dy}{y} q(x/y) + \mathcal{O}(m^4/\mu^4). \quad (30)$$

The target mass corrections in Eqs. (27) and (29) agree with those derived in Ref. [9] when expanded to $\mathcal{O}(m^2 v^2/(\mu v)^2)$. Interestingly, the target mass correction to the pPDF is suppressed as $\mathcal{O}(1-x)$ at $x \rightarrow 1$ and not enhanced $\mathcal{O}(1/(1-x))$ in contrast to that to the qPDFs (27) and the structure functions in DIS.

C. Borel transform and renormalons

The coefficient function H^A in Eq. (17) and the similar coefficient function H^A in the $\overline{\text{MS}}$ scheme have the perturbative expansion

$$H = \delta(1-\alpha) + \sum_{k=0}^{\infty} h_k \alpha^{k+1}, \quad \alpha_s = \frac{\alpha_s(\mu)}{4\pi}, \quad (31)$$

with factorially growing coefficients $h_k \sim k!$.

A convenient way to handle such a series is to consider the Borel transform

$$B[H](w) = \sum_{k=0}^{\infty} \frac{h_k}{k!} \left(\frac{w}{\beta_0} \right)^k \quad (32)$$

where powers of $\beta_0 = 11/3N_C - 2/3n_f$ are inserted for the later convenience. The Borel image can be used as a generating function for the fixed-order coefficients

$$h_k = \beta_0^k \left(\frac{d}{dw} \right)^k B[H](w)|_{w=0}. \quad (33)$$

Moreover, the sum of the series can be obtained as the integral over positive values of the Borel parameter w

$$H = \delta(1-\alpha) + \frac{1}{\beta_0} \int_0^\infty dw e^{-w/\beta_0} B[H](w). \quad (34)$$

As it stands, the integral is not defined because the Borel transform generally has singularities on the integration path, known as (infrared) renormalons. One can adopt a definition of the integral deforming the contour above or below the real axis, or as the principle value. These definitions are arbitrary, and their difference, which is exponentially small in the coupling, must be viewed as an intrinsic uncertainty of perturbation theory that has to be removed by adding power-suppressed nonperturbative corrections. Another potential problem concerns the convergence of the Borel integral at $w \rightarrow \infty$. Since the quantity of interest depends on the single hard scale $1/|zv|$, dimension counting requires that the Borel transform can be written as $(\mu^2 z^2 v^2)^w$ times a function $F(w)$

of the Borel parameter and dimensionless kinematic variables. Combining $(\mu^2 z^2 e^2)^w$ with $e^{-w/(\beta_0 \alpha_s(w))}$ one sees that the (principal value) integral is convergent, provided the distance between the quarks $|x|$ is sufficiently small compared to $1/\Lambda_{\text{QCD}}$ and $F(w)$ does not increase exponentially at $w \rightarrow \infty$, which is the case of all known examples.

Naturally, a full all-order calculation cannot be performed. Instead, we employ the approximation [28, 29] restricting ourselves to the perturbative series generated by the running-coupling effects in the one-loop diagrams, i.e. using QCD coupling at the scale of the gluon virtuality. Such contributions can be traced by computing the diagrams with the insertion of k fermion loops in the one-loop diagram and replacing $-\frac{5}{3}n_f \leftrightarrow \beta_0 = \frac{11}{3}N_c - \frac{2}{3}n_f$, see Appendix A. Another, equivalent technique [67] is based on the calculation of one-loop diagrams with an effective gluon mass.

The singularity structure of the Borel transform can be extracted separately without explicit evaluation of the bubble-chain. It can be done by replacing the running coupling constant in the loop diagrams by its effective form,

$$\beta_0 \alpha_s(-k^2) = \int_0^\infty dw e^{\frac{w}{\Lambda}} \left(\frac{\Lambda^2}{-k^2} \right)^w, \quad (35)$$

where $\Lambda \equiv \Lambda_{\text{QCD}}^{\overline{\text{MS}}}$ and the factor $e^{\frac{w}{\Lambda}}$ represents the $\overline{\text{MS}}$ scheme. Such replacement leads to the modified form of the gluon propagator

$$\frac{1}{-k^2 - i\epsilon} \mapsto \frac{(\Lambda^2)^w}{(-k^2 - i\epsilon)^{1+w}}. \quad (36)$$

At one-loop level, the pole structure in w reproduces the pole structure for the Borel plane. In our work, we have used both methods of calculation for the cross-check of the result.

III. LARGE- β_0 COEFFICIENT FUNCTIONS

The leading contributions to the renormalon singularities in the coefficient functions for qTIDs are shown in Fig. 1, see Appendix A for technical details. We obtain

$$\begin{aligned} B[H^1](w) = & \frac{2C_F}{w} \left[\left(\frac{1+\alpha^2}{1-\alpha} - \left(2\alpha_2 F_1(1, 2-w; 2+w, \alpha) \right. \right. \right. \\ & \left. \left. \left. + \alpha(1-w^2) \right) \alpha^w h_0(w, X) \right]_+ \\ & + \delta(\hat{\alpha}) \left[\frac{3(w^2-w-1)}{(w+2)(2w-1)} h_0(w, X) - \frac{3}{2} \right] \\ & + \tilde{R}(w), \end{aligned} \quad (37)$$

$$B[H^1](w) = B[H^1](w) - 4C_F \hat{\alpha}(1+w) \alpha^w h_0(w, X), \quad (38)$$

where $\hat{\alpha} = 1 - \alpha$,

$$\begin{aligned} h_0(w, X) = & X^w \frac{\Gamma(1-w)}{\Gamma(2+w)}, \\ X = & \frac{v^2 z^2 \mu^2 e^{2/\Lambda}}{4}, \end{aligned} \quad (39)$$

and the function $\tilde{R}(w)$ is defined as the series expansion in terms of another function

$$\begin{aligned} R(w) = & 2C_F \left[\left(\frac{1+\alpha^2}{1-\alpha} G_0(w) - 1 + \alpha^w \hat{\alpha}(2+w) G_0(w) \right)_+ \right. \\ & \left. + \frac{\delta(\hat{\alpha})}{w} \left[\frac{3}{2} - \frac{2w+3}{(w+2)(w+1)} G_0(w) \right] \right], \\ G_0(w) = & \frac{\Gamma(4+2w)}{6\Gamma(1-w)\Gamma(1+w)\Gamma^2(2+w)}, \end{aligned} \quad (40)$$

such that

$$R(w) = \sum_n w^n R_n, \quad \tilde{R}(w) = \sum_n \frac{w^n}{(n+1)!} R_n. \quad (41)$$

The Taylor expansion of the Borel transform at $w=0$ gives the perturbative expansion for the coefficient functions in terms of the coupling constant. The $\mathcal{O}(\alpha_s)$ term is

$$H^{(1)1}(\alpha) = 2C_F \left[\left(-L_\mu \frac{1+\alpha^2}{1-\alpha} + \frac{3-8\alpha+3\alpha^2-4\ln\hat{\alpha}}{1-\alpha} \right)_+ + \delta(\hat{\alpha}) \left(\frac{3}{2} L_\mu + \frac{7}{2} \right) \right], \quad (42)$$

$$H^{(1)3}(\alpha) = 2C_F \left[\left(-L_\mu \frac{1+\alpha^2}{1-\alpha} + \frac{1-4\alpha+\alpha^2-4\ln\hat{\alpha}}{1-\alpha} \right)_+ + \delta(\hat{\alpha}) \left(\frac{3}{2} L_\mu + \frac{5}{2} \right) \right], \quad (43)$$

and the n_f -part of the $\mathcal{O}(\alpha_s^2)$ correction reads

$$H^{(2)1}(\alpha) = -\frac{4C_F n_f}{3} \left[\left(-\frac{L_\mu^2}{2} \frac{1+\alpha^2}{1-\alpha} \right)_+ - L_\mu \left[\frac{1+\alpha^2}{1-\alpha} \left(\ln\alpha + \frac{2}{3} \right) + 4 \frac{\alpha + \ln\hat{\alpha}}{1-\alpha} \right] + \delta(\hat{\alpha}) \left(\frac{3L_\mu^2}{4} + \frac{19}{4} L_\mu + \frac{159}{16} \right) \right] \quad (44)$$

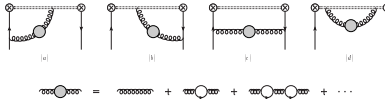


FIG. 1. Bubble-chain contribution to the coefficient function. The Wilson line factor is shown by the double dotted line.

$$H^{(2)}(\alpha) = H^{(1)}(\alpha) + \frac{4C_F g_f}{3} \bar{\alpha} \left(2L_p + 2 \ln \alpha + \frac{10}{3} \right), \quad (45)$$

$$+ \left[\frac{1 + \alpha^2}{1 - \alpha} \left(\frac{41}{18} + \frac{7}{6} \ln \alpha - \frac{\ln^2 \alpha}{4} \right) - \frac{4}{1 - \alpha} \left(\text{Li}_2(\alpha) + \ln^2 \bar{\alpha} + \frac{8}{3} \ln \bar{\alpha} + \ln \bar{\alpha} \ln \alpha \right) - \frac{\alpha(13 + 6 \ln \alpha)}{1 - \alpha} \right]_+.$$

where

$$L_p = \ln \left(\frac{-v^2 s^2 \mu^2}{4c^2 s^2 w} \right). \quad (46)$$

We have checked that the expressions in (42) reproduce after the Fourier transform (12) the one-loop correction to the qPDFs calculated in [38, 39].

A. Singularities of the Borel transform

The structure of singularities of the Borel transform of the coefficient function is illustrated in Fig. 2. There are an UV renormalon singularity at $w = 1/2$ and a series of IR renormalons at positive integer $w = 1, 2, \dots$. To the accuracy of our calculation (single bubble chain) all singularities are simple poles. These singularities obstruct the Borel integral in (34) (Borel-non-summable renormalons) and must be matched by the nonperturbative corrections. Note, that the $\tilde{R}(w)$ term in (37) is an analytic function of w . Thus, it is irrelevant for the discussion of singularities.

1. Ultraviolet renormalons at $w = 1/2$

The singularity at $w = 1/2$ is generated by the contribution of large momenta in the self-energy insertions in



FIG. 2. Singularity structure of the Borel transform

the Wilson line and is part of the renormalization factor

$$B[H] \stackrel{w \rightarrow 1/2}{\sim} \frac{-4C_F}{w - 1/2} \sqrt{X}. \quad (47)$$

This singularity is well-known [48] and is in the one-to-one correspondence to the linear UV divergence in the Wilson line's self energy (see also discussion prior to (13)). It can be removed by considering normalized qPDFs, (13) or (14), and will not be considered further in this work.

2. Infrared renormalons

The leading IR renormalon singularity is at $w = 1$. We obtain

$$B[H^+](w) \stackrel{w \rightarrow 1}{\sim} \frac{-4C_F}{1 - w} \left[\alpha + \bar{\alpha} \ln \bar{\alpha} \right] X,$$

$$B[H^-](w) \stackrel{w \rightarrow 1}{\sim} \frac{-4C_F}{1 - w} \left[\alpha + \bar{\alpha} \ln \bar{\alpha} + \alpha \bar{\alpha} \right] X. \quad (48)$$

We remind our notation $\bar{\alpha} = 1 - \alpha$. These expressions present our main result.

Renormalon singularities at $w = n$ ($n = 2, 3, \dots$) have a generic form

$$B[H](w) \sim \frac{2C_F}{n - w} \left[\alpha^n p_{n-1}(\alpha) + \frac{(-1)^n \delta(\bar{\alpha})}{n!(n-2) \ln^2(2n-1)} \right] X^w, \quad (49)$$

where $p_n(\alpha)$ is a polynomial of order n , e.g. $p_1(\alpha) = (5\alpha - 3)/6$, $p_2(\alpha) = (\alpha^2 - 25\alpha + 20)/180$, etc.

IV. POWER CORRECTIONS

A singularity on the integration path in Eq. (34) means that the perturbation theory is incomplete and the sum

of the series is ill-defined. It is customary [28] to estimate the corresponding ambiguity as

$$\delta H(w_0) = -\pi \frac{1}{\beta_0} e^{-w_0/(\beta_0 \mu^2)} \text{Res}_{w=w_0} [B[H](w)], \quad (50)$$

where w_0 is the position of the singularity and $\text{Res}_{w=w_0} [B[H](w)]$ is the corresponding residue. Note that $e^{-w_0/(\beta_0 \mu^2)} = (\Lambda^2/\mu^2)w_0$. Following the standard logic [28, 29] we assume that this ambiguity must be canceled by adding a non-perturbative correction of the same order of magnitude.

A. Ioffe time quasi-distributions

Considering $\delta H(1)$, we obtain the leading power correction to the qITDs as functions of the “Ioffe-time”

$$\tau = (p \cdot v)z \quad (51)$$

$$\begin{aligned} \mathcal{I}^1(\tau) &= I(\tau) + \kappa(v^2 \Lambda^2) \int_0^1 d\alpha (\alpha + \bar{\alpha} \ln \bar{\alpha}) I(\alpha\tau), \\ \mathcal{I}^2(\tau) &= I(\tau) + \kappa(v^2 \Lambda^2) \int_0^1 d\alpha (\alpha + \bar{\alpha} \ln \bar{\alpha} + \alpha \bar{\alpha}) I(\alpha\tau), \end{aligned} \quad (52)$$

where κ is a real number of order one. The renormalon ambiguity (48) corresponds to

$$|\kappa| = -\pi \frac{1}{\beta_0} \left(-\frac{e^{5/3}}{4e^{-2/3}\pi} \right) \approx \frac{\pi e^{5/3+2/3\pi}}{4\beta_0} \gtrsim 1.5, \quad (53)$$

but this number is only indicative. Alternatively, one can put $\kappa \sim 1$ and think of Λ as a certain nonperturbative parameter of the order of Λ_{QCD} that determines an overall normalization of the power correction and cannot be fixed in this approach. Note that also the sign of the correction is not determined.

For the normalized qITDs defined in Eq. (13) we obtain instead

$$\begin{aligned} \tilde{\mathcal{I}}^0(\tau) &= I(\tau) + \kappa(v^2 \Lambda^2) \int_0^1 d\alpha [\alpha + \bar{\alpha} \ln \bar{\alpha}] I(\alpha\tau), \\ \tilde{\mathcal{I}}^1(\tau) &= I(\tau) + \kappa(v^2 \Lambda^2) \int_0^1 d\alpha [\alpha + \bar{\alpha} \ln \bar{\alpha} + \alpha \bar{\alpha}] I(\alpha\tau), \end{aligned} \quad (54)$$

where the “plus” distribution is defined as usual,

$$[f(\alpha)]_+ = f(\alpha) - \delta(\alpha) \int_0^1 d\beta f(\beta). \quad (55)$$

The leading power corrections to the qITDs $\tilde{\mathcal{I}}^{(i)}$ (14) that are normalized to the vacuum correlator, are the same as for the un-normalized distributions (52). The expressions in (52) and (54) present the starting point for the following analysis.

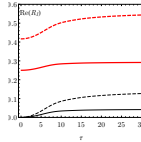


FIG. 3. Real parts of $\mathcal{R}_2(\tau)$ (thick, red) and $\mathcal{R}_3(\tau)$ (thin, black) for the simple quark PDF model in Eq. (60). Solid curves are for $\text{Re } R^i$, dashed curves for $\text{Re } R^i$.

In order to visualize the functional dependence of the power correction on the “Ioffe time” τ relative to the leading-twist result, we write

$$\begin{aligned} \mathcal{I} &= I(\tau) \left\{ 1 + \kappa(v^2 \Lambda^2) \mathcal{R}_2(\tau) \right\}, \\ \mathcal{I} &= I(\tau) \left\{ 1 + \kappa(v^2 \Lambda^2) \mathcal{R}_3(\tau) \right\}, \end{aligned} \quad (56)$$

where

$$\begin{aligned} \mathcal{R}_2^1(\tau) &= \frac{1}{I(\tau)} \int_0^1 d\alpha (\alpha + \bar{\alpha} \ln \bar{\alpha}) I(\alpha\tau), \\ \mathcal{R}_2^2(\tau) &= \frac{1}{I(\tau)} \int_0^1 d\alpha (\alpha + \bar{\alpha} \ln \bar{\alpha} + \alpha \bar{\alpha}) I(\alpha\tau). \end{aligned} \quad (57)$$

For the normalized qITDs one obtains

$$\mathcal{R}_2^1(\tau) = \mathcal{R}_2^1(\tau) - \frac{1}{4}, \quad \mathcal{R}_2^2(\tau) = \mathcal{R}_2^2(\tau) - \frac{5}{12}, \quad (58)$$

and, obviously,

$$\tilde{\mathcal{R}}_2^{(i)}(\tau) = \mathcal{R}_2^{(i)}(\tau), \quad (59)$$

so that we do not consider them separately.

In general, the qITDs $\mathcal{I}(\tau)$ and the higher-twist coefficients $\mathcal{R}_2(\tau)$ and $\mathcal{R}_3(\tau)$ are complex functions, but their imaginary parts appear to be small. The real parts, $\text{Re } \mathcal{R}_2(\tau)$ and $\text{Re } \mathcal{R}_3(\tau)$, for a simple model of the valence quark distribution

$$q(x) = x^{-1/2} (1-x)^3 \quad (60)$$

are plotted in Fig. 3. The power correction to the “transverse” qITD turns out to be roughly factor four larger than for the “longitudinal” qITD. In both cases the R -functions flatten out at large Ioffe times $\tau \gtrsim 10$ that are, however, hardly accessible in present day lattice calculations. The normalization to zero proton momentum corresponds to the subtraction of the value at $\tau = 0$, so that for $z \rightarrow 0$ there is no power correction by construction.

B. Parton quasi-distributions

Making the Fourier transformation of the above results for the qTTDs we obtain the qPDFs

$$\begin{aligned}\mathcal{Q}^l(x, p) &= q(x) - \frac{\kappa v^2 \Lambda^2}{(pv)^2} \left(\frac{d}{dx} \right)^2 \int_{|x|}^1 \frac{dy}{y} (y + \bar{y} \ln \bar{y}) q\left(\frac{x}{y}\right) \\ &= q(x) - \frac{\kappa v^2 \Lambda^2}{x^2 (pv)^2} \left\{ \int_{|x|}^1 \frac{dy}{y} \frac{y^2}{[1-y]} q\left(\frac{x}{y}\right) \right. \\ &\quad \left. + q(x) - |x|q'(x) \right\}\end{aligned}\quad (61)$$

and

$$\begin{aligned}\mathcal{Q}^t(x, p) &= q(x) - \frac{\kappa v^2 \Lambda^2}{(pv)^2} \left(\frac{d}{dx} \right)^2 \int_{|x|}^1 \frac{dy}{y} (y + \bar{y} \ln \bar{y} + y\bar{y}) q\left(\frac{x}{y}\right) \\ &= q(x) - \frac{\kappa v^2 \Lambda^2}{x^2 (pv)^2} \left\{ \int_{|x|}^1 \frac{dy}{y} \left[\frac{y^2}{[1-y]} - 2y \right] q\left(\frac{x}{y}\right) \right. \\ &\quad \left. + 2q(x) - |x|q'(x) \right\}\end{aligned}\quad (62)$$

Assuming for what follows $x > 0$, these expressions can be rewritten in the form

$$\mathcal{Q}^{l(t)}(x, p) = q(x) \left\{ 1 - \frac{v^2 \Lambda^2}{x^2 \bar{x} (pv)^2} \mathcal{R}_{\mathcal{Q}}^{l(t)}(x) \right\} \quad (63)$$

with

$$\begin{aligned}\mathcal{R}_{\mathcal{Q}}^l(x) &= \frac{\bar{x}}{q(x)} \left\{ \int_x^1 \frac{dy}{1-y} [y q\left(\frac{x}{y}\right) - q(x)] + q(x) - xq'(x) \right\}, \\ \mathcal{R}_{\mathcal{Q}}^t(x) &= \frac{\bar{x}}{q(x)} \left\{ \int_x^1 \frac{dy}{1-y} [y(2y-1)q\left(\frac{x}{y}\right) - q(x)] + 2q(x) \right. \\ &\quad \left. - xq'(x) \right\}.\end{aligned}\quad (64)$$

Note that we have extracted the prefactor $1/(x^2 \bar{x})$ for the power correction anticipating that it is enhanced as $1/x^2$ and $1/(1-x)$ in the regions of small $x \rightarrow 0$ and large $x \rightarrow 1$ Bjorken variable, respectively.

The normalized QPDFs $\mathcal{Q}^{l(t)}(x, p)$ are obtained by replacing the kernels in the first lines in Eqs. (61) and (62) by the plus distributions, cf. (54). Writing the result in the form

$$\mathcal{Q}^{l(t)}(x, p) = \hat{q}(x) \left\{ 1 - \frac{v^2 \Lambda^2}{x^2 \bar{x} (pv)^2} \mathcal{R}_{\mathcal{Q}}^{l(t)}(x) \right\}, \quad (65)$$

where $\hat{q}(x)$ is the quark PDF normalized to the unit integral, one obtains

$$\begin{aligned}\mathcal{R}_{\mathcal{Q}}^l(x) &= \mathcal{R}_{\mathcal{Q}}^l - \frac{1}{4} \frac{x \bar{x}^2}{q(x)} q''(x), \\ \mathcal{R}_{\mathcal{Q}}^t(x) &= \mathcal{R}_{\mathcal{Q}}^t - \frac{5}{12} \frac{x \bar{x}^2}{q(x)} q''(x).\end{aligned}\quad (66)$$

Note that the additional terms are proportional to the second derivative of the quark PDF and thus enhanced as $1/(1-x)^2$ at $x \rightarrow 1$. As already mentioned, the normalization to the vacuum correlator does not affect the $1/p^2$ power corrections so that $\mathcal{R}_{\mathcal{Q}}^{l(t)}(x) = \mathcal{R}_{\mathcal{Q}}^{l(t)}(x)$ and we do not need to consider this case separately.

For a numerical study we have used the MSTW NLO valence u - and d -quark distributions [68] at the scale 2 GeV and the simple model in Eq. (60). It turns out that the power corrections for (unnormalized) “longitudinal” and “transverse” qPDFs are similar in size so that we show $\mathcal{R}_{\mathcal{Q}}^l(x)$ on the left panel in Fig. 4, and the difference $\mathcal{R}_{\mathcal{Q}}^t(x) - \mathcal{R}_{\mathcal{Q}}^l(x)$ on the right panel.

The x -dependence of $\mathcal{R}_{\mathcal{Q}}(x)$ is similar for all quark PDF models: The power correction is small for $x \rightarrow 0$ (but non-zero, $\mathcal{R}_{\mathcal{Q}}^l(0) \sim 0.2$, $\mathcal{R}_{\mathcal{Q}}^t(0) \sim 0.4$) and increases steeply with x (almost linearly). For the u -quark, the result is very similar to the simple model in Eq. (60) whereas for the d -quark the power correction is, roughly, factor two larger. Note that the difference $\mathcal{R}_{\mathcal{Q}}^t(x) - \mathcal{R}_{\mathcal{Q}}^l(x)$ depends on the PDF model only very weakly.

Constructing the qPDFs from the qTTDs normalized to the value at zero momentum has a large effect. This is illustrated in Fig. 5 where in the upper panels we show the results for the “longitudinal” case and in the lower panels the difference between the “longitudinal” and “transverse” distributions. The three panels (from left to right) correspond to the MSTW valence u -quarks, d -quarks, and the simple model in Eq. (60), respectively. The red solid lines stand for $\mathcal{R}_{\mathcal{Q}}(x)$ (i.e. the same as in Fig. 4), and the result for the normalized qPDFs, $\mathcal{R}_{\mathcal{Q}}(x)$, is shown by the black dashed curves.

We see that the normalization to the qTTD at zero momentum significantly reduces the power correction at moderate values of $x \lesssim 0.5$ at the cost of dramatical increase at higher x values. This normalization procedure is, therefore, not suitable to access the large- x behavior of the PDFs, but apparently minimizes power corrections in the not-so-large x region.

C. Parton pseudo-distributions

Power corrections for the pPDF can be obtained easily from the corresponding expression for the “transverse” qTTDs (52), (54). Writing the result as

$$\mathcal{P}(x, z, \mu) = q(x) \left\{ 1 + (v^2 z^2 \Lambda^2) \theta(|x| < 1) \mathcal{R}_{\mathcal{P}}(x) \right\}, \quad (67)$$

and similarly for the pPDF normalized to zero momentum, $\mathbf{P}(x, z)$, we obtain

$$\begin{aligned}\mathcal{R}_{\mathcal{P}}(x) &= -\frac{1}{q(x)} \int_{|x|}^1 \frac{dy}{y} (y + \bar{y} \ln \bar{y} + y\bar{y}) q\left(\frac{x}{y}\right), \\ \mathbf{R}_{\mathcal{P}}(x) &= -\frac{1}{q(x)} \int_{|x|}^1 \frac{dy}{y} [y + \bar{y} \ln \bar{y} + y\bar{y}] \cdot q\left(\frac{x}{y}\right)\end{aligned}$$

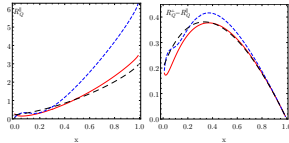


FIG. 4. $\mathcal{R}_u^V(x)$ (left panel) and $\mathcal{R}_d^V(x) - \mathcal{R}_u^V(x)$ (right panel) for MSTW valence u-quarks (long black dashes), d-quarks (short blue dashes) (at 2 GeV), and for the model in Eq. (60) (red solid curves).

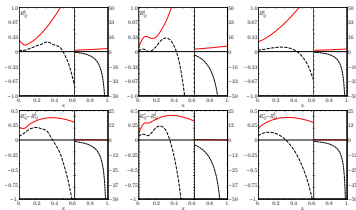


FIG. 5. Upper panels: $\mathcal{R}_u^V(x)$ (solid red curves) and $\mathcal{R}_d^V(x)$ (dashed black curves) for MSTW valence u-quarks (left), d-quarks (middle), both at 2 GeV, and for the model in Eq. (60) (right). Lower panels: the same for $\mathcal{R}_d^V(x) - \mathcal{R}_u^V(x)$ and $\mathcal{R}_d^V(x) - \mathcal{R}_u^V(x)$. Note split panels: the results for $0 < x < 0.6$ and $0.6 < x < 1$ are shown using a different scale on the vertical axis.

$$= \mathcal{R}_P(x) - 5/12. \quad (68)$$

The numerical results are shown in Fig. 6.

Note that $\mathcal{R}_P(x)$ is very similar for all considered models for the valence quark PDFs and decreases at $x \rightarrow 1$. Indeed, it is easy to see that $\mathcal{R}_P(x) \sim \mathcal{O}(1-x)$ in this limit, similar to the target mass correction, Eq. (30). This suppression is removed once pPDF is normalized to zero momentum [24] (which adds a negative constant), but can be upheld if normalized to the vacuum expectation value of the same operator.

V. CONCLUSIONS

We have presented an analysis of power-suppressed (higher-twist) contributions to qPDFs and pPDFs based on the study of factorial divergences (renormalons) in the corresponding coefficient functions within the bubble-chain approximation. Factorial asymptotic implies that the sum of the series is only defined to a power accuracy and therefore, the QCD perturbation theory must be corrected by nonperturbative power-suppressed con-

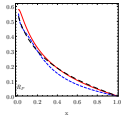


FIG. 6. Power correction to the pPDF $R_P(08)$ for MSTW valence u -quarks (long black dashes), d -quarks (short black dashes), both at 2 GeV, and for the simple model $q(x) = x^{-1/2}(1-x)^3$ (red solid curve).

tributions to produce unambiguous predictions. Our results have to be considered as a “minimal model” for the higher-twist corrections that captures effects that are necessary for the selfconsistency of the theory, but possibly misses other nonperturbative corrections that are, e.g. protected by symmetries and not “seen” through perturbative expansions. Our main conclusions are as follows:

- Position space PDFs (qTDFs) have flat power corrections at large loft times. Generally, power corrections are much larger for the “transverse” projection as compared to the “longitudinal” projection.
- Power corrections for qPDFs have a generic behavior

$$\mathcal{Q}(x, p) \sim q(x) \left(1 + \mathcal{O} \left(\frac{\Lambda^2}{p^2} \frac{1}{x^2(1-x)} \right) \right). \quad (69)$$

Note that the corresponding target mass corrections, (27) and (29), do not show up the $1/x^2$ enhancement. This behavior is commensurate to the suppression of Nachtmann’s target mass corrections $\sim x^2 m^2/Q^2$ at small x for the DIS structure functions. The normalization of the underlying qTDFs to unity at the zero momentum considerably reduces the size the power correction to the qPDFs at $x \lesssim 0.5$ at the cost of a strong enhancement at $x \gtrsim 0.5 - 0.6$. Thus, such a normalization procedure is not suitable to access the large- x behavior of the PDFs, but apparently minimizes power corrections in the intermediate x -region. Note that the above discussion is based on a normalization procedure different from the nonperturbative renormalization used in ref. [17], thus the power corrections for the latter might behave differently from what has been shown here.

- Power corrections for pPDFs have a generic behavior

$$\mathcal{P}(x, z) \sim q(x) \left\{ 1 + \mathcal{O} \left(z^2 \Lambda^2 (1-x) \right) \right\}. \quad (70)$$

The suppression at $x \rightarrow 1$ is lifted by the zero-momentum normalization factor. However, it can be upheld by the normalization to the vacuum matrix element of the same operator. We conclude that pPDFs can offer an interesting alternative to qPDFs for the study of large- x region.

As a byproduct of this study, we provide the results for the coefficient functions in the large- n_f approximation, terms $\sim n_f^2 \alpha_s^{k+1}$, to all orders in perturbation theory. These results can be useful to estimate the effects of uncalculated higher orders and scale-setting using the BLM-type procedure. The corresponding analysis goes beyond the scope of this paper and will be presented in a future publication.

ACKNOWLEDGEMENTS

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Appendix A: Leading-twist coefficient function in the bubble-chain approximation

In this appendix we present some details of the calculation of coefficient function H in the bubble-chain approximation. The calculation is performed by explicit evaluation and renormalization of diagrams with n -insertions of bubbles. The evaluation closely follows the one presented in details in refs.[30, 36], with the main difference being that it is made directly in the coordinate space.

To obtain the twist-two coefficient function in $\overline{\text{MS}}$ -scheme for the operator

$$O_F^a(z, 0) \sim \bar{q}(zv) \gamma^\mu [zv, 0] q(0), \quad (A1)$$

it is sufficient to calculate its free-quark matrix element (with massless on-shell quarks). Since there is only a single scale z^2 , the expression for diagrams have simple structure that contains poles in ϵ and $\ln(z^2 \mu^2)$. The poles in ϵ correspond to UV poles (to be renormalized) and collinear poles (to be incorporated into the definition of collinear PDF). Therefore, the coefficient function is given by the finite part of the (renormalized) diagrams (for a more detailed discussion see [36, 69]).

The diagrams contributing to the large- n_f limit are presented in Fig.1. The (bare) propagator with the insertion of n -fermion loops in the coordinate representation reads

$$\Delta_{F,0}^{\mu\nu} = -\frac{1}{8\pi^{d/2}} \bar{\Gamma}_n \frac{\Gamma(2 - (n+1)\epsilon)}{\Gamma(2 + n\epsilon)} \quad (A2)$$

$$\times \left[\frac{1 + 2n\epsilon}{1 - (n+1)\epsilon} g^{\mu\nu} + 2 \frac{x^\mu x^\nu}{x^2} \right] \frac{1}{(-x^2 + i0)^{1-(n+1)\epsilon}},$$

where $d = 4 - 2\epsilon$ is the dimension of space-time in the dimension regularization,

$$R_\epsilon = -\frac{2n_f}{3} \frac{a_\epsilon}{\epsilon} \frac{6\Gamma(1+\epsilon)\Gamma^2(2-\epsilon)}{4\Gamma(4-2\epsilon)}, \quad (A3)$$

and $a_\epsilon = g^2/(4\pi)^{d/2}$. One can check explicitly that this propagator satisfies the integral equation

$$\Delta_n^{\mu\nu}(x) = \int d^d z \Delta_{n-n}(x-z) \Delta_n^{\mu\nu}(z), \quad (A4)$$

with $1 \leq n \leq n-1$. This equation reflects the fact that the composition of two bubble-chains is again a bubble-chain.

The ϵ -poles in Δ_n correspond to UV poles of fermion loops, and to be renormalized later.

The leading large- n_f diagrams structurally reproduce the one-loop diagrams calculated in the Landau gauge. Due to the gauge invariance, the transverse part of the gluon propagator cancels in the sum of diagrams, and therefore, the complete result can be obtained by the use of only longitudinal part, which significantly simplifies the calculation. The reduced propagator reads

$$\tilde{\Delta}_n^{\mu\nu} = \frac{-1}{4\pi^{d/2}} \frac{\Gamma(1-(n+1)\epsilon)}{\Gamma(1+n\epsilon)} \frac{g^{\mu\nu} R_n^\epsilon}{(-x^2 + i0)^{1-(n+1)\epsilon}}. \quad (A5)$$

The expressions for diagrams evaluated with this propagator are (notation corresponds to Fig.1)

$$(a)^\mu = 2a_\epsilon C_F \frac{R_n^\epsilon}{4\epsilon} (-v^2 z^2)^{(n+1)\epsilon} \frac{\Gamma(-(n+1)\epsilon)}{\Gamma(2+n\epsilon)} \quad (A6)$$

$$\times \left\{ \int_0^1 (O_n^\epsilon(z, \alpha z) - O_n^\epsilon(z, 0))(1-\epsilon)^{\alpha^{1+n\epsilon}} {}_2F_1(1, 2-(n+2)\epsilon, 2+n\epsilon; \alpha) d\alpha - \frac{1}{2} O_n^\epsilon(z, 0) \right\},$$

$$(b)^\mu = 2a_\epsilon C_F \frac{R_n^\epsilon}{4\epsilon} (-v^2 z^2)^{(n+1)\epsilon} \frac{\Gamma(-(n+1)\epsilon)}{\Gamma(2+n\epsilon)} \quad (A7)$$

$$\times \left\{ \int_0^1 (O_n^\epsilon(\hat{\alpha} z, 0) - O_n^\epsilon(z, 0))(1-\epsilon)^{\alpha^{1+n\epsilon}} {}_2F_1(1, 2-(n+2)\epsilon, 2+n\epsilon; \alpha) d\alpha - \frac{1}{2} O_n^\epsilon(z, 0) \right\},$$

$$(c)^\mu = 2a_\epsilon C_F \frac{R_n^\epsilon}{4\epsilon} (-v^2 z^2)^{(n+1)\epsilon} \frac{\Gamma(-(n+1)\epsilon)}{\Gamma(1+n\epsilon)} \quad (A8)$$

$$\times \int_0^1 d\alpha \int_0^{\hat{\alpha}} d\beta (1-\alpha-\beta)^{n\epsilon} \left[(1+n\epsilon) O_n^\epsilon(\hat{\alpha} z, \beta z) - 2(n+1)\epsilon \frac{v^\mu v_\nu}{v^2} O_n^\epsilon(\hat{\alpha} z, \beta z) \right],$$

$$(d)^\mu = -2a_\epsilon C_F \frac{R_n^\epsilon}{4\epsilon} (-v^2 z^2)^{(n+1)\epsilon} \frac{\Gamma(-(n+1)\epsilon)}{\Gamma(1+n\epsilon)} \frac{O_n^\epsilon(z, 0)}{1-2(n+1)\epsilon}. \quad (A9)$$

The renormalization of these diagrams is straightforward and discussed in details e.g. in ref.[30]. Next, we present the diagram-by-diagram renormalized expression for the coefficient function. One obtains

$$(a)_{ren}^\mu = \frac{2C_F}{\beta_0} (\beta_0 a_\epsilon)^{n+1} \left\{ \int_0^1 \frac{d\alpha d\tau}{t} (O_n^\epsilon(z, t\alpha z) - O_n^\epsilon(z, 0)) \left[\frac{r_{n+1}(\hat{\alpha}) + r_n(\hat{\alpha})}{n+1} + (-1)^n n! y_0^{[n+1]}(\hat{\alpha} t^2 x^2) \right] \right. \\ \left. - \frac{1}{2} O_n^\epsilon(z, 0) \left[\frac{h_{n+1}^{[0]}}{n+1} + (-1)^n n! y_0^{[n+1]}(x^2) \right] \right\}, \quad (A10)$$

$$(b)_{ren}^\mu = \frac{2C_F}{\beta_0} (\beta_0 a_\epsilon)^{n+1} \left\{ \int_0^1 \frac{d\alpha d\tau}{t} (O_n^\epsilon((1-t\alpha)z, 0) - O_n^\epsilon(z, 0)) \left[\frac{r_{n+1}(\hat{\alpha}) + r_n(\hat{\alpha})}{n+1} + (-1)^n n! y_0^{[n+1]}(\hat{\alpha} t^2 x^2) \right] \right. \\ \left. - \frac{1}{2} O_n^\epsilon(z, 0) \left[\frac{h_{n+1}^{[0]}}{n+1} + (-1)^n n! y_0^{[n+1]}(x^2) \right] \right\}, \quad (A11)$$

$$(c)_{ren}^\mu = -\frac{2C_F}{\beta_0} (\beta_0 a_\epsilon)^{n+1} \int_0^1 d\alpha \int_0^{\hat{\alpha}} d\beta \left\{ \frac{r_{n+1}(\gamma) + 2r_n(\gamma) + r_{n-1}(\gamma)}{n+1} \right. \\ \left. + (-1)^n n! y_0^{[n+1]}(\gamma x^2) - (-1)^n n! y_0^{[n]}(\gamma x^2) \right\} O_n^\epsilon(\hat{\alpha} z, \beta z) + 2(-1)^n n! y_0^{[n]}(\gamma x^2) \frac{v^\mu v_\nu}{v^2} O_n^\epsilon(\hat{\alpha} z, \beta z) \Big\}, \quad (A12)$$

$$(d)_{ren}^\mu = \frac{2C_F}{\beta_0} (\beta_0 a_\epsilon)^{n+1} \left\{ \frac{y_{n+1}^{[0]}}{n+1} + (-1)^n n! y_0^{[n+1]}(x^2) \right\} O_n^\epsilon(z, 0), \quad (A13)$$

where $x^2 = -z^2 v^2 \mu^2/4$, $\gamma = 1 - \alpha - \beta$. In these expressions

we have promoted the factors $(-2n_f/3)$ to the β_0 coeffi-

cient of QCD, $\beta_0 = 11/3C_A - 2n_f/3$. The functions $g_j^{[n]}$, $h_j^{[n]}$, etc., are defined as coefficients of expansion

$$G(\epsilon, s, x^2) = \sum_{k,j=0}^{\infty} g_k^{[j]}(x^2) s^k \epsilon^j, \quad (A14)$$

$$\frac{G(\epsilon, s, x^2)}{1-s\epsilon} = \sum_{k,j=0}^{\infty} h_k^{[j]}(x^2) s^k \epsilon^j, \quad (A15)$$

$$\frac{G(\epsilon, s, x^2)}{1+2s} = \sum_{k,j=0}^{\infty} y_k^{[j]}(x^2) s^k \epsilon^j, \quad (A16)$$

$$x^2 G(\epsilon, 0, x^2) = \sum_{k=0}^{\infty} r_k(x) \epsilon^k, \quad (A17)$$

where

$$G(\epsilon, s, x^2) = (x^2)^{-s} \frac{\Gamma(1+s)}{\Gamma(1-s+\epsilon)} \times \left(\frac{6\Gamma(1-\epsilon)\Gamma^2(2+\epsilon)}{\Gamma(4+2\epsilon)} \right)^{s/\epsilon-1}.$$

Note, that operators on the r.h.s. of (A10-A13) are renormalized operators. Taking the hadronic matrix element of these diagrams and performing the Borel summation we obtain the final result given in Eq. (37).

The factorial divergences $\sim \beta_0^n n!$ that correspond to Borel non-summable renormalons can be inferred directly from the expressions in (A10-A13) by inspection. These terms give rise to the renormalon contribution in (37). The remaining terms are regular and contribute to the function $\tilde{R}(w)$ in Eq. (37). Finally, specializing to the particular n values one obtains the $\sim a_s^{n+1} n_f^n$ contributions to the coefficient function. At $n \rightarrow 0$ one obtains the NLO coefficient function, quoted in (42). The $n = 1$ result corresponds to the n_f part of the NNLO coefficient function, Eq. (44).

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P49: 伪部分子分布的单圈演化

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阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

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One-loop evolution of parton pseudo-distribution functions on the lattice

Anatoly Radyushkin^{1,2}

¹Physics Department, Old Dominion University, Norfolk, VA 23529, USA

²Thomas Jefferson National Accelerator Facility, Newport News, VA 23606, USA

We incorporate recent calculations of one-loop corrections for the reduced Ioffe-time pseudo-distribution $\mathfrak{M}(\nu, z_\perp^2)$ to extend the leading-logarithm analysis of lattice data obtained by Orginos et al. We observe that the one-loop corrections contain a large term reflecting the fact that effective distances involved in the most important diagrams are much smaller than the nominal distance z_3 . The large correction in this case may be absorbed into the evolution term, and the perturbative expansion used for extraction of parton densities at the $\mu \approx 2$ GeV scale is under control. The extracted parton distribution is rather close to global fits in the $x > 0.1$ region, but deviates from them for $x < 0.1$.

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I. INTRODUCTION

Feynman's parton distribution functions (PDFs) [1] $f(x)$ are the crucial building blocks in the description of hard inclusive processes in quantum chromodynamics (QCD). Accumulating nonperturbative information about the hadron structure, the PDFs are a natural subject for a lattice study. However, straightforward definitions of PDFs refer to matrix elements of bilocal operators on the light cone $z^2 = 0$, the intervals inaccessible on the Euclidean lattice.

The ideas of how to get information from space-like intervals date to the pioneering paper of W. Detmold and D. Lin [2] who proposed a lattice study of the deep-inelastic-type Euclidean correlators of heavy-light currents. Later, V. Braun and D. Müller [3] proposed to use Euclidean correlators to extract the pion distribution amplitude, another function [4] playing a fundamental role in perturbative QCD studies of hard exclusive processes. The use of correlators in the form of "lattice cross sections" was more recently advocated in the papers by Qin and collaborators [5, 6].

The current correlators involve a quark propagator connecting the current vertices. This factor is avoided in the proposal by X. Ji [7] to study the quasi-PDFs $Q(y, p_3)$ that describe the distribution of the spatial z_3 -component of the hadron momentum p_3 . While being different from the Feynman PDFs $f(y)$ describing the distribution of the hadron's "plus"-momentum $p_+ = p_0 + p_3$, they coincide with $f(y)$ in the infinite momentum limit $p_3 \rightarrow \infty$.

Both on the lattice and in the usual continuum space, the basic object for all types of PDFs is the matrix element $M(z, p)$ generically (i.e. ignoring the inessential spin complications) written as $\langle p | \phi(0) \phi(z) | p \rangle$. By Lorentz invariance, it is a function of the Ioffe time $(pz) = -\nu$ [8] and the interval z^2 , $M(z, p) \equiv \mathcal{M}(\nu, -z^2)$.

In the (formal) light-cone limit $z^2 = 0$, the Fourier transform of $\mathcal{M}(\nu, 0)$ with respect to ν gives $f(x)$. In this sense, the ν -dependence of the Ioffe-time distribution (ITD) $\mathcal{M}(\nu, -z^2)$ reflects the longitudinal structure of the PDFs. As shown in Ref. [9], the z^2 -dependence

of $\mathcal{M}(\nu, -z^2)$ determines the $k_\perp$ -dependence of the (straight-link in the case of QCD) transverse momentum dependent parton distributions (TMDs) $\mathcal{F}(x, k_\perp)$.

Since the quasi-PDFs $Q(y, p_3)$ are given by the Fourier transform of $\mathcal{M}(z_3 p_3, z_\perp^2)$ with respect to z_3 , their shape is distorted by nonperturbative transverse momentum effects entering through the second argument of $\mathcal{M}(z_3 p_3, z_\perp^2)$. While, in a general perspective, the $k_\perp$ -dependence of $\mathcal{F}(x, k_\perp)$ provides information about the three-dimensional structure of hadrons, in the case of the quasi-PDFs it is a nuisance responsible for the unwanted difference between $Q(y, p_3)$ and $f(y)$ that is very strong at momenta reached in existing lattice calculations of quasi-PDFs.

To decrease the impact of the z^2 -dependence of the ITD $\mathcal{M}(\nu, -z^2)$, it was proposed [10] to consider the reduced ITD $\mathfrak{M}(\nu, -z^2)$ given by the ratio of $\mathcal{M}(\nu, -z^2)$ and the rest-frame distribution $\mathcal{M}(0, -z^2)$. Though there are no first-principle grounds that the nonperturbative part of the z^2 -dependence disappears in this ratio, it is natural to expect that it is strongly reduced.

The ideal case when $\mathfrak{M}(\nu, -z^2)$ is just a function of ν corresponds to factorization of the x and $k_\perp$ dependencies of the TMD $\mathcal{F}(x, k_\perp)$. In fact, the idea that $\mathcal{F}(x, k_\perp) = f(x)K(k_\perp)$ in the soft region $k_\perp^2 \lesssim 1$ GeV² is a standard assumption of the TMD practitioners (see, e.g., Ref. [11]), with a Gaussian being the most popular form for $K(k_\perp)$.

An exploratory lattice study of the reduced ITD was performed in Ref. [12] (and also described in Ref. [13]). The results show that $\mathfrak{M}(\nu, z_\perp^2)$ is basically a universal function of ν , with small deviations from the common curve for the points corresponding to the smallest values of z_3 .

As demonstrated in Ref. [12], these deviations may be explained by perturbative evolution. While the leading logarithmic approximation (LLA) used in Ref. [12] is sufficient to analyze the $\ln z_\perp^2$ dependence, one needs to go beyond it to specify the scale μ which should be attributed to the extracted scale-dependent PDFs $f(x, \mu^2)$.

To this end, one needs complete expressions for one-

loop corrections to ITDs. Recently, such calculations have been reported in Refs. [14, 15]. Our goal here is to give a more detailed discussion of the LLA treatment of the evolution, and also to extend the analysis beyond the LLA. As we will show, the one-loop correction contains a large contribution that considerably changes the results obtained in the LLA.

To make this article self-contained, we outline in Sec. II the basics of the Ioffe-time distributions and pseudo-PDFs. In Sec. III, we discuss the structure of one-loop corrections. In Sec. IV, we describe the evolution effects revealed in the lattice study of Ref. [12], and convert the data for the reduced ITD $\mathfrak{M}(\nu, z_3^2)$ into the standard parton densities $f(x, \mu^2)$ defined in the $\overline{\text{MS}}$ scheme. The summary of the paper and conclusions are given in Sec. V.

II. IOFFE-TIME DISTRIBUTIONS AND PSEUDO-PDFs

The basic object for defining parton distributions is a matrix element of a bilocal operator that (skipping inessential details of its spin structure) may be written generically like $\langle p | \phi(0) \phi(z) | p \rangle$. Due to invariance under Lorentz transformations, it is given by a function of two scalars, the *Ioffe time* (pz) [8] (which will be denoted by $-\nu$) and the interval z^2

$$\langle p | \phi(0) \phi(z) | p \rangle = \mathcal{M}(-\nu, -z^2) = \mathcal{M}(\nu, -z^2) \quad (1)$$

(again, the sign for the second argument is chosen so as to have a positive value for spacelike z). One can demonstrate [9, 16] that, for all relevant Feynman diagrams, its Fourier transform $\mathcal{P}(x, -z^2)$ with respect to (pz) has $-1 \leq x \leq 1$ as support, i.e.,

$$\mathcal{M}(-\nu, -z^2) = \int_{-1}^1 dx e^{-ix\nu} \mathcal{P}(x, -z^2). \quad (2)$$

In this covariant definition of x , one does not need to assume that z is on the light cone $z^2 = 0$ or that p is light-like $p^2 = 0$.

On the light cone $z^2 = 0$, we formally have $\mathcal{P}(x, 0) = f(x)$. Hence, the function $\mathcal{P}(x, -z^2)$ may be treated as a generalization of the concept of PDFs onto non-lightlike intervals z^2 , and following [10], we will refer to it as the *pseudo-PDF*. In view of lattice applications, we will take the separation $z = \{0, 0, 0, z_3\}$ oriented in the direction specified by the hadron momentum $p = \{E, 0, 0, P\}$.

In renormalizable theories (including QCD), the function $\mathcal{M}(\nu, -z^2)$ has logarithmic $\sim \ln(-z^2)$ singularities. In deep inelastic scattering (DIS), they result in a logarithmic scaling violation with respect to the photon virtuality Q^2 . A wide-spread statement is that the Q^2 -dependent DIS structure functions $W(x_B, Q^2)$ probe the hadron structure at distances $\sim 1/Q$. In the case of

the pseudo-PDFs $\mathcal{P}(x, z_3^2)$, one may say that they literally describe the hadron structure at the distance z_3 .

Just like the DIS form factors $W(x_B, Q^2)$ are written in terms of the universal parton densities $f(x, Q^2)$, the pseudo-PDFs obtained from lattice calculations may be expressed through the usual parton distributions. The latter are defined by the operators on the light cone $z^2 = 0$, i.e., in a logarithmically singular limit. In the approach based on the operator product expansion (OPE), the standard procedure is to remove these singularities with the help of some prescription.

The most popular of them is the $\overline{\text{MS}}$ scheme based on the dimensional regularization. Consequently, the resulting PDFs have a dependence on the renormalization scale μ , and therefore one should write the PDFs as $f(x, \mu^2)$. Switching from x to the Ioffe time ν gives the functions

$$\mathcal{I}(\nu, \mu^2) = \int_{-1}^1 dx e^{ix\nu} f(x, \mu^2) \quad (3)$$

introduced in Ref. [17] and called there the *Ioffe-time distributions*. In this context, the functions $\mathcal{M}(\nu, -z^2)$ that are the Fourier transforms of pseudo-PDFs, should be called the Ioffe-time *pseudo-distributions* or *pseudo-ITDs*.

To get a relation between the pseudo-PDFs $\mathcal{P}(x, z_3^2)$ and the $\overline{\text{MS}}$ parton densities $f(x, \mu^2)$, one can use the nonlocal light-cone OPE [18, 19] (see also [15]) for the matrix element defining $\mathcal{P}(x, z_3^2)$, i.e., for the pseudo-ITD. The result

$$\mathcal{M}(\nu, -z^2) = \sum_i \int_{-1}^1 dw C_i(w, z^2 \mu^2, \alpha_s) \mathcal{I}_i(w\nu, \mu^2) + \mathcal{O}(z^2), \quad (4)$$

has the structure similar to that of the usual OPE for the DIS structure functions $W(x, Q^2)$. In this expression, the twist-2 coefficient functions C_i are given by an expansion in the strong coupling constant α_s , while $\mathcal{O}(z^2)$ symbolizes higher-twist terms.

However, the application of the OPE to the pseudo-ITDs and pseudo-PDFs in QCD faces complications related to the gauge link. Namely, when z is off the light cone, the link generates linear $\sim z_3/a$ and logarithmic $\sim \ln(1 + z_3^2/a^2)$ ultraviolet (UV) divergences, where a is an UV regulator with the dimension of length (it may be a finite lattice spacing). Though disappearing for $z_3 = 0$, these divergences require an additional UV regularization when z_3 is finite.

Fortunately, these divergences are multiplicative [20–24] (see also recent Refs. [25–27]), and cancel in the ratio, the reduced Ioffe-time distribution,

$$\mathfrak{M}(\nu, z_3^2) = \frac{\mathcal{M}(\nu, z_3^2)}{\mathcal{M}(0, z_3^2)}, \quad (5)$$

introduced in our paper [10], and partially motivated by this cancellation. The remaining $\ln z_3^2$ singularities,

present only in the numerator of the ratio, are described by the nonlocal light-cone OPE.

As stated in Ref. [12], for small spacelike intervals $z^2 = -z_3^2$, and at the leading logarithm level, the reduced pseudo-PDFs are related to the $\overline{\text{MS}}$ distributions by a simple rescaling of their second arguments, namely,

$$\mu^2 = 4e^{-\gamma_E} / z_3^2, \quad (6)$$

where γ_E is the Euler's constant (a more detailed discussion will be given later on). This rescaling factor is very close to 1, since $2e^{-\gamma_E} = 1.12$. However, this factor may be changed by the $\mathcal{O}(\alpha_s)$ terms present in the coefficient function.

III. STRUCTURE OF ONE-LOOP CORRECTIONS

A. Confinement and infrared cut-offs

There are several standard techniques to calculate gluon radiative corrections in QCD. Most of them are oriented to work in the region of absolute perturbative QCD (pQCD) applicability. A straightforward use of such methods, however, may need some care in applications involving energy scales that are not very large. For this reason, let us discuss some features of calculations on the border of applicability of perturbative methods.

To begin with, one should remember that quarks and gluons are confined, i.e. the propagators of all diagrams (even in a continuum case) are embedded in a finite volume whose size is determined by the hadron's radius. The confinement effects lead, in particular, to a rapid decrease of correlators like ITDs or pseudo-PDFs at distances z_3 larger than the hadronic radius R . Still, at short distances one can use asymptotic freedom and obtain, in particular, the $\ln z_3^2$ singularities.

Thus, it makes sense to treat pseudo-ITDs and pseudo-PDFs as sums of the soft and hard parts. The soft part basically reflects the size of the system and is assumed to be finite for $z_3 = 0$. The hard part is singular for $z_3 \rightarrow 0$, and is produced by perturbative interactions. The hard part may be visualized then as generated from the soft part through a hard exchange kernel $H(0, z; z_1, z_2)$,

$$\begin{aligned} \mathcal{M}^{\text{hard}}(\nu, -z^2 = z_3^2) \\ = \int d^4 z_1 d^4 z_2 H(0, z; z_1, z_2) \mathcal{M}^{\text{soft}}(z_1, z_2). \end{aligned} \quad (7)$$

In the standard pQCD factorization approaches, the soft part is mimicked by on-shell parton states, and the $\ln z_3^2$ -singularities appear either as $\ln(z_3^2 m^2)$, where m is the parton mass or $\ln(z_3^2 \mu_{\text{IR}}^2)$, where μ_{IR} is the scale used in dimensional regularization of infrared singularities in the case of massless partons.

Since $\mathcal{M}^{\text{soft}}(z_1, z_2)$ in Eq. (7) rapidly decreases for large separations $|z_1 - z_2|$, the hadronic size R provides an infrared cut-off for the integral, even when the quarks are

massless. While at short distances one gets the $\ln(z_3^2/R^2)$ behavior, the logarithmic form is just an approximation valid for $z_3 \ll R$. Such a restriction may be hard to implement on the lattice.

Of course, the exact form of the IR regularization imposed by confinement is not known. To get a feeling, let us take an infrared regularization by a mass term. A typical integral producing the $\ln z_3^2$ singularity then has the form

$$I_K(z_3^2) = \int_0^\infty \frac{d\alpha}{\alpha} e^{-z_3^2/4\alpha - \alpha m^2}, \quad (8)$$

where α is the Schwinger's α -parameter and m is the infrared regulator. One can see that

$$\begin{aligned} I_K(z_3^2) &= 2K_0(mz_3) \\ &= -\ln(m^2 z_3^2) + 2\ln(2e^{-\gamma_E}) + \mathcal{O}(z_3^2), \end{aligned} \quad (9)$$

where $K_0(mz_3)$ is the modified Bessel function. Its expansion for small z_3 explicitly shows the expected $\ln(z_3^2 m^2)$ singularity.

The usual pQCD factorization procedure is to split $\ln(z_3/R)$ into the short-distance part $\ln(z_3\mu)$ that is attributed to the coefficient function and the long-distance part $\ln(1/\mu R)$ that is absorbed into the "renormalized" PDF $f(x, \mu^2)$. Given the commonly used lattice spacing $a \sim 0.1$ fm and the hadron size $R \lesssim 1$ fm, the question is whether there is enough interval for the logarithmic part of the z_3 -dependence to be visible in the data at all.

An important feature of the Bessel function $K_0(mz_3)$ is that it exponentially decreases when z_3 exceeds the infrared cut-off $1/m$. Thus, if instead of the short-distance approximation of $I_K(z_3^2)$ by $\ln(1/z_3^2)$, one would use the "exact" $I_K(z_3^2)$ function for the evolution term, there will be no evolution corrections for large z_3 . In other words, the logarithmic evolution disappears at large distances.

B. Rescaling relation

To fix a relation between the pseudo-PDF scale z_3 and the $\overline{\text{MS}}$ scale μ , one should take into account constant terms, like $2\ln(2e^{-\gamma_E})$ in Eq. (9). In the $\overline{\text{MS}}$ -OPE approach, one takes $z^2 = 0$ and then applies the dimensional regularization which adds the α' factor into the integral (8) making it convergent. After that, one uses the $\overline{\text{MS}}$ -prescription, which is arranged to produce exactly $\ln(\mu^2/m^2)$ as the result in this case,

$$\begin{aligned} I_{Dm}(\mu^2) &= \int_0^\infty \frac{d\alpha}{\alpha} (\alpha\mu^2 e^{\gamma_E})^\epsilon e^{-\alpha m^2} = \Gamma(\epsilon) \left(\frac{\mu^2 e^{\gamma_E}}{m^2} \right)^\epsilon \\ &\rightarrow \frac{1}{\epsilon} + \ln(\mu^2/m^2). \end{aligned} \quad (10)$$

Thus, the constant term in Eq. (9) provides the leading-logarithm rescaling coefficient $2e^{-\gamma_E}$ between the pseudo-PDFs and $\overline{\text{MS}}$ parton distributions expressed by Eq. (6).

One may ask what happens if one uses another type of the IR regularization. In particular, the Gaussian models for TMDs suggest that the decrease for large z_3 is also Gaussian. One may expect that the hard correction should resemble, for large z_3 , the behavior of the soft part. Thus, the exponential $e^{-m|z_3|}$ fall-off of the modified Bessel function may look too slow. A Gaussian decrease can be easily provided by a sharp IR cut-off

$$I_G(z_3^2) = \int_0^{z_3^2/4} \frac{d\alpha}{\alpha} e^{-z_3^2/4\alpha} = \Gamma[0, z_3^2/z_0^2] \quad (11)$$

applied to Eq. (8). For small z_3^2 , the incomplete gamma-function $\Gamma(0, z_3^2/z_0^2)$ has a logarithmic singularity

$$\Gamma(0, z_3^2/z_0^2) = \ln(z_0^2/z_3^2) - \gamma_E + \mathcal{O}(z_3^2), \quad (12)$$

while for large z_3^2 , the function $I_G(z_3^2)$ has a Gaussian $e^{-z_3^2/z_0^2}$ fall-off. Again, we can calculate the $z_3 = 0$ version of Eq. (11) using the $\overline{\text{MS}}$ -scheme to obtain

$$\begin{aligned} I_{DG}(z_3^2) &= \int_0^{z_3^2/4} \frac{d\alpha}{\alpha} (\alpha\mu^2 e^{\gamma_E})^\epsilon = \frac{1}{\epsilon} \left(\frac{z_0^2\mu^2 e^{\gamma_E}}{4} \right)^\epsilon \\ &\rightarrow \frac{1}{\epsilon} + \ln(z_0^2\mu^2) - 2\ln(2e^{-\gamma_E}) - \gamma_E. \end{aligned} \quad (13)$$

One can see that the pseudo-PDF/PDF rescaling (6) remains intact. This is a natural result, because the relation between the finite- z_3 and $\overline{\text{MS}}$ cut-offs concerns only the short-distance properties of the bilocal operator.

C. One-loop correction

The discussion given in the previous section addresses only the overall rescaling between two regularization schemes (just like the relation between the values of the QCD scale Λ in, say, MOM and $\overline{\text{MS}}$ schemes). To establish a connection between the pseudo-PDFs and the $\overline{\text{MS}}$ -PDFs, we need, in addition, the constant part of the one-loop coefficient function in the nonlocal OPE of Eq. (4). It was given in Refs. [14] and [15], with some differences between them. After rechecking our calculation and fixing typos, we present our result in the form

$$\begin{aligned} \mathfrak{I}(\nu, z_3^2) &= \mathfrak{I}^{\text{soft}}(\nu, 0) - \frac{\alpha_s}{2\pi} C_F \int_0^1 dw \left\{ \frac{1+w^2}{1-w} \right. \\ &\quad \times \left[\ln \left(z_3^2 \mu^2 \frac{e^{2\gamma_E}}{4} \right) + 1 \right] \\ &\quad \left. + 4 \frac{\ln(1-w)}{1-w} \right\} [\mathfrak{I}^{\text{soft}}(w\nu, 0) - \mathfrak{I}^{\text{soft}}(\nu, 0)] . \end{aligned} \quad (14)$$

Turning to the PDF counterpart, we take $z^2 = 0$ and

using the $\overline{\text{MS}}$ scheme for the UV divergence, obtain

$$\begin{aligned} \mathcal{I}(\nu, \mu^2) &= \mathfrak{I}^{\text{soft}}(\nu, 0) \\ &- \frac{\alpha_s}{2\pi} C_F \int_0^1 dw [\mathfrak{I}^{\text{soft}}(w\nu, 0) - \mathfrak{I}^{\text{soft}}(\nu, 0)] \\ &\times \left\{ \frac{1+w^2}{1-w} \ln(m^2/\mu^2) + 2(1-w) \right\} . \end{aligned} \quad (15)$$

The logarithmic part here involves a convolution that may be symbolically written as $B \otimes \mathfrak{I}(\nu)$ where

$$B(w) = \left[\frac{1+w^2}{1-w} \right]_+ \quad (16)$$

is the Altarelli-Parisi kernel [28].

Combining Eqs. (14) and (15), we obtain the relation

$$\begin{aligned} \mathcal{I}(\nu, \mu^2) &= \mathfrak{I}(\nu, z_3^2) + \frac{\alpha_s}{2\pi} C_F \int_0^1 dw \mathfrak{I}(\nu, w z_3^2) \\ &\times \left\{ B(w) \left[\ln \left(z_3^2 \mu^2 \frac{e^{2\gamma_E}}{4} \right) + 1 \right] \right. \\ &\quad \left. + \left[4 \frac{\ln(1-w)}{1-w} - 2(1-w) \right] \right\} \end{aligned} \quad (17)$$

which is in agreement with a recent result of Ref. [29] (see also Ref. [30]). Eq. (17) allows one to convert the data points for $\mathcal{M}(\nu, z_3^2)$ into the “data” for $\mathcal{I}(\nu, \mu^2)$.

The first contribution in the second line is an obvious term reflecting the general multiplicative scale difference between the z^2 and $\overline{\text{MS}}$ cut-offs. If all the further terms are neglected, then the only difference between $\mathfrak{I}(\nu, z_3^2)$ and $\mathcal{I}(\nu, \mu^2)$ is just the rescaling $\mu^2 = 4e^{-2\gamma_E}/z_3^2$. In that case, one can evolve the $\mathfrak{I}(\nu, z_3^2)$ data to a particular z_3 value z_0 , and treat (in this approximation) the resulting function $\mathfrak{I}(\nu, z_0^2)$ as the $\overline{\text{MS}}$ ITD corresponding to the scale $\mu = 2e^{-\gamma_E}/z_0$, which is numerically close to $1/z_0$.

This simple rescaling relation (used in Ref. [12]) is modified when the further terms of Eq. (17) are included. In particular, the term proportional to the Altarelli-Parisi kernel $B(w)$ may be absorbed into the $\ln z_3^2$ term, which would just change the rescaling relation into $\mu = 2e^{-1/2-\gamma_E}/z_0$.

The term with $[\ln(1-w)]/(1-w)$ produces a large negative contribution. In Feynman gauge, according to Ref. [15], it comes from the evolution part of the vertex diagrams involving the gauge link (see Fig.1). The key point is that the gluon is attached there to a running t_{z_3} position on the link. After integration over t , etc., the net outcome is that the z_3 -dependence of these diagrams is generated by an effective scale smaller than z_3 . Indeed, let us combine the $[\ln(1-w)]/(1-w)$ term with the $\ln z_3^2$

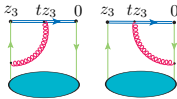


FIG. 1. Coordinate representation for diagrams producing a large one-loop correction.

logarithm by rewriting Eq. (17) as

$$\begin{aligned} \mathcal{I}(\nu, \mu^2) = & \mathfrak{M}(\nu, z_3^2) + \frac{\alpha_s}{\pi} C_F \int_0^1 dw \mathfrak{M}(w\nu, z_3^2) \\ & \times \left\{ \frac{1+w^2}{1-w} \ln \left[(1-w) z_3 \mu \frac{e^{\gamma_E+1/2}}{2} \right] \right. \\ & \left. + [(w+1) \ln(1-w) - (1-w)] \right\}_+ \quad (18) \end{aligned}$$

We see that z_3 enters now through a running $(1-w)z_3$ location. The remaining $(w+1) \ln(1-w)$ term is much less singular than $B(w)$ for $w=1$, and does not produce large contributions.

Thus, the magnitude of the one-loop correction is governed by the combined evolution logarithm. It cannot be made zero by a particular choice of μ because it depends on the integration variable w . Still, the w -integrated contribution will vanish for some μ that we may write as

$$\mu = \frac{2e^{-1/2-\gamma_E}}{(1-w)} \frac{1}{z_3}, \quad (19)$$

where $\langle 1-w \rangle$ is the “average” value of $1-w$. Since $B(w)$ is strongly enhanced for $w=1$, we should expect that $\langle 1-w \rangle$ is numerically small, leading to a $\mu \sim k/z_3$ rescaling with a rather large coefficient k . As we will see, $k \sim 4$ in this case.

Again, one may ask if the perturbative formula (17) involving the $\ln z_3^2$ logarithm may be applied to actual lattice data. In particular, our exercise with the mass-term IR regularization and the resulting Bessel function shows that the logarithmic behavior $\ln z_3^2$ of the hard term is valid only for z_3 values well below the IR cut-off R , which is given by the hadron size in our case. Hence, a practical question is whether the data really show a logarithmic evolution behavior in some region of small z_3 .

IV. EVOLUTION IN LATTICE DATA

A. General features

An exploratory lattice study of the reduced pseudo-ITD $\mathfrak{M}(\nu, z_3^2)$ for the valence $u_v - d_v$ parton distribution in the nucleon has been reported in Ref. [12]. An amazing observation made there was that, when plotted as functions of ν , the data both for real and imaginary parts lie close to respective universal curves. The data show no polynomial z_3 -dependence for large z_3 . Given that z_3^2/a^2 changes in the explored range from 1 to about 200, we interpret this result as the *total absence* of higher-twist terms in the reduced pseudo-ITD.

As explained in Refs. [10, 12] and in the Introduction, such an outcome corresponds to a factorization of the ν - and z_3^2 -dependences of the soft part of the Ioffe-time distribution $\mathcal{M}(\nu, z_3^2) = M(\nu)M(0, z_3^2)$. In terms of TMD $\mathcal{F}(x, k_\perp^2)$, this corresponds to factorization of its x - and $k_\perp^2$ -dependences in the region of soft $k_\perp$. However, as observed in Ref. [12], there is quite visible z_3 -dependence for small values of z_3 , namely, $z_3 \lesssim 6a$, that may be explained by perturbative evolution.

Let us consider first the real part. It corresponds to the cosine Fourier transform

$$\Re(\nu) = \text{Re} \mathfrak{M}(\nu) = \int_0^1 dx \cos(\nu x) q_v(x) \quad (20)$$

of the function $q_v(x)$ corresponding to the valence combination, i.e., the difference $q_v(x) = q(x) - \bar{q}(x)$ of quark and antiquark distributions. In our case, $q = u - d$.

In Ref. [12], it was found that the data for the real part are very close (see Fig. 2) to the curve $\Re f(\nu)$ generated by the function

$$f(x) = \frac{315}{32} \sqrt{x}(1-x)^3. \quad (21)$$

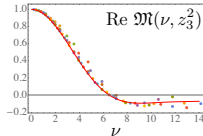
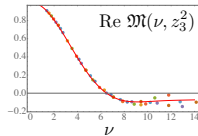
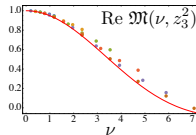


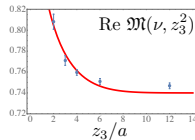
FIG. 2. Real part of $\mathfrak{M}(\nu, z_3^2)$ plotted as a function of $\nu = Pz_3$ and compared to the curve given by Eqs. (20), (21).

FIG. 3. Real part of $\Re(\nu, z_3^2)$ for z_3 ranging from $7a$ to $13a$.FIG. 4. Real part of $\Re(\nu, z_3^2)$ for z_3 ranging from a to $6a$.

This shape was obtained by forming cosine Fourier transforms of the normalized $x^a(1-x)^b$ -type functions and fixing the parameters a, b through fitting the data.

While all the data points have been used in the fit, the shape of the curve is obviously dominated by the points with smaller values of $\Re(\nu, z_3^2)$. To give a more detailed illustration, we show in Fig. 3 the points corresponding to z_3 values in the range $7a \leq z_3 \leq 13a$. As one can see, there is some scatter for the points with the largest values of ν in the region $\nu \geq 10$, where the finite-volume effects become important. Otherwise, practically all the points lie on the universal curve based on $f(x)$. In this sense, there is no z_3 -evolution visible in the large- z_3 data.

In Fig. 4, we show the points in the region $a \leq z_3 \leq 6a$ (note that, on the lattice, $z_3 = 0$ means that also $\nu = 0$, and $\Re(0, 0) = 1$ by definition). In this case, all the points lie higher than the universal curve. We recall that the perturbative evolution increases the real part of the pseudo-ITD when z_3 decreases. Thus, one may con-

FIG. 5. Dependence on z_3 for $\nu = 3\pi/4 \approx 2.3562$.

ture that the observed higher values of $\Re$ for smaller- z_3 points may be a consequence of the evolution.

A typical pattern of the z_3 -dependence of the lattice points is shown in Fig. 5 for a “magic” Ioffe-time value $\nu = 3\pi/4$ that may be obtained from five different combinations of z_3 and P values used in Ref. [12]. The shape of the eye-ball fit line is given by the incomplete gamma-function $\Gamma(0, z_3^2/30a^2)$. This function entirely conforms to the expectation that the z_3 -dependence has a “perturbative” logarithmic $\ln(1/z_3^2)$ behaviour for small z_3 , and rapidly vanishes for z_3 larger than $6a$.

As expected, $\Re(\nu, z_3^2)$ decreases when z_3 increases. We also see that the evolution “stops” for large z_3 . In this context, the overall curve based on Eq. (21) corresponds to the “low normalization point”, i.e., to the region, where the perturbative evolution is absent.

B. Building $\overline{\text{MS}}$ ITD

Thus, we see that the data of Fig. 5 show a logarithmic evolution behavior in the small z_3 region. Still, the z_3 -behavior starts to visibly deviate from a pure logarithmic $\ln z_3^2$ pattern for $z_3 \gtrsim 5a$. This sets the boundary $z_3 \lesssim 4a$ on the “logarithmic region”. So, let us try to use Eq. (17) in that region to construct the $\overline{\text{MS}}$ ITD.

It is instructive to split the contributions in Eq. (17), where we will denote $\Re \mathcal{I}(\nu, \mu^2) = \mathcal{I}_R(\nu, \mu^2)$. The first, “evolution” part, given by

$$\mathcal{I}_R^{\overline{\text{MS}}}(\nu, \mu^2) = \Re(\nu, z_3^2) + \frac{\alpha_s}{2\pi} C_F \int_0^1 dw \Re(w\nu, z_3^2) \times B(w) \ln \left(\frac{z_3^2 \mu^2 e^{\gamma_E w}}{4} \right) \quad (22)$$

(recall that $\Re(\nu, z_3^2) = \Re \mathcal{I}(\nu, z_3^2)$) corresponds to the leading logarithm approximation used in Ref. [12]. For

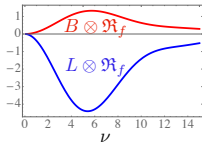


FIG. 6. Functions $B \otimes \mathfrak{R}_f$ (upper line) and $L \otimes \mathfrak{R}_f$ (lower line) of Eq. (24).

$z_3 = 2e^{-\gamma_\pi}/\mu$, the logarithm vanishes, and we have

$$\mathcal{I}_R^\pi(\nu, \mu^2) = \mathfrak{R}(\nu, (2e^{-\gamma_\pi}/\mu)^2) = \mathfrak{R}(\nu, (1.12/\mu)^2). \quad (23)$$

This happens, of course, only if, for an appropriately chosen α_s , the $\ln z_3^2$ -dependence of the one-loop correction cancels the actual z_3^2 -dependence of the data, visible as scatter in the data points in Fig. 4. In Ref. [12], it was found that this happens when $\alpha_s/\pi \approx 0.1$. Thus, Eq. (17) is accurate only in the region, where the data show a logarithmic dependence on z_3 , i.e., $z_3 \leq 4a$ in our case.

Since the difference between $\mathfrak{R}(w\nu, z_3^2)$ and $\mathfrak{R}_f(w\nu)$ is $\mathcal{O}(\alpha_s)$, we may replace $\mathfrak{R}(w\nu, z_3^2)$ by $\mathfrak{R}_f(w\nu)$ in Eq. (22) (recall that $\mathfrak{R}_f(\nu)$ corresponds to the PDF of Eq. (21)). The remaining part of $\mathcal{I}(\nu, \mu^2)$ (where we have already substituted $\mathfrak{R}(w\nu, z_3^2)$ by $\mathfrak{R}_f(w\nu)$)

$$\begin{aligned} \mathcal{I}_R^{\text{NL}}(\nu) &= \frac{\alpha_s}{2\pi} C_F \int_0^1 dw \mathfrak{R}_f(w\nu) \\ &\times \left\{ B(w) + \left[4 \frac{\ln(1-w)}{1-w} - 2(1-w) \right]_+ \right\} \\ &= \frac{\alpha_s}{2\pi} C_F [B \otimes \mathfrak{R}_f + L \otimes \mathfrak{R}_f] \end{aligned} \quad (24)$$

is due to corrections beyond the leading logarithm approximation.

As we have discussed, the $L \otimes \mathfrak{R}_f$ term reflects the fact that the actual scale in the evolution part of the vertex diagrams is less than z_3 . To illustrate its impact, we show, in Fig. 6, the functions $B \otimes \mathfrak{R}_f$ and $L \otimes \mathfrak{R}_f$. One can see that the last one is negative and rather large. Its ν -dependence is similar to that of the $B \otimes \mathfrak{R}_f$ function. In fact, in the $\nu < 5$ region, we have $L \otimes \mathfrak{R}_f \approx -3.5 B \otimes \mathfrak{R}_f$. Thus, the combined effect of these two terms is close to that of $-2.5 B \otimes \mathfrak{R}_f$. As a result, the inclusion of these terms may be approximately treated as a LLA evolution

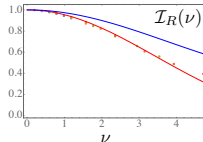


FIG. 7. Function $\mathcal{I}_R(\nu, \mu^2)$ for $\mu = 1/a$ calculated using the data with z_3 from a to $4a$. The upper curve corresponds to the ITD of the CJ15 global fit PDF.

with a modified rescaling factor. Specifically, we may write

$$\mathcal{I}_R(\nu, \mu^2) \approx \mathfrak{R}(\nu, (2e^{1.25-\gamma_\pi}/\mu)^2) \approx \mathfrak{R}(\nu, (4/\mu)^2). \quad (25)$$

Thus, the rescaling factor has changed by a factor of 4 compared to the original LLA value!

We may use $\mu \approx 4/z_3$ as a guide, but the actual numerical calculations should, of course, be done using the ‘exact’ Eq. (17). To proceed, we choose the value $\mu = 1/a$ which, at the lattice spacing of 0.093 fm used in Ref. [12] is approximately 2.15 GeV. The estimate (25) tells us that the ITD $\mathcal{I}_R(\nu, \mu^2)$ at this scale should be close to the pseudo-ITD $\mathfrak{R}(\nu, z_3^2)$ for $z_3 \approx 4a$, a distance that is on the border of the $z_3 \leq 4a$ region. Taking the value $\alpha_s/\pi = 0.1$ used in Ref. [12] and applying the full one-loop relation (17) to the data with $z_3 \leq 4a$, we generate the points for $\mathcal{I}_R(\nu, (1/a)^2)$.

As seen from Fig. 7, all the points are close to some universal curve with a rather small scatter. The curve itself was obtained by fitting the points by the cosine transform of a normalized $Nx^a(1-x)^b$ distribution, which gave $a = 0.35$ and $b = 3$. The magnitude of the scatter illustrates the error of the fit for the ITD in the $\nu \leq 4$ region. In Fig. 7, we compare our $\mu = 1/a$ ITD with the ITD obtained from the global fit PDFs corresponding to the CJ15 [31] global fit. One can see that our ITD is systematically below the curve based on the global fit PDFs.

The reason for the discrepancy may be understood from Fig. 8, where we compare the normalized $Nx^{0.35}(1-x)^3 = q_\pi(x, \mu = 2.15 \text{ GeV})$ distribution to CJ15 [31] and MMHT 2014 [32] global fit PDFs, taken at the scale $\mu = 2.15 \text{ GeV}$. Unlike the $\sim x^{0.35}$ function, these PDFs are singular for small x , which leads to the enhancement of ITDs for large and moderate values of ν .

To fit the points for $\mathcal{I}_R(\nu, \mu^2)$, we have used the same simplest $Nx^a(1-x)^b$ Ansatz for the PDF as in Ref. [12].

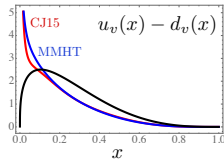


FIG. 8. Curve for $u_v(x) - d_v(x)$ at $\mu = 2.15$ GeV built from the data shown in Fig. 7 and compared to CJ15 and MMHT global fits.

In principle, one may use more complicated models for PDFs and get practically the same fitted curve for the ITD in the $\nu \lesssim 4$ region, while a somewhat different curve for PDF $q_v(x)$. The reason is simple: the inverse cosine Fourier transform is unique only when one exactly knows the ITD in the whole $0 \leq \nu < \infty$ region. Performing such a transform from a limited $\nu \lesssim 4$ region, one needs to add some assumptions either about the behavior of the ITD outside this region or about a functional form of the PDF $q_v(x)$. We fixed our choice by taking $q_v(x) \sim x^a(1-x)^b$. The study of how the shape of $q_v(x)$ varies if one uses more complicated forms, in particular, those used in the global fits [31, 32] is an interesting problem that, however, goes beyond the scope of the present paper.

Comparing to the LLA results of Ref. [12], we observe that the large negative one-loop correction in Eq. (17) has visibly changed the extracted PDF, which is now further from the global fit PDFs. The main reason is that the $z_0 = 2a$ pseudo-ITD constructed in Ref. [12] was treated there as corresponding to the $\mu \approx 1$ GeV scale, while according to the modified rescaling relation (25), it should correspond to $\mu \approx 4$ GeV. Hence, to get the $\mu \approx 2$ GeV curve, one needs to evolve it down in μ .

Still, the guiding idea of Ref. [12], that the $\overline{\text{MS}}$ ITDs $\mathcal{I}_R(\nu, \mu^2)$ can be obtained from the reduced pseudo-ITDs $\mathcal{R}(\nu, z_3^2)$ by an appropriate rescaling $\mu = k/z_3$, works with a rather good accuracy for all $z_3 \lesssim 6a$ if one takes $k \approx 4$. By this rescaling relation, the $\mu = 1/2a \approx 1$ GeV ITD corresponds to the $z_3 \approx 8a$ reduced pseudo-ITD. As we discussed, a boundary point beyond which the evolution stops, is $z_3 \approx 6a$. Hence, the pseudo-ITD at this distance is given by the ITD $\mathcal{R}_l(\nu)$ corresponding to the universal fit function $f(x)$ of Eq. (21). This result may be also obtained by a direct numerical calculation based on Eq. (17).

Using Eq. (17) one may also evolve the $\overline{\text{MS}}$ ITD below $\mu = 1/2a$, and the resulting functions will be changing with μ . On the other hand, the pseudo-ITDs do not change with z_3 when $z_3 \gtrsim 6a$. Hence, the rescaling connection $\mathcal{I}_R(\nu, \mu^2) \approx \mathcal{R}_l(\nu, (4/\mu)^2)$ in this region becomes less and less accurate when μ decreases, and eventually makes no sense.

C. Imaginary part

Imaginary part of the pseudo-ITD may be considered in a similar way. It corresponds to the sine Fourier transform

$$\text{Im } \mathfrak{M}(\nu) = \int_0^1 dx \sin(\nu x) [q(x) + \bar{q}(x)] \quad (26)$$

of the function given by the sum $q(x) + \bar{q}(x)$ of quark and antiquark distributions. This function differs from the valence combination $q_v(x) = q(x) - \bar{q}(x)$ by $2\bar{q}(x) = 2[u(x) - d(x)]$. In Fig. 9, we show the data for large z_3 values $z_3 \gtrsim 7a$. Just like in the case of the real part (see Fig. 3), the points with $\nu \lesssim 10$ are close to a universal curve. Representing $q(x) + \bar{q}(x) = q_v(x) + 2\bar{q}(x)$ and taking $f(x)$ of Eq. (21) as $q_v(x)$, we find

$$\bar{q}(x) \approx 0.1 [20x(1-x)^3]. \quad (27)$$

Note that in Ref. [12], the fit was made for all the z_3 points (i.e. the points with $z_3 \lesssim 6a$ have been also included), and the overall coefficient for $\bar{q}(x)$ was obtained to be 0.07 rather than 0.1.

In Fig. 10, we show data with $z_3 \lesssim 4a$. As one can see, all these points are below the curve obtained by fitting the $z_3 \gtrsim 7a$ data. This is in agreement with the fact that, in the region $\nu \lesssim 6$, the perturbative evolution decreases

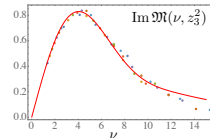


FIG. 9. Imaginary part of $\mathfrak{M}(\nu, z_3^2)$ for z_3 ranging from $7a$ to $13a$. The curve corresponds to $\bar{q}(x) + q(x) = f(x) + 2\bar{q}(x)$, with $f(x)$ given by Eq. (21) and $\bar{q}(x)$ given by Eq. (27).

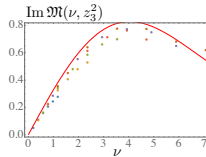


FIG. 10. Imaginary part of $\mathfrak{M}(\nu, z_3^2)$ for z_3 ranging from a to $6a$. The curve is the same as in Fig. 9.

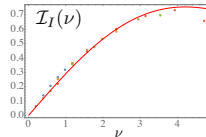


FIG. 11. Function $\mathcal{I}_I(\nu, \mu^2)$ for $\mu = 1/a$ calculated using the data with z_3 from a to $4a$. The curve is described in the text.

the imaginary part of the pseudo-ITD when z_3 decreases. Note that the 1-loop relation holds for the whole function $\mathfrak{M} = \text{Re}\mathfrak{M} + i\text{Im}\mathfrak{M}$. So, we should just separate there real and imaginary parts, and the construction of the $\overline{\text{MS}}$ function $\text{Im}\mathcal{I}(\nu, \mu^2) = \mathcal{I}_I(\nu, \mu^2)$ proceeds in the same way as for the real part.

The results are shown in Fig. 11. Again, all the points are rather close to a universal curve with a rather small scatter. The curve shown corresponds to the sine Fourier transform of the sum of the valence distribution $q_v(x, \mu = 1/a) = N x^{0.35} (1-x)^3$ obtained from the study of the real part, and the antiquark contribution $2q(x, \mu = 1/a)$. The latter was found from the fit to be given by $q(x, \mu = 1/a = 2.15 \text{ GeV}) = 0.07[20x(1-x)^7]$.

V. SUMMARY AND CONCLUSIONS

In this paper, we have extended the leading-logarithm analysis of lattice data for parton pseudo-distributions and reduced pseudo-ITDs performed in Ref. [12]. To this end, we incorporated recent results for the reduced pseudo-ITDs at the one-loop level [15] (see also [14, 29]).

It was found that the correction contains a large term resulting in essential numerical changes compared to the LLA. The large correction appears since effective distances involved in the most important diagrams are much smaller than the nominal distance z_3 . This leads to a change (from $k_{\text{LLA}} \approx 1$ to $k \approx 4$ in the case of our particular ITDs) of the coefficient k in the rescaling relation $\mu = k/z_3$ that allows to (approximately) convert the pseudo-PDFs $\mathcal{P}(x, z_3^2)$ into the $\overline{\text{MS}}$ PDFs $f(x, \mu^2)$.

While the rescaling relation serves as an instructive guide for quick estimates and semi-quantitative analysis, the $\overline{\text{MS}}$ ITDs may be directly constructed applying the exact one-loop formula. Using it, we have obtained the ITD $\mathcal{I}(\nu, \mu^2)$ at the $\mu = 1/a \approx 2.15 \text{ GeV}$ $\overline{\text{MS}}$ scale using the data in the $0 \leq z_3 \leq 4a$ region.

We found that $\mathcal{I}(\nu, \mu^2)$ at this scale is close to the reduced pseudo-ITD $\mathfrak{M}(\nu, z_3^2)$ for $z_3 \sim 4a$. Since all the data in the $a \leq z_3 \leq 4a$ region do not differ much from the $z_3 = 4a$ ones (see Fig. 4), the conversion of the $\mathfrak{M}(\nu, z_3^2)$ data into $\mathcal{I}(\nu, 1/a^2)$ does not involve large changes, i.e., the perturbative expansion for $\overline{\text{MS}}$ ITD $\mathcal{I}(\nu, \mu^2)$ in terms of the reduced pseudo-ITDs $\mathfrak{M}(\nu, z_3^2)$ is under control. A formal reason is that the large correction in this case can be absorbed into the z_3^2 -dependent evolution term, with remaining corrections being small.

Phenomenologically, the PDF extracted in this way follows the trend of those given by the global fits in the $x > 0.1$ region, but does not reproduce their singular behavior in the $x < 0.1$ region. The latter is usually related to the $x^{-0.5}$ pattern of the ρ -meson Regge trajectory. Since the ρ -meson is essentially a rather narrow resonance in the $\pi\pi$ system, one should not expect to accurately reproduce the ρ -meson properties in a lattice simulation in which the pions are as heavy as 600 MeV. Thus, one may hope that using simulations at physical pion mass would produce a better agreement with the global fits in the small- x region. This hope is supported by recent extractions [33, 34] of $q_v(x)$ using the quasi-PDF lattice simulations at physical pion mass.

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P50: 同位旋矢量分布的重正化匹配系统分析

主题归属	LaMET、quasi/pseudo 分布与因子化
底稿	arXiv:1807.06566
arXiv / DOI	1807.06566
完整原始 PDF	17 页; 来源 PyQCD; SHA-256 460780f4eed9...
转排索引	EN 24 页; ZH 22 页 (只作书目对照)
备注	索引未记录额外备注。

阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

页码范围：P50-p001 – P50-p017

Unpolarized isovector quark distribution function from Lattice QCD: A systematic analysis of renormalization and matching (Lattice Parton Collaboration (LPC))

Yu-Sheng Liu,¹ Jiunn-Wei Chen,^{2,3} Yi-Kai Huo,^{4,5} Luchang Jin,^{6,7} Maximilian Schlemmer,⁸ Andreas Schäfer,⁸ Peng Sun,^{9,*} Wei Wang,^{4,†} Yi-Bo Yang,^{10,11} Jian-Hui Zhang,^{8,12} Qi-An Zhang,¹ Kuan Zhang,^{13,14} and Yong Zhao^{3,14}

¹*Fang-Dao Lee Institute, Shanghai Jiao Tong University, Shanghai 200240, China*

²*Department of Physics, Center for Theoretical Physics,
and Leung Center for Cosmology and Particle Astrophysics,
National Taiwan University, Taipei, Taiwan 106*

³*Center for Theoretical Physics, Massachusetts Institute of Technology, Cambridge, MA 02139, USA*

⁴*INFAC, SKLPPC, MOE KLPPC, School of Physics and Astronomy,
Shanghai Jiao Tong University, Shanghai 200240, China*

⁵*Zhuzhen College, Shanghai Jiao Tong University, Shanghai 200240, China*

⁶*Physics Department, University of Connecticut, Storrs, Connecticut 06269-3046, USA*

⁷*RIKEN BNL Research Center, Brookhaven National Laboratory, Upton, NY 11973, USA*

⁸*Institut für Theoretische Physik, Universität Regensburg, D-93040 Regensburg, Germany*

⁹*Nanjing Normal University, Nanjing, Jiangsu, 210023, China*

¹⁰*Department of Physics and Astronomy, Michigan State University, East Lansing, MI 48824, USA*

¹¹*CAS Key Laboratory of Theoretical Physics, Institute of Theoretical Physics,
Chinese Academy of Sciences, Beijing 100190, China*

¹²*Center of Advanced Quantum Studies, Department of Physics,
Beijing Normal University, Beijing 100875, China*

¹³*University of Chinese Academy of Sciences, School of Physical Sciences, Beijing 100049, China*

¹⁴*Physics Department, Brookhaven National Laboratory, Bldg. 510A, Upton, NY 11973, USA
(Dated: December 5, 2019)*

We present a detailed Lattice QCD study of the unpolarized isovector quark Parton Distribution Function (PDF) using large-momentum effective theory framework. We choose a quasi-PDF defined by a spatial correlator which is free from mixing with other operators of the same dimension. In the lattice simulation, we use a Gaussian-momentum-smeared source at $M_\pi = 356$ MeV and $P_z \in \{1.8, 2.3\}$ GeV. To control the systematics associated with the excited states, we explore five different source-sink separations. The nonperturbative renormalization is conducted in a regularization-independent momentum subtraction scheme, and the matching between the renormalized quasi-PDF and $\overline{\text{MS}}$ PDF is calculated based on perturbative QCD up to one-loop order. Systematic errors due to renormalization and perturbative matching are also analyzed in detail. Our results for lightcone PDF are in reasonable agreement with the latest phenomenological analysis.

I. INTRODUCTION

Parton distribution functions (PDFs) of nucleons are not only important quantities characterizing the internal hadron structures but are also key ingredients to make predictions for high-energy scattering processes [1–3]. Thus calculating PDFs from first principles has been a holy grail in nuclear and particle physics. Since PDFs are embedded with the low-energy quark and gluon degrees of freedom in the hadron, they involve infrared (IR) dynamics of strong interactions and can only be determined by nonperturbative methods such as Lattice QCD.

Within QCD factorization [4], the quark PDF is defined as

$$q(x, \mu) = \int \frac{d\xi^-}{4\pi} e^{-ixP^+ \xi^-} \langle P | \bar{\psi}(\xi^-) \gamma^+ U(\xi^-, 0) \psi(0) | P \rangle, \quad (1)$$

where $|P\rangle$ denotes the nucleon state with momentum $P_\mu = (P, 0, 0, P_z)$. x is the quark momentum fraction, μ is the renormalization scale in the $\overline{\text{MS}}$ scheme. $\xi^\pm = (t \pm z)/\sqrt{2}$ are the lightcone coordinates. The light-like Wilson line is introduced to maintain the gauge invariance:

$$U(\xi^-, 0) = P \exp \left(-ig \int_0^{\xi^-} d\eta^- A^+(\eta^-) \right). \quad (2)$$

PDFs are defined with lightcone coordinates, but the Lattice simulation can only be conducted in Euclidean space with no proper treatment for lightcone quantities which involves real time. Thus simulating PDFs on a Euclidean Lattice is an extremely difficult task. Early studies based on operator product expansion (OPE) were only able to derive the lowest few moments of the PDFs [5–8].

Recently, a novel approach that allows to directly access the x -dependence of PDFs from Lattice QCD was proposed in Ref. [9], now formulated as large-momentum effective theory (LaMET) [10]. Within this framework, one can extract PDFs—as well as other lightcone

* Corresponding author: 06260@njnu.edu.cn

† Corresponding author: wei.wang@sjtu.edu.cn

quantities—from the correlations of certain static operators in a nucleon state. On the one hand, the static correlations, often referred to as quasi observables, can be directly calculated on a Euclidean Lattice and depend dynamically on the nucleon momentum. On the other hand, at large momentum, the quasi observables can be factorized into the parton observable and a perturbative matching coefficient, up to corrections suppressed by powers of the large nucleon momentum. Equating the results from the two sides provides a straightforward way to determine the lightcone PDFs.

To calculate the quark PDF in LaMET, one starts with a “quasi-PDF” which is defined as an equal-time correlation of quarks along the z direction [9]:

$$\bar{q}_T(x, P_z) = \int_{-\infty}^{\infty} \frac{dz}{4\pi} e^{izP_z} \langle P | O_T(z) | P \rangle. \quad (3)$$

In the above, $O_T(z) = \bar{\psi}(z)\Gamma U(z,0)\psi(0)$ with $\Gamma = \gamma^z$ or $\Gamma = \gamma^t$, and the space-like Wilson line is:

$$U(z,0) = P \exp \left(-ig \int_0^z dz' A_z(z') \right). \quad (4)$$

For finite but large momentum P_z , $\bar{q}_T(x, P_z)$ has support in $-\infty < x < \infty$. Unlike the lightcone PDF that is boost invariant, the quasi-PDF has a nontrivial dependence on the nucleon momentum P_z . After renormalizing the quasi-PDF in a scheme such as the regularization-independent momentum subtraction (RI/MOM) scheme, one can match the renormalized quasi-PDF to the $\overline{\text{MS}}$ PDF through the factorization theorem [9–14]:

$$\bar{q}(x, P_z, p_\mu^R, \mu_R) = \int_{-1}^1 \frac{dy}{|y|} C \left(\frac{x}{y}, r, \frac{yP_z}{\mu}, \frac{yP_z}{P_z^2} \right) q(y, \mu) + \mathcal{O} \left(\frac{M^2}{P_z^2}, \frac{\Lambda_{\text{QCD}}^2}{x^2 P_z^2} \right), \quad (5)$$

where p_μ^R and μ_R are introduced in RI/MOM scheme: p_μ^R is the momentum of the involved parton and μ_R is renormalization scale. $r = \mu_R^2/(p_\mu^R)^2$, C is the perturbative matching coefficient, and $\mathcal{O}(M^2/P_z^2, \Lambda_{\text{QCD}}^2/x^2 P_z^2)$ denotes nucleon mass and higher-twist contributions suppressed by powers of the large nucleon momentum. The flavor indices in q , $\bar{q}$ and C are implied. When $-1 < y < 0$, the distributions refer to the antiquark distributions.

Since the proposal of LaMET, remarkable progress has been made in both theoretical aspect and Lattice calculations. It should be pointed out that these developments are achieved in an interactive way. The LaMET was first used to calculate the proton isovector quark distribution f_{u-d} [15–20], including the unpolarized, polarized and transversity cases, and subsequently to the meson distribution amplitudes [21, 22]. The first Lattice studies used the matching coefficients at one-loop order in a transverse-momentum cutoff scheme [23–25]. However, as was found in Ref. [23], the original quasi-PDF suffers from an ultraviolet (UV) linear divergence

which might pose a severe problem for the renormalization of its Lattice matrix elements [26–28]. Then many attentions have been paid to the renormalization property [29–37], and finally the multiplicative renormalizability of quasi-PDF in coordinate space in the continuum was proven to all orders in strong coupling constant α_s [34, 35]. This finding has further motivated the Lattice analysis of nonperturbative renormalization (NPR) of the quasi-PDF [36, 38, 39] in the RI/MOM scheme [40], and the calculation of the matching coefficients between the RI/MOM quasi-PDFs and $\overline{\text{MS}}$ PDFs [11]. Besides the renormalization, the finite nucleon mass corrections were also worked out to all orders of M^2/P_z^2 [17], and higher-twist $\mathcal{O}(\Lambda_{\text{QCD}}^2/x^2 P_z^2)$ effects were numerically removed by extrapolating the results at several P_z values to infinite momentum [15, 17]. Based on these studies, calculations of the isovector quark PDF at physical pion mass have become available [20, 41–43]. Potential operator mixing in the Lattice renormalization of the quasi-PDF has also been investigated [33, 36, 38, 39], and the mixing pattern classified in Ref. [44]. Ways to reduce the systematic uncertainties from Fourier transforming the spatial correlation at long distance were proposed in Refs. [41, 45]. The LaMET was also attempted to study transverse-momentum-dependent distributions [46–53], as well as the gluon PDF [54–59].

In addition to LaMET, other interesting approaches have been proposed in recent years to calculate the PDFs from Lattice QCD. For example, one can extract the PDFs from a class of “Lattice cross sections” [13, 14], while a smeared quasi-PDF in the gradient flow method was proposed to sweep the power divergence in the Lattice calculation [60, 61]. One can also study a pseudo distribution [62], related to the quasi-PDF through Fourier transforms. While this method shows interesting renormalization features [63, 64], it coincides with LaMET regarding the factorization into PDFs [12, 65, 66]. Moreover there are proposals using current-current correlators to compute the hadronic tensor [67, 68], or the higher moments of the PDF, lightcone distribution amplitudes, etc. [68–73]. These different approaches are subject to their own systematics, but they can be compared to each other.

It was argued that the power divergent mixing between local moment operators may spoil the renormalization of quasi-PDFs [27, 28], however such problem dissolves in LaMET since one first needs to take the continuum limit of the quasi-PDF after renormalization on the Lattice, and then match it to obtain the x -dependence of the PDF. The factorization has been derived rigorously [12, 13] in the continuum, and one only needs to focus on the renormalization of the non-local spatial correlator only. Thus the renormalization of local moment operators is irrelevant to quasi-PDF. Besides, there are also confusions on the LaMET matching between Minkowskian and Euclidean matrix elements of the quasi-PDF [74], which have been clarified in Refs. [65, 75].

Most of the available Lattice calculations have used $\Gamma = \gamma^2$ (except [42, 43]) for the unpolarized quasi-PDF, which is now known to mix with the scalar quasi-PDF operator O_I at $O(a^2)$ [33, 36, 44]. This operator mixing introduces an additional systematic uncertainty in nonperturbative renormalization [36, 38, 39, 41], thus limiting the accuracy of the extracted PDF. On the contrary, the $\Gamma = \gamma^4$ case is free from operator mixing with O_I at $O(a^2)$ [33, 36, 44]. Therefore, it is highly desirable to start from the quasi-PDF with $\Gamma = \gamma^4$. This is one main motif of this study.

In this work, we will carry out a Lattice calculation of the unpolarized isovector quark distribution from the quasi-PDF with $\Gamma = \gamma^4$ with the same nonperturbative renormalization procedure as for the $\Gamma = \gamma^2$ case in Ref. [39]. The calculation is performed using clover fermions on a CLS ensemble of gauge configurations with $N_f = 2 + 1$ (degenerate up/down, and strange) flavors under open boundary condition [76] with pion mass $M_\pi = 356$ MeV and Lattice spacing $a = 0.086$ fm [77]. We will examine the dependence on the nucleon momentum P_z and the RI/MOM scales p_s^R , μ_R , as well as on choices of the projection operator for the amputated Green's function in RI/MOM renormalization. Due to large uncertainties, it is hard to see the sea quark asymmetry observed in early studies which were performed without Lattice renormalization [15–18]. In the future we plan to analyze CLS ensembles with a better accuracy and down to physical masses and $a < 0.04$ fm, both for $(m_s + m_u + m_d)$ fixed to its physical value and for physical m_s using flavour SU(3) and SU(2) extrapolations.

The rest of this paper is organized as follows. In Sec. II, we briefly review the procedure of nonperturbative renormalization and matching of the quasi-PDF in the RI/MOM scheme, in particular the explicit one-loop matching coefficient for the $\Gamma = \gamma^4$ case. In Sec. III, we describe the details of Lattice simulation of the hadronic matrix elements as well as its nonperturbative renormalization. Systematic errors in the calculation are also discussed in this section. In Sec. IV, we present our results on the x -dependence of the unpolarized isovector quark PDF with the statistical and systematic uncertainties, and the last section contains the summary of our work.

II. NONPERTURBATIVE RENORMALIZATION AND MATCHING

To recover the continuum limit of a quasi-PDF matrix element, nonperturbative renormalization on the Lattice is required to deal with linear and logarithmic UV divergences. In this work, we follow the RI/MOM scheme elaborated in Refs. [11, 39], and match the result to the $\overline{\text{MS}}$ PDF with the one-loop matching coefficient [11].

A. RI/MOM renormalization on the Lattice

The spatial correlator $O_T(z)$ has been proven to be multiplicatively renormalizable in coordinate space in the continuum [34, 35], which enables the renormalization in RI/MOM scheme [40].

For each value of z , the RI/MOM renormalization factor Z is obtained by requiring loop corrections for the matrix element of a quasi-PDF operator vanish in an off-shell quark state at a given momentum:

$$Z(z, p_z^R, a^{-1}, \mu_R) = \frac{\sum_s \langle p, s | O_T(z) | p, s \rangle}{\sum_s \langle p, s | O_T(z) | p, s \rangle_{\text{tree}}} \Big|_{p^2 = -p_z^R} \quad (6)$$

The bare matrix element $\sum_s \langle p, s | O_T(z) | p, s \rangle$ will be calculated on the Lattice from the amputated Green's function $\Lambda_{\gamma^4}(p, z)$ of $O_{\gamma^4}(z)$, with a projection operator $\mathcal{P}$ for the Dirac matrix:

$$\sum_s \langle p, s | O_{\gamma^4}(z) | p, s \rangle = \text{Tr} [\Lambda_{\gamma^4}(z, p) \mathcal{P}] \quad (7)$$

Due to the breaking of Lorentz covariance in $O_T(z)$, the RI/MOM subtraction depends on two scales μ_R and p_z^R . As a result, the renormalization factor $Z(z, p_z^R, a^{-1}, \mu_R)$ depends on the Lattice spacing as well as on the two RI/MOM scales μ_R and p_z^R .

Based on the symmetry of $O_T(z)$ on the Lattice, the amputated Green's function $\Lambda_{\gamma^4}(p, z)$ is not only proportional to the tree-level result γ^4 , but also includes two other independent Lorentz structures:

$$\Lambda_{\gamma^4}(p, z) = \tilde{F}_1(p, z) \gamma^4 + \tilde{F}_2(p, z) \frac{p_z \gamma^2}{p_z} + \tilde{F}_p(p, z) \frac{p_z \not{p}}{p^2} \quad (8)$$

In the above $\tilde{F}_i$ s are independent form factors that are invariant under the hyper cubic group $H(4)$. According to Eq. (8) the RI/MOM renormalization factor Z will also depend on the projection operator $\mathcal{P}$. One can choose to single out $\tilde{F}_1$ only [11], which we call the minimal projection. This projection has the simplest form but captures all the UV divergence in $\Lambda_{\gamma^4}(p, z)$. Optionally, one can choose $\mathcal{P} = \not{p}/(4p^4)$, which we call the $\not{p}$ projection. The renormalization factors Z with the minimal and $\not{p}$ projections are defined as:

$$Z_{\text{min}}(z, p_z^R, a^{-1}, \mu_R) = \tilde{F}_1(p, z) \Big|_{p^2 = -p_z^R} \quad (9)$$

$$Z_{\not{p}}(z, p_z^R, a^{-1}, \mu_R) = \left[\tilde{F}_1(p, z) + \tilde{F}_2(p, z) + \tilde{F}_p(p, z) \right] \Big|_{p^2 = -p_z^R} \quad (10)$$

The bare nucleon matrix element from a Lattice calculation in coordinate space

$$\tilde{h}(z, P_z, a^{-1}) = \frac{1}{2P^0} \langle P | O_{\gamma^4}(z) | P \rangle$$

is renormalized according to

$$\begin{aligned} \tilde{h}_R(z, P_z, p_z^R, \mu_R) \\ = Z^{-1}(z, p_z^R, a^{-1}, \mu_R) \tilde{h}(z, P_z, a^{-1}) \Big|_{|a \rightarrow 0}. \end{aligned} \quad (11)$$

Here $\tilde{h}_R(z, P_z, p_z^R, \mu_R)$ is the continuum limit of the renormalized matrix element. Consequently, the quasi-PDF $\tilde{q}_R(x, P_z, p_z^R, \mu_R)$ in RI/MOM scheme is obtained through the Fourier transformation of $\tilde{h}_R(z, P_z, p_z^R, \mu_R)$:

$$\tilde{q}_R(x, P_z, p_z^R, \mu_R) = P_z \int \frac{dz}{2\pi} e^{ixP_z} \tilde{h}_R(z, P_z, p_z^R, \mu_R). \quad (12)$$

In RI/MOM, $\tilde{h}_R(z, P_z, p_z^R, \mu_R)$ and $\tilde{q}_R(x, P_z, p_z^R, \mu_R)$ are independent of the UV regulator, and the one-step matching between the quasi-PDF and $\overline{\text{MS}}$ PDF can be carried out in the continuum theory with dimensional regularization [11].

The quasi-PDFs will eventually be matched to the same $\overline{\text{MS}}$ PDF, and the two projections with Z_{up} and Z_{p} should generate the same result. However the matching coefficient can only be calculated at a fixed loop order,

At one-loop order, it is

$$\tilde{q}^{(1)}(x, p, \rho) = \text{Tr} \left[\left(\left[\tilde{f}(x, \rho) \right]_+ \gamma^t + \left[\tilde{f}_s(x, \rho) \right]_+ \frac{p_t}{p_z} \gamma^t + \left[\tilde{f}_p(x, \rho) \right]_+ \frac{p_t \not{p}}{p^2} \right) \mathcal{P} \right]. \quad (14)$$

The $\tilde{f}_i$ s are

$$\tilde{f}_t(x, \rho) = \frac{\alpha_s C_F}{2\pi} \begin{cases} \frac{8x^2(1-x) - x\rho(13-10x) + 3\rho^2}{2(1-x)(1-\rho)(4x-4x^2-\rho)} + \frac{4x(2-x) - x\rho - 3\rho}{2(1-x)(1-\rho)^2} \ln \frac{2x-1+\sqrt{1-4\rho}}{2x-1-\sqrt{1-\rho}} & x > 1 \\ \frac{4(1-x)(1-\rho)}{2(1-x)(1-\rho)} + \frac{4(2-x) - x\rho - 3\rho}{2(1-x)(1-\rho)^2} \ln \frac{1+\sqrt{1-4\rho}}{1-\sqrt{1-\rho}} & 0 < x < 1, \\ -\frac{8x^2(1-x) - x\rho(13-10x) + 3\rho^2}{2(1-x)(1-\rho)(4x-4x^2-\rho)} - \frac{4x(2-x) - x\rho - 3\rho}{2(1-x)(1-\rho)^2} \ln \frac{2x-1+\sqrt{1-4\rho}}{2x-1-\sqrt{1-\rho}} & x < 0 \end{cases}, \quad (15)$$

$$\tilde{f}_s(x, \rho) = \frac{\alpha_s C_F}{2\pi} \begin{cases} \frac{-32x^2(1-x)^2(2x-1) - 4x\rho(8-43x+65x^2-38x^3+8x^4)x\rho^2(5-41x+42x^2-8x^3)+2\rho^3(2-x)}{2(1-x)^2(1-\rho)^2(4x-4x^2-\rho)^2} + \frac{4-8x+8x^2+\rho(3-13x+4x^2+2\rho)}{4(1-x)(1-\rho)^{3/2}} \ln \frac{2x-1+\sqrt{1-4\rho}}{2x-1-\sqrt{1-\rho}} & x > 1 \\ \frac{-3+13x-12x^2-2\rho(2-3x)}{2(1-x)(1-\rho)^2} + \frac{4-8x+8x^2+\rho(3-13x+4x^2+2\rho)}{4(1-x)(1-\rho)^{3/2}} \ln \frac{1+\sqrt{1-4\rho}}{1-\sqrt{1-\rho}} & 0 < x < 1, \\ -\frac{32x^2(1-x)^2(2x-1) - 4x\rho(8-43x+65x^2-38x^3+8x^4)x\rho^2(5-41x+42x^2-8x^3)+2\rho^3(2-x)}{2(1-x)(1-\rho)^2(4x-4x^2-\rho)^2} - \frac{4-8x+8x^2+\rho(3-13x+4x^2+2\rho)}{4(1-x)(1-\rho)^{3/2}} \ln \frac{2x-1+\sqrt{1-4\rho}}{2x-1-\sqrt{1-\rho}} & x < 0 \end{cases}, \quad (16)$$

$$\tilde{f}_p(x, \rho) = \frac{\alpha_s C_F}{2\pi} \begin{cases} \frac{16x\rho(1-x)^2(1-6x) - 2\rho^2(1-22x+26x^2-4x^3) - \rho^3(7-6x)}{2(1-\rho)^2(4x-4x^2-\rho)^2} + \frac{-\rho(8-12x+\rho)}{4(1-\rho)^{3/2}} \ln \frac{2x-1+\sqrt{1-4\rho}}{2x-1-\sqrt{1-\rho}} & x > 1 \\ \frac{2-4x+\rho(7-8x)}{2(1-\rho)^2} + \frac{-\rho(8-12x+\rho)}{4(1-\rho)^{3/2}} \ln \frac{1+\sqrt{1-4\rho}}{1-\sqrt{1-\rho}} & 0 < x < 1, \\ -\frac{16x\rho(1-x)^2(1-6x) - 2\rho^2(1-22x+26x^2-4x^3) - \rho^3(7-6x)}{2(1-\rho)^2(4x-4x^2-\rho)^2} - \frac{-\rho(8-12x+\rho)}{4(1-\rho)^{3/2}} \ln \frac{2x-1+\sqrt{1-4\rho}}{2x-1-\sqrt{1-\rho}} & x < 0 \end{cases}, \quad (17)$$

and

$$\rho = \frac{-p^2 - i\epsilon}{p_z^2} \quad (18)$$

hence remanent dependence on the projection operator is inevitable.

Using the same logic one reaches the conclusion that the quasi-PDF's dependence on the RI/MOM scales μ_R and p_z^R should also be fully cancelled by the matching coefficient. Any fixed-order matching calculation will inevitably lead to a residual μ_R , p_z^R , and P_z dependence of the final result for the PDF. These dependencies should be carefully studied and included in the systematic uncertainties.

B. One-loop matching for quasi-PDF and PDF

To obtain the matching coefficient between the quasi-PDF $\tilde{q}_R(x, P_z, p_z^R, \mu_R)$ and lightcone PDF $q(x, \mu)$, one can calculate the off-shell quark matrix elements in perturbation theory. In the following, the calculation will be conducted in Landau gauge for both minimal and $\not{p}$ projections. See Appendix 1 for the results in a general covariant gauge with a general Lorentz structure.

The lowest order quark quasi-PDF is

$$\tilde{q}^{(0)}(x) = \delta(1-x). \quad (13)$$

with $i\epsilon$ giving the prescription to analytically extrapolate ρ from $\rho < 1$ (Minkowski) to $\rho > 1$ (Euclidean). No-

tion that the vector current conservation guarantees that vertex corrections and wave function contributions can be combined into generalized plus functions [11]. These functions are defined with two arbitrary functions $h(x)$ and $g(x)$:

$$\int dx [h(x)]_+ g(x) = \int dx h(x) [g(x) - g(1)] . \quad (19)$$

For the lightcone PDF with the same off-shell IR regulation in Landau gauge, the tree level contribution is

$$q^{(0)}(x) = \delta(1-x) , \quad (20)$$

and the one-loop correction in the $\overline{\text{MS}}$ scheme is

$$q^{(1)}(x, p, \mu) = -\text{Tr} \left[\left(\left[f_+ \left(x, \frac{\mu^2}{p^2} \right) \right]_+ \gamma^+ + \left[f_p(x) \frac{p^\mu \not{p}}{p^2} \right]_+ \right) \not{p} \right] . \quad (21)$$

Here

$$f_+ \left(x, \frac{\mu^2}{p^2} \right) = \frac{\alpha_s C_F}{2\pi} \theta(x) \theta(1-x) \left[\frac{-5 + 10x - 6x^2}{2(1-x)} + \frac{1+x^2}{1-x} \ln \frac{\mu^2}{-x(1-x)p^2} \right] , \quad (22)$$

$$f_p(x) = \frac{\alpha_s C_F}{2\pi} (1-2x) \theta(x) \theta(1-x) . \quad (23)$$

To match the quasi-PDF to lightcone PDF, one needs to take the on shell limit ($p^2 \rightarrow 0$ or $p \rightarrow 0$) and the large momentum limit ($p_t \rightarrow p_z$) for the bare quasi-PDF

$$\tilde{q}_B^{(1)}(x, \rho) = \tilde{q}^{(1)}(x, p_t \rightarrow p_z, \vec{p}_\perp, p_z, \rho \rightarrow 0) . \quad (24)$$

One can observe that both terms proportional to γ^t and γ^z in Eq. (14) approach lightcone operators in the large momentum limit and the combination of them captures the correct collinear behavior. Therefore the bare quasi-PDF in minimal projection is defined to pick up the coefficient of γ^t and γ^z in Eq. (14):

$$\tilde{q}_B^{(1)}(x, \rho) \Big|_{\text{mp}} = \left[\tilde{f}_t(x, \rho) + \tilde{f}_z(x, \rho) \right] \Big|_{\rho \rightarrow 0} . \quad (25)$$

For the lightcone PDF, the coefficient of γ^+ in Eq. (21) is used for minimal projection:

$$q^{(1)}(x, p, \mu) \Big|_{\text{mp}} = f_+ \left(x, \frac{\mu^2}{p^2} \right) . \quad (26)$$

The bare matching coefficient is then derived as

$$f_{1,\text{mp}} \left(x, \frac{p_z}{\mu} \right) \Big|_+ = \tilde{q}_B^{(1)}(x, \rho) \Big|_{\text{mp}} - q^{(1)}(x, p, \mu) \Big|_{\text{mp}} , \quad (27)$$

where

$$f_{1,\text{mp}} \left(x, \frac{p_z}{\mu} \right) = \frac{\alpha_s C_F}{2\pi} \times \begin{cases} \frac{1+x^2}{1-x} \ln \frac{x}{x-1} + 1 & x > 1 \\ \frac{1+x^2}{1-x} \ln \frac{4x(1-x)p_z^2}{\mu^2} - \frac{x(1+x)}{1-x} & 0 < x < 1 \\ -\frac{1+x^2}{1-x} \ln \frac{x}{x-1} - 1 & x < 0 \end{cases} . \quad (28)$$

In RI/MOM, the quasi-PDF is renormalized with an additional counterterm. We find that in the $|x| \rightarrow \infty$ limit, only $\tilde{f}_t(x, \rho)$ behaves as $1/|x|$. When integrating over x , this term recovers UV divergence in the local limit $z = 0$. Therefore, it is a natural choice to pick up the γ^t term in Eq. (14) as a counterterm:

$$\tilde{q}_{CT}^{(1)} \left(x, r, \frac{p_z}{p_z^*} \right) \Big|_{\text{mp}} = \left[\frac{p_z}{p_z^*} \right] f_{2,\text{mp}} \left(1 + \frac{p_z}{p_z^*} (x-1), r \right) \Big|_+ . \quad (29)$$

Here $r = \mu_R^2/(p_z^*)^2$, and

$$f_{2,\text{mp}}(x, r) = \tilde{f}_t(x, r) = \frac{\alpha_s C_F}{2\pi} \begin{cases} \frac{-3r^2+13rx-8x^2-10rx^2+8x^3}{2(r-1)(x-1)(r-4x+4x^2)} + \frac{-3r+8x-rx-4x^2}{2(r-1)^{3/2}(x-1)} \tan^{-1} \frac{\sqrt{r-1}}{2x-1} & x > 1 \\ \frac{-3r^2+13rx-8x^2-10rx^2+8x^3}{2(r-1)(1-x)} + \frac{3r-8x-rx-4x^2}{2(r-1)^{3/2}(1-x)} \tan^{-1} \sqrt{r-1} & 0 < x < 1 \\ -\frac{3r^2+13rx-8x^2-10rx^2+8x^3}{2(r-1)(x-1)(r-4x+4x^2)} - \frac{3r+8x-rx-4x^2}{2(r-1)^{3/2}(x-1)} \tan^{-1} \frac{\sqrt{r-1}}{2x-1} & x < 0 \end{cases} . \quad (30)$$

Finally, the matching coefficient C in the factorization formula given in Eq.(5) is derived as

$$C\left(x, r, \frac{p_z}{\mu}, \frac{p_z}{p_R^2}\right) = \delta(1-x) + \left[\bar{q}_B^{(1)}(x, \rho) - q^{(1)}(x, p, \mu) - \bar{q}_{CT}^{(1)}\left(x, r, \frac{p_z}{p_R^2}\right) \right] \Big|_{\text{mp}} + \mathcal{O}(\alpha_s^2) \\ = \delta(1-x) + \left[f_{1,\text{mp}}\left(x, \frac{p_z}{\mu}\right) - \left| \frac{p_z}{p_R^2} \right| f_{2,\text{mp}}\left(1 + \frac{p_z}{p_R^2}(x-1), r\right) \right]_+ + \mathcal{O}(\alpha_s^2). \quad (31)$$

Here the coupling $\alpha_s(\mu)$ is in the standard $\overline{\text{MS}}$ scheme. Note that the antiquark distribution is mapped into the region $-1 < y < 0$ by setting $q(y) = -q(-y)$.

For $\not{p}$ projection, one has the bare quasi-PDF

$$\bar{q}_B^{(1)}(x, \rho) \Big|_{\not{p}} = \left[\bar{f}_t(x, \rho) + \bar{f}_z(x, \rho) + \bar{f}_p(x, \rho) \right]_+. \quad (32)$$

The lightcone PDF with a similar projection is

$$q^{(1)}(x, p, \mu) \Big|_{\not{p}} = \left[f_+ \left(x, \frac{\mu^2}{p^2} \right) + f_p(x) \right]_+. \quad (33)$$

Under this projection, the matching coefficient for the bare quasi-PDF coincides with Eq. (27):

$$f_{1,\not{p}}\left(x, \frac{p_z}{\mu}\right)_+ = \bar{q}_B^{(1)}(x, \rho) \Big|_{\not{p}} - q^{(1)}(x, p, \mu) \Big|_{\not{p}} \\ = f_{1,\text{mp}}\left(x, \frac{p_z}{\mu}\right)_+. \quad (34)$$

The counter-term can be obtained by calculating with $\mathcal{P} = \not{p}/(4p^2)$:

$$f_{2,\not{p}}(x, r) = \frac{\alpha_s C_F}{2\pi} \begin{cases} \frac{3-3(r-2s)}{2(r-1)(x-1)} + \frac{4(x-s)^2+3s^2}{(r-4x+4r^2)^2} + \frac{2-2r-rx+2s^2}{(r-1)^2s^2(x-1)} \tan^{-1} \frac{\sqrt{r-1}}{2s-1} & x > 1 \\ \frac{3-3r-2s+4x^2}{2(r-1)(1-x)} + \frac{-2+2r+rx-2s^2}{(r-1)^2s^2(1-x)} \tan^{-1} \frac{\sqrt{r-1}}{2s-1} & 0 < x < 1 \\ \frac{3-3r-2s}{2(r-1)(x-1)} - \frac{4(x-s)^2+3s^2}{(r-4x+4r^2)^2} - \frac{2-2r-rx+2s^2}{(r-1)^2s^2(x-1)} \tan^{-1} \frac{\sqrt{r-1}}{2s-1} & x < 0 \end{cases}. \quad (35)$$

The corresponding RI/MOM matching coefficient is obtained by replacing “mp” with “p” in Eq. (31), and the difference between $f_{2,\not{p}}$ and $f_{2,\text{mp}}$ vanishes in the $p_z^R=0$ limit. The matching coefficient with $\Gamma = \gamma^z$ is also given in Appendix 2.

III. LATTICE CALCULATION OF PDF

A. Lattice Matrix Elements

In this subsection, we give the results of a Lattice-QCD calculation using clover valence fermions on the CLS $32^3 \times 96$ 2+1 flavor clover fermion ensemble H102 with Lattice spacing $a = 0.086$ fm, pion mass $M_\pi = 356$ MeV and box size $L \approx 2.7$ fm ($M_\pi L \approx 4.9$) [77]. We use $\kappa_l = 0.136865$ and $C_{\text{SW}} = 1.98625$ for the valence clover fermion. We apply APE smearing [78] with size=2.5a twice in the source/sink smearing and also in the quasi-PDF operator O_T , but not in the fermion propagators.

First of all, we will explore the nonperturbative renor-

TABLE I. Momentum modes used in the NPR analysis. The four digits (in units of $2\pi/L$) in brackets correspond to three spatial momentum components and the energy component. The z direction can be selected by setting the Wilson link along any of the three spatial directions. Thus these momentum modes approximately cover three choices of $\mu_R = \sqrt{-(p^R)^2}$ and several sets of $p_R^z = 2\pi i/L (i = 0, 1, 2, \dots)$.

$\alpha^2 \mu_R^2$	Momentum modes p^R
[1.109, 1.118]	(5,2,0,0) (4,3,1,5)
[1.957, 1.966]	(5,5,0,3) (6,3,2,4) (5,4,1,9)
[2.814, 2.832]	(6,5,1,10) (7,4,2,6) (6,3,0,16)

malization in RI/MOM scheme. Following Ref. [39], we use Landau gauge fixed wall sources (while limiting the source in the time slice range $t \in [32, 64]$ to avoid the boundary effect from open boundary condition at $t=0$), and generate the propagators with the momentum modes listed in Table. I. The four digits in brackets correspond to three spatial component and the energy component in units of $2\pi/L$. The z direction can be selected by setting

the Wilson link along any of the three spatial directions. Thus these results approximately cover three values of $\mu_R = \sqrt{-(p^R)^2}$ (2.4, 3.2 and 3.9 GeV, corresponding to $a^3\mu_R^3=1.1, 2.0$ and 2.8), and $p_z^R = \{0, 1, 2, \dots\} \times 2\pi/L$ up to the upper limit $p_z^R < \mu_R$. Note that in deriving these momentum modes we have required the spatial components of a given momentum mode different with each other, and adjusted p_z^R to ensure μ_R invariant (within 2%). These choices allow us to explore the dependence on each component of p , but one should be cautious that the results may suffer from sizable discretization errors since the normal constraint $\frac{\sum_{\mu} p_{\mu}^2}{\sum_{\mu} a^2 p_{\mu}^2} < 0.3$ is not respected. These discretization errors will be investigated in the future.

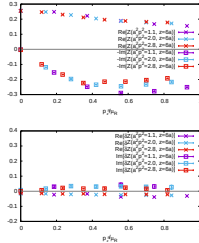


FIG. 1. The NPR $Z = Z_{top}$ (top) and $\delta Z = Z_g - Z_{top}$ (bottom) at $z = 6a$ (≈ 0.5 fm) as a function of $p_z^R/\mu_R = 1/\sqrt{r}$, with various μ_R . At $p_z^R = 0$, Z is real and $\delta Z/Z$ is less than 5%.

As shown in Eq. (31), the one-loop matching formula primarily depends on the combination $r = \mu_R^2/(p_z^R)^2$ but is independent of p_z^R . In Fig. 1, the Z_{top} and $Z_g - Z_{top}$ at fixed $z \sim 0.5$ fm are plotted as a function of $p_z^R/\mu_R = 1/\sqrt{r}$. From this figure, one can see that the NPR factors, both real and imaginary parts, only show the dependence on r regardless of the values of p_z^R or μ_R , with $p_z^R/\mu_R < 0.4$.

In Fig. 2, we show $1/Z_{top}$ and $1/Z_g - 1/Z_{top}$ as functions of the Wilson link length z , with the same $\mu_R=3.2$ GeV and $p_z^R = 1.4$ GeV and two different values of

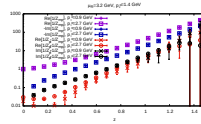


FIG. 2. The inverse of the minimum projection renormalization factor $1/Z_{top}$ and the difference $1/Z_g - 1/Z_{top}$, as a function of z with the same p_z^R and μ_R , but different p_z^R . The crosses correspond to $p_z^R = 0.9$ GeV and the open boxes (circles) are for $p_z^R = 2.7$ GeV. Most results show mild dependence on p_z^R .

$p_z^R=0.9$ GeV and 2.7 GeV. As shown in this figure, the $1/Z_{top}$ and $1/Z_g - 1/Z_{top}$ with the two different p_z^R 's are close to each other for all z (the curves with the same color). This is consistent with the 1-loop matching formula. At $z < 0.3$ fm, the real part of $1/Z_g - 1/Z_{top}$ with the two different p_z^R 's can be slightly nonzero, but it is still smaller than $1/Z_g$ by two orders of magnitude.

In the following we will take p_z^R to be zero and estimate the systematic uncertainty from the p_z^R dependence by varying the p_z^R .

In the calculation of nucleon matrix element, we use Gaussian momentum smearing [79] for the quark field

$$\psi(x) \rightarrow S_{\text{smear}} \psi(x) = \frac{1}{1 + 6\alpha} \left[\psi(x) + \alpha \sum_j U_j^{APE}(x) e^{ikj} \psi(x + \hat{e}_j) \right], \quad (36)$$

where k is the desired momentum, $U_j^{APE}(x)$ are the APE smeared gauge links in the j direction, and α is a tunable parameter as in traditional Gaussian smearing.

Such a momentum source is designed to increase the overlap with nucleus of the desired boost momentum and we are able to reach higher-boosted momentum for the nucleon states than in the previous work [39]. Although in the exploratory study, we varied the Gaussian smearing radius to better overlap with the largest momentum used in the calculation, the field smearing is still centered around zero momentum in momentum space. When we switch to the momentum smearing, the smearing center will be shifted to momentum $O(k)$, which will immediately allow us to reach higher boost momenta with better signal-to-noise ratios in the matrix elements. In this work, we use two values of nucleon boost momenta, $P_z = n \frac{\mu_R}{2}$, with $n \in \{4, 5\}$, which corresponds to 1.8 and 2.3 GeV.

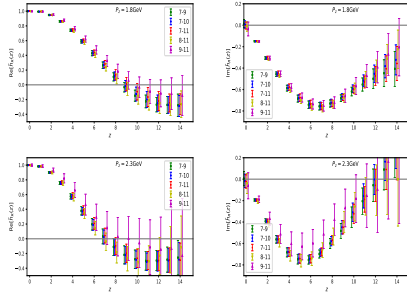


FIG. 3. The real (left) and imaginary (right) parts of the isovector nucleon matrix elements for unpolarized PDFs as functions of z at different momenta, with $P_z = \frac{M_N}{2} \approx 1.8$ GeV (top) and $\frac{M_N}{2} \approx 2.3$ GeV (bottom) respectively. The RI/MOM renormalization factors with $[\mu, p_0^2] = [3.2, 0]$ GeV and the normalization at $z = 0$ are applied on the bare matrix elements to improve the visibility at large z . At a given positive z value, the data is slightly offset to show the ground-state matrix element from the fits using different ranges: from left to right they are: $t_{\text{sep}} \in [7, 9]$, $[7, 10]$, $[7, 11]$, $[8, 11]$, and $[9, 11]$. Different analyses are consistent within statistical errors while the fits with separation 7 and 8 have smaller uncertainties compared to other cases.

On the Lattice, we calculate the time-independent and nonlocal in z direction correlators of a nucleon with finite- P_z boost

$$\tilde{h}_{\text{bkt}}(z, P_z, \Gamma; a^{-1}) = \langle 0; \vec{P} | O_T(z) | 0; \vec{P} \rangle. \quad (37)$$

Here the state $|0; \vec{P}\rangle$ represents the ground (nucleon) state with momentum $\vec{P} = \{0, 0, P_z\}$. $\Gamma = \gamma^t$ is used for the unpolarized parton distribution.

As the nucleon boost momentum increases, one anticipates that excited-state contributions are more severe; therefore, a careful study of the excited-state contamination is necessary. To do so, we calculate the nucleon matrix element $\tilde{h}_{\text{bkt}}$ at five source-sink separations $t_{\text{sep}} \in \{7, 8, 9, 10, 11\} \times 0.086$ fm, with $\{4, 4, 8, 8, 16\}$ measurements on each of 2005 gauge configurations respectively in the $P_z = 1.8$ GeV case, and of 2,000 configurations in the $P_z = 2.3$ GeV case. We use a multi-

grid algorithm [80, 81] with the Chroma software package [82] to speed up the inversion of the quark propagator. Following Ref. [83], each three-point (3pt) correlator $C^{\text{3pt}}(t, t_{\text{sep}})$ can be decomposed as (assuming the source is at $t = 0$)

$$\begin{aligned} C^{\text{3pt}}(t, t_{\text{sep}}; P_z, \Gamma) &= \mathcal{A}_0^2(0) \langle O_T | 0 \rangle e^{-E_0 t t_{\text{sep}}} \\ &+ \mathcal{A}_1^2(1) \langle O_T | 1 \rangle e^{-E_1 t t_{\text{sep}}} \\ &+ \mathcal{A}_1 \mathcal{A}_0(1) \langle O_T | 0 \rangle e^{-E_1(t_{\text{sep}}-t)} e^{-E_0 t} \\ &+ \mathcal{A}_0 \mathcal{A}_1(0) \langle O_T | 1 \rangle e^{-E_0(t_{\text{sep}}-t)} e^{-E_1 t} + \dots, \end{aligned} \quad (38)$$

where $|n\rangle$ with $n > 0$ represents the excited states. The operator is inserted at time t , and the nucleon state is annihilated at the sink time t_{sep} (which is also the source-sink separation). The spectrum weights $\mathcal{A}_{0,1}$ and energies $E_{0,1}$ in Eq. (39) can be obtained from the two-point

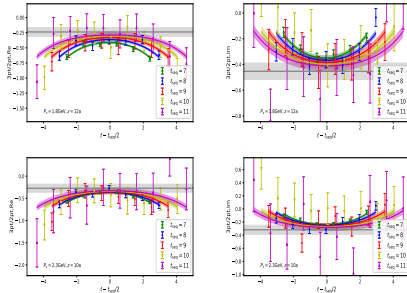


FIG. 4. The real (left) and imaginary (right) parts of the renormalized isovector nucleon matrix elements for unpolarized PDFs with $P_z, z = 8\pi/L, 12$ (top) and $10\pi/L, 10$ (bottom) which correspond to $zP_z \sim 9.5$. The data points and the band predicted by the fit using $t_{\text{sep}} \in [7, 11]$ agree with each other well.

(2pt) correlator:

$$C^{2pt}(t_{\text{sep}}; P_z) = A_0^2 e^{-E_0 t_{\text{sep}}} + A_1^2 e^{-E_1 t_{\text{sep}}} + \dots \quad (39)$$

Eventually we apply the joint fit with the 3pt functions at several t_{sep} and 2pt function using the following form [83]:

$$\begin{aligned} \frac{C^{3pt}(t, t_{\text{sep}})}{C^{2pt}(t_{\text{sep}})} &= \\ &= \frac{\tilde{h}_{\text{int}} + C_2(e^{-\Delta E t} + e^{-\Delta E(t_{\text{sep}}-t)}) + C_3 e^{-\Delta E t_{\text{sep}}}}{1 + C_1 e^{-\Delta E t_{\text{sep}}}}, \end{aligned} \quad (40)$$

with $\Delta E = E_1 - E_0$. $C_{0,1,2,3}$ and $E_{0,1}$ are free parameters. We limit the range of t as $t \in [1, t_{\text{sep}} - 1]$ for 3pt/2pt ratio and $t \in [7, 11]$ for the 2pt to make the $\chi^2/d.o.f.$ of the fit to be $\mathcal{O}(1)$. Using the ratio of 3pt/2pt instead of the 3pt function itself can improve the stability of the fit, especially when $C^{2pt}(t)$ with $t < 7$ is included in the fit.

In Fig. 3, we show the ground-state nucleon matrix elements $h_{\text{int}}(z, P_z, \gamma_t)$ obtained from five fits: using the

separations $t_{\text{sep}} \in [7, 9], [7, 10], [7, 11], [8, 11]$, and $[9, 11]$ (The data points correspond to the same z but are shifted horizontally to enhance the visibility). The data are further normalized by multiplying the renormalization factor with $(\mu_R, p^R) = \{3.2, 0\}$ GeV and the real part normalized to 1 at $z = 0$. From this figure, one can see that there is no clear signal for excited-state contributions in any of these analyses. If the data with smallest two separations are dropped, uncertainties are getting much larger. In the fit, we keep the C_3 term to make a moderate estimate of the uncertainty even when this term is not statistically significant.

For a comparison between data and the fit, we show our results at large z like $(P_z, z) = (8\pi/L, 12a)$ and $(10\pi/L, 10a)$ with $t_{\text{sep}} \in [7, 11]$ in Fig. 4. In these spatial separations, the real part of matrix element seems to be negative. The ground-state contribution obtained from the fit is shown as the black band. As one can see, most data can be well described in the fit and thereby we use the two-state fits and the interval $t_{\text{sep}} \in [7, 11]$ to obtain the results in the rest part of this paper.

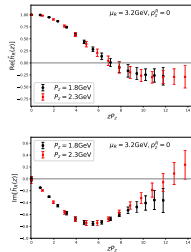


FIG. 5. The renormalized quasi-PDF matrix elements with $P_z = 1.8$ and 2.3 GeV, using the minimal projection with $p_z^R = 0$ and $\mu_R = 3.2$ GeV, as function of zP_z .

The renormalized quasi-PDF matrix elements with two values of P_z are plotted in Fig. 5, as function of zP_z for $p_z^R = 0$ and $\mu_R = 3.2$ GeV. The results with different P_z are consistent with each other within statistical uncertainties. This indicates that power corrections due to higher-twist effects might not be sizable.

B. Systematic Uncertainties

In this subsection, we will consider four systematic uncertainties from: Fourier transformation (FT), unphysical scales p_z^R and μ_R , projection used in the RI/MOM scheme, and inversion of the matching coefficient.

In the following, we explain the details to include these systematic uncertainties.

1) Fourier transformation. As shown in Fig. 5, the $\tilde{h}_R(z)$ with $P_z = 2.3$ GeV is consistent with zero when $z > 12a$. Thus in the standard matching from quasi-PDF to PDF, it is reasonable to truncate the results at $z = 12a$. With this spirit, the quasi-PDF and matched PDF using the standard FT are shown in the upper panel of Fig. 7, from which one can see the matched PDF shows an oscillatory behavior. A “derivative” method was proposed in Ref. [41] to cure this oscillatory behavior. To

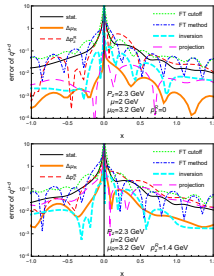


FIG. 6. Different contributions to the systematic errors. See the text for detailed information.

be concrete, one takes the derivative of the renormalized nucleon matrix elements $\partial_z \tilde{h}_R(z)$, whose Fourier transform differs from the original matrix element in a known way:

$$\tilde{q}_R(x) = \int_{-\infty}^{\infty} \frac{dz}{2\pi} \frac{ie^{ixP_z z}}{x} \partial_z \tilde{h}_R(z), \quad (41)$$

provided that $\tilde{h}_R(z)$ goes to zero as $|z| \rightarrow \infty$. With the same truncation, the result is shown in the lower panel of Fig. 7 and apparently the oscillatory behavior is less severe. Besides, results obtained using the derivative method is consistent with the standard FT method in most kinematics region except at small x . This is anticipated as two methods only differ at the large z region where we have made the truncation. We show the difference as the dot-dashed-blue line in Fig. 6, together with the error from varying the truncation from $z = 10a$ to $14a$ (dotted-green line).

2) Unphysical scales p_z^R and μ_R . There are two unphysical scales p_z^R and μ_R introduced in RI/MOM. In principle, when matching the quasi-PDF matrix element onto lightcone PDF, the dependence on these two scales in the matrix element should exactly cancel with that in the matching kernel. However, since the quasi-PDF matrix element is non-perturbatively renormalized on the

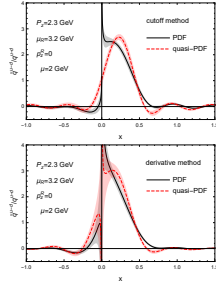


FIG. 7. The quasi-PDF (dashed-red) with nucleon boost momentum 2.3 GeV and matched PDF (solid-black) at $\mu = 2$ GeV using minimal projection, with the RI/MOM parameters $p_z^R = 0$ and $\mu_R = 3.2$ GeV. The upper and lower figures are obtained using the derivative and cutoff methods to perform Fourier transformation. The matching strategy has an important impact on the final results for PDFs.

Lattice, while the matching coefficient is calculated at one-loop order in perturbation theory, there will be residual dependence on these two scales after the perturbative matching. To estimate the residual p_z^R and μ_R dependence, we choose $p_z^R = 0$ GeV and $\mu_R = 3.2$ GeV as the central value, and vary p_z^R from -1.4 to 1.4 GeV (dashed-red line) and μ_R from 2.4 to 3.9 GeV (thick-solid-orange line). The difference between these matched PDFs is treated as the systematics of the residual dependence on unphysical scales, in Fig. 6. As shown in the figure, the systematic uncertainty due to the μ_R dependence is small compared to the other sources, but the residual p_z^R dependence could be sizable.

3) Dependencies on the projection. There are two projections discussed in this work: the minimum and “p” projections. With $p_z^R = 0$, the projection dependence in both the NPR factor are less than 5% for all the z , and vanishes in the 1-loop perturbative matching. Thus one can expect that the difference due to the projections is also small, as depicted in the long-dashed-magenta lines

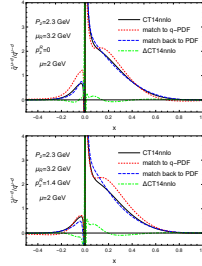


FIG. 8. Effects of inversion matching formula using minimal projection: The solid-black, dotted-red, dotted-blue, and dot-dashed-green lines represent CT14nnlo PDF, applying inverse matching from CT14nnlo PDF [84] to quasi-PDF, applying matching again to get back to the PDF, the difference between PDF with iterative matching and the original CT14nnlo PDF. The upper (lower) figure corresponds to $p_z^R = 0$ (1.4 GeV). These plots show that the method we used to invert the matching formula is less reliable for small $|x|$. The difference shown by the dot-dashed-green curve is taken into account into our systematic error.

in Fig. 6.

4) Inversion of matching. To extract the PDF from the quasi-PDF, one needs to invert the factorization formula Eq. (5). This could be done by changing the sign of α_s in C , and convoluting the new matching coefficient with $\bar{q}$. More explicitly, we have

$$q(x, \mu) = \int_{-\infty}^{\infty} \frac{dy}{|y|} C' \left(\frac{x}{y}, r, \frac{yP_z}{\mu}, \frac{yP_z}{p_z^R} \right) \bar{q}(y, P_z, p_z^R, \mu_R) + \mathcal{O} \left(\frac{M^2}{P_z^2}, \frac{\Lambda_{\overline{\text{MS}}}^2}{x^2 P_z^2}, \alpha_s^2 \right), \quad (42)$$

where $C' = C(\alpha_s \rightarrow -\alpha_s)$. We estimate the error due to inverting the factorization formula by starting from the PDF from a global analysis [84], applying Eq.(5) and then Eq. (42) to return to the PDF. This manipulation should give the same lightcone PDF. However, since the matching is only accurate up to $\mathcal{O}(\alpha_s)$, the two results would

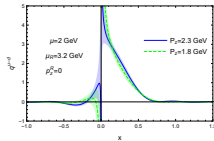


FIG. 9. Nucleon boost momentum dependence of the matched unpolarized isovector PDFs: the dotted-green and solid-blue lines correspond to the nucleon momentum P_z to be 1.8 and 2.3 GeV, respectively. The cutoff of Fourier transformation is chosen to be $zP^+ \sim 12$ ($z = 15a$ for $P_z = 1.8$ GeV and $z = 12a$ for $P_z = 2.3$ GeV).

differ, and the difference gives a good estimate of the systematic error coming from the inversion and higher order corrections. This is shown in Fig. 8, from which one can find that the error only becomes sizable when $|x|$ is small. This is expected because the relevant momentum scale is xP^+ such that higher order corrections become large at small x .

There are more sophisticated methods to invert the factorization, such as using a recursion procedure. However, as we can see in Fig. 6, the systematic error caused by the matching procedure (thick-dashed-cyan line) is also smaller than those from the first two sources in most regions.

As shown in Fig. 6, one can find that the dominant uncertainties arise from the p_z^R dependence and from FT. With $p_z^R = 0$, the uncertainties from different projections, and inversion and the matching are typically less than 10% except in the region with very small $|x|$. The uncertainty from the μ_R dependence is even smaller. With $p_z^R = 1.4$ GeV, uncertainties from these three sources are getting larger in magnitude, but still smaller than those from the two major sources.

IV. FINAL RESULTS FOR PDF

With the derivative method of FT and the matching using $p_z^R = 0$ GeV and $\mu_R = 3.2$ GeV, we show the dependence on the nucleon boosted momentum in Fig. 9 with the statistical uncertainties. They are consistent with each other as we can expect from the consistency of the quasi-PDF matrix element results in Fig. 5.

Finally, we show our results for PDF and a comparison with global-analysis [84–86] in Fig. 10. As can be seen from the plot, our results show a reasonable agreement

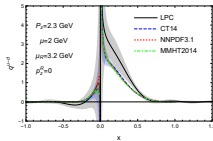


FIG. 10. Results for PDF at $\mu = 2$ GeV calculated from RI/MOM quasi-PDF at nucleon momentum $P_z = 2.3$ GeV: Comparing with CT14nnlo (90CL) [84], NNPDF3.1 (68CL) [85], and MMHT2014 (68CL) [86]. Our results agree with the global-analysis within uncertainties.

in the large- x region, but at small- x region there exists notable difference majorly due to the systematic uncertainties from the FT truncation method and also the p_z^R dependence.

V. SUMMARY

In this paper, we have studied the quasi-PDF defined with γ^t which is free from mixing at $O(a^0)$. We have used $M_\pi = 356$ MeV Lattice data to demonstrate the matching procedure and show that the excited state contamination is well under control. The one-loop matching coefficient is calculated and we have discussed the sources of systematic errors as well as the choice of the projection in detail.

We have found that the systematic uncertainties from the FT truncation method and also the p_z^R dependence are sizable. But those uncertainties from μ_R , inversion of matching and choice of projection are relatively minor with $p_z^R = 0$. At the same time, the significant change from quasi-PDF to matched PDF suggests that higher-loop corrections are needed as exhibited in Fig. 7.

Controlling systematic uncertainty from the excited state is very challenging since the relative uncertainty grows very fast when either source-sink separation t_{sep} or nucleon momentum P_z become large. The two-state fit with smaller separation provides a possibility to obtain a precise result in small $t_{\text{sep}} < 1\text{fm}$ region, while for an accurate measurement at large separation using very high statistics, estimating the systematic uncertainty of such a fit is still needed.

Besides the uncertainties that we have studied, in the future we plan to investigate other systematics such as Lattice discretization and finite volume effects [87] as well as higher twist contributions that affect the small- x re-

sult. The latter can be improved with larger nucleon momentum and estimated by extrapolating to infinite nucleon momentum.

Our final result for lightcone PDF agrees with the global analysis in the large- x region, which gives an encouraging signal that LaMET may allow us to precisely access parton physics in the future.

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APPENDIX

1. One-loop quasi-PDF with γ^α in general covariant gauge

The gluon propagator in the general covariant gauge is

$$iD_{\mu\nu}^g(k) = -\frac{i}{k^2} \left[g^{\mu\nu} - (1-\tau) \frac{k^\mu k^\nu}{k^2} \right]. \quad (43)$$

For general $\Gamma = \gamma^\alpha$, the one-loop result can be expressed as

$$\bar{q}^{(1)}(x, p, \rho) = \text{Tr} \left[\left(\left[\bar{f}_\alpha(x, \rho) \right]_+ \gamma^\alpha + \left[\bar{f}_z(x, \rho) \right]_+ \frac{p_\alpha \not{p}}{p^2} + \left[\bar{f}_\rho(x, \rho) \right]_+ \frac{p_\alpha \not{p}}{p^2} \right) \mathcal{P} \right], \quad (44)$$

where

$$\begin{aligned} \bar{f}_\alpha(x, \rho) = & \frac{\alpha_s C_F}{2\pi} \begin{cases} \frac{x-\rho}{(1-x)(1-\rho)} + \frac{2x(2-x)-\rho(1+x)}{2(1-x)(1-\rho)^{3/2}} \ln \frac{2x-1+\sqrt{1-\rho}}{2x-1-\sqrt{1-\rho}} & x > 1 \\ \frac{-3x+2x^2+\rho}{(1-x)(1-\rho)} + \frac{2(2-x)-\rho(1+x)}{2(1-x)(1-\rho)^{3/2}} \ln \frac{2x-1+\sqrt{1-\rho}}{2x-1-\sqrt{1-\rho}} & 0 < x < 1 \\ -\frac{x-\rho}{(1-x)(1-\rho)} - \frac{2x(2-x)-\rho(1+x)}{2(1-x)(1-\rho)^{3/2}} \ln \frac{2x-1+\sqrt{1-\rho}}{2x-1-\sqrt{1-\rho}} & x < 0 \end{cases} \\ & + \frac{\alpha_s C_F}{2\pi} (1-\tau) \begin{cases} \frac{\rho(-3x+2x^2+\rho)}{2(1-x)(1-\rho)(4x-4x^2-\rho)} + \frac{-\rho}{4(1-\rho)^{3/2}} \ln \frac{2x-1+\sqrt{1-\rho}}{2x-1-\sqrt{1-\rho}} & x > 1 \\ \frac{-x+\rho}{2(1-x)(1-\rho)} + \frac{-\rho}{4(1-\rho)^{3/2}} \ln \frac{2x-1+\sqrt{1-\rho}}{2x-1-\sqrt{1-\rho}} & 0 < x < 1 \\ -\frac{\rho(-3x+2x^2+\rho)}{2(1-x)(1-\rho)(4x-4x^2-\rho)} - \frac{-\rho}{4(1-\rho)^{3/2}} \ln \frac{2x-1+\sqrt{1-\rho}}{2x-1-\sqrt{1-\rho}} & x < 0 \end{cases}, \quad (45) \end{aligned}$$

$$\begin{aligned}
\tilde{f}_2(x, \rho) &= \frac{\alpha_s C_F}{2\pi} \begin{cases} -\frac{2\rho(1-7x+6x^2)-\rho^2(1+2x)}{(1-\rho)^2(4x-4x^2-\rho)} g_{2\alpha} + \frac{4x(1-3x+2x^2)-\rho(2-11x+12x^2-4x^3)-\rho^2}{(1-x)(1-\rho)^2(4x-4x^2-\rho)} \ln \frac{2x-1+\sqrt{1-\rho}}{2x-1-\sqrt{1-\rho}} & x > 1 \\ +\frac{\rho(4-6x-\rho)}{2(1-\rho)^{3/2}} g_{2\alpha} + \frac{2-4x+4x^2-5x\rho+2x^2\rho+\rho^2}{2(1-x)(1-\rho)^{3/2}} \ln \frac{1+\sqrt{1-\rho}}{1-\sqrt{1-\rho}} & 0 < x < 1 \\ -\frac{2+2x-\rho(1-4x)}{1-\rho} g_{2\alpha} + \frac{1-(1-x)\rho^2}{2(1-x)(1-\rho)^2} \ln \frac{1+\sqrt{1-\rho}}{1-\sqrt{1-\rho}} & x < 0 \end{cases} \\
+ \frac{\alpha_s C_F}{2\pi} (1-\tau) &\begin{cases} -\frac{\rho(4-6x-\rho)}{2(1-\rho)^2} g_{2\alpha} + \frac{2-4x+4x^2-5x\rho+2x^2\rho+\rho^2}{2(1-x)(1-\rho)^2} \ln \frac{1+\sqrt{1-\rho}}{1-\sqrt{1-\rho}} & x > 1 \\ -\frac{2\rho(1-7x+6x^2)-\rho^2(1+2x)}{(1-\rho)^2(4x-4x^2-\rho)} g_{2\alpha} - \frac{4x(1-3x+2x^2)-\rho(2-11x+12x^2-4x^3)-\rho^2}{(1-x)(1-\rho)^2(4x-4x^2-\rho)} \ln \frac{2x-1+\sqrt{1-\rho}}{2x-1-\sqrt{1-\rho}} & 0 < x < 1 \\ -\frac{\rho(4-6x-\rho)}{2(1-\rho)^2} g_{2\alpha} + \frac{2-4x+4x^2-5x\rho+2x^2\rho+\rho^2}{2(1-x)(1-\rho)^2} \ln \frac{1+\sqrt{1-\rho}}{1-\sqrt{1-\rho}} & x < 0 \end{cases} \\
\tilde{f}_p(x, \rho) &= \frac{\alpha_s C_F}{2\pi} \begin{cases} -\frac{\rho(1-2x)}{2(1-\rho)^2(4x-4x^2-\rho)^2} g_{2\alpha} + \frac{\rho(4x(2-9x+6x^2)+\rho(1-10x+2\rho))}{2(1-\rho)^2(4x-4x^2-\rho)^2} \ln \frac{1+\sqrt{1-\rho}}{1-\sqrt{1-\rho}} & x > 1 \\ +\frac{\rho(2+\rho)(q_{\alpha}+3)}{4(1-\rho)^{3/2}} \ln \frac{1+\sqrt{1-\rho}}{1-\sqrt{1-\rho}} & 0 < x < 1 \\ -\frac{3q_{\alpha}+1-2\rho}{2(1-\rho)^2} + \frac{\rho(2+\rho)(q_{\alpha}+3)}{4(1-\rho)^{3/2}} \ln \frac{1+\sqrt{1-\rho}}{1-\sqrt{1-\rho}} & x < 0 \end{cases} \\
\tilde{f}_p(x, \rho) &= \frac{\alpha_s C_F}{2\pi} \begin{cases} -\frac{4x\rho(3-5x+2x^2)+\rho^2(4-3x+4x^2-4x^3)-\rho^3}{(1-x)(1-\rho)^2(4x-4x^2-\rho)} g_{2\alpha} + \frac{-2\rho\rho(5-6x)+\rho^2(3-2x)}{(1-\rho)^2(4x-4x^2-\rho)} \ln \frac{2x-1+\sqrt{1-\rho}}{2x-1-\sqrt{1-\rho}} & x > 1 \\ +\left[\frac{\rho(2+\rho)(q_{\alpha}+3)}{2(1-\rho)^2} + \frac{-2x+3\rho-4x\rho}{2(1-x)(1-\rho)} g_{2\alpha} + \frac{-\rho(2-6x+\rho)}{2(1-\rho)^2} \ln \frac{1+\sqrt{1-\rho}}{1-\sqrt{1-\rho}}\right] & 0 < x < 1 \\ +\left[\frac{-2\rho(3-5x+2x^2)+\rho^2(4-3x+4x^2-4x^3)-\rho^3}{(1-x)(1-\rho)^2(4x-4x^2-\rho)} g_{2\alpha} - \frac{2\rho\rho(5-6x)+\rho^2(3-2x)}{(1-\rho)^2(4x-4x^2-\rho)} \ln \frac{2x-1+\sqrt{1-\rho}}{2x-1-\sqrt{1-\rho}}\right] & x < 0 \end{cases} \\
+ \frac{\alpha_s C_F}{2\pi} (1-\tau) &\begin{cases} -\frac{16\rho\rho(1-3x+2x^2)+4x\rho^2(3-2x)-\rho^3(5-2x)+2\rho}{2(1-\rho)^2(4x-4x^2-\rho)^2} g_{2\alpha} + \frac{\rho(1-2x)[16x(1-x)-2\rho(1+2x-2x^2)-\rho^2]}{2(1-\rho)^2(4x-4x^2-\rho)^2} \ln \frac{1+\sqrt{1-\rho}}{1-\sqrt{1-\rho}} & x > 1 \\ +\frac{\rho(5-2\rho)(q_{\alpha}+2+2\rho)}{2(1-\rho)^2} + \frac{\rho(4-\rho)(q_{\alpha}+1)}{4(1-\rho)^{3/2}} \ln \frac{1+\sqrt{1-\rho}}{1-\sqrt{1-\rho}} & 0 < x < 1 \\ -\frac{16\rho\rho(1-3x+2x^2)+4x\rho^2(3-2x)-\rho^3(5-2x)+2\rho}{2(1-\rho)^2(4x-4x^2-\rho)^2} g_{2\alpha} - \frac{\rho(1-2x)[16x(1-x)-2\rho(1+2x-2x^2)-\rho^2]}{2(1-\rho)^2(4x-4x^2-\rho)^2} \ln \frac{1+\sqrt{1-\rho}}{1-\sqrt{1-\rho}} & x < 0 \end{cases}
\end{aligned}
\tag{46}$$

2. One-loop quasi-PDF with $\Gamma = \gamma^*$ in Landau gauge

For $\Gamma = \gamma^*$, we can obtain the matching coefficient Eq. (31) using the general formula with similar definition of the minimal and $\not{p}$ projections in Sec. II B. The bare matching coefficients are

$$f_{1,p} \left(x, \frac{p_+}{\mu} \right) = f_{1,\not{p}} \left(x, \frac{p_+}{\mu} \right) = \frac{\alpha_s C_F}{2\pi} \begin{cases} \frac{1+x^2}{1-x} \ln \frac{x}{x-1} + 1 & x > 1 \\ \frac{1+x^2}{4x(1-x)p^2} + \frac{2-5x+x^2}{1-x} & 0 < x < 1 \\ -\frac{1+x^2}{1-x} \ln \frac{x}{x-1} - 1 & x < 0 \end{cases} \tag{48}$$

and the corresponding counterterms are

$$f_{2,np}(x, r) = \frac{\alpha_s C_F}{2\pi} \begin{cases} \frac{3r-(1-2x)^2}{2(r-1)(1-x)} - \frac{4x^2(2-3r+2x+4rx-12x^2+8x^3)}{(r-1)(r-4x+4x^2)} + \frac{2-3r+2x^2}{(r-1)^{1/2}(x-1)} \tan^{-1} \frac{\sqrt{r-1}}{2x-1} & x > 1 \\ \frac{1-3r+4x^2}{2(r-1)(1-x)} + \frac{2+3r+2x^2}{(r-1)^{1/2}(1-x)} \tan^{-1} \sqrt{r-1} & 0 < x < 1 \\ -\frac{3r-(1-2x)^2}{2(r-1)(1-x)} + \frac{8x^2(2-3r+2x+4rx-12x^2+8x^3)}{(r-1)(r-4x+4x^2)} - \frac{2-3r+2x^2}{(r-1)^{1/2}(x-1)} \tan^{-1} \frac{\sqrt{r-1}}{2x-1} & x < 0 \end{cases} \tag{49}$$

$$f_{2,p}(x, r) = \frac{\alpha_s C_F}{2\pi} \begin{cases} 1 + \frac{r(3-4x)-8x(x-1)^2}{2(x-1)(r-4x+4x^2)} + \frac{2+3r-2x^2}{\sqrt{r-1}(x-1)} \tan^{-1} \frac{\sqrt{r-1}}{2x-1} & x > 1 \\ \frac{1+3r}{2(1-x)} + \frac{2+3r+2x^2}{\sqrt{r-1}(1-x)} \tan^{-1} \sqrt{r-1} & 0 < x < 1 \\ -1 - \frac{r(3-4x)-8x(x-1)^2}{2(x-1)(r-4x+4x^2)} - \frac{2+3r-2x^2}{\sqrt{r-1}(x-1)} \tan^{-1} \frac{\sqrt{r-1}}{2x-1} & x < 0 \end{cases} \tag{50}$$

The result was also calculated in Ref. [11]. Although the matching coefficient with $\Gamma = \gamma^2$ is not useful for isovector unpolarized PDF because it suffers from operator mixing in renormalization procedure, it can be used for isovector helicity PDF due to different symmetry properties.

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